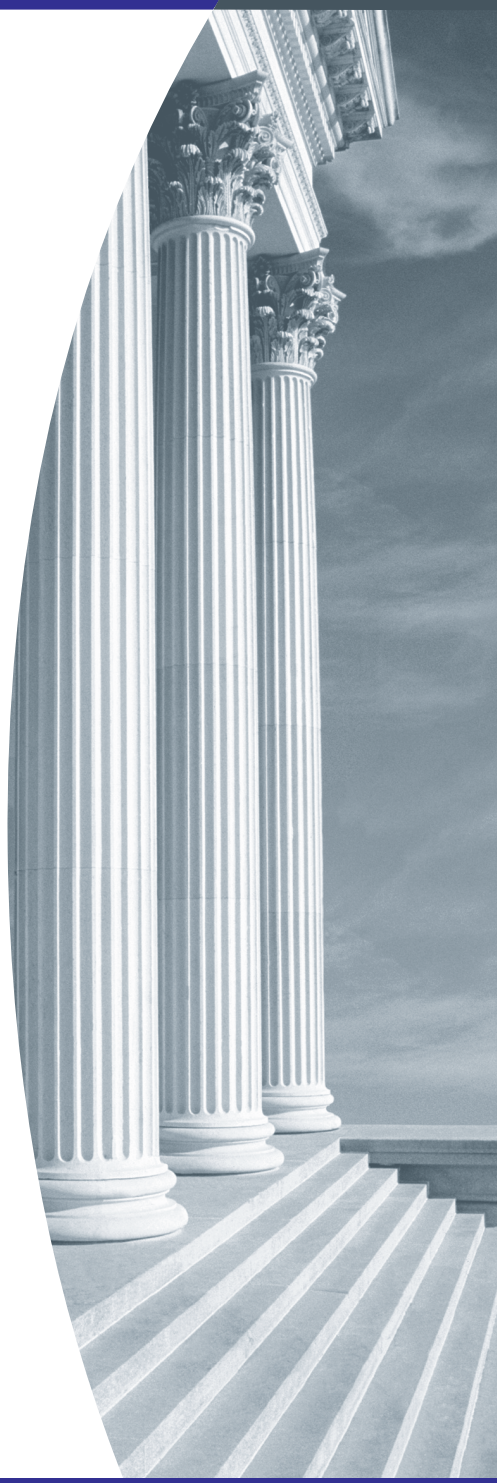
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Pure and Applied UNDERGRADUATE TEXTS 17



Invitation to Classical Analysis

Peter Duren

American Mathematical Society

Invitation to Classical Analysis



Pure and Applied
UNDERGRADUATE TEXTS · 17

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Peter Duren



American Mathematical Society
Providence, Rhode Island

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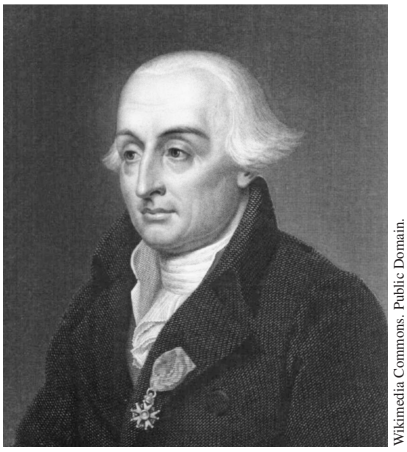
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Karl Weierstrass

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Preface

This is a book for undergraduates. To be more precise, it is designed for students who have learned the basic principles of analysis, as taught to undergraduates in advanced calculus courses, and are prepared to explore substantial topics in classical analysis. And there is much to explore: Fourier series, orthogonal polynomials, Stirling's formula, the gamma function, Bernoulli numbers, elliptic integrals, Bessel functions, Tauberian theorems, etc. Yet the modern undergraduate curriculum typically does not encompass such topics, except perhaps by way of physical applications. In effect the student struggles to master abstract concepts and general theorems of analysis, then is left wondering what to do with them.

It was not always so. Around 1950 the typical advanced calculus course in American colleges contained a selection of concrete topics such as those just mentioned. However, the development could not be entirely rigorous because the underlying theory of calculus had been deferred to graduate courses. To remedy this unsatisfactory state of affairs, the theory of calculus was moved to the undergraduate level. Textbooks by Walter Rudin and Creighton Buck helped transform advanced calculus to a theoretical study of basic principles. Certainly much was gained in the process, but also much was lost. Various concrete topics, natural sequels to the abstract theory, were crowded out of the curriculum.

The purpose of this book is to recover the lost topics and introduce others, making them accessible at the undergraduate level by building on the theoretical foundation provided in modern advanced calculus courses. My aim has been to develop the mathematics in a rigorous way while holding the prerequisites to a minimum. The exposition probes rather deeply into each topic and is at times intellectually demanding, but every effort has

been made to include the background necessary for full comprehension. It is hoped that undergraduate students (and other readers) will find the material exciting and will be inspired to make further studies in the realm of classical analysis.

The book evolved from a collection of notes prepared over the years for students in my own advanced calculus courses, as supplements to the text, presenting concrete topics such as the Wallis product formula, Stirling's formula, the Weierstrass approximation theorem, Euler's sum, and the convergence of Fourier series. Similar handouts for students in differential equations treated the method of Frobenius and the Sturm comparison theorem, both with applications to Bessel functions. Versions of the chapters on infinite products and the gamma function served as background material for students in my complex analysis courses. Feedback from students in all of those courses helped to improve the presentation.

Although the reader is assumed to have acquired a good command of basic principles, a preliminary chapter is included by way of review, and to provide a convenient reference when the principles are invoked in later chapters. The second and third chapters also contain some standard material. The book is designed for individual study, but may serve also as a text for a second-semester course in advanced calculus. There is some progression of material throughout the book, but each chapter is largely self-contained.

The desire to maintain an elementary level, excluding techniques of complex analysis and Lebesgue integration, sometimes led to awkward problems of exposition. In the end I am especially impressed by the power of Fourier analysis. The convergence theorem for Fourier series often turns out to be a viable substitute for the more systematic methods of complex analysis. On the other hand, some topics do call for more advanced techniques. For instance, Lebesgue integrals are the natural setting for Fourier transforms, and the functional equation for the Riemann zeta function can be fully understood only in the context of complex function theory. Nevertheless, I felt compelled to discuss both topics in the book.

Classical analysis is a vast area that defies comprehensive treatment. Any book on the subject must make a selection of topics. With its focus on functions of a single real variable, this book automatically excludes a great many attractive topics. Otherwise, the choices partly reflect my own background and interests.

Many exercises are offered. Mathematics is not a spectator sport; it is learned by participation. My conviction too is that abstract principles are best appreciated when applied to good problems. Therefore, while some exercises are straightforward, I have tried to make others more interesting,

more challenging, and consequently more rewarding. Hints are often provided. A few of the problems, not really exercises, come with an asterisk and an invitation to consult the literature. References for each topic are grouped at the end of the relevant chapter.

Historical notes are sprinkled throughout the book. To put a human face on the mathematics, the book includes capsule scientific biographies of the major players and a gallery of portraits. Some historical notes also shed light on the origin and evolution of mathematical ideas. A few discussions of physical applications serve the same purpose.

Many friends and colleagues helped to shape the book. I am especially indebted to Dick Askey, Martin Chuaqui, Dima Khavinson, Jeff Lagarias, Hugh Montgomery, Brad Osgood, Bill Ross, and Harold Shapiro for mathematical suggestions and encouragement as the writing progressed. Alex Lapanowski, a student in the undergraduate honors program at Michigan, read large parts of the manuscript and made valuable suggestions. Dragan Vukotić and his students Irina Arévalo and Diana Giraldo also read portions of the manuscript and spotted a number of small errors. I am enormously grateful to David Ullrich, who read the manuscript carefully, checked many of the exercises, pointed out errors and inaccuracies, and suggested important improvements in the exposition. In particular, he devised the relatively simple proof of Hilbert's inequality presented in Chapter 4. Thanks to all of these readers the book is better, but any remaining faults are the author's responsibility.

In the final stages of preparation the AMS production staff made expert contributions to the book. Special thanks go to the editor Ina Mette for her continual encouragement and willingness, for better or worse, to accommodate my peculiar wishes in regard to content and format. Finally, I must acknowledge the important role of my wife Gay. Without her support the book would not have been written.

Peter Duren

Chapter 1
Basic Principles

Classical analysis is rooted in differential and integral calculus. Calculus began with Newton and Leibniz as a collection of ideas not precisely formulated, but highly effective in scientific work. As the years progressed, however, mathematicians recognized the limitations of intuitive concepts and sought to place calculus on a firm theoretical foundation. Their efforts led to a framework of definitions, theorems, and proofs that has become part of the standard curriculum for students of advanced calculus.

This preliminary chapter offers a rapid overview of the basic theory of calculus. Definitions and major theorems are stated for later reference, and proofs are generally included. It is assumed that the reader is already acquainted with most of this material but may feel a need for review. The coverage is not complete, as some relevant topics are omitted. More extensive treatments can be found in introductory texts such as Rudin [**8**], Ross [**7**], or Mattuck [**6**].

1.1. Mathematical induction

Before turning to the calculus, we want to give a brief review of mathematical induction. This is an important method for verifying relations that depend on positive integers. The positive integers, also known as the natural numbers, are simply the numbers $n = 1, 2, 3, \ldots$. The symbol $\mathbb{N}$ will denote the set of all positive integers.

Suppose now that P_n is a proposition expressed in terms of an integer n, which is to be proved for all $n \in \mathbb{N}$. The first step is to verify that P_1 is true. This is usually quite trivial. Next comes the inductive step. Under

the inductive hypothesis that P_n is true for some $n \in \mathbb{N}$, it is deduced that P_{n+1} must also be true. Since P_1 is true, we conclude that P_2 is true, which implies that P_3 is true, and so on. In this manner we are able to prove that P_n is true for every $n \in \mathbb{N}$.

As a simple illustration, suppose that P_n is the proposition that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}\,.$$

Then P_1 is clearly true, since $\frac{1(1+1)}{2} = 1$. Suppose now that P_n is true for some $n \in \mathbb{N}$. Then

$$1 + 2 + 3 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + n + 1 = \frac{(n+1)(n+2)}{2}\,,$$

which says that P_{n+1} is true. Thus the formula holds for all $n \in \mathbb{N}$.

To give another example, let P_n be the inequality $n^2 \le 2^n$. The inequality is easily verified for $n = 1$ and $n = 2$, but it is false for $n = 3$, since $3^2 = 9 > 8 = 2^3$. However, it is again true for $n = 4$, and an inductive argument shows that it remains true for all $n \ge 4$. Indeed, suppose that $n^2 \le 2^n$ for some $n \ge 4$. Then

$$(n+1)^2 = \left(\frac{n+1}{n}\right)^2 n^2 \le \left(1 + \frac{1}{n}\right)^2 2^n \le 2^{n+1}\,,$$

which shows that the truth of P_n implies that of P_{n+1}, provided that $n \ge 4$. Therefore, since P_4 is true, it follows that P_n is true for all $n \ge 4$.

Other applications of induction appear in exercises at the end of this chapter.

1.2. Real numbers

The mathematician Leopold Kronecker is reputed to have said, "God created the positive integers. All the rest is the work of man." We will not attempt a formal construction of the real number system here but will need to review some basic principles.

The set of all integers (positive, negative, or zero) is customarily denoted by $\mathbb{Z}$, presumably because "Zahlen" is the German word for "numbers". The *rational numbers* are the quotients m/n of integers $m, n \in \mathbb{Z}$ with $n \ne 0$. A rational number m/n is said to be in *lowest terms* if m and n have no

common divisor; that is, if there is no integer $k \geq 2$ such that m/k and n/k are integers. The set of all rational numbers is denoted by $\mathbb{Q}$.

The symbol $\mathbb{R}$ denotes the set of all real numbers. We will take for granted the existence of the system of real numbers with its familiar properties of addition and multiplication. It is well known that the rational numbers are everywhere dense in $\mathbb{R}$. In other words, if $a, b \in \mathbb{R}$ and $a < b$, then $a < r < b$ for some $r \in \mathbb{Q}$.

However, not every real number is rational. To confirm this, let us show that $\sqrt{2}$ is irrational; more precisely, there is no rational number r such that $r^2 = 2$. Suppose, for purpose of contradiction, that some rational number r has the property $r^2 = 2$. We may take $r > 0$ and express it in lowest terms by $r = m/n$, where $m, n \in \mathbb{N}$. Then $r^2 = 2$ implies $m^2 = 2n^2$, so that m^2 is even. Therefore, m is even and so $m^2 = (2k)^2 = 4k^2$ for some $k \in \mathbb{N}$. Hence $n^2 = 2k^2$ and n^2 is even, so that n is also even. Since m and n are both even, they have a common divisor 2, contrary to our assumption that $r = m/n$ is in lowest terms. This contradiction shows that no rational number r can have the property $r^2 = 2$. Thus $\sqrt{2}$ is irrational.

Like the rationals, the irrational numbers are everywhere dense in $\mathbb{R}$. In fact, it follows from the density of the rationals that the rational multiples of $\sqrt{2}$ are everywhere dense.

A number $x_0 \in \mathbb{R}$ is said to be *algebraic* if it satisfies an equation

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = 0$$

for some polynomial P whose coefficients $a_0, a_1, \ldots, a_n$ are integers, not all zero. It is easily seen that every rational number is algebraic. The number $\sqrt{2}$ is also algebraic, since it satisfies the equation $x^2 - 2 = 0$. Real numbers that are not algebraic are called *transcendental*. It is known, for instance, that the numbers π and e are transcendental. (See Chapter 2, Sections 2.1 and 2.2 for further discussion.)

It is an important fact that the rational numbers form a countable set. An infinite set S is said to be *countable* or *denumerable* if it can be put into one-to-one correspondence with the set $\mathbb{N}$ of positive integers. This amounts to saying that the elements of S can be arranged in a sequence $x_1, x_2, x_3, \ldots$ with no repetitions. To show that the set $\mathbb{Q}$ has this property, it suffices to show that the set of all positive rational numbers is countable. To see this, arrange the numbers $r = m/n$, where $m, n \in \mathbb{N}$, in an infinite rectangular array with m/n in the mth row and nth column. Then by weaving diagonally through the array, as indicated in the diagram below, the set of all positive rationals can be placed in a sequence without repetition. Whenever the path encounters a rational number already recorded in different form, it passes

over that number without including it in the sequence.

$$\begin{array}{ccccccccc}
\frac{1}{1} & \rightarrow & \frac{1}{2} & & \frac{1}{3} & \rightarrow & \frac{1}{4} & & \frac{1}{5} & \cdots \\
 & \swarrow & & \nearrow & & \swarrow & & & & \\
\frac{2}{1} & & \frac{2}{2} & & \frac{2}{3} & & \frac{2}{4} & & \frac{2}{5} & \cdots \\
\downarrow & \nearrow & & \swarrow & & & & & & \\
\frac{3}{1} & & \frac{3}{2} & & \frac{3}{3} & & \frac{3}{4} & & \frac{3}{5} & \cdots \\
 & \swarrow & & & & & & & & \\
\frac{4}{1} & & \frac{4}{2} & & \frac{4}{3} & & \frac{4}{4} & & \frac{4}{5} & \cdots \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots &
\end{array}$$

Thus the resulting sequence is

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}, \frac{5}{1}, \frac{1}{5}, \frac{1}{6}, \ldots .$$

Every positive rational number must appear exactly once in the sequence. This proves that $\mathbb{Q}$ is a countable set.

On the other hand, the set $\mathbb{R}$ of all real numbers is not countable. To see this, it suffices to show that the real numbers in the interval $0 \leq x < 1$ do not form a countable set. The idea of the proof is quite ingenious and is due to Georg Cantor (1845–1918), the founder of modern set theory. (More of Cantor's ideas will be discussed in Chapter 12.) Cantor's proof proceeds as follows. Suppose, for purpose of contradiction, that the set $[0, 1) = \{x \in \mathbb{R} : 0 \leq x < 1\}$ is countable, so that all of its members can be arranged in a sequence

$$\begin{aligned}
x_1 &= 0.d_{11}d_{12}d_{13}\ldots \\
x_2 &= 0.d_{21}d_{22}d_{23}\ldots \\
x_3 &= 0.d_{31}d_{32}d_{33}\ldots \\
&\ \ .\ \ .\ \ .
\end{aligned}$$

Here $x_n = 0.d_{n1}d_{n2}d_{n3}\ldots$ is the decimal expansion of the number x_n, chosen to end in a sequence of 0's (not 9's) in cases of ambiguity. We will reach a contradiction by displaying a number in $[0, 1)$ that is not on the list. If $0 \leq d_{nn} \leq 4$, set $b_n = 7$. If $5 \leq d_{nn} \leq 9$, set $b_n = 2$. Then the number $y = 0.b_1b_2b_3\ldots$ lies in $[0, 1)$ but does not appear in the purported enumeration. Indeed, $y \neq x_n$ for each n because the nth digit of its decimal expansion differs from that of x_n by at least 3, so that $|y - x_n| > 10^{-n}$. This shows that no sequence of numbers in $[0, 1)$ can exhaust the entire interval, and so $[0, 1)$ is uncountable. Therefore, the set of all real numbers is uncountable.

The real number system has a very important property, known as completeness, that manifests itself in various ways and lies at the very foundation of analysis. One form of completeness is the *least upper bound principle*, whose statement requires us first to recall some terminology. A nonempty set $S \subset \mathbb{R}$ is said to be *bounded above* if there is a number $y \in \mathbb{R}$ such that $x \leq y$ for all $x \in S$. Such a number y is called an *upper bound* of S. A number M is called the *least upper bound* or *supremum* of S if M is an upper bound and $M \leq y$ for every upper bound y. This is indicated by writing $M = \sup S$. Similarly, a set $S \subset \mathbb{R}$ is *bounded below* if there is a number $y \in \mathbb{R}$, called a *lower bound* of S, such that $y \leq x$ for all $x \in S$. A number m is called the *greatest lower bound* or *infimum* of S if $y \leq m$ for every lower bound y. For this we write $m = \inf S$. A set $S \subset \mathbb{R}$ is said to be *bounded* if it is bounded both above and below.

Least Upper Bound Principle. *If a set of real numbers is bounded above, then it has a least upper bound.*

Equivalently, every set of real numbers that is bounded below has a greatest lower bound. We will take the least upper bound principle as an axiom. If the real number system is actually constructed, for instance by the method of "Dedekind cuts", the least upper bound principle can be deduced as a theorem. (See for instance Rudin [**8**].)

It is instructive to note that the system of rational numbers is not complete, since it fails to have the least upper bound property. Consider, for instance, the set $S = \{r \in \mathbb{Q} : r^2 < 2\}$. Clearly, the set S has a (rational) upper bound, but it does not have a smallest rational upper bound, essentially because $\sqrt{2}$ is irrational.

Finally, we need to set some notation for intervals of real numbers. For $a, b \in \mathbb{R}$ with $a < b$, we define $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$ and $(a, b) = \{x \in \mathbb{R} : a < x < b\}$. The intervals $a \leq x < b$ and $a < x \leq b$ are denoted by $[a, b)$ and $(a, b]$ respectively. The unbounded intervals $x \geq a$ and $x \leq b$ are denoted respectively by $[a, \infty)$ and $(-\infty, b]$. The intervals $x > a$ and $x < b$ are denoted by (a, ∞) and $(-\infty, b)$. We will usually write $\mathbb{R}$ instead of $(-\infty, \infty)$. The empty set is denoted by $\emptyset$.

1.3. Completeness principles

We now develop some further principles, basic to analysis, that can be regarded as expressing the completeness of the real number system. These are the monotone boundedness theorem, the Bolzano–Weierstrass theorem, the Cauchy criterion for convergence, and the Heine–Borel theorem. Each will be derived ultimately from the least upper bound principle, and all are

in some sense equivalent to it. We begin with sequences of real numbers and the concept of convergence.

A sequence of real numbers can be defined as a mapping from $\mathbb{N}$ into $\mathbb{R}$, a real-valued function defined on the positive integers. It is customarily written as $\{x_n\} = \{x_1, x_2, x_3, \dots\}$, where $x_n \in \mathbb{R}$ for each $n \in \mathbb{N}$. The sequence is said to *converge* to a *limit* $L \in \mathbb{R}$ if to each $\varepsilon > 0$ there corresponds a number N such that $|x_n - L| < \varepsilon$ for all $n \geq N$. This is tantamount to saying that for each prescribed discrepancy ε the numbers x_n are within distance ε of L for all but a finite number of indices n. It is easy to see that a sequence can have at most one limit. The *range* of a sequence is its set of values $\{x_n : n \in \mathbb{N}\}$. A sequence is said to be bounded if its range is a bounded set. Every convergent sequence is bounded, but the converse is false. The convergence of a sequence $\{x_n\}$ to a limit L is indicated by writing $L = \lim_{n\to\infty} x_n$ or simply $x_n \to L$ as $n \to \infty$. A sequence that does not converge is called *divergent*.

If $x_n \to L$ and $y_n \to M$, then $(x_n + y_n) \to L + M$ and $x_n y_n \to LM$. If also $M \neq 0$, then $y_n \neq 0$ for all n sufficiently large and $x_n/y_n \to L/M$.

More generally, the notation $x_n \to +\infty$ (or simply $x_n \to \infty$) as $n \to \infty$ indicates that to each $a \in \mathbb{R}$ there corresponds a number N such that $x_n > a$ for all $n \geq N$. This is expressed by saying that the sequence $\{x_n\}$ tends to $+\infty$. Thus it diverges in a specific way. Similarly, the notation $x_n \to -\infty$ means that for each $a \in \mathbb{R}$ there is a number N such that $x_n < a$ for all $n \geq N$.

A sequence is called *monotonic* if it is either nondecreasing or nonincreasing; that is, if either

$$x_1 \leq x_2 \leq x_3 \leq \dots \qquad \text{or} \qquad x_1 \geq x_2 \geq x_3 \geq \dots .$$

Monotone Boundedness Theorem. *Every monotonic and bounded sequence of real numbers is convergent.*

Proof. It suffices to suppose that the sequence $\{x_n\}$ is nondecreasing. Then by hypothesis, its range $S = \{x_n : n \in \mathbb{N}\}$ is bounded above. Therefore, by the least upper bound principle, the set S has a least upper bound $\lambda = \sup S$. We will show that $\{x_n\}$ converges to λ. Since λ is an upper bound for S, we have $x_n \leq \lambda$ for all $n \in \mathbb{N}$. On the other hand, for each $\varepsilon > 0$ the number $\lambda - \varepsilon$ is not an upper bound for S, since λ is the least upper bound. Consequently, $x_N > \lambda - \varepsilon$ for some $N \in \mathbb{N}$. Since the sequence $\{x_n\}$ is nondecreasing, it follows that $x_n > \lambda - \varepsilon$ for all $n \geq N$. Combining this with the inequality $x_n \leq \lambda$, we see that $-\varepsilon < 0 \leq \lambda - x_n \leq \varepsilon$, or $|x_n - \lambda| < \varepsilon$ for all $n \geq N$. This proves that $x_n \to \lambda$ as $n \to \infty$. □

With the monotone boundedness theorem in hand, we can now prove the Bolzano–Weierstrass theorem. To state the latter result, we need a definition. A number ξ is a *cluster point* of a set $S \subset \mathbb{R}$ if each deleted neighborhood of ξ contains a point of S. A *deleted neighborhood* of ξ is an interval (a, b) with the point ξ ($a < \xi < b$) removed. Thus ξ is a cluster point of S if for each $\varepsilon > 0$ there exists a point $x \in S$ such that $0 < |x - \xi| < \varepsilon$. Note that a finite set cannot have any cluster points. Also an unbounded infinite set need not have any cluster points; consider for example the set $\mathbb{N}$ of positive integers. A cluster point of a set need not belong to the set. For instance, 0 is a cluster point of the set $S = \{\frac{1}{n} : n \in \mathbb{N}\}$ but $0 \notin S$. Observe also that if ξ is a cluster point of a set S, then every neighborhood of ξ contains infinitely many points of S.

Bolzano–Weierstrass Theorem. *Every bounded infinite set of real numbers has a cluster point.*

Proof. Let $S \subset \mathbb{R}$ be a bounded infinite set. Then S is contained in some closed bounded interval $[a, b]$. The strategy of proof is to "divide and conquer". The midpoint of the interval is $\frac{a+b}{2}$. At least one of the intervals $[a, \frac{a+b}{2}]$ and $[\frac{a+b}{2}, b]$ must contain infinitely many points of S. If the left-hand interval qualifies, let $a_1 = a$ and $b_1 = \frac{a+b}{2}$. Otherwise, let $a_1 = \frac{a+b}{2}$ and $b_1 = b$. In either case the subinterval $[a_1, b_1]$ contains infinitely many points of S. Now repeat the process. Divide the interval $[a_1, b_1]$ in half and choose a subinterval $[a_2, b_2]$ of the form $[a_1, \frac{a_1+b_1}{2}]$ or $[\frac{a_1+b_1}{2}, b_1]$ that again contains infinitely many points of S. Proceeding inductively, construct a nested sequence of intervals $[a_n, b_n]$ with lengths $2^{-n}(b-a)$, each containing infinitely many points of S.

Since $a_1 \le a_2 \le a_3 \le \cdots \le b$ and $b_1 \ge b_2 \ge b_3 \ge \cdots \ge a$, it follows from the monotone boundedness theorem that both of the sequences $\{a_n\}$ and $\{b_n\}$ are convergent. Thus $a_n \to \alpha \le b$ and $b_n \to \beta \ge a$ as $n \to \infty$. Since $a_n < b_n$ for all n, it is clear that $\alpha \le \beta$. But in fact $\alpha = \beta$, since for each $\varepsilon > 0$ we have

$$\begin{aligned} |\alpha - \beta| &\le |\alpha - a_n| + |a_n - b_n| + |b_n - \beta| \\ &= |\alpha - a_n| + 2^{-n}(b-a) + |b_n - \beta| < \varepsilon \end{aligned}$$

for all n sufficiently large. Let ξ denote the common value of α and β, so that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \xi\,.$$

The final step in the proof is to show that ξ is a cluster point of S. But for each given $\varepsilon > 0$ we see that $[a_n, b_n] \subset [\xi - \varepsilon, \xi + \varepsilon]$ for all n sufficiently large. Indeed, since $\{a_n\}$ increases to ξ and $\{b_n\}$ decreases to ξ, there exists

a number N such that $\xi - \varepsilon < a_n \le \xi$ and $\xi \le b_n < \xi + \varepsilon$ for all $n \ge N$. But by construction each interval $[a_n, b_n]$ contains an infinite number of points in S. Therefore, for each $\varepsilon > 0$ the deleted neighborhood

$$\{x \in \mathbb{R} : 0 < |x - \xi| < \varepsilon\}$$

contains a point of S, in fact infinitely many. This says that ξ is a cluster point of S, which proves the theorem. □

It is now a short step to the sequential form of the theorem. A *subsequence* of a sequence $\{x_n\}$ is a sequence of the form $\{x_{n_k}\}$, where $n_k \in \mathbb{N}$ and $1 \le n_1 < n_2 < n_3 < \dots$.

Bolzano–Weierstrass Theorem (sequential form). *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let $\{x_n\}$ be a bounded sequence and let $S = \{x_n : n \in \mathbb{N}\}$ be its set of values. Then S is a bounded set. If S is a finite set, then some value is repeated infinitely often and so $\{x_n\}$ has a constant subsequence, which is certainly convergent. If S is an infinite set, the earlier form of the Bolzano–Weierstrass theorem says that S has a cluster point ξ. Therefore, each deleted neighborhood of ξ contains a point x_n for some n. Having selected a point x_{n_1} for which $0 < |\xi - x_{n_1}| < 1$, we can choose an index $n_2 > n_1$ such that $0 < |\xi - x_{n_2}| < \frac{1}{2}$, and then an index $n_3 > n_2$ such that $0 < |\xi - x_{n_3}| < \frac{1}{3}$, and so forth. Then the subsequence $\{x_{n_k}\}$ converges to ξ. □

We are now ready to establish the Cauchy criterion for convergence of a numerical sequence. First let us record the key definition. A *Cauchy sequence* of real numbers is a sequence $\{x_n\}$ with the property that for each $\varepsilon > 0$ there exists an integer N depending on ε such that $|x_n - x_m| < \varepsilon$ for all $n, m \ge N$. It is easy to see that every convergent sequence has this property. The converse, however, is not so obvious.

Theorem (Cauchy criterion). *Every Cauchy sequence of real numbers is convergent.*

Proof. The proof will consist of three main steps. The first step is to show that every Cauchy sequence is bounded. To prove this, choose for instance $\varepsilon = 1$ and refer to the definition of a Cauchy sequence to conclude that for some integer N the inequality $|x_n - x_N| < 1$ holds for all $n > N$. Therefore,

$$|x_n| \le |x_N| + |x_n - x_N| \le |x_N| + 1$$

for all $n > N$. But this implies that the sequence is bounded, since it follows

that

$$|x_n| \le \max\bigl\{|x_1|, |x_2|, \dots, |x_N|, |x_N| + 1\bigr\}$$

for all $n \in \mathbb{N}$.

The second step is to apply the sequential form of the Bolzano–Weierstrass theorem. Since a Cauchy sequence is bounded, it must have a convergent subsequence $\{x_{n_k}\}$.

The final step is to observe that if a Cauchy sequence has a convergent subsequence, then in fact the full sequence is convergent. Indeed, if $x_{n_k} \to L$ as $k \to \infty$, then for each $\varepsilon > 0$ we see that

$$|x_n - L| \le |x_n - x_{n_k}| + |x_{n_k} - L| < |x_n - x_{n_k}| + \frac{\varepsilon}{2}$$

for all indices n_k sufficiently large. But because $\{x_n\}$ is a Cauchy sequence, it then follows that

$$|x_n - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

for all n sufficiently large. This shows that $x_n \to L$ as $n \to \infty$, which completes the proof. □

Next we apply the Bolzano–Weierstrass theorem to prove the Heine–Borel theorem. The proof will require two preliminary results, known as the nested sets theorem and the countable basis theorem, which will now be discussed.

First recall that a set G of real numbers is said to be *open* if each point $x_0 \in G$ has a neighborhood

$$N_r(x_0) = \{x \in \mathbb{R} : |x - x_0| < r\}$$

that lies entirely in G. A set $F \subset \mathbb{R}$ is said to be *closed* if its complement

$$\widetilde{F} = \mathbb{R} \setminus F = \{x \in \mathbb{R} : x \notin F\}$$

is open. The union of any collection of open sets is open. The intersection of any finite collection of open sets is open. Similarly, an arbitrary intersection of closed sets is closed, and every finite union of closed sets is closed. It can be proved (*cf.* Exercise 11) that a set is closed if and only if it contains all of its cluster points.

Nested Sets Theorem. *Let $F_1, F_2, \dots$ be a sequence of nonempty closed bounded sets of real numbers which are nested in the sense that $F_{n+1} \subset F_n$ for every $n \in \mathbb{N}$. Then the intersection $\bigcap_{n=1}^{\infty} F_n$ is nonempty.*

The theorem says that under the stated hypotheses the sets must contain some common element. This may seem obvious, but if either of the

hypotheses "closed" or "bounded" is relaxed the conclusion is false. For instance, the *open* intervals $I_n = (0, 1/n)$, for $n = 1, 2, \ldots$, are nonempty, bounded, and nested: $I_1 \supset I_2 \supset \ldots$. However, their intersection $\bigcap_{n=1}^{\infty} I_n$ is empty. The intervals $[n, \infty)$ are closed and nested but not bounded, and again their intersection is empty.

Proof of theorem. The theorem is trivial if some set F_m is finite, because then F_n is finite for every $n \geq m$. Since the sets F_n are nonempty and nested, this clearly implies that some point belongs to all of the sets. Thus we may suppose that each set F_n is infinite.

An infinite sequence of distinct points x_n will now be constructed inductively as follows. Let x_1 be any point in F_1. Choose $x_2 \in F_2$ with $x_2 \neq x_1$. At the nth stage of the construction, having chosen distinct points $x_k \in F_k$ for $k = 1, 2, \ldots, n-1$, choose $x_n \in F_n$ different from all of its predecessors. Since every set F_n is infinite, it is always possible to make such a choice. Let $S = \{x_n : n \in \mathbb{N}\}$ be the set of all points chosen in this process. Note that S is a bounded set, since $S \subset F_1$ and F_1 is bounded. Also S is an infinite set, since all of the numbers x_n were chosen to be distinct. Thus by the Bolzano–Weierstrass theorem, S has a cluster point ξ. Then ξ is a cluster point of each set F_m, since the points x_n belong to F_m for all $n \geq m$. But F_m is closed and therefore contains all of its cluster points. Hence $\xi \in F_n$ for every $n \in \mathbb{N}$. In other words, ξ belongs to the intersection of the sets F_n, which is therefore not empty. This proves the nested sets theorem. □

Countable Basis Theorem. *There exists a countable collection of bounded open intervals $I_1, I_2, \cdots \subset \mathbb{R}$ with the following property. For each $x \in \mathbb{R}$ and for each open set $G \subset \mathbb{R}$ with $x \in G$, there is an interval I_n such that $x \in I_n \subset G$.*

Proof. Consider the family $\mathcal{F}$ of all open neighborhoods

$$N_r(p) = \{x \in \mathbb{R} : |x - p| < r\}$$

with rational centers p and rational radii r. In other words,

$$\mathcal{F} = \{N_r(p) : p \in \mathbb{Q}\,, r \in \mathbb{Q}\}\,.$$

The family $\mathcal{F}$ is in one-to-one correspondence with the Cartesian product $\mathbb{Q} \times \mathbb{Q}$ and is therefore countable. It can be enumerated as $I_1, I_2, \ldots$. Now if x belongs to an open set G, then by the definition of open set, $N_\varepsilon(x) \subset G$ for some $\varepsilon > 0$. But then by the density of the rational numbers, there exist $p, r \in \mathbb{Q}$ with $x \in N_r(p) \subset N_\varepsilon(x)$. Thus $x \in I_n \subset G$ for some $I_n \in \mathcal{F}$, as required. □

In order to state the Heine–Borel theorem, we need to make some definitions. We say that a collection $\mathcal{C}$ of subsets of $\mathbb{R}$ is a *covering* of a set $S \subset \mathbb{R}$ if S is contained in the union of all of the sets in $\mathcal{C}$. Equivalently, $\mathcal{C}$ is a covering of S if each point $x \in S$ belongs to some set in the collection $\mathcal{C}$. An *open covering* is a covering that consists of open sets. A *finite covering* is a covering that consists of finitely many sets. A *subcovering* of a covering $\mathcal{C}$ is a subcollection of the sets in $\mathcal{C}$ (in other words, a subset of $\mathcal{C}$) which again constitutes a covering of the set S.

Heine–Borel Theorem. *Every open covering of a closed bounded set of real numbers contains a finite subcovering.*

The theorem says that whenever a closed bounded set of real numbers is contained in the union of some collection of open sets, then it is actually contained in the union of finitely many of those sets. The statement becomes false if either of the adjectives "closed" or "bounded" is omitted. For instance, the collection

$$\mathcal{C} = \{(\tfrac{1}{n}, 1) : n \in \mathbb{N}\}$$

is an open covering of the open set $(0, 1)$, and in fact $\bigcup_{n=1}^{\infty}(\frac{1}{n}, 1) = (0, 1)$, yet no finite subcollection of $\mathcal{C}$ is a covering of $(0, 1)$. To give another example, the collection

$$\mathcal{C} = \{(-n, n) : n \in \mathbb{N}\}$$

is an open covering of any set $S \subset \mathbb{R}$, but no finite subcollection of $\mathcal{C}$ can be a covering of any unbounded set S.

Proof of theorem. We will show first that every countable open covering of a closed bounded set contains a finite subcovering. Thus we assume that a closed bounded set S is contained in the union of a countable family of open sets $G_1, G_2, \ldots$. In other words,

$$S \subset \bigcup_{k=1}^{\infty} G_k \,.$$

We want to show that $S \subset \bigcup_{k=1}^{n} G_k$ for some $n \in \mathbb{N}$. Let $H_n = \bigcup_{k=1}^{n} G_k$. Then each set H_n is open and we see that $H_1 \subset H_2 \subset H_3 \subset \ldots$ and $S \subset \bigcup_{n=1}^{\infty} H_n$. Now define the sets $F_n = \widetilde{H_n} \cap S$, where $\widetilde{H_n}$ is the complement of H_n. Then each set F_n is closed and bounded, and $F_1 \supset F_2 \supset F_3 \ldots$. We want to prove that $S \subset H_n$ for some n, which is equivalent to saying that F_n is empty. Suppose on the contrary that $F_n \neq \emptyset$ for every $n \in \mathbb{N}$. Then by the nested sets theorem there is a point $x \in \bigcap_{n=1}^{\infty} F_n$. In other words, $x \in F_n$ for every $n \in \mathbb{N}$. According to the definition of F_n, this says that $x \in S$ and $x \notin H_n$ for all n, which contradicts the hypothesis that $S \subset \bigcup_{n=1}^{\infty} H_n$. The

contradiction shows that in fact some set F_n is empty, or that $S \subset H_n$ for some $n \in \mathbb{N}$, which was to be shown.

It remains to prove that a countable subcovering can be extracted from an arbitrary open covering of S. Suppose, then, that

$$S \subset \bigcup_{\alpha \in A} G_\alpha\,, \qquad \text{where } \ \mathcal{C} = \{G_\alpha : \alpha \in A\}$$

is a collection, not necessarily countable, of open sets G_α indexed by a set A. By assumption, each point $x \in S$ belongs to G_α for some $\alpha \in A$. Choose one of the sets G_α that contains x and write it as $G_{\alpha(x)}$. Now let $\mathcal{F} = \{I_n\}$ be the countable collection of bounded open intervals I_n supplied by the countable basis theorem. For each $x \in S$, that theorem allows us to choose an interval $I_n = I_{n(x)}$ such that $x \in I_{n(x)} \subset G_{\alpha(x)}$. Note that the collection of sets $\{I_{n(x)} : x \in S\}$ is a subcollection of $\mathcal{F}$ and is therefore a countable covering of S. Now with each interval I_n in this covering we can associate a set G_α in the original covering $\mathcal{C}$ such that $I_n \subset G_\alpha$. This gives the required countable subcovering of $\mathcal{C}$. □

To summarize the proof, a countable covering is first extracted from an arbitrary open covering of S. In this part of the argument it is not necessary to make any special assumptions about S. However, in the next step, where a further finite subcovering is extracted from the countable subcovering, it is essential to assume that S is closed and bounded.

In the general context of a metric space, any set with the Heine–Borel property is said to be *compact*. Thus the content of the Heine–Borel theorem is that a set of real numbers is compact if (and only if) it is closed and bounded.

1.4. Numerical sequences

At this point it is convenient to discuss some standard examples of sequences whose limits can be found with help of the binomial theorem

$$(1+x)^n = 1 + \sum_{k=1}^{n} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots + x^n\,,$$

where $n \in \mathbb{N}$ and

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}$$

denotes a binomial coefficient. Here are the examples.

Example 1. For each fixed number $r \in \mathbb{R}$ with $|r| < 1$,

$$\lim_{n\to\infty} r^n = 0\,.$$

Proof. Since $(-r)^n = (-1)^n r^n$, it suffices to suppose that $0 < r < 1$. Write $r = \frac{1}{1+a}$, where $a > 0$. Then $na \le (1+a)^n$ and it follows that

$$0 < r^n = \frac{1}{(1+a)^n} \le \frac{1}{na} \to 0 \qquad \text{as} \quad n \to \infty\,.$$

By the "squeeze principle", this shows that $r^n \to 0$.

Example 2. For each constant $a > 0$,

$$\lim_{n\to\infty} \sqrt[n]{a} = 1\,.$$

Proof. Since $\sqrt[n]{1/a} = 1/\sqrt[n]{a}$, it suffices to consider $a > 1$. Then $\sqrt[n]{a} > 1$, and we can write $\sqrt[n]{a} = 1 + h_n$ for some $h_n > 0$. It follows that

$$a = (1+h_n)^n \ge n\,h_n\,, \qquad \text{so that} \quad 0 < h_n < \frac{a}{n} \to 0\,.$$

This shows that $h_n \to 0$ as $n \to \infty$, which says that $\sqrt[n]{a} \to 1$ as $n \to \infty$.

Example 3.

$$\lim_{n\to\infty} \sqrt[n]{n} = 1\,.$$

Proof. Again $\sqrt[n]{n} > 1$, so we can write $\sqrt[n]{n} = 1 + h_n$ for some $h_n > 0$. Then

$$n = (1+h_n)^n \ge 1 + nh_n + \frac{n(n-1)}{2!}\,h_n^2 > \frac{n(n-1)}{2}\,h_n^2\,, \qquad \text{for} \quad n \ge 2\,,$$

which shows that

$$0 < h_n < \sqrt{\frac{2}{n-1}} \to 0 \qquad \text{as} \quad n \to \infty\,.$$

This proves that $h_n \to 0$, so that $\sqrt[n]{n} \to 1$ as $n \to \infty$.

Example 4. For each pair of numbers $p > 0$ and r with $|r| < 1$,

$$\lim_{n\to\infty} n^p r^n = 0\,.$$

Proof. This is a stronger form of Example 1. Again it is enough to take $0 < r < 1$. Let $a_n = n^p r^n$ and observe that the ratios

$$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^p r \to r \qquad \text{as} \quad n \to \infty\,.$$

Therefore, for any fixed R in the interval $r < R < 1$, we can infer that

$$0 < \frac{a_{n+1}}{a_n} < R$$

for all n sufficiently large, say for all $n \geq N$. Then

$$a_{N+1} < R\,a_N\,, \qquad a_{N+2} < R\,a_{N+1} < R^2 a_N\,,$$

and in general

$$0 < a_{N+k} < a_N R^k\,, \qquad k = 1, 2, \ldots\,.$$

Thus $a_{N+k} \to 0$ as $k \to \infty$, since $0 < R < 1$ and so $R^k \to 0$. In other words, $a_n \to 0$ as $n \to \infty$, as claimed.

Example 5. For each number $p > 0$,

$$\lim_{n\to\infty} \frac{\log n}{n^p} = 0\,.$$

Proof. Consider first what happens as $n \to \infty$ over the subsequence $\{2^k\}$. For $n = 2^k$, we have

$$\frac{\log n}{n^p} = \frac{\log(2^k)}{(2^k)^p} = \frac{k \log 2}{(2^p)^k} \to 0 \qquad \text{as} \quad k \to \infty\,,$$

by Example 4, since $r = 1/2^p < 1$ for $p > 0$.

Now observe that each integer $n \geq 2$ lies in an interval $2^k \leq n < 2^{k+1}$, for some uniquely determined integer $k = 1, 2, \ldots$. Thus

$$0 < \log n < \log(2^{k+1}) = (k+1)\log 2\,,$$

whereas

$$n^p \geq (2^k)^p = (2^p)^k\,.$$

Combining the two estimates, we see that

$$0 < \frac{\log n}{n^p} \leq \frac{(k+1)\log 2}{(2^p)^k}\,.$$

But $k \to \infty$ as $n \to \infty$, and

$$\lim_{k\to\infty} \frac{(k+1)\log 2}{(2^p)^k} = 0\,,$$

by Example 4, again because $r = 1/2^p < 1$ for $p > 0$. An application of the "squeeze principle" completes the proof.

For an arbitrary sequence $\{x_n\}$ of real numbers, the upper and lower limits

$$\limsup_{n\to\infty} x_n \qquad \text{and} \qquad \liminf_{n\to\infty} x_n$$

will now be defined and characterized in three equivalent ways. We will see that

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \lim_{n\to\infty} x_n$$

if the sequence $\{x_n\}$ converges. With suitable interpretation, the same will be true if $x_n \to +\infty$ or $x_n \to -\infty$. However, the "lim sup" and "lim inf" are more general quantities that help to specify the eventual behavior of divergent sequences that do not tend to $+\infty$ or $-\infty$.

First some conventions for unbounded sequences. If $\{x_n\}$ is not bounded above, we write

$$\limsup_{n\to\infty} x_n = +\infty\,.$$

Similarly, if $\{x_n\}$ is not bounded below, we write

$$\liminf_{n\to\infty} x_n = -\infty\,.$$

If $x_n \to -\infty$, we say that $\limsup_{n\to\infty} x_n = -\infty$. If $x_n \to +\infty$, we say that $\liminf_{n\to\infty} x_n = +\infty$.

Suppose now that $\{x_n\}$ is bounded above and does not tend to $-\infty$. Let S denote the set of subsequential limits of $\{x_n\}$. This is the set of numbers λ for which some subsequence $\{x_{n_k}\} = \{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$ converges to λ. We then define $\limsup_{n\to\infty} x_n$ to be $\sup S$. Observe that S is nonempty, in view of the Bolzano–Weierstrass theorem. Indeed, if $\{x_n\}$ is bounded above and does not tend to $-\infty$, then it has a bounded subsequence, and so a further subsequence is convergent. Similarly, if $\{x_n\}$ is bounded below and does not tend to $+\infty$, we define $\liminf_{n\to\infty} x_n = \inf S$.

It may be remarked that the number $\sup S$ actually belongs to S, so that $\limsup x_n$ is in fact the largest subsequential limit. Similarly, $\liminf x_n$ is the smallest subsequential limit if the sequence is bounded below and does not tend to $+\infty$. (See Exercise 21.)

With these definitions, we see that

$$\liminf_{n\to\infty} x_n \le \limsup_{n\to\infty} x_n$$

for every sequence $\{x_n\}$, with the usual interpretation of the symbol "$\le$" if either of the quantities is infinite. Equality occurs if and only if the sequence $\{x_n\}$ converges or tends to $+\infty$ or $-\infty$. For bounded sequences, equality

occurs if and only if the set S contains only one number, which must then be the limit of the sequence.

Another simple consequence of the definitions is that

$$\limsup_{n\to\infty}(-x_n) = -\liminf_{n\to\infty} x_n\,,$$

with the standard interpretation if either quantity is infinite. This relation shows that the upper and lower limits are in some sense reflections of each other, so that properties of one will transfer automatically to the other.

The preceding definitions of "lim sup" and "lim inf" were given in terms of the set S of subsequential limits. Although this description is perhaps the easiest to comprehend, it is awkward to apply. We now give an equivalent formulation that is better suited to manipulation.

Theorem. *If a sequence $\{x_n\}$ is bounded above and does not tend to $-\infty$, then the number*

$$\beta = \limsup_{n\to\infty} x_n = \sup S$$

is characterized by the following property. For each $\varepsilon > 0$,

(i) $x_n < \beta + \varepsilon$ *for all but a finite number of indices* n

and

(ii) $x_n > \beta - \varepsilon$ *for infinitely many indices* n*.*

Similarly, if $\{x_n\}$ is bounded below and does not tend to $+\infty$, the number

$$\alpha = \liminf_{n\to\infty} x_n = \inf S$$

is characterized by the property that for each $\varepsilon > 0$,

(i) $x_n > \alpha - \varepsilon$ *for all but a finite number of indices* n

and

(ii) $x_n < \alpha + \varepsilon$ *for infinitely many indices* n*.*

Proof. We need only discuss the description of "lim sup". Let us show first that $\beta = \limsup_{n\to\infty} x_n$ satisfies the conditions (i) and (ii). If (i) fails for some $\varepsilon > 0$, then the inequality $x_n \geq \beta + \varepsilon$ holds for infinitely many indices n. This implies that $x_{n_k} \geq \beta + \varepsilon$ for some subsequence $\{x_{n_k}\}$. But by hypothesis $\{x_n\}$ is bounded above, so the subsequence $\{x_{n_k}\}$ is also bounded above. Hence by the Bolzano–Weierstrass theorem it has a further subsequence that converges to a limit $\lambda \geq \beta + \varepsilon$. Thus λ belongs to the set S of subsequential limits of $\{x_n\}$ and $\lambda > \beta$, which contradicts the fact that $\beta = \sup S$ is an upper bound for S. This shows that β satisfies (i). If (ii) fails to hold for some $\varepsilon > 0$, then $x_n \leq \beta - \varepsilon$ for all but a finite number of indices n, and so $\beta - \varepsilon$ is an upper bound for S, which violates

the definition of β as the least upper bound of S. Thus we have proved that $\beta = \limsup_{n\to\infty} x_n$ satisfies both (i) and (ii). Conversely, it is clear that at most one number β can satisfy both (i) and (ii) for each $\varepsilon > 0$, so that a number with this property must be $\limsup_{n\to\infty} x_n$.

The proof for "lim inf" can be carried out in a similar way, or the result can be deduced from the relation $\liminf_{n\to\infty} x_n = -\limsup_{n\to\infty} (-x_n)$. □

The condition (i) for "lim sup" is equivalent to saying that for each $\varepsilon > 0$ there exists a number N (depending on ε) such that $x_n < \beta + \varepsilon$ for all $n \geq N$. A similar remark applies to "lim inf".

For a third formulation of upper and lower limits, one can show that

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \left(\sup_{k\geq n} x_k \right) \qquad \text{and} \qquad \liminf_{n\to\infty} x_n = \lim_{n\to\infty} \left(\inf_{k\geq n} x_k \right),$$

with appropriate interpretation for unbounded sequences. To see that the limits exist for bounded sequences $\{x_n\}$, let

$$y_n = \sup_{k\geq n} x_k \qquad \text{and} \qquad z_n = \inf_{k\geq n} x_k\,,$$

and observe that $y_{n+1} \leq y_n$, whereas $z_{n+1} \geq z_n$. Thus both limits exist by the monotone boundedness theorem. It is left as an exercise to identify the limits as $\limsup x_n$ and $\liminf x_n$, respectively.

The following theorem is useful in the study of infinite series.

Theorem. *For any sequence $\{r_n\}$ of positive numbers,*

$$(i) \quad \limsup_{n\to\infty} \sqrt[n]{r_n} \leq \limsup_{n\to\infty} \frac{r_{n+1}}{r_n}$$

and

$$(ii) \quad \liminf_{n\to\infty} \frac{r_{n+1}}{r_n} \leq \liminf_{n\to\infty} \sqrt[n]{r_n}\,.$$

Proof. We will treat only the inequality (i). (The proof of the inequality (ii) is similar and is left as an exercise.) We may suppose that $\beta = \limsup_{n\to\infty} r_{n+1}/r_n$ is finite. Then for each $\varepsilon > 0$ there is a number N such that $0 < r_{n+1}/r_n < \beta + \varepsilon$ for all $n \geq N$. For $n > N$ it follows that

$$r_n = r_N \cdot \frac{r_{N+1}}{r_N} \cdot \frac{r_{N+2}}{r_{N+1}} \cdots \frac{r_n}{r_{n-1}} < r_N(\beta+\varepsilon)^{n-N}\,, \quad \text{and so}$$

$$\sqrt[n]{r_n} < \sqrt[n]{r_N(\beta+\varepsilon)^{-N}}\,(\beta+\varepsilon) < \beta + 2\varepsilon$$

for all n sufficiently large, since $\sqrt[n]{a} \to 1$ for each fixed $a > 0$. It follows that

$$\limsup_{n\to\infty} \sqrt[n]{r_n} \leq \beta + 2\varepsilon \qquad \text{for each } \varepsilon > 0\,,$$

which implies that $\limsup_{n\to\infty} \sqrt[n]{r_n} \leq \beta$. □

1.5. Infinite series

A formal infinite series $\sum_{k=1}^{\infty} a_k$ is said to be *convergent* to a *sum* s if its sequence $\{s_n\}$ of *partial sums* $s_n = \sum_{k=1}^{n} a_k$ converges to s. One very important example is the *geometric series* $\sum_{k=0}^{\infty} r^k$, with partial sums

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}\,, \qquad r \neq 1\,.$$

The series converges if and only if $|r| < 1$, in which case its sum is $s = \frac{1}{1-r}$.

If a series converges, then $a_n = s_n - s_{n-1} \to s - s = 0$ as $n \to \infty$. In other words, a necessary condition for convergence of an infinite series $\sum a_k$ is that its general term a_k tend to zero as $k \to \infty$. However, this condition is far from sufficient.

Infinite series with terms of constant sign are easiest to analyze. If $a_k \geq 0$ for all k, then $s_n \leq s_{n+1}$ and the partial sums form a nondecreasing sequence. By the monotone boundedness theorem, then, the series converges if and only if its partial sums are bounded above. For an example, let $a_k = 1/k$ and observe that

$$\begin{aligned} s_{2^m} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \dots \\ &\qquad + \left(\frac{1}{2^{m-1}+1} + \dots + \frac{1}{2^m}\right) \\ &\geq 1 + \frac{1}{2} + 2 \cdot \frac{1}{4} + 4 \cdot \frac{1}{8} + \dots + 2^{m-1} \cdot \frac{1}{2^m} = 1 + \frac{m}{2} \to \infty \end{aligned}$$

as $m \to \infty$, which shows that the series $\sum_{k=1}^{\infty} 1/k$ diverges. On the other hand, if $a_k = 1/k^2$, then

$$\begin{aligned} s_{2^m} &= 1 + \left(\frac{1}{2^2} + \frac{1}{3^2}\right) + \left(\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2}\right) + \dots \\ &\qquad + \left(\frac{1}{2^{2(m-1)}} + \dots + \frac{1}{(2^m - 1)^2}\right) + \frac{1}{2^{2m}} \\ &\leq 1 + 2 \cdot \frac{1}{2^2} + 4 \cdot \frac{1}{4^2} + \dots + 2^{m-1} \cdot \frac{1}{2^{2(m-1)}} + \frac{1}{2^{2m}} \\ &= 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{m-1}} + \frac{1}{2^{2m}} < 2\,, \end{aligned}$$

so the sequence $\{s_n\}$ of partial sums is bounded and the series $\sum_{k=1}^{\infty} 1/k^2$ converges. In similar manner it can be seen that the series $\sum_{k=1}^{\infty} 1/k^p$ converges if $p > 1$ and diverges if $p \leq 1$.

Series with positive decreasing terms are often analyzed most easily by comparison with an integral. To be more specific, suppose $f(x)$ is a positive

continuous nonincreasing function on the interval $[1, \infty)$. Then by comparison of areas it is apparent that

$$\sum_{k=2}^{n} f(k) \le \int_1^n f(x)\,dx \le \sum_{k=1}^{n-1} f(k)\,, \qquad n = 2, 3, \dots\,,$$

which shows that the sum $\sum_{k=1}^{\infty} f(k)$ and the integral $\int_1^{\infty} f(x)\,dx$ converge or diverge together.

A series $\sum a_k$ is said to be *absolutely convergent* if $\sum |a_k|$ is convergent.

Theorem. *Every absolutely convergent series is convergent.*

Proof. Suppose that a series $\sum a_k$ is absolutely convergent. Let

$$s_n = a_1 + a_2 + \cdots + a_n \qquad \text{and} \qquad t_n = |a_1| + |a_2| + \cdots + |a_n|\,.$$

By hypothesis, the sequence $\{t_n\}$ is convergent, and so it is a Cauchy sequence. Thus for each $\varepsilon > 0$ there exists a number N such that $|t_n - t_m| < \varepsilon$ whenever $n > m \ge N$. But

$$|s_n - s_m| = |a_{m+1} + a_{m+2} + \cdots + a_n| \le |a_{m+1}| + |a_{m+2}| + \cdots + |a_n| = t_n - t_m\,,$$

so it follows that $\{s_n\}$ is also a Cauchy sequence, hence is convergent. This says that the series $\sum a_k$ is convergent, as claimed. □

The theorem just proved is essentially equivalent to the *comparison test* for convergence of an infinite series. Specifically, if $|a_k| \le b_k$ and $\sum b_k$ is convergent, then $\sum a_k$ is (absolutely) convergent.

Comparison with a geometric series is facilitated by two special devices known as the *ratio test* and the *root test*.

Ratio Test. *If $a_k \ne 0$ for all k, the infinite series $\sum a_k$ is convergent if*

$$\limsup_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| < 1 \quad \textit{and divergent if} \quad \liminf_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right| > 1\,.$$

Root Test. *The infinite series $\sum a_k$ is convergent if*

$$\limsup_{k\to\infty} \sqrt[k]{|a_k|} < 1 \quad \textit{and divergent if} \quad \limsup_{k\to\infty} \sqrt[k]{|a_k|} > 1\,.$$

Proofs. In view of the theorem at the end of Section 1.4, the validity of the ratio test follows from that of the root test. More specifically, the condition for convergence in the ratio test implies that in the root test, and the conditions for divergence are similarly related. Therefore, it will suffice to prove the validity of the root test.

Let $\beta = \limsup_{k\to\infty} \sqrt[k]{|a_k|}$ and suppose that $\beta < 1$. Then for any number r with $\beta < r < 1$ there is an index N such that $\sqrt[k]{|a_k|} \le r$ for all $k \ge N$. Thus $|a_k| \le r^k$ for all $k \ge N$ and the series $\sum a_k$ is absolutely convergent, hence convergent, by comparison with the geometric series $\sum r^k$. If $\beta > 1$, choose any number ρ with $1 < \rho < \beta$ and note that $\sqrt[k]{|a_k|} > \rho$ for infinitely many indices k, by the definition of "lim sup". Consequently, $|a_k| > \rho^k \to \infty$ as $k \to \infty$ through some subsequence. In particular, the general term of the series $\sum a_k$ does not tend to zero, so the series is divergent. □

Although the root test is more powerful than the ratio test, the latter is often more convenient to apply, especially when factorials are involved.

The question of convergence becomes more delicate when the series is not absolutely convergent. The basic theorem on series with alternating signs goes back to Leibniz and is a useful test for convergence. Gottfried Wilhelm Leibniz (1646–1716) is known as a founder of calculus, along with Isaac Newton (1643–1727).

Leibniz' Alternating Series Theorem. *Suppose that* $a_1 \ge a_2 \ge a_3 \ge \cdots \ge 0$ *and* $\lim_{k\to\infty} a_k = 0$. *Then the infinite series*

$$\sum_{k=1}^{\infty} (-1)^{k+1} a_k \;=\; a_1 - a_2 + a_3 - a_4 + \dots$$

is convergent.

Proof. Let

$$s_n = \sum_{k=1}^{n} (-1)^{k+1} a_k$$

denote the partial sums. The strategy is to apply the monotonicity to show that each of the subsequences $\{s_{2n}\}$ and $\{s_{2n+1}\}$ is convergent, then to use the hypothesis that $a_k \to 0$ to see that the two limits agree.

Observe first that

$$s_{2n+2} = s_{2n} + (a_{2n+1} - a_{2n+2}) \ge s_{2n}\,,$$

since $a_{2n+1} \ge a_{2n+2}$ by hypothesis. This shows that

$$s_2 \le s_4 \le s_6 \le s_8 \le \dots\,.$$

Similarly,

$$s_{2n+3} = s_{2n+1} - (a_{2n+2} - a_{2n+3}) \le s_{2n+1}\,,$$

so that

$$s_1 \ge s_3 \ge s_5 \ge s_7 \ge \dots\,.$$

In other words, the sequence $\{s_{2n}\}$ is nondecreasing, whereas the sequence $\{s_{2n+1}\}$ is nonincreasing. To see that both subsequences are bounded, note that

$$s_2 \le s_{2n} \le s_{2n} + a_{2n+1} = s_{2n+1} \le s_1\,.$$

Therefore, by the monotone boundedness theorem, each of the subsequences $\{s_{2n}\}$ and $\{s_{2n+1}\}$ is convergent. Let

$$s = \lim_{n\to\infty} s_{2n} \qquad \text{and} \qquad t = \lim_{n\to\infty} s_{2n+1}\,.$$

Now invoke the hypothesis that $a_k \to 0$ to show that the two limits agree:

$$t = \lim_{n\to\infty} s_{2n+1} = \lim_{n\to\infty} (s_{2n} + a_{2n+1}) = s + 0 = s\,.$$

From this it is easy to deduce that the full sequence $\{s_n\}$ converges to the same limit s. □

As a simple application of the theorem, we can infer that the infinite series

$$1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \dots$$

is convergent. We will see later, in several different ways, that its sum is $\log 2$. This is an example of a *conditionally convergent* series, convergent but not absolutely convergent.

We now record a less familiar result known as Cauchy's double series theorem, an important tool that will be applied to power series in Chapter 3. By way of orientation, recall that an area integral of a function of two real variables can be iterated in two ways, and an interchange of the order of integration is often productive, although it is not always permissible and must be justified. Cauchy's theorem deals with a discrete analogue, a double series

$$\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm}\,,$$

where the a_{nm} are real numbers. Such a double series is said to *converge to the sum s* if for each fixed n the series $\sum_{m=1}^{\infty} a_{nm}$ converges to a sum A_n and $\sum_{n=1}^{\infty} A_n = s$. The double series is *absolutely convergent* if $\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} |a_{nm}|$ is convergent.

Cauchy's Double Series Theorem. *If either of the double series*

$$\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm} \qquad \textit{or} \qquad \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{nm}$$

is absolutely convergent, then both are convergent and their sums are equal.

Without the hypothesis of absolute convergence, the statement is false. Even if both series converge, their sums need not be equal (*cf.* Exercise 34).

Proof of theorem. Suppose first that $a_{nm} \geq 0$ for all $n, m = 1, 2, \ldots$, and assume that $\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm}$ is (absolutely) convergent. In other words, $\sum_{m=1}^{\infty} a_{nm} = A_n < \infty$ for each n, and $\sum_{n=1}^{\infty} A_n$ converges. Then $a_{nm} \leq A_n$ and so

$$B_m = \sum_{n=1}^{\infty} a_{nm} \leq \sum_{n=1}^{\infty} A_n < \infty \qquad \text{for each } m\,.$$

Also, $a_{n1}+a_{n2}+\cdots+a_{nm} \leq A_n$, so that $B_1+B_2+\cdots+B_m \leq \sum_{n=1}^{\infty} A_n < \infty$. It follows that

$$B = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{nm} = \sum_{m=1}^{\infty} B_m \leq \sum_{n=1}^{\infty} A_n = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm} = A\,.$$

In particular, the left-hand double series is convergent, so by interchanging the roles of n and m we can deduce by a similar argument that $A \leq B$. Hence $A = B$ and the theorem is proved in the special case where all of the numbers a_{nm} have the same sign.

In the general case where the numbers a_{nm} have variable signs, we write $a_{nm} = \alpha_{nm} + \beta_{nm}$, where $\alpha_{nm} = a_{nm}$ if $a_{nm} > 0$ and $\alpha_{nm} = 0$ otherwise. Thus $\alpha_{nm} \geq 0$ and $\beta_{nm} \leq 0$, so we infer from the special case already proved that

$$\begin{aligned}\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm} &= \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} \alpha_{nm} + \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} \beta_{nm} \\ &= \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \alpha_{nm} + \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \beta_{nm} = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{nm}\,,\end{aligned}$$

as claimed. □

1.6. Continuous functions and derivatives

A real-valued function f defined on a set $S \subset \mathbb{R}$ is said to be *continuous* at a point $x_0 \in S$ if for each $\varepsilon > 0$ there exists a number $\delta > 0$ such that

$|f(x) - f(x_0)| < \varepsilon$ for all $x \in S$ with $|x - x_0| < \delta$. It is equivalent to require that $f(x_n) \to f(x_0)$ for each sequence of points $x_n \in S$ such that $x_n \to x_0$. A function is said to be continuous on the set S if it is continuous at each point of S.

It is easy to prove that if f and g are defined on a common set S and are continuous at a point $x_0 \in S$, then their sum $f + g$ and product fg are continuous at x_0, and the quotient f/g is continuous there if $g(x_0) \neq 0$. This allows us to conclude that every polynomial is continuous on $\mathbb{R}$, since it is trivially true that the function $f(x) = x$ is continuous. More generally, every rational function (the quotient of two polynomials) is continuous except at zeros of the denominator.

An important property of continuous functions is that they map connected sets to connected sets. To put it in more prosaic terms, if a continuous function on an interval assumes two different values, then it must assume every value in between. Although this fact is intuitively obvious, the proof is not trivial.

Intermediate Value Theorem. *If a function f is continuous on an interval $[a, b]$ and if $f(a) < \lambda < f(b)$, then $f(\xi) = \lambda$ for some $\xi \in (a, b)$.*

Of course, the same conclusion holds if $f(b) < \lambda < f(a)$.

Proof of theorem. Let $S = \{x \in [a, b] : f(x) < \lambda\}$ and define $\xi = \sup S$. Then $\xi \in (a, b)$ since $f(x) < \lambda$ in some interval $[a, a + \delta)$ and $f(x) > \lambda$ in some interval $(b - \delta, b]$. If $f(\xi) < \lambda$, then $f(x) < \lambda$ throughout some neighborhood of ξ, which contradicts the definition of ξ as an upper bound of S. If $f(\xi) > \lambda$, then $f(x) > \lambda$ in some neighborhood of ξ, and so ξ cannot be the smallest upper bound of S. Hence $f(\xi) = \lambda$. □

A function f is *bounded above* on a set S if its range of values $f(S) = \{f(x) : x \in S\}$ has an upper bound. Similarly, f is *bounded below* on S if $f(S)$ has a lower bound. The function is said to be *bounded* on S if it is bounded above and below; that is, if $|f(x)| \leq B$ for some constant B and all $x \in S$. We say that f has a *maximum* at a point $x_0 \in S$ if $f(x) \leq f(x_0)$ for all $x \in S$. Similarly, f has a *minimum* at x_1 if $f(x) \geq f(x_1)$ for all $x \in S$.

Theorem. *If a function is continuous on a closed bounded set $S \subset \mathbb{R}$, then it attains a maximum and a minimum on S.*

Proof. Let f be continuous on S, and let $M = \sup\{f(x) : x \in S\}$, where $M = +\infty$ if f is not bounded above on S. Let $\{x_n\}$ be a sequence of points in S for which $f(x_n) \to M$ as $n \to \infty$. By the Bolzano–Weierstrass theorem, some subsequence $\{x_{n_k}\}$ converges to a point $x_0 \in S$. Since f is continuous

at x_0, it follows that $f(x_{n_k}) \to f(x_0)$ as $k \to \infty$. But $f(x_{n_k}) \to M$, so this proves that $M = f(x_0) < \infty$. Hence f attains a maximum value at x_0. A similar argument shows that f attains a minimum on S. □

A function f defined on a set $S \subset \mathbb{R}$ is said to be *uniformly continuous* on S if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - f(t)| < \varepsilon$ for all pairs of points $x, t \in S$ with $|x - t| < \delta$. Every uniformly continuous function is pointwise continuous, but the converse is false. For instance, the function $f(x) = 1/x$ is continuous at each point of the interval $(0, 1]$ but is not uniformly continuous there. Similarly, the function $f(x) = x^2$ is pointwise continuous but not uniformly continuous on $[0, \infty)$. However, the converse is true if the set S is closed and bounded, as we now proceed to prove.

Theorem. *If a function is continuous at each point of a closed bounded set $S \subset \mathbb{R}$, then it is uniformly continuous on S.*

Proof. The result can be obtained with help of the Bolzano–Weierstrass theorem, but it is more natural to base a proof on the Heine–Borel theorem. Suppose a function f is pointwise continuous on the closed bounded set S. Given $\varepsilon > 0$, we want to produce a number $\delta > 0$ such that $|f(x) - f(t)| < \varepsilon$ for all pairs of points $x, t \in S$ with $|x - t| < \delta$. The continuity of f at each point $x \in S$ says that there is a number $\delta(x) > 0$ depending on x such that

$$|f(t) - f(x)| < \frac{\varepsilon}{2} \qquad \text{for all } \ t \in S \cap I(x)\,,$$

where $I(x)$ is the interval $(x - \delta(x), x + \delta(x))$. Define the smaller interval

$$J(x) = (x - \tfrac{1}{2}\delta(x), x + \tfrac{1}{2}\delta(x)) \subset I(x)\,.$$

The collection $\mathcal{C} = \{J(x) : x \in S\}$ of all such intervals $J(x)$ is an open covering of S, since each $x \in S$ belongs to its associated set $J(x)$. Thus by the Heine–Borel theorem, $\mathcal{C}$ contains a finite subcovering of S. In other words, there is a finite set of points $x_1, x_2, \ldots, x_n \in S$ for which

$$S \subset J(x_1) \cup J(x_2) \cup \cdots \cup J(x_n)\,.$$

Now set $\delta = \frac{1}{2}\min\{\delta(x_1), \delta(x_2), \ldots, \delta(x_n)\}$, and observe that $\delta > 0$. Choose arbitrary points $x, t \in S$ with $|x - t| < \delta$. Then $x \in J(x_k)$ for some k, since the sets $J(x_1), \ldots, J(x_n)$ form a covering of S. Hence $x \in I(x_k)$. But the point t also belongs to the interval $I(x_k)$, in view of the inequality

$$|t - x_k| \le |t - x| + |x - x_k| < \delta + \tfrac{1}{2}\delta(x_k) \le \delta(x_k)\,.$$

Because the points x and t belong to the same interval $I(x_k)$, we conclude that

$$|f(x) - f(t)| \le |f(x) - f(x_k)| + |f(x_k) - f(t)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon\,,$$

as desired. This proves that f is uniformly continuous on S. □

Now let $f(x)$ be a real-valued function defined on an arbitrary set $S \subset \mathbb{R}$, and suppose that a is a cluster point of S. Then $f(x)$ is said to have a *limit* L at a if for each $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in S$ with $0 < |x - a| < \delta$. This is indicated by writing

$$\lim_{x \to a} f(x) = L \qquad \text{or} \qquad f(x) \to L \ \text{ as } \ x \to a\,.$$

It is equivalent to require that $f(x_n) \to L$ for every sequence of points $x_n \in S$ such that $x_n \to a$ as $n \to \infty$.

It is sometimes convenient to speak of one-sided limits. If a is a right-hand cluster point of S, meaning that every neighborhood of a contains a point $x \in S$ with $x > a$, then the function $f(x)$ has a *right-hand limit* L if for each $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in S$ with $0 < x - a < \delta$. This is indicated by writing

$$\lim_{x \to a+} f(x) = L \qquad \text{or} \qquad f(x) \to L \ \text{ as } \ x \to a+\ .$$

Similarly, if the point a is a left-hand cluster point of S, then $f(x)$ is said to have a *left-hand limit* L, written

$$\lim_{x \to a-} f(x) = L \qquad \text{or} \qquad f(x) \to L \ \text{ as } \ x \to a-\ .$$

if to each $\varepsilon > 0$ there corresponds a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in S$ with $0 < a - x < \delta$.

A limit of special importance is the *derivative*

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}\,,$$

where f is a function defined in an open interval containing the point x_0. If the derivative exists, the function f is said to be *differentiable* at the point x_0. A function is said to be differentiable on an open set S if it has a derivative at each point of S. One-sided derivatives can be defined as one-sided limits of the difference quotient.

The familiar rules of differentiation are easily deduced. The derivative of the sum of two differentiable functions is the sum of their derivatives,

the product $f(x)g(x)$ has derivative $f'(x)g(x) + f(x)g'(x)$, and the quotient $f(x)/g(x)$ has derivative

$$\frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

wherever $g(x) \neq 0$. The "chain rule" for differentiation of the composition $h(x) = f(g(x))$ is more subtle but is found to be $h'(x) = f'(g(x))g'(x)$.

Every student of calculus learns that the maximum and minimum of a function are likely to occur at a *critical point*; that is, at a point where the derivative vanishes. This principle has a more precise statement, and it applies more generally to local maxima and minima. If a function f is defined on an open set S containing a point x_0, we say that it has a *local maximum* at x_0 if $f(x) \leq f(x_0)$ for all x in some neighborhood

$$N_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\} \subset S\,.$$

A *local minimum* is defined similarly. Here is the precise statement.

Theorem. *If a function f is defined on an open set S and has a local maximum or local minimum at a point $x_0 \in S$, and if f is differentiable at x_0, then $f'(x_0) = 0$.*

Proof. It suffices to consider a local minimum. By hypothesis, $f(x) \geq f(x_0)$ for all x in some neighborhood $N_\delta(x_0)$. Consequently, the difference quotient

$$g(x) = \frac{f(x) - f(x_0)}{x - x_0}\,, \qquad x \neq x_0\,,$$

defined in $N_\delta(x_0)$ except at x_0, satisfies the inequalities $g(x) \geq 0$ for $x > x_0$ and $g(x) \leq 0$ for $x < x_0$. Since f is differentiable at x_0, it follows that

$$f'(x_0) = \lim_{x \to x_0+} g(x) \geq 0 \qquad \text{and} \qquad f'(x_0) = \lim_{x \to x_0-} g(x) \leq 0\,.$$

Hence $f'(x_0) = 0$, as claimed. □

It is now a short step to a time-honored result.

Rolle's Theorem. *If a function f is continuous in an interval $[a, b]$ and differentiable in the open interval (a, b), and if $f(a) = f(b)$, then $f'(\xi) = 0$ for some point $\xi \in (a, b)$.*

Proof. If f is constant in $[a, b]$, then $f'(x) \equiv 0$ in (a, b) and the conclusion is trivial. If f is not constant, then f attains a maximum or a minimum at some point $\xi \in (a, b)$, and it follows from the previous theorem that $f'(\xi) = 0$. □

Rolle's theorem can be generalized as follows.

Mean Value Theorem. *If a function f is continuous in an interval $[a, b]$ and differentiable in the open interval (a, b), then*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} \qquad \textit{for some } \ \xi \in (a, b)\,.$$

Proof. Define the linear function

$$L(x) = \frac{f(b) - f(a)}{b - a}(x - a) + f(a)\,,$$

so that $L(a) = f(a)$ and $L(b) = f(b)$. Then apply Rolle's theorem to the function $g(x) = f(x) - L(x)$. □

The mean value theorem can be applied to show that a function with zero derivative throughout an interval must be constant. More generally, if the derivatives of two functions coincide in some interval, the two functions must differ by a constant. Another corollary of the mean value theorem is that a function with positive derivative in an interval is strictly increasing. Similarly, a function with negative derivative is strictly decreasing.

1.7. The Riemann integral

Loosely speaking, the integral of a positive function is the area under its graph. The problem is to make that notion precise, and to extend it to a wider class of functions. Such an effort leads naturally to the definition of the Riemann integral. It is often possible to calculate an integral directly from the definition, but the most effective method comes via the fundamental theorem of calculus, which allows calculation of an integral by a process of antidifferentiation. In fact, the deepest results emerge when both methods are applied to the same problem, as in the derivation of Stirling's formula to be given in the next chapter.

For a formal development of the Riemann integral, consider a function f defined and bounded on an interval $[a, b]$. Let P be an arbitrary partition $a = x_0 < x_1 < x_2 < \cdots < x_n = b$ of the interval $[a, b]$. For $k = 1, 2, \ldots, n$, let

$$M_k = \sup\{f(x) : x_{k-1} \le x \le x_k\} \quad \text{and} \quad m_k = \inf\{f(x) : x_{k-1} \le x \le x_k\}\,,$$

and form the upper and lower sums

$$U(f, P) = \sum_{k=1}^{n} M_k(x_k - x_{k-1}) \qquad \text{and} \qquad L(f, P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1})$$

of f with respect to the partition P. Clearly, $L(f,P) \le U(f,P)$ for every partition P. If the partition P is refined to a partition $Q \supset P$ by adjoining extra points of subdivision, it can be seen that $L(f,P) \le L(f,Q)$ and $U(f,Q) \le U(f,P)$. In particular, if Q is a common refinement of two partitions P_1 and P_2, then

$$L(f,P_1) \le L(f,Q) \le U(f,Q) \le U(f,P_2)\,.$$

Thus $L(f,P_1) \le U(f,P_2)$ for every pair of partitions P_1 and P_2, and so it follows that

$$\sup_P L(f,P) \le \inf_P U(f,P)\,.$$

In other words, the supremum of the lower sums taken over all partitions is less than or equal to the infimum of the upper sums. If equality occurs, the function f is said to be *Riemann integrable* over the interval $[a,b]$, and its integral is defined to be the common value, written as

$$\int_a^b f(x)\,dx = \sup_P L(f,P) = \inf_P U(f,P)\,.$$

Equivalently, we can say that f is Riemann integrable if and only if for each $\varepsilon > 0$ there exists a partition P such that $U(f,P) - L(f,P) < \varepsilon$.

Not every bounded function is Riemann integrable. Consider, for instance, the function defined by $f(x) = 1$ if x is rational and $f(x) = 0$ if x is irrational. Then $L(f,P) = 0$ and $U(f,P) = b - a$ for every partition P, since both the rationals and the irrationals are everywhere dense. Thus f is not Riemann integrable.

On the other hand, every continuous function on an interval $[a,b]$ is integrable. To see this, we apply the theorem (proved in the previous section) that every continuous function on a closed bounded interval is uniformly continuous there. Thus the pointwise continuity of a function f implies that for each $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - f(t)| < \varepsilon/(b-a)$ for all pairs of points $x, t \in [a,b]$ with $|x - t| < \delta$. Now choose a partition P with n points and all subintervals of length less than δ. Then $M_k - m_k \le \varepsilon/(b-a)$ and so

$$U(f,P) - L(f,P) = \sum_{k=1}^{n} (M_k - m_k)(x_k - x_{k-1}) \le \varepsilon\,,$$

which shows that f is integrable over $[a,b]$.

The argument can be modified to show that f is integrable if it is continuous except for a finite number of jump discontinuities. It can also be seen that every monotonic bounded function is integrable. Suppose, for instance,

that f is nondecreasing and $|f(x)| \le B$ on $[a,b]$. Given $\varepsilon > 0$, choose a partition P comprised of n subintervals of length less than $\varepsilon/2B$. Then

$$U(f,P)-L(f,P) = \sum_{k=1}^{n} \big[f(x_k)-f(x_{k-1})\big](x_k-x_{k-1}) < \frac{\varepsilon}{2B}\big[f(b)-f(a)\big] \le \varepsilon\,.$$

Hence f is integrable over $[a,b]$.

Standard properties of the integral can now be derived. For instance, if f and g are integrable over $[a,b]$, then so is $c_1 f + c_2 g$ for any constants c_1 and c_2, and

$$\int_a^b \big[c_1 f(x) + c_2 g(x)\big]\,dx = c_1 \int_a^b f(x)\,dx + c_2 \int_a^b g(x)\,dx\,.$$

The product fg is also integrable. If f and g are integrable and $f(x) \le g(x)$ on $[a,b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx\,.$$

In particular, the inequality

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$$

holds whenever f is integrable. (Here the main difficulty is to show that $|f|$ is integrable.) If $a < c < b$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx\,.$$

Proofs are omitted here, but can be found in introductory texts such as Ross [**7**].

It is often possible to calculate an integral directly from the definition. The technique was known to Archimedes, who essentially applied a limiting process to calculate the volumes of geometric solids such as the cone and sphere. As a simple illustration, let us calculate the integral of $f(x) = x^2$ over the interval $[0,1]$. Let P_n denote the partition of $[0,1]$ with n subintervals of equal length $1/n$. Then

$$L(f,P_n) = \sum_{k=1}^{n} f(x_{k-1})(x_k - x_{k-1}) = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{k-1}{n}\right)^2 = \frac{(n-1)(2n-1)}{6n^2}$$

and

$$U(f,P_n) = \sum_{k=1}^{n} f(x_k)(x_k - x_{k-1}) = \frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^2 = \frac{(n+1)(2n+1)}{6n^2}\,,$$

where the formula for the sum of squares (*cf.* Exercise 1) has been invoked. Since both $L(f, P_n)$ and $U(f, P_n)$ converge to $1/3$ as $n \to \infty$, we conclude that $\int_0^1 x^2\, dx = 1/3$.

Riemann's original definition of the integral was based not on upper and lower sums but on "Riemann sums", as they are now called. Given a partition P of $[a, b]$ in the form $a = x_0 < x_1 < x_2 < \cdots < x_n = b$, select an arbitrary point ξ_k in each subinterval $[x_{k-1}, x_k]$ and let $\xi = (\xi_1, \xi_2, \ldots, \xi_n)$. Then the corresponding *Riemann sum* is

$$S(f, P, \xi) = \sum_{k=1}^{n} f(\xi_k)(x_k - x_{k-1})\,.$$

It is clear that $L(f, P) \le S(f, P, \xi) \le U(f, P)$ for every partition P and for each choice ξ of intermediate points. With $\|P\| = \max_k(x_k - x_{k-1})$ denoting the *norm* of the partition P, it can be shown that the Riemann sums converge to the integral as $\|P\| \to 0$. More precisely, we have the following theorem (proof omitted).

Theorem. *A bounded function f is Riemann integrable on an interval $[a, b]$ if and only if there is a number I with the following property. For each $\varepsilon > 0$ there exists a $\delta > 0$ such that $|S(f, P, \xi) - I| < \varepsilon$ for each partition P with $\|P\| < \delta$ and all choices $\xi = (\xi_1, \xi_2, \ldots, \xi_n)$ of intermediate points. If such a number I exists, then $I = \int_a^b f(x)dx$.*

The intermediate value theorem (*cf.* Section 1.6) for continuous functions can be applied to establish a corresponding theorem for integrals.

Intermediate Value Theorem for Integrals. *If a function f is continuous on an interval $[a, b]$, then*

$$\frac{1}{b-a}\int_a^b f(x)\,dx = f(x_0) \qquad \textit{for some} \quad x_0 \in [a, b]\,.$$

Proof. As previously shown (*cf.* Section 1.6), the continuous function f attains a minimum value m and a maximum value M on the interval $[a, b]$. Since $m \le f(x) \le M$ for all $x \in [a, b]$, it follows that

$$m \le \frac{1}{b-a}\int_a^b f(x)\,dx \le M\,.$$

An appeal to the intermediate value theorem for continuous functions now gives the desired result. □

We now turn to the fundamental theorem of calculus, which says that integration and differentiation are inverse processes. It is convenient to adopt the convention that

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx\,.$$

Fundamental Theorem of Calculus.

(*a*) *Let f be an integrable function on an interval $[a,b]$, and define*

$$F(x) = \int_a^x f(t)\,dt\,, \qquad a \le x \le b\,.$$

Then F is continuous on $[a,b]$. Furthermore, F is differentiable at each point $x_0 \in (a,b)$ where f is continuous, and $F'(x_0) = f(x_0)$.

(*b*) *Let g be a function continuous on $[a,b]$ and differentiable on (a,b), and suppose that g' is integrable on $[a,b]$. Then*

$$\int_a^b g'(x)\,dx = g(b) - g(a)\,.$$

Part (b) of the theorem provides the standard device of antidifferentiation to compute an integral. It is also the basis for the well-known formula of *integration by parts*:

$$\int_a^b u(x)v'(x)\,dx = u(b)v(b) - u(a)v(a) - \int_a^b v(x)u'(x)\,dx\,.$$

Proof of theorem. (a). By definition, an integrable function is bounded. Suppose that $|f(x)| \le B$ for all $x \in [a,b]$. Then

$$|F(y) - F(x)| = \left|\int_x^y f(t)\,dt\right| \le B|y-x|\,, \qquad x,y \in [a,b]\,,$$

which shows that F is (uniformly) continuous in $[a,b]$. For $x_0 \in (a,b)$, write

$$\frac{F(x)-F(x_0)}{x-x_0} - f(x_0) = \frac{1}{x-x_0}\int_{x_0}^x \big[f(t)-f(x_0)\big]\,dt\,, \qquad x \in (a,b),\ x \ne x_0\,.$$

Let $\varepsilon > 0$ be given. If f is continuous at x_0, then for some $\delta > 0$ the inequality $|f(x) - f(x_0)| < \varepsilon$ holds for all $x \in [a,b]$ with $|x - x_0| < \delta$. Therefore,

$$\left|\frac{F(x)-F(x_0)}{x-x_0} - f(x_0)\right| < \varepsilon \qquad \text{if } \ 0 < |x-x_0| < \delta\,,$$

which proves that F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

(b). Since g' is integrable, for each $\varepsilon > 0$ there is a partition P such that

$$U(g', P) - L(g', P) < \varepsilon\,.$$

On the other hand, an application of the mean value theorem to the subinterval $[x_{k-1}, x_k]$ of this partition P gives

$$g'(\xi_k)(x_k - x_{k-1}) = g(x_k) - g(x_{k-1}) \qquad \text{for some } \ \xi_k \in (x_{k-1}, x_k)\,.$$

Addition then produces the Riemann sum

$$S(g', P, \xi) = \sum_{k=1}^{n} g'(\xi_k)(x_k - x_{k-1}) = \sum_{k=1}^{n} \big[g(x_k) - g(x_{k-1})\big] = g(b) - g(a)\,.$$

Since $L(g', P) \le S(g', P, \xi) \le U(g', P)$, we see that both of the numbers $[g(b) - g(a)]$ and $\int_a^b g'(x)dx$ lie between $L(g', P)$ and $U(g', P)$, so that

$$\left| \int_a^b g'(x)\,dx - \big[g(b) - g(a)\big] \right| < \varepsilon\,.$$

Because $\varepsilon > 0$ was prescribed arbitrarily, this gives the desired result. □

Integrals over unbounded intervals are defined by limiting processes. If a function f is bounded and Riemann integrable over each bounded interval $[a, b]$, we define

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx \qquad \text{and} \qquad \int_{-\infty}^b f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx\,,$$

provided the limits exist. Then we define

$$\int_{-\infty}^\infty f(x)\,dx = \lim_{a\to-\infty} \int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_{-\infty}^b f(x)\,dx\,,$$

assuming again that the relevant limits exist.

Integrals are defined in a similar way for integrands with a singularity at an endpoint of the interval of integration. For instance, if f is defined in a half-open interval $(a, b]$ and is bounded and integrable over each subinterval $[c, b]$ for $a < c < b$, we define

$$\int_a^b f(x)\,dx = \lim_{c\to a+} \int_c^b f(x)\,dx$$

if the limit exists. For example,

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{c\to 0+} \big[2\sqrt{x}\big]_c^1 = 2\,.$$

1.8. Uniform convergence

Suppose that $f_1, f_2, \ldots$ are functions defined on a set $S \subset \mathbb{R}$. The sequence $\{f_n(x)\}$ is said to be *pointwise convergent* on S to a function $f(x)$ if for each point $x_0 \in S$ the numerical sequence $\{f_n(x_0)\}$ converges to $f(x_0)$. Simple examples show that desirable properties of the functions f_n such as continuity or integrability need not be preserved under pointwise convergence. However, under a stronger mode of convergence known as uniform convergence, the situation is more satisfactory.

By definition, a sequence of functions $\{f_n\}$ is *uniformly convergent* to f on a set S if for each $\varepsilon > 0$ there exists a number N such that $|f_n(x) - f(x)| < \varepsilon$ for all $x \in S$ whenever $n \geq N$. Uniform convergence implies pointwise convergence, but it imposes the stronger requirement that the number N depend only on ε and not on any particular point $x \in S$.

Some examples will illustrate the distinction. The functions $f_n(x) = x/n$ converge pointwise to $f(x) \equiv 0$ but the convergence is not uniform on $\mathbb{R}$. The functions $f_n(x) = x^n$ converge pointwise but not uniformly in the interval $[0, 1]$ to the limit $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. Note that each of the functions x^n is continuous on $[0, 1]$ but their limit function is not. The functions

$$g_n(x) = \begin{cases} nx\,, & 0 \leq x \leq 1/n \\ 2 - nx\,, & 1/n \leq x \leq 2/n \\ 0\,, & 2/n \leq x \leq 1 \end{cases}$$

converge pointwise to $g(x) \equiv 0$ on $[0, 1]$, but the convergence is not uniform.

On a more positive note, we now show that continuity is preserved under uniform convergence.

Theorem. *Suppose that the functions $f_1, f_2, \ldots$ are defined and continuous on a set $S \subset \mathbb{R}$, and that $f_n(x) \to f(x)$ uniformly on S as $n \to \infty$. Then f is continuous on S.*

Proof. Let $\varepsilon > 0$ be given. For any point $x_0 \in S$, the inequality

$$|f(x) - f(x_0)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)|$$

holds for all $x \in S$. By the uniform convergence, we infer that there exists a number N (depending only on ε) such that

$$|f(x) - f(x_0)| < \frac{\varepsilon}{3} + |f_n(x) - f_n(x_0)| + \frac{\varepsilon}{3} \qquad \text{for all } x \in S$$

whenever $n \geq N$. Choose any $n \geq N$ and apply the continuity of f_n to conclude that for some $\delta > 0$ the inequality $|f(x) - f(x_0)| < \varepsilon$ holds for all $x \in S$ with $|x - x_0| < \delta$. Thus f is continuous at x_0. □

An infinite series $\sum_{n=1}^{\infty} u_n(x)$ of functions defined on a set $S \subset \mathbb{R}$ is understood to be uniformly convergent on S to a sum $s(x)$ if its sequence of partial sums $s_n(x) = \sum_{k=1}^{n} u_k(x)$ converges uniformly to $s(x)$ on S. The previous theorem has an equivalent expression for series.

Corollary. *If an infinite series $\sum_{n=1}^{\infty} u_n(x)$ of continuous functions converges uniformly on a set S, then its sum is continuous on S.*

When a sequence of integrable functions $f_n(x)$ converges pointwise to $f(x)$ on an interval $[a, b]$, the limit function f need not be integrable. If f is integrable, it may well happen that the integrals $\int_a^b f_n(x)\,dx$ do not converge to $\int_a^b f(x)\,dx$. Consider, for example, $f_n(x) = ng_n(x)$, where $g_n(x)$ is the "triangle function" defined above. The sequence $\{f_n(x)\}$ converges pointwise (but not uniformly) on $[0, 1]$ to $f(x) \equiv 0$, yet

$$\int_0^1 f_n(x)\,dx = 1 \qquad \text{for every } n, \text{ whereas} \qquad \int_0^1 f(x)\,dx = 0\,.$$

A similar example is given in Exercise 31. The next theorem shows, however, that integrals behave as expected under uniform convergence.

Theorem. *If the functions $f_1, f_2, \ldots$ are integrable over an interval $[a, b]$ and $f_n(x) \to f(x)$ uniformly on $[a, b]$, then f is integrable and*

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \int_a^b f_n(x)\,dx\,.$$

Proof. The main difficulty is to show that f is integrable. Observe that

$$\begin{aligned} &U(f, P) - L(f, P) \\ &\quad \le |U(f, P) - U(f_n, P)| + |U(f_n, P) - L(f_n, P)| + |L(f_n, P) - L(f, P)| \end{aligned}$$

for any partition P of $[a, b]$. Let $\varepsilon > 0$. Then by the uniform convergence there exists an N such that the inequalities

$$|U(f, P) - U(f_n, P)| < \frac{\varepsilon}{3} \qquad \text{and} \qquad |L(f_n, P) - L(f, P)| < \frac{\varepsilon}{3}$$

hold for every partition P and all $n \ge N$. Now for any fixed $n \ge N$ the integrability of f_n allows the selection of a partition P for which $U(f_n, P) - L(f_n, P) < \varepsilon/3$. Hence $U(f, P) - L(f, P) < \varepsilon$, and f is integrable. Finally, another appeal to the uniform convergence shows that

$$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx \right| \le \int_a^b |f(x) - f_n(x)|\,dx < \varepsilon$$

for all n sufficiently large, which gives the desired result. □

Corollary. *If the functions $u_1, u_2, \dots$ are integrable over an interval $[a,b]$ and the infinite series $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly to $s(x)$ on $[a,b]$, then s is integrable and*

$$\int_a^b s(x)\,dx = \sum_{n=1}^{\infty} \int_a^b u_n(x)\,dx\,.$$

On the other hand, the uniform limit of a sequence of differentiable functions need not be differentiable, and its sequence of derivatives need not converge. For example, the functions $f_n(x) = \frac{1}{n}\sin(n^2x)$ converge to $f(x) \equiv 0$ uniformly on the whole real line, but their derivatives $f_n'(x) = n\cos(n^2x)$ do not even stay bounded as $n \to \infty$. Nevertheless, we have the following theorem.

Theorem. *Let the functions $f_1, f_2, \dots$ be continuous on an interval $[a,b]$ and have continuous derivatives on (a,b). Suppose that $f_n(x) \to f(x)$ for each $x \in [a,b]$ and that the sequence of derivatives $f_n'(x)$ is uniformly convergent on (a,b). Then f is differentiable on (a,b) and $f_n'(x) \to f'(x)$.*

Proof. Let $g(x)$ denote the uniform limit of $\{f_n'(x)\}$. Then by the two preceding theorems, g is continuous on (a,b) and

$$\int_a^x g(t)\,dt = \lim_{n\to\infty} \int_a^x f_n'(t)\,dt = \lim_{n\to\infty} \big[f_n(x) - f_n(a)\big] = f(x) - f(a)$$

for each $x \in (a,b)$. Thus by the fundamental theorem of calculus, f is differentiable and $f'(x) = g(x)$ on (a,b). □

Corollary. *Let the functions $u_1, u_2, \dots$ be continuous on an interval $[a,b]$ and have continuous derivatives on (a,b). Suppose that the infinite series $\sum_{n=1}^{\infty} u_n(x)$ converges to $s(x)$ for each $x \in [a,b]$, and that the series $\sum_{n=1}^{\infty} u_n'(x)$ is uniformly convergent on (a,b). Then s is differentiable on (a,b) and $s'(x) = \sum_{n=1}^{\infty} u_n'(x)$.*

The corollary provides a condition that allows an infinite series to be differentiated term by term. The continuous analogue of this process, differentiation under the integral sign, can be justified by a similar argument. In the simplest case the integral extends over a bounded interval.

Theorem. *Let $f(x,y)$ be continuous together with its partial derivative $\frac{\partial f}{\partial x}(x,y)$ on a rectangle*

$$R = [a,b] \times [c,d] = \{(x,y) : a \le x \le b\,,\ c \le y \le d\}\,, \qquad a < b\,,\ c < d\,.$$

Then

$$\frac{d}{dx}\int_c^d f(x,y)\,dy = \int_c^d \frac{\partial f}{\partial x}(x,y)\,dy\,.$$

In particular, the left-hand integral is differentiable.

Proof. The continuity of f and $\frac{\partial f}{\partial x}$ on the compact set $R \subset \mathbb{R}^2$ implies that both functions are uniformly continuous there, so that the integrals

$$\Phi(x) = \int_c^d f(x,y)\,dy \qquad \text{and} \qquad \Psi(x) = \int_c^d \frac{\partial f}{\partial x}(x,y)\,dy$$

are continuous on the interval $[a,b]$. Therefore, for each point $x \in [a,b]$ we have

$$\begin{aligned}\int_a^x \Psi(t)\,dt &= \int_a^x \int_c^d \frac{\partial f}{\partial t}(t,y)\,dy\,dt = \int_c^d \int_a^x \frac{\partial f}{\partial t}(t,y)\,dt\,dy \\ &= \int_c^d \big[f(x,y) - f(a,y)\big]\,dy = \Phi(x) - \Phi(a)\,.\end{aligned}$$

Thus by the fundamental theorem of calculus, Φ is differentiable and $\Phi'(x) = \Psi(x)$ for each $x \in (a,b)$. □

Essentially the same proof works for improper integrals

$$\Phi(x) = \int_c^\infty f(x,y)\,dy \qquad \text{and} \qquad \Psi(x) = \int_c^\infty \frac{\partial f}{\partial x}(x,y)\,dy\,, \qquad a \le x \le b\,,$$

under the additional hypotheses that the integral $\int_c^\infty f(x,y)\,dy$ converges pointwise and the integral $\int_c^\infty \frac{\partial f}{\partial x}(x,y)\,dy$ converges uniformly on $[a,b]$. Then $\Psi(t)$ is continuous on $[a,b]$, since it is the uniform limit of the continuous functions $\int_c^r (\partial f/\partial t)(t,y)\,dy$ as $r \to \infty$. The uniform convergence also implies that

$$\begin{aligned}\int_a^x \Psi(t)\,dt &= \lim_{r\to\infty} \int_a^x \int_c^r \frac{\partial f}{\partial t}(t,y)\,dy\,dt = \lim_{r\to\infty} \int_c^r \int_a^x \frac{\partial f}{\partial t}(t,y)\,dt\,dy \\ &= \Phi(x) - \Phi(a)\,,\end{aligned}$$

so that $\Phi'(x) = \Psi(x)$. Further details and related results can be found, for instance, in the book by Burkill and Burkill [**2**].

The remainder of this section is devoted to some important conditions for uniform convergence. We begin with the observation that the Cauchy criterion for convergence of a numerical sequence can be adapted to a criterion for uniform convergence. Let $f_1, f_2, \ldots$ be functions defined on a set $S \subset \mathbb{R}$. Then the sequence $\{f_n(x)\}$ is called a *uniform Cauchy sequence* on S if for each $\varepsilon > 0$ there is a number N such that

$$|f_n(x) - f_m(x)| < \varepsilon \quad \text{for all } x \in S$$

if $n, m \ge N$.

Theorem. *Every uniform Cauchy sequence is uniformly convergent.*

Proof. If $\{f_n(x)\}$ is a uniform Cauchy sequence on S, then clearly $\{f_n(x_0)\}$ is a numerical Cauchy sequence for each $x_0 \in S$. Thus by the Cauchy criterion, the sequence $\{f_n(x)\}$ converges pointwise on S to a function $f(x)$. To see that the convergence is uniform, apply the uniform Cauchy property:

$$|f_n(x) - f_m(x)| < \varepsilon \quad \text{for all } x \in S$$

when $n, m \geq N$. Hold $n \geq N$ fixed and let $m \to \infty$ to deduce that

$$|f_n(x) - f(x)| \leq \varepsilon \quad \text{for all } x \in S$$

if $n \geq N$. This proves the theorem. □

As an application of the uniform Cauchy criterion, we can now establish a very useful condition for uniform convergence of an infinite series, traditionally known as the *Weierstrass M-test*.

Theorem (Weierstrass M-test). *Let the functions $u_1, u_2, \ldots$ be defined on a set $S \subset \mathbb{R}$, and suppose that $|u_n(x)| \leq M_n$ for all $x \in S$, for $n = 1, 2, \ldots$. Suppose further that the numerical series $\sum_{n=1}^{\infty} M_n$ is convergent. Then the series $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly on S.*

Proof. Let $\varepsilon > 0$. Denote the partial sums of the series by

$$s_n(x) = \sum_{k=1}^{n} u_k(x)\,, \qquad n = 1, 2, \ldots.$$

Then for $m < n$ we see that

$$\begin{aligned} |s_n(x) - s_m(x)| = \left| \sum_{k=m+1}^{n} u_k(x) \right| &\leq \sum_{k=m+1}^{n} |u_k(x)| \\ &\leq \sum_{k=m+1}^{n} M_k < \varepsilon \qquad \text{for all } x \in S \end{aligned}$$

if m is sufficiently large, since the series $\sum_{k=1}^{\infty} M_k$ converges. Thus $\{s_n(x)\}$ is a uniform Cauchy sequence, and so it converges uniformly on S. □

We conclude this section with a special result, known as Dini's theorem, which provides another useful criterion for uniform convergence.

Dini's Theorem. *Suppose that the functions $f_1(x), f_2(x), \ldots$ are continuous on a closed bounded interval $[a,b]$, and $f_1(x) \geq f_2(x) \geq \cdots \geq 0$ for all $x \in [a,b]$. If $f_n(x) \to 0$ pointwise for each $x \in [a,b]$, then $f_n(x) \to 0$ uniformly in $[a,b]$.*

Proof. It is convenient to begin by extending each function f_n continuously to the entire real line by defining $f_n(x) = f_n(a)$ for $x < a$ and $f_n(x) = f_n(b)$ for $x > b$. Let $\varepsilon > 0$ be given. Then for each fixed $x_0 \in [a,b]$ there is an index $N = N(x_0)$ such that $0 \leq f_n(x_0) < \varepsilon/2$ for all $n \geq N$. By continuity it follows that $0 \leq f_N(x) < \varepsilon$ for all x in the interval $I(x_0) = (x_0 - \delta, x_0 + \delta)$ for some radius $\delta = \delta(x_0) > 0$. In view of the hypothesis that $0 \leq f_{n+1}(x) \leq f_n(x)$, this implies that $0 \leq f_n(x) < \varepsilon$ for all $n \geq N(x_0)$ and all $x \in I(x_0)$. Now observe that the collection $\{I(x_0) : x_0 \in [a,b]\}$ of all such intervals forms an open covering of $[a,b]$, and apply the Heine–Borel theorem to extract a finite subcollection $\{I(x_1), I(x_2), \ldots, I(x_m)\}$ whose union contains $[a,b]$. Taking N^* to be the maximum of the indices $N(x_1), N(x_2), \ldots, N(x_m)$, we conclude that $0 \leq f_n(x) < \varepsilon$ for all $x \in [a,b]$ whenever $n \geq N^*$. This proves the uniform convergence. □

Corollary 1. *If a sequence of continuous functions $f_n(x)$ converges monotonically to $f(x)$ for each $x \in [a,b]$ and if $f(x)$ is continuous, then the convergence is uniform.*

Proof. By assumption, either $f_n(x) \leq f_{n+1}(x) \leq f(x)$ or $f_n(x) \geq f_{n+1}(x) \geq f(x)$. Define $g_n(x) = f(x) - f_n(x)$ or $g_n(x) = f_n(x) - f(x)$, respectively, to obtain a sequence of continuous functions $g_n(x)$ that decreases pointwise to 0. The theorem then says that $g_n(x) \to 0$ uniformly in $[a,b]$ as $n \to \infty$. □

Corollary 2. *Let $f_n(x)$ and $f(x)$ be nonnegative continuous functions that are integrable over the real line. Suppose that either*

$$0 \leq f_n(x) \leq f_{n+1}(x) \leq f(x) \qquad \text{or} \qquad 0 \leq f(x) \leq f_{n+1}(x) \leq f_n(x)\,,$$

and that $f_n(x) \to f(x)$ pointwise in $\mathbb{R}$ as $n \to \infty$. Then

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f_n(x)\,dx = \int_{-\infty}^{\infty} f(x)\,dx\,.$$

Proof. Let $\varepsilon > 0$ be given. Suppose first that $0 \leq f_n(x) \leq f(x)$. Since f is integrable, we can choose a number $R > 0$ for which

$$0 \leq \int_{-\infty}^{-R} f(x)\,dx + \int_{R}^{\infty} f(x)\,dx < \frac{\varepsilon}{2}\,,$$

which implies that

$$0 \le \int_{-\infty}^{-R} [f(x) - f_n(x)]\,dx + \int_{R}^{\infty} [f(x) - f_n(x)]\,dx < \frac{\varepsilon}{2}$$

for all n. On the other hand, $f_n(x) \to f(x)$ uniformly on $[-R, R]$ by Dini's theorem, so that

$$0 \le \int_{-R}^{R} [f(x) - f_n(x)]\,dx < \frac{\varepsilon}{2}$$

for all n sufficiently large. Addition of these inequalities now shows that

$$0 \le \int_{-\infty}^{\infty} [f(x) - f_n(x)]\,dx < \varepsilon$$

for sufficiently large n, which is the desired result. The case where $0 \le f(x) \le f_n(x)$ is treated in similar fashion, with the integrability of f_1 allowing the reduction to a finite interval $[-R, R]$. □

Corollary 3. *If the functions $u_n(x)$ are continuous, nonnegative, and integrable over $\mathbb{R}$, and if the infinite series $\sum_{n=1}^{\infty} u_n(x)$ converges pointwise in $\mathbb{R}$ to a continuous sum $s(x)$ that is integrable over $\mathbb{R}$, then*

$$\int_{-\infty}^{\infty} s(x)\,dx = \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} u_n(x)\,dx\,.$$

Proof. Since $u_n(x) \ge 0$, the partial sums $s_n(x)$ increase to $s(x)$, so the conclusion follows from Corollary 2. □

1.9. Historical remarks

Sometime around the year 1670, Isaac Newton (1642–1727) and Gottfried Wilhelm Leibniz (1646–1716) discovered the basic principles of calculus. The techniques were effective in solving a whole host of scientific problems, notably problems of mechanics, but the logical foundations were insecure. Concepts such as limits were understood intuitively at best and were the focus of great controversy in early years. In fact, two centuries would elapse before those concepts were clearly formulated and the subject acquired a sound logical structure.

Leonhard Euler (1707–1783) was the greatest mathematician of the 18th century and the most prolific of all time. His work spanned virtually every aspect of mathematics and its applications. He wrote a series of three calculus texts that introduced the now-standard notation and served as models

for later treatments. In those texts Euler made functions, not curves, the primary objects of study. However, his definition of a function was rather restrictive and he accepted without question the notion of an "infinitely small" quantity known as a differential. His proofs and derivations of formulas tended to be nothing more than plausible arguments, not at all rigorous by modern standards. Joseph-Louis Lagrange (1736–1813), best known for his fundamental work in mechanics, tried to free calculus from differentials and limits by placing heavy emphasis on power series. His attempt was not successful, but he did give the first clear analysis of Taylor series expansion.

In 1821, Augustin-Louis Cauchy (1789–1857) published the first rigorous textbook in calculus, a *Cours d'analyse* written for his students at the École Royale Polytechnique in Paris. There he gave clear definitions of limits and continuity, essentially the definitions that are commonly accepted today. He formulated the notion of a Cauchy sequence and he introduced the definition of integral that was later refined and completed by Riemann. The book is a landmark in the history of mathematics. However, Cauchy failed to distinguish pointwise continuity from uniform continuity, and pointwise convergence from uniform convergence, making some of his arguments fallacious. Cauchy, like Euler, published a huge volume of work in many areas of mathematical science. Perhaps his greatest achievement was to develop the theory of functions of a complex variable, including the calculus of residues.

The dominant figure in the second half of the 19th century was Karl Weierstrass (1815–1897). His career was truly remarkable. For some years he taught at a Gymnasium, a German secondary school, carrying out high-level mathematical research in obscurity. Then at age 39 he published a groundbreaking paper in *Crelle's Journal*. It was an immediate sensation. Two years later, Weierstrass had moved to a professorship at the University of Berlin. There his lectures contained many original results and attracted students from all over the world, many of whom became outstanding mathematicians in later years. His lectures on introductory analysis perfected the program that Cauchy had initiated, bringing new rigor and clarity to calculus while introducing further results.

Bernard Bolzano (1781–1848), a Bohemian monk in Prague, was a contemporary of Cauchy who independently obtained many of the same results in the foundations of analysis. In 1817 he published a proof of the intermediate value theorem, often called "Bolzano's theorem". In the same long paper, Bolzano introduced the least upper bound principle and "Cauchy's criterion" for convergence of a sequence. However, his paper appeared in an obscure journal published in Prague, far from the scientific centers of the day, and there is no evidence that Cauchy knew of Bolzano's ideas before writing his own *Cours d'analyse* a few years later. As suggested by

Bolzano's preoccupation with the foundations of analysis, he was a philosopher and a free thinker. His sermons in Prague attracted large audiences. The Habsburg regime regarded him as a threat and issued an order in 1819 that placed Bolzano under house arrest and prohibited him from publishing even the results of his mathematical research. At his death he left a large number of manuscripts that were edited and published only after long delay, starting around 1920. It then came to light that Bolzano had independently discovered the Bolzano–Weierstrass theorem, had proved that a continuous function on a closed bounded interval attains a maximum and minimum value, and had constructed a continuous nowhere differentiable function.

Bernhard Riemann (1826–1866) gave his definition of an integral in his Habilitationschrift of 1853, using "Riemann sums". The approach in terms of upper and lower sums is due to Gaston Darboux (1842–1917) and appeared in 1875.

The history of the Heine–Borel theorem is rather complicated. It is named for Eduard Heine (1821–1881) and Émile Borel (1871–1956), but the attribution is considered inappropriate. Heine obtained a variant of the result in 1872, and in 1895 Borel stated and proved what is now called the Heine–Borel theorem, but only for countable coverings. Several years later, Lebesgue and others generalized the result to arbitrary coverings. As early as 1862, however, Dirichlet had used a form of the theorem implicitly in his lectures, to prove that a continuous function on a closed bounded interval is uniformly continuous. The theorem had also been known to Weierstrass and others.

The books by Bottazzini [**1**], Edwards [**3**], Grabiner [**4**], and Jahnke [**5**] are good sources of information about the historical development of analysis.

1.10. Metric spaces

In later chapters of this book, we will make occasional remarks that refer to the relatively modern concept of a metric space. For the sake of completeness, we include here a brief discussion of this important topic, which unifies certain aspects of classical analysis.

A *metric space* is an ordered pair (S, ρ), where S is an arbitrary set and ρ is a real-valued function defined on the Cartesian product $S \times S$ with the properties

(i) $\rho(x, y) \geq 0$ for all $x, y \in S$, with equality if and only if $x = y$,

(ii) $\rho(x, y) = \rho(y, x)$ for all $x, y \in S$, and

(iii) $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$ for all $x, y, z \in S$.

A function ρ with these three properties is called a *metric*. Property (iii) is known as the triangle inequality.

The leading example is $(\mathbb{R}, d)$, the set of real numbers equipped with the usual metric, or distance function, $d(x, y) = |x - y|$. A more general example is $(\mathbb{R}^n, d_n)$, the n-dimensional Euclidean space with metric

$$d_n(x, y) = \left\{ \sum_{k=1}^{n} (x_k - y_k)^2 \right\}^{1/2}, \qquad x = (x_1, \ldots, x_n), \quad y = (y_1, \ldots, y_n).$$

A more artificial example consists of an arbitrary set endowed with the metric

$$\rho(x, y) = \begin{cases} 1, & \text{if } x \neq y \\ 0, & \text{if } x = y. \end{cases}$$

Many basic concepts of analysis generalize naturally to the abstract setting of a metric space (S, ρ). For instance, a sequence $\{x_n\}$ of elements $x_n \in S$ is said to converge to a limit $x \in S$ if for each $\varepsilon > 0$ there is a number N such that $\rho(x_n, x) < \varepsilon$ for all $n \geq N$. A sequence $\{x_n\}$ is called a Cauchy sequence if for each $\varepsilon > 0$ there exists an N such that $\rho(x_n, x_m) < \varepsilon$ whenever $n, m \geq N$. A metric space is said to be *complete* if every Cauchy sequence is convergent. Thus the space $(\mathbb{R}, d)$ is complete, whereas $(\mathbb{Q}, d)$ is not. (Recall that $\mathbb{Q}$ denotes the set of rational numbers.)

For applications to analysis, it is especially useful to consider metric spaces whose elements are functions. For instance, the set S can be taken to consist of all continuous functions on a fixed interval $[a, b]$, with metric

$$\rho(f, g) = \sup_{x \in [a,b]} |f(x) - g(x)|, \qquad f, g \in S.$$

Convergence in this metric is equivalent to uniform convergence on $[a, b]$. Another space is given by the same set of functions with metric

$$\rho(f, g) = \int_a^b |f(x) - g(x)| \, dx.$$

The first of these two metric spaces is complete, but the second is not.

An extensive theory of metric spaces has been developed, and there are many examples important in analysis. Further information can be found in the book by Rudin [**8**] and in other texts.

1.11. Complex numbers

Although this book deals primarily with real-valued functions of a real variable and makes no use of complex function theory, it is often convenient to work with complex arithmetic. Hence we conclude this preliminary chapter with a brief account of the complex number system.

The need for an extension of the real number system is already apparent in the problem of solving a quadratic equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$ and $a \neq 0$. According to the quadratic formula, the solutions have the form

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\,.$$

Of course, it may well happen that $b^2 - 4ac < 0$, in which case the square-root is undefined and the equation has no solutions in the algebra of real numbers. In this case, however, it is fruitful to interpret the quadratic formula as providing a pair of solutions

$$-\frac{b}{2a} \pm i\frac{\sqrt{4ac - b^2}}{2a}\,, \qquad \text{where } \; i = \sqrt{-1}\,.$$

In effect, this amounts to extending the algebra of real numbers to the algebra of complex numbers $z = x + iy$, where x and y are real numbers.

It should be emphasized that in the process of enlarging the set $\mathbb{R}$ of real numbers to the set $\mathbb{C}$ of complex numbers, the algebraic operations of addition and multiplication are also extended to $\mathbb{C}$. The sum and product of two complex numbers $z = x + iy$ and $w = u + iv$ are defined by

$$z + w = (x + u) + i(y + v) \qquad \text{and} \qquad zw = (xu - yv) + i(xv + yu)\,.$$

These expressions result from formal manipulations, assuming that the imaginary number i is subject to the usual rules of arithmetic. In this way the algebra of real numbers is extended to the algebra of complex numbers, preserving the basic properties of the arithmetic operations. Specifically, addition and multiplication are both commutative and associative, and multiplication is distributive over addition: $z(w_1 + w_2) = zw_1 + zw_2$.

A complex number $z = x + iy$ is said to be real if $y = 0$, and purely imaginary if $x = 0$. The *real part* of z is $\mathrm{Re}\{z\} = x$ and the *imaginary part* is $\mathrm{Im}\{z\} = y$. The *complex conjugate* of z is $\overline{z} = x - iy$. Conjugation commutes with addition and multiplication: $\overline{z + w} = \overline{z} + \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$. The *modulus* of a complex number z is $|z| = \sqrt{x^2 + y^2}$, a generalization of the absolute value of a real number. Observe that $z\,\overline{z} = |z|^2$ and that $|\overline{z}| = |z|$. Two useful formulas are

$$\mathrm{Re}\{z\} = \frac{1}{2}(z + \overline{z}) \qquad \text{and} \qquad \mathrm{Im}\{z\} = \frac{1}{2i}(z - \overline{z})\,.$$

It may be noted that $|\mathrm{Re}\{z\}| \leq |z|$ and $|\mathrm{Im}\{z\}| \leq |z|$. A simple calculation shows that

$$|z + w|^2 = (z + w)(\overline{z} + \overline{w}) = |z|^2 + 2\,\mathrm{Re}\{z\overline{w}\} + |w|^2\,.$$

Complex numbers have a natural geometric interpretation. Just as real numbers can be identified with points on the real line, complex numbers can be identified with points in the complex plane. Specifically, a complex number $z = x + iy$ corresponds to the point in the plane with Cartesian coordinates (x, y). The x-axis is then called the real axis, and the y-axis is called the imaginary axis. The real and imaginary parts of z are its orthogonal projections onto the real and imaginary axes, respectively. The complex conjugate of z is its reflection across the real line. The modulus of z is its distance from the origin. More generally, $|z - w|$ is the distance between the points z and w, and so $|z - \alpha| = R$ is the equation of a circle with center $\alpha \in \mathbb{C}$ and radius $R > 0$. Representing points in the plane by polar coordinates (r, θ), we can write $x = r\cos\theta$ and $y = r\sin\theta$, so that $z = x + iy = r(\cos\theta + i\sin\theta)$. The angle θ is called the *argument* of z and is denoted $\theta = \arg\{z\}$. Of course, the argument is determined only up to an additive multiple of 2π.

From the geometric viewpoint of the complex plane, the sum of two complex numbers is given by the parallelogram law, as in vector addition. For a geometric interpretation of the product, it is useful to consider polar representations

$$z = r(\cos\theta + i\sin\theta) \qquad \text{and} \qquad w = \rho(\cos\varphi + i\sin\varphi)\,.$$

Then by the standard addition formulas for sine and cosine,

$$\begin{aligned} zw &= r\rho[(\cos\theta\cos\varphi - \sin\theta\sin\varphi) + i(\cos\theta\sin\varphi + \sin\theta\cos\varphi)] \\ &= r\rho[\cos(\theta+\varphi) + i\sin(\theta+\varphi)]\,. \end{aligned}$$

In other words, the product of two complex numbers is the number whose modulus is the product of their moduli and whose argument is the sum of their arguments. In symbols,

$$|zw| = |z||w| \qquad \text{and} \qquad \arg\{zw\} = \arg\{z\} + \arg\{w\}\,.$$

One corollary of the relation $|zw| = |z||w|$ is the triangle inequality $|z + w| \le |z| + |w|$. For a proof, observe that

$$\begin{aligned} |z+w|^2 &= |z|^2 + 2\,\mathrm{Re}\{z\overline{w}\} + |w|^2 \le |z|^2 + 2\,|z\overline{w}| + |w|^2 \\ &= |z|^2 + 2\,|z||w| + |w|^2 = (|z| + |w|)^2\,. \end{aligned}$$

For a geometric interpretation that justifies the name "triangle inequality", rewrite it in the form

$$|\alpha - \beta| \le |\alpha - \gamma| + |\gamma - \beta|\,, \qquad \text{where } \alpha, \beta, \gamma \in \mathbb{C}\,.$$

Then the inequality says that the distance from α to β is no greater than the distance from α to γ plus the distance from γ to β. The three points α, β, γ can be viewed as vertices of a triangle.

In fact the formula

$$|\alpha - \beta|^2 = |\alpha|^2 + |\beta|^2 - 2\,\mathrm{Re}\{\alpha\overline{\beta}\}$$

is none other than the law of cosines. To see this, view the three points 0, α, and β as vertices of a triangle. Then the sides have lengths $|\alpha|$, $|\beta|$, and $|\alpha - \beta|$. Let $\theta = \arg\{\alpha\}$ and $\varphi = \arg\{\beta\}$, where $0 < \theta - \varphi < \pi$. Then the angle at the origin, the vertex opposite the side of length $|\alpha - \beta|$, is $\psi = \theta - \varphi$. But $\mathrm{Re}\{\alpha\overline{\beta}\} = |\alpha||\beta|\cos(\theta - \varphi)$, so the formula does reduce to the law of cosines: $c^2 = a^2 + b^2 - 2ab\cos\psi$.

Another consequence of the product formula is the *de Moivre formula*

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta\,,$$

which is a compact expression for a family of trigonometric identities. In view of the binomial theorem it gives, for instance

$$\cos 5\theta = \cos^5\theta - 10\cos^3\theta\sin^2\theta + 5\cos\theta\sin^4\theta\,.$$

The product formula appears more natural when the polar form of a complex number is expressed in exponential form. Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$ results from a formal comparison of Taylor series expansions. (See Chapter 3, Exercise 23.) If we write $z = re^{i\theta}$ and $w = \rho e^{i\varphi}$, the product formula is $zw = r\rho e^{i(\theta+\varphi)}$. It is possible to define the exponential of a complex number as a power series, but for the moment we may simply regard $e^{i\theta}$ as a notation for $\cos\theta + i\sin\theta$, with the property $e^{i\theta}e^{i\varphi} = e^{i(\theta+\varphi)}$ that is characteristic of an exponential function. Observe also that $\overline{e^{i\theta}} = e^{-i\theta}$ and $\left|e^{i\theta}\right| = 1$, and that

$$\frac{d}{d\theta}\left\{e^{i\theta}\right\} = -\sin\theta + i\cos\theta = ie^{i\theta}\,.$$

The quotient of two complex numbers $a + ib$ and $c + id \neq 0$ can be defined as the solution z of the equation $(c + id)z = a + ib$. To see that this equation has a unique solution, let $z = x + iy$ and write the complex equation in real form as a pair of linear equations

$$cx - dy = a\,, \qquad dx + cy = b\,.$$

This system has the unique solution

$$x = \frac{ac + bd}{c^2 + d^2}\,, \qquad y = \frac{bc - ad}{c^2 + d^2}\,.$$

The quotient can be calculated by writing

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{ac + bd}{c^2 + d^2} + i\,\frac{bc - ad}{c^2 + d^2}\,.$$

In more concise form, $z/w = z\,\overline{w}/|w|^2$.

Exercises

1. Use mathematical induction to prove that

$$(a) \qquad 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}\,, \qquad n = 1, 2, \ldots\,,$$

$$(b) \qquad 1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2\,, \qquad n = 1, 2, \ldots\,.$$

2. (a) Verify the identity of Pascal's triangle:

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}\,, \qquad k = 1, 2, \ldots, n\,.$$

(b) Use mathematical induction to prove the binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}\,, \qquad n = 1, 2, \ldots\,.$$

3. (a) Use the binomial theorem to show that

$$(1+x)^n \geq 1 + nx \qquad \text{for all } \; x \geq 0\,.$$

(b) Prove that the inequality holds more generally for all $x > -1$.

4. Prove the inequality $|\sin n\theta| \leq n|\sin\theta|\,,\ n = 1, 2, \ldots$.

5. The *Fibonacci sequence* $\{x_n\}$ is defined by

$$x_0 = 0,\ x_1 = 1,\ x_2 = 1,\ x_3 = 2,\ x_4 = 3,\ x_5 = 5,\ x_6 = 8, \ldots\,.$$

Here the general rule is that each number in the sequence is the sum of the two previous numbers: $x_{n+2} = x_{n+1} + x_n$ for $n = 0, 1, 2, \ldots$.

(a) Show that the recurrence relation $x_{n+2} = x_{n+1} + x_n$ has solutions of the form $x_n = \alpha^n$ for exactly two real constants α. Find this pair of numbers.

(b) Show that the Fibonacci sequence has the form

$$x_n = A\,\alpha^n + B\,\beta^n\,, \qquad n = 0, 1, 2, \ldots\,,$$

for some real constants A, B, α, and β. Determine these numbers.

(c) Check your formula for x_n by induction.

6. If X and Y are arbitrary countable sets, show that their Cartesian product

$$X \times Y = \{(x, y) \ : \ x \in X\,,\ y \in Y\}$$

is countable.

7. Prove that the collection of all infinite sequences of 0's and 1's is uncountable.

8. Prove that the collection of all sets of positive integers is uncountable.

9. Prove that the set of all algebraic numbers is countable and deduce the existence of transcendental numbers.

Hint. Use the fact that a polynomial of degree n has at most n real roots.

10. (a) Show that the reciprocal of each nonzero algebraic number is algebraic.

(b) Show that the square root of each positive algebraic number is algebraic.

11. Show that a set of real numbers is closed if and only if it contains all of its cluster points.

12. For any set S of real numbers, the *derived set* S' is defined as the set all cluster points of S. (It may be empty.) Show that for every set $S \subset \mathbb{R}$ the set S' is closed.

13. Let $\{s_n\}$ be a sequence of real numbers and let

$$\sigma_n = \frac{s_1 + s_2 + \cdots + s_n}{n}$$

be its sequence of arithmetic means.

(a) If $\{s_n\}$ converges to s, show that $\{\sigma_n\}$ also converges to s.

(b) Give an example to show that the converse is false: the sequence $\{\sigma_n\}$ may converge although the sequence $\{s_n\}$ does not.

14. Without appeal to Stirling's formula, calculate

$$\lim_{n\to\infty} \frac{1}{n}(n!)^{1/n}\,.$$

Suggestion. Apply the last theorem in Section 1.4.

15. Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots > 0$. Prove that the series

$$\sum_{n=1}^{\infty} a_n \qquad \text{converges if and only if} \qquad \sum_{k=0}^{\infty} 2^k a_{2^k} \qquad \text{converges.}$$

16. Show that if

$$\lim_{n\to\infty} (a_{n+1} - a_n) = \lambda\,, \qquad \text{then} \qquad \lim_{n\to\infty} \frac{a_n}{n} = \lambda\,.$$

17. If $x_1 = 1$ and

$$x_{n+1} = \sqrt{1 + x_n}\,, \qquad n = 1, 2, \ldots\,,$$

prove that the sequence $\{x_n\}$ converges and find its limit.

18. Given numbers a_1 and b_1 with $0 < a_1 < b_1$, define the sequences $\{a_n\}$ and $\{b_n\}$ inductively by

$$a_{n+1} = \sqrt{a_n b_n} \qquad \text{and} \qquad b_{n+1} = \frac{a_n + b_n}{2}\,, \qquad n = 1, 2, \ldots .$$

Show that $\{a_n\}$ and $\{b_n\}$ both converge, and to the same limit.

Hint. First observe that $\sqrt{ab} \le \frac{a+b}{2}$ for all $a > 0$ and $b > 0$.

19. Using the results of Section 1.4, verify the unrestricted limits

$$(a) \quad \lim_{x\to\infty} x^p r^x = 0 \qquad \text{for } \; p > 0 \;\text{ and }\; 0 < r < 1\,,$$

$$(b) \quad \lim_{x\to\infty} x^{-p} \log x = 0 \qquad \text{for } \; p > 0\,,$$

$$(c) \quad \lim_{x\to 0+} x^p \log x = 0 \qquad \text{for } \; p > 0\,,$$

$$(d) \quad \lim_{x\to 0+} x^x = 1\,.$$

20. Without appeal to Stirling's formula, verify the asymptotic relation

$$\binom{n}{k} \sim \frac{1}{k!}\, n^k \qquad \text{for each fixed } k \in \mathbb{N}, \text{ as } n \to \infty\,.$$

In other words, show that the ratio of the two expressions tends to 1 as $n \to \infty$.

21. For a sequence $\{x_n\}$ that is bounded above and does not tend to $-\infty$, let S be the set of all subsequential limits. Show that S contains the number $\sup S$, so that $\{x_n\}$ has a largest subsequential limit.

22. For bounded sequences $\{a_n\}$ and $\{b_n\}$ prove that

$$\limsup_{n\to\infty} (a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n \qquad \text{and}$$

$$\liminf_{n\to\infty} (a_n + b_n) \ge \liminf_{n\to\infty} a_n + \liminf_{n\to\infty} b_n\,.$$

Give examples to show that strict inequality may occur.

23. For all bounded sequences $\{x_n\}$ verify that

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \left(\sup_{k\ge n} x_k \right) \qquad \text{and} \qquad \liminf_{n\to\infty} x_n = \lim_{n\to\infty} \left(\inf_{k\ge n} x_k \right)\,.$$

24. Prove the second inequality in the theorem at the end of Section 1.4:

$$\liminf_{n\to\infty} \frac{r_{n+1}}{r_n} \le \liminf_{n\to\infty} \sqrt[n]{r_n}\,.$$

25. Suppose a function $f(x)$ is continuous and has the property $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Prove that $f(x) = Cx$, where $C = f(1)$.

Hint. Prove it first for rational numbers x.

26. Define a function f on the interval $(0,1)$ as follows. If x is rational and $x = p/q$ in lowest terms, let $f(x) = 1/q$. If x is irrational, let $f(x) = 0$. Show that f is discontinuous at each rational point of $(0,1)$ but is continuous at every irrational point.

27. Generalize the mean value theorem by proving that if f and g are continuous on $[a,b]$ and differentiable on (a,b), then

$$[f(b) - f(a)]g'(\xi) = [g(b) - g(a)]f'(\xi) \qquad \text{for some } \xi \in (a,b)\,.$$

28. Apply the generalized mean value theorem of the preceding exercise to establish *l'Hôpital's rule* in the following form. If f and g are differentiable in an open interval (a,b) and $f'(x)/g'(x) \to L$ as $x \to a+$, and if $f(x) \to 0$ and $g(x) \to 0$ as $x \to a+$. then $f(x)/g(x) \to L$ as $x \to a+$.

29. Verify that the function defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}\,, & x \neq 0 \\ 0\,, & x = 0\,. \end{cases}$$

is differentiable everywhere on $\mathbb{R}$, but its derivative is not continuous at the origin.

30. The preceding example shows that a derivative may exist but fail to be continuous. Show nevertheless that a derivative always has the intermediate value property. Specifically, prove that if f is differentiable on an interval $[a,b]$ and $f'(a) < \lambda < f'(b)$, then $f'(\xi) = \lambda$ for some $\xi \in (a,b)$.

Hint. Show that $g(x) = f(x) - \lambda x$ attains its minimum value at a point $\xi \in (a,b)$.

31. For $0 \le x \le 1$, define the functions $f_n(x) = n^2 x^n (1-x)$, where $n = 1, 2, \ldots$.

(a) Show that $\lim_{n\to\infty} f_n(x) = 0$ for each $x \in [0,1]$.

(b) Show directly, by consideration of $f_n(\frac{n}{n+1})$, that the convergence is not uniform on $[0,1]$.

(c) Show that $\lim_{n\to\infty} \int_0^1 f_n(x)dx = 1$, and conclude again that the convergence of $f_n(x)$ to 0 is not uniform on $[0,1]$.

32. Let the functions $f_1(x), f_2(x), \ldots$ be continuous on the interval $[0,1]$ and suppose that $f_n(x) \to f(x)$ uniformly on $[0,1]$ as $n \to \infty$. Show that $f_n(1/n) \to f(0)$.

33. Apply the Bolzano–Weierstrass theorem instead of the Heine–Borel theorem to show that a function continuous at each point of a closed bounded set $S \subset \mathbb{R}$ is uniformly continuous on S.

34. Let the numbers a_{nm} be defined by

$$a_{nm} = \begin{cases} 0\,, & \text{for } 1 \le n < m \\ -1\,, & \text{for } n = m \\ 2^{m-n}\,, & \text{for } 1 \le m < n\,. \end{cases}$$

Show that

$$\sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{nm} = -2 \qquad \text{but} \qquad \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{nm} = 0\,.$$

Explain why this example does not contradict Cauchy's double series theorem (see Section 1.5).

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Chapter 2
Special Sequences

This chapter is devoted to a study of some special sequences that arise often in analysis. We begin with two familiar sequences which converge to e, the base of natural logarithms. As a byproduct of the analysis, the number e is found to be irrational. This prompts an elementary proof that π is irrational as well. Further topics include Euler's constant γ and the infinite product formulas of Vieta and Wallis. The chapter concludes with a crown jewel of classical analysis, Stirling's approximation to the factorial function.

2.1. The number e

The number $e = 2.71828182845904\ldots$, known as the base of natural logarithms, arises through the limit

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$$

and is found to have the form

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots .$$

Our aim is to reconcile the two expressions and to show in particular that the first limit exists. We will also show that e is an irrational number.

First of all, let us prove that the infinite series converges, so that we can legitimately define the number e as its sum. Let

$$s_n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}$$

be the nth partial sum, and observe that $s_n < s_{n+1}$, since all terms are positive. On the other hand, the inequality $k! \geq 2^{k-1}$ shows that

$$s_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} < 3$$

for all n. Thus the sequence $\{s_n\}$ is monotonic and bounded, and is therefore convergent. We denote its limit, or the sum of the infinite series, by e.

Now consider the expressions

$$x_n = \left(1 + \frac{1}{n}\right)^n , \qquad n = 1, 2, 3, \ldots .$$

In order to study the behavior of the sequence $\{x_n\}$, we can use a hand calculator to compute

$$x_1 = 2\,,\ x_2 = 2.25\,,\ x_3 = 2.370\ldots\,,\ x_4 = 2.441\ldots\,,\ x_5 = 2.488\ldots\,,\text{etc.}$$

The data suggest that $\{x_n\}$ is an increasing sequence. Our next step will be to show that this is actually true.

According to the binomial theorem, we can write

$$\begin{aligned} x_n = \left(1 + \frac{1}{n}\right)^n &= \sum_{k=0}^{n} \binom{n}{k} \left(\frac{1}{n}\right)^k \\ &= 1 + n\left(\frac{1}{n}\right) + \frac{n(n-1)}{2!}\left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{1}{n}\right)^3 \\ &\quad + \cdots + \frac{n(n-1)(n-2)\cdots 1}{n!}\left(\frac{1}{n}\right)^n \\ &= 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \\ &\quad + \cdots + \frac{1}{n!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{n-1}{n}\right) . \end{aligned}$$

A similar expansion of x_{n+1} gives

$$x_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} \left(\frac{1}{n+1}\right)^k .$$

Observe now that for each fixed k in the range $2 \leq k \leq n$, the term

$$\begin{aligned} \binom{n+1}{k}\left(\frac{1}{n+1}\right)^k &= \frac{1}{k!}\left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1}\right)\cdots\left(1 - \frac{k-1}{n+1}\right) \\ &\geq \frac{1}{k!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-1}{n}\right) = \binom{n}{k}\left(\frac{1}{n}\right)^k . \end{aligned}$$

Furthermore, the expansion for x_{n+1} contains an extra term $\left(\frac{1}{n+1}\right)^{n+1} > 0$, corresponding to $k = n+1$. These two inequalities combine to show that $x_{n+1} > x_n$ for $n = 1, 2, 3, \ldots$.

The above expansion for x_n also shows that

$$x_n \le 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} = s_n < 3\,.$$

Therefore, $x_1 < x_2 < x_3 < \cdots < 3$, and so the sequence $\{x_n\}$ is monotonic and bounded, hence convergent. We are going to prove that

$$\lim_{n\to\infty} x_n = e\,,$$

the limit of the sequence $\{s_n\}$. Since $x_n \le s_n$ for all n, it follows that

$$\lim_{n\to\infty} x_n \le \lim_{n\to\infty} s_n = e\,.$$

On the other hand, for each fixed $m \le n$ the sum in the previous calculation can be truncated to give

$$\begin{aligned} x_n \ge 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \\ + \cdots + \frac{1}{m!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{m-1}{n}\right)\,. \end{aligned}$$

Now hold m fixed and let $n \to \infty$ to infer that

$$\lim_{n\to\infty} x_n \ge 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{m!} = s_m$$

for each index m. Letting $m \to \infty$, we see from this that

$$\lim_{n\to\infty} x_n \ge \lim_{m\to\infty} s_m = e\,.$$

Combining this with the earlier inequality $\lim_{n\to\infty} x_n \le e$, we conclude that

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} s_n = e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots\,.$$

The infinite series

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots$$

converges quite rapidly. For instance $s_7 = 2.71825396\ldots$ already gives the correct value of e to four decimal places. To estimate the rate of convergence, we can write

$$\begin{aligned} e - s_n &= \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \ldots \\ &< \frac{1}{(n+1)!}\left\{1 + \frac{1}{n+1} + \left(\frac{1}{n+1}\right)^2 + \ldots\right\} = \frac{1}{n!\,n}\,, \end{aligned}$$

after summing the geometric series. Therefore,

$$0 < e - s_n < \frac{1}{n!\,n} \qquad \text{for} \quad n = 1, 2, 3, \ldots .$$

It is now easy to show that e is irrational. Suppose on the contrary that e is rational, so that $e = \frac{m}{n}$ for some positive integers m and n. By the estimate just given,

$$0 < n!(e - s_n) < \frac{1}{n} \le 1\,.$$

But

$$n!s_n = n!\left(1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!}\right)$$

is an integer, and $n!e$ is also an integer under the assumption that $e = \frac{m}{n}$. Hence that assumption has led to the conclusion that $n!(e - s_n)$ is an integer between 0 and 1, which is impossible. The contradiction shows that e is irrational.

In fact, it is known that e is a transcendental number. Recall that a real number x is said to be *algebraic* if it satisfies some polynomial equation

$$a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = 0$$

with integer coefficients a_0, a_1, … , a_n. All rational numbers are algebraic, and so are many irrational numbers such as $\sqrt{2}$. A real number that is not algebraic is said to be *transcendental*. Both e and π are known to be transcendental, but those assertions are not easy to prove. The transcendence of e was proved by Charles Hermite [**2**] in 1873. Then in 1882 Ferdinand von Lindemann [**4**] adapted Hermite's method to establish the transcendence of π. A simpler version of the Hermite–Lindemann proof can be found in the book by Ivan Niven [**5**]. In the next section we present Niven's proof of the more elementary fact that π is irrational.

2.2. Irrationality of π

We now digress from the theme of this chapter to prove that the number π is irrational. This fact lies intrinsically deeper than the irrationality of e, and was proved by more sophisticated methods before Ivan Niven [**5**] found the remarkably elementary proof that will be presented here. In fact, the proof yields the stronger result that π^2 is irrational.

Consider the polynomial

$$f(x) = \frac{1}{n!}\, x^n (1-x)^n \,,$$

where n is a positive integer to be specified later. This function satisfies

$$0 < f(x) < \frac{1}{n!} \qquad \text{for } \ 0 < x < 1 \,.$$

It is easy to see that each of the derivatives $f^{(k)}(0)$ is an integer. Indeed, $f^{(k)}(0) = 0$ for $0 \le k < n$, and for $k \ge n$ a calculation shows that the kth derivative of $x^n(1-x)^n$ at the origin is an integer divisible by $n!$. (This remains true if the factor $(1-x)^n$ is replaced by any other polynomial with integer coefficients.) By the symmetry relation $f(1-x) = f(x)$, it follows that every derivative $f^{(k)}(1)$ is also an integer. Observe finally that $f^{(k)}(x) \equiv 0$ for all $k > 2n$, since f is a polynomial of degree $2n$.

Now suppose, for purpose of contradiction, that π^2 is rational, so that $\pi^2 = p/q$ for some positive integers p and q. Define the polynomial

$$g(x) = q^n \left[\pi^{2n} f(x) - \pi^{2n-2} f^{(2)}(x) + \pi^{2n-4} f^{(4)}(x) - \cdots + (-1)^n f^{(2n)}(x) \right] ,$$

and note that both $g(0)$ and $g(1)$ are integers under the supposition that $\pi^2 = p/q$. Because of the structure of g, we see that

$$g''(x) + \pi^2 g(x) = q^n \pi^{2n+2} f(x) \,.$$

In view of this relation, a simple calculation gives

$$\begin{aligned} \frac{d}{dx}\big\{ g'(x) \sin \pi x - \pi g(x) \cos \pi x \big\} &= \big[g''(x) + \pi^2 g(x) \big] \sin \pi x \\ &= q^n \pi^{2n+2} f(x) \sin \pi x = \pi^2 p^n f(x) \sin \pi x \,. \end{aligned}$$

Consequently,

$$\pi p^n \int_0^1 f(x) \sin \pi x \, dx = \left[\frac{1}{\pi}\, g'(x) \sin \pi x - g(x) \cos \pi x \right]_0^1 = g(1) + g(0) \,,$$

which is an integer. On the other hand, since $0 < f(x) < 1/n!$ for $0 < x < 1$, we find that

$$0 < \pi p^n \int_0^1 f(x) \sin \pi x \, dx < \frac{\pi p^n}{n!} < 1$$

if n is chosen sufficiently large. But this is impossible, because there is no integer between 0 and 1. Thus the assumption that π^2 is rational has led to a contradiction, and so we conclude that π^2 is irrational, which implies that π is irrational.

The transcendence of π has an interesting application to classical geometry. It settles once and for all the ancient problem of "squaring the circle" with straight edge and compass. Given an arbitrary circle, the problem is to construct a square of the same area. Since the circle has area πr^2, this amounts to starting with a line segment of unit length and constructing a segment of length $\sqrt{\pi}$. A segment of length π could then be constructed, since from any segment of length ℓ it is possible to construct a segment of length ℓ^2. But it can be shown that the length of every segment constructible with straight edge and compass, starting with a segment of unit length, is an algebraic number. (A good reference for this fact is the book by Courant and Robbins [**1**].) Therefore, if it were possible to square the circle with straight edge and compass, the number π would have to be algebraic. But π is transcendental, so it is impossible to square the circle.

2.3. Euler's constant

Euler's constant is

$$\gamma = \lim_{n\to\infty} \left\{ \sum_{k=1}^{n} \frac{1}{k} - \log n \right\} .$$

It is named for Leonhard Euler, who first discussed it in 1734. The number γ is an important constant that occurs frequently in mathematical formulas. The existence of the limit is not obvious. Our aim is to prove that the limit exists and to determine its approximate numerical value.

Consider the curve $y = 1/x$ for $1 \le x \le n$, where $n = 2, 3, \ldots$. The area under the curve is given by

$$A_n = \int_1^n \frac{1}{x} \, dx = \log n \, .$$

Now construct rectangular boxes of heights $1/k$ over the intervals $[k, k+1]$, as shown in Figure 1.

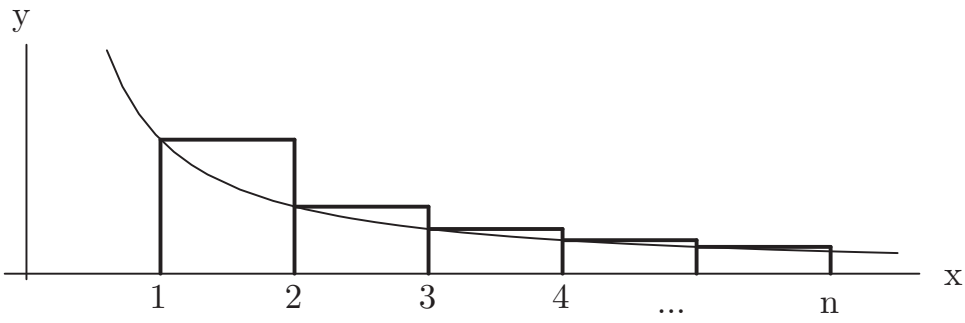


Figure 1. The curve $y = 1/x$ and rectangular boxes.

Since

$$\frac{1}{k+1} \le \frac{1}{x} \le \frac{1}{k} \qquad \text{for} \quad k \le x \le k+1\,,$$

it follows that

$$\frac{1}{k+1} = \int_k^{k+1} \frac{1}{k+1}\,dx \le \int_k^{k+1} \frac{1}{x}\,dx \le \int_k^{k+1} \frac{1}{k}\,dx = \frac{1}{k}$$

for $k = 1, 2, \ldots$. Adding these inequalities over $k = 1, 2, \ldots, n-1$, we have

$$\sum_{k=1}^{n-1} \frac{1}{k+1} \le \int_1^n \frac{1}{x}\,dx \le \sum_{k=1}^{n-1} \frac{1}{k}\,.$$

With the notation

$$S_n = \sum_{k=1}^{n} \frac{1}{k}\,,$$

this says that $S_n - 1 \le A_n \le S_{n-1}$. The two inequalities can be rearranged to give

$$0 \le S_{n-1} - A_n \le 1 - S_n + S_{n-1} = 1 - \frac{1}{n}\,.$$

This shows that the sequence $\{S_{n-1} - A_n\}$ is positive and is bounded above by 1.

Geometrically, the quantity $S_{n-1} - A_n$ is the sum of areas of those portions of the boxes that lie above the curve $y = 1/x$ from $x = 1$ to n. In order to estimate this total area, imagine that all of these boxes are slid to the left until they lie inside the first box, as shown in Figure 2, where the shaded regions have total area $S_{n-1} - A_n$. Since the regions are nonoverlapping and lie inside a square of area 1, this conceptual exercise gives a geometric interpretation of the inequality $S_{n-1} - A_n \le 1$.

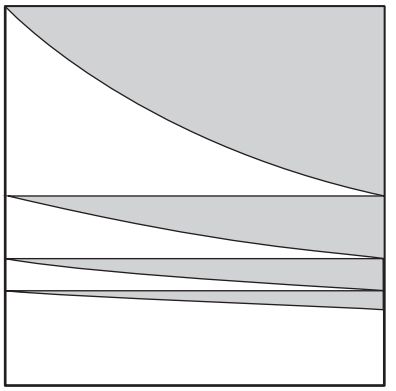


Figure 2. Geometric estimate of $S_{n-1} - A_n$.

Next observe that

$$A_{n+1} - A_n = \int_n^{n+1} \frac{1}{x}\, dx \le \frac{1}{n} = S_n - S_{n-1}\,,$$

or $S_{n-1} - A_n \le S_n - A_{n+1}$, which says that the sequence $\{S_{n-1} - A_n\}$ is nondecreasing. An appeal to the monotone boundedness theorem now shows that the sequence converges. Denoting its limit by γ, we have

$$\gamma = \lim_{n\to\infty} (S_{n-1} - A_n) = \lim_{n\to\infty} \left(S_n - \tfrac{1}{n} - A_n\right) = \lim_{n\to\infty} (S_n - A_n)\,.$$

The existence of Euler's constant has now been established, and the proof shows that $0 \le \gamma \le 1$.

In fact, it is visually clear from Figure 2 that γ is slightly larger than $\frac{1}{2}$. Our next task is to derive quantitative bounds on γ by estimating the area

$$\alpha_k = \frac{1}{k} - \int_k^{k+1} \frac{1}{x}\, dx$$

of the region in the kth box that lies above the curve $y = 1/x$. Since the curve is convex, that region contains a triangle of area

$$\frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+1}\right)$$

and is contained in a trapezoid of area

$$\frac{1}{k} - \frac{1}{k+\frac{1}{2}}\,,$$

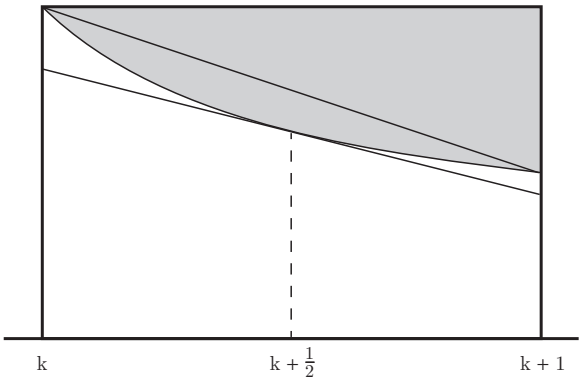


Figure 3. Estimation of the area α_k.

constructed by drawing the tangent line to the curve at the point where $x = k + \frac{1}{2}$. (See Figure 3 and note that the trapezoid above the tangent line is obtained by removing the lower trapezoid from the entire rectangle.)

Thus a comparison of areas shows that

$$\frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+1}\right) \le \alpha_k \le \frac{1}{k} - \frac{1}{k+\frac{1}{2}} = 2\left(\frac{1}{2k} - \frac{1}{2k+1}\right).$$

Summing these inequalities, we find

$$\frac{1}{2}\sum_{k=1}^{n-1}\left(\frac{1}{k} - \frac{1}{k+1}\right) \le \sum_{k=1}^{n-1}\alpha_k \le 2\sum_{k=1}^{n-1}\left(\frac{1}{2k} - \frac{1}{2k+1}\right),$$

or

$$\frac{1}{2}\left(1 - \frac{1}{n}\right) \le S_{n-1} - A_n \le 2\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \cdots - \frac{1}{2n-1}\right).$$

Letting $n \to \infty$, we infer that

$$0.5 \le \gamma \le 2\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \ldots\right) = 2(1 - \log 2) = 0.6137\ldots,$$

since (see Exercise 7)

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \log 2\,.$$

The actual value of Euler's constant is

$$\gamma = 0.577215664901532\ldots\,.$$

It has been computed to thousands of decimal places and no periodicities have been detected, so it is strongly suspected to be an irrational number. In fact, it is generally conjectured to be a transcendental (or nonalgebraic) number, like the constants π and e. However, no one has ever been able to prove that γ is irrational!

2.4. Vieta's product formula

The infinite product formula

$$\sqrt{\tfrac{1}{2}}\,\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}}}\,\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}}}}\cdots = \frac{2}{\pi}$$

was discovered in 1593 by the French lawyer and mathematician François Viète (1540–1603), whose name was rendered into Latin as Franciscus Vieta. The formula derives special interest from the unexpected appearance of the number π. It can be written more explicitly as

$$\lim_{n\to\infty}\,(b_1 b_2\cdots b_n) = \frac{2}{\pi}\,,$$

where

$$b_1 = \sqrt{\tfrac{1}{2}} \qquad \text{and} \qquad b_{n+1} = \sqrt{\tfrac{1}{2}+\tfrac{1}{2}\,b_n}\,, \qquad n = 1, 2, \ldots\,.$$

The proof is based on a simple trigonometric identity. We begin with the double-angle formula

$$\sin x = 2\,\cos\frac{x}{2}\,\sin\frac{x}{2}\,, \qquad x > 0\,.$$

Introducing the same identity in the form

$$\sin\frac{x}{2} = 2\,\cos\frac{x}{4}\,\sin\frac{x}{4}\,,$$

we see that

$$\sin x = 4\,\cos\frac{x}{2}\,\cos\frac{x}{4}\,\sin\frac{x}{4}\,.$$

After n iterations, the process leads to the formula

$$\sin x = 2^n\,\cos\frac{x}{2}\,\cos\frac{x}{4}\,\cos\frac{x}{8}\cdots\cos\frac{x}{2^n}\,\sin\frac{x}{2^n}\,,$$

or

$$\cos\frac{x}{2}\,\cos\frac{x}{4}\,\cos\frac{x}{8}\cdots\cos\frac{x}{2^n} = \frac{\sin x}{2^n\,\sin(x/2^n)}\,,$$

provided that $x/2^n$ is not an integer multiple of π. In particular, the formula holds for any fixed $x > 0$ whenever n is sufficiently large. Observe now that

$$\lim_{n\to\infty}\,2^n\,\sin\frac{x}{2^n} = x\,, \qquad \text{since} \qquad \lim_{\theta\to 0}\,\frac{\sin\theta}{\theta} = 1\,.$$

Consequently, we have derived the infinite product formula

$$\lim_{n\to\infty} \cos\frac{x}{2}\cos\frac{x}{4}\cos\frac{x}{8}\cdots\cos\frac{x}{2^n} = \frac{\sin x}{x}\,, \qquad 0 < x < \pi\,,$$

written in standard notation as

$$\prod_{n=1}^{\infty} \cos\frac{x}{2^n} = \frac{\sin x}{x}\,.$$

With the choice $x = \pi/2$, the formula reduces to

$$\prod_{n=2}^{\infty} \cos\frac{\pi}{2^n} = \frac{2}{\pi}\,.$$

But $\cos\frac{\pi}{4} = \sqrt{\frac{1}{2}}$, and the half-angle formula

$$\cos\tfrac{\theta}{2} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\cos\theta}\,, \qquad 0 < \theta < \pi\,,$$

shows that

$$\cos\frac{\pi}{2^{n+1}} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\cos\frac{\pi}{2^n}}\,, \qquad n = 1, 2, 3, \ldots\,.$$

Therefore, $b_n = \cos(\pi/2^{n+1})$ and the last infinite product is simply an equivalent statement of Vieta's formula.

2.5. Wallis product formula

The remarkable *Wallis product formula* is

$$\lim_{n\to\infty} \frac{2\cdot 2}{1\cdot 3}\,\frac{4\cdot 4}{3\cdot 5}\,\frac{6\cdot 6}{5\cdot 7}\cdots\frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{\pi}{2}\,. \tag{1}$$

It was first recorded by John Wallis (1616–1703) as early as 1656. Equivalent formulations are

$$\lim_{n\to\infty} \left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{4^2}\right)\cdots\left(1 - \frac{1}{(2n)^2}\right) = \frac{2}{\pi} \tag{2}$$

and

$$\lim_{n\to\infty} \frac{2^{2n}(n!)^2}{(2n)!\sqrt{n}} = \sqrt{\pi}\,. \tag{3}$$

The factors in (2) are simply the reciprocals of those in (1). To see that (1) and (3) are equivalent, observe first that because $\frac{2n+1}{2n} \to 1$, the formula (1) implies

$$\lim_{n\to\infty} \frac{2^2 4^2 6^2 \cdots (2n-2)^2}{3^2 5^2 7^2 \cdots (2n-1)^2}(2n) = \frac{\pi}{2}\,.$$

Taking square roots, we deduce that

$$\begin{aligned}\sqrt{\frac{\pi}{2}} &= \lim_{n\to\infty} \frac{2\cdot 4\cdot 6\cdots(2n-2)}{3\cdot 5\cdot 7\cdots(2n-1)}\sqrt{2n} \\ &= \lim_{n\to\infty} \frac{2^2 4^2 6^2 \cdots (2n-2)^2 (2n)^2}{(2n)!\sqrt{2n}} \\ &= \frac{1}{\sqrt{2}} \lim_{n\to\infty} \frac{2^{2n}(n!)^2}{(2n)!\sqrt{n}}\,,\end{aligned}$$

which shows that (1) implies (3). Since all steps are reversible, a similar argument shows that (3) implies (1).

Taken in the version (3), the Wallis product formula may be regarded as a weak form of Stirling's formula $n! \sim n^n e^{-n}\sqrt{2\pi n}$, or

$$\lim_{n\to\infty} \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} = 1\,,$$

which will be proved in the next section.

For a proof of the product formula (1), consider the integrals

$$I_n = \int_0^{\pi/2} \sin^n x\, dx\,, \qquad n = 0, 1, 2, \ldots.$$

Observe that $I_0 = \frac{\pi}{2}$ and $I_1 = 1$. To compute I_n for $n \geq 2$, write

$$\sin^n x = \sin^{n-1} x \sin x$$

and integrate by parts to obtain $I_n = (n-1)(I_{n-2} - I_n)$, which gives the *reduction formula*

$$I_n = \frac{n-1}{n}\, I_{n-2}\,, \qquad n = 2, 3, \ldots.$$

Iterating the reduction formula and using the values of I_0 and I_1, we find

$$I_{2n} = \frac{2n-1}{2n}\,\frac{2n-3}{2n-2} \cdots \frac{1}{2}\cdot\frac{\pi}{2}$$

and

$$I_{2n+1} = \frac{2n}{2n+1}\,\frac{2n-2}{2n-1} \cdots \frac{2}{3}\cdot 1\,.$$

Dividing one equation by the other, we arrive at the expression

$$\frac{\pi}{2} = \frac{2\cdot 2}{1\cdot 3}\frac{4\cdot 4}{3\cdot 5}\frac{6\cdot 6}{5\cdot 7}\cdots\frac{(2n)(2n)}{(2n-1)(2n+1)}\frac{I_{2n}}{I_{2n+1}}\,.$$

It is now clear that a proof of the Wallis product formula reduces to showing that $I_{2n}/I_{2n+1} \to 1$ as $n \to \infty$. But the inequalities

$$0 < \sin^{2n+1} x < \sin^{2n} x < \sin^{2n-1} x$$

hold for $0 < x < \frac{\pi}{2}$, so that $0 < I_{2n+1} < I_{2n} < I_{2n-1}$. It follows that

$$1 < \frac{I_{2n}}{I_{2n+1}} < \frac{I_{2n-1}}{I_{2n+1}} = \frac{2n+1}{2n} \to 1\,,$$

which shows that $I_{2n}/I_{2n+2} \to 1$ as $n \to \infty$. This completes the proof of the Wallis product formula (1).

The formula is actually a special case of a much more general relation. The sine function has the infinite product representation

$$\sin \pi x = \pi x \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right)\,, \qquad x \in \mathbb{R}\,,$$

as will be proved in Chapter 8. Set $x = \frac{1}{2}$ to obtain the Wallis product formula (2).

2.6. Stirling's formula

The asymptotic formula

$$n! \sim n^n e^{-n}\sqrt{2\pi n}\,, \qquad n \to \infty\,,$$

is known as *Stirling's formula*. It is of basic importance for instance in probability theory and combinatorics, because it gives precise information about the growth of the factorial function. The symbol "$\sim$" means that

$$\lim_{n\to\infty} \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} = 1\,.$$

The formula was discovered in 1730, and the Scottish mathematician James Stirling (1692–1770) included it in his book *Methodus Differentialis* (London, 1730). However, most of the credit actually belongs to Abraham de Moivre (1667–1754), a mathematician born in France who spent his adult life in England, where he became an intellectual companion to Isaac Newton. Although de Moivre was never able to obtain a suitable position in England,

he made fundamental discoveries in the early theory of probability, including a preliminary version of the central limit theorem. In 1730, he found that $n! \sim C\, n^{n+\frac{1}{2}} e^{-n}$ for some constant C, and Stirling determined that $C = \sqrt{2\pi}$. In fact, as we shall see, the value of the constant is implicit in the Wallis product formula.

We propose to prove Stirling's formula by showing that

$$1 < \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} < 1 + \frac{1}{4n}\,, \qquad n = 1, 2, \ldots\,.$$

The error estimates are important in quantitative applications of the formula.

With the observation that

$$\log(n!) = \log 1 + \log 2 + \cdots + \log n\,,$$

it is natural to base a proof on a careful study of the area under the logarithmic curve $y = \log x$ from $x = 1$ to n. (See Figure 4.) An integration by parts calculates this area as

$$A_n = \int_1^n \log x\, dx = \Big[x \log x - x\Big]_1^n = n \log n - n + 1\,.$$

On the other hand, the area can be estimated geometrically. Since the logarithm is a concave function, the curve $y = \log x$ lies above each of its chords connecting successive points $(k, \log k)$, for $k = 1, 2, \ldots, n$. Thus A_n is larger than the sum of areas of the trapezoids under those line segments. The total area of the trapezoids is

$$\begin{aligned} T_n &= \tfrac{1}{2}\,\log 2 + \tfrac{1}{2}\,(\log 2 + \log 3) + \cdots + \tfrac{1}{2}\,(\log(n-1) + \log n) \\ &= \log 2 + \log 3 + \cdots + \log(n-1) + \tfrac{1}{2}\,\log n \\ &= \log(n!) - \tfrac{1}{2}\,\log n\,. \end{aligned}$$

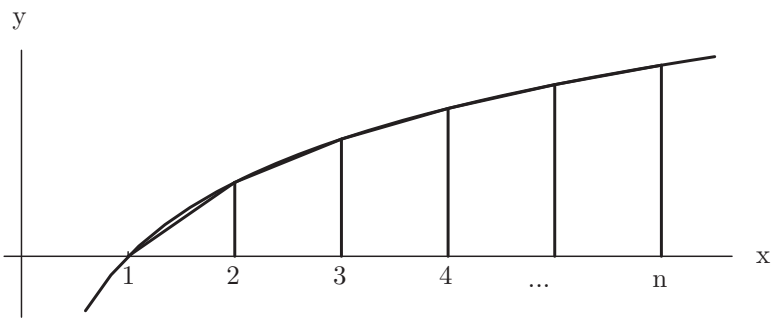


Figure 4. The logarithmic curve and inscribed trapezoids.

Now let α_k denote the area of the small region bounded by the curve $y = \log x$ and the line segment joining the two points $(k, \log k)$ and $(k+1, \log(k+1))$, for $k = 1, 2, \ldots, n-1$. Then the total area under the curve is

$$A_n = T_n + E_n\,, \qquad \text{where} \quad E_n = \alpha_1 + \alpha_2 + \cdots + \alpha_{n-1}\,.$$

Inserting the expressions for A_n and T_n, we can write this relation in the form

$$\log(n!) = \left(n + \tfrac{1}{2}\right)\log n - n + 1 - E_n\,,$$

or

$$n! = C_n\, n^{n+\frac{1}{2}} e^{-n}\,, \qquad \text{where} \quad C_n = e^{1-E_n}\,.$$

The sequence $\{E_n\}$ is increasing, since each term α_k is positive. We now show that the sequence $\{E_n\}$ has an upper bound and is therefore convergent. In order to estimate α_k, we construct the tangent line to the curve $y = \log x$ at the point where $x = k + \frac{1}{2}$ (see Figure 5) and compare areas:

$$\begin{aligned}
\alpha_k &< \log\left(k + \frac{1}{2}\right) - \frac{1}{2}\left(\log k + \log(k+1)\right) \\
&= \frac{1}{2}\log\left(\frac{k+\frac{1}{2}}{k}\right) - \frac{1}{2}\log\left(\frac{k+1}{k+\frac{1}{2}}\right) \\
&= \frac{1}{2}\log\left(1 + \frac{1}{2k}\right) - \frac{1}{2}\log\left(1 + \frac{1}{2k+1}\right) \\
&< \frac{1}{2}\log\left(1 + \frac{1}{2k}\right) - \frac{1}{2}\log\left(1 + \frac{1}{2(k+1)}\right)\,.
\end{aligned}$$

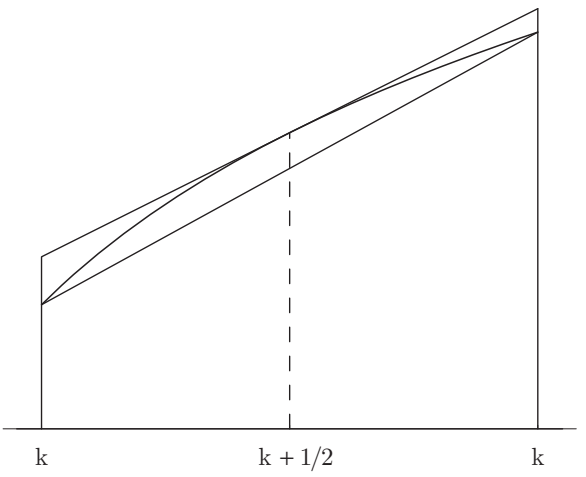


Figure 5. Estimation of the area α_k.

Adding these inequalities, we find that

$$E_n = \sum_{k=1}^{n-1} \alpha_k < \sum_{k=1}^{n-1} \left\{ \frac{1}{2} \log\left(1 + \frac{1}{2k}\right) - \frac{1}{2} \log\left(1 + \frac{1}{2(k+1)}\right) \right\}$$
$$= \frac{1}{2} \log \frac{3}{2} - \frac{1}{2} \log\left(1 + \frac{1}{2n}\right) < \frac{1}{2} \log \frac{3}{2}\,,$$

since the dominant series telescopes. Thus E_n increases to a finite limit $E = \sum_{k=1}^{\infty} \alpha_k$, and so $C_n = e^{1-E_n}$ decreases to a limit $C = e^{1-E} > 0$. In particular, $C_n > C$, and so $1 < C_n/C = e^{E-E_n}$. But

$$E - E_n = \sum_{k=n}^{\infty} \alpha_k < \sum_{k=n}^{\infty} \left\{ \frac{1}{2} \log\left(1 + \frac{1}{2k}\right) - \frac{1}{2} \log\left(1 + \frac{1}{2(k+1)}\right) \right\}$$
$$= \frac{1}{2} \log\left(1 + \frac{1}{2n}\right)\,,$$

again because the dominant series telescopes. Therefore,

$$1 < C_n/C = e^{E-E_n} < \sqrt{1 + \frac{1}{2n}} < 1 + \frac{1}{4n}\,.$$

In summary, we have shown that

$$0 < C < C_n = \frac{n!}{n^{n+\frac{1}{2}} e^{-n}} < C\left(1 + \frac{1}{4n}\right)\,.$$

In order to finish the proof of Stirling's formula with error estimates, it now remains only to show that $C = \sqrt{2\pi}$. This is where we invoke the Wallis product formula. It gives

$$\sqrt{\pi} = \lim_{n\to\infty} \frac{2^{2n}(n!)^2}{(2n)!\sqrt{n}} = \lim_{n\to\infty} \frac{2^{2n}\left(C_n\, n^{n+\frac{1}{2}} e^{-n}\right)^2}{\left(C_{2n}(2n)^{2n+\frac{1}{2}} e^{-2n}\right)\sqrt{n}}$$
$$= \frac{1}{\sqrt{2}} \lim_{n\to\infty} \frac{C_n^2}{C_{2n}} = \frac{C^2}{\sqrt{2}C} = \frac{C}{\sqrt{2}}\,.$$

Thus $C = \sqrt{2\pi}$ and the proof is complete.

Exercises

1. (*a*) Verify the inequality

$$x - \frac{x^2}{2} < \log(1+x) < x\,, \qquad x > 0\,,$$

and use it to show that

$$\lim_{n\to\infty}\left(1+\frac{1}{n^2}\right)^n = 1 \qquad \text{and} \qquad \lim_{n\to\infty}\left(1-\frac{1}{n^2}\right)^n = 1\,.$$

Conclude that

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n = \frac{1}{e}\,.$$

(*b*) Find

$$\lim_{n\to\infty}\left(1+\frac{1}{n^2}\right)\left(1+\frac{2}{n^2}\right)\cdots\left(1+\frac{n}{n^2}\right)\,.$$

2. Sharpen the inequality of the previous problem to

$$x-\frac{x^2}{2} < \log(1+x) < x-\frac{x^2}{2}+\frac{x^3}{3}\,, \qquad x>0\,.$$

(This is an improvement for small x.) Apply the result to calculate

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} e^{-n}\,.$$

3. (*a*) Evaluate

$$\lim_{n\to\infty}\frac{1}{n}\left\{\frac{(2n)!}{n!}\right\}^{1/n}$$

by showing, directly from the definition of an integral, that the logarithm of the expression tends to $\int_1^2 \log x\,dx$. Calculate the integral.

(*b*) Check your result with the help of Stirling's formula.

4. Prove that the infinite series

$$\sum_{n=1}^{\infty}\frac{1}{(n!)^2}$$

converges, and that its sum is an irrational number.

5. Write $e = s_n + r_n$, where $s_n = 1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{n!}$.

(*a*) Show that $\qquad \dfrac{1}{n+1} < n!r_n < \dfrac{1}{n}\,.$

(*b*) Calculate $\qquad \lim\limits_{n\to\infty} n\sin(2\pi n!e)\,.$

(*c*) Conclude that e is irrational.

6. Test the following infinite series for convergence and absolute convergence. Explain your reasoning.

$$(a) \quad \sum_{n=1}^{\infty} \frac{n!e^n}{n^{n+p}}, \qquad p \in \mathbb{R}.$$

$$(b) \quad \sum_{n=1}^{\infty} (-1)^n \sin\frac{a}{n}, \qquad a > 0.$$

$$(c) \quad \sum_{n=1}^{\infty} \left(1 - \cos\frac{1}{n}\right).$$

Note. These facts about the sine function may be useful:

$$\frac{2}{\pi}x \le \sin x \le x \qquad \text{for } 0 \le x \le \frac{\pi}{2}; \qquad \lim_{x\to 0} \frac{\sin x}{x} = 1.$$

7. Use the existence of Euler's constant γ to verify that

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \log 2.$$

8. Show by a similar method that

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots = \frac{3}{2}\log 2.$$

9. Show that

$$\lim_{n\to\infty} \left\{ \sum_{k=0}^{n} \frac{1}{(2k+1)} - \frac{1}{2}\log n \right\} = \frac{\gamma}{2} + \log 2.$$

10. Prove the existence of the limit

$$\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right),$$

and calculate its value.

11. Let

$$x_n = \sum_{k=1}^{n} \frac{1}{k} - \log n \qquad \text{and} \qquad y_n = x_n - \frac{1}{2n}.$$

(*a*) Show that the sequence $\{x_n\}$ is decreasing and $\{y_n\}$ is increasing, so that $y_n < \gamma < x_n$ for $n = 1, 2, \ldots$.

(*b*) Use a hand calculator to compute $x_{10} = 0.62638\ldots$. Conclude that $0.576 < \gamma < 0.627$.

(*c*) Use *Mathematica*® or a similar computer software system to calculate x_n for $n = 10,000$. Use the result to give the bounds $0.57721 < \gamma < 0.57726$ and thus to calculate $\gamma = 0.5772\ldots$ to 4 decimal places.

***12.** With x_n defined as in the preceding exercise, it can be shown by an elementary geometric (or graphical) argument due to Robert Young [**10**] that

$$\frac{1}{2(n+1)} < x_n - \gamma < \frac{1}{2n}\,, \qquad n = 1, 2, \ldots.$$

This indicates that the convergence of $\{x_n\}$ to Euler's constant is very slow.

13. Prove that

$$\sum_{k=1}^{\infty} (-1)^{k+1} \frac{\log k}{k} = \frac{1}{2}(\log 2)^2 - \gamma \log 2\,,$$

where γ is Euler's constant. Note that Leibniz' alternating series theorem implies that the series converges. Evaluate its sum in the following two steps.

(a) Integrate by parts to calculate

$$\int_1^x \frac{\log t}{t}\, dt = \frac{1}{2}(\log x)^2\,, \qquad x \geq 1\,,$$

and apply the result to show that the sequence $\{d_n\}$ given by

$$d_n = \sum_{k=1}^{n} \frac{\log k}{k} - \frac{1}{2}(\log n)^2$$

is convergent. (Imitate the argument used to prove existence of Euler's constant.)

(b) Write the partial sums in the form

$$\sum_{k=1}^{2n} (-1)^{k+1} \frac{\log k}{k} = \sum_{k=1}^{2n} \frac{\log k}{k} - 2\sum_{k=1}^{n} \frac{\log 2k}{2k}\,,$$

express the right-hand side in terms of d_{2n} and d_n, and let $n \to \infty$.

Note. See also Chapter 9, Exercise 18. The proof is due to H. F. Sandham [**8**].

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14. Verify the formula

$$\tfrac{1}{2}\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\cdot\tfrac{1}{2}}\,\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\cdot\tfrac{1}{2}}}\,\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\sqrt{\tfrac{1}{2}+\tfrac{1}{2}\cdot\tfrac{1}{2}}}}\cdots=\frac{3\sqrt{3}}{4\pi}\,.$$

15. Show that the integrals

$$I_n=\int_0^{\pi/2}\sin^n x\,dx\,,\qquad n=1,2,\dots\,,$$

are evaluated by the formulas

$$I_{2n}=\frac{(2n)!\,\pi}{2^{2n+1}(n!)^2}\qquad\text{and}\qquad I_{2n+1}=\frac{2^{2n}(n!)^2}{(2n+1)!}\,.$$

16. Derive the Wallis product formula from Stirling's formula.

17. Use Stirling's formula to verify the asymptotic relation

$$\binom{2n}{n}\sim\frac{2^{2n}}{\sqrt{\pi n}}\qquad\text{as } n\to\infty\,.$$

18. Integrate the inequalities

$$\sin^{2n+1}x<\sin^{2n}x<\sin^{2n-1}x\,,\qquad 0<x<\frac{\pi}{2}\,,$$

from 0 to $\pi/2$ to prove that

$$\frac{2}{2n+1}\left\{\frac{(2n)!!}{(2n-1)!!}\right\}^2<\pi<\frac{1}{n}\left\{\frac{(2n)!!}{(2n-1)!!}\right\}^2,\qquad n=1,2,\dots\,,$$

where

$$(2n)!!=2\cdot4\cdot6\cdots(2n)\qquad\text{and}\qquad(2n-1)!!=1\cdot3\cdot5\cdots(2n-1)\,.$$

Obtain the Wallis product formula as a corollary.

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Chapter 3
Power Series and Related Topics

This chapter begins with a review of the basic properties of power series. The more delicate problem of boundary behavior is then addressed with a discussion of Abel's theorem. In the converse direction, Taylor's formula with remainder is developed and applied to the expansion of certain functions in power series. The results then allow an elementary evaluation of Euler's sum $\sum 1/k^2 = \pi^2/6$. The chapter concludes with Weierstrass's famous example of a continuous function that is not differentiable at any point.

3.1. General properties of power series

A *power series* is an infinite series of the form

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots .$$

The number a is called the *center* of the power series, and the numbers c_n are its coefficients. The first problem is to find the set of numbers $x \in \mathbb{R}$ for which the series converges.

Obviously, the series converges for $x = a$ since every term then vanishes except possibly for $c_0 = f(a)$. It need not converge anywhere else, but if it does converge at some point $x_0 \neq a$, then it will converge in the interval $|x-a| < |x_0-a|$, uniformly in each closed subinterval $|x-a| \leq \rho < |x_0-a|$.

To see this, observe first that the convergence at x_0 implies that the terms $c_n(x_0 - a)^n$ tend to zero as $n \to \infty$, so that

$$|c_n(x_0 - a)^n| \le A\,, \qquad n = 0, 1, 2, \dots\,,$$

for some constant A. Therefore,

$$|c_n(x - a)^n| = |c_n(x_0 - a)^n| \left|\frac{x - a}{x_0 - a}\right|^n \le A\left(\frac{\rho}{|x_0 - a|}\right)^n\,, \qquad |x - a| \le \rho\,.$$

But the geometric series $\sum(\rho/|x_0 - a|)^n$ converges, so the series $\sum c_n(x-a)^n$ converges uniformly in the interval $|x - a| \le \rho$ by the Weierstrass M-test. As a consequence, we see that if the series diverges at some point x_1, then it must diverge at every point x with $|x - a| > |x_1 - a|$. The conclusion is that the series converges throughout some symmetric interval $|x - a| < R$ and it diverges everywhere outside that interval, except possibly at the endpoints. The number R is known as the *radius of convergence* of the power series. It can happen that $R = \infty$, meaning that the series converges for all $x \in \mathbb{R}$, or that $R = 0$ if it converges only for $x = a$.

These conclusions are summarized as follows.

Theorem. *A power series $\sum_{n=0}^{\infty} c_n(x - a)^n$ has a radius of convergence R, where $0 \le R \le \infty$. If $R = \infty$, the series converges for all x and converges uniformly in each interval $|x - a| \le \rho < \infty$. If $R = 0$, it converges only at $x = a$. If $0 < R < \infty$,the series converges in the open interval $|x - a| < R$ and converges uniformly in each closed subinterval $|x - a| \le \rho < R$, whereas it diverges everywhere outside the closed interval $|x - a| \le R$.*

Here are some examples. The geometric series $\sum_{n=0}^{\infty} x^n$ has radius of convergence $R = 1$. It converges in the interval $(-1, 1)$ but diverges at both endpoints. The series $\sum_{n=1}^{\infty} \frac{1}{n}x^n$ again has $R = 1$, and it diverges at $x = 1$ but converges at $x = -1$. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}x^n$ has $R = 1$ and converges at both endpoints. The series $\sum_{n=1}^{\infty} n!x^n$ converges only for $x = 0$; its radius of convergence is $R = 0$. At the other extreme, the series $\sum_{n=1}^{\infty} \frac{1}{n!}x^n$ converges for all x, so $R = \infty$.

The radius of convergence of an arbitrary power series $\sum c_n(x - a)^n$ is given by the formula

$$R = \frac{1}{\beta}\,, \qquad \text{where} \qquad \beta = \limsup_{n\to\infty} \sqrt[n]{|c_n|}\,.$$

Here it is understood that $R = \infty$ if $\beta = 0$ and $R = 0$ if $\beta = \infty$. For a proof, we need only appeal to the root test for convergence of an infinite series.

In view of the uniform convergence within the interval of convergence, the sum function

$$f(t) = \sum_{n=0}^{\infty} c_n (t-a)^n$$

is continuous in the interval $a - R < t < a + R$ and the series can be integrated term by term to give

$$\int_a^x f(t)\,dt = \sum_{n=0}^{\infty} c_n \int_a^x (t-a)^n\,dt = \sum_{n=0}^{\infty} \frac{1}{n+1}\, c_n (x-a)^{n+1}$$

for $x \in (a-R, a+R)$. As an illustration, the geometric series

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n\,, \qquad |t| < 1\,,$$

is integrated to produce the series expansion

$$\log \frac{1}{1-x} = \int_0^x \frac{1}{1-t}\,dt = \sum_{n=0}^{\infty} \frac{1}{n+1} x^{n+1} = x + \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 + \dots\,, \qquad |x| < 1\,.$$

For another important example, start with the series

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-1)^n t^{2n}\,, \qquad |t| < 1\,,$$

to obtain the expansion

$$\begin{aligned} \tan^{-1} x &= \int_0^x \frac{1}{1+t^2}\,dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{2n}\,dt \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} = x - \tfrac{1}{3}x^3 + \tfrac{1}{5}x^5 - \tfrac{1}{7}x^7 + \dots\,, \qquad |x| < 1\,. \end{aligned}$$

A power series can also be differentiated term by term within its interval of convergence. We have the following theorem.

Theorem. *The sum of a power series*

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n\,, \qquad |x-a| < R\,,$$

is differentiable and its derivative is

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}\,, \qquad |x-a| < R\,.$$

Proof. Observe that the formally differentiated power series

$$g(x) = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$$

has the same radius of convergence R, since $\sqrt[n]{n} \to 1$. Term-by-term integration gives

$$G(x) = \int_a^x g(t)\,dt = \sum_{n=1}^{\infty} c_n(x-a)^n = f(x) - c_0\,, \qquad |x-a| < R\,.$$

But the integrand is continuous, so the fundamental theorem of calculus tells us that $G(x)$ is differentiable for $|x-a| < R$ and $G'(x) = g(x)$. Since $G(x) = f(x) - c_0$, this proves the theorem. □

Because the derivative $f'(x)$ is given by another power series with the same radius of convergence, the theorem shows that $f''(x)$ exists and is expressed by the twice-differentiated power series. Continuing in this way, we see that the sum of a power series has derivatives of all orders, calculated by successive term-by-term differentiation.

3.2. Abel's theorem

The short life of Niels Henrik Abel (1802–1829) was filled with remarkable discoveries in a wide variety of mathematical areas, ranging from the insolubility of quintic equations by radicals to the theory of elliptic functions. From an obscure village in southern Norway he rose to international prominence but was never able to obtain an appropriate scientific position. His posthumous statue (1908) by the famous Norwegian sculptor Gustav Vigeland now stands in a small park near the center of Oslo, adjacent to the Royal Palace.

The theorem of Abel to be presented here is relatively elementary but quite useful in evaluating the sums of certain infinite series. It is the prototype for a class of theorems now known as "Abelian theorems", and its proof involves a basic device now called "Abel summation".

Abel's Theorem. *If an infinite series $\sum_{n=0}^{\infty} a_n$ converges to the sum s, then*

$$\lim_{x\to 1-} \sum_{n=0}^{\infty} a_n x^n = s\,.$$

Note that the convergence of the series $\sum_{n=0}^{\infty} a_n$ says that the power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges at the point $x = 1$. It then follows that the power series converges for all x in the interval $(-1, 1]$, and Abel's theorem asserts that the sum function f is continuous at the point 1. Since the sum of a power series is known to be continuous within its interval of convergence $(-R, R)$, the theorem as stated is of interest only when $R = 1$.

Here are some examples. The function $f(x) = \log(1 + x)$ has the power series expansion

$$\log(1 + x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \tfrac{1}{4}x^4 + \dots , \qquad |x| < 1 .$$

For $x = 1$, the series reduces to $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$, which converges by Leibniz' alternating series theorem. Thus Abel's theorem tells us that

$$1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots = \lim_{x \to 1-} \log(1 + x) = \log 2 .$$

Another example is provided by the power series expansion

$$\tan^{-1} x = x - \tfrac{1}{3}x^3 + \tfrac{1}{5}x^5 - \tfrac{1}{7}x^7 + \dots , \qquad |x| < 1 .$$

Again the series converges for $x = 1$, so Abel's theorem says that

$$1 - \tfrac{1}{3} + \tfrac{1}{5} - \tfrac{1}{7} + \cdots = \lim_{x \to 1-} \tan^{-1} x = \tan^{-1} 1 = \tfrac{\pi}{4} .$$

On the other hand, the converse of Abel's theorem is false in the absence of additional hypotheses. In other words, the existence of the limit of $f(x)$ as $x \to 1-$ does not imply the convergence of the power series at $x = 1$. For instance,

$$1 - x + x^2 - x^3 + \cdots = \frac{1}{1 + x} \to \tfrac{1}{2}$$

as $x \to 1-$, but the series $1 - 1 + 1 - 1 + \dots$ is divergent.

The proof of Abel's theorem makes use of a device known as *Abel summation* or *summation by parts*, the discrete analogue of integration by parts. It can be expressed as follows.

Lemma. *Let $\{a_k\}$ and $\{b_k\}$ be arbitrary sequences of real numbers, and let $s_n = \sum_{k=0}^{n} a_k$. Then for $0 \le m < n$,*

$$\sum_{k=m}^{n} a_k b_k = \sum_{k=m}^{n-1} s_k (b_k - b_{k+1}) + s_n b_n - s_{m-1} b_m ,$$

where $s_{-1} = 0$.

Proof.

$$\sum_{k=m}^{n} a_k b_k = \sum_{k=m}^{n} (s_k - s_{k-1}) b_k = \sum_{k=m}^{n} s_k b_k - \sum_{k=m-1}^{n-1} s_k b_{k+1}$$
$$= \sum_{k=m}^{n-1} s_k (b_k - b_{k+1}) + s_n b_n - s_{m-1} b_m \,. \qquad \square$$

Proof of Abel's theorem. Again let $s_n = \sum_{k=0}^{n} a_k$ denote the partial sums of the series, so that $s_n \to s$ as $n \to \infty$. For $|x| < 1$, let

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \,, \qquad \text{and let} \qquad f_n(x) = \sum_{k=0}^{n} a_k x^k$$

denote the partial sums of the power series. Then by the lemma,

$$f_n(x) = \sum_{k=0}^{n-1} s_k (1-x) x^k + s_n x^n \,.$$

Fixing x in the interval $(0, 1)$ and letting $n \to \infty$, we arrive at the expression

$$f(x) = (1-x) \sum_{k=0}^{\infty} s_k x^k \,, \qquad 0 < x < 1 \,,$$

since $s_n \to s$ and $x^n \to 0$. But

$$(1-x) \sum_{k=0}^{\infty} x^k = 1 \,, \qquad |x| < 1 \,,$$

so we can write

$$f(x) - s = (1-x) \sum_{n=0}^{\infty} (s_n - s) x^n \,, \qquad 0 < x < 1 \,.$$

Given $\varepsilon > 0$, choose N such that $|s_n - s| < \frac{\varepsilon}{2}$ for all $n \geq N$. Then

$$|f(x) - s| \leq (1-x) \left\{ \sum_{n=0}^{N-1} |s_n - s| x^n + \sum_{n=N}^{\infty} |s_n - s| x^n \right\}$$
$$< (1-x) \sum_{n=0}^{N-1} |s_n - s| + \tfrac{\varepsilon}{2} < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon$$

if x is sufficiently close to 1. In other words, there is a number $\delta > 0$ such that

$$|f(x) - s| < \varepsilon \qquad \text{when} \qquad 0 < 1 - x < \delta \,.$$

This proves the theorem. $\square$

It is now easy to deduce a more general version of Abel's theorem.

Corollary. *If a power series*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

has radius of convergence R, where $0 < R < \infty$, and if the series converges for $x = R$, then $\lim_{x \to R-} f(x) = f(R)$. In other words, the function $f(x)$ is left-continuous at $x = R$.

For a proof, we need only consider the function $g(x) = f(Rx)$, whose power series development has radius of convergence 1.

Abel's theorem is much easier to prove in the special case where all of the coefficients a_n are of one sign. Then the result is a direct consequence of the Weierstrass M-test for uniform convergence. To see this, observe that $|a_n x^n| \leq a_n$ for $|x| \leq 1$ if $a_n \geq 0$, so the convergence of the series $\sum_{n=0}^{\infty} a_n$ implies the uniform convergence of the power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

in the closed interval $[-1, 1]$. Therefore, the sum $f(x)$ is continuous in $[-1, 1]$ and is in particular continuous at the point 1, as Abel's theorem asserts.

The technique of Abel summation has many applications. For instance, it can be used to obtain the following result.

Theorem. *If an infinite series $\sum_{n=1}^{\infty} a_n$ has bounded partial sums, and if $\{b_n\}$ is a sequence of positive numbers that decrease to zero, then the series $\sum_{n=1}^{\infty} a_n b_n$ is convergent.*

Proof. By hypothesis, the partial sums $s_n = \sum_{k=1}^{n} a_k$ have the property $|s_n| \leq M$ for some constant M and all n. An Abel summation gives

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} s_k (b_k - b_{k+1}) + s_n b_n \,.$$

Since $s_n b_n \to 0$ as $n \to \infty$ and

$$\sum_{k=1}^{n-1} |s_k (b_k - b_{k+1})| \leq M \sum_{k=1}^{n-1} (b_k - b_{k+1}) = M(b_1 - b_n) \leq M b_1 \,,$$

it follows that the series $\sum_{k=1}^{\infty} a_k b_k$ converges. □

Taking for instance $a_n = (-1)^{n+1}$, we see that $|s_n| \leq 1$, and so we have a new proof of Leibniz' alternating series theorem.

There is an integral analogue of Abel's theorem that involves Laplace transforms instead of power series. A bounded function is said to be integrable over the positive half-line $[0,\infty)$ if its Riemann integral exists over each bounded interval $[0,R]$ and tends to a finite limit as $R\to\infty$. Such a function need not be absolutely integrable over $[0,\infty)$. For instance, the function $g(t)=\frac{\sin t}{t}$ is integrable, by Leibniz' alternating series theorem, but $|g(t)|$ is not integrable over $[0,\infty)$.

On the other hand, if $g(t)$ is any bounded locally integrable function, then $e^{-xt}g(t)$ is absolutely integrable over $[0,\infty)$ for each $x>0$, so the *Laplace transform*

$$f(x)=\int_0^\infty e^{-xt}g(t)\,dt\,,\qquad x>0\,,$$

is well defined. Here is the integral analogue of Abel's theorem.

Theorem. *If $g(t)$ is bounded and integrable over $[0,\infty)$, then*

$$\lim_{x\to 0+}\int_0^\infty e^{-xt}g(t)\,dt=\int_0^\infty g(t)\,dt\,.$$

Proof. The strategy is to apply integration by parts essentially as in the proof of Abel's theorem. Let

$$G(t)=\int_0^t g(s)\,ds\,,\qquad 0\le t<\infty\,,$$

denote the indefinite integral of g, so that

$$\lim_{t\to\infty}G(t)=I=\int_0^\infty g(s)\,ds\,.$$

We want to show that the Laplace transform $f(x)$ converges to I as $x\to 0+$. For this purpose we integrate by parts to obtain

$$f(x)=\int_0^\infty e^{-xt}g(t)\,dt=x\int_0^\infty e^{-xt}G(t)\,dt\,,\qquad x>0\,.$$

Hence

$$f(x)-I=x\int_0^\infty e^{-xt}\big[G(t)-I\big]\,dt\,.$$

Given $\varepsilon>0$, choose R large enough that $|G(t)-I|<\varepsilon/2$ for all $t\ge R$, and observe that

$$\begin{aligned}|f(x)-I|&\le x\int_0^R e^{-xt}|G(t)-I|\,dt+x\int_R^\infty e^{-xt}|G(t)-I|\,dt\\&\le xR\max_{0\le t\le R}|G(t)-I|+\frac{\varepsilon}{2}<\varepsilon\end{aligned}$$

for all $x>0$ sufficiently small. This proves that $f(x)\to I$ as $x\to 0+$. □

3.3. Cauchy products and Mertens' theorem

The *Cauchy product* of two infinite series

$$\sum_{n=0}^{\infty} a_n \qquad \text{and} \qquad \sum_{n=0}^{\infty} b_n$$

is the series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_{n-k} b_k = a_n b_0 + a_{n-1} b_1 + \cdots + a_0 b_n \,, \qquad n = 0, 1, 2, \ldots .$$

This definition is suggested by power series considerations. Multiplication of two power series

$$\sum_{n=0}^{\infty} a_n x^n \qquad \text{and} \qquad \sum_{n=0}^{\infty} b_n x^n$$

gives

$$\begin{aligned}
&\left(\sum_{n=0}^{\infty} a_n x^n\right)\left(\sum_{n=0}^{\infty} b_n x^n\right) \\
&\quad = \left(a_0 + a_1 x + a_2 x^2 + \ldots\right)\left(b_0 + b_1 x + b_2 x^2 + \ldots\right) \\
&\quad = a_0 b_0 + (a_1 b_0 + a_0 b_1) x + (a_2 b_0 + a_1 b_1 + a_0 b_2) x^2 + \ldots \\
&\quad = c_0 + c_1 x + c_2 x^2 + \ldots ,
\end{aligned}$$

where the coefficients c_n are those in the Cauchy product.

Such a calculation requires justification, which will be given presently in the form of Mertens' theorem (*cf.* Exercise 14). Meanwhile, the result of formal manipulation will serve to motivate the notion of a Cauchy product. Observe that this operation of multiplication is commutative in the sense that

$$c_n = \sum_{k=0}^{n} a_{n-k} b_k = \sum_{k=0}^{n} b_{n-k} a_k \,.$$

The Cauchy product of two convergent series need not converge. Consider for example the series

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \ldots ,$$

which converges by Leibniz' alternating series theorem. We will show that the Cauchy product of this series with itself is divergent. Here we have

$$c_n = \sum_{k=0}^{n} a_{n-k}a_k = (-1)^n \sum_{k=0}^{n} \frac{1}{\sqrt{n-k+1}\sqrt{k+1}}\,.$$

But

$$(n-k+1)(k+1) = \left(\frac{n}{2}+1\right)^2 - \left(\frac{n}{2}-k\right)^2 \le \left(\frac{n}{2}+1\right)^2\,,$$

so that

$$|c_n| \ge \sum_{k=0}^{n} \frac{2}{n+2} = \frac{2(n+1)}{n+2} \ge 1\,, \qquad n = 0, 1, 2, \ldots\,.$$

Hence the series $\sum c_n$ diverges, since its terms do not tend to 0.

The preceding example involves a series $\sum a_n$ that is convergent but not absolutely convergent. If $\sum a_n$ and $\sum b_n$ are both absolutely convergent, it is permissible to rearrange terms to show that their Cauchy product is convergent. A theorem of Mertens says that absolute convergence of only one of the series allows the same conclusion.

Mertens' Theorem. *Suppose the series $\sum a_n$ and $\sum b_n$ converge to sums A and B, respectively. Let $\sum c_n$ be the Cauchy product of the two series. If $\sum a_n$ is absolutely convergent, then $\sum c_n$ converges to sum $C = AB$.*

Proof. Let

$$A_n = \sum_{k=0}^{n} a_k\,, \qquad B_n = \sum_{k=0}^{n} b_k\,, \qquad \text{and} \quad C_n = \sum_{k=0}^{n} c_k$$

denote the partial sums, and let $R_n = B - B_n$. Then we can write

$$\begin{aligned} C_n &= a_0b_0 + (a_0b_1 + a_1b_0) + \cdots + (a_0b_n + a_1b_{n-1} + \cdots + a_nb_0) \\ &= a_0B_n + a_1B_{n-1} + \cdots + a_nB_0 \\ &= A_nB - a_0R_n - a_1R_{n-1} - \cdots - a_nR_0\,. \end{aligned}$$

Since $A_n \to A$, it will suffice to show that

$$\sigma_n = a_0R_n + a_1R_{n-1} + \cdots + a_nR_0 \to 0 \qquad \text{as} \quad n \to \infty\,.$$

But $R_n \to 0$, so for each $\varepsilon > 0$ there is a number N such that $|R_n| < \varepsilon$ for all $n \ge N$. Then for $n > N$ we have

$$\begin{aligned} |\sigma_n| &\le |a_0R_n + a_1R_{n-1} + \cdots + a_{n-N}R_N| + |a_{n-N+1}R_{N-1} + \cdots + a_nR_0| \\ &\le \varepsilon \sum_{k=0}^{n} |a_k| + |a_{n-N+1}R_{N-1} + \cdots + a_nR_0|\,. \end{aligned}$$

Since $a_n \to 0$ as $n \to \infty$, we conclude that

$$|\sigma_n| \le \varepsilon \left(1 + \sum_{k=0}^{\infty} |a_k| \right)$$

for all n sufficiently large. This proves that $\sigma_n \to 0$, and it follows that $C_n \to AB$ as $n \to \infty$, which proves the theorem. □

Franz Josef Mertens (1840–1927) is best known for his work in number theory. As a student at the University of Berlin, he attended lectures by Kummer, Kronecker, and Weierstrass. Later he held professorships at universities in Krakow, Graz, and Vienna.

3.4. Taylor's formula with remainder

Suppose a function $f(x)$ is given as the sum of a power series with positive radius of convergence:

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n , \qquad |x-a| < R ,$$

where a is some real number. Then, as we have seen, $f \in C^\infty$. In other words, f has derivatives of all orders in the interval $(a-R, a+R)$. Moreover, the derivatives of f can be calculated by successive term-by-term differentiation. Thus

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1} , \qquad f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n (x-a)^{n-2} , \quad \text{etc.}$$

Proceeding in this manner, we see that $f'(a) = c_1$, $f''(a) = 2c_2$, and in general the kth derivative of f at the point a is $f^{(k)}(a) = k! c_k$. Thus the coefficients c_n can be expressed in terms of f by the simple formula

$$c_n = \frac{1}{n!} f^{(n)}(a) , \qquad n = 0, 1, 2, \ldots . \tag{1}$$

The problem now arises to describe the class of functions f that can be expressed as the sum of a power series in some open interval $(a-R, a+R)$. We know that any such function must be of class C^∞, and if it has a power series expansion at the point a, the coefficients must have the form (1). But is the converse true? In other words, if $f \in C^\infty$ in some interval $(a-R, a+R)$, must it have a power series expansion

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a)(x-a)^n \tag{2}$$

in that interval, or at least in some smaller open interval about the point a? This is a much more delicate question, and the answer in general is no. A counterexample is outlined in Exercise 6.

The formal expansion (2) of a function f into power series is known as the *Taylor series* of f at the point a. It is so named after Brooke Taylor (1685–1731), an English mathematician who recorded it as early as 1715, without considering the question of convergence. The problem of convergence of the Taylor series to a given function was first addressed by Colin Maclaurin (1698–1746) in 1742 and was successfully analyzed by Joseph-Louis Lagrange (1736–1813) in his book *Théorie des fonctions analytiques*, published in 1797.

Although the problem of Taylor series representation is fully understood only in the context of functions of a complex variable, sufficient conditions for validity of the expansion are available through an explicit formula for the difference between $f(x)$ and the nth partial sum of its Taylor series. This expression is commonly called Taylor's formula with remainder, but its first clear statement appears in the work of Lagrange.

Here is a derivation of Taylor's formula. Let $f(x)$ have continuous derivatives of all orders on some open interval containing a given point $a \in \mathbb{R}$. By the fundamental theorem of calculus,

$$f(x) = f(a) + \int_a^x f'(t)\,dt \qquad \text{for } x \text{ near } a.$$

Now hold x fixed and integrate by parts, letting $u(t) = f'(t)$ and $v(t) = -(x-t)$ in the usual notation. This leads to

$$f(x) = f(a) + \int_a^x f'(t)v'(t)\,dt = f(a) + f'(a)(x-a) + \int_a^x f''(t)(x-t)\,dt\,.$$

Another integration by parts produces

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{1}{2}f''(a)(x-a)^2 + \tfrac{1}{2}\int_a^x f'''(t)(x-t)^2\,dt\,.$$

Continuing in this manner, we obtain

$$f(x) = S_n(x) + R_n(x)\,, \qquad n = 1, 2, \ldots\,,$$

where

$$S_n(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(a)(x-a)^k$$

and

$$R_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,dt\,.$$

This is *Taylor's formula*. The polynomial $S_n(x)$ is the nth partial sum of the *Taylor series* of f at the point a, and $R_n(x)$ is called the *remainder*.

In order to prove that $S_n(x)$ converges to $f(x)$ as $n \to \infty$, or that f has the Taylor series expansion

$$f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(a)(x-a)^k \,,$$

we have to show that $R_n(x)$ tends to 0 as $n \to \infty$. This is equivalent to proving that the formal Taylor series of f actually converges to the sum $f(x)$.

For this purpose, the *Lagrange form* of the remainder is often convenient. It is

$$R_n(x) = f^{(n+1)}(\xi) \frac{(x-a)^{n+1}}{(n+1)!} \,,$$

where ξ is some point between a and x. (We are allowing either $x > a$ or $x < a$.) A derivation of the Lagrange formula makes use of the following lemma, which can be proved by appeal to the intermediate value theorem for continuous functions.

Lemma. *Let $g(t)$ and $h(t)$ be continuous functions on an interval $[a, b]$, and suppose that $h(t) \geq 0$. Then*

$$\int_a^b g(t)h(t)\,dt = g(\xi) \int_a^b h(t)\,dt$$

for some $\xi \in (a, b)$.

To derive the Lagrange form of the remainder from the integral representation, we take $g(t) = f^{(n+1)}(t)$ and $h(t) = (x-t)^n$ in the lemma to conclude that

$$R_n(x) = \frac{1}{n!} f^{(n+1)}(\xi) \int_a^x (x-t)^n \,dt = f^{(n+1)}(\xi) \frac{(x-a)^{n+1}}{(n+1)!}$$

for some ξ between a and x. If instead we take $h(t) = 1$, an application of the lemma leads to the *Cauchy form* of the remainder:

$$R_n(x) = f^{(n+1)}(\xi) \frac{(x-\xi)^n}{n!}(x-a)$$

for some ξ between a and x. Specifically, this means that $a < \xi < x$ or $x < \xi < a$.

Here are two examples.

Example 1. Let $f(x) = \sin x$ and let $a = 0$. We are going to show that the familiar Taylor series of the sine function actually converges to $\sin x$ for every $x \in \mathbb{R}$. Since it is a power series, we can then conclude that the convergence is uniform on each bounded subset of $\mathbb{R}$, but this will also follow directly from our estimate of the remainder in Taylor's formula. We find that

$$f(0) = 0\,, \quad f'(0) = 1\,, \quad f''(0) = 0\,, \quad f'''(0) = -1\,, \quad f^{(4)}(0) = 0\,,$$
$$f^{(5)}(0) = 1\,, \ \ldots\,, \text{ so the formal Taylor series of } \sin x \text{ is}$$

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\,.$$

It is easy to see (by the ratio test, for instance) that this infinite series converges for all $x \in \mathbb{R}$, but convergence alone does not imply that its sum is $\sin x$. To prove this, we have to verify that $R_n(x) \to 0$ as $n \to \infty$ for each fixed $x \in \mathbb{R}$. Referring to the Lagrange form of the remainder and noting that $f^{(n+1)}(\xi)$ is equal to $\pm\sin\xi$ or $\pm\cos\xi$, we see that $|f^{(n+1)}(\xi)| \le 1$, so that

$$|R_n(x)| \le \frac{|x|^{n+1}}{(n+1)!} \to 0 \qquad \text{as } n \to \infty\,.$$

A similar analysis verifies the expansions

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \qquad \text{and} \qquad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

for all $x \in \mathbb{R}$.

Example 2. Next let $f(x) = \log(1 + x)$ for $x > -1$, and again take $a = 0$. A calculation gives

$$f^{(n)}(x) = (-1)^{n+1}(n-1)!\,(1+x)^{-n}\,, \qquad n = 1, 2, \ldots\,.$$

Thus the formal Taylor series of $\log(1 + x)$ about the origin is found to be

$$x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \tfrac{1}{4}x^4 + \ldots\,.$$

We invoke Taylor's formula to write $f(x) = S_n(x) + R_n(x)$, where

$$S_n(x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \cdots + (-1)^{n+1}\tfrac{1}{n}x^n$$

and

$$R_n(x) = \frac{1}{n!}\int_0^x f^{(n+1)}(t)\,(x-t)^n\,dt = (-1)^n \int_0^x \left(\frac{x-t}{1+t}\right)^n \frac{1}{1+t}\,dt\,.$$

We now apply the inequality $\left|\frac{x-t}{1+t}\right| \le |x|$, which can be seen to hold for all t between 0 and x. (See Exercise 4.) It provides the estimate

$$|R_n(x)| \le |x|^n \left| \int_0^x \frac{1}{1+t}\, dt \right| = |x|^n \log(1+|x|)\,,$$

which tends to 0 as $n \to \infty$ if $-1 < x < 1$. This confirms that

$$\log(1+x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \tfrac{1}{4}x^4 + \dots\,, \qquad |x| < 1\,,$$

a result previously obtained by direct integration of a geometric series.

For $x = 1$ the Lagrange form of the remainder is more effective. It yields

$$R_n(1) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) = (-1)^n \frac{(1+\xi)^{-n+1}}{n+1}$$

for some ξ in the interval $0 < \xi < 1$. Thus $|R_n(1)| \le \frac{1}{n+1} \to 0$, which shows that the power series converges to $f(1)$ when $x = 1$. Observe that we have given another derivation, without appeal to Abel's theorem, of the formula

$$1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots = \log 2\,.$$

3.5. Newton's binomial series

This important generalization of the binomial theorem was formulated by Isaac Newton (1642–1727) as early as 1665. Recall that the binomial theorem gives the formula

$$(1+x)^n = 1 + \sum_{k=1}^{n} \binom{n}{k} x^k\,,$$

where n is a positive integer and

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)\cdots(n-k+1)}{k!}$$

is a binomial coefficient. Newton's generalization is

$$(1+x)^\alpha = 1 + \sum_{k=1}^{\infty} \binom{\alpha}{k} x^k\,, \qquad |x| < 1\,,$$

where α is an arbitrary real number and

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}$$

is a generalized binomial coefficient. Observe that if α is a positive integer, then $\binom{\alpha}{k} = 0$ for all $k \geq \alpha + 1$ and Newton's binomial series reduces to the standard binomial theorem.

We can derive Newton's formula as a Taylor series expansion about the origin of the function $f(x) = (1+x)^\alpha$, where $\alpha \neq 0, 1, 2, \ldots$ and $x > -1$. Our assumption that α is not a positive integer or zero implies that $\binom{\alpha}{k} \neq 0$ for all k. Successive differentiations produce the formula

$$\frac{1}{k!} f^{(k)}(x) = \binom{\alpha}{k} (1+x)^{\alpha-k}, \qquad k = 1, 2, \ldots .$$

Thus

$$S_n(x) = 1 + \sum_{k=1}^{n} \binom{\alpha}{k} x^k$$

and

$$R_n(x) = r_n \int_0^x \left(\frac{x-t}{1+t}\right)^n (1+t)^{\alpha-1}\, dt\,, \quad \text{where } r_n = \frac{\alpha(\alpha-1)\cdots(\alpha-n)}{n!}\,.$$

To estimate the factors r_n, observe that

$$|r_{n+1}/r_n| = \left|\frac{\alpha - n + 1}{n+1}\right| \to 1 \qquad \text{as } n \to \infty\,,$$

which implies that for each fixed number $\rho > 1$, the inequality $|r_{n+1}/r_n| \leq \rho$ holds for all $n \geq N$. Thus $|r_n| \leq C\rho^n$ for all n, where C is a constant. Appealing to the elementary inequality $\left|\frac{x-t}{1+t}\right| \leq |x|$ used in the previous section, we can estimate $R_n(x)$ by

$$|R_n(x)| \leq |r_n|\, |x|^n \left| \int_0^x (1+t)^{\alpha-1}\, dt \right| .$$

Given $x \in (-1, 1)$, choose $\rho > 1$ such that $\rho|x| < 1$. Then

$$|r_n|\, |x^n| \leq C(\rho|x|)^n \to 0 \qquad \text{as } n \to \infty\,.$$

It follows that $R_n(x) \to 0$ in the interval $-1 < x < 1$ as $n \to \infty$, which proves Newton's binomial series expansion

$$(1+x)^\alpha = 1 + \sum_{k=1}^{\infty} \binom{\alpha}{k} x^k\,, \qquad -1 < x < 1\,.$$

In fact, the proof shows that for $\alpha > 0$ the series converges uniformly to $(1+x)^\alpha$ in the closed interval $-1 \leq x \leq 1$. To see this, note that

$r_n \to 0$ for every $\alpha \in \mathbb{R}$ (*cf.* Chapter 5, Exercise 16), whereas the integral $\int_{-1}^{1}(1+t)^{\alpha-1}dt$ converges for $\alpha > 0$. Thus $R_n(x) \to 0$ uniformly in $[-1,1]$.

To give an explicit example of Newton's series, let $\alpha = -\frac{1}{2}$. Then

$$\begin{aligned}\binom{-\frac{1}{2}}{k} &= \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2k-1}{2}\right)}{k!} \\ &= (-1)^k \frac{1\cdot 3\cdot 5\cdots(2k-1)}{2^k k!} = \frac{(-1)^k(2k)!}{2^{2k}(k!)^2}\,,\end{aligned}$$

and the series reduces to

$$(1+x)^{-1/2} = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k(2k)!}{2^{2k}(k!)^2}\, x^k\,, \qquad |x| < 1\,.$$

3.6. Composition of power series

Taylor's formula offers a systematic way to verify that a given function has a power series expansion and to find the expansion. However, the calculation of successive derivatives is often a daunting prospect and other methods are likely to be more practical. In particular, if two functions f and g have power series expansions about the origin, it is useful to know that the same is true for their product fg and quotient f/g (if $g(0) \neq 0$), and to be able to calculate at least the first few terms of those expansions conveniently from the power series for f and g.

Suppose now that

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} a_n x^n \qquad \text{for} \quad |x| < R_1 \qquad \text{and} \\ g(x) &= \sum_{n=0}^{\infty} b_n x^n \qquad \text{for} \quad |x| < R_2\,. \end{aligned}$$

Then, as we saw in Section 3.3, Mertens' theorem can be used to prove that the product has the power series expansion

$$f(x)g(x) = \sum_{n=0}^{\infty} c_n x^n\,, \qquad \text{where} \quad c_n = \sum_{k=0}^{n} a_{n-k}b_k\,,$$

convergent at least in the interval $|x| < R = \min\{R_1, R_2\}$. In fact, the radius of convergence R may well be larger than both R_1 and R_2. For instance, consider the pair of functions

$$f(x) = \frac{1+x}{1-x} \qquad \text{and} \qquad g(x) = \frac{1-x}{1+x}\,.$$

Although f and g have power series expansions of radii $R_1 = R_2 = 1$, their product $f(x)g(x) \equiv 1$ has a (trivial) expansion with radius of convergence $R = \infty$.

Similarly, we will be able to show that the quotient f/g has a power series development in some interval $|x| < R$ where $g(x) \neq 0$, but the radius of convergence R may be smaller than both R_1 and R_2. To give a simple example, let $f(x) \equiv 1$ and $g(x) = 1 + x^2$. Then $R_1 = R_2 = \infty$ and $g(x) > 0$ for all $x \in \mathbb{R}$, yet f/g has the expansion

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots ,$$

with radius of convergence $R = 1$.

In order to show in general that f/g can be represented by a power series, it will suffice to give a proof for the reciprocal $1/g$, since $f/g = f \cdot (1/g)$. Supposing for convenience that $g(0) = 1$, we can view $1/g$ as the postcomposition of $g - 1$ with the function

$$\varphi(y) = \frac{1}{1+y} = 1 - y + y^2 - y^3 + \dots , \qquad |y| < 1 .$$

Thus, if $|x|$ is sufficiently small we find

$$\begin{aligned} \frac{1}{g(x)} = \frac{1}{1 + (b_1 x + b_2 x^2 + \dots)} &= 1 - (b_1 x + b_2 x^2 + \dots) \\ &\quad + (b_1 x + b_2 x^2 + \dots)^2 - \dots \\ &= 1 - b_1 x + (b_1^2 - b_2) x^2 + (-b_1^3 + 2 b_1 b_2 - b_3) x^3 + \dots . \end{aligned}$$

In this way we can calculate, for instance,

$$\frac{x}{\sin x} = \frac{1}{1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 - \dots} = 1 + \tfrac{1}{6}x^2 + \tfrac{7}{360}x^4 + \tfrac{31}{15120}x^6 + \dots .$$

Of course, these formal manipulations need to be justified. In doing so, we will consider composition of arbitrary power series, since the extra generality does not complicate the argument. For convenience we will assume that both functions vanish at the origin. Suppose, then, that

$$f(x) = \sum_{n=1}^{\infty} a_n x^n \qquad \text{and} \qquad g(x) = \sum_{n=1}^{\infty} b_n x^n$$

are power series developments in some interval $|x| < r$. Then $|g(x)| < r$ if x is sufficiently small, say $|x| < \rho \leq r$, and the composition

$$h(x) = f(g(x)) = \sum_{n=1}^{\infty} a_n [g(x)]^n$$

is defined. Each of the functions $[g(x)]^n$ is a product of power series and so has its own expansion

$$[g(x)]^n = \sum_{m=1}^{\infty} b_{nm} x^m , \qquad n = 1, 2, \dots ,$$

where $b_{nm} = 0$ for all $m < n$. Therefore, the composite function has the form

$$h(x) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} a_n b_{nm} x^m . \tag{3}$$

Interchanging the order of summation, we find

$$h(x) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} a_n b_{nm} x^m = \sum_{m=1}^{\infty} \left(\sum_{n=1}^{m} a_n b_{nm} \right) x^m ,$$

since $b_{nm} = 0$ for all $n > m$. This is the desired representation of h as a power series. Moreover, it validates the formal manipulation previously used to calculate the coefficients in the expansion.

However, we have to justify the interchange of summation. For this purpose we can apply Cauchy's double series theorem (see Chapter 1, Section 1.5), provided we are able to show that the double series (3) is absolutely convergent. This is not clear, but the proof can be completed by the following argument. Recall that a power series is absolutely convergent within its interval of convergence. Let

$$F(x) = \sum_{n=1}^{\infty} |a_n| x^n \qquad \text{and} \qquad G(x) = \sum_{m=1}^{\infty} |b_m| |x|^m ,$$

and choose a positive number $\rho_1 \le \rho$ small enough that $|G(x)| < r$ whenever $|x| < \rho_1$. Then the composition

$$H(x) = F(G(x)) = \sum_{n=1}^{\infty} |a_n| [G(x)]^n$$

is defined for $|x| < \rho_1$. Writing

$$[G(x)]^n = \sum_{m=1}^{\infty} B_{nm} |x|^m ,$$

we infer in particular that the double series

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} |a_n| B_{nm} |x|^m$$

is convergent. But since $|b_{nm}| \le B_{nm}$, this implies that the double series (3) is absolutely convergent for $|x| < \rho_1$, which completes the proof that the composite function $h(x) = f(g(x))$ can be expanded in power series in some neighborhood of the origin, and that the formal method for calculating the coefficients is valid.

Finally, it must be acknowledged that from the viewpoint of complex function theory, some paradoxes encountered in the real domain are readily explained, and the theory becomes much more transparent. In particular, the tools of complex analysis provide easy proofs of the existence of power series developments for the product, quotient, and composition of two functions known to have power series expansions. However, the actual calculation of coefficients in those developments involves essentially the same tricks discussed here for real power series. Thus the argument presented in this section, justifying the formal calculation of coefficients in the composition of power series on the basis of Cauchy's double series theorem, is also of interest for complex power series and is readily adapted to that setting.

3.7. Euler's sum

Our purpose is to evaluate the sum of the infinite series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}\,.$$

It is easy to see that the series converges, but more difficult to show that its sum is $\pi^2/6$. The problem goes back at least to the year 1690, when Jacob Bernoulli (1654–1705) determined that the series converges to a sum less than 2 but was unable to evaluate it precisely. He and his younger brother Johann Bernoulli (1667–1748), his eventual successor to a chair at the University of Basel (Switzerland), drew attention to the problem, which became known as the *Basel problem*. Johann Bernoulli was the principal teacher of Leonhard Euler (1707–1783), who entered the University of Basel at the age of 14.

Euler became interested in the Basel problem and sought to calculate the sum numerically. The series converges too slowly for accurate numerical evaluation, but in 1731 Euler found a clever transformation (see Exercise 17) to express it in the form

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = (\log 2)^2 + 2\sum_{k=1}^{\infty} \frac{1}{k^2 2^k}\,,$$

thereby gaining a series that converges much more rapidly. To calculate $\log 2$ he took the Taylor series expansion

$$-\log(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots$$

and set $x = \frac{1}{2}$ to obtain

$$\log 2 = \frac{1}{2} + \frac{1}{8} + \frac{1}{24} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k2^k} ,$$

a series that converges rapidly enough for effective calculation. As a result, Euler was able to compute

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = 1.644934\ldots ,$$

accurate to 6 decimal places. Later he used what is now called the "Euler-Maclaurin summation formula" (developed in Chapter 11 of this book) to compute the sum to 20 decimal places. Then in 1735 Euler solved the Basel problem, making the sensational discovery that the sum of the series is $\pi^2/6$.

Euler's first derivation of the sum $\pi^2/6$ was not at all rigorous. Other mathematicians criticized his method, but they could hardly question the result because $\pi^2/6$ matched the true sum numerically to 20 decimal places! (Euler's "proof" is described at the end of Section 8.5.)

Soon after his initial triumph, Euler evaluated the more general sums

$$\sum_{k=1}^{\infty} \frac{1}{k^{2n}} = (-1)^{n+1} \frac{2^{2n-1}\pi^{2n}}{(2n)!} B_{2n} , \qquad n = 1, 2, 3, \ldots ,$$

in terms of the *Bernoulli numbers* B_n, defined by the power-series expansion

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{1}{n!} B_n \, x^n , \qquad |x| < 1 .$$

A calculation gives the values

$$B_2 = \tfrac{1}{6} , \quad B_4 = -\tfrac{1}{30} , \quad B_6 = \tfrac{1}{42} , \quad B_8 = -\tfrac{1}{30} , \quad B_{10} = \tfrac{5}{66} .$$

Thus

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} , \qquad \sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90} , \qquad \sum_{k=1}^{\infty} \frac{1}{k^6} = \frac{\pi^6}{945} , \quad \text{etc.}$$

We will evaluate these more general sums in Chapter 11, which is devoted to Bernoulli numbers and their applications.

Many proofs of Euler's basic relation $\sum 1/k^2 = \pi^2/6$ have been found, but none is really simple. The standard proofs use advanced methods such

as Fourier series or complex function theory. Here we present two relatively elementary derivations. The first is a simplified version of a proof given by Boo Rim Choe [**4**], which appeals to Newton's binomial series and the Wallis product formula. The second method is an ingenious argument based on an elementary trigonometric sum. It was found by the Russian twins Yaglom and Yaglom ([**18**]; see [**19**], Problem 145), and rediscovered independently by Holme [**8**] and Papadimitriou [**11**]. The paper by Stark [**14**] gives references to other elementary proofs.

First Proof. We begin with the observation that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{k=1}^{\infty} \frac{1}{(2k)^2} + \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}\,,$$

or

$$\frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}\,,$$

so it will be equivalent to show that

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}\,.$$

The proof will use Newton's binomial series

$$(1+x)^{\alpha} = 1 + \sum_{n=1}^{\infty} \binom{\alpha}{n} x^n\,, \qquad |x| < 1\,,$$

in the special case where $\alpha = -1/2$. Our previous calculation (*cf.* Section 3.5) led to the expression

$$\binom{-\frac{1}{2}}{n} = \frac{(-1)^n (2n)!}{2^{2n}(n!)^2}\,, \qquad n = 1, 2, \ldots\,.$$

Therefore, after replacing x by $-t^2$ we arrive at the expansion

$$(1-t^2)^{-\frac{1}{2}} = 1 + \sum_{n=1}^{\infty} c_n t^{2n}\,, \qquad |t| < 1\,, \qquad \text{with } \; c_n = \frac{(2n)!}{2^{2n}(n!)^2}\,.$$

Now recall the Wallis product formula in the form

$$\lim_{n\to\infty} \frac{2^{2n}(n!)^2}{(2n)!\sqrt{n}} = \sqrt{\pi}\,.$$

As a consequence, we have the asymptotic formula

$$c_n \sim \frac{1}{\sqrt{\pi n}}\,, \qquad n \to \infty\,.$$

The next step is to integrate the power series term by term to obtain

$$\sin^{-1} x = \int_0^x (1-t^2)^{-\frac{1}{2}}\,dt = x + \sum_{n=1}^{\infty} \frac{1}{2n+1}\,c_n\,x^{2n+1}\,, \qquad |x| < 1\,.$$

Since $c_n \sim 1/\sqrt{\pi n}$, this last power series converges at $x = 1$. In other words,

$$\sum_{n=1}^{\infty} \frac{1}{2n+1}\,c_n < \infty\,.$$

Thus by the Weierstrass M-test, the series converges uniformly in the interval $[-1, 1]$, and so the sum is continuous there. We may therefore infer that

$$\sin^{-1} x = x + \sum_{n=1}^{\infty} \frac{1}{2n+1}\,c_n\,x^{2n+1}$$

throughout the closed interval $[-1, 1]$.

Now substitute $x = \sin\theta$ to arrive at the identity

$$\theta = \sin\theta + \sum_{n=1}^{\infty} \frac{1}{2n+1}\,c_n \sin^{2n+1}\theta\,, \qquad |\theta| \le \tfrac{\pi}{2}\,.$$

The last series converges uniformly in the interval $[-\pi/2, \pi/2]$, so we may integrate term by term:

$$\int_0^{\pi/2} \theta\,d\theta = \int_0^{\pi/2} \sin\theta\,d\theta + \sum_{n=1}^{\infty} \frac{1}{2n+1}\,c_n \int_0^{\pi/2} \sin^{2n+1}\theta\,d\theta\,.$$

Now recall the formula

$$I_{2n+1} = \int_0^{\pi/2} \sin^{2n+1}\theta\,d\theta = \frac{2\cdot 4\cdot 6\cdots(2n)}{1\cdot 3\cdot 5\cdots(2n+1)} = \frac{2^{2n}(n!)^2}{(2n+1)!}\,,$$

obtained from the reduction formula and used in the proof of the Wallis product formula (see Chapter 2, Section 2.5). Comparing this with the formula for c_n, we see that

$$c_n\,I_{2n+1} = \frac{1}{2n+1}\,,$$

so that the result of the term by term integration reduces to

$$\frac{\pi^2}{8} = 1 + \sum_{k=1}^{\infty} \frac{1}{(2k+1)^2}\,.$$

But as we noted at the outset, this relation is equivalent to

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}\,. \qquad \square$$

Second Proof. Here is a totally different approach, based upon the curious identity

$$\sum_{k=1}^{n} \cot^2\left(\frac{k\pi}{2n+1}\right) = \frac{n(2n-1)}{3}\,, \qquad n = 1, 2, \ldots\,. \tag{4}$$

Deferring a proof of (4), let us show first how it leads to an evaluation of Euler's sum. Start with the inequality

$$\sin x < x < \tan x\,, \qquad 0 < x < \tfrac{\pi}{2}\,.$$

(See Chapter 4, Section 4.1.) Take reciprocals and squares to infer that

$$\cot^2 x < 1/x^2 < \csc^2 x = 1 + \cot^2 x\,, \qquad 0 < x < \tfrac{\pi}{2}\,.$$

Insert the numbers $x = k\pi/(2n+1)$, for $k = 1, 2, \ldots, n$, and take sums to arrive at the inequality

$$\sum_{k=1}^{n} \cot^2\left(\frac{k\pi}{2n+1}\right) < \frac{(2n+1)^2}{\pi^2} \sum_{k=1}^{n} \frac{1}{k^2} < n + \sum_{k=1}^{n} \cot^2\left(\frac{k\pi}{2n+1}\right)\,.$$

In view of (4), this reduces to

$$\frac{n(2n-1)}{3} < \frac{(2n+1)^2}{\pi^2} \sum_{k=1}^{n} \frac{1}{k^2} < n + \frac{n(2n-1)}{3}\,.$$

Now multiply the inequality by $\pi^2/(4n^2)$ and let $n \to \infty$ to conclude that

$$\frac{\pi^2}{6} \le \sum_{k=1}^{\infty} \frac{1}{k^2} \le \frac{\pi^2}{6}\,, \qquad \text{or} \qquad \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}\,.$$

A proof of the identity (4) can be based on the de Moivre formula

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i \sin n\theta\,,$$

where $i = \sqrt{-1}$. The de Moivre formula can be recast in the form

$$\cos n\theta + i \sin n\theta = \sin^n\theta (\cot\theta + i)^n = \sin^n\theta \sum_{k=0}^{n} \binom{n}{k} i^k \cot^{n-k}\theta\,.$$

Equating imaginary parts, we find that

$$\sin n\theta = \sin^n\theta \left\{ \binom{n}{1} \cot^{n-1}\theta - \binom{n}{3} \cot^{n-3}\theta + \binom{n}{5} \cot^{n-5}\theta - \ldots \right\}\,.$$

If $n = 2m + 1$ is an odd integer, this relation takes the more specific form

$$\sin(2m+1)\theta = \sin^{2m+1}\theta\, P_m(\cot^2\theta)\,, \qquad 0 < \theta < \tfrac{\pi}{2}\,, \tag{5}$$

where P_m is the polynomial of degree m defined by

$$P_m(x) = \binom{2m+1}{1} x^m - \binom{2m+1}{3} x^{m-1} + \binom{2m+1}{5} x^{m-2} - \cdots + (-1)^m\,.$$

Now observe that $\sin^{2m+1}\theta \neq 0$ for $0 < \theta < \pi/2$, whereas $\sin(2m+1)\theta$ vanishes at the m distinct points

$$\theta = \frac{k\pi}{2m+1}\,, \qquad k = 1, 2, \ldots, m\,.$$

In view of the relation (5), we conclude that $P_m(x) = 0$ at the m distinct points

$$x_k = \cot^2\left(\frac{k\pi}{2m+1}\right)\,, \qquad k = 1, 2, \ldots, m\,.$$

But from the coefficients of the polynomial $P_m(x)$, we see that

$$\sum_{k=1}^{m} \cot^2\left(\frac{k\pi}{2m+1}\right) = \sum_{k=1}^{m} x_k = \frac{\binom{2m+1}{3}}{\binom{2m+1}{1}} = \frac{m(2m-1)}{3}\,,$$

which proves (4). □

The last step of the argument used the fact that a monic polynomial of degree m with roots $x_1, x_2, \ldots, x_m$ has the factorization

$$x^m + a_{m-1}x^{m-1} + \cdots + a_1 x + a_0 = (x - x_1)(x - x_2)\cdots(x - x_m)\,.$$

By expanding the product and equating coefficients of x^{m-1}, one sees that the sum of the roots is

$$\sum_{k=1}^{m} x_k = -a_{m-1}\,.$$

Apostol [**1**] generalized this proof of (4) to show that

$$\sum_{k=1}^{m} \cot^{2n}\left(\frac{k\pi}{2m+1}\right) = (-1)^{n+1}\,\frac{2^{4n-1}}{(2n)!}\,B_{2n}\,m^{2n} + O\big(m^{2n-1}\big)\,, \qquad m \to \infty\,.$$

In the same way as before, this formula then leads to the general result

$$\sum_{k=1}^{\infty} \frac{1}{k^{2n}} = (-1)^{n+1}\,\frac{2^{2n-1}\pi^{2n}}{(2n)!}\,B_{2n}\,, \qquad n = 1, 2, 3, \ldots\,.$$

Yaglom and Yaglom [**19**] indicate a similar generalization of the argument.

Leonhard Euler (1707–1783) was the greatest mathematician of the 18th century and one of the greatest ever. His work spanned all branches of mathematics, pure and applied. The sheer quantity of his output is astounding. A project to publish his collected works (*Opera Omnia*) was started in 1911 and has run to over 70 volumes but is still far from complete. Euler made lasting contributions to many areas of mathematics, including differential equations, elliptic integrals, infinite series and products, number theory, algebra, hydrodynamics, optics, and classical geometry. His work initiated major fields of investigation such as calculus of variations, graph theory, and combinatorial topology. Born and educated in Basel, he worked for most of his life in St. Petersburg (where he is buried) and Berlin. At age 31 he suffered a severe illness that left him blind in one eye, and eventually he became totally blind. His solution of the Basel problem was his first work to gain international recognition.

The recent book by Varadarajan [**15**] gives a full account of Euler's life and mathematical work. Euler's studies of the Riemann zeta function

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}\,, \qquad x > 1\,,$$

as it came to be called, are described in an article by Ayoub [2]. We will develop some of this material in Chapter 10.

3.8. Continuous nowhere differentiable functions

In the early days of calculus, the prevailing wisdom was that a continuous function must have a derivative at "almost all" points. It was hard to imagine otherwise. Thus it came as a rude shock when Karl Weierstrass (1815–1897) produced an example of a continuous function that is not differentiable at any point! Writing in 1872, Weierstrass [**17**] reported having met a former student of Bernhard Riemann(1826–1866) who told him Riemann had claimed as early as 1861 that the continuous function

$$f(x) = \sum_{k=1}^{\infty} \frac{\sin(k^2 x)}{k^2} \tag{6}$$

is nowhere differentiable. However, Riemann had not published a proof and Weierstrass was unable to verify it, so he produced his own example. Perhaps inspired by Riemann's claim, Weierstrass [**17**] constructed the function

$$f(x) = \sum_{k=0}^{\infty} b^k \cos(a^k \pi x)\,, \tag{7}$$

where a is an odd integer and b is a number in the interval $0 < b < 1$ such that $ab > 1 + 3\pi/2$. Since $0 < b < 1$, it follows from the Weierstrass M-test that this series of continuous functions converges uniformly on the whole real line, so that its sum is everywhere continuous. The assumption that $ab > 1 + 3\pi/2$ is needed for the more technical proof that the function is nowhere differentiable. We will come to the details presently.

After Weierstrass published his construction, many other examples were devised. One idea was to look at a general function of the form

$$f(x) = \sum_{k=0}^{\infty} b^k \varphi(a^k x)\,,$$

where φ is a suitable periodic function. In a scholarly paper written in 1918, Knopp [**9**] discussed this approach among others, and gave many references to earlier literature. A few years later, van der Waerden [**16**] discovered a relatively simple example of this type. In modified form presented by Rudin [**12**], van der Waerden's example is

$$f(x) = \sum_{k=0}^{\infty} (3/4)^k \varphi(4^k x)\,, \tag{8}$$

where the function φ is defined by $\varphi(x) = |x|$ for $-1 \le x \le 1$ and is extended periodically to the whole line so that $\varphi(x + 2) = \varphi(x)$.

It would come to light, however, that Bernard Bolzano (1781–1848) had already constructed an example in 1834, although his discussion was not entirely precise and he claimed the nonexistence of a derivative only at a countable dense set of points. Bolzano's work remained obscure and was not properly published until 1921, at which point his function was found to be nowhere differentiable. His construction is geometric. A curve undergoes successive modifications to produce in the limit a curve that does not have a well determined tangent direction at any point. A more precise exposition was given by Kowalewski [**10**].

Presumably unaware of Bolzano's construction, Riemann and Weierstrass used much the same idea. Each new term of the series has smaller amplitude but much higher frequency than the preceding terms, so the graphs of the partial sums become increasingly "wiggly" and it seems plausible that the limiting function will have an "infinitely wiggly" graph at each point. The problem is to make these vague notions precise.

To prove that the Weierstrass function (7) is nowhere differentiable, write $f(x) = s_n(x) + r_n(x)$, where

$$s_n(x) = \sum_{k=0}^{n-1} b^k \cos(a^k \pi x)$$

is the nth partial sum of the series and $r_n(x)$ is the remainder. For an arbitrarily chosen point x, let

$$\phi_n(h) = \frac{s_n(x+h) - s_n(x)}{h} \qquad \text{and} \qquad \psi_n(h) = \frac{r_n(x+h) - r_n(x)}{h}$$

denote the respective difference quotients. By the mean value theorem,

$$\phi_n(h) = s_n'(t) = -\sum_{k=0}^{n-1} a^k \pi b^k \sin(a^k \pi t)$$

for some point t between x and $x+h$. It follows that

$$|\phi_n(h)| \le \pi \sum_{k=0}^{n-1} (ab)^k = \pi \frac{(ab)^n - 1}{ab-1} < \frac{\pi (ab)^n}{ab-1}\,. \tag{9}$$

Now hold n fixed and let m be the integer closest to $a^n x$, so that $|a^n x - m| \le \frac{1}{2}$. (In case of ambiguity, when $a^n x$ is a half-integer, take for instance $m = a^n x + \frac{1}{2}$.) Let $\lambda_n = a^n x - m$ and define the increment

$$h_n = \frac{1-\lambda_n}{a^n} = \frac{m+1}{a^n} - x\,,$$

so that $0 < h_n \le \frac{3}{2} a^{-n}$. Since a is an odd integer, we see that

$$\cos\bigl(a^k \pi (x+h_n)\bigr) = \cos\bigl(a^{k-n}(m+1)\pi\bigr) = (-1)^{m+1} \qquad \text{for } k \ge n\,.$$

Also, we can write

$$\begin{aligned} \cos\bigl(a^k \pi x\bigr) &= \cos\bigl(a^{k-n}(m+\lambda_n)\pi\bigr) \\ &= \cos\bigl(a^{k-n} m\pi\bigr) \cos\bigl(a^{k-n}\lambda_n \pi\bigr) \\ &= (-1)^m \cos\bigl(a^{k-n}\lambda_n \pi\bigr)\,, \qquad k \ge n\,. \end{aligned}$$

Combining the last two formulas, we have

$$\begin{aligned} \psi_n(h_n) &= \frac{1}{h_n} \sum_{k=n}^{\infty} b^k \left[\cos\Bigl(a^k \pi (x+h_n)\Bigr) - \cos\Bigl(a^k \pi x\Bigr)\right] \\ &= \frac{(-1)^{m+1}}{h_n} \sum_{k=n}^{\infty} b^k \left[1 + \cos(a^{k-n}\lambda_n \pi)\right]\,. \end{aligned}$$

Every term in this infinite series is nonnegative, so we can discard all but the first term to obtain the estimate

$$|\psi_n(h_n)| \ge \frac{b^n}{h_n}[1 + \cos(\lambda_n \pi)] \ge \frac{b^n}{h_n} \ge \tfrac{2}{3}(ab)^n\,,$$

since $|\lambda_n| \le \frac{1}{2}$ and $0 < h_n \le \frac{3}{2}a^{-n}$.

The final step is to invoke the estimate (9) and conclude that

$$\begin{aligned}\left|\frac{f(x+h_n)-f(x)}{h_n}\right| &= |\phi_n(h_n)+\psi_n(h_n)| \\ &\ge |\psi_n(h_n)| - |\phi_n(h_n)| \\ &\ge (ab)^n\left\{\frac{2}{3} - \frac{\pi}{ab-1}\right\}.\end{aligned}$$

Because $ab > 1 + 3\pi/2$, the lower bound is positive and we see that

$$\left|\frac{f(x+h_n)-f(x)}{h_n}\right| \to \infty \qquad \text{as } n \to \infty\,.$$

But $h_n > 0$ and $h_n \to 0$ as $n \to \infty$, so this shows that the Weierstrass function f has unbounded right-hand difference quotients at the point x. Since x was chosen arbitrarily, we conclude in particular that Weierstrass's function is not differentiable at any point.

G. H. Hardy [**7**] showed that the requirement $ab > 1 + 3\pi/2$ can be replaced by $ab \ge 1$, but then the proof of nondifferentiability is more difficult.

A similar argument applies to van der Waerden's function (8), but easier estimates suffice to show that its difference quotients are unbounded at each point. The details are outlined in Exercise 20.

And what about Riemann's function? Is it nowhere differentiable as Riemann claimed? In 1916, Hardy [**7**] showed that Riemann's function (6) is not differentiable at any irrational multiple of π or at any rational multiple $\xi\pi$ with ξ of the form $2p/(4q+1)$ or $(2p+1)/(4q+2)$, where p and q are positive integers. There the problem lay until 1970, when Joseph Gerver [**5**] made the unexpected discovery that Riemann's function actually has a derivative, whose value is always $-1/2$, at every point $\xi\pi$ where $\xi = (2p+1)/(2q+1)$; in other words, at every multiple of π by a rational number whose numerator and denominator are both odd. Gerver [**6**] then went on to show that Riemann's function is not differentiable at any point $\xi\pi$ with $\xi = 2p/(2q+1)$ or $(2p+1)/(2q)$, thereby arriving at a full solution of the problem.

A. Smith [**13**] gave a simpler proof of Gerver's result that Riemann's function has a derivative at the specified points.

Finally, it should be mentioned that the existence of continuous nowhere differentiable functions can be proved without actually constructing an example. On the basis of a result from metric space theory known as the Baire category theorem, it can be shown that many such functions exist. In fact,

it turns out to be very unusual, in a certain precise sense, for a continuous function to have bounded one-sided difference quotients at any point! Details may be found for instance in the book by Boas [**3**].

Exercises

1. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{n!}{n^n} x^n$.

2. Directly from the geometric series $\sum_{n=0}^{\infty} x^n = 1/(1-x)$, calculate the sums

$$(a) \qquad \sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2},$$

$$(b) \qquad \sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3},$$

for all $|x| < 1$.

3. (a) If the power series $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ have respective radii of convergence R_1 and R_2, show that the power series $\sum_{n=0}^{\infty} a_n b_n x^n$ has radius of convergence $R \geq R_1 R_2$. Give an example where $R_1 = R_2 = 0$ but $R = \infty$.

(b) If a power series $\sum_{n=0}^{\infty} a_n x^n$ has only a finite number of distinct coefficients a_n and is not a polynomial, show that it has radius of convergence $R = 1$.

4. For $|x| \leq 1$, verify the inequality

$$\left| \frac{x-t}{1+t} \right| \leq |x| \qquad \text{for all } t \text{ between } 0 \text{ and } x.$$

5. Show that

$$\tfrac{1}{2} \log \frac{1+x}{1-x} = x + \tfrac{1}{3} x^3 + \tfrac{1}{5} x^5 + \dots, \qquad |x| < 1.$$

Use the result to calculate $\log \frac{11}{9} = 0.20067\dots$ by choosing only the first 3 nonzero terms. Estimate the remainder in the Taylor series expansion to show that this numerical value is correct to 5 decimal places. Finally, check the number on a calculator or computer and record it to 8 or 10 decimal places.

6. Consider the function $f(x) = e^{-1/x}$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. Show that f is of class C^∞ on $\mathbb{R}$, and that $f^{(n)}(0) = 0$ for $n = 0, 1, 2, \dots$. Conclude that f cannot be expanded into Taylor series at the origin.

7. Here is another way to prove the validity of Newton's binomial series expansion

$$(1+x)^\alpha = 1 + \sum_{k=1}^{\infty} \binom{\alpha}{k} x^k\,, \qquad |x| < 1\,.$$

First prove that the series converges for $|x| < 1$, and let $f(x)$ denote its sum. Verify the identity

$$\binom{\alpha-1}{k} + \binom{\alpha-1}{k-1} = \binom{\alpha}{k}$$

and use it to show that $(1+x)f'(x) = \alpha f(x)$. Then solve the differential equation to conclude that $f(x) = (1+x)^\alpha$.

8. Specialize Newton's binomial series to verify the expansion

$$\sqrt{1+x} = 1 + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}(2k)!}{2^{2k}(k!)^2(2k-1)} x^k\,, \qquad |x| < 1\,.$$

9. Apply Abel's theorem to show that

$$1 + \sum_{k=1}^{\infty} \frac{(-1)^k (2k)!}{2^{2k}(k!)^2} = 1/\sqrt{2}\,.$$

10. Show that

$$(a) \qquad 1 + \sum_{k=1}^{\infty} \frac{(2k)!}{2^{2k}(k!)^2(2k+1)} = \frac{\pi}{2} \qquad \text{and}$$

$$(b) \qquad 1 + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}(2k)!}{2^{2k}(k!)^2(2k-1)} = \sqrt{2}\,.$$

11. Suppose that the functions $u_1(x), u_2(x), \ldots$ are continuous on a set $E \subset \mathbb{R}$, and there is a constant M such that

$$\left| \sum_{k=1}^{n} u_k(x) \right| \le M\,, \qquad n = 1, 2, \ldots\,,$$

for all $x \in E$. Let $\{c_n\}$ be a sequence of positive numbers that decrease to zero: $c_1 \ge c_2 \ge \cdots \ge c_n > 0$ and $c_n \to 0$ as $n \to \infty$. Prove that the series $\sum_{n=1}^{\infty} c_n u_n(x)$ converges uniformly on E.

12. (*a*) Use the identity

$$\cos(k-\tfrac{1}{2})x - \cos(k+\tfrac{1}{2})x = 2\sin\tfrac{1}{2}x\,\sin kx$$

to calculate the sum

$$\sum_{k=1}^{n} \sin kx = \frac{\cos\frac{1}{2}x - \cos(n+\frac{1}{2})x}{2\sin\frac{1}{2}x}\,.$$

(*b*) If $c_1 \geq c_2 \geq \cdots \geq c_n > 0$ and $c_n \to 0$ as $n \to \infty$, show that the series

$$\sum_{n=1}^{\infty} c_n \sin nx$$

is uniformly convergent in the interval $[\delta, 2\pi - \delta]$ for each $\delta > 0$.

13. If the infinite series $\sum_{k=1}^{\infty} a_k$ converges and the sequence $\{b_k\}$ is monotonic and bounded, prove that the series $\sum_{k=1}^{\infty} a_k b_k$ converges.

14. If two functions f and g have power series expansions

$$f(x) = \sum_{n=0}^{\infty} a_n x^n \qquad \text{and} \qquad g(x) = \sum_{n=0}^{\infty} b_n x^n\,, \qquad |x| < R\,,$$

apply Mertens' theorem to prove that their product $h = fg$ is given by the power series

$$h(x) = \sum_{n=0}^{\infty} c_n x^n\,, \qquad \text{where} \quad c_n = \sum_{k=0}^{n} a_{n-k} b_k\,,$$

convergent in the interval $|x| < R$.

15. (*a*) Suppose the series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ converge to sums A and B, respectively. Let $c_n = \sum_{k=0}^{n} a_{n-k} b_k$. If $\sum_{n=0}^{\infty} c_n$ converges to sum C, use Abel's theorem to prove that $C = AB$.

(*b*) If the series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ are absolutely convergent, prove that $\sum_{n=0}^{\infty} c_n$ is absolutely convergent and

$$\sum_{n=0}^{\infty} |c_n| \leq \left(\sum_{n=0}^{\infty} |a_n|\right)\left(\sum_{n=0}^{\infty} |b_n|\right)\,.$$

16. The *Catalan numbers* C_n are defined recursively by $C_0 = 1$ and

$$C_n = \sum_{k=0}^{n-1} C_{n-1-k} C_k\,, \qquad n = 1, 2, \ldots\,.$$

Thus $C_1 = C_0^2 = 1$, $C_2 = 2C_0C_1 = 2$, $C_3 = 2C_0C_2 + C_1^2 = 5$, $C_4 = 14$, $C_5 = 42$, etc. Derive the formula

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)(n!)^2}\,, \qquad n = 1, 2, \ldots\,,$$

and check it by induction.

Suggestion. Consider the generating function $G(x) = \sum_{n=0}^{\infty} C_n x^n$ and show that

$$xG(x)^2 = G(x) - 1\,, \qquad \text{so that} \quad xG(x) = \frac{1 - \sqrt{1-4x}}{2}\,, \qquad |x| < \frac{1}{4}\,.$$

17. Carry out the following steps to derive the formula that Euler used for numerical calculation of the sum $\sum 1/k^2$.

(a) Observe that

$$-\int_0^1 \frac{\log(1-x)}{x}\,dx = \int_0^1 \left\{\sum_{k=1}^{\infty} \frac{x^{k-1}}{k}\right\} dx = \sum_{k=1}^{\infty} \frac{1}{k^2}\,.$$

Justify the term-by-term integration.

(b) Split the integral and change variables to write

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = -\int_0^{1/2} \frac{\log(1-x)}{x}\,dx - \int_0^{1/2} \frac{\log t}{1-t}\,dt\,.$$

(c) Integrate by parts to obtain

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{1}{k^2} &= (\log 2)^2 - 2\int_0^{1/2} \frac{\log(1-x)}{x}\,dx \\ &= (\log 2)^2 + 2\sum_{k=1}^{\infty} \frac{1}{k^2 2^k}\,.\end{aligned}$$

18. The function

$$f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}\,, \qquad |x| \le 1\,,$$

is known as the *dilogarithm*. Generalize the formula of Exercise 17 by showing that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = (\log x)\log(1-x) + f(x) + f(1-x)\,, \qquad 0 < x < 1\,,$$

a result also known to Euler.

Hint. Compute the derivative of the right-hand side.

19. Calculate the integral

$$\int_0^\infty \frac{x}{e^x - 1}\, dx = \frac{\pi^2}{6}\,.$$

Hint. Expand the integrand into a geometric series.

***20.** (*a*) Show that the van der Waerden function f defined by (8) is continuous on the whole real line.

(*b*) Show that the function φ in (8) satisfies the Lipschitz condition $|\varphi(s) - \varphi(t)| \le |s - t|$.

(*c*) For any fixed number $x \in \mathbb{R}$ and any integer $n > 0$, define the number $\delta_n = \pm\frac{1}{2}\,4^{-n}$, where the sign is chosen so that no integer lies strictly between $4^n x$ and $4^n(x + \delta_n)$. Next define the difference quotients

$$Q_k = \frac{\varphi(4^k(x + \delta_n)) - \varphi(4^k x)}{\delta_n}\,, \qquad k = 0, 1, 2, \ldots,$$

and show that $Q_k = 0$ for every $k > n$.

(*d*) Show that $|Q_k| \le 4^k$ for $0 \le k \le n$ and that $|Q_n| = 4^n$.

(*e*) Show that the difference quotients of f satisfy

$$\left|\frac{f(x + \delta_n) - f(x)}{\delta_n}\right| \ge 3^n \left|4^{-n} Q_n\right| - \left|\sum_{k=0}^{n-1} 3^k \left(4^{-k} Q_k\right)\right| \ge \tfrac{1}{2}(3^n + 1)\,,$$

and conclude that f is nowhere differentiable.

Reference. Additional details can be found in Rudin [**12**].

21. Use the identity $(1 + x)^n (1 + x)^n = (1 + x)^{2n}$ to prove that

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}\,, \qquad n = 1, 2, \ldots\,.$$

22. Compare power series to prove that

$$e^{2x} \le \frac{1 + x}{1 - x}\,, \qquad 0 \le x < 1\,.$$

23. The theory of power series can be extended to the complex plane. For any complex coefficients c_n the power series $\sum_{n=0}^{\infty} c_n z^n$ in a complex variable z can be shown to converge for every point in some disk $|z| < R$

and to diverge whenever $|z| > R$. (Here, as for real power series, we adopt the convention that $0 \le R \le \infty$.) For instance, this analysis allows us to define the exponential function

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

for all complex numbers z. Use this definition to derive Euler's formula

$$e^{ix} = \cos x + i \, \sin x \,, \qquad x \in \mathbb{R} \,.$$

Note in particular that $e^{i\pi} = -1$.

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Chapter 4
Inequalities

Inequalities are essential tools for analysis. Proofs of theorems or derivations of formulas often depend on estimates that are conveniently performed by appeal to general inequalities. The primary goal of this chapter is to develop some basic inequalities, involving sums and integrals, that are commonly applied in analysis.

4.1. Elementary inequalities

We begin by recalling three elementary but extremely useful inequalities. The first is

$$1 + x \le e^x\,, \qquad -\infty < x < \infty\,,$$

or equivalently

$$\log x \le x - 1\,, \qquad 0 < x < \infty\,.$$

The graphs in Figure 1 illustrate the two forms of the inequality.

The proof is quite easy. The function $f(x) = e^x - x - 1$ has the value $f(0) = 0$ and derivative $f'(x) = e^x - 1$, with $f'(0) = 0$. But e^x is an increasing function, so $f'(x) > 0$ for $x > 0$ and $f'(0) < 0$ for $x < 0$. Thus $f(x)$ decreases to zero for $x < 0$ and increases from zero for $x > 0$, so that $f(x) > 0$ for all $x \neq 0$. In other words, $1 + x < e^x$ for all $x \neq 0$.

The next inequality is

$$\frac{2}{\pi}\,x < \sin x < x\,, \qquad 0 < x < \frac{\pi}{2}\,,$$

as illustrated by Figure 2.

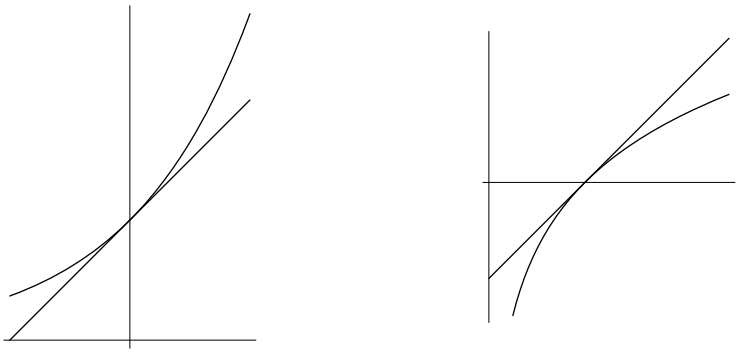


Figure 1. The inequalities $1 + x \leq e^x$ and $\log x \leq x - 1$.

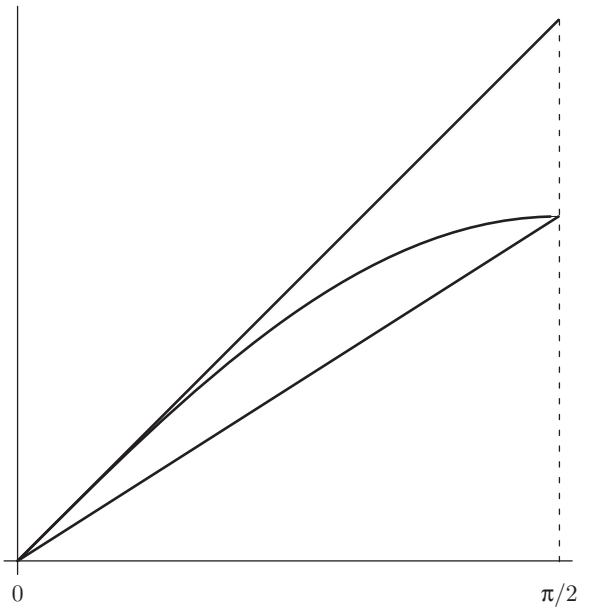


Figure 2. The inequality $2x/\pi < \sin x < x$.

We will give a proof based directly on the geometric definitions of sine and cosine, not assuming the differentiation formula $\dfrac{d}{dx}\{\sin x\} = \cos x$ but deriving it along the way. The proof proceeds by comparing areas of the two right triangles and the circular sector in Figure 3.

This shows that

$$\tfrac{1}{2}\cos\theta\sin\theta < \tfrac{1}{2}\theta < \tfrac{1}{2}\tan\theta\,,$$

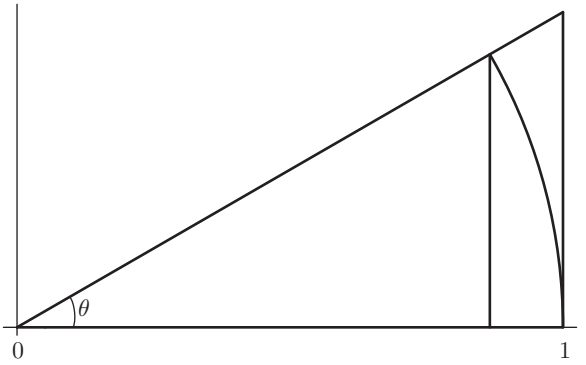


Figure 3. The inequality $\cos\theta \sin\theta < \theta < \tan\theta$.

so that $\theta < \tan\theta$ and

$$\cos\theta < \frac{\sin\theta}{\theta} < \frac{1}{\cos\theta}\,, \qquad 0 < \theta < \frac{\pi}{2}\,.$$

It now follows from the squeeze principle that

$$\lim_{x\to 0} \frac{\sin x}{x} = 1\,,$$

which implies, in view of the half-angle formula $1 - \cos x = 2\sin^2(x/2)$, that

$$\lim_{x\to 0} \frac{1-\cos x}{x} = 0\,.$$

Hence the addition formula for the sine function gives

$$\frac{\sin(x+h) - \sin x}{h} = \sin x \left(\frac{\cos h - 1}{h}\right) + \cos x \left(\frac{\sin h}{h}\right) \to \cos x$$

as $h \to 0$. In other words, $\dfrac{d}{dx}\{\sin x\} = \cos x$. Now observe that

$$\frac{d}{dx}\left\{\frac{\sin x}{x}\right\} = \frac{x\cos x - \sin x}{x^2} < 0\,, \qquad 0 < x < \frac{\pi}{2}\,,$$

since $x < \tan x$. Thus the function $g(x) = \dfrac{\sin x}{x}$ decreases from $g(0) = 1$ to $g(\pi/2) = 2/\pi$, so that

$$\frac{2}{\pi} < \frac{\sin x}{x} < 1\,, \qquad 0 < x < \frac{\pi}{2}\,,$$

which was to be proved.

Our third inequality is

$$(1+x)^p \geq 1+px\,, \qquad p \geq 1\,,\ x > -1\,.$$

If p is a positive integer and $x \geq 0$, the inequality is a simple consequence of the binomial theorem. An inductive proof (see Exercise 1) extends its validity to $x > -1$ when $p = 1, 2, \ldots$. In the general case where p need not be an integer, the inequality can be proved by calculus. Take $p > 1$ and define the function

$$h(x) = (1+x)^p - px - 1\,,$$

with $h(0) = 0$ and derivative

$$h'(x) = p\left[(1+x)^{p-1} - 1\right]\,.$$

Then $h'(x) < 0$ for $-1 < x < 0$, whereas $h'(x) > 0$ for $x > 0$. Since $h(0) = 0$, it follows that $h(x) \geq 0$ for all $x > -1$, which is the desired result.

4.2. Cauchy's inequality

Cauchy's inequality is well known and is one of the most widely used of all inequalities. It can be stated as follows.

Cauchy's Inequality. *For arbitrary real numbers $a_1, a_2, \ldots, a_n$ and b_1, $b_2, \ldots, b_n$,*

$$\left|\sum_{k=1}^{n} a_k b_k\right| \leq \left(\sum_{k=1}^{n} a_k^2\right)^{1/2} \left(\sum_{k=1}^{n} b_k^2\right)^{1/2}\,.$$

Unless $a_k \equiv 0$ or $b_k \equiv 0$, equality occurs if and only if $a_k = \lambda b_k$ for some real number λ and all $k = 1, 2, \ldots, n$.

This famous result was recorded by Augustin-Louis Cauchy (1789–1857) in his text *Cours d'analyse* for students at the École Royale Polytechnique in Paris. The book, published in 1821, was a landmark in the history of mathematics because it was the first serious attempt at a rigorous development of calculus.

Cauchy's inequality can be recast in vector form. With the notation $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ for vectors in $\mathbb{R}^n$, one defines their *dot product* (or *scalar product*) by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n\,.$$

The *norm* of $\mathbf{a}$ is defined to be $\|\mathbf{a}\| = \left\{a_1^2 + a_2^2 + \cdots + a_n^2\right\}^{1/2}$, so that $\mathbf{a}\cdot\mathbf{a} = \|\mathbf{a}\|^2$. Cauchy's inequality then says that $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\|\|\mathbf{b}\|$.

For a proof, let t be a real parameter and write

$$0 \le \|\mathbf{a} - t\mathbf{b}\|^2 = (\mathbf{a} - t\mathbf{b}) \cdot (\mathbf{a} - t\mathbf{b}) = \|\mathbf{a}\|^2 - 2t\,\mathbf{a} \cdot \mathbf{b} + t^2\,\|\mathbf{b}\|^2 .$$

But if a quadratic polynomial $At^2 + Bt + C$ is nonnegative for all values of the variable t, then $B^2 - 4AC \le 0$. Thus we conclude that $|\mathbf{a} \cdot \mathbf{b}| \le \|\mathbf{a}\|\,\|\mathbf{b}\|$, which is Cauchy's inequality. If $\mathbf{b} \ne \mathbf{0}$, equality occurs if and only if $\mathbf{a} - t\mathbf{b} = \mathbf{0}$, or $\mathbf{a} = t\mathbf{b}$ for some real number t.

A more illuminating proof comes from *Lagrange's identity*

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 = \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right) - \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n}(a_j b_k - a_k b_j)^2 ,$$

which can be viewed as a quantitative form of Cauchy's inequality since it gives an explicit formula, obviously nonnegative, for the discrepancy $\|\mathbf{a}\|^2\|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2$. Once discovered, Lagrange's identity can be verified by straightforward algebra.

Lagrange's identity plays an important role in vector analysis. For vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$, the *cross product* (or *vector product*) is defined by

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1) ,$$

so that

$$\|\mathbf{a} \times \mathbf{b}\|^2 = \sum_{j<k}(a_j b_k - a_k b_j)^2 = \frac{1}{2}\sum_{j=1}^{3}\sum_{k=1}^{3}(a_j b_k - a_k b_j)^2 .$$

Hence Lagrange's identity reduces to $(\mathbf{a} \cdot \mathbf{b})^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \|\mathbf{a} \times \mathbf{b}\|^2$ when $n = 3$. But it follows from the law of cosines that $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$, where θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$. Therefore, Lagrange's identity gives the formula $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$.

An important generalization of Cauchy's inequality was discovered in 1884 by Otto Hölder (1859–1937), who is known primarily for his work in abstract algebra. Hölder stated his inequality in different form, but here is the modern formulation.

Hölder's Inequality. *Let p and q be positive numbers such that $\frac{1}{p} + \frac{1}{q} = 1$. Then for arbitrary real numbers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$,*

$$\left|\sum_{k=1}^{n} a_k b_k\right| \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p}\left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q} .$$

Unless $a_k \equiv 0$ or $b_k \equiv 0$, equality occurs if and only if $a_k b_k$ has constant sign and $|a_k|^p = \lambda |b_k|^q$ for some number $\lambda > 0$ and all $k = 1, 2, \ldots, n$.

Two positive numbers p and q such that $\frac{1}{p} + \frac{1}{q} = 1$ are called *conjugate indices*. The standard proof of Hölder's inequality makes use of *Young's inequality*

$$ab \le \frac{1}{p}\, a^p + \frac{1}{q}\, b^q\,, \qquad a > 0\,,\ b > 0\,,$$

where p and q are conjugate indices. For $p = q = 2$ it is simply the inequality $(a - b)^2 \ge 0$. Equality occurs in Young's inequality if and only if $a^p = b^q$.

Here is a geometric proof of Young's inequality. The area under the curve $y = x^{p-1}$, for $0 \le x \le a$, is given by the integral

$$\int_0^a x^{p-1}\, dx = \frac{1}{p}\, a^p\,.$$

The inverse function is found to be $x = y^{1/(p-1)} = y^{q-1}$, since $\frac{1}{p} + \frac{1}{q} = 1$. The area under the curve $x = y^{q-1}$, for $0 \le y \le b$, is

$$\int_0^b y^{q-1}\, dy = \frac{1}{q}\, b^q\,.$$

But it is clear geometrically (see Figure 4) that the sum of these two areas is greater than or equal to the area of the rectangle $0 \le x \le a$, $0 \le y \le b$. Hence

$$ab \le \frac{1}{p}\, a^p + \frac{1}{q}\, b^q\,,$$

with equality if and only if $a^{p-1} = b$, so that the curve $y = x^{p-1}$ passes through the vertex (a, b) of the rectangle. The condition for equality reduces to $a^p = b^q$, as claimed.

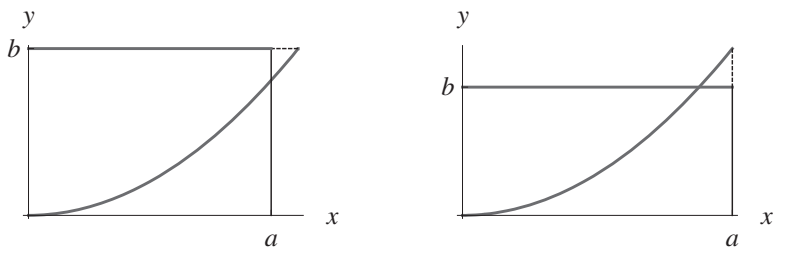


Figure 4. Young's inequality: $a^{p-1} < b$ and $a^{p-1} > b$.

For a proof of Hölder's inequality, it is enough to consider the case where

$$\sum_{k=1}^{n} |a_k|^p = \sum_{k=1}^{n} |b_k|^q = 1\,,$$

because the general case can then be deduced by normalization. After this reduction, Young's inequality gives

$$\begin{aligned}\left|\sum_{k=1}^{n} a_k b_k\right| &\le \frac{1}{p}\sum_{k=1}^{n} |a_k|^p + \frac{1}{q}\sum_{k=1}^{n} |b_k|^q = \frac{1}{p} + \frac{1}{q} = 1 \\ &= \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q},\end{aligned}$$

which is Hölder's inequality. A close examination of the proof shows that equality occurs if and only if $a_k b_k$ has constant sign and $|a_k|^p = \lambda |b_k|^q$ for all k. The details are left as an exercise.

In terms of a parameter $p \ge 1$, the notion of *norm* of a vector $\mathbf{a} = (a_1, a_2, \ldots a_n)$ can be generalized to

$$\|\mathbf{a}\|_p = \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p}.$$

Previously we wrote $\|\mathbf{a}\|$ for $\|\mathbf{a}\|_2$. Hölder's inequality is compactly stated as $|\mathbf{a}\cdot\mathbf{b}| \le \|\mathbf{a}\|_p \, \|\mathbf{b}\|_q$, where p and q are conjugate indices. With the definition

$$\|\mathbf{b}\|_\infty = \max_{1\le k\le n} |b_k|\,,$$

Hölder's inequality extends to the case $p = 1$, $q = \infty$, where the proof is trivial. (For motivation of the notation $\|\mathbf{b}\|_\infty$, see Exercise 15.)

The term "norm" is justified by the relation

$$\|\mathbf{a} + \mathbf{b}\|_p \le \|\mathbf{a}\|_p + \|\mathbf{b}\|_p\,, \qquad p \ge 1\,,$$

known as the *triangle inequality* because of its geometric interpretation when $p = 2$ and $n = 2$ or 3. For $p = 2$ the triangle inequality is a simple consequence of Cauchy's inequality:

$$\begin{aligned}\|\mathbf{a} + \mathbf{b}\|^2 &= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = \|\mathbf{a}\|^2 + 2\mathbf{a}\cdot\mathbf{b} + \|\mathbf{b}\|^2 \\ &\le \|\mathbf{a}\|^2 + 2\,\|\mathbf{a}\|\|\mathbf{b}\| + \|\mathbf{b}\|^2 = (\|\mathbf{a}\| + \|\mathbf{b}\|)^2.\end{aligned}$$

For $p \ne 2$ the result is more subtle and is known as *Minkowski's inequality.*

Minkowski's Inequality. *For $1 \le p < \infty$ and for arbitrary real numbers $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$,*

$$\left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{1/p} \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} + \left(\sum_{k=1}^{n} |b_k|^p\right)^{1/p}.$$

Hermann Minkowski (1864–1909) is famous for his 4-dimensional "space-time" non-Euclidean model of the universe, which served as a foundation for the mathematical theory of relativity. He also did important work in number theory, where he developed a well-known theory of "geometry of numbers". Minkowski's inequality appeared in his book *Geometrie der Zahlen*, first published in 1896.

The inequality is trivial for $p = 1$, since $|a_k + b_k| \le |a_k| + |b_k|$ for all k. To prove it for $1 < p < \infty$, observe that

$$\sum_{k=1}^{n} |a_k + b_k|^p \le \sum_{k=1}^{n} |a_k||a_k + b_k|^{p-1} + \sum_{k=1}^{n} |b_k||a_k + b_k|^{p-1},$$

and apply Hölder's inequality to each of the sums on the right-hand side. Since the conjugate index of p is $q = p/(p-1)$, we find

$$\sum_{k=1}^{n} |a_k||a_k + b_k|^{p-1} \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{(p-1)/p},$$

and similarly

$$\sum_{k=1}^{n} |b_k||a_k + b_k|^{p-1} \le \left(\sum_{k=1}^{n} |b_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{(p-1)/p}.$$

Thus we have arrived at the inequality

$$\|\mathbf{a} + \mathbf{b}\|_p^p \le (\|\mathbf{a}\|_p + \|\mathbf{b}\|_p)\, \|\mathbf{a} + \mathbf{b}\|_p^{p-1}.$$

Assuming that $\|\mathbf{a} + \mathbf{b}\|_p \ne 0$ (otherwise there is nothing to prove), we now divide both sides by $\|\mathbf{a} + \mathbf{b}\|_p^{p-1}$ to obtain Minkowski's inequality.

All of the foregoing inequalities have immediate extensions to infinite sequences, or to Euclidean spaces of infinite dimension. For $0 < p < \infty$, a sequence $\mathbf{a} = (a_1, a_2, \dots)$ is said to belong to the space ℓ^p if

$$\|\mathbf{a}\|_p^p = \sum_{k=1}^{\infty} |a_k|^p < \infty.$$

In this terminology, for instance, the extended form of Hölder's inequality says that if $\mathbf{a} \in \ell^p$ and $\mathbf{b} \in \ell^q$, where p and q are conjugate indices, then the sequence $\mathbf{c} = (c_1, c_2, \dots)$ with $c_k = a_k b_k$ belongs to ℓ^1 and $\|\mathbf{c}\|_1 \le \|\mathbf{a}\|_p \|\mathbf{b}\|_q$.

4.3. Arithmetic–geometric mean inequality

For any set of positive numbers $a_1, a_2, \ldots, a_n$, the *arithmetic mean* A_n and the *geometric mean* G_n are defined by

$$A_n = \frac{a_1 + a_2 + \cdots + a_n}{n} \qquad \text{and} \qquad G_n = \sqrt[n]{a_1 a_2 \cdots a_n}\,.$$

The *arithmetic–geometric mean inequality* says that the arithmetic mean is larger unless all of the numbers are equal.

Theorem. *For any positive numbers $a_1, a_2, \ldots, a_n$, the inequality $G_n \le A_n$ holds, with equality if and only if $a_1 = a_2 = \cdots = a_n$.*

This fact has been known for a long time, and many clever proofs have been devised. We will give Cauchy's proof, which is particularly ingenious. The first step is to apply an inductive argument to establish the inequality in the special case where n is a power of 2. The result for intervening values of n is then deduced by a method of descent.

For $n = 1$ there is nothing to prove, since $G_1 = A_1 = a_1$. For $n = 2$ the inequality $G_2 \le A_2$ takes the form

$$a_1 a_2 \le \left(\frac{a_1 + a_2}{2}\right)^2$$

and is equivalent to the trivial relation $(\sqrt{a_1} - \sqrt{a_2})^2 \ge 0$, where equality occurs only for $a_1 = a_2$. The proof for $n = 4$ now follows by writing $a_1 a_2 a_3 a_4 = (a_1 a_2)(a_3 a_4)$ and applying the result for $n = 2$:

$$\begin{aligned}(a_1 a_2)(a_3 a_4) &\le \left(\frac{a_1 + a_2}{2}\right)^2 \left(\frac{a_3 + a_4}{2}\right)^2 = \left\{\left(\frac{a_1 + a_2}{2}\right)\left(\frac{a_3 + a_4}{2}\right)\right\}^2 \\ &\le \left\{\frac{1}{2}\left[\left(\frac{a_1 + a_2}{2}\right) + \left(\frac{a_3 + a_4}{2}\right)\right]\right\}^4 = \left(\frac{a_1 + a_2 + a_3 + a_4}{4}\right)^4 .\end{aligned}$$

This last argument can be generalized to prove the inequality for all indices of the form $n = 2^k$, $k = 1, 2, \ldots$. Proceeding inductively, suppose it has been established for some index $n = 2^k$, so that

$$a_1 a_2 \cdots a_{2^k} \le \left(\frac{a_1 + a_2 + \cdots + a_{2^k}}{2^k}\right)^{2^k} .$$

Then the inequality for $n = 2^{k+1}$ is deduced by writing

$$\begin{aligned}a_1 a_2 \cdots a_{2^{k+1}} &= \left(a_1 a_2 \cdots a_{2^k}\right)\left(a_{2^k+1} a_{2^k+2} \cdots a_{2^{k+1}}\right) \\ &\le \left(\frac{a_1 + a_2 + \cdots + a_{2^k}}{2^k}\right)^{2^k} \left(\frac{a_{2^k+1} + a_{2^k+2} + \cdots + a_{2^{k+1}}}{2^k}\right)^{2^k} \\ &\le \left(\frac{a_1 + a_2 + \cdots + a_{2^{k+1}}}{2^{k+1}}\right)^{2^{k+1}} ,\end{aligned}$$

where the last step again used the result for $n = 2$. This proves the inequality for all indices $n = 2^k$.

If n is not a power of 2, then $2^{k-1} < n < 2^k$ for some integer k. Apply the inequality for index 2^k to the numbers

$$a_1, a_2, \dots, a_n, A_n, A_n, \dots, A_n\,,$$

where the arithmetic mean A_n appears $m = 2^k - n$ times. Then

$$\begin{aligned} a_1 a_2 \cdots a_n \, {A_n}^m &\le \left(\frac{a_1 + a_2 + \cdots + a_n + mA_n}{2^k} \right)^{2^k} \\ &= \left(\frac{(n+m)A_n}{2^k} \right)^{2^k} = A_n^{2^k}\,, \end{aligned}$$

or ${G_n}^n \, {A_n}^m \le {A_n}^{2^k}$, which reduces to the desired inequality $G_n \le A_n$. An inspection of the proof shows that equality occurs at each step precisely when all of the numbers a_j are equal.

4.4. Integral analogues

The inequalities of Hölder and Minkowski have clear analogues for integrals. We will operate under the standing assumption that our functions are Riemann integrable over the real line. Functions defined on a fixed subinterval can then be regarded as a special case. We will say that a function f belongs to the class L^p, where $0 < p < \infty$, if it has finite L^p norm:

$$\|f\|_p = \left(\int_{-\infty}^{\infty} |f(x)|^p \, dx \right)^{1/p} < \infty\,.$$

If f is bounded, we say that $f \in L^\infty$ and define the norm

$$\|f\|_\infty = \sup_{x \in \mathbb{R}} |f(x)|\,.$$

In reality, the space L^p is customarily defined to consist of the larger class of Lebesgue measurable functions with finite L^p norm. The Hölder and Minkowski inequalities remain valid in that more general context, but the theory of Lebesgue integrals is beyond the scope of this book and we will confine ourselves to Riemann integrals.

In the terminology of L^p spaces, the continuous forms of Hölder's and Minkowski's inequalities can be stated as follows.

Hölder's Inequality (continuous form). *Let p and q be conjugate indices: $\frac{1}{p} + \frac{1}{q} = 1$. If $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and $\|fg\|_1 \le \|f\|_p \, \|g\|_q$.*

Minkowski's Inequality (continuous form). *Let $1 \le p \le \infty$. If $f \in L^p$ and $g \in L^p$, then $(f+g) \in L^p$ and $\|f+g\|_p \le \|f\|_p + \|g\|_p$.*

The proofs are essentially the same as for sums and are left as exercises. Equality occurs in Hölder's inequality if and only if $f(x)g(x)$ has constant sign and $|f(x)|^p = \lambda |g(x)|^q$.

The integral analogue of Cauchy's inequality, or Hölder's inequality for $p = q = 2$, is

$$\left| \int_{-\infty}^{\infty} f(x)g(x)\,dx \right| \le \left(\int_{-\infty}^{\infty} |f(x)|^2\,dx \right)^{1/2} \left(\int_{-\infty}^{\infty} |g(x)|^2\,dx \right)^{1/2} .$$

It is known as the *Schwarz inequality*, or the *Cauchy–Schwarz inequality*, after Hermann Amandus Schwarz (1843–1921), who recorded it in a paper on minimal surfaces published in 1885. However, the result had already been formulated as early as 1859 by a Russian mathematician, Victor Yacovlevich Bunyakovski (1804–1889), and so it often bears his name as well. In any case, as Bunyakovski himself observed, the integral form is a simple consequence of Cauchy's inequality, by passage from Riemann sums to integrals.

There is also an integral analogue of the arithmetic–geometric mean inequality, which becomes apparent if the geometric mean is written in the form

$$(a_1 a_2 \cdots a_n)^{1/n} = \exp\left\{ \frac{1}{n} \sum_{k=1}^{n} \log a_k \right\} .$$

This suggests the inequality

$$\exp\left\{ \frac{1}{b-a} \int_a^b \log f(x)\,dx \right\} \le \frac{1}{b-a} \int_a^b f(x)\,dx$$

for positive functions $f(x)$ on the interval $[a, b]$. Under the assumption that f and $\log f$ are Riemann integrable, a proof can be based on the classical arithmetic–geometric mean inequality for sums, approximating the integrals by their corresponding Riemann sums. However, as we will see presently, the result depends only on the convexity of the exponential function and is a special case of a much more general inequality, known as Jensen's inequality.

4.5. Jensen's inequality

The statement of Jensen's inequality will involve the notion of a convex function. A function $\varphi(x)$ is said to be *convex* on an interval $I \subset \mathbb{R}$ if for each pair of points $x_1, x_2 \in I$ it satisfies the inequality

$$\varphi(tx_1 + (1-t)x_2) \le t\varphi(x_1) + (1-t)\varphi(x_2)\,, \qquad 0 \le t \le 1\,.$$

The function is *strictly convex* if for each pair of distinct points $x_1, x_2 \in I$,

$$\varphi(tx_1 + (1-t)x_2) < t\varphi(x_1) + (1-t)\varphi(x_2)\,, \qquad 0 < t < 1\,.$$

Geometrically, these inequalities say that the graph of the function lies on or below every chord. Jensen's inequality, in most primitive form, asserts that a similar inequality then holds for more general means. Here, then, is the discrete form of the inequality.

Jensen's Inequality. *Let $\varphi(x)$ be a convex function on an interval $I \subset \mathbb{R}$, and let $t_1, t_2, \ldots, t_n$ be any collection of positive numbers with sum $t_1 + t_2 + \cdots + t_n = 1$. Then for each choice of points $x_1, x_2, \ldots, x_n \in I$ the inequality*

$$\varphi(t_1x_1 + t_2x_2 + \cdots + t_nx_n) \le t_1\varphi(x_1) + t_2\varphi(x_2) + \cdots + t_n\varphi(x_n)$$

holds. If φ is strictly convex, equality occurs only for $x_1 = x_2 = \cdots = x_n$.

The inequality is named for the Danish mathematician J. L. W. V. Jensen (1859–1925), who recognized the importance of convexity and published a more restrictive result in 1906. Jensen was an amateur mathematician who received no formal training in research-level mathematics and never held an academic position, supporting himself as an employee of the telephone company in Copenhagen.

Before turning to a proof of Jensen's inequality, let us observe that the arithmetic–geometric mean inequality is an immediate corollary. For this we need only choose the function $\varphi(x) = e^x$ and take $t_1 = t_2 = \cdots = t_n = \frac{1}{n}$. Writing $a_k = e^{x_k}$, we observe that

$$t_1x_1 + t_2x_2 + \cdots + t_nx_n = \tfrac{1}{n}\,\log(a_1a_2\cdots a_n) = \log\left(\sqrt[n]{a_1a_2\cdots a_n}\right)\,.$$

Since the exponential function is convex, Jensen's theorem gives

$$\sqrt[n]{a_1a_2\cdots a_n} \le \tfrac{1}{n}(a_1 + a_2 + \cdots + a_n)\,,$$

which is the arithmetic–geometric mean inequality. Moreover, because the exponential function is strictly convex, equality occurs only when all of the numbers a_k are equal.

Jensen's inequality will now be proved by induction. For $n = 2$ it is simply a restatement of the definition of convexity. Suppose the inequality holds for some $n \ge 2$ and write

$$\sum_{k=1}^{n+1} t_kx_k = t_{n+1}x_{n+1} + (1 - t_{n+1})\sum_{k=1}^{n} \frac{t_k}{1 - t_{n+1}}\,x_k$$

for arbitrary points $x_1, \dots, x_{n+1} \in I$ and positive weights t_k with sum $t_1 + t_2 + \cdots + t_{n+1} = 1$. Then by the convexity of φ and the inductive hypothesis,

$$\varphi\Big(\sum_{k=1}^{n+1} t_k x_k\Big) \le t_{n+1}\varphi(x_{n+1}) + (1-t_{n+1})\,\varphi\Big(\sum_{k=1}^{n} \frac{t_k}{1-t_{n+1}}\, x_k\Big)$$
$$\le t_{n+1}\varphi(x_{n+1}) + (1-t_{n+1}) \sum_{k=1}^{n} \frac{t_k}{1-t_{n+1}}\, \varphi(x_k) = \sum_{k=1}^{n+1} t_k \varphi(x_k)\,,$$

which is the desired result for $n+1$. This proves Jensen's inequality. The case of equality is left as an exercise.

It is not difficult to formulate an appropriate analogue for integrals. In continuous form, Jensen's inequality may be expressed as follows.

Jensen's Inequality (continuous form). *Let $\varphi(y)$ be a convex function on an interval $J \subset \mathbb{R}$. Suppose the function $y = f(x)$ is integrable over an interval $I \subset \mathbb{R}$, with $f(I) \subset J$. Let $w(x) \ge 0$ be a weight function on I with integral $\int_I w(x)\,dx = 1$. Then*

$$\varphi\left(\int_I f(x)w(x)\,dx\right) \le \int_I \varphi(f(x))w(x)\,dx\,.$$

If φ is strictly convex, $w(x) > 0$, and f is continuous, then equality holds only when f is constant.

The continuous form can be deduced from the discrete form by the usual process of approximating integrals by their Riemann sums, but it is more interesting to give an independent proof. For simplicity we will make the assumption that φ has a nondecreasing derivative on J. It then follows that the graph of φ lies above each of its tangent lines. More precisely, for each point $y_0 \in J$ the inequality

$$\varphi(y_0) + \varphi'(y_0)(y - y_0) \le \varphi(y)$$

holds for all $y \in J$, with strict inequality for all $y \ne y_0$ if φ is strictly convex. Setting $y = f(x)$, we have

$$\varphi(y_0) + \varphi'(y_0)(f(x) - y_0) \le \varphi(f(x))\,, \qquad x \in I\,.$$

Now multiply the inequality by $w(x)$ and integrate over I to obtain

$$\varphi(y_0) + \varphi'(y_0)\left(\int_I f(x)w(x)\,dx - y_0\right) \le \int_I \varphi(f(x))w(x)\,dx\,,$$

since w is a unit weight. With the choice $y_0 = \int_I f(x)w(x)dx$, this gives the continuous form of Jensen's inequality. The case of equality is again left as an exercise.

The proof is essentially the same without the simplifying assumption that φ is differentiable and has a nondecreasing derivative, because it is possible to find a suitable substitute for the tangent line to the graph of a general convex function φ. Details may be found in the classic book of Hardy, Littlewood, and Pólya [**2**] or in the recent book by Steele [**6**].

Specific applications of Jensen's inequality, in either discrete or continuous form, will depend upon knowing that a given function φ is convex. The most useful criterion for convexity is that φ have a second derivative and $\varphi''(x) \geq 0$ on an interval (a, b). Similarly, φ is strictly convex if $\varphi''(x) > 0$ on (a, b). Although it is intuitively clear from graphical considerations that the property $\varphi''(x) \geq 0$ implies convexity, a proof is more elusive.

Here is one approach. Given that $\varphi''(x) \geq 0$, or that $\varphi'(x)$ is nondecreasing on (a, b), it is to be shown that

$$\varphi(tx + (1-t)y) \leq t\varphi(x) + (1-t)\varphi(y)$$

whenever $a < x < y < b$ and $0 < t < 1$. To see this, let $s = 1 - t$ and write

$$\begin{aligned}\varphi(tx + sy) - t\varphi(x) - s\varphi(y) &= t\int_x^{tx+sy} \varphi'(u)\,du - s\int_{tx+sy}^{y} \varphi'(u)\,du \\ &\leq ts(y-x)\varphi'(tx+sy) - st(y-x)\varphi'(tx+sy) = 0\,,\end{aligned}$$

by the monotonicity of φ'. The same argument shows that φ is strictly convex if φ' is strictly increasing.

4.6. Hilbert's inequality

This chapter concludes with a special inequality, due to David Hilbert (1862–1943), sometimes called *Hilbert's double series theorem.*

Hilbert's Inequality. *For arbitrary square-summable sequences $\{a_j\}$ and $\{b_k\}$, the inequality*

$$\left|\sum_{j=1}^{\infty}\sum_{k=1}^{\infty}\frac{a_j b_k}{j+k-1}\right| \leq \pi\left(\sum_{j=1}^{\infty} a_j^2\right)^{1/2}\left(\sum_{k=1}^{\infty} b_k^2\right)^{1/2} \tag{1}$$

holds, with strict inequality unless one of the sequences $\{a_j\}$ or $\{b_k\}$ is identically zero. The constant π cannot be replaced by any smaller number.

In other words, the bound is best possible but is attained only in trivial cases.

Hilbert's inequality has found applications to real and complex analysis and to analytic number theory. Hilbert originally included the result in

lectures at Göttingen on integral equations, but his proof was first published in the 1908 dissertation of his student Hermann Weyl. Actually, Hilbert obtained a weaker inequality with constant 2π, and several years later Issai Schur constructed another proof that led to the sharp constant π. Since that time many proofs have been found, but none is entirely simple. (See, for instance, Hardy–Littlewood–Pólya [**4**], Oleszkiewicz [**5**], and Steele [**6**].) The proof presented here was discovered by David Ullrich [**7**] and seems relatively simple and straightforward. The strategy is to begin with a continuous form of the theorem and to deduce the discrete form from it.

Hilbert's Inequality (continuous form). *For any functions f and g in the space $L^2(0,\infty)$, the inequality*

$$\left| \int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\, dxdy \right| \le \pi \, \|f\|_2 \, \|g\|_2 \tag{2}$$

holds, with strict inequality unless f or g is the zero-function. The constant π cannot be replaced by any smaller number.

Proof of continuous form. The change of variables $x = u^2, y = v^2$ gives

$$\int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\, dxdy = 4 \int_0^\infty \int_0^\infty \frac{f(u^2)g(v^2)}{u^2+v^2}\, uv\, dudv\,.$$

Now introduce polar coordinates $u = r\cos\theta$, $v = r\sin\theta$ and apply the Cauchy–Schwarz inequality to obtain

$$\begin{aligned}
&\left| \int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\, dxdy \right| \\
&\le 4 \int_0^{\pi/2} \cos\theta \sin\theta \int_0^\infty \left| f(r^2\cos^2\theta)\, g(r^2\sin^2\theta) \right| r\, drd\theta \\
&\le 4 \int_0^{\pi/2} \cos\theta \sin\theta \left\{ \int_0^\infty f(r^2\cos^2\theta)^2\, r\, dr \right\}^{1/2} \left\{ \int_0^\infty g(r^2\sin^2\theta)^2\, r\, dr \right\}^{1/2} d\theta \\
&= 2 \int_0^{\pi/2} \left\{ \int_0^\infty f(s)^2\, ds \right\}^{1/2} \left\{ \int_0^\infty g(t)^2\, dt \right\}^{1/2} d\theta = \pi \, \|f\|_2 \, \|g\|_2\,,
\end{aligned}$$

with the substitution $s = r^2\cos^2\theta$, $t = r^2\sin^2\theta$. Thus we have arrived at the inequality (2). We defer the proofs that the constant π is best possible and is never attained, passing first to the discrete form of the inequality. □

Proof of discrete form. The discrete form (1) of Hilbert's inequality is essentially a special case of the continuous form (2). In order to deduce (1) from (2), we define the functions ψ_j on $(0,\infty)$ by $\psi_j(x) = 1$ for $x \in (j-1, j]$

and $\psi_j(x) = 0$ elsewhere, $j = 1, 2, \ldots$. Given a pair of sequences $\{a_j\}$ and $\{b_k\}$ in ℓ^2, let

$$f(x) = \sum_{j=1}^{\infty} a_j \psi_j(x) \qquad \text{and} \qquad g(y) = \sum_{k=1}^{\infty} b_k \psi_k(y)\,.$$

Then

$$\|f\|_2^2 = \sum_{j=1}^{\infty} a_j^2\,, \qquad \|g\|_2^2 = \sum_{k=1}^{\infty} b_k^2\,, \qquad \text{and}$$
$$\int_0^{\infty}\int_0^{\infty} \frac{f(x)g(y)}{x+y}\,dxdy = \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} w_{jk}\, a_j b_k\,, \quad \text{where}$$
$$w_{jk} = \int_{k-1}^{k}\int_{j-1}^{j} \frac{1}{x+y}\,dxdy\,.$$

It is obvious that $w_{jk} > 1/(j+k)$, and in fact the convexity of the integrand implies that $w_{jk} > 1/(j+k-1)$. Specifically, for each fixed $y > 0$, we see that

$$\int_{j-1}^{j} \frac{1}{x+y}\,dx > \frac{1}{y+j-\frac{1}{2}}\,, \qquad j = 1, 2, \ldots\,,$$

by comparing the area under the curve with the area under the tangent line at the midpoint $x = j - \frac{1}{2}$. A similar estimate then shows that

$$w_{jk} > \int_{k-1}^{k} \frac{1}{y+j-\frac{1}{2}}\,dy > \frac{1}{(k-\frac{1}{2})+(j-\frac{1}{2})} = \frac{1}{j+k-1}\,. \tag{3}$$

Therefore, if neither $\{a_j\}$ nor $\{b_k\}$ is the zero-sequence, it follows from (2) that

$$\left|\sum_{j=1}^{\infty}\sum_{k=1}^{\infty} \frac{a_j b_k}{j+k-1}\right| < \sum_{j=1}^{\infty}\sum_{k=1}^{\infty} w_{jk}\,|a_j b_k| = \int_0^{\infty}\int_0^{\infty} \frac{|f(x)g(y)|}{x+y}\,dxdy$$
$$\leq \pi\,\|f\|_2\,\|g\|_2 = \pi\left(\sum_{j=1}^{\infty} a_j^2\right)^{1/2}\left(\sum_{k=1}^{\infty} b_k^2\right)^{1/2},$$

which proves (1) with strict inequality. Note that (3) gives strict inequality because $|a_j b_k| > 0$ for some pair of indices j and k. □

Proof of sharpness. The next step is to show that the constant π is best possible in the continuous form of Hilbert's inequality, then to adapt the calculations to deduce that π is best possible in the discrete form as well. For the first purpose it is useful to choose the functions

$$f(x) = g(x) = h_R(x) = \begin{cases} 1/\sqrt{x} & \text{for } 1 \le x \le R \\ 0 & \text{elsewhere in } (0,\infty)\,, \end{cases} \tag{4}$$

where R is a large parameter. Then $\|f\|_2\,\|g\|_2 = \log R$ and

$$\begin{aligned} \int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\,dxdy &= 4\int_0^\infty \int_0^\infty \frac{f(u^2)g(v^2)}{u^2+v^2}\,uv\,dudv \\ &= \int_1^{\sqrt{R}} \int_1^{\sqrt{R}} \frac{4}{u^2+v^2}\,dudv\,. \end{aligned}$$

The problem is then to obtain an effective lower bound for the last integral in terms of R. We propose to show that

$$\int_1^{\sqrt{R}} \int_1^{\sqrt{R}} \frac{4}{u^2+v^2}\,dudv > \pi\log R - \pi\log 2 - 8\,. \tag{5}$$

Because $\|f\|_2\,\|g\|_2 = \log R$, it will then follow, by letting R tend to infinity, that the constant π in the inequality (2) cannot be replaced by any smaller number.

To derive the estimate (5), it is convenient to define the disjoint rectangular regions

$$\begin{aligned} Q_R &= \left\{(u,v) : 1 \le u \le \sqrt{R}\,,\ 1 \le v \le \sqrt{R}\right\}\,, \\ E_R &= \left\{(u,v) : 0 \le u < 1\,,\ 1 \le v \le \sqrt{R}\right\}\,, \\ F_R &= \left\{(u,v) : 1 \le u \le \sqrt{R}\,,\ 0 \le v < 1\right\} \end{aligned}$$

and the quarter-annulus

$$A_R = \left\{(u,v) : \sqrt{2} \le (u^2+v^2)^{1/2} \le \sqrt{R}\,,\ u \ge 0\,,\ v \ge 0\right\}\,.$$

Then $A_R \subset Q_R \cup E_R \cup F_R$ and we have

$$\int_1^{\sqrt{R}} \int_1^{\sqrt{R}} \frac{4}{u^2+v^2}\,dudv = \iint\limits_{Q_R} > \iint\limits_{A_R} - \iint\limits_{E_R} - \iint\limits_{F_R}\,. \tag{6}$$

But a simple calculation yields

$$\iint\limits_{F_R} \frac{4}{u^2+v^2}\,dudv = \iint\limits_{E_R} \frac{4}{u^2+v^2}\,dudv$$
$$= \int_1^{\sqrt{R}} \int_0^1 \frac{4}{u^2+v^2}\,dudv < \int_1^\infty \frac{4}{v^2}\,dv = 4\,,$$

and a change to polar coordinates gives

$$\iint\limits_{A_R} \frac{4}{u^2+v^2}\,dudv = \int_0^{\pi/2} \int_{\sqrt{2}}^{\sqrt{R}} \frac{4}{r}\,dr = \pi(\log R - \log 2)\,.$$

Putting these results into (6), we arrive at the inequality (5), which shows that the constant π is sharp in the continuous form (2) of Hilbert's inequality.

In order to show that π is also best possible in the discrete form (1), we choose

$$a_j = b_j = \begin{cases} 1/\sqrt{j} & \text{for } j = 1, 2, \ldots, N \\ 0 & \text{for } j > N\,, \end{cases}$$

where N is a large integer, and let $f(x) = g(x) = h_{N+1}(x)$ as defined in (4). Then

$$f(x) \le \sum_{j=1}^\infty a_j \psi_{j+1}(x) \qquad \text{and} \qquad g(y) \le \sum_{k=1}^\infty b_k \psi_{k+1}(y)$$

for all $x, y \in (0, \infty)$, so that

$$\int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\,dxdy \le \sum_{j=1}^\infty \sum_{k=1}^\infty w_{j+1,k+1}\, a_j b_k\,,$$

where

$$w_{j+1,k+1} = \int_k^{k+1} \int_j^{j+1} \frac{1}{x+y}\,dxdy < \frac{1}{j+k} < \frac{1}{j+k-1}\,.$$

Consequently, in view of the lower bound (5), we have shown that

$$\sum_{j=1}^\infty \sum_{k=1}^\infty \frac{a_j b_k}{j+k-1} > \sum_{j=1}^\infty \sum_{k=1}^\infty w_{j+1,k+1}\, a_j b_k$$
$$\ge \int_0^\infty \int_0^\infty \frac{f(x)g(y)}{x+y}\,dxdy > \pi \log(N+1) - \pi \log 2 - 8 \sim \pi \log N$$

as $N \to \infty$. On the other hand,

$$\left(\sum_{j=1}^{\infty} a_j^2\right)^{1/2}\left(\sum_{k=1}^{\infty} b_k^2\right)^{1/2} = \left(\sum_{j=1}^{N} 1/j\right)^{1/2}\left(\sum_{k=1}^{N} 1/k\right)^{1/2} \sim \log N\,.$$

Letting N tend to infinity, we conclude that the constant π in (1) is best possible. □

In summary, the three steps of proof combine to establish the discrete form of Hilbert's inequality in its entirety and the continuous form except for the assertion of strict inequality. Reviewing the proof of the continuous form, we see that the key to strict inequality is to examine the conditions for equality in our application of the Cauchy–Schwarz inequality. The details are technical, however, and will be omitted.

Exercises

1. (*a*) Use mathematical induction to prove that

$$(1+x)^n \geq 1 + nx\,, \qquad x > -1\,,$$

for $n = 1, 2, \ldots$.

(*b*) Generalize the result by proving that

$$(1+a_1)(1+a_2)\cdots(1+a_n) \geq 1 + \sum_{k=1}^{n} a_k$$

whenever $a_k > -1$ for $k = 1, 2, \ldots, n$ and all a_k have the same sign.

2. For $a_k > 0$ and $a_1 + a_2 + \cdots + a_n < 1$, prove:

$$(a) \quad 1 + \sum_{k=1}^{n} a_k < (1+a_1)(1+a_2)\cdots(1+a_n) < \frac{1}{1 - \sum_{k=1}^{n} a_k}\,,$$

$$(b) \quad 1 - \sum_{k=1}^{n} a_k < (1-a_1)(1-a_2)\cdots(1-a_n) < \frac{1}{1 + \sum_{k=1}^{n} a_k}\,.$$

Hint: Observe that $1 + a_k < 1/(1 - a_k)$.

3. For positive numbers $a_1, a_2, \ldots, a_n$, the *harmonic mean* H_n is defined by

$$H_n = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}\,.$$

Prove that $H_n \leq G_n$, with equality if and only if $a_1 = a_2 = \cdots = a_n$.

4. Let $a_1, a_2, \dots, a_n$ be positive numbers with sum $a_1 + a_2 + \cdots + a_n = n$. Show that $a_1 a_2 \cdots a_n \le 1$, and that the maximum occurs only when all of the numbers are equal to 1.

5. Prove without calculus that among all 3-dimensional rectangular boxes of fixed volume, the cube has the smallest surface area.

6. Show that

$$\sqrt{n} \le \sqrt[n]{n!} \le \frac{n+1}{2}, \qquad n = 1, 2, \dots .$$

7. Let $x_n = (1 + \frac{1}{n})^n$ and $y_n = (1 + \frac{1}{n})^{n+1}$. Observe that $x_n < y_n$. Use the arithmetic–geometric mean inequality to show that $x_n < x_{n+1}$ and $y_{n+1} < y_n$, and deduce that the sequences $\{x_n\}$ and $\{y_n\}$ both converge to a common limit. Recall that the common limit is e, the base of natural logarithms. Conclude that

$$\left(1 + \frac{1}{n}\right)^n < e < \left(1 + \frac{1}{n}\right)^{n+1}, \qquad n = 1, 2, \dots .$$

Taking $n = 10,000$, use a calculator to show that $2.7181 < e < 2.7185$.

Hint: Consider the numbers $a_1 = a_2 = \cdots = a_n = \frac{n+1}{n}$, $a_{n+1} = 1$. Then consider the numbers $a_1 = a_2 = \cdots = a_{n+1} = \frac{n}{n+1}$, $a_{n+2} = 1$.

8. Show that

$$\sum_{k=1}^{n} |a_k| \le \sqrt{n} \left(\sum_{k=1}^{n} a_k^2 \right)^{1/2} \le \sqrt{n} \sum_{k=1}^{n} |a_k|, \qquad n = 1, 2, \dots .$$

9. Verify the condition for equality in Hölder's inequality.

10. Prove the continuous forms of the Hölder and Minkowski inequalities.

11. Verify the conditions for equality in both forms of Jensen's inequality, discrete and continuous.

12. Use Jensen's inequality to prove Minkowski's inequality.

13. If $f \in L^r \cap L^s$ and $\frac{1}{p} = \frac{\lambda}{r} + \frac{1-\lambda}{s}$ for $0 < \lambda < 1$, show that $\|f\|_p \le \|f\|_r{}^{\lambda} \, \|f\|_s{}^{1-\lambda}$. As a special case, show that if $f \in L^1 \cap L^2$. then $f \in L^p$ for $1 < p < 2$.

14. If p, q, and r are positive numbers satisfying $\dfrac{1}{p} + \dfrac{1}{q} + \dfrac{1}{r} = 1$, show that

$$\|fgh\|_1 \le \|f\|_p \, \|g\|_q \, \|h\|_r .$$

15. (*a*) Show that if a sequence $\mathbf{a} = (a_1, a_2, \dots)$ belongs to the space ℓ^p for some $p < \infty$, then $\mathbf{a} \in \ell^q$ for all q with $p < q < \infty$.

(*b*) If $\mathbf{a} \in \ell^p$ for some $p < \infty$, show that $\|\mathbf{a}\|_p \to \|\mathbf{a}\|_\infty$ as $p \to \infty$.

Hint. If $1 \le x_n \le C$, then $\sqrt[n]{x_n} \to 1$ as $n \to \infty$.

16. Prove the following generalization of Hilbert's inequality. Let p and q lie in the interval $(1, \infty)$ and be conjugate indices, so that $\frac{1}{p} + \frac{1}{q} = 1$. Then for arbitrary real sequences $\{a_j\} \in \ell^p$ and $\{b_k\} \in \ell^q$, the inequality

$$\left| \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{a_j b_k}{j+k-1} \right| \le \frac{\pi}{\sin(\pi/p)} \left(\sum_{j=1}^{\infty} |a_j|^p \right)^{1/p} \left(\sum_{k=1}^{\infty} |b_k|^q \right)^{1/q}$$

holds, with strict inequality unless one of the sequences $\{a_j\}$ or $\{b_k\}$ is identically zero. The constant $\pi / \sin(\pi/p)$ cannot be replaced by any smaller number.

17. Let $a_1, a_2, \ldots, a_n$ be arbitrary real numbers, and suppose that M and m are the maximum and minimum, respectively, of their partial sums $s_k = a_1 + a_2 + \cdots + a_k$ for $k = 1, 2, \ldots, n$. If $b_1 \ge b_2 \ge \cdots \ge b_n \ge 0$, prove that

$$m b_1 \le a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \le M b_1 \,.$$

Hint. Apply an Abel summation.

18. By successive integrations of the inequality $\cos x \le 1$, show that for all $x \ge 0$,

$$\sin x \le x\,, \quad \cos x \ge 1 - \frac{x^2}{2!}\,, \quad \sin x \ge x - \frac{x^3}{3!}\,, \quad \cos x \le 1 - \frac{x^2}{2!} + \frac{x^4}{4!}\,,$$

and in general that

$$\sin x \le x - \frac{x^3}{3!} + \cdots + \frac{x^{4k+1}}{(4k+1)!}\,, \qquad \cos x \ge 1 - \frac{x^2}{2!} + \cdots - \frac{x^{4k+2}}{(4k+2)!}\,,$$
$$\sin x \ge x - \frac{x^3}{3!} + \cdots - \frac{x^{4k+3}}{(4k+3)!}\,, \qquad \cos x \le 1 - \frac{x^2}{2!} + \cdots + \frac{x^{4k+4}}{(4k+4)!}$$

for each $k = 0, 1, 2, \ldots$ and all $x \ge 0$. Prove by induction.

19. Let $f(x) \ge 0$ on the interval $[0, 1]$, and suppose it has a finite integral $A = \int_0^1 f(x)\,dx$. Prove that

$$\sqrt{1 + A^2} \le \int_0^1 \sqrt{1 + f(x)^2}\,dx \le 1 + A\,.$$

Interpret the inequalities geometrically by taking $f = g'$, where g is a non-decreasing continuously differentiable function on $[0, 1]$.

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Chapter 5
Infinite Products

Infinite products, like infinite series, arise often and are natural objects in mathematical analysis. The convergence theory of infinite products is closely related to that of infinite series but is more subtle. In this chapter we develop standard criteria for convergence and uniform convergence of infinite products.

5.1. Basic concepts

Infinite products are analogous to infinite series, and the two are closely related. The convergence of an infinite product

$$\prod_{k=1}^{\infty}(1+a_k) = (1+a_1)(1+a_2)\cdots,$$

where a_k are real numbers, comes down to the behavior of the sequence of partial products

$$p_n = \prod_{k=1}^{n}(1+a_k) = (1+a_1)(1+a_2)\cdots(1+a_n)$$

as n tends to infinity. The reason for writing the factors in the form $(1+a_k)$ will become apparent shortly.

It turns out that the "right" notion of convergence is not the obvious one. It would seem natural to declare an infinite product $\prod_{k=1}^{\infty}(1+a_k)$ convergent if its partial products p_n form a convergent sequence, but that

convention has several drawbacks. First of all, any reasonable definition of convergence should depend only on the "eventual behavior" of the factors, not on the values of finitely many factors. But if a single factor $1 + a_m = 0$, then $p_n = 0$ for all $n \geq m$, and so $p_n \to 0$ as $n \to \infty$. To avoid this trivial complication, we make the standing assumption (to be modified later) that the product contains no zero factors. In other words, $1 + a_k \neq 0$ for all k. An infinite product is then said to be *convergent* if its sequence of partial products converges to a finite, nonzero limit. Thus in the above notation the requirement is that

$$\lim_{n\to\infty} p_n = p \neq 0\,,$$

and the infinite product is then assigned the *value* p. The infinite product is declared to be *divergent* if it is not convergent; that is, if the sequence $\{p_n\}$ either diverges or tends to zero. If $p_n \to 0$, the product is said to *diverge to zero*.

One advantage of insisting on a nonzero limit is that convergence of an infinite product implies convergence of the corresponding product of reciprocal factors, since the sequence $\{1/p_n\}$ then also converges to a finite, nonzero limit. Another important consequence is that the factors of a convergent product tend to 1, just as the terms of a convergent series tend to 0. Specifically, if an infinite product converges to p, then

$$1 + a_n = p_n/p_{n-1} \to p/p = 1 \qquad \text{as} \quad n \to \infty\,,$$

or equivalently, $a_n \to 0$. However, this is only a necessary condition for convergence; it is far from sufficient.

Here are two examples.

Example 1. If $a_k = \frac{1}{k}$, then $1 + a_k = \frac{k+1}{k}$, and the partial product "telescopes" to give

$$p_n = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = n + 1\,.$$

Thus the infinite product $\prod_{k=1}^{\infty}(1 + \frac{1}{k})$ is divergent.

Example 2. Suppose $a_k = -\frac{1}{k^2}$ for $k = 2, 3, \ldots$. Then

$$1 + a_k = 1 - \frac{1}{k^2} = \frac{k^2 - 1}{k^2} = \frac{(k-1)(k+1)}{k^2}\,,$$

and again the partial product telescopes to give

$$p_n = \prod_{k=2}^{n}\left(1 - \frac{1}{k^2}\right) = \frac{1 \cdot 3}{2^2} \cdot \frac{2 \cdot 4}{3^2} \cdot \frac{3 \cdot 5}{4^2} \cdots \frac{(n-1)(n+1)}{n^2} = \frac{1}{2} \cdot \frac{n+1}{n} \to \frac{1}{2}$$

as $n \to \infty$. Thus the infinite product converges, and

$$\prod_{k=2}^{n} \left(1 - \frac{1}{k^2}\right) = \frac{1}{2}\,.$$

Our two examples suggest the possibility that convergence of an infinite product $\prod(1+a_k)$ is equivalent to convergence of the corresponding infinite series $\sum a_k$. This is actually true if the numbers a_k are all of the same sign, as we now proceed to show.

Theorem 1. *If $a_k \geq 0$ for all k, then the product $\prod_{k=1}^{\infty}(1+a_k)$ converges if and only if the series $\sum_{k=1}^{\infty} a_k$ converges. The same is true if $a_k \leq 0$ for all k.*

Proof. Again let $p_n = \prod_{k=1}^{n}(1+a_k)$ denote the partial products, and let $s_n = \sum_{k=1}^{n} a_k$ denote the partial sums of the related series. Suppose first that $a_k \geq 0$ for all k. Then $1 \leq p_n \leq p_{n+1}$ and $s_n \leq s_{n+1}$, so by the monotone boundedness theorem, either of the sequences $\{p_n\}$ or $\{s_n\}$ is convergent if and only if it is bounded above. But in view of the inequality

$$1 + x \leq e^x\,,$$

it is easy to see that

$$1 + s_n \leq p_n \leq e^{s_n}\,, \qquad n = 1, 2, \ldots\,.$$

Therefore, the boundedness of $\{p_n\}$ is equivalent to that of $\{s_n\}$, and the theorem is proved for the case where $a_k \geq 0$.

Suppose next that $a_k \leq 0$, and assume without loss of generality that $a_k > -1$ for all k. Now $0 < p_{n+1} \leq p_n$, and so the infinite product converges if and only if $p_n \geq b$ for some constant $b > 0$. Observe first that

$$1 \leq 1 - a_k \leq \frac{1}{1 + a_k}\,,$$

so the convergence of $\prod(1 + a_k)$ implies that of $\prod(1 - a_k)$, which implies, by what we have just proved, that the series $\sum a_k$ converges. Conversely, suppose the series $\sum a_k$ is convergent. This implies that $a_k \to 0$, so after removing a finite number of factors we may assume that $-\frac{1}{2} \leq a_k \leq 0$. Then an application of the elementary inequality

$$1 \leq (1 + x)(1 - 2x)\,, \qquad -\tfrac{1}{2} \leq x \leq 0\,,$$

shows that

$$\frac{1}{1 - 2a_k} \leq 1 + a_k\,.$$

However, the infinite product $\prod(1-2a_k)$ converges, since $-2a_k \geq 0$ and the series $\sum a_k$ converges. Thus the above inequality shows that the product $\prod(1+a_k)$ is also convergent. This completes the proof. □

The question now arises whether the theorem remains true under more general conditions. Do the infinite product $\prod(1+a_k)$ and the infinite series $\sum a_k$ always converge and diverge together? The answer is no. Consider for instance the following example.

Example 3. The infinite product

$$\prod_{k=2}^{\infty}\left(1-\frac{1}{k}\right)$$

diverges to zero, as does the product

$$\left(1+\frac{1}{\sqrt{2}}\right)\left(1-\frac{1}{\sqrt{2}}\right)\left(1+\frac{1}{\sqrt{3}}\right)\left(1-\frac{1}{\sqrt{3}}\right)\left(1+\frac{1}{\sqrt{4}}\right)\left(1-\frac{1}{\sqrt{4}}\right)\cdots.$$

However, the first product corresponds to a divergent series, whereas the second corresponds to a convergent series.

It is more difficult to find a convergent product whose associated series diverges. Here is one example.

Example 4. Consider the infinite product

$$\left(1-\frac{1}{\sqrt{2}}\right)\left(1+\frac{1}{\sqrt{2}}+\frac{1}{2}\right)\left(1-\frac{1}{\sqrt{3}}\right)\left(1+\frac{1}{\sqrt{3}}+\frac{1}{3}\right)$$
$$\cdot\left(1-\frac{1}{\sqrt{4}}\right)\left(1+\frac{1}{\sqrt{4}}+\frac{1}{4}\right)\cdots.$$

Note that

$$\left(1-\frac{1}{\sqrt{k}}\right)\left(1+\frac{1}{\sqrt{k}}+\frac{1}{k}\right)=1-\frac{1}{k^{3/2}} \qquad \text{and} \qquad \left(1-\frac{1}{\sqrt{k}}\right)\to 1\,,$$

so the product converges although its associated series

$$-\frac{1}{\sqrt{2}}+\left(\frac{1}{\sqrt{2}}+\frac{1}{2}\right)-\frac{1}{\sqrt{3}}+\left(\frac{1}{\sqrt{3}}+\frac{1}{3}\right)-\cdots=\frac{1}{2}+\frac{1}{3}+\ldots$$

is divergent. More precisely, the series has a subsequence of partial sums $s_{2n}=1+\frac{1}{2}+\cdots+\frac{1}{n}$, and so it diverges.

5.2. Absolute convergence

A useful necessary and sufficient condition for convergence of infinite products can be adapted from the Cauchy criterion for convergence of a sequence. Here is the result.

Theorem 2. *An infinite product* $\prod(1 + a_k)$ *is convergent if and only if its factors have the following property. For each* $\varepsilon > 0$ *there is a number* N *such that*

$$|(1 + a_m)(1 + a_{m+1}) \cdots (1 + a_n) - 1| < \varepsilon \tag{1}$$

for all indices m *and* n *with* $n > m \geq N$.

Proof. Suppose first that the product converges. Then the partial products $p_n \to p \neq 0$ as $n \to \infty$. In particular, the partial products are bounded away from zero: $|p_n| \geq b$ for some $b > 0$ and all n. For each $\varepsilon > 0$, the Cauchy condition says that

$$|p_n - p_m| < \varepsilon b\,, \qquad n > m \geq N\,,$$

for some N sufficiently large. But $|p_m| \geq b$, so this implies that

$$|p_n/p_m - 1| < \varepsilon\,, \qquad n > m \geq N\,,$$

which is essentially the condition (1).

Conversely, if the factors have the property (1), then for each $\varepsilon > 0$ the inequality $|p_n/p_m - 1| < \varepsilon$ holds whenever $n > m \geq N$. Choose $\varepsilon = \frac{1}{2}$ to infer that

$$|p_n - p_M| < \tfrac{1}{2}\,|p_M|\,, \qquad n > M\,,$$

for some specific index M. This implies that

$$\tfrac{1}{2}\,|p_M| < |p_n| < \tfrac{3}{2}\,|p_M|\,, \qquad n > M\,.$$

In other words, the partial products are bounded away from both zero and infinity. Therefore, for each $\varepsilon > 0$ it follows from (1) that

$$|p_n - p_m| < \varepsilon\,|p_m| < \tfrac{3}{2}\,\varepsilon\,|p_M|$$

for all m and n greater than some $N \geq M$. This shows that $\{p_n\}$ is a Cauchy sequence, and so $p_n \to p$ as $n \to \infty$. Finally, $p \neq 0$ since the partial products p_n are bounded away from zero. Thus the infinite product is convergent. □

We can now discuss the question of absolute convergence. An infinite product $\prod(1 + a_k)$ is said to be *absolutely convergent* if the product

$\prod(1+|a_k|)$ is convergent. In view of Theorem 1, a product is absolutely convergent if and only if its associated series $\sum a_k$ is absolutely convergent. A convergent product need not be absolutely convergent, as simple examples show. For instance, the product

$$\left(1+\frac{1}{2}\right)\left(1-\frac{1}{2}\right)\left(1+\frac{1}{3}\right)\left(1-\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\left(1-\frac{1}{4}\right)\cdots$$

is not absolutely convergent but it is convergent, because its partial products tend to the same limit as do those of the convergent product

$$\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\cdots.$$

On the other hand, absolute convergence implies convergence.

Theorem 3. *Every absolutely convergent product is convergent.*

Proof. The inequality

$$|(1+a_m)(1+a_{m+1})\cdots(1+a_n)-1| \le (1+|a_m|)(1+|a_{m+1}|)\cdots(1+|a_n|)-1$$

can be verified by multiplying out the left-hand product, subtracting 1, applying the triangle inequality, and regrouping the factors. Thus the desired result is an immediate consequence of Theorem 2. □

5.3. Logarithmic series

We now come to a very natural necessary and sufficient condition for convergence of an infinite product. The formal identity

$$\log\left(\prod_{k=1}^{\infty}(1+a_k)\right) = \sum_{k=1}^{\infty}\log(1+a_k)$$

suggests that the product and its associated series of logarithms will converge or diverge together. Under an assumption to guarantee that the logarithms are well-defined, this turns out to be correct.

Theorem 4. *If $a_k > -1$ for all k, then the infinite product $\prod_{k=1}^{\infty}(1+a_k)$ converges if and only if the infinite series $\sum_{k=1}^{\infty}\log(1+a_k)$ converges.*

Proof. If the product converges, so that $p_n \to p \neq 0$, then it follows from the continuity of the logarithm function that $\log p_n \to \log p$. But $\log p_n$ is simply the nth partial sum of the infinite series, so this shows that the series converges. Conversely, the convergence of the series says that $\log p_n \to L$, so

$p_n \to e^L \neq 0$, and the infinite product converges. The proof shows that the value of the product is the exponentiated sum of the series, as expected. □

From Theorem 4 it is a short step to a condition under which convergence of the product $\prod_{k=1}^{\infty}(1 + a_k)$ is equivalent to that of the sum $\sum_{k=1}^{\infty} a_k$. Cauchy attributes the following theorem to Gaspard-Gustave Coriolis (1792–1843), who is better known for discovering the *Coriolis force* on the surface of a rotating body.

Theorem 5. *If $\sum_{k=1}^{\infty} a_k^2 < \infty$, then the product $\prod_{k=1}^{\infty}(1 + a_k)$ converges if and only if the series $\sum_{k=1}^{\infty} a_k$ converges.*

Proof. The deduction from Theorem 4 will use the inequality

$$\tfrac{1}{3}x^2 < x - \log(1 + x) < x^2\,, \qquad |x| < \delta\,, \tag{2}$$

for some positive constant $\delta < 1$ sufficiently small. This is a simple consequence of the limit

$$\lim_{x\to 0} \frac{x - \log(1 + x)}{x^2} = \frac{1}{2}\,.$$

If the series $\sum a_k^2$ converges, then $a_k \to 0$ and so $|a_k| < \delta$ for all indices k larger than some number N. The inequality (2) then gives

$$|a_k - \log(1 + a_k)| \le a_k^2\,, \qquad k > N\,.$$

Thus the series

$$\sum_{k=1}^{\infty} \big(a_k - \log(1 + a_k)\big)$$

converges (absolutely). It follows that if either of the series

$$\sum_{k=1}^{\infty} a_k \qquad \text{or} \qquad \sum_{k=1}^{\infty} \log(1 + a_k)$$

converges, then so does the other. In view of Theorem 4, this completes the proof. □

If the divergent product of Example 3 is written in the first form, with $a_k = -1/k$, then $\sum a_k^2 < \infty$, so the simultaneous divergence of $\prod(1+a_k)$ and $\sum a_k$ is in accordance with Theorem 5. On the other hand, if the product is written in the second form, then $\sum a_k^2 = \infty$ and so the convergence of $\sum a_k$ and divergence of $\prod(1 + a_k)$ is beyond the jurisdiction of Theorem 5. For Example 4, where the $\prod(1 + a_k)$ converges but the $\sum a_k$ diverges, we see similarly that $\sum a_k^2 = \infty$.

If the product $\prod(1+a_k)$ and the sum $\sum a_k$ both converge, then $\sum a_k^2$ also converges (*cf.* Exercise 5). However, the convergence of $\sum a_k^2$ is by no means necessary for the simultaneous divergence of product and sum. For example, let $a_k = 1/\sqrt{k}$. Then the product $\prod(1+a_k)$, the sum $\sum a_k$, and the sum $\sum a_k^2$ all diverge.

5.4. Uniform convergence

Up to this point, we have maintained the standing assumption that an infinite product has no zero-factors, or equivalently that $a_k \neq -1$ for all k. In various applications, however, especially those dealing with infinite products of functions, it is essential to allow zero-factors and to modify the definition of convergence accordingly. An infinite product with finitely many zero-factors is said to be *convergent* if after removal of those factors the product of remaining factors is convergent. In this case the product is said to converge and have value 0, or to *converge to zero*. A product with infinitely many zero-factors is regarded as divergent.

One of the most important uses of infinite products is to give explicit representations of special functions. For instance, it will be shown in Chapter 8 that the sine function has the expression

$$\sin \pi x = \pi x \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2}\right) \tag{3}$$

for all real numbers x. Note that the infinite product converges for each fixed x, by Theorem 1. It vanishes precisely for $x = 0, \pm 1, \pm 2, \ldots$, which we know are the zeros of the function $\sin \pi x$. An infinite product representation of this type can be viewed as a generalization of the factorization of a polynomial according to its (real) zeros.

Consider now a general infinite product of the form

$$\prod_{k=1}^{\infty} \big(1 + g_k(x)\big)\,, \tag{4}$$

where the functions g_k are defined on a common set $E \subset \mathbb{R}$. The product is said to converge *pointwise on* E if it converges for each point $x \in E$. We can then write

$$f(x) = \prod_{k=1}^{\infty} \big(1 + g_k(x)\big)\,, \qquad x \in E\,.$$

The product is said to be *uniformly convergent* on E if its sequence of partial products

$$f_n(x) = \prod_{k=1}^{n} \big(1 + g_k(x)\big)$$

converges to $f(x)$ uniformly for $x \in E$. Note that $f(x) \neq 0$ for all $x \in E$ if each $g_k(x) \neq -1$ in E.

The tests for convergence of numerical products, as given in Theorems 1, 2, and 4, can be adapted to provide criteria for uniform convergence of products of functions. We begin with a uniform version of the Cauchy criterion.

Theorem 6. *Let $g_1(x), g_2(x), \ldots$ be bounded functions on a set $E \subset \mathbb{R}$, with $g_k(x) \neq -1$ for $x \in E$. Then the infinite product* (4) *converges uniformly on E if its partial products $f_n(x)$ have the following property. For each $\varepsilon > 0$ there is a number N such that*

$$|f_n(x)/f_m(x) - 1| < \varepsilon \qquad \text{for all } \ x \in E \tag{5}$$

and for all indices m and n with $n > m \geq N$. Conversely, the partial products have this property if the infinite product (4) *converges uniformly on E to a function $f(x)$ that is bounded below by a positive constant.*

Proof. Suppose first that the partial products have the property (5). Then corresponding to $\varepsilon = \frac{1}{2}$ there is an index M for which the uniform inequality

$$|f_n(x) - f_M(x)| < \tfrac{1}{2}\,|f_M(x)|\,, \qquad x \in E\,,$$

holds for all indices $n > M$. Consequently,

$$\tfrac{1}{2}\,|f_M(x)| < |f_n(x)| < \tfrac{3}{2}\,|f_M(x)|\,, \qquad x \in E\,,\ n > M\,. \tag{6}$$

Then since each of the factors $1 + g_k(x)$ is bounded, the condition (5) and its consequence (6) imply that

$$|f_n(x) - f_m(x)| < \varepsilon |f_m(x)| < \tfrac{3}{2}\,\varepsilon |f_M(x)| \leq C\,\varepsilon\,, \qquad x \in E\,,$$

for all m and n greater than some index N, where C is an absolute constant. This says that $\{f_n(x)\}$ is a uniform Cauchy sequence, so that $f_n(x) \to f(x)$ uniformly on E as $n \to \infty$. The inequality (6) shows that $|f(x)| \geq \frac{1}{2}|f_M(x)| > 0$ for each $x \in E$, so we can conclude that the infinite product (4) is uniformly convergent on E.

Next suppose that the product (4) converges uniformly on E to a function $f(x)$ with the property $|f(x)| \geq \delta$ for some $\delta > 0$ and all $x \in E$. Then $\{f_n(x)\}$ is a uniform Cauchy sequence, so for each $\varepsilon > 0$ there is an index N for which

$$|f_n(x) - f_m(x)| < \tfrac{1}{2}\,\varepsilon\delta\,, \qquad x \in E\,,$$

whenever $n > m \geq N$. But the uniform convergence implies that $|f_m(x)| > \frac{1}{2}|f(x)| \geq \frac{1}{2}\,\delta$ for all $x \in E$ and all m sufficiently large. This shows that (5) holds if N is sufficiently large, so the proof is complete. □

Theorem 7. *If the series $\sum_{k=1}^{\infty} |g_k(x)|$ converges uniformly on a set $E \subset \mathbb{R}$, then the product $\prod_{k=1}^{\infty}\big(1 + g_k(x)\big)$ converges uniformly on E.*

Proof. Let

$$s_n(x) = \sum_{k=1}^{n} |g_k(x)| \qquad \text{and} \qquad F_n(x) = \prod_{k=1}^{n} \big(1 + |g_k(x)|\big)\,.$$

Then for $n > m$,

$$1 \le F_n(x)/F_m(x) \le e^{s_n(x) - s_m(x)}\,, \qquad x \in E\,.$$

Therefore, because $\{s_n(x)\}$ is a uniform Cauchy sequence on E, there corresponds to each $\varepsilon > 0$ a number N such that

$$0 \le F_n(x)/F_m(x) - 1 < \varepsilon\,, \qquad x \in E\,,$$

whenever $n > m \ge N$. As in the proof of Theorem 3, it follows that

$$|f_n(x)/f_m(x) - 1| < \varepsilon\,, \qquad x \in E\,,$$

for $n > m \ge N$. The desired result can now be inferred from Theorem 6, since the uniform convergence of $\sum |g_k(x)|$ implies that $g_k(x) \to 0$ uniformly on E. □

Theorem 8. *If $g_k(x) > -1$ and the series $\sum_{k=1}^{\infty} \log\big(1 + g_k(x)\big)$ converges uniformly on a set $E \subset \mathbb{R}$, then the product $\prod_{k=1}^{\infty}\big(1 + g_k(x)\big)$ converges uniformly on E.*

Proof. The partial sums

$$t_n(x) = \sum_{k=1}^{n} \log\big(1 + g_k(x)\big)$$

form a uniform Cauchy sequence on E. Thus to each $\varepsilon > 0$ there corresponds a number N such that

$$1 - \varepsilon < 1 + t_n(x) - t_m(x) \le e^{t_n(x) - t_m(x)} < 1 + \varepsilon\,, \qquad x \in E\,,$$

whenever $n > m \ge N$. In other words, the partial products

$$f_n(x) = \prod_{k=1}^{n} \big(1 + g_k(x)\big)$$

have the property (5), so Theorem 6 ensures that the infinite product converges uniformly on E. □

Exercises

1. (a) Calculate partial products and show directly that both infinite products in Example 3 diverge to zero.

(b) Carry out the details to show that the infinite product in Example 4 is convergent.

2. Calculate partial products and show that

$$\prod_{k=2}^{\infty}\left(1-\frac{2}{k(k+1)}\right)=\frac{1}{3}\,.$$

3. Show that

$$\prod_{k=1}^{\infty}\left(1+\frac{(-1)^{k+1}}{k}\right)=1\,.$$

4. Let $\{x_k\}$ be a sequence of real numbers for which $\sum_{k=1}^{\infty} x_k^2<\infty$. Prove that the product $\prod_{k=1}^{\infty}\cos x_k$ converges.

5. Adapt the proof of Theorem 5 to show that the sum $\sum a_k^2$ converges whenever the sum $\sum a_k$ and the product $\prod(1+a_k)$ both converge.

6. Prove that

$$\prod_{k=3}^{\infty}\left(1-\frac{4}{k^2}\right)=\frac{1}{6}\qquad\text{and}\qquad\prod_{k=4}^{\infty}\left(1-\frac{9}{k^2}\right)=\frac{1}{20}\,.$$

7. Generalize the results of Exercise 6 by showing that

$$\prod_{k=m+1}^{\infty}\left(1-\frac{m^2}{k^2}\right)=\frac{(m!)^2}{(2m)!}\,,\qquad m=1,2,3,\ldots\,.$$

8. Conclude from Exercise 7 that

$$\prod_{k=m+1}^{\infty}\frac{k^2}{k^2-m^2}=\binom{2m}{m}\,,\qquad m=1,2,3,\ldots\,,$$

and deduce that

$$\prod_{\substack{k=1\\k\neq m}}^{\infty}\frac{k^2}{k^2-m^2}=(-1)^{m+1}\,2\,,\qquad m=2,3,4,\ldots\,.$$

9. (a) Suppose a regular n-gon is circumscribed about a circle of radius r and another circle is circumscribed about the n-gon. Show that the second circle has radius $R=r\,\sec\frac{\pi}{n}$.

(b) Suppose now that an equilateral triangle is circumscribed about a circle of radius 1, then a circle C_3 is circumscribed about the triangle, then a square is circumscribed about C_3, another circle C_4 is circumscribed about the square, a regular pentagon is circumscribed about C_4, and so on. If C_n has radius R_n, show that

$$\lim_{n\to\infty} R_n = \prod_{n=3}^{\infty} \sec\frac{\pi}{n}\,,$$

and prove that the infinite product converges. (Its numerical value is known to be approximately 8.700 .)

10. Show that the infinite product

$$\prod_{k=1}^{\infty}\left(1-\frac{x^2}{k^2}\right)$$

converges uniformly in every bounded subset $E \subset \mathbb{R}$. Conclude that

$$s(x) = \pi x \prod_{k=1}^{\infty}\left(1-\frac{x^2}{k^2}\right)$$

is continuous on the whole real line, without using the fact that $s(x) = \sin \pi x$.

11. Recall that $\sinh x = \frac{1}{2}\big(e^x - e^{-x}\big)$ and $\sin x = \frac{1}{2i}\big(e^{ix} - e^{-ix}\big)$. Use the formula (3) for $\sin \pi x$ to conclude, at least formally, that

$$\sinh \pi x = \pi x \prod_{k=1}^{\infty}\left(1+\frac{x^2}{k^2}\right)\,.$$

12. Show that

$$\prod_{k=1}^{\infty}\left(1+\frac{1}{k^2}\right) = \frac{\sinh \pi}{\pi} \qquad \text{and} \qquad \prod_{k=2}^{\infty}\frac{k^2-1}{k^2+1} = \frac{\pi}{\sinh \pi}\,.$$

13. Find a general formula for the partial products and show that

$$\prod_{k=2}^{\infty}\frac{k^3-1}{k^3+1} = \frac{2}{3}\,.$$

Hint. Factor the polynomials $k^3 - 1$ and $k^3 + 1$.

14. Verify the formula

$$(1+x)(1+x^2)(1+x^4)(1+x^8)\cdots = \frac{1}{1-x}\,, \qquad |x|<1\,.$$

15. Use the identity $\sin 2x = 2\sin x\cos x$ to show that

$$\prod_{k=1}^{\infty} \cos\frac{x}{2^{k+1}} = \frac{\sin x}{x}\,, \qquad x\neq 0\,.$$

This is a generalization of Vieta's product formula (see Chapter 2, Section 2.4).

16. For each real number $\alpha > 0$, show that the generalized binomial coefficients

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$$

tend to zero as $n\to\infty$.

Reference

[1] Konrad Knopp, *Theory and Application of Infinite Series*, Second English edition, Blackie & Son, London and Glasgow, 1951; reprinted by Dover Publications, Mineola, NY, 1990.

Chapter 6
Approximation by Polynomials

The Weierstrass approximation theorem is one of the gems of classical analysis. In this chapter we give three different proofs of the basic theorem, plus a proof of Stone's far-reaching generalization. The discussion begins with interpolation theory and leads naturally to Chebyshev polynomials, which solve a particular problem of best approximation. Later in the chapter we consider the general problem (still unsolved) of finding the best uniform approximation to an arbitrary continuous function by polynomials of given degree.

6.1. Interpolation

The main focus of this chapter is a theorem of Weierstrass [**15**] that every continuous function on a closed bounded interval can be approximated uniformly by algebraic polynomials; that is, by functions of the form

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \,,$$

where n is a positive integer and the coefficients a_k are real numbers. The result is important because it often reduces a problem about continuous functions to the corresponding problem for polynomials. Broadly speaking, the theorem is a prototype for a large collection of results in approximation theory.

To put the approximation problem in better perspective, we begin with

a discussion of interpolation. Suppose a set of points

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

is specified with distinct abcissas x_k. Then it is possible to construct a polynomial of degree at most n whose graph passes through all of the points, and this can be done in exactly one way. In other words, there is a unique polynomial P of degree $\le n$ such that $P(x_k) = y_k$ for $k = 0, 1, \ldots, n$. To see this, observe that the problem amounts to solving the system of linear equations

$$\begin{aligned} a_0 + a_1 x_0 + a_2 x_0^2 + \ldots + a_n x_0^n &= y_0 \\ a_0 + a_1 x_1 + a_2 x_1^2 + \ldots + a_n x_1^n &= y_1 \\ &\cdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \ldots + a_n x_n^n &= y_n \end{aligned}$$

for the variables $a_0, a_1, \ldots, a_n$. The problem has a unique solution because the matrix of coefficients is nonsingular; that is, its determinant does not vanish. The determinant is known as a *Vandermonde determinant* and it is calculated by the formula

$$\begin{vmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & \ldots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^n \end{vmatrix} = \prod_{j<k} (x_k - x_j)\,.$$

(See Exercise 1 for a derivation.) In particular, a polynomial of degree $\le n$ cannot vanish at $n+1$ points unless it vanishes identically.

In fact, the required polynomial is given by the *Lagrange interpolation formula*

$$P(x) = \sum_{j=0}^{n} \frac{y_j\, \ell(x)}{\ell'(x_j)(x - x_j)}\,,$$

where $\ell(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$. More explicitly,

$$P(x) = \sum_{j=0}^{n} y_j \, \frac{(x - x_0) \cdots (x - x_{j-1})(x - x_{j+1}) \cdots (x - x_n)}{(x_j - x_0) \cdots (x_j - x_{j-1})(x_j - x_{j+1}) \cdots (x_j - x_n)}\,.$$

It is readily verified that P is a polynomial of degree less than $n+1$, and that $P(x_k) = y_k$ for $k = 0, 1, \ldots, n$. Thus by uniqueness it is the solution to the interpolation problem.

The Lagrange interpolation formula is customarily used as a practical method of *extrapolation*, or approximate reconstruction of a function whose

values are known (or have been measured) at a finite set of points. If a function f is continuous over an interval $[a, b]$, and $x_0, x_1, \ldots, x_n$ are distinct points in that interval, the Lagrange interpolation

$$P(x) = \sum_{j=0}^{n} \frac{f(x_j)\,\ell(x)}{\ell'(x_j)(x - x_j)}\,,$$

can be viewed as an approximation to $f(x)$ over the interval $[a, b]$. But how good is the approximation? Certainly it is exact if f is itself a polynomial of degree $\leq n$, but in the absence of further information there will be no control away from the nodes x_k, and the error

$$\|f - P\|_\infty = \max_{a \leq x \leq b} |f(x) - P(x)|$$

as measured in the uniform norm may be quite large. The error can be controlled, however, if f is known to be sufficiently smooth.

Theorem 1. *Suppose a function f has $n+1$ continuous derivatives on the interval $[a, b]$, and let P be its Lagrange interpolation polynomial with nodes $x_0, x_1, \ldots, x_n$. Then*

$$|f(x) - P(x)| \leq \frac{1}{(n+1)!}\,\|f^{(n+1)}\|_\infty\,|\ell(x)|\,,$$

where $\ell(x) = (x - x_0)(x - x_1)\cdots(x - x_n)$.

Proof. Fix an arbitrary point x in $[a, b]$, not equal to any of the nodes x_k. Define the function

$$g(t) = f(t) - P(t) - c\,\ell(t)\,,$$

where $c = [f(x) - P(x)]/\ell(x)$, so that $g(x) = 0$. It is clear that $g(x_k) = 0$ for $k = 0, 1, \ldots, n$. Thus g has at least $n + 2$ distinct zeros in $[a, b]$. By Rolle's theorem, the derivative $g'(t)$ vanishes at least once between each pair of zeros, so g' has at least $n + 1$ zeros. Continuing the argument by taking successive derivatives, we find ultimately that $g^{(n+1)}(\xi) = 0$ at some point $\xi \in (a, b)$. But P is a polynomial of degree $\leq n$ and ℓ is a monic polynomial of degree $n + 1$, so $P^{(n+1)}(t) \equiv 0$ and $\ell^{(n+1)}(t) \equiv (n + 1)!$. Therefore, the conclusion is that $f^{(n+1)}(\xi) = c(n + 1)!$, or

$$f(x) - P(x) = \frac{1}{(n+1)!}\,f^{(n+1)}(\xi)\,\ell(x)\,,$$

which yields the desired result. □

The question now arises as to how the nodes should be situated to minimize the error of the Lagrange approximation, as measured by the uniform

norm $\|F - P\|_\infty$. Theorem 1 suggests that the key is to minimize $\|\ell\|_\infty$. A more general formulation is to find the monic polynomial of fixed degree that has smallest norm over a given interval. It will be convenient to work with the interval $[-1, 1]$ and to consider monic polynomials

$$Q(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$$

of fixed degree n (instead of $n + 1$).

We begin by recalling that $\cos n\theta$ can be expressed as a polynomial of degree n in $\cos\theta$. This follows most easily from the de Moivre formula

$$\cos n\theta + i \sin n\theta = (\cos\theta + i \sin\theta)^n .$$

Expanding the right-hand side by the binomial theorem and taking real parts, one finds

$$\cos n\theta = \cos^n\theta - \binom{n}{2} \cos^{n-2}\theta \sin^2\theta + \binom{n}{4} \cos^{n-4}\theta \sin^4\theta - \ldots .$$

But $\sin^2\theta = 1 - \cos^2\theta$, so we can write $\cos n\theta = T_n(\cos\theta)$ for a uniquely determined polynomial

$$T_n(x) = c_n x^n + c_{n-1}x^{n-1} + \ldots + c_1 x + c_0$$

of degree n, known as the nth *Chebyshev polynomial*. The "T" is for "Tchebicheff", a French transliteration of "Chebyshev".

P. L. Chebyshev (1821–1894) was a primary founder of the Russian school of analysis. In addition to his work in approximation theory, he made seminal contributions to number theory, probability theory, and mechanics. He introduced the Chebyshev polynomials in 1854.

The first few Chebyshev polynomials can be calculated explicitly from the preceding formulas. They are

$$T_0(x) = 1 , \quad T_1(x) = x , \quad T_2(x) = 2x^2 - 1 , \quad T_3(x) = 4x^3 - 3x ,$$
$$T_4(x) = 8x^4 - 8x^2 + 1 , \quad T_5(x) = 16x^5 - 20x^3 + 5x , \quad \ldots .$$

It is easier to calculate the Chebyshev polynomials recursively by the relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) , \qquad n = 2, 3, \ldots .$$

To derive the recurrence relation, introduce the definition $\cos n\theta = T_n(\cos\theta)$ into the identity

$$\cos(n + 1)\theta + \cos(n - 1)\theta = 2 \cos\theta \cos n\theta$$

and set $x = \cos\theta$. An inductive argument now shows that $T_n(x)$ has leading coefficient $c_n = 2^{n-1}$ for $n > 0$. Additional properties are collected in the following lemma.

Lemma 1. *For $n > 0$ the Chebyshev polynomial $T_n(x)$ has simple zeros at the points*

$$x_k = \cos\left(\frac{2k-1}{2n}\pi\right), \qquad k = 1, 2, \ldots, n.$$

In the interval $[-1, 1]$, the local maxima of $|T_n(x)|$ occur at the points

$$\xi_k = \cos(k\pi/n), \qquad k = 0, 1, \ldots, n,$$

and $T_n(\xi_k) = (-1)^k$.

Proof. Note that $T_n(x) = \cos n\theta = 0$ for

$$\theta = \frac{2k-1}{2n}\pi, \qquad \text{or} \quad x = x_k = \cos\left(\frac{2k-1}{2n}\pi\right), \qquad k = 1, 2, \ldots, n.$$

A simple calculation shows that

$$T_n'(x) = \frac{d}{dx}\left\{\cos\left(n\cos^{-1}x\right)\right\} = \frac{n}{\sqrt{1-x^2}}\sin\left(n\cos^{-1}x\right) = 0$$

for

$$\cos^{-1}x = k\pi/n, \qquad \text{or} \quad x = \xi_k = \cos(k\pi/n), \qquad k = 0, 1, \ldots, n.$$

In particular, $T_n'(x_k) \neq 0$, so all of the zeros are simple. Finally,

$$T_n(\xi_k) = \cos\left(n\cos^{-1}\xi_k\right) = \cos(k\pi) = (-1)^k \qquad \square$$

Figure 1 shows graphs of the Chebyshev polynomials of degrees 10 and 15 over the interval $[-1, 1]$.

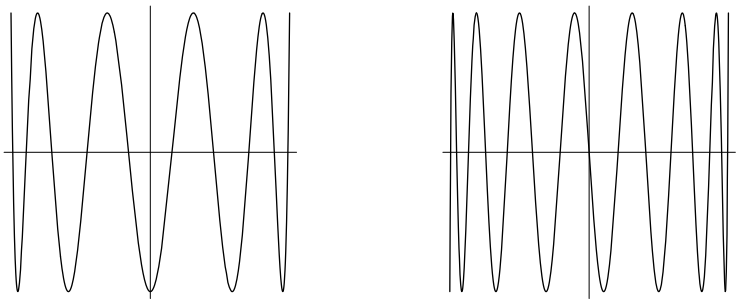


Figure 1. Chebyshev polynomials $T_n(x)$ for $n = 10, 15$.

We can now show that among all monic polynomials of fixed degree, the normalized Chebyshev polynomials have smallest uniform norm.

Theorem 2. *Every monic polynomial*

$$Q(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$$

of degree $n > 0$ satisfies the inequality

$$\|Q\|_\infty = \max_{-1\le x\le 1} |Q(x)| \ge 2^{1-n} = \|2^{1-n}T_n\|_\infty \,,$$

where T_n is the nth Chebyshev polynomial.

The theorem says in particular that for all choices of points $x_1, x_2, \ldots, x_n$ on the real line, the function

$$\ell(x) = (x - x_1)(x - x_2)\cdots(x - x_n)$$

has smallest norm for

$$x_k = \cos\left(\frac{2k-1}{2n}\pi\right), \qquad k = 1, 2, \ldots, n\,.$$

It is clear *a priori* that for minimum norm all of the points x_k must lie in the interval $[-1, 1]$, since $|\ell(x)|$ is a product of distances. Another interpretation of the theorem is that among all polynomials of degree $\le n-1$, the polynomial

$$P_n(x) = x^n - 2^{1-n}T_n(x)$$

gives the best uniform approximation to the function x^n over the interval $[-1, 1]$.

Proof of Theorem 2. Let $\widehat{T}_n = 2^{1-n}T_n$ and suppose that $\|Q\|_\infty < \|\widehat{T}_n\|_\infty$ for some monic polynomial Q of degree n. According to Lemma 2, the maximum of $|\widehat{T}_n(x)|$ occurs at the points $\xi_k = \cos(k\pi/n)$, with alternating signs. Specifically,

$$(-1)^k\widehat{T}_n(\xi_k) = 2^{1-n} = \|\widehat{T}_n\|_\infty \,, \qquad k = 0, 1, \ldots, n\,,.$$

Therefore, $Q(\xi_k) < \widehat{T}_n(\xi_k)$ when k is even, and $Q(\xi_k) > \widehat{T}_n(\xi_k)$ when k is odd. This implies that the difference $\widehat{T}_n(x) - Q(x)$ vanishes at some point in each of the n intervals (ξ_1, ξ_0), (ξ_2, ξ_1), $\ldots$, (ξ_n, ξ_{n-1}). But this is impossible, because $\widehat{T}_n$ and Q are both monic polynomials of degree n, so $\widehat{T}_n - Q$ is a polynomial of degree $\le n-1$ and can have at most $n-1$ zeros. Thus we have arrived at a contradiction, which proves the theorem. □

6.2. Weierstrass approximation theorem

It can be truly said that the theory of approximation was launched in 1885 with Weierstrass's discovery that every continuous function can be approximated uniformly by polynomials. Here is a precise statement.

Theorem 3 (Weierstrass Approximation Theorem). *If a function $f(x)$ is continuous on a closed bounded interval $[a, b]$, then for each $\varepsilon > 0$ there exists a polynomial $P(x)$ such that $|f(x) - P(x)| < \varepsilon$ for all x in $[a, b]$.*

Karl Weierstrass (1815–1897) was 70 years old at the time of his discovery. In fact, the theorem was discovered independently at almost the same time by Carl Runge (1856–1927), who had been a student of Weierstrass five years earlier. Runge obtained the result as part of his groundbreaking work on approximation theory in the complex domain.

The theorem of Weierstrass is remarkable because continuous functions can have a much more complicated behavior than polynomials. For instance, in 1872 Weierstrass had constructed examples of continuous functions that are not differentiable at any point. If the function f is sufficiently smooth, it can be approximated locally by partial sums of its Taylor series, but in general these polynomials will not provide the global approximation of the Weierstrass theorem. A more promising idea is to construct from f its Lagrange interpolation polynomials at n equally spaced points in the interval $[a, b]$. As $n \to \infty$, it is reasonable to expect this sequence of polynomials to converge uniformly to f. However, these interpolation polynomials need not converge pointwise to f, even if f has derivatives of all orders. In fact, the interpolation polynomials can fail to be uniformly bounded. This surprising state of affairs makes the Weierstrass theorem appear all the more remarkable.

Runge gave a striking example in 1901 that illustrates the failure of approximation by Lagrange interpolation polynomials. For the function $f(x) = 1/(1+x^2)$, Runge considered the interpolation polynomials at equally spaced nodes over the interval $[-5, 5]$ and showed that they are unbounded for $c < |x| < 5$, where $c = 3.63\ldots$. Sergei Bernstein showed in 1912 that for the function $f(x) = |x|$ in the interval $[-1, 1]$ the interpolation polynomials at equally spaced nodes converge only at the points -1, 0, and 1. One might suspect, in light of the results discussed in the preceding section, that the bad behavior in these examples is the fault of equally spaced nodes, and that nodes at the scaled roots of Chebyshev polynomials would produce better results. Indeed, this is true when the target function f is sufficiently smooth; for instance, when it has a continuous derivative. However, it is a theorem of Georg Faber, proved in 1914, that for *any* preassigned sequence of nodes

in a given interval, there is some continuous function whose interpolation polynomials are unbounded in that interval.

Many proofs of the Weierstrass theorem are known. We shall present three of them, all quite elegant, which shed light on various aspects of the problem. The first of these proofs, due to Lebesgue [**5**], is ingenious but quite simple. The second proof, due to Landau [**4**], proceeds by convolution of the given continuous function f with a polynomial "peaking kernel". Landau's proof closely resembles the original argument of Weierstrass (details to be discussed later), but is more direct. The third proof, due to Sergei Bernstein [**1**], is motivated by the theory of probability. It uses the values of f at equally spaced points to construct a sequence of polynomials, known as the Bernstein polynomials, that are shown to converge uniformly to f. The proofs by Landau and Bernstein are constructive in the sense that they produce explicit sequences of polynomials which converge uniformly to the given continuous function.

Henri Lebesgue (1875–1941) is famous for developing what is now called the Lebesgue integral. It extends the Riemann integral to a wider class of functions and has become an essential tool in mathematical analysis. Here is Lebesgue's proof of the Weierstrass theorem.

Lebesgue's proof. This proof has a striking feature. It reduces the approximation of an arbitrary continuous function f to the approximation of a single function, namely $f(x) = |x|$. The reduction proceeds as follows. If a function f is continuous at each point of the closed bounded interval $[a, b]$, it is *uniformly* continuous there. Therefore, f can be approximated uniformly by a continuous piecewise linear function g whose graph is a polygonal path connecting finitely many points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ on the curve $y = f(x)$, where

$$a = x_1 < x_2 < \cdots < x_n = b\,.$$

Observe that the function

$$\varphi_k(x) = |x - x_k| + (x - x_k)$$

is equal to 0 for $x \le x_k$ and is equal to $2(x - x_k)$ for $x \ge x_k$. Consider the function

$$h(x) = c_0 + \sum_{j=1}^{n-1} c_j\, \varphi_j(x)\,,$$

where the constants c_j are to be determined. The function h varies linearly in each of the intervals $[x_k, x_{k+1}]$, so it will coincide with the function g if the constants c_j can be chosen such that $h(x_k) = y_k$ for $k = 1, 2, \ldots, n$.

These conditions can be written in the form

$$c_0 + 2\sum_{j=1}^{k-1} c_j\,(x_k - x_j) = y_k\,, \qquad k = 1, 2, \ldots, n\,,$$

and so a unique solution $(c_0, c_1, \ldots, c_n)$ can be calculated recursively. With this choice of constants, $h(x) = g(x)$ in $[a, b]$. Thus the uniform approximation of $g(x)$ by polynomials reduces to the uniform approximation of each function $\varphi_k(x)$, hence to that of $|x - x_k|$, and finally to the uniform approximation of $|x|$ by polynomials in $[a, b]$. This last problem is trivial unless $a < 0 < b$, and then it suffices to consider a symmetric interval, say $[-b, b]$. After the substitution $x = bt$, the problem reduces to uniform approximation of $|t|$ by polynomials in the interval $[-1, 1]$.

To perform the last approximation, begin with the binomial expansion

$$\sqrt{1 - x} = 1 - \tfrac{1}{2}x - \tfrac{1}{8}x^2 - \ldots\,,$$

recalling that the series converges uniformly in the interval $[-1, 1]$. (See Chapter 3, Section 3.5.) Now replace x by $1 - t^2$ to see that

$$|t| = 1 - \tfrac{1}{2}(1 - t^2) - \tfrac{1}{8}(1 - t^2)^2 - \ldots\,,$$

and the convergence is again uniform in $[-1, 1]$. This shows that $|t|$ can be approximated uniformly by partial sums of the series, which are polynomials in t, and the proof is complete. □

6.3. Landau's proof

Edmund Landau (1877–1938) was primarily an analytic number theorist who discovered many new results on the distribution of prime numbers. His two-volume work *Handbuch der Lehre von der Verteilung der Primzahlen* (1909) was the first systematic presentation of analytic number theory. Landau was a perfectionist known for the beauty and clarity of his writing. His little books *Grundlagen der Analysis* (1930) and *Darstellung und Begründung einiger neuerer Ergebnisse der Funktionentheorie* (1916) are classics. In addition to his original discoveries, he was noted for finding simpler and more elegant proofs of known results. His proof of the Weierstrass approximation theorem is no exception.

Landau's proof. The following streamlined version of Landau's proof is adapted from the book of Rudin [**11**]. It is sufficient to take the interval $[a, b]$ to be $[0, 1]$, since the general result then follows by translation and change of scale. It is convenient to assume also that $f(0) = f(1) = 0$. This

assumption entails no loss of generality, because the general result is then deduced by first subtracting a linear polynomial to produce a function

$$g(x) = f(x) - f(0) - [f(1) - f(0)]\,x$$

continuous on $[0,1]$ with the property $g(0) = g(1) = 0$.

After these reductions, consider the functions

$$K_n(x) = c_n(1-x^2)^n\,, \qquad n = 1, 2, 3, \ldots\,,$$

where the positive constants c_n are chosen so that

$$\int_{-1}^{1} K_n(x)\,dx = 1\,.$$

Note that $K_n(x)$ is a polynomial of degree $2n$ and is an even function, meaning that $K_n(-x) = K_n(x)$. It is also evident that $0 < K_n(x) \le K_n(0) = c_n$ for $-1 < x < 1$. The functions $K_n(x)$ peak more and more sharply at the origin as n increases. Some sample graphs are shown in Figure 2.

One further property of the functions $K_n(x)$, suggested by Figure 2, is that $K_n(x) \to 0$ uniformly in the set $\delta \le |x| \le 1$ for each fixed $\delta > 0$. To see this, we will need an estimate on the rate of growth of the constant factors c_n. Direct calculation of the integral and an appeal to Stirling's formula (*cf.* Chapter 9, Exercise 24) shows in fact that

$$c_n = \frac{(2n+1)!}{2^{2n+1}(n!)^2} \sim \sqrt{n/\pi}\,, \qquad n \to \infty\,,$$

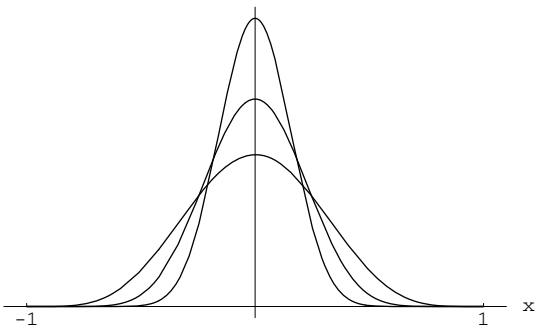


Figure 2. The peaking kernel $K_n(x)$ for $n = 5, 10, 20$.

but the more elementary estimate $c_n < \sqrt{n}$ will suffice for our purpose. Recall first that the inequality $(1+t)^n \geq 1+nt$ holds for all $t > -1$, a fact easily verified by induction. Hence

$$(1-x^2)^n \geq 1-nx^2\,, \qquad |x|<1\,.$$

Using this inequality, we find that

$$\begin{aligned}\frac{1}{c_n} = \int_{-1}^{1}(1-x^2)^n\,dx &= 2\int_0^1 (1-x^2)^n\,dx \geq 2\int_0^{1/\sqrt{n}}(1-x^2)^n\,dx \\ &\geq 2\int_0^{1/\sqrt{n}}(1-nx^2)\,dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}}\,,\end{aligned}$$

which shows that $c_n < \sqrt{n}$ for $n = 1, 2, 3, \ldots$. It follows that for $0 < \delta < 1$,

$$K_n(x) = c_n(1-x^2)^n \leq \sqrt{n}(1-\delta^2)^n\,, \qquad \delta \leq |x| \leq 1\,.$$

Thus $K_n(x) \to 0$ as $n \to \infty$, uniformly in each set $\delta \leq |x| \leq 1$.

Now define the convolution

$$P_n(x) = \int_0^1 K_n(x-t)f(t)\,dt\,, \qquad 0 \leq x \leq 1\,.$$

For each fixed x the number $P_n(x)$ may be viewed as a "weighted average" of the numbers $f(t)$ as t ranges over the interval $[0,1]$, with greatest weight attached (when n is large) to the values of t near x. Intuitively, then, it is to be expected that $P_n(x) \to f(x)$ as $n \to \infty$. Because K_n is a polynomial, it is easy to see that P_n is also a polynomial.

In order to prove that $P_n(x) \to f(x)$ uniformly in $[0,1]$, it is convenient first to extend the given function by setting $f(x) = 0$ for all x outside the interval $[0,1]$. The extended function is continuous on $\mathbb{R}$ because of our initial assumption that $f(0) = f(1) = 0$. Since K_n is an even function, the interval of integration can be shifted to give

$$P_n(x) = \int_0^1 K_n(t-x)f(t)\,dt = \int_{-x}^{1-x} K_n(t)f(x+t)\,dt\,.$$

But $f(x+t) = 0$ for $t \leq -x$ and for $t \geq 1-x$, so we can write

$$P_n(x) = \int_{-1}^{1} K_n(t)f(x+t)\,dt\,.$$

This shows that

$$P_n(x) - f(x) = \int_{-1}^{1} K_n(t)[f(x+t) - f(x)]\,dt\,,$$

since $\int_{-1}^{1} K_n(t)\,dt = 1$. Now write $|f(x)| \le M$ in $[0,1]$, and invoke the basic theorem that a function continuous at each point of a closed bounded interval is *uniformly* continuous there. Thus to each $\varepsilon > 0$ there corresponds a number $\delta > 0$ such that

$$|f(x+t) - f(x)| < \varepsilon \qquad \text{whenever} \quad x \in [0,1] \ \text{ and } \ |t| < \delta\,.$$

Therefore, recalling the estimate

$$K_n(t) \le \sqrt{n}(1-\delta^2)^n\,, \qquad \delta \le |t| \le 1\,,$$

we find

$$\begin{aligned}
|P_n(x) - f(x)| &= \left| \int_{-1}^{1} K_n(t)[f(x+t) - f(x)]\,dt \right| \\
&\le \int_{-1}^{1} K_n(t)|f(x+t) - f(x)|\,dt \\
&\le 2M \int_{-1}^{-\delta} K_n(t)\,dt + \int_{-\delta}^{\delta} K_n(t)|f(x+t) - f(x)|\,dt \\
&\quad + 2M \int_{\delta}^{1} K_n(t)\,dt \\
&\le 4M\sqrt{n}(1-\delta^2)^n + \varepsilon \int_{-\delta}^{\delta} K_n(t)\,dt \\
&< 4M\sqrt{n}(1-\delta^2)^n + \varepsilon < 2\varepsilon
\end{aligned}$$

for all n sufficiently large. Thus $P_n(x) \to f(x)$ as $n \to \infty$, uniformly in $[0,1]$. This completes Landau's proof of the theorem. □

The main idea of Landau's proof, convolution with a peaking kernel, was already present (as Landau [**4**] acknowledged) in the original proof given by Weierstrass [**15**]. But instead of a polynomial peaking kernel, Weierstrass used the kernel $e^{-(x/a)^2}$ with $a > 0$, for which

$$\int_{-\infty}^{\infty} e^{-(x/a)^2}\,dx = a\sqrt{\pi}\,.$$

By a similar argument he could show that the integrals

$$\frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(\frac{x-t}{a}\right)^2} f(t)\,dt$$

converge uniformly to $f(x)$ as $a \to 0$. He then approximated the integrals uniformly by taking partial sums of the Taylor series of the kernel $e^{-(x/a)^2}$, and so approximated $f(x)$ uniformly by polynomials.

6.4. Bernstein polynomials

Sergei Bernstein (1880–1968) was born in the Ukraine but got his early mathematical training in Paris. He then returned to the Ukraine and eventually held positions in Leningrad and Moscow. He did basic work in the area of interpolation and approximation and made important contributions to the theory of probability. In his proof of the Weierstrass theorem he introduced the Bernstein polynomials, as they are now called, which turned out to have many interesting properties.

Bernstein's proof. Again we may assume that the underlying interval is $[0,1]$. Given a function $f(x)$ continuous on $[0,1]$, we propose to show that the *Bernstein polynomials*

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} f(k/n)\, x^k\, (1-x)^{n-k}$$

converge uniformly to $f(x)$. Here

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is a binomial coefficient and $0! = 1$.

We now digress to discuss the heuristic source of the Bernstein polynomials. As indicated by the title of Bernstein's paper [**1**], the construction is motivated by probability theory, although the proof of uniform convergence makes no essential reference to these concepts. Let x be a fixed number with $0 < x < 1$. If a point is chosen at random from the interval $[0,1]$, it will lie in the interval $[0,x]$ with probability x and in the complementary subinterval $[x,1]$ with probability $1-x$. If n points are chosen independently, the probability that exactly k of them lie in the interval $[0,x]$ is given by the binomial distribution

$$b(k;n,x) = \binom{n}{k} x^k (1-x)^{n-k}\,, \qquad k = 0, 1, \ldots, n\,.$$

Consider now a random variable Y_n that has the value $f(k/n)$ if k of the chosen points are in $[0,x]$. The *expectation* of Y_n is

$$E(Y_n) = \sum_{k=0}^{n} f(k/n) b(k;n,x) = B_n(x)\,.$$

It is plausible (and can be proved on probabilistic grounds, by appeal to the "strong law of large numbers") that $E(Y_n) \to f(x)$ as $n \to \infty$.

For an analytic proof, we will need some standard facts about the binomial distribution, as stated in the following lemma.

Lemma 2.

$$(a) \quad \sum_{k=0}^{n} b(k;n,x) = 1\,;$$

$$(b) \quad \sum_{k=0}^{n} (k-nx)^2\, b(k;n,x) = nx(1-x)\,.$$

The proof of (a) is the simple observation that by the binomial theorem, the sum is equal to $[x+(1-x)]^n = 1$. The sum in (b) is a calculation of the variance of the binomial distribution; the proof is deferred for the moment. In view of (a), we have

$$B_n(x) - f(x) = \sum_{k=0}^{n} [f(k/n) - f(x)]\, b(k;n,x)\,.$$

The proof that $B_n(x)$ tends uniformly to $f(x)$ appeals again to the *uniform* continuity of f on $[0,1]$. For each $\varepsilon > 0$, there is a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all pairs of points $x, y \in [0,1]$ with $|x-y| < \delta$. Again suppose $|f(x)| \le M$ for all x in the interval $[0,1]$, and write

$$|B_n(x) - f(x)| \le \sum_{k=0}^{n} |f(k/n) - f(x)|\, b(k;n,x) = S_1 + S_2\,,$$

where the sum S_1 extends over all integers k $(0 \le k \le n)$ with $\left|\frac{k}{n} - x\right| < \delta$ and the sum S_2 is taken over those integers k with $\left|\frac{k}{n} - x\right| \ge \delta$. By the uniform continuity of f and part (a) of the lemma, we see that $S_1 < \varepsilon$. On the other hand,

$$\begin{aligned} S_2 &\le 2M \sum_{|\frac{k}{n}-x|\ge\delta} b(k;n,x) \le 2M \sum_{k=0}^{n} \frac{(k/n-x)^2}{\delta^2}\, b(k;n,x) \\ &= \frac{2M}{\delta^2 n^2} \sum_{k=0}^{n} (k-nx)^2\, b(k;n,x) = \frac{2Mx(1-x)}{\delta^2 n} \le \frac{2M}{\delta^2 n} < \varepsilon \end{aligned}$$

for all $n \ge N > 2M/(\delta^2\varepsilon)$, where part (b) of the lemma has been used. Observe that N depends only on ε, so we have shown that

$$|B_n(x) - f(x)| \le 2\varepsilon \qquad \text{for all} \quad x \in [0,1]$$

whenever $n \ge N$. This shows that $B_n(x) \to f(x)$ uniformly in $[0,1]$, which completes Bernstein's proof of the Weierstrass approximation theorem. □

Proof of Lemma 2. It was already observed that

$$1 = [x + (1-x)]^n = \sum_{k=0}^{n} \binom{n}{k} x^k (1-x)^{n-k} = \sum_{k=0}^{n} b(k; n, x)\,,$$

which proves (a). To prove (b), we first calculate

$$\begin{aligned}\mu = \sum_{k=0}^{n} k\, b(k; n, x) &= nx \sum_{k=1}^{n} \binom{n-1}{k-1} x^{k-1}(1-x)^{n-k} \\ &= nx \sum_{j=0}^{n-1} b(j; n-1, x) = nx\,,\end{aligned}$$

by (a), after the shift of index $j = k-1$. Next observe that with $\mu = nx$

$$\begin{aligned}\sum_{k=0}^{n} (k - nx)^2\, b(k; n, x) &= \sum_{k=0}^{n} (k^2 - 2k\mu + \mu^2) b(k; n, x) \\ &= \sum_{k=0}^{n} k^2\, b(k; n, x) - \mu^2\,.\end{aligned}$$

But

$$\begin{aligned}\sum_{k=0}^{n} k^2\, b(k; n, x) &= \mu + \sum_{k=2}^{n} k(k-1) b(k; n, x) \\ &= \mu + n(n-1)x^2 \sum_{k=2}^{n} \binom{n-2}{k-2} x^{k-2}(1-x)^{n-k} \\ &= \mu + n(n-1)x^2 \sum_{j=0}^{n-2} b(j; n-2, x) \\ &= \mu + n(n-1)x^2 = nx(1-x) + \mu^2\,,\end{aligned}$$

which combines with the previous formula to yield the desired result. □

The Bernstein polynomials have the remarkable property that for functions f with continuous νth derivative, $B_n^{(\nu)}(x) \to f^{(\nu)}(x)$ uniformly in $[0, 1]$. (See Exercise 14.) The books by Cheney [**2**], Davis [**3**], and Lorentz [**6**] may be consulted for this and other facts about Bernstein polynomials.

6.5. Best approximation

The theorem of Weierstrass is a qualitative statement that uniform approximation by polynomials is possible. For any given function f continuous over an interval $[a, b]$, it says that the uniform distance

$$\|f - P\|_\infty = \max_{a \le x \le b} |f(x) - P(x)|$$

can be made arbitrarily small with some choice of polynomial P. The theorem suggests the problem of finding the best approximation by polynomials of fixed degree. More precisely, it is of interest to determine the quantity

$$E_n(f) = \inf_P \|f - P\|_\infty \,,$$

where the infimum is taken over all polynomials of degree $\le n$. Is the infimum attained, and if so what are the polynomials of best approximation?

We will see presently that an extremal polynomial of degree $\le n$ always exists and is unique. On the other hand, no explicit formula is known for this polynomial of best approximation to an arbitrarily prescribed function f that is continuous on a specified interval $[a, b]$. To find such a formula is an open problem of long standing. It turns out, however, that the polynomial of best approximation can be characterized by the alternation property already encountered in the special case of Chebyshev polynomials. Deferring a precise statement, we begin with some general remarks about polynomials.

According to the Lagrange interpolation formula, any polynomial P of degree $\le n$ can be written as

$$P(x) = \sum_{k=0}^{n} \frac{P(x_k)\,\ell(x)}{\ell'(x_k)(x - x_k)} \,, \qquad \ell(x) = \prod_{j=0}^{n} (x - x_j) \,,$$

for any system $\{x_0, x_1, \dots, x_n\}$ of distinct points. The representation has the form

$$P(x) = \sum_{k=0}^{n} P(x_k)\, q_k(x) \,,$$

where the functions $q_k(x)$ are fixed polynomials of degree $\le n$, depending only on the points x_j. This formula makes clear that if a sequence of polynomials of degree $\le n$ is bounded at $n+1$ specified points (in particular, if the polynomials are uniformly bounded on an interval), then the corresponding sequences of coefficients are also bounded.

Theorem 4. *For any function f continuous over an interval $[a, b]$, and for any positive integer n, there exists a polynomial P of degree $\le n$ for which the distance $\|f - P\|_\infty$ attains a minimum value. Furthermore, this polynomial of best approximation is unique.*

Proof of existence. Since $E_n(f)$ is the infimum of $\|f - P\|_\infty$, there is a sequence of polynomials

$$P_j(x) = \sum_{k=0}^{n} a_{jk}\, x^k\,, \qquad j = 1, 2, \ldots\,,$$

of degree $\le n$ such that $\|f - P_j\|_\infty \to E_n(f)$ as $j \to \infty$. Then the norms $\|P_j\|_\infty$ are bounded, so by the preceding remarks the coefficients a_{jk} are also bounded. Thus by the Bolzano–Weierstrass theorem, some subsequence $\{a_{j_\nu k}\}$ of the coefficients converges, say to a_k, for each k. Then the corresponding polynomials $P_{j_\nu}(x)$ converge to

$$P(x) = \sum_{k=0}^{n} a_k\, x^k\,,$$

uniformly in the interval $[a, b]$. Therefore,

$$E_n(f) = \lim_{\nu\to\infty} \|f - P_{j_\nu}\|_\infty = \|f - P\|_\infty\,,$$

which shows that P is a polynomial of best approximation. □

The uniqueness property will be derived from the following theorem.

Theorem 5. *Let f be a continuous function on an interval $[a, b]$, and suppose P is a polynomial of degree $\le n$ that gives the best uniform approximation to f on $[a, b]$, so that $\|f - P\|_\infty = E_n(f)$. Then there are $n + 2$ points x_k in $[a, b]$,*

$$a \le x_1 < x_2 < \cdots < x_{n+2} \le b\,,$$

at which $f(x_k) - P(x_k) = \pm\, E_n(f)$ with alternating signs; that is,

$$f(x_{k+1}) - P(x_{k+1}) = -[f(x_k) - P(x_k)]\,, \qquad k = 1, 2, \ldots, n + 1\,.$$

Before embarking on the proof, let us remark that the Chebyshev polynomials provide a special case of this alternating property. As previously noted, Theorem 2 can be interpreted as saying that among all polynomials of degree $\le n - 1$,

$$P(x) = x^n - 2^{1-n} T_n(x)$$

gives the best uniform approximation to the function $f(x) = x^n$ over the interval $[-1, 1]$. Lemma 1 then confirms that

$$f(x) - P(x) = 2^{1-n} T_n(x)$$

attains its maximum absolute value in the interval $[-1, 1]$, with alternating signs at $n + 1$ points $\xi_k = \cos(k\pi/n)$, for $k = 0, 1, \ldots, n$. Note that n is replaced here by $n - 1$.

Proof of Theorem 5. For convenience, set $\varphi(x) = f(x) - P(x)$ and $\delta = E_n(f)$. Suppose that $\delta > 0$, excluding the trivial case where f is itself a polynomial of degree $\leq n$. If $\varphi(x)$ has the same sign wherever $|\varphi(x)| = \delta$, then the approximation can be improved by adding to P a small constant of that sign. But this is impossible if P is a polynomial of best approximation. Thus there must be some variation of signs at points where $|\varphi(x)| = \delta$.

Because the function φ is uniformly continuous, the interval $[a, b]$ can be divided into a finite number of subintervals in such a way that $\varphi(x)$ does not vanish in any interval where $|\varphi|$ attains its maximum value δ. Let

$$I_1 = [a_1, b_1]\,, \quad I_2 = [a_2, b_2]\,, \quad \ldots\,, \quad I_m = [a_m, b_m]$$

denote the subintervals in which $|\varphi|$ attains the value δ, with numbering chosen so that

$$a \leq a_1 < b_1 \leq a_2 < b_2 \leq \cdots \leq a_m < b_m \leq b\,.$$

Let σ_k denote the sign of $\varphi(x)$ in I_k; in other words, $\sigma_k = +1$ if $\varphi(x) > 0$ in I_k, and $\sigma_k = -1$ if $\varphi(x) < 0$ in I_k. As previously noted, the sequence $\{\sigma_1, \sigma_2, \ldots, \sigma_m\}$ must display some variation of sign. If $\sigma_{k+1} = -\sigma_k$, then $b_k < a_{k+1}$ and $|\varphi(x)| < \delta$ in the interval (b_k, a_{k+1}). Thus after slight enlargement of the intervals I_k and I_{k+1} if necessary, we may assume that $|\varphi(b_k)| < \delta$ and $|\varphi(a_{k+1})| < \delta$ while maintaining the inequality $b_k < a_{k+1}$ together with the property that $\varphi(x) \neq 0$ in I_k and I_{k+1}. After these adjustments have been made, we can assert that $|\varphi(x)| \leq \delta' < \delta$ for all x in the complementary set

$$S = [a, b] \setminus (I_1 \cup I_2 \cup \cdots \cup I_m)\ .$$

We are to prove that the sequence $\{\sigma_1, \sigma_2, \ldots, \sigma_m\}$ exhibits at least $n + 1$ changes of sign from an entry σ_k to its successor σ_{k+1}. Suppose, on the contrary, that there are only $\nu \leq n$ changes of sign, and let $k_1, k_2, \ldots, k_\nu$ denote the indices where these changes occur. Thus

$$\sigma_{k_j+1} = -\sigma_{k_j}\,, \qquad j = 1, 2, \ldots, \nu\,.$$

Then we can choose points $\xi_j \in S$ so that

$$b_{k_j} < \xi_j < a_{k_j+1}\,, \qquad j = 1, 2, \ldots, \nu\,,$$

and form the function

$$q(x) = \sigma_{k_1}(\xi_1 - x)(\xi_2 - x)\cdots(\xi_\nu - x)\,,$$

which is a polynomial of degree $\nu \le n$. By construction, $q(x)$ has the same sign as $\varphi(x)$ in each of the intervals $I_1, I_2, \dots, I_m$.

We now claim that for sufficiently small $\varepsilon > 0$, the polynomial $Q = P + \varepsilon q$ provides a better approximation:

$$\|f - Q\|_\infty < \|f - P\|_\infty = \delta\,.$$

This will contradict the extremal property of P, thus proving the theorem. To establish the claim, choose $\varepsilon > 0$ so small that

$$\varepsilon|q(x)| < \delta - \delta' \qquad \text{for all } \ x \in [a,b]\,.$$

Then

$$|f(x) - Q(x)| = |\varphi(x) - \varepsilon q(x)| < \delta$$

for all $x \in (I_1 \cup I_2 \cup \cdots \cup I_m)$, since $\varphi(x)$ and $q(x)$ have the same sign there. On the other hand, if $x \in S$, then $|\varphi(x)| \le \delta'$ and so

$$\begin{aligned} |f(x) - Q(x)| = |\varphi(x) - \varepsilon q(x)| &\le |\varphi(x)| + \varepsilon|q(x)| \\ &< \delta' + (\delta - \delta') = \delta\,. \end{aligned}$$

Combining the two estimates, we see that

$$\|f - Q\|_\infty < \|f - P\|_\infty\,,$$

which contradicts the hypothesis that P is a polynomial of degree $\le n$ of best approximation to f. This shows that the sequence $\{\sigma_1, \sigma_2, \dots, \sigma_m\}$ must have at least $n+1$ changes of sign, as the theorem asserts. □

As a consequence of Theorem 5, we can now infer that the polynomial of best approximation is unique. Suppose that two polynomials P and Q of degree $\le n$ both give the best approximation to f:

$$\|f - P\|_\infty = \|f - Q\|_\infty = E_n(f)\,.$$

Then the polynomial $R = \frac{1}{2}(P+Q)$ also gives the best approximation, since

$$\|f - R\|_\infty \le \tfrac{1}{2}\|f - P\|_\infty + \tfrac{1}{2}\|f - Q\|_\infty = E_n(f)\,.$$

In view of Theorem 5, it follows that $|f(x) - R(x)|$ attains its maximum value $E_n(f)$ at $n+2$ distinct points of the interval $[a,b]$. But since

$$f(x) - R(x) = \tfrac{1}{2}\big(f(x) - P(x)\big) + \tfrac{1}{2}\big(f(x) - Q(x)\big)\,,$$

an application of the triangle inequality shows that

$$f(x) - P(x) = f(x) - Q(x) = f(x) - R(x)$$

at any point where $|f(x) - R(x)| = E_n(f)$. Thus $P(x) = Q(x)$ at $n+2$ distinct points, which implies that $P(x) \equiv Q(x)$, since both are polynomials of degree $\le n$. This proves the uniqueness.

Finally, we can show that the alternation property of Theorem 5 is not only necessary for P to be the polynomial of best approximation, but sufficient as well.

Theorem 6. *The alternation property of Theorem* 5 *characterizes the polynomial P of degree $\le n$ of best approximation. More precisely, if f is continuous on $[a,b]$ and Q is a polynomial of degree $\le n$ such that $f(x)-Q(x)$ attains its maximum absolute value at $n+2$ distinct points in $[a,b]$ with alternating signs, then $Q=P$.*

Proof. Let $d=\|f-Q\|_\infty$ and suppose that

$$|f(x_k)-Q(x_k)|=d \qquad \text{for} \quad a\le x_1<x_2<\dots x_{n+2}\le b\,.$$

Suppose also that the signs alternate, so that

$$f(x_{k+1})-Q(x_{k+1})=-[f(x_k)-Q(x_k)]\,, \qquad k=1,2,\dots,n+1\,.$$

Since P is the polynomial of best approximation, it is clear that $d\ge\delta=\|f-P\|_\infty$. If $d>\delta$, then the polynomial

$$P-Q=(f-Q)-(f-P)$$

has the sign of $f-Q$ at the points x_k. This implies that $P-Q$ changes sign at least $n+1$ times, and so it has at least $n+1$ distinct zeros. But $P-Q$ is a polynomial of degree $\le n$, so it follows that $P(x)-Q(x)\equiv 0$, or $Q=P$. This is not possible, however, if $d>\delta$. Thus $d=\delta$ and $Q=P$ by uniqueness of the polynomial of best approximation. □

The alternation results of Theorems 5 and 6, which together characterize the polynomial of best approximation, go back at least to Émile Borel (1905) and Charles de la Vallée Poussin (1910). The exposition of this section is adapted from the book by de la Vallée Poussin [**14**], which contains much more information. Cheney [**2**] presents the same material in terms of "Haar systems", which are more general than polynomials. The alternation criterion is the basis for an algorithm, developed by Evgeny Remez in 1934, for the numerical calculation of the polynomial of best approximation. For a full description the reader is referred to the book by Cheney [**2**].

6.6. Stone–Weierstrass theorem

The Weierstrass approximation theorem can be generalized to several variables. Weierstrass showed that a continuous function $f(x_1,x_2,\dots,x_n)$ can be approximated uniformly on any closed bounded subset of $\mathbb{R}^n$ by linear

combinations of monomials $x_1^{k_1} x_2^{k_2} \cdots x_n^{k_n}$, where the exponents $k_1, k_2, \ldots, k_n$ are nonnegative integers. Landau also extended the theorem to several variables, by a simple inductive argument.

The extension to several variables is actually a special case of a far-reaching generalization of Weierstrass's theorem discovered in 1937 by the American mathematician Marshall Stone (1903–1989). Ten years later, Stone wrote an expository article [**12**] giving a simpler and more elementary proof. The theorem in generalized form, now known as the *Stone–Weierstrass theorem*, has a wide range of applications. In order to state it, we need to define a few terms. A family $\mathcal{A}$ of real-valued functions defined on an abstract set E is said to be an *algebra* if $f, g \in \mathcal{A}$ and $c \in \mathbb{R}$ imply that $f + g \in \mathcal{A}$, $fg \in \mathcal{A}$, and $cf \in \mathcal{A}$. A family $\mathcal{A}$ of functions defined on E is said to *separate points* if for each pair of distinct points $x, y \in E$ there is a function $f \in \mathcal{A}$ for which $f(x) \neq f(y)$. The family $\mathcal{A}$ is said to *vanish at a point* $x \in E$ if $f(x) = 0$ for every $f \in \mathcal{A}$.

Theorem 7 (Stone–Weierstrass Theorem). *Let $\mathcal{A}$ be an algebra of real-valued continuous functions on a compact subset E of a metric space. If $\mathcal{A}$ separates points and vanishes at no point of E, then every continuous function can be approximated uniformly on E by functions in $\mathcal{A}$.*

It is clear that the set of all polynomials $P(x)$ in one real variable is an algebra that separates points in any interval $[a, b]$ and does not vanish at any point, so the classical approximation theorem of Weierstrass is a special case of the Stone–Weierstrass theorem. The set of polynomials $P(F(x))$ of a fixed function F is a more general example of an algebra that does not vanish at any point, but it separates points only if F is univalent on E. As a special case, the set of all *even* polynomials is an algebra that does not separate points in any symmetric interval $[-a, a]$.

Recall that a subset of a metric space is said to be compact if it has the Heine–Borel property: each open covering contains a finite subcovering. For the sake of simplicity, we will prove the Stone–Weierstrass theorem under the special assumption that E is a closed bounded subset of the real line, but the argument generalizes with little change to arbitrary metric spaces. A simple algebraic lemma will be needed.

Lemma 3. *Let $\mathcal{A}$ be an algebra of real-valued functions on a set E that separates points and does not vanish at any point. Then for each pair of distinct points $x_1, x_2 \in E$ and each pair of numbers $y_1, y_2 \in \mathbb{R}$, there is a function $f \in \mathcal{A}$ for which $f(x_1) = y_1$ and $f(x_2) = y_2$.*

Proof of lemma. The algebra $\mathcal{A}$ separates points, so it contains a function g for which $g(x_1) \neq g(x_2)$. Because $\mathcal{A}$ vanishes at no point of E, there are

functions h_1 and h_2 in $\mathcal{A}$ with $h_1(x_1) \neq 0$ and $h_2(x_2) \neq 0$. Define

$$\varphi = [g - g(x_1)]\, h_2\,, \qquad \psi = [g - g(x_2)]\, h_1\,.$$

Then $\varphi, \psi \in \mathcal{A}$ and $\varphi(x_1) = \psi(x_2) = 0$, whereas $\varphi(x_2) \neq 0$ and $\psi(x_1) \neq 0$. Thus the function

$$f = \frac{y_1}{\psi(x_1)}\,\psi + \frac{y_2}{\varphi(x_2)}\,\varphi$$

belongs to $\mathcal{A}$ and has the properties $f(x_1) = y_1$, $f(x_2) = y_2$. □

Proof of theorem. Let $\mathcal{B}$ denote the uniform closure of $\mathcal{A}$; that is, the set of all functions uniformly approximable by functions in $\mathcal{A}$. More precisely, $g \in \mathcal{B}$ if for each $\varepsilon > 0$ there exists $f \in \mathcal{A}$ such that $|g(x) - f(x)| < \varepsilon$ for all $x \in E$. It is easy to see that $\mathcal{B}$ is an algebra of continuous functions. We are to prove that $\mathcal{B}$ is the algebra of *all* continuous functions on E.

The first step is to show that $|f| \in \mathcal{B}$ whenever $f \in \mathcal{B}$. To see this, let $M = \sup\{|f(x)| : f \in E\}$ and let $\varepsilon > 0$. By the Weierstrass approximation theorem, there is a polynomial P such that

$$|P(y) - |y|| < \varepsilon\,, \qquad y \in [-M, M]\,.$$

This implies that

$$|P(f(x)) - |f(x)|| < \varepsilon\,, \qquad x \in E\,.$$

But $\mathcal{B}$ is an algebra, so $P \circ f \in \mathcal{B}$ if $f \in \mathcal{B}$. Thus $|f| \in \mathcal{B}$, since $\mathcal{B}$ is uniformly closed.

It now follows that

$$\max(f, g) = \tfrac{1}{2}(f + g) + \tfrac{1}{2}|f - g| \qquad \text{and}$$
$$\min(f, g) = \tfrac{1}{2}(f + g) - \tfrac{1}{2}|f - g|$$

belong to $\mathcal{B}$ whenever $f, g \in \mathcal{B}$. Iteration gives the more general result that $\max\{f_1, f_2, \ldots, f_n\}$ and $\min\{f_1, f_2, \ldots, f_n\}$ are in $\mathcal{B}$ if $f_1, f_2, \ldots, f_n \in \mathcal{B}$.

Next we claim that for each function f continuous on E, for each point $s \in E$, and for each $\varepsilon > 0$, there exists a function $g_s \in \mathcal{B}$ such that $g_s(s) = f(s)$ and $g_s(x) > f(x) - \varepsilon$ for all $x \in E$. To see this, observe that because $\mathcal{A}$ is a subalgebra of $\mathcal{B}$ that separates points and does not vanish at any point, $\mathcal{B}$ has the same properties. For an arbitrarily specified point $t \in E$, the lemma then shows that some function $h_t \in \mathcal{B}$ satisfies

$$h_t(s) = f(s) \qquad \text{and} \qquad h_t(t) = f(t)\,.$$

By the continuity of h_t and f, it follows that $h_t(x) > f(x) - \varepsilon$ for all x in some open set U_t containing t. But the collection of sets $\{U_t : t \in E\}$ forms an open covering of the compact set E, so the Heine–Borel theorem says that

$$E \subset U_{t_1} \cup U_{t_2} \cup \cdots \cup U_{t_n}$$

for some finite set of points $t_1, t_2, \ldots, t_n \in E$. Now define

$$h = \max\{h_{t_1}, h_{t_2}, \ldots, h_{t_n}\}\,.$$

Then $h \in \mathcal{B}$ and it has the properties $h(s) = f(s)$ and $h(x) > f(x) - \varepsilon$ for all $x \in E$, by construction of the functions h_t and the neighborhoods U_t. Thus with $g_s = h$ our claim is established.

The final step is to show that every continuous function is uniformly approximable by functions in $\mathcal{B}$, or equivalently that $f \in \mathcal{B}$. The proof involves another appeal to the Heine–Borel theorem. Given a function f continuous on E, a point $s \in E$, and a number $\varepsilon > 0$, we have constructed a function $g_s \in \mathcal{B}$ for which $g_s(s) = f(s)$ and $g_s(x) > f(x) - \varepsilon$ for all $x \in E$. By continuity, the inequality $g_s(x) < f(x) + \varepsilon$ persists in some open set V_s containing s. The sets V_s form an open covering of E, so there is a finite subcovering:

$$E \subset V_{s_1} \cup V_{s_2} \cup \cdots \cup V_{s_m}$$

for some finite collection of points $s_1, s_2, \ldots, s_m$. Now define

$$g = \min\{g_{s_1}, g_{s_2}, \ldots, g_{s_m}\}\,.$$

Then $g \in \mathcal{B}$ and $g(x) < f(x) + \varepsilon$ for all $x \in E$, by the construction of g and the sets V_s. On the other hand, $g(x) > f(x) - \varepsilon$ for all $x \in E$ because each function g_s has this property. Combination of the two inequalities gives

$$|f(x) - g(x)| < \varepsilon \qquad \text{for all } \; x \in E\,,$$

which completes the proof of the Stone–Weierstrass theorem. □

The theorem is not true as stated for algebras of complex-valued functions, but it does hold for self-conjugate algebras. In other words, the theorem remains true for complex algebras under the additional hypothesis that $\overline{f} \in \mathcal{A}$ whenever $f \in \mathcal{A}$, where $\overline{f}(x) = \overline{f(x)}$ is the complex conjugate function. For further details the reader may consult Davis [**3**] or Rudin [**11**].

References to additional proofs of the Weierstrass approximation theorem and further discussion may be found in Cheney [**2**] and Davis [**3**]. The article by Pincus [**10**] discusses the historical evolution of ideas leading to the discovery of the theorem.

6.7. Refinements of Weierstrass theorem

We conclude this chapter with some sharpened versions of the Weierstrass approximation theorem. The primary result, found in 1914 by Julius Pál [**8**], asserts that the approximation can be achieved by polynomials with prescribed initial coefficients. As Pál [**8**, **9**] observed, this comparatively simple result has some astonishing implications.

Theorem 8 (Pál's Theorem). *Let f be a continuous function on an interval $[-r, r]$ for some $r > 0$, and let $a_0 = f(0)$. Then for each $\varepsilon > 0$ and arbitrarily prescribed numbers $a_1, a_2, \ldots, a_n$, there exists a polynomial $P(x)$ with initial coefficients $a_0, a_1, \ldots, a_n$ such that $|f(x) - P(x)| < \varepsilon$ for all $x \in [-r, r]$.*

Proof. Without loss of generality, we may take $r = 1$. For convenience, set

$$a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = g(x)\,.$$

Then $g(0) = a_0 = f(0)$, and by continuity we see that $|f(x) - g(x)| < \varepsilon/3$ throughout an interval $[-\delta, \delta]$ for some $\delta > 0$. Define the function

$$\varphi(x) = \frac{f(x) - g(x)}{x^{n+1}} \qquad \text{for } \ \delta \le |x| \le 1\,.$$

Set $\varphi(0) = 0$ and define φ to be linear in each of the intervals $[0, \delta]$ and $[-\delta, 0]$. This function φ is continuous on $[-1, 1]$, so by the Weierstrass approximation theorem there is a polynomial Q for which

$$|\varphi(x) - Q(x)| < \frac{\varepsilon}{3} \qquad \text{for all } \ x \in [-1, 1]\,.$$

Now define the polynomial $P(x) = g(x) + x^{n+1}Q(x)$ and conclude that

$$|f(x) - P(x)| < \frac{\varepsilon}{3}\,|x|^{n+1} \le \frac{\varepsilon}{3} \qquad \text{for } \ \delta \le |x| \le 1\,.$$

For $0 \le x \le \delta$ we have $|\varphi(x) - Q(x)| < \varepsilon/3$ and $|\varphi(x)| \le |\varphi(\delta)|$, so that $|Q(x)| \le \varepsilon/3 + |\varphi(\delta)|$. Therefore,

$$\begin{aligned} |f(x) - P(x)| &< |f(x) - g(x)| + |x|^{n+1}|Q(x)| \\ &< \frac{\varepsilon}{3} + \delta^{n+1}|Q(x)| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \delta^{n+1}|\varphi(\delta)| \\ &= \frac{2\varepsilon}{3} + |f(\delta) - g(\delta)| < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon \end{aligned}$$

for $0 \le x \le \delta$. A similar estimate applies to the interval where $-\delta \le x \le 0$. Together with the earlier estimate for $\delta \le |x| \le 1$, this shows that $|f(x) - P(x)| < \varepsilon$ for all $x \in [-1, 1]$. By construction, the polynomial $P(x)$ has initial coefficients $a_0, a_1, \ldots, a_n$, so the proof is complete. □

There is an interesting application to power-series "expansion" of an arbitrary continuous function. As Pál [**8**] pointed out, Theorem 8 has the following immediate corollary.

Corollary 1. *Corresponding to any continuous function f on an interval $[-r, r]$ there is a power series $\sum_{k=0}^{\infty} a_k x^k$ whose partial sums $s_n(x) = \sum_{k=0}^{n} a_k x^k$ have a subsequence $\{s_{n_j}\}$ that converges uniformly to f in $[-r, r]$.*

Of course, the power series need not converge at any point $x_0 \neq 0$, since that would make its sum f infinitely differentiable in the interval $(-x_0, x_0)$, whereas f is not required to have a derivative at any point.

One particular case of Theorem 8 is worthy of note. Taking $a_0 = a_1 = \cdots = a_n = 0$, we infer that each continuous function f on an interval $[-r, r]$ with $f(0) = 0$ can be approximated uniformly by polynomials of the form

$$P(x) = a_{n+1}x^{n+1} + a_{n+2}x^{n+2} + \cdots + a_{n+k}x^{n+k}\,,$$

where n is any prescribed positive integer. This special case leads to a striking refinement of the Weierstrass approximation theorem, also due to Pál [**9**].

Corollary 2. *If $0 < r < 1$, then each function f continuous in the interval $[-r, r]$ with $f(0) = 0$ can be approximated uniformly in $[-r, r]$ by a polynomial all of whose coefficients are integers.*

Proof. Given $\varepsilon > 0$, choose n so large that

$$r^{n+1} + r^{n+2} + \cdots = \frac{r^{n+1}}{1-r} < \frac{\varepsilon}{2}\,.$$

Next invoke the special case of Theorem 8 to produce a polynomial of the form

$$Q(x) = a_{n+1}x^{n+1} + a_{n+2}x^{n+2} + \cdots + a_{n+k}x^{n+k}$$

such that $|f(x) - Q(x)| < \varepsilon/2$ for all $x \in [-r, r]$. Finally, define the polynomial

$$P(x) = c_{n+1}x^{n+1} + c_{n+2}x^{n+2} + \cdots + c_{n+k}x^{n+k}\,,$$

where the coefficients c_{n+j} are integers with $|c_{n+j} - a_{n+j}| < 1$. Then

$$\begin{aligned} |f(x) - P(x)| &\leq |f(x) - Q(x)| + |Q(x) - P(x)| \\ &< \frac{\varepsilon}{2} + \left(r^{n+1} + r^{n+2} + \cdots + r^{n+k}\right) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

for all $x \in [-r, r]$, as desired. □

Note that Corollary 2 fails unless $r < 1$. Consider for instance the function $f(x) = x/2$ on the interval $[-1, 1]$.

Finally, Corollary 1 can be sharpened to produce a single power series with the property that some subsequence of its partial sums converges uniformly to any given continuous function that vanishes at the origin. This astounding result, the existence of a *universal* power series that generates every continuous function, is the prototype for what has become a large body of similar results involving infinite series of functions. It was first presented by Pál [**9**] but was attributed to his Hungarian colleague Mihály Fekete.

Corollary 3 (Fekete's theorem). *There exists a single power series* $\sum_{k=1}^{\infty} a_k x^k$ *whose partial sums* $s_n(x) = \sum_{k=1}^{n} a_k x^k$ *approximate every function* f *continuous on an arbitrary interval* $[-r, r]$*, provided only that* $f(0) = 0$*. More precisely, to each such function* f *there corresponds an increasing sequence* $\{n_1, n_2, \dots\}$ *of positive integers such that* $s_{n_j}(x) \to f(x)$ *as* $j \to \infty$*, uniformly in* $-r \le x \le r$*.*

Proof. It follows directly from the Weierstrass theorem that each function f continuous on an interval $[a, b]$ can be approximated uniformly with any prescribed accuracy by polynomials with rational coefficients (*cf.* Exercise 19). The set of polynomials with rational coefficients is countable. Let $\{Q_1(x), Q_2(x), \dots\}$ be an enumeration of all polynomials with rational coefficients that vanish at the origin: $Q_1(0) = Q_2(0) = \cdots = 0$. Choose a polynomial $P_1(x)$ with $P_1(0) = 0$ such that $|P_1(x) - Q_1(x)| < 1$ for all $x \in [-1, 1]$. Appealing to Pál's theorem (Theorem 8), next choose a polynomial $P_2(x)$ that begins with $P_1(x)$ and has the approximation property

$$|P_2(x) - Q_2(x)| < \frac{1}{2} \qquad \text{for all } \ x \in [-2, 2]\,.$$

Now proceed inductively. Having chosen $P_1, P_2, \dots, P_{n-1}$, choose a polynomial $P_n(x)$ that begins with $P_{n-1}(x)$ and satisfies

$$|P_n(x) - Q_n(x)| < \frac{1}{n} \qquad \text{for all } \ x \in [-n, n]\,.$$

Then $P_1(x), P_2(x), \dots$ are partial sums of a power series with the desired property. To see this, suppose $f(x)$ is continuous on an interval $[-r, r]$ and $f(0) = 0$. Choose an increasing sequence $\{n_k\}$ of indices such that $Q_{n_k}(x) \to f(x)$ uniformly on $[-r, r]$ as $k \to \infty$. Then by construction, $[P_{n_k}(x) - Q_{n_k}(x)] \to 0$ and so $P_{n_k}(x) \to f(x)$ uniformly on $[-r, r]$. □

Needless to say, a universal power series, as described in Corollary 3, cannot converge at any point $x \neq 0$.

Pál's theorem strengthens the Weierstrass theorem by showing that finite linear combinations of the monomials $1, x^n, x^{n+1}, \dots$ are sufficient for

uniform approximation of any continuous function on a closed bounded interval. His result suggests the problem of characterizing the sets of monomials $1, x^{n_1}, x^{n_2}, \dots$ whose finite linear combinations are uniformly dense in the space of all continuous functions. Sergei Bernstein studied the problem in 1912 and found a partial solution. The full solution, at least on the interval $[0, 1]$, is quite elegant and was found around 1915 by Chaim Müntz [**7**] and Otto Szász [**13**].

Müntz–Szász Theorem. *Let $\{n_1, n_2, \dots\}$ be an increasing sequence of positive integers. In order that every function continuous on the interval $[0, 1]$ be uniformly approximable by polynomials of the form*

$$P(x) = c_0 + c_1 x^{n_1} + \cdots + c_k x^{n_k} ,$$

it is necessary and sufficient that the series $\sum_{k=1}^{\infty} 1/n_k$ diverge.

In fact, the exponents n_k need not be integers. The theorem remains valid for any increasing sequence of positive exponents that tend to infinity. A similar theorem holds for mean-square approximation of functions that are square-integrable over the interval $[0, 1]$. There the natural method of proof (introduced by Szász) involves Gram determinants and a classical formula of Cauchy to evaluate the particular determinants that arise. The uniform version of the theorem (as stated here) then follows from the mean-square version by simple estimates. The details will not be pursued here, but may be found in books on approximation theory, for instance in Cheney [**2**] or Davis [**3**].

It may be remarked that the Müntz–Szász theorem extends readily to any interval $[0, r]$, but not to an interval $[-r, r]$ that contains the origin in its interior. For example, the exponents $n_k = 2k$ satisfy the divergence criterion but linear combinations of the monomials x^{2k} will approximate only the even functions.

Exercises

1. Use induction to calculate the Vandermonde determinant

$$\begin{vmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{vmatrix} = \prod_{j<k} (x_k - x_j) .$$

Suggestion. Moving from right to left, subtract x_n times the kth column from the $(k+1)$st column, then remove common factors and reduce the problem to the calculation of a Vandermonde determinant with n replaced by $n - 1$.

2. (*a*) For $m, n = 1, 2, \ldots$, derive the trigonometric identity

$$\cos(n+m)\theta + \cos(n-m)\theta = 2\cos m\theta \cos n\theta\,.$$

(*b*) Show that

$$\int_{-\pi}^{\pi} \cos m\theta\, \cos n\theta\, d\theta = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n > 0\,. \end{cases}$$

(*c*) Show that the Chebyshev polynomials have the orthogonality property

$$\int_{-1}^{1} T_m(x)\, T_n(x)\, \frac{dx}{\sqrt{1-x^2}} = \begin{cases} 0 & \text{if } m \neq n \\ \pi/2 & \text{if } m = n\,. \end{cases}$$

3. (*a*) Show that the Chebyshev polynomials satisfy the differential equation

$$(1-x^2)T_n''(x) - xT_n'(x) + n^2 T_n(x) = 0\,.$$

(*b*) Set $W(x) = T_n'(x)T_m(x) - T_m'(x)T_n(x)$ and show that

$$(1-x^2)W'(x) - xW(x) + (n^2 - m^2)T_n(x)T_m(x) = 0\,.$$

(*c*) Conclude that

$$\int_{-1}^{1} T_n(x)\, T_m(x)\, \frac{dx}{\sqrt{1-x^2}} = 0 \qquad \text{if} \quad n \neq m\,.$$

4. Verify the explicit formula

$$T_n(x) = \frac{1}{2}\left\{\left(x + i\sqrt{1-x^2}\right)^n + \left(x - i\sqrt{1-x^2}\right)^n\right\}$$

for $-1 \leq x \leq 1$ and $n = 0, 1, 2, \ldots$.

5. (*a*) Show that $\sin(n+1)\theta / \sin\theta$ is a polynomial of degree n in $\cos\theta$. Denoted by $U_n(x)$, it is called a *Chebyshev polynomial of second kind*. Thus

$$\frac{\sin(n+1)\theta}{\sin\theta} = U_n(\cos\theta)\,.$$

Suggestion. Base an inductive argument on the identity

$$\sin(n+1)\theta = \sin n\theta \cos\theta + \cos n\theta \sin\theta\,.$$

(*b*) Calculate $U_0(x) = 1$, $U_1(x) = 2x$, $U_2(x) = 4x^2 - 1$. Verify the identity

$$\sin(n+1)\theta + \sin(n-1)\theta = 2\cos\theta \sin n\theta$$

and derive the recurrence relation

$$U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)\,, \qquad n = 1, 2, \ldots\,,$$

which is the same as for the Chebyshev polynomials $T_n(x)$.

(*c*) Verify the trigonometric identity

$$\cos(n-m)\theta - \cos(n+m)\theta = 2\sin m\theta \sin n\theta$$

and use it to show that

$$\int_{-\pi}^{\pi} \sin m\theta \sin n\theta \, d\theta = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n > 0. \end{cases}$$

(*d*) Derive the orthogonality relation

$$\int_{-1}^{1} U_m(x)U_n(x)\sqrt{1-x^2}\, dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi/2 & \text{if } m = n > 0. \end{cases}$$

6. (*a*) Show that the Chebyshev polynomials of second kind satisfy the differential equation

$$(1-x^2)U_n''(x) - 3xU_n'(x) + n(n+2)U_n(x) = 0\,.$$

(*b*) Use the differential equation to derive the orthogonality relation

$$\int_{-1}^{1} U_m(x)U_n(x)\sqrt{1-x^2}\, dx = 0\,, \qquad m \neq n\,.$$

7. (*a*) Show that $U_n(1) = n+1$, $n = 0, 1, 2, \ldots$.

(*b*) Show that $T_n'(x) = nU_{n-1}(x)$, $n = 1, 2, \ldots$.

(*c*) For any polynomial P of degree $\leq n$, *Markov's inequality* asserts that

$$\max_{-1\leq x\leq 1} |P'(x)| \leq n^2 \max_{-1\leq x\leq 1} |P(x)|\,.$$

Give an example to show that the constant n^2 cannot be replaced by any smaller constant.

8. Show that the polynomials $T_n(x)$ and $U_n(x)$ are even functions when n is even and odd functions when n is odd.

9. Apply the Weierstrass approximation theorem to show that if a function f has a continuous derivative of order n on an interval $[a, b]$, then for each $\varepsilon > 0$ there is a polynomial P such that

$$|f(x) - P(x)| < \varepsilon\,, \quad |f'(x) - P'(x)| < \varepsilon\,, \quad \ldots\,, \quad |f^{(n)}(x) - P^{(n)}(x)| < \varepsilon$$

for all $x \in [a, b]$.

10. Suppose f is continuous on $[a,b]$ and has the property

$$\int_a^b x^n f(x)\,dx = 0\,, \qquad n = 0,1,2,\ldots\,.$$

Prove that $f(x) \equiv 0$ on $[a,b]$.

Hint. Show by the Weierstrass approximation theorem that $\int_a^b f(x)^2\,dx = 0$.

11. Let $f(x)$ be continuous on the interval $[0,1]$. Show that

$$\lim_{n\to\infty} \sqrt{n} \int_0^1 \frac{f(x)}{1+nx^2}\,dx = \frac{\pi}{2} f(0)\,.$$

Suggestion: First let $f(x) = x^k$, $k = 0,1,2,\ldots$.

12. If f is continuous in the interval $[0,1]$, show that its Bernstein polynomials satisfy

$$\max_{0\le x\le 1} |B_n(x)| \le \max_{0\le x\le 1} |f(x)|\,, \qquad n = 1,2,\ldots\,.$$

13. For $m = 0,1,2,\ldots$ let

$$B_n(x;m) = \sum_{k=0}^{n} \binom{n}{k} \left(\frac{k}{n}\right)^m x^k (1-x)^{n-k}$$

denote the nth Bernstein polynomial of the function $f(x) = x^m$.

(a) Show that

$$B_n(x;0) = 1\,, \quad B_n(x;1) = x\,, \quad B_n(x;2) = x^2 + \frac{1}{n}x(1-x)\,.$$

In particular, $B_1(x;2) = x$.

(b) Show that

$$B_n(x;3) = \frac{1}{n^2}\left\{(n-1)(n-2)x^3 + 3(n-1)x^2 + x\right\}\,.$$

In particular, $B_1(x;3) = x$ and $B_2(x;3) = \frac{1}{4}(3x^2 + x)$.

Hint. Write $k^3 = k(k-1)(k-2) + 3k^2 - 2k$.

(c) Show by induction that $B_n(x;m)$ has degree n for $n \le m$ and degree m for $n > m$.

(d) Generalize (c) by showing that if f is any polynomial of degree m, then its Bernstein polynomial $B_n(x)$ has degree n for $n \le m$ and degree $\le m$ for $n > m$.

14. Let $B_n(x;f)$ be the nth Bernstein polynomial of a function f defined on $[0,1]$.

(a) Perform a calculation to derive the formula

$$B_n'(x;f) = n\sum_{k=0}^{n-1}\binom{n-1}{k}\left\{f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right)\right\}x^k(1-x)^{n-1-k},$$

so that $B_n'(x;f) = B_{n-1}(x;g_n)$, where g_n is the function defined by

$$g_n(x) = n\left\{f\left(\frac{n-1}{n}\,x + \frac{1}{n}\right) - f\left(\frac{n-1}{n}\,x\right)\right\}.$$

(b) If f has a continuous derivative in $[0,1]$, show that $g_n(x)$ converges uniformly to $f'(x)$ as $n \to \infty$.

(c) Deduce that $|B_{n-1}(x;g_n) - B_{n-1}(x;f')| \to 0$ uniformly in $[0,1]$.

Hint. Apply the result of Exercise 12.

(d) Conclude that if f is continuously differentiable, then $B_n'(x;f) \to f'(x)$ uniformly in $[0,1]$.

*(e) By a similar method, prove more generally that if f has a continuous derivative of order ν, then $B_n^{(\nu)}(x;f) \to f^{(\nu)}(x)$ as $n \to \infty$, uniformly in $[0,1]$.

Reference: Lorentz [**6**].

15. Let I^n denote the closed unit cube in $\mathbb{R}^n$, consisting of all points $(x_1, x_2, \ldots, x_n)$ with $0 \le x_j \le 1$ for $j = 1, 2, \ldots, n$. If a function $f(x_1, x_2, \ldots, x_n)$ is continuous on I^n, show that it can be uniformly approximated on I^n by polynomials; that is, by finite linear combinations of the monomials $x_1^{k_1}x_2^{k_2}\cdots x_n^{k_n}$.

16. Apply the Stone–Weierstrass theorem to prove the trigonometric form of the Weierstrass theorem. Specifically, prove that if $f(x)$ is continuous on $\mathbb{R}$ and $f(x+2\pi) = f(x)$ for all $x \in \mathbb{R}$, then for each $\varepsilon > 0$ there is a trigonometric polynomial

$$P(x) = a_0 + \sum_{k=1}^{n}(a_k\cos kx + b_k \sin kx)$$

such that $|f(x) - P(x)| < \varepsilon$ for all $x \in \mathbb{R}$.

17. Let f be continuous on the interval $[0, 2\pi]$ with $f(0) = f(2\pi)$, and form the *Poisson integral*

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi} \frac{1-r^2}{1-2r\cos(\theta-t)+r^2}\, f(t)\, dt\,, \qquad 0 \le r < 1\,.$$

Show that $u(r,\theta) \to f(\theta)$ as $r \to 1$, uniformly for $\theta \in [0, 2\pi]$. Give two proofs, one patterned after Landau's proof of Weierstrass's theorem, the other invoking the trigonometric form of Weierstrass's theorem.

18. Carry out the details in the deduction of Corollary 1 from Pál's theorem (Theorem 8).

19. For each function f continuous on a closed bounded interval $[a, b]$ and for each $\varepsilon > 0$, show that there exists a polynomial Q with rational coefficients such that $|Q(x) - f(x)| < \varepsilon$ for all $x \in [a, b]$.

20. For each function f continuous on a closed bounded interval $[a, b]$ and for each $\varepsilon > 0$, show that there exist polynomials P and Q such that $P(x) < f(x) < Q(x)$ and $Q(x) - P(x) < \varepsilon$ for all $x \in [a, b]$.

21. Show directly, without appeal to the Müntz–Szász theorem, that for each integer $m > 0$, every continuous function can be approximated uniformly on the interval $[0, 1]$ by polynomials of the form

$$P(x) = c_0 + c_1 x^{n_1} + \cdots + c_k x^{n_k}\,, \qquad \text{where} \quad n_k = km \quad \text{for} \quad k = 1, 2, \ldots.$$

Hint. If $f(x)$ is continuous on $[0, 1]$, so is $g(t) = f(t^{1/m})$.

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Chapter 7
Tauberian Theorems

It makes no sense to speak of the sum of a divergent infinite series. Nevertheless, a series that is not "too badly divergent" can be assigned a generalized sum in a variety of natural ways. We have already encountered this notion of summability in connection with Abel's theorem, which asserts that every convergent series is Abel summable to its ordinary sum. More generally, an Abelian theorem is any statement to the effect that a method of summability assigns to each convergent series its ordinary sum. A Tauberian theorem goes in the opposite direction and asserts that every summable series which is not too badly divergent is actually convergent. In this chapter we develop some elegant and important Tauberian theorems that will find application to other topics later in the book.

7.1. Summation of divergent series

According to Abel's theorem, if an infinite series $\sum_{n=0}^{\infty} a_n$ converges and has sum s, then the power series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges for $|x| < 1$ and $f(x) \to s$ as $x \to 1-$. The theorem was originally viewed as a device for evaluating the sums of convergent series such as

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \log 2 \qquad \text{and}$$
$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}\,.$$

However, the possible existence of the Abel limit when a series diverges suggests a natural way to assign it a generalized sum. For example, the series $\sum_{n=0}^{\infty}(-1)^n$ is divergent, but

$$f(x) = \sum_{n=0}^{\infty}(-1)^n x^n = \frac{1}{1+x} \to \frac{1}{2} \qquad \text{as} \quad x \to 1-\,,$$

so we may say that the series is *Abel summable* to the sum $\frac{1}{2}$. Observe that the extended notion of sum retains its linearity. In other words, if $\sum a_n$ is Abel summable to A and $\sum b_n$ is Abel summable to B, then $\sum(a_n + b_n)$ is summable to $A + B$ and $\sum ca_n$ is summable to cA for any constant c. Also, Abel's theorem guarantees that the Abel sum of a convergent series exists and is equal to the ordinary sum.

A similar technique for summation of divergent series can be based on the averages of partial sums. Let

$$s_n = a_0 + a_1 + \cdots + a_n$$

denote the partial sums of the series, and let

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1}$$

denote their arithmetic means, also known as the *Cesàro means*. The series $\sum a_n$ is said to be *Cesàro summable* to the sum σ if if $\sigma_n \to \sigma$ as $n \to \infty$. Recall (*cf.* Chapter 1, Exercise 13) that $\sigma_n \to s$ whenever $s_n \to s$, but the sequence $\{\sigma_n\}$ may converge when $\{s_n\}$ does not. For instance, the divergent series $\sum_{n=0}^{\infty}(-1)^n$ has partial sums $s_n = 1$ when n is even and $s_n = 0$ when n is odd, so it is Cesàro summable to the sum $\frac{1}{2}$, which is the same as its Abel sum. The series $\sum_{n=1}^{\infty}(-1)^{n+1}n$ is not Cesàro summable, but is Abel summable to $\frac{1}{4}$. (See Exercise 1.)

With this meager evidence in hand, one may suspect that Abel summation is more powerful than Cesàro summation, which is indeed the case. In 1880, Georg Frobenius [**2**] strengthened Abel's theorem by showing that a series is Abel summable if it is Cesàro summable. Here is a more precise statement, in notation introduced above.

Frobenius' Theorem. *If $\sigma_n \to \sigma$ as $n \to \infty$, then the power series $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges for $|x| < 1$ and $f(x) \to \sigma$ as $x \to 1-$.*

Proof. Two summations by parts give

$$\begin{aligned}\sum_{n=0}^{\infty} a_n x^n &= a_0 + \sum_{n=1}^{\infty}(s_n - s_{n-1})x^n = (1-x)\sum_{n=0}^{\infty} s_n x^n \\ &= (1-x)^2 \sum_{n=0}^{\infty}(s_0 + s_1 + \cdots + s_n)x^n = (1-x)^2 \sum_{n=0}^{\infty}(n+1)\sigma_n x^n\,.\end{aligned}$$

In view of the identity

$$\sum_{n=0}^{\infty}(n+1)x^n = \frac{1}{(1-x)^2}\,,$$

this may be written as

$$\sum_{n=0}^{\infty} a_n x^n - \sigma = (1-x)^2 \sum_{n=0}^{\infty}(n+1)(\sigma_n - \sigma)x^n\,.$$

But $\sigma_n \to \sigma$, so for each $\varepsilon > 0$ there is an integer N such that

$$|\sigma_n - \sigma| < \varepsilon \qquad \text{for all} \quad n > N\,.$$

Then for $0 < x < 1$,

$$\begin{aligned}\left|\sum_{n=0}^{\infty}(n+1)(\sigma_n-\sigma)x^n\right| &\le \sum_{n=0}^{N}(n+1)|\sigma_n-\sigma|x^n + \sum_{n=N+1}^{\infty}(n+1)|\sigma_n-\sigma|x^n \\ &\le \sum_{n=0}^{N}(n+1)|\sigma_n-\sigma| + \frac{\varepsilon}{(1-x)^2}\,,\end{aligned}$$

and so

$$\left|\sum_{n=0}^{\infty} a_n x^n - \sigma\right| \le (1-x)^2 \sum_{n=0}^{N}(n+1)|\sigma_n-\sigma| + \varepsilon < 2\varepsilon$$

when x is sufficiently close to 1. In other words, $f(x) \to \sigma$ as $x \to 1-$. □

Many other summation procedures have been introduced, and the corresponding "Abelian theorems" proved, asserting that whenever a series is convergent or summable by some method, it must be summable to the same sum by another more powerful method. For instance, a series is said to be *Borel summable* if the limit

$$\lim_{x\to\infty}\ e^{-x}\sum_{n=0}^{\infty}\frac{1}{n!}\,s_n x^n$$

exists. The method is named for Émile Borel, who introduced it in 1899 and pointed out the corresponding Abelian theorem, that a convergent series is Borel summable to its ordinary sum (*cf.* Exercise 2). Borel summability arises naturally in complex function theory, especially in problems of analytic continuation.

To give one more example, a series is said to be *Lambert summable* if the limit

$$\lim_{x\to 1-} \sum_{n=0}^{\infty} a_n L(x^n)$$

exists, where

$$L(x) = \frac{x \log(1/x)}{1-x} \qquad \text{for} \quad 0 < x < 1\,, \quad \text{and} \quad L(1) = 1\,,$$

is known as the *Lambert kernel*. Note that $L(x) \to 1$ as $x \to 1-$. Lambert summability has important applications to number theory and is involved in proofs of the prime number theorem.

7.2. Tauber's theorem

An infinite series may be summable by various methods and yet fail to converge. We have seen elementary examples of divergent series that are Cesàro or Abel summable. In the converse direction, however, there is a general principle that if a series is summable by some method and is not "too badly divergent", then it is actually convergent. The first result of this type was found in 1897 by Alfred Tauber (1866–1942), an Austrian mathematician who later specialized in actuarial mathematics. Tauber [**12**] showed that the converse of Abel's theorem is valid under the additional hypothesis that $na_n \to 0$.

Tauber's Theorem. *Suppose $na_n \to 0$ as $n \to \infty$, so that the power series*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges in the interval $(-1, 1)$. If $f(x) \to A$ as $x \to 1-$, then the infinite series $\sum_{n=0}^{\infty} a_n$ converges to the sum A.

Briefly, Tauber's theorem asserts that if a series is Abel summable and its terms satisfy the additional condition $na_n \to 0$, then the series is convergent. The theorem uncovers a remarkable phenomenon. Not only does the Abel method fail to sum series that diverge too rapidly, but it also fails to sum series whose divergence is too slow. For example, if $|a_n| \le 1/(n \log n)$, Tauber's theorem tells us that the series $\sum a_n$ is not Abel summable unless it is convergent. The same is true if $|a_n| \le 1/n$, but the proof is based on a stronger form of Tauber's theorem requiring only that the sequence $\{na_n\}$ be bounded. More about this later.

Proof of Tauber's theorem. With the notation $s_n = \sum_{k=0}^{n} a_k$ for the partial sums, we can write

$$s_n - f(x) = \sum_{k=1}^{n} a_k(1 - x^k) - \sum_{k=n+1}^{\infty} a_k x^k \,.$$

Taking $0 < x < 1$, we apply the inequality

$$1 - x^k = (1 - x)(1 + x + \cdots + x^{k-1}) \le k(1 - x)$$

to conclude that

$$|s_n - f(x)| \le (1 - x) \sum_{k=1}^{n} k|a_k| + \sum_{k=n+1}^{\infty} |a_k| x^k \,.$$

Now let $\varepsilon > 0$ be given. By hypothesis, we may choose n large enough that $k|a_k| < \varepsilon$ for all $k > n$. Then

$$\sum_{k=n+1}^{\infty} |a_k| x^k < \varepsilon \sum_{k=n+1}^{\infty} \frac{1}{k} x^k < \frac{\varepsilon}{n} \sum_{0}^{\infty} x^k = \frac{\varepsilon}{n(1 - x)} \,.$$

Now choose $x = x_n = 1 - \frac{1}{n}$, so that $1 - x_n = \frac{1}{n}$ and $x_n \to 1$ as $n \to \infty$. This gives the estimate

$$|s_n - f(x_n)| \le \frac{1}{n} \sum_{k=1}^{n} k|a_k| + \varepsilon$$

for n sufficiently large. But since $k|a_k| \to 0$, it follows that the arithmetic means

$$\frac{1}{n} \sum_{k=1}^{n} k|a_k|$$

also tend to 0 as $n \to \infty$. Therefore, $|s_n - f(x_n)| < 2\varepsilon$ for all n sufficiently large. In view of the hypothesis that $f(x) \to A$ as $x \to 1-$, this proves that $s_n \to A$. □

7.3. Theorems of Hardy and Littlewood

Beginning around 1910, the two British analysts G. H. Hardy (1877–1947) and J. E. Littlewood (1885–1977) began a series of investigations initially inspired by Tauber's theorem. First Hardy [**4**] proved an analogue of Tauber's theorem by postulating the more restrictive Cesàro summability instead of Abel summability but requiring only that the sequence $\{na_n\}$ be bounded. His result can be stated as follows.

Hardy's Theorem. *If the infinite series $\sum_{n=0}^{\infty} a_n$ is Cesàro summable and $\{na_n\}$ is bounded, then the series is convergent.*

Hardy conjectured that Tauber's theorem remains valid if the condition $na_n \to 0$ is weakened to the boundedness of $\{na_n\}$. His Cambridge colleague Littlewood [**11**] then verified the conjecture by proving the following theorem, which strengthens the theorems of both Tauber and Hardy.

Littlewood's Theorem. *Suppose the sequence $\{na_n\}$ is bounded, so that the power series*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges in the interval $(-1, 1)$. If $f(x) \to A$ as $x \to 1-$, then the infinite series $\sum_{n=0}^{\infty} a_n$ converges to the sum A.

This work marked the beginning of the famous Hardy–Littlewood collaboration, which extended over a period of 35 years and was one of the most fruitful collaborations in the history of mathematics. Together these two mathematicians made seminal contributions to a variety of subjects including number theory, Fourier series, and functions of a complex variable. They also developed a series of results analogous to Tauber's theorem, which they called "Tauberian theorems". A *Tauberian theorem* is the converse of an Abelian theorem under a supplementary growth condition called a *Tauberian condition*. For instance, Littlewood's Tauberian theorem asserts that the converse of Abel's theorem is true under the Tauberian condition that $\{na_n\}$ is bounded.

In 1914, Hardy and Littlewood together found a Tauberian theorem that goes from Abel to Cesàro summability and therefore provides a partial converse of Frobenius' theorem.

Hardy–Littlewood Theorem. *If $f(x) = \sum_{n=0}^{\infty} a_n x^n \to A$ as $x \to 1-$ and $s_n \geq 0$ for all n, then $\sigma_n \to A$ as $n \to \infty$.*

It is easy to see that the Hardy–Littlewood theorem remains valid more generally if $s_n \geq -C$ for some constant C, which is certainly true if the partial sums s_n are bounded. Hardy and Littlewood also found that the Tauberian condition in Littlewood's theorem, the boundedness of $\{na_n\}$, can be relaxed to a one-sided condition $a_n \geq -C/n$.

The first proofs of Littlewood's theorem and of the Hardy–Littlewood theorem were similar and were much more difficult than those of Tauber's or Hardy's theorem. For a long time no simplifications were found and these results were considered to be quite deep. Then in 1930, Karamata [**8**] found a clever and surprisingly simple approach based on the Weierstrass

approximation theorem. Karamata's elegant proof will be discussed in the next section. His strategy was to prove the Hardy–Littlewood theorem, then to deduce Littlewood's theorem from it. Littlewood's original approach was to prove Cesàro summability and then appeal to Hardy's theorem to conclude that the series converges. Some years later, Wielandt [**14**] found a simple refinement of Karamata's method that avoids the detour through Cesàro summability and proves Littlewood's theorem directly. The details are described in the next section.

7.4. Karamata's proof

We now turn to Karamata's proof of the Hardy–Littlewood and Littlewood Tauberian theorems. Our treatment is adapted from the books of Titchmarsh [**13**] and Korevaar [**10**]. Karamata's method is based on the Weierstrass approximation theorem, which can be invoked to establish the following lemma.

Lemma. *Let $g(x)$ be continuous on the interval $[0,1]$ except for a possible jump-discontinuity at a point $c \in (0,1)$. Then for each $\varepsilon > 0$ there exist polynomials $P(x)$ and $Q(x)$ such that $P(x) < g(x) < Q(x)$ and*

$$\int_0^1 \big[g(x) - P(x)\big]\,dx < \varepsilon \qquad \int_0^1 \big[Q(x) - g(x)\big]\,dx < \varepsilon\,.$$

Proof. Suppose that $g(x)$ has a jump-discontinuity at $c \in (0,1)$, and let $g(c+)$ and $g(c-)$ denote the right- and left-hand limits. Suppose without loss of generality that $g(c-) \le g(c+)$. For fixed $\delta \in (0,c)$, let $\ell(x)$ be the linear function such that

$$\ell(c-\delta) = g(c-\delta) + \varepsilon/2\,, \qquad \ell(c) = g(c+) + \varepsilon/2\,.$$

Define

$$\phi(x) = \begin{cases} g(x) + \varepsilon/2 & \text{if } \ 0 \le x < c-\delta \ \text{ or } \ c < x \le 1 \\ \max\{\ell(x),\, g(x) + \varepsilon/2\} & \text{if } \ c-\delta \le x \le c\,. \end{cases}$$

Then $\phi(x)$ is continuous in $[0,1]$ and $\phi(x) \ge g(x) + \varepsilon/2$. Choose a polynomial $Q(x)$ such that $|Q(x) - \phi(x)| < \varepsilon/2$ for all $x \in [0,1]$. Then $Q(x) > g(x)$ and

$$\int_0^1 \big[Q(x) - g(x)\big]\,dx < \varepsilon$$

if δ is sufficiently small. A similar construction produces a polynomial $P(x) < g(x)$ with $\int_0^1 [g(x) - P(x)]\,dx < \varepsilon$. □

With the lemma in hand, we first prove the Hardy–Littlewood theorem, which is a somewhat easier than Littlewood's theorem.

Proof of Hardy–Littlewood theorem. By hypothesis,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = (1-x)\sum_{n=0}^{\infty} s_n x^n \to A \qquad \text{as } x \to 1- \ .$$

We claim that this implies, more generally, that

$$\lim_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, p(x^n) = A\int_0^1 p(t)\,dt \tag{1}$$

for every polynomial $p(x)$. To see this, it suffices to consider monomials $p(x) = x^k$. Then the left-hand side is

$$\begin{aligned}(1-x)\sum_{n=0}^{\infty} s_n\, x^{n+kn} &= \frac{1-x}{1-x^{k+1}}\left\{(1-x^{k+1})\sum_{n=0}^{\infty} s_n (x^{k+1})^n\right\} \\ &\to \frac{A}{k+1} = A\int_0^1 p(t)\,dt \qquad \text{as } x \to 1-,\end{aligned}$$

as claimed. Invoking the lemma, we can now infer from (1) that

$$\lim_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, g(x^n) = A\int_0^1 g(t)\,dt \tag{2}$$

if g is continuous in $[0,1]$ except for a possible jump-discontinuity. It is here that the hypothesis $s_n \geq 0$ is needed. To carry out the details, let $\varepsilon > 0$ be given and choose polynomials P and Q as in the lemma. Then

$$\sum_{n=0}^{\infty} s_n\, x^n\, P(x^n) \leq \sum_{n=0}^{\infty} s_n\, x^n\, g(x^n) \leq \sum_{n=0}^{\infty} s_n\, x^n\, Q(x^n)\,, \qquad 0 < x < 1\,.$$

Since $s_n \geq 0$ implies $A \geq 0$, it follows that

$$\begin{aligned}\limsup_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, g(x^n) &\leq \lim_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, Q(x^n) \\ &= A\int_0^1 Q(t)\,dt \leq A\int_0^1 g(t)\,dt + A\varepsilon\,.\end{aligned}$$

Letting $\varepsilon \to 0$, we conclude that

$$\limsup_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, g(x^n) \leq A\int_0^1 g(t)\,dt\,.$$

In a similar way we find that

$$\liminf_{x\to 1-} (1-x)\sum_{n=0}^{\infty} s_n\, x^n\, g(x^n) \ge A\int_0^1 g(t)\,dt\,.$$

Combining the last two inequalities, we arrive at a proof of (2).

Now choose

$$g(t) = \begin{cases} 0\,, & 0 \le t < 1/e \\ 1/t\,, & 1/e \le t \le 1\,. \end{cases}$$

Then

$$\int_0^1 g(t)\,dt = \int_{1/e}^1 1/t\,dt = 1\,.$$

Let $x_N = e^{-1/N}$ and observe that $x_N^n \ge 1/e$ if and only if $n \le N$, so that

$$\sum_{n=0}^{\infty} s_n\, x_N^n\, g(x_N^n) = \sum_{n=0}^{N} s_n = (N+1)\sigma_N\,.$$

The relation (2) now shows that

$$(N+1)(1-x_N)\sigma_N = (1-x_N)\sum_{n=0}^{\infty} s_n\, x_N^n\, g(x_N^n) \to A\int_0^1 g(t)\,dt = A$$

as $N\to\infty$, since $x_N \to 1$. But

$$(N+1)(1-x_N) = (N+1)(1-e^{-1/N}) \to 1 \qquad \text{as } N\to\infty\,,$$

so this proves that $\sigma_N \to A$. □

Proof of Littlewood's theorem. For each polynomial

$$P(x) = \sum_{k=1}^{m} b_k x^k \qquad \text{with } \ P(0) = 0\,,$$

the hypothesis that $f(x)\to A$ implies

$$\sum_{n=0}^{\infty} a_n P(x^n) = \sum_{k=1}^{m} b_k \sum_{n=0}^{\infty} a_n x^{kn} \to A\sum_{k=1}^{m} b_k = P(1)A \tag{3}$$

as $x\to 1-$. Now choose the "cutoff function"

$$g(t) = \begin{cases} 0\,, & 0 \le t < 1/e \\ 1\,, & 1/e \le t \le 1\,, \end{cases}$$

so that

$$s_N = \sum_{n=0}^{N} a_n = \sum_{n=0}^{\infty} a_n\, g(x_N^n)\,, \qquad \text{where} \quad x_N = e^{-1/N}\,.$$

To prove that $s_N \to A$ as $N \to \infty$, it will suffice to show that

$$\sum_{n=0}^{\infty} a_n\, g(x^n) \to g(1)A = A \qquad \text{as} \quad x \to 1-\ .$$

In order to show that

$$\limsup_{x\to 1-} \sum_{n=0}^{\infty} a_n\, g(x^n) \le A\,, \tag{4}$$

we will apply the relation (3) to a polynomial P with the properties $P(0) = 0$, $P(1) = 1$, and $P(t) \ge g(t)$ for $0 \le t \le 1$. For this purpose we choose a polynomial Q with

$$Q(t) \ge \frac{g(t) - t}{t(1-t)} = h(t)\,, \qquad 0 < t < 1\,,$$

which gives a polynomial $P(t) = t + t(1-t)Q(t)$ with the required properties. The Tauberian condition $|na_n| \le C$ implies that $na_n \ge -C$, so that

$$\begin{aligned}
&\sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n) = -\sum_{n=1}^{\infty} a_n \big[P(x^n) - g(x^n)\big] \\
&\le C \sum_{n=1}^{\infty} \frac{1}{n}\big[P(x^n) - g(x^n)\big] \ \le\ C \sum_{n=1}^{\infty} \frac{1-x}{1-x^n}\big[P(x^n) - g(x^n)\big] \\
&= C(1-x)\sum_{n=1}^{\infty} \phi(x^n)\,, \qquad \text{where} \quad \phi(t) = \frac{P(t) - g(t)}{1-t}\,.
\end{aligned}$$

Here we have used the elementary inequality

$$\frac{1-x^n}{1-x} = 1 + x + \cdots + x^{n-1} \le n \qquad \text{for} \quad 0 \le x < 1\,.$$

We claim now that

$$\lim_{x\to 1-} (1-x) \sum_{n=1}^{\infty} \phi(x^n) = \int_0^1 \frac{\phi(t)}{t}\, dt\,,$$

since $\phi(t)$ has a continuous extension to the interval $[0,1]$ except for the jump-discontinuity at $t = 1/e$. Indeed, for any fixed $x \in (0,1)$, the integral is approximated by its Riemann sum

$$\sum_{n=1}^{\infty} \frac{\phi(x^n)}{x^n}(x^n - x^{n+1}) = (1-x)\sum_{n=1}^{\infty} \phi(x^n)\,,$$

which tends to the integral as $x \to 1-$. But

$$\int_0^1 \frac{\phi(t)}{t}\,dt = \int_0^1 \frac{\big[P(t)-t\big] - \big[g(t)-t\big]}{t(1-t)}\,dt = \int_0^1 \big[Q(t)-h(t)\big]\,dt\,.$$

Since $Q(t) \geq h(t)$, the last integral is positive, and in view of the lemma it can be made arbitrarily small by suitable choice of the polynomial Q. Putting everything together and recalling from (3) that

$$\lim_{x\to 1-} \sum_{n=0}^{\infty} a_n P(x^n) = A\,,$$

we conclude that (4) holds.

A similar argument shows that

$$\liminf_{x\to 1-} \sum_{n=0}^{\infty} a_n g(x^n) \geq A\,. \tag{5}$$

For this we choose $P(t) = t + t(1-t)Q(t)$, where Q is a polynomial with

$$Q(t) \leq h(t) = \frac{g(t)-t}{t(1-t)} \qquad \text{for} \quad 0 < t < 1\,.$$

Then $P(0) = 0$, $P(1) = 1$, and $P(t) \leq g(t)$ for $0 \leq t \leq 1$. Hence the Tauberian condition $na_n \geq -C$ gives

$$\sum_{n=0}^{\infty} a_n g(x^n) - \sum_{n=0}^{\infty} a_n P(x^n) \geq -C\sum_{n=1}^{\infty} \frac{1}{n}\big[g(x^n) - P(x^n)\big] \geq C(1-x)\sum_{n=1}^{\infty} \phi(x^n)\,,$$

since $1 - x^n \leq n(1-x)$ and now

$$\phi(t) = \frac{P(t)-g(t)}{1-t} \leq 0 \qquad \text{for} \quad 0 < t < 1\,.$$

Consequently, we may conclude as before that

$$\liminf_{x\to 1-} \sum_{n=0}^{\infty} a_n\, g(x^n) \geq A + C\int_0^1 \big[Q(t)-h(t)\big]\,dt\,.$$

Another appeal to the lemma shows that the last integral can be made arbitrarily close to zero with suitable choice of the polynomial Q, which gives (5). Combining (4) and (5), we see that

$$\lim_{x\to 1-} \sum_{n=0}^{\infty} a_n\, g(x^n) = A\,.$$

As previously noted, this implies that $s_n \to A$, and so the theorem is proved. It should be observed that the proof used only the weaker Tauberian condition $na_n \geq -C$. □

Jovan Karamata (1902–1967) was a Yugoslavian, born in Zagreb (Croatia) and trained in mathematics at the University of Novi Sad (Serbia) and at Belgrade University, where he received a doctoral degree in 1926. Four years later his two-page paper [**8**] appeared, outlining his totally new, relatively simple approach to Tauberian theorems. The paper created an immediate sensation and brought him international recognition. Later he made other significant contributions to Tauberian theory (*cf.* Korevaar [**10**]). He was a professor at Belgrade University until 1951, then spent the rest of his career in Switzerland at the University of Geneva.

7.5. Hardy's power series

The following problem appeared in the puzzles column of *Emissary*, a newsletter of the Mathematical Sciences Research Institute [**1**].

Problem. *For positive real x less than 1, define*

$$f(x) = x - x^2 + x^4 - x^8 + x^{16} - \dots\,.$$

Does $f(x)$ have a limit as x approaches 1 from below? If so, what is the limit?

Before considering the problem in earnest, let us make a few simple observations. First of all, by pairing terms one sees that

$$f(x) = (x - x^2) + (x^4 - x^8) + \cdots > 0$$

and

$$f(x) = x - (x^2 - x^4) - (x^8 - x^{16}) - \cdots < x$$

for $0 < x < 1$. In particular, $f(x)$ is bounded in the interval $[0, 1]$. The identity $f(x) = x - f(x^2)$ shows that if $f(x)$ has a limit as $x \to 1-$, the limit must be $\frac{1}{2}$. Iteration of the identity shows that $f(x^4) < f(x)$, which suggests that $f(x)$ is an increasing function in the interval $0 < x < 1$. If

so, then by the monotone boundedness theorem $f(x)$ does have a limit, and $f(x) \to \frac{1}{2}$ as $x \to 1-$. Figure 1 displays the graph of

$$\sum_{k=0}^{30} (-1)^k x^{2^k}, \qquad 0 < x < 1,$$

as generated by *Mathematica*. The figure appears to confirm that $f(x)$ increases to $\frac{1}{2}$ as $x \to 1-$.

However, the limit does not exist! Hardy [**3**] studied the series in 1907 and found by intricate analysis that as x tends to 1, the function $f(x)$ undergoes miniscule oscillations about the value $\frac{1}{2}$ and does not converge. A closer inspection of Figure 1 actually reveals the small oscillations. Figure 2 displays a graph of the same partial sum (up to $k = 30$), magnified to clarify the oscillations near the point $(1, \frac{1}{2})$.

In fact, the nonexistence of the limit is a direct consequence of a result of Hardy and Littlewood [**7**] known as the "high-indices" theorem, a remarkable Tauberian theorem for lacunary power series. Here is the statement.

High-Indices Theorem. *If* $f(x) = \sum_{k=1}^{\infty} a_k x^{n_k}$ *for* $0 < x < 1$, *where the exponents* n_k *are positive integers satisfying a condition of the form* $n_{k+1}/n_k \geq q > 1$, *and if* $f(x) \to A$ *as* $x \to 1-$, *then the infinite series* $\sum_{k=1}^{\infty} a_k$ *converges to* A.

Hardy and Littlewood obtained this result in 1926, relatively late in the development of basic Tauberian theory. Their discovery appears to have been inspired in part by the special power series (discussed above)

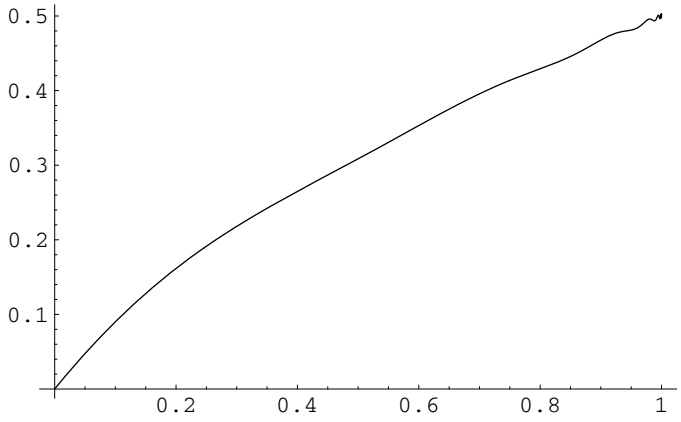


Figure 1. Hardy's sum for $0 \leq x \leq 1$.

that Hardy had studied in 1907. The most astounding feature of the high-indices theorem is that lacunarity alone serves as a Tauberian condition. For Hardy's power series we see that $n_k = 2^k$ is sufficiently lacunary, and the partial sums of the series $\sum a_k$ are alternately equal to 0 and 1, so the high-indices theorem shows that $f(x)$ cannot tend to a limit as $x \to 1-$.

A proof of the high-indices theorem is beyond the scope of this book, but can be found for instance in the book by Korevaar [**10**], p. 50 ff. Alternatively, we can appeal to the more elementary theorem of Hardy and Littlewood, as stated at the end of Section 7.3, to show that Hardy's sum does not tend to a limit as $x \to 1-$. For the function

$$f(x) = \sum_{n=1}^{\infty} a_n\, x^n = \sum_{k=0}^{\infty} (-1)^k x^{2^k}\,,$$

it is seen that all partial sums $s_n = a_1 + \cdots + a_n$ are either 0 or 1, so the hypothesis $s_n \geq 0$ of the Hardy-Littlewood theorem is clearly satisfied. Therefore, if we can show that the sequence $\{\sigma_n\}$ of Cesàro means does not converge, it will follow that $f(x)$ does not converge as $x \to 1-$. In fact, a direct calculation (*cf.* Exercise 3) reveals that

$$\liminf_{n\to\infty} \sigma_n = \tfrac{1}{3} \qquad \text{and} \qquad \limsup_{n\to\infty} \sigma_n = \tfrac{2}{3}\,,$$

so the series $\sum a_n$ is not Cesàro summable. Thus by the Hardy-Littlewood theorem, it is not Abel summable. In other words, $f(x)$ does not tend to a limit as $x \to 1-$.

A recent paper by Keating and Reade [**9**] offers further analysis of Hardy's power series and similar examples. There the authors apply the

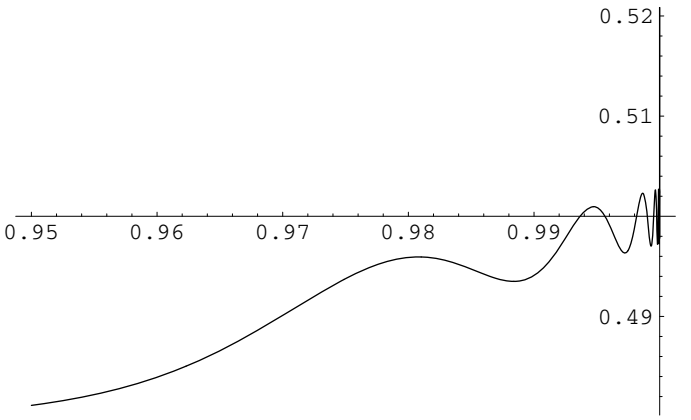


Figure 2. Hardy's sum for $0.95 \leq x \leq 1$.

Poisson summation formula of Fourier analysis (see Chapter 8) to prove that as $x \to 1-$, Hardy's function undergoes persistent oscillations about the value $\frac{1}{2}$ with approximate amplitude 0.00275, which is consistent with Figure 2.

Hardy's classic book [**5**] contains a wealth of information about Abelian and Tauberian theorems. Korevaar's more recent book [**10**] is a comprehensive reference for Tauberian theorems and their applications.

Exercises

1. Show that the series

$$\sum_{n=1}^{\infty}(-1)^{n+1}n = 1 - 2 + 3 - 4 + \dots$$

is not Cesàro summable, but is Abel summable to $\frac{1}{4}$.

2. Show that every convergent series is Borel summable to its ordinary sum. Specifically, show that if $s_n \to s$, then

$$e^{-x}\sum_{n=0}^{\infty}\frac{1}{n!}\,s_n\,x^n \to s \qquad \text{as} \quad x \to \infty\,.$$

3. Consider Hardy's power series

$$\sum_{n=1}^{\infty}a_n\,x^n = \sum_{k=0}^{\infty}(-1)^k x^{2^k}\,.$$

Thus $a_n = (-1)^k$ if $n = 2^k$ and $a_n = 0$ otherwise. Show directly that the series $\sum a_n$ is not Cesàro summable by computing

$$\liminf_{n\to\infty}\,\sigma_n = \tfrac{1}{3} \qquad \text{and} \qquad \limsup_{n\to\infty}\,\sigma_n = \tfrac{2}{3}\,.$$

4. Show that

$$\sum_{k=0}^{\infty}(-1)^k x^{k^2} \to \frac{1}{2} \qquad \text{as} \quad x \to 1-\,.$$

Suggestion. Show that the series $\sum a_n$ is Cesàro summable, where $a_n = (-1)^k$ if $n = k^2$ and $a_n = 0$ otherwise.

5. (*a*) Show directly, without appeal to a Tauberian theorem, that the infinite series $\sum_{n=1}^{\infty}\frac{1}{n}$ is not Cesàro summable.

(*b*) Show directly that the same series is not Abel summable.

6. Show that for $a_n \geq 0$, the series $\sum a_n$ is Abel summable only if it is convergent.

7. (*a*) Prove the Abelian theorem for Lambert summability. In other words, show that if

$$\sum_{n=0}^{\infty} a_n = s\,, \qquad \text{then} \qquad \lim_{x\to 1-} \sum_{n=0}^{\infty} a_n L(x^n) = s\,,$$

where $L(x)$ is the Lambert kernel (*cf.* Section 7.1).

Hint. Apply a summation by parts and observe that $L(x)$ increases from 0 to 1 in the interval $[0, 1]$.

(*b*) Make the change of variables $x = e^{-t}$ to recast the conclusion in the form

$$a_0 + \lim_{t\to 0+} \sum_{n=1}^{\infty} \frac{n a_n t}{e^{nt} - 1} = s\,.$$

8. Show that the series

$$\sum_{n=1}^{\infty} \cos n\theta \qquad \text{and} \qquad \sum_{n=1}^{\infty} \sin n\theta\,, \qquad 0 < \theta < 2\pi\,,$$

are Abel summable to $-\frac{1}{2}$ and $\frac{1}{2}\cot\frac{\theta}{2}$, respectively.

Hint. Consider real and imaginary parts of the geometric series $\sum_{n=1}^{\infty} z^n$, where $z = re^{i\theta}$.

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Chapter 8
Fourier Series

Power series expansions of functions are known to play a vital role in analysis. In this chapter we turn to another kind of expansion, a sum of sines and cosines known as a Fourier series. It is fundamentally different from a Taylor series expansion in that the coefficients are determined not by differentiation but by integration, exploiting a property of orthogonality; and the function to be represented need not be differentiable and may even have points of discontinuity. Here we develop criteria for convergence and Cesàro summability of Fourier series, with applications to specific examples. The discussion then shifts to a continuous analogue, the Fourier transform and its inversion, leading ultimately to the remarkable Poisson summation formula.

8.1. Physical origins

The theory of Fourier series has its roots in problems of mathematical physics, classical problems of wave propagation, heat conduction, electrostatic potentials, and the like. Consider for example the problem of calculating the steady-state distribution of temperature $u(r, \theta)$ in a circular metal plate of unit radius, given a boundary temperature $u(1, \theta) = f(\theta)$, where r and θ are standard polar coordinates. Here f is an "arbitrary" function, not necessarily continuous, with the property $f(\theta + 2\pi) = f(\theta)$. According to the laws of physics, the solution $u(r, \theta)$ will be a harmonic function; that is, a solution of the Laplace equation, which takes the form

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

in polar coordinates.

Now if $f(\theta) = \sin n\theta$ or $\cos n\theta$ for $n = 1, 2, \ldots$, this boundary-value problem can be solved by inspection. The solutions are $u(r, \theta) = r^n \sin n\theta$ and $r^n \cos n\theta$, respectively. But the Laplace equation is linear, so a principle of superposition applies. This means that for any constant c, the solution with boundary function $cf(\theta)$ is $cu(r, \theta)$, and if $v(r, \theta)$ is the solution corresponding to another boundary function $g(\theta)$, then the $u + v$ solves the problem for boundary function $f + g$.

All of this suggests that the boundary-value problem will be solved for an arbitrary periodic function $f(\theta)$ if that function can be expanded into an infinite series of the form

$$f(\theta) = \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n \cos n\theta + b_n \sin n\theta)\,, \tag{1}$$

for some coefficients a_n and b_n. Setting aside questions of convergence, we will then arrive at the solution

$$u(r, \theta) = \tfrac{1}{2}a_0 + \sum_{n=1}^{\infty}(a_n r^n \cos n\theta + b_n r^n \sin n\theta)\,.$$

An expansion of the form (1) is called a *Fourier series*, after the French mathematician and physicist Joseph Fourier (1768–1830), who conceived the method and developed it in *Théorie analytique de la chaleur*, a revolutionary monograph on heat conduction published in 1822. Fourier's arguments were not rigorous, and his ideas were not readily accepted. In particular, the notion that a discontinuous function could be represented as a sum of sines and cosines seemed inconceivable and was greeted with skepticism. Ultimately the paradoxes inherent in Fourier series played an important role in leading Cauchy and others to develop more precise concepts in mathematical analysis, thereby placing the entire apparatus on a more secure foundation.

Fourier series are the prototype for a variety of orthogonal expansions that arise in similar manner from problems of mathematical physics. For example, expansions into series of Bessel functions and Legendre polynomials will be encountered in Chapter 13 of this book.

Joseph Fourier had a colorful career that combined science with public service. His early work on the theory of algebraic equations was interrupted by the French Revolution of 1789. He then became embroiled in politics and was actually condemned to the guillotine at one time. After the Revolution he taught for a while at the new École Polytechnique in Paris, then was appointed as a scientific advisor in Napoleon's expedition to Egypt. Upon return to France, he became a civil administrator in Grenoble, during which

time (around 1805) he began the work on heat diffusion that would culminate in his famous book of 1822.

Since the time of Fourier, an extensive mathematical theory of Fourier series has emerged, with broad applications not only to mathematical physics, but to such fields as coding theory, signal transmission, data storage and retrieval, crystallography, medical imaging, and pure mathematics itself. The mathematical theory will be our focus in this chapter.

8.2. Orthogonality relations

The theory of Fourier series is based on orthogonality relations for sines and cosines. By analogy with vectors in Euclidean spaces, two functions f and g are said to be *orthogonal* over an interval $[a, b]$ if

$$\int_a^b f(x)g(x)\,dx = 0\,.$$

The fundamental relations are

$$\begin{aligned}
&\int_{-\pi}^{\pi} \cos nx\,dx = \int_{-\pi}^{\pi} \sin nx\,dx = 0\,, \qquad n = 1, 2, \dots\,,\\
&\int_{-\pi}^{\pi} \cos nx \sin mx\,dx = 0\,, \qquad n, m = 1, 2, \dots\,,\\
&\int_{-\pi}^{\pi} \cos nx \cos mx\,dx = \int_{-\pi}^{\pi} \sin nx \sin mx\,dx = 0\,, \qquad n \neq m\,,\\
&\int_{-\pi}^{\pi} \cos^2 nx\,dx = \int_{-\pi}^{\pi} \sin^2 nx\,dx = \pi\,, \qquad n = 1, 2, \dots\,.
\end{aligned} \tag{2}$$

They follow easily from the trigonometric identities

$$\begin{aligned}
2\cos nx \sin mx &= \sin(n+m)x - \sin(n-m)x\,,\\
2\cos nx \cos mx &= \cos(n+m)x + \cos(n-m)x\,,\\
2\sin nx \sin mx &= \cos(n-m)x - \cos(n+m)x\,.
\end{aligned}$$

Suppose now that for some coefficients a_k and b_k the trigonometric series

$$f(x) = \tfrac{1}{2}a_0 + \sum_{k=1}^{\infty}(a_k \cos kx + b_k \sin kx) \tag{3}$$

converges uniformly in the interval $[-\pi, \pi]$ to a sum $f(x)$. (The reason for the $\frac{1}{2}$ will become apparent shortly.) Because of the uniform convergence, the function f is continuous and we can integrate term by term to see that

$$\int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi}^{\pi} \tfrac{1}{2}a_0 = \pi a_0\,.$$

If we multiply by $\cos nx$ for some index $n = 1, 2, \dots$ and integrate term by term, we find

$$\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \sum_{k=1}^{\infty} a_k \int_{-\pi}^{\pi} \cos kx \cos nx\,dx + \sum_{k=1}^{\infty} b_k \int_{-\pi}^{\pi} \sin kx \cos nx\,dx$$
$$= \pi a_n + 0 = \pi a_n\,,$$

in view of the orthogonality relations (2). Similarly, multiplication by $\sin nx$ and integration gives

$$\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \pi b_n\,.$$

The numbers

$$(4)\qquad \begin{aligned} a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx\,, \qquad n = 0, 1, 2, \dots\,, \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx\,, \qquad n = 1, 2, \dots \end{aligned}$$

are called the *Fourier coefficients* of the function f, and the trigonometric series of the form (3) with these coefficients is known as the *Fourier series* of f. As our calculations show, the Fourier series of a continuous function f is the only trigonometric series of the form (3) that can converge uniformly to f in the interval $[-\pi, \pi]$. In particular, no function can have more than one uniformly convergent trigonometric series expansion.

A much more delicate question is whether the Fourier series of a given continuous periodic function will actually converge to the function, uniformly or even pointwise. Before turning to this important question, we shall consider a problem of best "mean-square" approximation.

8.3. Mean-square approximation

A function f is said to be *square-integrable* over an interval $[a, b]$ if it is Riemann integrable over $[a, b]$ and the integral $\int_a^b f(x)^2\,dx$ is finite. Suppose f is square-integrable over $[-\pi, \pi]$, and consider the following question. Among all trigonometric polynomials

$$T(x) = \tfrac{1}{2}c_0 + \sum_{k=1}^{n}(c_k \cos kx + d_k \sin kx)$$

of degree at most n, which one gives the best mean-square approximation to f? In other words, how should the coefficients c_k and d_k be chosen to minimize the integral

$$\int_{-\pi}^{\pi} [f(x) - T(x)]^2\,dx\ ?$$

To answer this question, we let a_k and b_k be the Fourier coefficients of f, as defined by (4), and write

(5)
$$\int_{-\pi}^{\pi}[f(x)-T(x)]^2\,dx = \int_{-\pi}^{\pi} f(x)^2\,dx - 2\int_{-\pi}^{\pi} f(x)T(x)\,dx + \int_{-\pi}^{\pi} T(x)^2\,dx$$
$$= \int_{-\pi}^{\pi} f(x)^2\,dx - \pi a_0 c_0 - 2\pi\sum_{k=1}^{n}(c_k a_k + d_k b_k) + \tfrac{1}{2}\pi c_0^2 + \pi\sum_{k=1}^{n}(c_k^2 + d_k^2)\,,$$

where the orthogonality relations (2) have been used to calculate the integral $\int_{-\pi}^{\pi} T(x)^2\,dx$. Now let

$$s_n(x) = \tfrac{1}{2}a_0 + \sum_{k=1}^{n}(a_k\cos kx + b_k\sin kx) \tag{6}$$

denote the nth partial sum of the Fourier series of f. With the choice of trigonometric polynomial $T(x) = s_n(x)$, the expression (5) reduces to

$$\int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx = \int_{-\pi}^{\pi} f(x)^2\,dx - \tfrac{1}{2}\pi a_0^2 - \pi\sum_{k=1}^{n}(a_k^2 + b_k^2)\,. \tag{7}$$

A straightforward calculation now leads to the identity

$$\int_{-\pi}^{\pi}[f(x)-T(x)]^2\,dx = \int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx + \tfrac{1}{2}\pi(a_0-c_0)^2 + \pi\sum_{k=1}^{n}\left\{(a_k-c_k)^2 + (b_k-d_k)^2\right\}\,.$$

From this it follows that

$$\int_{-\pi}^{\pi}[f(x)-T(x)]^2\,dx \geq \int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx\,,$$

with equality if and only if $c_0 = a_0$ and $c_k = a_k$, $d_k = b_k$ for $k = 1, 2, \ldots, n$. In other words, the Fourier polynomial s_n gives the best mean-square approximation to f among all trigonometric polynomials of degree less than or equal to n.

The identity (7) also shows that

$$\tfrac{1}{2}\pi a_0^2 + \pi\sum_{k=1}^{n}(a_k^2 + b_k^2) \leq \int_{-\pi}^{\pi} f(x)^2\,dx\,,$$

since $\int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx \geq 0$. Letting $n \to \infty$, one sees that the infinite series converges and that

$$\frac{1}{2}\pi a_0^2 + \pi\sum_{k=1}^{\infty}(a_k^2+b_k^2) \leq \int_{-\pi}^{\pi} f(x)^2\,dx\,. \tag{8}$$

The relation (8) is known as *Bessel's inequality.* An immediate corollary is that the Fourier coefficients a_k and b_k of a square-integrable function tend to zero as $n \to \infty$.

Actually, Bessel's inequality is valid for more general systems of orthogonal functions; no special properties of sines and cosines were used in its derivation. However, for the trigonometric system considered here, we can actually assert that

$$\frac{1}{2}\pi a_0^2 + \pi\sum_{k=1}^{\infty}(a_k^2+b_k^2) = \int_{-\pi}^{\pi} f(x)^2\,dx\,,$$

an identity known as *Parseval's relation.* This will follow from the equation (7) if we can show that

$$\lim_{n\to\infty}\int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx = 0\,. \tag{9}$$

For this purpose we will appeal to the trigonometric form of the Weierstrass approximation theorem, stated here for easy reference.

Weierstrass Approximation Theorem. *Let $f(x)$ be continuous on the interval $[-\pi,\pi]$, with $f(-\pi)=f(\pi)$. Then for each $\varepsilon>0$ there is a trigonometric polynomial $T(x)$ such that $|f(x)-T(x)|<\varepsilon$ for $-\pi\leq x\leq\pi$.*

The trigonometric form of the Weierstrass approximation theorem is closely related to the algebraic form, as presented in Chapter 6. A proof is outlined in Exercise 6. Another proof will be given later in this chapter as a corollary of Fejér's theorem. To show that the relation (9) holds for every function f that is square-integrable over $[-\pi,\pi]$, we note first that such a function can be approximated in mean by a function g that is continuous on $[-\pi,\pi]$ and has the property $g(-\pi)=g(\pi)$. Combining this fact with the Weierstrass approximation theorem, we see that for each $\varepsilon>0$ there is a trigonometric polynomial T such that

$$\int_{-\pi}^{\pi}[f(x)-T(x)]^2\,dx < \varepsilon\,.$$

But if T has degree m, it then follows that

$$\int_{-\pi}^{\pi}[f(x)-s_n(x)]^2\,dx < \varepsilon \qquad \text{for all } \ n\geq m\,,$$

since we have shown that the Fourier polynomial s_n gives the best mean-square approximation to f among all trigonometric polynomials of degree n or lower. This verifies (9), which proves the Parseval relation.

8.4. Convergence of Fourier series

Our goal is now to develop conditions that ensure the convergence of a Fourier series to its associated function. In order to obtain a suitable expression for the partial sums, we begin with an elementary calculation.

Lemma 1.

$$\tfrac{1}{2} + \sum_{k=1}^{n} \cos kx = \frac{\sin(n+\frac{1}{2})x}{2\sin\frac{1}{2}x}\,, \qquad n = 1, 2, \dots\,.$$

Proof. Multiply the sum by $2\sin\frac{1}{2}x$ and use the identity

$$2\sin\tfrac{1}{2}x\,\cos kx = \sin(k+\tfrac{1}{2})x - \sin(k-\tfrac{1}{2})x\,.$$

The resulting sum then telescopes and gives the stated result. □

The function

$$D_n(x) = \frac{\sin(n+\frac{1}{2})x}{2\sin\frac{1}{2}x} = \tfrac{1}{2} + \sum_{k=1}^{n} \cos kx$$

is known as the *Dirichlet kernel*. It has the property

$$\frac{1}{\pi}\int_{-\pi}^{\pi} D_n(x)\,dx = 1\,, \qquad n = 1, 2, \dots\,, \tag{10}$$

since $\int_{-\pi}^{\pi} \cos kx\,dx = 0$ for $k = 1, 2, \dots$. The Dirichlet kernel is an even function, periodic with period 2π. Thus $D_n(-x) = D_n(x)$ and $D_n(x+2\pi) = D_n(x)$. A graph of the Dirichlet kernel for $n = 10$ is shown in Figure 1.

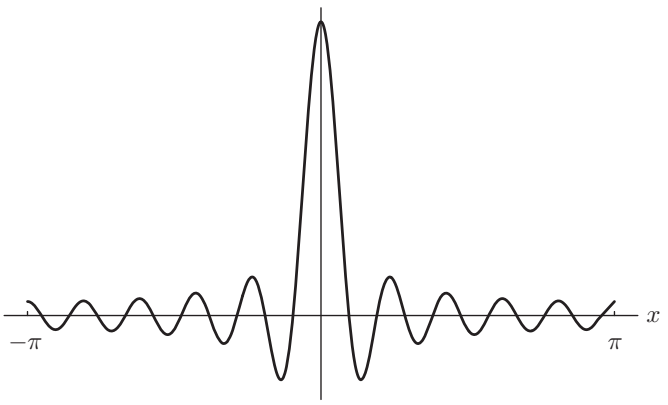


Figure 1. The Dirichlet kernel $D_{10}(x)$.

With the help of Lemma 1, we may now derive a useful formula for the nth partial sum $s_n(x)$ of the Fourier series of a function f, as defined by (6). We take f to be integrable over the interval $[-\pi, \pi]$, with the property $f(-\pi) = f(\pi)$, and we extend it to the whole real line as a periodic function with period 2π. Introducing the formulas (4) for the Fourier coefficients of f, we can write

$$\begin{aligned} s_n(x) &= \frac{1}{\pi}\int_{-\pi}^{\pi}\Big\{\tfrac{1}{2} + \sum_{k=1}^{n}\big(\cos kx\cos kt + \sin kx \sin kt\big)\Big\} f(t)\,dt \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi}\Big\{\tfrac{1}{2} + \sum_{k=1}^{n}\cos k(x-t)\Big\} f(t)\,dt \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} D_n(x-t) f(t)\,dt = \frac{1}{\pi}\int_{-\pi}^{\pi} D_n(t) f(x+t)\,dt\,, \end{aligned}$$

since the Dirichlet kernel D_n is even and both D_n and f have period 2π. This relation

$$s_n(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} D_n(t) f(x+t)\,dt \tag{11}$$

is known as the *Dirichlet formula* for the nth partial sum of a Fourier series.

In view of the property (10), we have deduced that

$$s_n(x) - f(x) = \frac{1}{\pi}\int_{-\pi}^{\pi} D_n(t)[f(x+t) - f(x)]\,dt\,. \tag{12}$$

In order to prove that $s_n(x) \to f(x)$, as $n \to \infty$, we have only to show that the integral tends to zero. Were it true that $D_n(x) \geq 0$, an argument similar to Landau's proof of the Weierstrass approximation theorem (see Chapter 6) could be given for any point x where f is continuous. However, it is apparent that $D_n(t)$ alternates in sign in the interval $[-\pi, \pi]$. In fact, it can be shown (see Exercise 7) that

$$\int_{-\pi}^{\pi} |D_n(t)|\,dt \sim C\log n\,, \qquad n \to \infty$$

for some constant $C > 0$, so this method of proof cannot succeed. What is worse, it is actually possible for the Fourier series of a continuous periodic function to diverge at some points. Thus in order to prove that the Fourier series converges to the function at a given point, we shall have to impose hypotheses stronger than continuity. We will assume that the function satisfies a smoothness condition near the point where convergence is to occur. The precise statement is as follows.

Convergence Theorem. *Let a function f be square-integrable over the interval $[-\pi, \pi]$, and let s_n be the nth partial sum of its Fourier series. Let f be extended to the real line by the periodicity condition $f(x + 2\pi) = f(x)$. Suppose that at some point $x \in \mathbb{R}$ the extended function satisfies a Lipschitz condition:*

$$|f(x + t) - f(x)| \le C\,|t|\,, \qquad |t| < \delta\,,$$

for some positive constants C and δ. Then $s_n(x) \to f(x)$ as $n \to \infty$.

Before passing to the proof, let us observe one very remarkable feature of the theorem. The Fourier coefficients a_k and b_k are determined by the values of f throughout an interval of length 2π, yet the convergence of the resulting Fourier series at any particular point x is governed only by the behavior of f in an arbitrarily small neighborhood of x. This phenomenon is known as *Riemann's localization principle.*

Proof of theorem. Begin by recalling the corollary of Bessel's inequality (8), that the Fourier coefficients of any square-integrable function g tend to zero:

$$\lim_{n\to\infty} \int_{-\pi}^{\pi} g(t) \cos nt\, dt = \lim_{n\to\infty} \int_{-\pi}^{\pi} g(t) \sin nt\, dt = 0\,.$$

Now write the formula (12) as

$$\begin{aligned} s_n(x) - f(x) &= \frac{1}{\pi} \int_{-\pi}^{\pi} D_n(t)[f(x + t) - f(x)]\, dt \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} \varphi_x(t) \sin(n + \tfrac{1}{2})t\ dt\,, \end{aligned} \tag{13}$$

where

$$\varphi_x(t) = \frac{f(x + t) - f(x)}{2 \sin \frac{1}{2}t}\,.$$

Then by the smoothness hypothesis on f,

$$|\varphi_x(t)| = \frac{|f(x + t) - f(x)|}{2\,|\sin \frac{1}{2}t|} \le \frac{C\,|t|}{2\,|\sin \frac{1}{2}t|} \le C$$

for $|t| \le \delta$. On the other hand, $|\sin \frac{1}{2}t| \ge |\sin(\delta/2)| > 0$ elsewhere in the interval $[-\pi, \pi]$, and f is a square-integrable function. Therefore, the function $\varphi_x(t)$ is square-integrable over the interval $[-\pi, \pi]$. But

$$\sin(n + \tfrac{1}{2})t = \cos \tfrac{1}{2}t\, \sin nt + \sin \tfrac{1}{2}t\, \cos nt\,,$$

so the integral

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \varphi_x(t) \sin(n + \tfrac{1}{2})t\ dt \tag{14}$$

can be regarded as a sum of Fourier sine and cosine coefficients of the square-integrable functions $\varphi_x(t)\cos\frac{1}{2}t$ and $\varphi_x(t)\sin\frac{1}{2}t$, respectively. Thus by the corollary to Bessel's inequality, the integral (14) tends to zero as $n\to\infty$. It therefore follows from (13) that $s_n(x)\to f(x)$ as $n\to\infty$, which completes the proof. □

It may be remarked that the proof remains valid under the weaker smoothness hypothesis

$$|f(x+t)-f(x)|\le C\,|t|^\alpha\,,\qquad |t|<\delta\,,$$

for some exponent $\alpha>\frac{1}{2}$. That still ensures the square-integrablity of $\varphi_x(t)$, so its Fourier coefficients tend to zero.

There is a more general form of the convergence theorem that applies to functions with jump discontinuities. If the function f has one-sided limits $f(x+)$ and $f(x-)$ at a point x, and it is sufficiently smooth in both of the one-sided neighborhoods of x, then its Fourier series converges to the average of the two limits. This may be viewed as a corollary of the convergence theorem. Here is a more precise statement.

Corollary. *Let f be periodic with period 2π and square-integrable over $[-\pi,\pi]$. At some point $x\in\mathbb{R}$, suppose f has one-sided limits*

$$f(x+)=\lim_{t\to0,\,t>0}f(x+t)\,,\qquad f(x-)=\lim_{t\to0,\,t>0}f(x-t)\,.$$

Suppose further that

$$|f(x+t)-f(x+)|\le Ct\quad\text{and}\quad|f(x-t)-f(x-)|\le Ct\,,\qquad 0<t<\delta\,,$$

for some positive constants C and δ. Then $s_n(x)\to\frac{1}{2}[f(x+)+f(x-)]$ as $n\to\infty$.

Proof. The result can be deduced from the more special form of the convergence theorem by subtracting from f the function g that has the same jump as f and is constant on both sides of the jump-point. Then $f-g$ has the properties required in the convergence theorem, and a direct calculation shows that the partial sums of g are all equal to $\frac{1}{2}[g(x+)+g(x-)]$ at the point x. Rather than pursue this line of proof, however, we will simply adapt the earlier proof of the convergence theorem to the more general situation.

Dirichlet's formula (11) allows us to write

$$\begin{aligned}s_n(x)-\tfrac{1}{2}[f(x+)+f(x-)]&=\frac{1}{\pi}\int_{-\pi}^{0}D_n(t)[f(x+t)-f(x-)]\,dt\\&\quad+\frac{1}{\pi}\int_{0}^{\pi}D_n(t)[f(x+t)-f(x+)]\,dt\,.\end{aligned}$$

By considering now the two square-integrable functions

$$\varphi_x(t) = \begin{cases} \dfrac{f(x+t) - f(x-)}{2\sin\frac{1}{2}t}\,, & -\pi \le t < 0 \\ 0\,, & 0 \le t \le \pi \end{cases}$$

and

$$\psi_x(t) = \begin{cases} 0\,, & -\pi \le t \le 0 \\ \dfrac{f(x+t) - f(x+)}{2\sin\frac{1}{2}t}\,, & 0 < t \le \pi\,, \end{cases}$$

we conclude as before that $s_n(x) \to \frac{1}{2}[f(x+) + f(x-)]$ as $n \to \infty$. □

Most of the preceding analysis is due to Peter Gustav Lejeune Dirichlet (1805–1859), who gave the first rigorous proof of convergence of Fourier series in 1829. Dirichlet was a major analyst and number theorist, regarded as the founder of analytic number theory. Born in the town of Düren (now in Germany) to a family of Belgian origin, he entered the Collège de France in Paris at age 17. Among his teachers there were Fourier, Laplace, Legendre, and Poisson. By 1828 he held a professorship in Berlin, and later moved to Göttingen as successor to Gauss.

The convergence theorem as presented here is not the most general, but it is general enough to cover most functions encountered in applications. We will see that it is also a very effective mathematical tool. Refinements and further information can be found in many books on Fourier series; for instance, in those of Jackson [**7**], Körner [**8**], Rogosinski [**9**], Widder [**10**], and Zygmund [**11**].

8.5. Examples

In calculating the Fourier coefficients of a given function f it is often useful to take advantage of special symmetries. For instance, if f is an even function, then $f(x)\sin nx$ is odd, and it is clear without calculation that the Fourier sine coefficients of f must vanish: $b_n = 0$ for all n. Also, the Fourier cosine coefficients take the form

$$a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx \qquad n = 0, 1, 2, \ldots\,.$$

Similarly, if f is an odd function, or if it differs by a constant from an odd function, then $a_n = 0$ for all $n \ge 1$ and the formula for b_n may be correspondingly simplified.

Example 1. For a first example, consider the function $f(x) = |x|$ for $-\pi \le x \le \pi$, and let it be extended periodically to the whole line so that

$f(x+2\pi) \equiv f(x)$. The result is the "sawtooth function" whose graph is shown in Figure 2. Here f is an even function in the interval $[-\pi, \pi]$, so $b_n = 0$ for all n; while

$$a_0 = \frac{2}{\pi}\int_0^\pi x\,dx = \pi$$

and an integration by parts gives

$$a_n = \frac{2}{\pi}\int_0^\pi x\cos nx\,dx = \frac{2}{n^2\pi}\left[(-1)^n - 1\right] = \begin{cases} -\dfrac{4}{n^2\pi}\,, & n \text{ odd} \\ 0\,, & n \text{ even.} \end{cases}$$

The function f satisfies a uniform Lipschitz condition (with $C = 1$), so the convergence theorem applies and gives

$$\begin{aligned} |x| &= \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2}\cos(2k+1)x \\ &= \frac{\pi}{2} - \frac{4}{\pi}\left\{\cos x + \frac{1}{9}\cos 3x + \frac{1}{25}\cos 5x + \dots\right\} \end{aligned}$$

for $-\pi \le x \le \pi$. By the Weierstrass M-test, the convergence is uniform. With the choice $x = 0$, the identity reduces to

$$\sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}\,,$$

from which it is easy to deduce Euler's sum

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}\,.$$

The uniform convergence of the cosine series to the sum $|x|$, periodically extended to the sawtooth function $f(x)$, is shown graphically in Figure 2, where *Mathematica* plots compare the function with the partial sum

$$s_5(x) = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{2}\frac{1}{(2k+1)^2}\cos(2k+1)x$$

of its Fourier series. Evidently the convergence is quite rapid, since the two graphs are barely distinguishable.

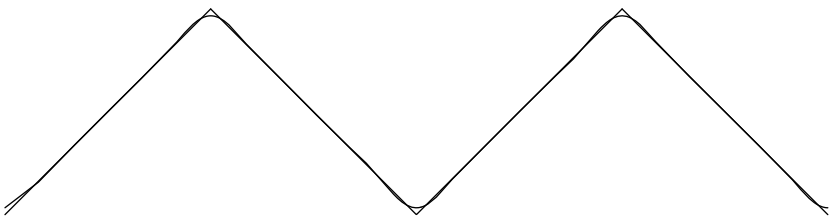


Figure 2. Fourier series of $|x|$ extended periodically.

Example 2. Another example worth exploring is the "square wave" function defined as the periodic extension of

$$f(x) = \begin{cases} -1\,, & -\pi < x < 0 \\ 1\,, & 0 < x < \pi\,. \end{cases}$$

This is an odd function whose Fourier coefficients are $a_n = 0$ and

$$b_n = \frac{2}{\pi}\int_0^\pi \sin nx\,dx = \frac{2}{n\pi}\big[1-(-1)^n\big] = \begin{cases} \dfrac{4}{n\pi}\,, & n \text{ odd} \\ 0\,, & n \text{ even.} \end{cases}$$

Except at its points of discontinuity, the function f clearly satisfies the requirements of the convergence theorem, so we infer that

$$\frac{4}{\pi}\sum_{k=0}^{\infty}\frac{1}{2k+1}\sin(2k+1)x = \begin{cases} -1\,, & -\pi < x < 0 \\ 1\,, & 0 < x < \pi\,. \end{cases}$$

Figure 3 illustrates the convergence by displaying plots of the partial sums of order 10 and 20, containing 5 and 10 terms, respectively.

Example 3. For a third example, consider the periodic extension of the linear function

$$f(x) = \tfrac{1}{2}(\pi - x)\,, \qquad 0 < x < 2\pi\,.$$

This is a sawtooth function with discontinuities at 0, $\pm 2\pi$, $\pm 4\pi, \ldots$. Because it is an odd function, it has Fourier coefficients $a_n = 0$ and

$$b_n = \frac{1}{\pi}\int_0^\pi (\pi - x)\sin nx\,dx = \frac{1}{n}\,, \qquad n = 1, 2, \ldots\,,$$

as an integration by parts shows. Another appeal to the convergence theorem now verifies that

$$\sum_{k=1}^{\infty}\frac{\sin kx}{k} = \begin{cases} -\tfrac{1}{2}(\pi + x)\,, & -\pi \le x < 0 \\ \tfrac{1}{2}(\pi - x)\,, & 0 < x \le \pi\,. \end{cases}$$

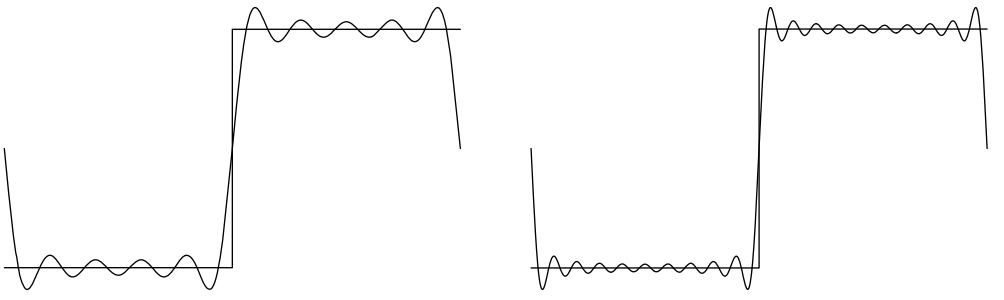


Figure 3. Fourier series of square wave, $n = 10$ and 20.

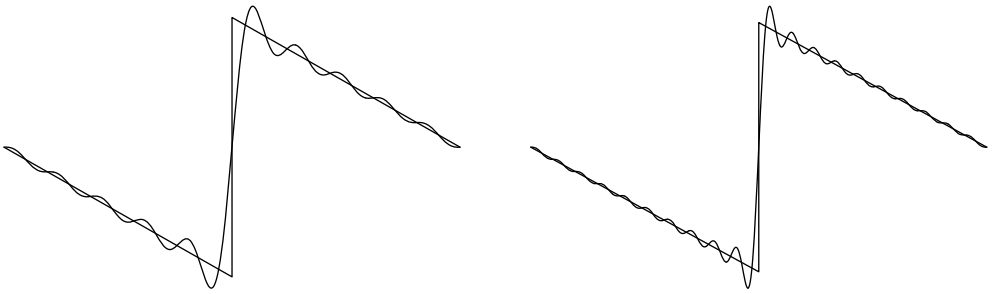


Figure 4. Fourier series of discontinuous sawtooth, $n = 10$ and 20.

Figure 4 shows graphs of the partial sums of order $n = 10$ and 20.

Example 4. We turn now to a special example, the Fourier series expansion of the function $\cos cx$, where the constant c is not an integer. The resulting identity will be seen to have surprising implications.

For $c \neq 0, \pm 1, \pm 2, \ldots$, the function $\cos cx$ does not have period 2π, but it has a continuous 2π-periodic extension $f(x)$ beyond the interval $[-\pi, \pi]$, since it is an even function and so $\cos(-c\pi) = \cos c\pi$. The periodic extension $f(x)$ need not be differentiable at the points $x = \pm\pi$, but it satisfies a Lipschitz condition there. The basic convergence theorem therefore applies and guarantees that the Fourier series converges to $f(x)$ at every point $x \in \mathbb{R}$. In particular, it guarantees the validity of the formula

$$\cos cx = \frac{2c}{\pi} \sin c\pi \left\{ \frac{1}{2c^2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{c^2 - n^2} \cos nx \right\}, \qquad -\pi \leq x \leq \pi\,. \tag{15}$$

In order to verify (15), we have only to check that the right-hand side is indeed the formal Fourier series of the function f. We may use the trigonometric identity

$$2 \cos cx \cos nx = \cos(c + n)x + \cos(c - n)x$$

to calculate the Fourier coefficients

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos cx \cos nx \, dx = \frac{2c}{\pi} \sin c\pi \frac{(-1)^n}{c^2 - n^2}$$

for $n = 1, 2, \ldots$, and

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos cx \, dx = \frac{2}{c\pi} \sin c\pi\,.$$

All of the Fourier sine-coefficients b_n vanish, since $\cos cx$ is an even function. Thus the infinite series in the formula (15) is the formal Fourier series of $f(x)$, and the basic convergence theorem shows that it converges to the sum $\cos cx$ throughout the interval $[-\pi, \pi]$.

Now comes an audacious trick. The equation (15) holds for every non-integer constant c and for all x in the interval $[-\pi, \pi]$. Let us fix $x = \pi$ and regard c as the variable. Changing notation from c to t, we deduce from (15) that

$$\pi \cot \pi t - \frac{1}{t} = \sum_{n=1}^{\infty} \frac{2t}{t^2 - n^2}\,, \qquad t \neq 0, \pm 1, \pm 2, \ldots\,. \tag{16}$$

Since the series in (16) converges uniformly in each interval $[-a, a]$ with $0 < a < 1$, we may integrate term by term over an interval $[\varepsilon, x]$, where $0 < \varepsilon < x < 1$, to obtain

$$\log\left(\frac{\sin \pi x}{\pi x}\right) - \log\left(\frac{\sin \pi \varepsilon}{\pi \varepsilon}\right) = \sum_{n=1}^{\infty} \log\left(\frac{x^2 - n^2}{\varepsilon^2 - n^2}\right)\,.$$

Letting ε tend to zero, we conclude that

$$\log\left(\frac{\sin \pi x}{\pi x}\right) = \sum_{n=1}^{\infty} \log\left(1 - \frac{x^2}{n^2}\right)\,, \qquad 0 < x < 1\,.$$

Exponentiating, we arrive at the infinite product representation

$$\sin \pi x = \pi x \left(1 - x^2\right) \left(1 - \frac{x^2}{2^2}\right) \left(1 - \frac{x^2}{3^2}\right) \cdots\,, \qquad -1 < x < 1\,. \tag{17}$$

In fact, the formula (17) holds for all $x \in \mathbb{R}$. To see this, we need only show that the right-hand side is periodic with period 2. In other words, if $p(x)$ denotes the infinite product (which converges for every $x \in \mathbb{R}$), we want to show that $p(x + 2) = p(x)$. But the partial product

$$p_n(x) = \pi x \left(1 - x^2\right) \left(1 - \frac{x^2}{2^2}\right) \cdots \left(1 - \frac{x^2}{n^2}\right)$$

can be rewritten as

$$p_n(x) = \frac{(-1)^n \pi}{(n!)^2} (x - n) \cdots (x - 1) x (x + 1) \cdots (x + n)\,,$$

which gives the relation

$$p_n(x + 2) = \frac{(x + n + 1)(x + n + 2)}{(x - n + 1)(x - n)}\, p_n(x)\,, \qquad n = 1, 2, \ldots\,.$$

Letting n tend to infinity, we conclude that $p(x + 2) = p(x)$.

Setting $x = \frac{1}{2}$ in (17), we obtain another proof of the Wallis product formula

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{4^2}\right)\left(1 - \frac{1}{6^2}\right)\cdots = \frac{2}{\pi}\,.$$

By comparing the coefficient of x^3 in the product (17) with that of the Taylor series expansion of $\sin \pi x$, we are led to Euler's sum $\sum 1/n^2 = \pi^2/6$. This was Euler's original derivation, although his discovery of the product formula for the sine function amounted only to a plausible argument.

8.6. Gibbs' phenomenon

For Examples 2 and 3, the graphs of partial sums converge nicely to the function away from the points of discontinuity, but they exhibit a persistent "overshoot" near the jump points. This strange behavior is known as the *Gibbs phenomenon*. Its discovery is generally attributed to the American mathematical physicist J. W. Gibbs in 1898, but it goes back at least to the English mathematician Henry Wilbraham in 1848. For this reason it is sometimes called the *Gibbs–Wilbraham phenomenon*. Wilbraham made painstaking hand calculations of partial sums for Example 2 and plotted graphs much like those in Figure 3.

In order to give a precise formulation of the Gibbs phenomenon, we shall focus on Example 3. Let

$$s_n(x) = \sum_{k=1}^{n} \frac{\sin kx}{k}$$

denote the partial sums of that Fourier series. Let ξ_n denote the first positive point where the function $s_n(x)$ attains a local maximum. An inspection of Figure 4 suggests the conjecture that the sequence $\{s_n(\xi_n)\}$ tends to a limit larger than $\pi/2$ as $n \to \infty$. In other words, the overshoot persists in the limit.

To prove the conjecture, first calculate the derivative

$$s_n'(x) = \sum_{k=1}^{n} \cos kx = D_n(x) - \tfrac{1}{2}\,,$$

where

$$D_n(x) = \frac{\sin(n + \frac{1}{2})x}{2 \sin \frac{1}{2}x}$$

is the Dirichlet kernel, as found in Lemma 1. Thus the equation $s_n'(x) = 0$ is equivalent to

$$\sin(n + \tfrac{1}{2})x = \sin \tfrac{1}{2}x\,.$$

This last equation implies that either

$$\tfrac{1}{2}x + 2k\pi = (n + \tfrac{1}{2})x \qquad \text{or} \qquad \pi - \tfrac{1}{2}x + 2k\pi = (n + \tfrac{1}{2})x$$

for $k = 0, \pm 1, \pm 2, \ldots$. Hence the positive critical points of $s_n(x)$ in the interval $(0, \pi]$ have the form

$$x = \frac{2k\pi}{n}\,, \qquad k = 1, 2, \ldots, [n/2]\,,$$

and

$$x = \frac{(2k+1)\pi}{n+1}\,, \qquad k = 0, 1, 2, \ldots, [n/2]\,,$$

where $[x]$ denotes the integer part of x. The critical points in the first group are found to be local minima and those in the second group are local maxima. Thus the first positive local maximum of $s_n(x)$ occurs at the point $x = \xi_n = \dfrac{\pi}{n+1}$.

Let us now investigate the behavior of

$$s_n(\xi_n) = \sum_{k=1}^{n} \frac{1}{k} \sin\left(\frac{k\pi}{n+1}\right)$$

as $n \to \infty$. A little manipulation brings the sum to the form

$$s_n(\xi_n) = \sum_{k=1}^{n+1} \frac{n+1}{k\pi} \sin\left(\frac{k\pi}{n+1}\right) \frac{\pi}{n+1} = \sum_{k=1}^{n+1} f(x_k)\,\Delta x_k\,,$$

where

$$f(x) = \frac{\sin x}{x}\,, \qquad x_k = \frac{k\pi}{n+1}\,, \qquad \text{and} \quad \Delta x_k = \frac{\pi}{n+1}\,.$$

The last sum is none other than a Riemann sum for the integral

$$\int_0^{\pi} f(x)\,dx = \int_0^{\pi} \frac{\sin x}{x}\,dx$$

with respect to a partition of the interval $[0, \pi]$ by $n + 1$ equally spaced points. With the understanding that $f(0) = 1$, the function f is continuous on the interval $[0, \pi]$. Therefore, the Riemann sums tend to the integral as $n \to \infty$, and we have proved that

$$\lim_{n\to\infty} s_n(\xi_n) = \int_0^{\pi} \frac{\sin x}{x}\,dx\,.$$

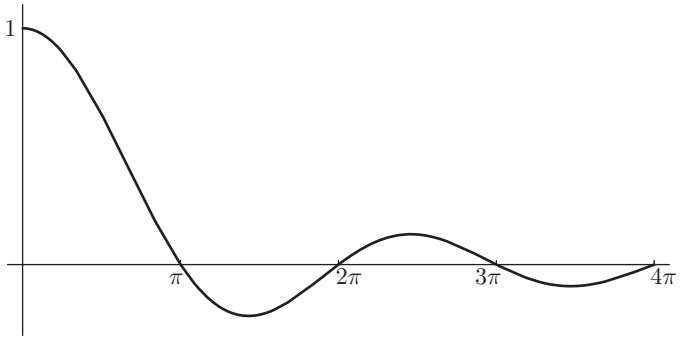


Figure 5. Graph of $\frac{\sin x}{x}$.

A graph of the function $(\sin x)/x$ is displayed in Figure 5. The graph suggests that

$$\frac{\sin x}{x} > 1 - \frac{x}{\pi}\,, \qquad 0 < x < \pi\,,$$

and this is not difficult to confirm (see Exercise 8). It follows that

$$\int_0^\pi \frac{\sin x}{x}\,dx > \int_0^\pi \left(1 - \frac{x}{\pi}\right)\,dx = \frac{\pi}{2}\,.$$

Numerical evaluation of the (nonelementary) integral actually gives

$$\int_0^\pi \frac{\sin x}{x}\,dx = 1.8519\cdots = (1.1789\ldots)\frac{\pi}{2}\,.$$

In the limit, then, the overshoot amounts to about 9% of the total jump π.

Although a particular example has served to illustrate the Gibbs phenomenon, it is important to emphasize that the phenomenon is quite general. It is true that our analysis has taken advantage of special features of the sawtooth function, and a similar analysis applies to the square wave of Example 2 (see Exercise 10). However, the phenomenon is not peculiar to special examples and will manifest itself at any jump-discontinuity. To be more precise, suppose that a periodic function g has a jump-discontinuity at a point x_0, so that $g(x_0+) - g(x_0-) = c > 0$, where $g(x_0+)$ and $g(x_0-)$ denote the right-hand and left-hand limits of $g(x)$, respectively, at x_0. Then, if f is the sawtooth function of Example 3, the function

$$h(x) = g(x) - \frac{c}{\pi}\,f(x - x_0)$$

has a continuous extension to x_0 and the partial sums of its Fourier series are

$$u_n(x) = t_n(x) - \frac{c}{\pi}\,s_n(x - x_0)\,,$$

where s_n and t_n are respective partial sums for f and g. If g is sufficiently well behaved elsewhere, it can be shown that $u_n(x) \to h(x)$ pointwise throughout some open neighborhood of x_0, and the convergence is uniform at each such point. In particular, the convergence is uniform at x_0, which is to say that for each $\varepsilon > 0$ there exist a $\delta > 0$ and an integer N such that $|u_n(x) - h(x)| < \varepsilon$ for $|x - x_0| < \delta$ and $n > N$. Equivalently, $u_n(x_n) \to h(x_0)$ whenever $x_n \to x_0$. Therefore, since the sums $s_n(x - x_0)$ have been shown to exhibit Gibbs' phenomenon at x_0, the sums $t_n(x)$ must also exhibit Gibbs' phenomenon there, with the same relative overshoot $(0.1789\ldots)c/2$, again about 9% of the total jump c. Further details will not be pursued here, but can be found for instance in Zygmund's book [**11**].

There is a very interesting account of the Gibbs–Wilbraham phenomenon and its history in the paper by Hewitt and Hewitt [**4**].

8.7. Arithmetic means of partial sums

The convergence theorem of Section 3 asserts that the Fourier series of a continuous periodic function converges wherever the function is sufficiently smooth. In Section 8.8 an example will be constructed to show that continuity alone does not guarantee the convergence. More precisely, the partial sums $s_n(x)$ of a continuous periodic function $f(x)$ need not converge and can even be unbounded. However, we now proceed to show that the sequence of arithmetic means

$$\sigma_n(x) = \frac{1}{n+1} \sum_{k=0}^{n} s_k(x)\,, \qquad n = 1, 2, \ldots\,, \tag{18}$$

always converges uniformly to the given function. This important result is known as Fejér's theorem, after the Hungarian mathematician Leopold Fejér (1880–1959), who discovered it in 1904. In the language of summability theory, the theorem says that the Fourier series of every continuous function is uniformly Cesàro summable to the function. The trigonometric polynomials $\sigma_n(x)$ are often called the *Fejér means* of f.

Fejér's Theorem. *Let $f(x)$ be continuous on $\mathbb{R}$ and periodic with period 2π. Let $\sigma_n(x)$ be the arithmetic means of the partial sums of the Fourier series of f, as defined in* (18)*. Then $\sigma_n(x) \to f(x)$ uniformly on $\mathbb{R}$ as $n \to \infty$.*

Corollary. *Every continuous periodic function $f(x)$ can be approximated uniformly on $\mathbb{R}$ by trigonometric polynomials. In other words, for each $\varepsilon > 0$ there exists a trigonometric polynomial $T(x)$ such that $|f(x) - T(x)| < \varepsilon$ for all $x \in \mathbb{R}$.*

The corollary is the trigonometric form of the Weierstrass approximation theorem. The proof of the theorem will require the calculation of another trigonometric sum.

Lemma 2.

$$\sum_{k=0}^{n} \sin(k+\tfrac{1}{2})x = \frac{\sin^2 \tfrac{1}{2}(n+1)x}{\sin \tfrac{1}{2}x}\,, \qquad n = 1, 2, \dots\,.$$

Proof of lemma. In view of the identity

$$2\sin \tfrac{1}{2}x\, \sin(k+\tfrac{1}{2})x = \cos kx - \cos(k+1)x\,,$$

the sum telescopes to give

$$2\sin \tfrac{1}{2}x \sum_{k=0}^{n} \sin(k+\tfrac{1}{2})x = 1 - \cos(n+1)x = 2\sin^2\big(\tfrac{1}{2}(n+1)x\big)\,. \qquad \square$$

Preparatory to a proof of the theorem, we will use Lemma 2 to derive a representation formula for the arithmetic means $\sigma_n(x)$ that is analogous to Dirichlet's formula (11) for the partial sums $s_n(x)$. Recalling that the Dirichlet kernel is

$$D_k(t) = \frac{\sin(k+\tfrac{1}{2})t}{2\sin \tfrac{1}{2}t}\,, \qquad k = 0, 1, 2, \dots\,,$$

and appealing to the Dirichlet formula and Lemma 2, we find

$$\begin{aligned} \sigma_n(x) &= \frac{1}{n+1}\sum_{k=0}^{n} s_k(x) = \frac{1}{n+1}\,\frac{1}{\pi}\int_{-\pi}^{\pi}\Big\{\sum_{k=0}^{n} D_k(t)\Big\} f(x+t)\,dt \\ &= \frac{1}{\pi}\int_{-\pi}^{\pi} K_n(t) f(x+t)\,dt\,, \end{aligned} \tag{19}$$

where

$$K_n(x) = \frac{1}{n+1}\frac{\sin^2 \tfrac{1}{2}(n+1)x}{2\sin^2 \tfrac{1}{2}x}\,.$$

The function $K_n(x)$ is called the *Fejér kernel*. Figure 6 shows a plot for $n = 10$.

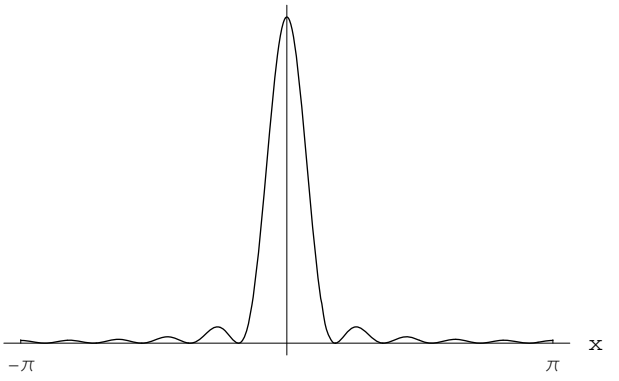


Figure 6. The Fejér kernel $K_{10}(x)$.

Because the Fejér kernel has the form

$$K_n(x) = \frac{1}{n+1} \sum_{k=0}^{n} D_k(x)\,,$$

the property (10) of the Dirichlet kernel implies that

$$\frac{1}{\pi} \int_{-\pi}^{\pi} K_n(x)\,dx = 1\,, \qquad n = 1, 2, \ldots\,. \tag{20}$$

Like the Dirichlet kernel, the Fejér kernel peaks at the origin as n increases, but it has the crucial advantage that $K_n(x) \geq 0$ for all x. Since $\sin^2 \frac{1}{2}x \geq (x/\pi)^2$, we see that

$$0 \leq K_n(x) \leq \frac{\pi^2}{2(n+1)\delta^2}\,, \qquad \delta \leq |x| \leq \pi\,, \tag{21}$$

for each δ in the interval $0 < \delta < \pi$.

Proof of Fejér's theorem. Since f is continuous and periodic, it is uniformly bounded: $|f(x)| \leq M$ for all $x \in \mathbb{R}$. Moreover, f is uniformly continuous. Thus for each $\varepsilon > 0$ there is a number $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for each pair of points $x, y \in \mathbb{R}$ with $|x - y| < \delta$. In view of (19) and (20), we can write

$$\sigma_n(x) - f(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} K_n(t)[f(x+t) - f(x)]\,dt\,.$$

Given $\varepsilon > 0$, choose δ as above. Then because $K_n(x) \geq 0$, we can appeal to the properties (20) and (21) of the Fejér kernel to infer that

$$\begin{aligned}|\sigma_n(x) - f(x)| &\leq \frac{1}{\pi}\int_{-\pi}^{\pi} K_n(t)|f(x+t) - f(x)|\,dt \leq \frac{2M}{\pi}\int_{-\pi}^{-\delta} K_n(t)\,dt \\ &\quad + \frac{1}{\pi}\int_{-\delta}^{\delta} K_n(t)|f(x+t) - f(x)|\,dt + \frac{2M}{\pi}\int_{\delta}^{\pi} K_n(t)\,dt \\ &\leq \frac{2M\pi^2}{(n+1)\delta^2} + \varepsilon < 2\varepsilon \qquad \text{for all } \ x \in \mathbb{R}\,,\end{aligned}$$

for every n sufficiently large. This proves that $\sigma_n(x) \to f(x)$ uniformly on $\mathbb{R}$ as $n \to \infty$. □

The representation formula (19) for the Fejér means, together with the property (20), shows at once that

$$|\sigma_n(x)| \leq \max_{x\in[-\pi,\pi]} |f(x)| \qquad \text{for all } \ x \in \mathbb{R}\,. \tag{22}$$

Hence there can be no Gibbs phenomenon for the Fejér means, no overshooting near points of discontinuity. The dramatic difference in mode of convergence is illustrated in Figure 7, where the Fejér means $\sigma_n(x)$ of the discontinuous sawtooth function

$$f(x) = \begin{cases} -\frac{1}{2}(\pi + x)\,, & -\pi \leq x < 0 \\ \frac{1}{2}(\pi - x)\,, & 0 < x \leq \pi \end{cases} \tag{23}$$

are shown for $n = 10$ and 20. This should be compared with the Gibbs phenomenon displayed by the partial sums $s_n(x)$, as shown in Figure 4.

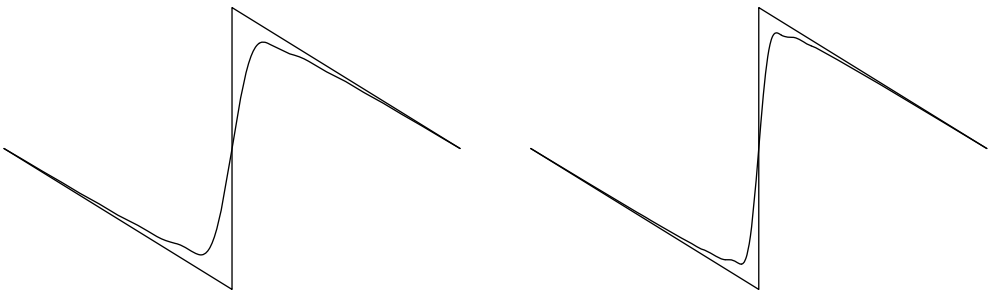


Figure 7. Fejér means of discontinuous sawtooth, $n = 10$ and 20.

With the help of the property (22) of the Fejér means, we can now prove that the partial sums $s_n(x)$ of the function (23) are uniformly bounded. The estimate (22) gives $|\sigma_n(x)| \le \pi/2$. On the other hand, a calculation shows that

$$|s_n(x) - \sigma_n(x)| = \frac{1}{n+1} \left| \sum_{k=1}^{n} \sin kx \right| < 1 \,.$$

Thus $|s_n(x)| \le 1 + \pi/2$ for all $x \in \mathbb{R}$. The details are left as an exercise.

8.8. Continuous functions with divergent Fourier series

We have seen that a Fourier series must converge to its associated function at every point where that function is sufficiently smooth. The question remains whether continuity alone will ensure the convergence. The answer is negative. The first example of a continuous function with divergent Fourier series was given by Paul du Bois-Reymond in 1873, and was rather complicated. Much later, in 1911, Fejér found a remarkably simple example, which will now be presented. The construction is reminiscent of Weierstrass's example of a continuous nowhere differentiable function.

First define the trigonometric polynomials

$$\begin{aligned} g_m(x) &= \frac{1}{m} + \frac{1}{m-1}\cos x + \frac{1}{m-2}\cos 2x + \cdots + \cos(m-1)x \\ &\qquad - \cos(m+1)x - \frac{1}{2}\cos(m+2)x - \cdots - \frac{1}{m}\cos 2mx \\ &= \sum_{k=1}^{m} \frac{1}{k}\big[\cos(m-k)x - \cos(m+k)x\big] = 2\sin mx \sum_{k=1}^{m} \frac{\sin kx}{k} \end{aligned}$$

for $m = 1, 2, \ldots$. Now let $\{m_k\}$ be a rapidly increasing sequence of positive integers, to be specified later. Define the function

$$F(x) = \sum_{k=1}^{\infty} \frac{1}{k^2}\, g_{m_k}(x) \,. \tag{24}$$

Recall that the partial sums of the series $\sum \frac{1}{k}\sin kx$ are uniformly bounded, as we saw at the end of the preceding section. Thus the functions $g_m(x)$ are also uniformly bounded, and the Weierstrass M-test shows that the infinite series (24) converges uniformly on $\mathbb{R}$. Therefore, the sum $F(x)$ is a continuous function on the real line, periodic with period 2π. With suitable choice of the sequence $\{m_k\}$, we will see that the Fourier series of F is divergent at the origin; in fact, its partial sums are unbounded there.

Let $s_{mn}(x)$ denote the nth partial sum of the Fourier series of $g_m(x)$. Then $s_{mn}(x) = g_m(x)$ for all $n \geq 2m$, since g_m is a trigonometric polynomial of degree $2m$. In view of the uniform convergence, the Fourier coefficients of F can be calculated by integrating the relevant infinite series term by term. This says that each Fourier coefficient of F is the sum of corresponding Fourier coefficients of the terms of the series (24). It follows that the nth partial sum $S_n(x)$ of the Fourier series of F is the sum of the nth partial sums of the terms. In other words,

$$S_n(x) = \sum_{k=1}^{\infty} \frac{1}{k^2}\, s_{m_k n}(x)\,. \tag{25}$$

In particular,

$$S_n(0) = \sum_{k=1}^{\infty} \frac{1}{k^2}\, s_{m_k n}(0)\,.$$

Observe now that $s_{mn}(0) \geq 0$ for every pair of indices m and n, and that

$$s_{mm}(0) = 1 + \frac{1}{2} + \cdots + \frac{1}{m} > \int_1^m \frac{dx}{x} = \log m\,.$$

Consequently, the relation (25) shows that for each index k

$$S_{m_k}(0) = \sum_{j=1}^{\infty} \frac{1}{j^2}\, s_{m_j\, m_k}(0) \geq \frac{1}{k^2}\, s_{m_k\, m_k}(0) > \frac{1}{k^2} \log(m_k)\,.$$

Now choose $m_k = 2^{k^3}$ to infer that

$$S_{m_k}(0) > \frac{1}{k^2} k^3 \log 2 = k \log 2\,,$$

so that the partial sums of F are unbounded at the origin.

The function $F(x)$ can be translated to obtain a function $F(x-r)$ whose Fourier series diverges at a specified point r. Then by forming a suitable weighted sum of the functions $F(x - r_k)$ corresponding to an arbitrary sequence r_k, we can construct a continuous function whose Fourier series diverges at every point r_k. For instance, the sequence r_k can be taken to be an enumeration of the rational numbers, and we then obtain a continuous function whose Fourier series diverges at a countable dense set of points.

On the other hand, Lennart Carleson proved in 1966 that the Fourier series of every continuous function converges almost everywhere. In other words, the points of divergence (if any) constitute a set of measure zero.

8.9. Fourier transforms

We now turn to the continuous analogue of a Fourier series, the Fourier transform and its inversion. It is customary and convenient to express the Fourier transform in the notation of complex exponentials $e^{it} = \cos t + i \sin t$. By way of motivation, we begin by recasting Fourier series expansions in complex form. To fix ideas, suppose that $f(t)$ is a real-valued Lipschitz-smooth function defined on the real line, periodic with period 2π. We claim that its Fourier series expansion can be expressed in the form

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{int}\,, \tag{26}$$

where the coefficients c_n are defined by

$$c_n = \frac{1}{2\pi}\int_0^{2\pi} f(t)e^{-int}\,dt\,, \qquad n = 0, \pm 1, \pm 2, \ldots. \tag{27}$$

To see this, observe that

$$c_n = \frac{1}{2\pi}\int_0^{2\pi} f(t)(\cos nt - i \sin nt)\,dt = \frac{1}{2}(a_n - ib_n)\,,$$

where

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(t)\cos nt\,dt \qquad \text{and} \qquad b_n = \frac{1}{\pi}\int_0^{2\pi} f(t)\sin nt\,dt$$

are the Fourier coefficients as previously defined for $n \geq 0$. Then $c_0 = \frac{1}{2}a_0$ and $c_{-n} = \overline{c_n} = \frac{1}{2}(a_n + ib_n)$, which shows that

$$c_n e^{int} + c_{-n}e^{-int} = a_n \cos nt + b_n \sin nt\,, \qquad n = 1, 2, \ldots.$$

Therefore,

$$\sum_{k=-n}^{n} c_k e^{ikt} = \frac{1}{2}a_0 + \sum_{k=1}^{n}(a_k \cos kt + b_k \sin kt) = s_n(t)$$

is the nth partial sum of the Fourier series of f, expressed in complex notation.

Recall now that $e^{i(s+t)} = e^{is}e^{it}$, by the addition formulas for sine and cosine. Differentiation gives

$$\frac{d}{dt}\left\{e^{iat}\right\} = -a \sin at + ia \cos at = iae^{iat}\,.$$

The complex form of a Fourier series can be obtained more directly through the orthogonality relations

$$\int_0^{2\pi} e^{int} e^{-imt}\, dt = \int_0^{2\pi} e^{i(n-m)t}\, dt = \begin{cases} 0 & \text{for } m \neq n \\ 2\pi & \text{for } m = n\,. \end{cases}$$

Formally, if a complex-valued function has an expansion of the form (26), multiplication by e^{-imt} and integration shows that the coefficients c_n are given by (27).

In what follows we will be dealing with complex-valued functions $f(t)$ defined on the real line $\mathbb{R}$ and Riemann integrable over each bounded interval. For $p > 0$ we will write $f \in L^p$ to indicate that $|f(t)|^p$ is integrable over $\mathbb{R}$. We will denote by L^∞ the set of all bounded locally integrable functions on $\mathbb{R}$.

The *Fourier transform*

$$\widehat{f}(x) = \int_{-\infty}^{\infty} e^{-ixt} f(t)\, dt\,, \qquad -\infty < x < \infty\,,$$

of a function $f \in L^1$ plays the role of a sequence of Fourier coefficients. In some sense, to be made precise later, the function can be recovered from its Fourier transform by the formula

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ixt} \widehat{f}(x)\, dx\,,$$

the continuous analogue of a Fourier series expansion. It is clear that

$$|\widehat{f}(x)| \leq \int_{-\infty}^{\infty} |f(t)|\, dt < \infty \qquad \text{if } \ f \in L^1\,,$$

so that the Fourier transform is always a bounded function. Simple examples will show, however, that the transform need not be integrable.

It is important to note that Fourier transformation is a linear operator. In other words, if $f, g \in L^1$ and $\alpha, \beta \in \mathbb{C}$, then $\widehat{\alpha f + \beta g} = \alpha \widehat{f} + \beta \widehat{g}$. It may also be observed that if $g(t) = f(at + b)$ for some real constants $a \neq 0$ and b, then

$$\widehat{g}(x) = \frac{1}{|a|}\, e^{ibx/a}\, \widehat{f}(x/a)\,.$$

Here are some examples.

Example 1. For some constant $R > 0$, consider the function

$$f(t) = \begin{cases} 1 & \text{for } |t| \leq R \\ 0 & \text{for } |t| > R\,. \end{cases}$$

Then for $x \neq 0$ we have

$$\widehat{f}(x) = \int_{-R}^{R} e^{-ixt}\,dt = 2\int_0^R \cos xt\,dt = 2\Big[\frac{\sin xt}{x}\Big]_0^R = \frac{2\sin Rx}{x}\,.$$

For $x = 0$ the result is $\widehat{f}(0) = 2R = \lim_{x\to 0}\widehat{f}(x)$. Thus $\widehat{f}$ is continuous and $\widehat{f}(x) \to 0$ as $|x| \to \infty$, but $\widehat{f} \notin L^1$.

Example 2. Suppose now that

$$f(t) = \begin{cases} 1 - \frac{|t|}{R} & \text{for } |t| \le R \\ 0 & \text{for } |t| > R\,. \end{cases}$$

Then for $x \neq 0$ an integration by parts gives

$$\begin{aligned}\widehat{f}(x) &= \int_{-R}^{R} e^{-ixt}\left(1 - \frac{|t|}{R}\right)\,dt = 2\int_0^R \left(1 - \frac{t}{R}\right)\cos xt\,dt \\ &= \frac{2}{Rx^2}(1 - \cos Rx) = R\left(\frac{\sin(Rx/2)}{Rx/2}\right)^2\,.\end{aligned}$$

For $x = 0$ we see that $\widehat{f}(0) = R = \lim_{x\to 0}\widehat{f}(x)$.

Example 3. Next let $f(t) = e^{-a|t|}$ for some constant $a > 0$. Then since f is an even function, we see that

$$\widehat{f}(x) = \int_{-\infty}^{\infty} e^{-ixt}e^{-a|t|}\,dt = \int_{-\infty}^{\infty} e^{-a|t|}\cos xt\,dt = 2\int_0^\infty e^{-at}\cos xt\,dt\,.$$

The last expression is a Laplace transform that can be calculated through two integrations by parts. The result is

$$\widehat{f}(x) = \frac{2a}{a^2 + x^2}\,.$$

Example 4. Now, for some constant $a > 0$, consider the function

$$f(t) = \begin{cases} e^{-at} & \text{for } t \ge 0 \\ 0 & \text{for } t < 0. \end{cases}$$

Then

$$\widehat{f}(x) = \int_0^\infty e^{-ixt}e^{-at}\,dt = \int_0^\infty e^{-at}(\cos xt\ - i\sin xt)\,dt = \frac{a - ix}{a^2 + x^2}\,.$$

Observe again that $\widehat{f} \notin L^1$.

Two more examples can be deduced from Examples 2 and 3 by applying the inversion formula for Fourier transforms, to be proved in the next section.

Example 5. For some constant $R > 0$, let

$$f(t) = \frac{2}{Rt^2}(1 - \cos Rt) = R\left(\frac{\sin(Rt/2)}{Rt/2}\right)^2 .$$

Then

$$\widehat{f}(x) = \begin{cases} 2\pi\left(1 - \frac{|x|}{R}\right) & \text{for } |x| \le R \\ 0 & \text{for } |x| > R\,. \end{cases}$$

Example 6. If $f(t) = \frac{a}{a^2+t^2}$ for some constant $a > 0$, then $\widehat{f}(x) = \pi e^{-a|x|}$.

Finally, it is interesting to note that the function $f(t) = e^{-t^2/2}$, which plays a central role in probability theory, is essentially its own Fourier transform. Here is a slight generalization.

Example 7. If $f(t) = e^{-at^2/2}$ for some constant $a > 0$, then

$$\widehat{f}(x) = \sqrt{2\pi/a}\; e^{-x^2/2a}\,.$$

We will be able to calculate this transform in an elementary way after having established some general properties of Fourier transforms, collected in the following theorem.

Theorem. *Let $f \in L^1$ and let $\widehat{f}(x)$ denote its Fourier transform. Then*

(*a*) *$\widehat{f}(x)$ is uniformly continuous on $\mathbb{R}$.*

(*b*) *$\widehat{f}(x) \to 0$ as $|x| \to \infty$.*

(*c*) *If $g(t) = tf(t)$ and $g \in L^1$, then $\widehat{f}$ is differentiable and $\widehat{f}\,'(x) = -i\widehat{g}(x)$.*

(*d*) *If f is differentiable and $f' \in L^1$, then f' has Fourier transform $\widehat{f'}(x) = ix\widehat{f}(x)$.*

Preparatory to the proof, we record the simple inequality

$$\left|e^{i\theta} - 1\right| \le |\theta|\,, \qquad \theta \in \mathbb{R}\,. \tag{28}$$

To verify this, write

$$\left|e^{i\theta} - 1\right| = \left|e^{i\theta/2} - e^{-i\theta/2}\right| = 2\left|\frac{e^{i\theta/2} - e^{-i\theta/2}}{2i}\right| = 2\,|\sin(\theta/2)| \le |\theta|\,.$$

Proof of theorem. (a). Given $\varepsilon > 0$, choose a constant $R > 0$ so large that

$$\int_{-\infty}^{-R} |f(t)|\,dt + \int_{R}^{\infty} |f(t)|\,dt < \frac{\varepsilon}{3}\,.$$

Then for arbitrary real numbers x and y, we can estimate

$$\begin{aligned}
|\widehat{f}(x) - \widehat{f}(y)| &= \left| \int_{-\infty}^{\infty} \left(e^{-ixt} - e^{-iyt}\right) f(t)\,dt \right| \\
&\leq 2\int_{-\infty}^{-R} |f(t)|\,dt + \int_{-R}^{R} \left|e^{-ixt} - e^{-iyt}\right| |f(t)|\,dt + 2\int_{R}^{\infty} |f(t)|\,dt \\
&< \frac{2\varepsilon}{3} + \int_{-R}^{R} \left|e^{-i(x-y)t} - 1\right| |f(t)|\,dt.
\end{aligned}$$

But by (28) we see that

$$\left|e^{-i(x-y)t} - 1\right| \leq |(x-y)t| \leq R\,|x-y| \qquad \text{for } \ |t| \leq R\,,$$

which shows that for some $\delta > 0$, the inequality

$$|\widehat{f}(x) - \widehat{f}(y)| \leq \frac{2\varepsilon}{3} + R\,|x-y| \int_{-R}^{R} |f(t)|\,dt < \varepsilon$$

holds whenever $|x - y| < \delta$. This proves the uniform continuity of $\widehat{f}$.

(b). Consider first the special case where f is the characteristic function of a bounded interval $[a, b]$. In other words,

$$f(t) = \begin{cases} 1 & \text{for } a \leq t \leq b \\ 0 & \text{for } t < a \text{ or } t > b. \end{cases}$$

Then we see from Example 1, by translation and change of scale, that $\widehat{f}(x) \to 0$ as $|x| \to \infty$. From this it follows more generally that $\widehat{f}(x) \to 0$ for any step-function f, by which we mean a finite linear combination of characteristic functions of bounded intervals. Now consider an arbitrary function $f \in L^1$. Given $\varepsilon > 0$, let $R > 0$ be chosen as in part (a). Then observe that since f is Riemann integrable over $[-R, R]$, it can be approximated in L^1 norm by step-functions. In other words, there exists a step-function $\varphi(t)$ supported on the interval $[-R, R]$ for which

$$\int_{-R}^{R} |f(t) - \varphi(t)|\,dt < \frac{\varepsilon}{3}\,.$$

Combining these estimates, we see that

$$\begin{aligned}|\widehat{f}(x)| &\le |\widehat{f}(x) - \widehat{\varphi}(x)| + |\widehat{\varphi}(x)| \le \int_{-\infty}^{\infty} |f(t) - \varphi(t)|\, dt + |\widehat{\varphi}(x)| \\ &= \int_{-\infty}^{-R} |f(t)|\, dt + \int_{-R}^{R} |f(t) - \varphi(t)|\, dt + \int_{R}^{\infty} |f(t)|\, dt + |\widehat{\varphi}(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + |\widehat{\varphi}(x)| < \varepsilon\end{aligned}$$

for $|x|$ sufficiently large, since $\widehat{\varphi}(x) \to 0$ as $|x| \to \infty$. This proves that $\widehat{f}(x) \to 0$ as $|x| \to \infty$.

(c). To show that $\widehat{f}$ has a derivative $\widehat{f}\,'(x) = -i\widehat{g}(x)$, write

$$\frac{\widehat{f}(x+h) - \widehat{f}(x)}{h} + i\widehat{g}(x) = \int_{-\infty}^{\infty} \left(\frac{e^{-iht} - 1}{ht} + i \right) e^{-ixt}\, tf(t)\, dt\,.$$

Given $\varepsilon > 0$, choose $R > 0$ such that

$$\int_{-\infty}^{-R} |tf(t)|\, dt + \int_{R}^{\infty} |tf(t)|\, dt < \frac{\varepsilon}{3}\,.$$

Then apply the inequality (28) to obtain

$$\begin{aligned}\left| \frac{\widehat{f}(x+h) - \widehat{f}(x)}{h} + i\widehat{g}(x) \right| &\le \int_{-\infty}^{\infty} \left| \frac{e^{-iht} - 1}{ht} + i \right| |tf(t)|\, dt \\ &< \int_{-R}^{R} \left| \frac{e^{-iht} - 1}{ht} + i \right| |tf(t)|\, dt + \frac{2\varepsilon}{3}\,.\end{aligned}$$

But $\frac{d}{d\theta}\left\{e^{-i\theta}\right\} = -i$ at $\theta = 0$, which says that for some $\delta > 0$ the inequality

$$\left| \frac{e^{-i\theta} - 1}{\theta} + i \right| \int_{-R}^{R} |tf(t)|\, dt < \frac{\varepsilon}{3}$$

holds for all θ with $|\theta| < \delta$. It follows that

$$\left| \frac{\widehat{f}(x+h) - \widehat{f}(x)}{h} + i\widehat{g}(x) \right| < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

when $|h| < \delta/R$, which proves that $\widehat{f}\,'(x) = -i\widehat{g}(x)$.

(d). The proof that f' has Fourier transform $ix\widehat{f}(x)$ will use integration by parts. The hypothesis $f \in L^1$ alone does not guarantee that $f(t) \to 0$ as $|t| \to \infty$, but with the additional assumption that $f' \in L^1$ we can write

$$f(t) = f(0) + \int_0^t f'(s)\, ds \to f(0) + \int_0^{\infty} f'(s)\, ds \qquad \text{as } \ t \to \infty\,,$$

by the fundamental theorem of calculus. Thus $f(t)$ tends to a limit as $t \to \infty$, and that limit must be zero since $f \in L^1$. The same argument shows that $f(t) \to 0$ as $t \to -\infty$. Consequently, integration by parts gives

$$\widehat{f'}(x) = \int_{-\infty}^{\infty} e^{-ixt} f'(t)\, dt = \Big[e^{-ixt} f(t)\Big]_{-\infty}^{\infty} + ix \int_{-\infty}^{\infty} e^{-ixt} f(t)\, dt = ix\widehat{f}(x)\,,$$

as desired. □

We are now prepared to calculate the Fourier transform of $f(t) = e^{-at^2/2}$, as given in Example 7. Since $f'(t) = -at\, e^{-at^2/2}$ is also integrable, Parts (c) and (d) of the preceding theorem both apply and show that

$$ix\widehat{f}(x) = \widehat{f'}(x) = -ia\widehat{f}\,'(x)\,.$$

Thus $\widehat{f}$ satisfies the differential equation $a\widehat{f}\,'(x) + x\widehat{f}(x) = 0$, which implies that

$$\frac{d}{dx}\left\{e^{x^2/2a}\widehat{f}(x)\right\} = 0\,, \qquad \text{and so} \qquad e^{x^2/2a}\widehat{f}(x) = C$$

for some constant C. But

$$C = \widehat{f}(0) = \int_{-\infty}^{\infty} e^{-at^2/2}\, dt = \sqrt{2/a} \int_{-\infty}^{\infty} e^{-s^2}\, ds = \sqrt{2\pi/a}\,,$$

which shows that $\widehat{f}(x) = \sqrt{2\pi/a}\; e^{-x^2/2a}$, as claimed. (For calculation of the last integral, see Section 9.1.)

The *convolution* $f * g$ of two functions $f, g \in L^1$ is defined by the integral

$$(f * g)(t) = \int_{-\infty}^{\infty} f(s) g(t-s)\, ds\,.$$

It is easily seen that the integral converges absolutely if one of the functions f or g is bounded. The same is true, by the Cauchy-Schwarz inequality, if f and g are both in the space L^2. The operation of convolution behaves much like multiplication of numbers. It is commutative and associative, and distributive over addition:

$$f * g = g * f\,, \qquad f * (g * h) = (f * g) * h\,, \qquad f * (g + h) = f * g + f * h\,.$$

Perhaps the most important property of the Fourier transform is that it converts convolution to ordinary pointwise multiplication: $\widehat{f * g}(x) = \widehat{f}(x)\widehat{g}(x)$. For a proof, we can interchange the order of integration to write

$$\begin{aligned}\widehat{f * g}(x) &= \int_{-\infty}^{\infty} e^{-ixt} \int_{-\infty}^{\infty} f(s) g(t-s)\, ds\, dt \\ &= \int_{-\infty}^{\infty} e^{-ixs} f(s) \int_{-\infty}^{\infty} e^{-ix(t-s)} g(t-s)\, dt\, ds = \widehat{f}(x)\widehat{g}(x)\,.\end{aligned}$$

On the other hand, the operation of convolution does not admit a multiplicative identity element. In other words, there is no function $g \in L^1$ with the property that $f * g = f$ for every function $f \in L^1$. If there were such a function, Fourier transformation would give in particular $\widehat{f}(x)\widehat{g}(x) = \widehat{f}(x)$ for the function $f(t) = e^{-t^2/2}$, with transform $\widehat{f}(x) = \sqrt{2\pi}\, e^{-x^2/2} \neq 0$. But that implies $\widehat{g}(x) \equiv 1$, contradicting the fact that $\widehat{g}(x) \to 0$ as $|x| \to \infty$, as shown in the preceding theorem. To circumvent the difficulty, it is often useful to implement an "approximate identity", a sequence of functions $g_n \in L^1$ such that $\int_{-\infty}^{\infty} g_n(t)dt = 1$ and $f * g_n \to f$ in some sense for every $f \in L^1 \cap L^\infty$. Indeed, this will be our underlying strategy for inversion of a Fourier transform.

8.10. Inversion of Fourier transforms

We propose to interpret and justify the formula

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ixt} \widehat{f}(x)\, dx\,,$$

which recovers a function from its Fourier transform. This is the continuous analogue of a Fourier series. As for Fourier series, however, the inversion problem is by no means straightforward. First of all, the transform $\widehat{f}$ need not be integrable, as we have seen by examples, so in general the formula does not make sense as stated. Even in cases where $\widehat{f}$ is integrable, the formula cannot be proved simply by interchanging the order of integration, since a complex exponential is not integrable over the entire real line. In order to deal with these problems, it will be necessary to introduce a truncating factor to allow the interchange of integration, then to remove the effect of truncation by passage to a limit. The process will involve the continuous analogue of Cesàro summation.

For motivation, we begin by recalling the Cesàro means of an infinite series and their relation to convergence. Given a formal infinite series $\sum_{k=-\infty}^{\infty} \alpha_k$ of complex numbers, we form the partial sums $s_n = \sum_{k=-n}^{n} \alpha_k$ and their Cesàro means

$$\sigma_n = \frac{1}{n+1}(s_0 + s_1 + \cdots + s_n) = \sum_{k=-n}^{n} \left(1 - \frac{|k|}{n+1}\right) \alpha_k\,.$$

If the series $\sum \alpha_k$ converges to a sum s, or equivalently if $s_n \to s$ as $n \to \infty$, then the Cesàro means σ_n also converge to s as $n \to \infty$. We will require a continuous version of this fact, now stated as a lemma.

Lemma 1. *Let $\alpha(t)$ be a complex-valued function that is Riemann integrable over the real line. Then*

$$\lim_{R\to\infty} \int_{-R}^{R} \left(1 - \frac{|t|}{R}\right) \alpha(t)\, dt = \int_{-\infty}^{\infty} \alpha(t)\, dt\,.$$

Proof. By symmetry, it suffices to show that

$$\lim_{R\to\infty}\int_0^R \left(1-\frac{t}{R}\right)\alpha(t)\,dt = \int_0^\infty \alpha(t)\,dt\,.$$

Integrate by parts to obtain

$$\int_0^R \left(1-\frac{t}{R}\right)\alpha(t)\,dt = \frac{1}{R}\int_0^R A(t)\,dt\,, \qquad \text{where } A(t)=\int_0^t \alpha(s)\,ds\,.$$

Let $I=\lim_{t\to\infty} A(t)$. Given $\varepsilon>0$, choose M large enough that $|A(t)-I|<\varepsilon/2$ for all $t>M$. Then for every $R>M$ sufficiently large, we have

$$\left|\frac{1}{R}\int_0^R A(t)\,dt - I\right| \le \frac{1}{R}\int_0^M |A(t)-I|\,dt + \frac{1}{R}\int_M^R |A(t)-I|\,dt < \varepsilon\,,$$

which was to be proved. □

Another lemma, a result on approximate identities under convolution, will be the key to our proof of the inversion theorem for Fourier transforms. It is stated here in general form.

Lemma 2. *Let $g\in L^1$ with integral $\int_{-\infty}^{\infty} g(s)ds = 1$, and define $g_R(s) = R\,g(Rs)$ for $R>0$. Let $f\in L^1\cap L^\infty$, and suppose that f is continuous at some point $t\in\mathbb{R}$. Then the convolution $(f*g_R)(t)$ converges to $f(t)$ as $R\to\infty$. If f is uniformly continuous on $\mathbb{R}$, then $(f*g_R)(t)$ converges to $f(t)$ uniformly for all $t\in\mathbb{R}$.*

Proof. Note first that $\int_{-\infty}^{\infty} g_R(s)ds = \int_{-\infty}^{\infty} g(s)ds = 1$. By hypothesis, $|f(s)|\le M$ for some constant $M>0$ and all $s\in\mathbb{R}$. Given $\varepsilon>0$, choose $\delta>0$ small enough that $|f(t+s)-f(t)|<\varepsilon/2$ if $|s|<\delta$. Then write

$$(f*g_R)(t)-f(t) = \int_{-\infty}^{\infty}[f(t-s)-f(t)]\,g_R(s)ds = \int_{-\infty}^{-\delta} + \int_{-\delta}^{\delta} + \int_{\delta}^{\infty}$$

and make the estimate

$$|(f*g_R)(t)-f(t)| \le 2M\int_{-\infty}^{-\delta}|g_R(s)|\,ds + \frac{\varepsilon}{2}\int_{-\delta}^{\delta}|g_R(s)|\,ds + 2M\int_{\delta}^{\infty}|g_R(s)|\,ds\,.$$

But

$$\int_{-\delta}^{\delta}|g_R(s)|\,ds = R\int_{-\delta}^{\delta}|g(Rs)|\,ds \le \int_{-\infty}^{\infty}|g(s)|\,ds<\infty\,, \qquad \text{and}$$
$$\int_{\delta}^{\infty}|g_R(s)|\,ds = \int_{\delta R}^{\infty}|g(s)|\,ds \to 0 \qquad \text{as } R\to\infty\,.$$

Similarly, $\int_{-\infty}^{-\delta} |g_R(s)|\,ds \to 0$ as $R \to \infty$. It follows that $|(f*g_R)(t) - f(t)| \le \varepsilon$ for all R sufficiently large, and so $(f * g_R)(t) \to f(t)$ as $R \to \infty$. If f is uniformly continuous on the whole real line, the same argument shows that the convergence is uniform. □

Remark. If g is an even function and f is only piecewise continuous, essentially the same argument shows that $(f*g_R)(t) \to \frac{1}{2}[f(t+) + f(t-)]$ for every $t \in \mathbb{R}$. A function f is said to be piecewise continuous on a bounded interval if it is continuous there except perhaps for a finite number of jump-discontinuities, where the right-hand and left-hand limits $f(t+)$ and $f(t-)$ exist but are not equal. To say that a function is piecewise continuous on $\mathbb{R}$ means that it is piecewise continuous on each bounded subinterval.

We turn now to the inversion formula for Fourier transforms.

Inversion Theorem. *Let $f \in L^1 \cap L^\infty$. Then*

$$\lim_{R\to\infty} \frac{1}{2\pi} \int_{-R}^{R} e^{ixt} \left(1 - \frac{|x|}{R}\right) \widehat{f}(x)\,dx = f(t)$$

for each point $t \in \mathbb{R}$ where f is continuous. If f is uniformly continuous on $\mathbb{R}$, the integrals converge to $f(t)$ uniformly on $\mathbb{R}$.

Proof. By the definition of the Fourier transform $\widehat{f}(x)$, we have

$$\int_{-R}^{R} e^{ixt} \left(1 - \frac{|x|}{R}\right) \widehat{f}(x)\,dx = \int_{-R}^{R} e^{ixt} \left(1 - \frac{|x|}{R}\right) \int_{-\infty}^{\infty} e^{-ixs} f(s)\,ds\,dx$$
$$= \int_{-\infty}^{\infty} f(s) \int_{-R}^{R} e^{-ix(s-t)} \left(1 - \frac{|x|}{R}\right) dx\,ds = \int_{-\infty}^{\infty} f(s) g_R(t-s)\,ds\,,$$

where

$$g(s) = \frac{2}{s^2}(1 - \cos s) \qquad \text{and} \qquad g_R(s) = R\,g(Rs)\,.$$

Here the interchange of the order of integration is justified by the absolute integrability of the integrand, and Example 2 in the last section has been applied to calculate the Fourier transform of the function $\left(1 - \frac{|x|}{R}\right)$.

We are now in position to apply Lemma 2. Integration by parts gives

$$\int_{-\infty}^{\infty} g(s)\,ds = 2\int_{-\infty}^{\infty} \frac{1}{s^2}(1 - \cos s)\,ds = 4\int_{0}^{\infty} \frac{1}{s^2}(1 - \cos s)\,ds$$
$$= 4\int_{0}^{\infty} \frac{\sin s}{s}\,ds = 2\pi\,.$$

(See Exercise 12 for calculation of the last integral.) Thus Lemma 2, adjusted by the factor 2π, shows that

$$\int_{-\infty}^{\infty} f(s)g_R(t-s)\,ds = (f * g_R)(t) \to 2\pi f(t) \qquad \text{as} \quad R \to \infty\,,$$

wherever f is continuous, and the convergence is uniform if f is uniformly continuous on $\mathbb{R}$. □

The preceding statement about uniform convergence may be taken as a continuous analogue of Fejér's theorem for Fourier series, since it says that the Cesàro means of the Fourier integral of $\widehat{f}(x)$ converge uniformly to $f(t)$ when f is uniformly continuous. On the other hand, when the Fourier transform is actually integrable, the inversion formula requires no special interpretation.

Corollary. *Let $f \in L^1 \cap L^\infty$ and suppose that $\widehat{f} \in L^1$. Then if f is piecewise continuous on $\mathbb{R}$, it is uniformly continuous on $\mathbb{R}$ and*

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ixt}\widehat{f}(x)\,dx \qquad \textit{for every} \quad t \in \mathbb{R}\,. \tag{29}$$

Proof. Since $\widehat{f} \in L^1$, the integral converges. By a slight generalization of the inversion theorem (see the remark following the proof of Lemma 2), the integral is Cesàro summable to $\frac{1}{2}[f(t+) + f(t-)]$ for every $t \in \mathbb{R}$. Hence by Lemma 1 the integral converges to $\frac{1}{2}[f(t+) + f(t-)]$ for every $t \in \mathbb{R}$. But the integral is the Fourier transform of a function $\widehat{f} \in L^1$, so as a function of t it is uniformly continuous on $\mathbb{R}$. Thus $f(t+) = f(t-) = f(t)$ and f has the representation (29). □

The corollary says in particular that a piecewise continuous function $f \in L^1 \cap L^\infty$ cannot have a transform $\widehat{f}$ in L^1 unless f is continuous. Observe that in Examples 1 and 4 of the last section, the functions f are piecewise continuous but not continuous, and indeed $\widehat{f} \notin L^1$ in both cases.

By appeal to the inversion theorem we can easily calculate the Fourier transforms of Examples 5 and 6. Because in Examples 2 and 3 the functions f are bounded and continuous, with transforms $\widehat{f} \in L^1$, the formulas of Examples 5 and 6 follow at once from the corollary.

We conclude this section with a version of Parseval's relation for Fourier transforms. In general form this says that for functions $f, g \in L^1 \cap L^\infty$ and continuous on $\mathbb{R}$, with transforms $\widehat{f}, \widehat{g} \in L^1$, the formula

$$2\pi \int_{-\infty}^{\infty} f(t)\,\overline{g(t)}\,dt = \int_{-\infty}^{\infty} \widehat{f}(x)\,\overline{\widehat{g}(x)}\,dx$$

holds. Observe that $L^1 \cap L^\infty \subset L^2$, and so the Cauchy–Schwarz inequality ensures that both integrals converge. To verify the formula, we can apply (29) to write

$$2\pi \int_{-\infty}^{\infty} f(t)\,\overline{g(t)}\,dt = \int_{-\infty}^{\infty} f(t) \int_{-\infty}^{\infty} \overline{e^{ixt}\,\widehat{g}(x)}\,dx\,dt$$
$$= \int_{-\infty}^{\infty} \overline{\widehat{g}(x)} \int_{-\infty}^{\infty} e^{-ixt} f(t)\,dt\,dx = \int_{-\infty}^{\infty} \widehat{f}(x)\,\overline{\widehat{g}(x)}\,dx\,.$$

In particular, if f is continuous on $\mathbb{R}$ and $f \in L^1 \cap L^\infty$, and if $\widehat{f} \in L^1$, then

$$2\pi \int_{-\infty}^{\infty} |f(t)|^2\,dt = \int_{-\infty}^{\infty} |\widehat{f}(x)|^2\,dx\,.$$

It may be noted that the more general form of Parseval's relation follows easily from the particular form just stated, by means of the polarization identity (*cf.* Exercise 28).

8.11. Poisson summation formula

This chapter concludes with an important relation between a function and its Fourier transform. The Poisson summation formula resembles a discrete version of Parseval's relation and is very easy to prove, yet it has profound applications. The leading idea is to form a "periodization" of a given function on the real line and thus to introduce the discrete mechanism of Fourier series. Here is a statement of the result.

Poisson Summation Formula. *Let $f \in L^1$. Suppose that f is continuous on $\mathbb{R}$ and that the infinite series*

$$g(t) = \sum_{k=-\infty}^{\infty} f(t+k)$$

converges uniformly in each bounded interval. Suppose also that the series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(2\pi n)$$

is absolutely convergent. Then

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \widehat{f}(2\pi n)\,. \tag{30}$$

Proof. Observe first that under the hypothesis of uniform convergence the sum $g(t)$ is continuous and is periodic with period 1; that is, $g(t+1) = g(t)$ for all $t \in \mathbb{R}$. The Fourier coefficients of g, adapted to the change of scale, are

$$c_n = \int_0^1 g(t)\, e^{-2\pi int}\, dt = \sum_{k=-\infty}^{\infty} \int_0^1 f(t+k)\, e^{-2\pi int}\, dt$$
$$= \sum_{k=-\infty}^{\infty} \int_k^{k+1} f(t)\, e^{-2\pi int}\, dt = \int_{-\infty}^{\infty} f(t)\, e^{-2\pi int}\, dt = \widehat{f}(2\pi n)\,,$$

where the term-by-term integration is justified by uniform convergence of the series $\sum f(t+k)$. But the hypothesis $\sum |\widehat{f}(2\pi n)| < \infty$ implies that the Fourier series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(2\pi n)\, e^{2\pi int}\,.$$

converges absolutely and uniformly to a continuous sum $h(t)$ for which $h(t+1) \equiv h(t)$. Since both h and g are continuous periodic functions with the same Fourier coefficients, it follows that $h(t) \equiv g(t)$. (See Exercise 21.) Therefore,

$$g(t) = \sum_{n=-\infty}^{\infty} f(t+n) = \sum_{n=-\infty}^{\infty} \widehat{f}(2\pi n)\, e^{2\pi int}\,. \tag{31}$$

Now set $t = 0$ to obtain the Poisson summation formula (30). It may be remarked that the formula (31) is really no more general than (30), since it follows from (30) by a simple translation of the function f. □

As a corollary, we see that if $\widehat{f}(x) = 0$ for all x outside the open interval $(-2\pi, 2\pi)$, then

$$\int_{-\infty}^{\infty} f(t)\, dt = \widehat{f}(0) = \sum_{n=-\infty}^{\infty} f(n)\,.$$

For a first application, consider Jacobi's theta function

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}\,, \qquad t > 0\,.$$

For fixed $t > 0$ we can write

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} f(n)\,, \qquad \text{where } \; f(s) = e^{-\pi t s^2}\,.$$

But by Example 7 in Section 8.9, the Fourier transform of f is

$$\widehat{f}(x) = (1/\sqrt{t})e^{-x^2/4\pi t}\,,$$

and so

$$\sum_{n=-\infty}^{\infty} \widehat{f}(2\pi n) = (1/\sqrt{t}) \sum_{n=-\infty}^{\infty} e^{-\pi n^2/t} = (1/\sqrt{t})\,\vartheta(1/t)\,.$$

Consequently, the remarkable identity

$$\vartheta(1/t) = \sqrt{t}\;\vartheta(t)\,,$$

known as Jacobi's inversion formula, follows from the Poisson summation formula.

For another application, let

$$f(t) = \left(\frac{\sin \pi t}{\pi t}\right)^2 .$$

According to Example 5, this function has Fourier transform

$$\widehat{f}(x) = \begin{cases} \left(1 - \frac{|x|}{2\pi}\right) & \text{for } |x| \le 2\pi \\ 0 & \text{for } |x| > 2\pi\,. \end{cases}$$

In particular, $\widehat{f}(2\pi n) = 0$ for all n except $n = 0$, and $\widehat{f}(0) = 1$. Hence the Poisson summation formula in the form (31) evaluates the sum

$$\sum_{n=-\infty}^{\infty} \left(\frac{\sin \pi(n+t)}{\pi(n+t)}\right)^2 = 1\,, \qquad t \in \mathbb{R}\,.$$

But $\sin^2 \pi(n+t) = \sin^2 \pi t$ for all n, so the expansion reduces to

$$\left(\frac{\sin \pi t}{\pi}\right)^2 \sum_{n=-\infty}^{\infty} \frac{1}{(n+t)^2} = 1\,,$$

which implies that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+t)^2} = \Big(\frac{\pi}{\sin \pi t}\Big)^2$$

if t is not an integer. For $t = \frac{1}{2}$ this says that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}\,,$$

which is equivalent to Euler's sum $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.

The Poisson summation formula can also be applied to prove the sampling theorem, a result of great importance in signal analysis. Here a signal is understood to be a continuous function $f(t)$ of time t, where $-\infty < t < \infty$. It is said to be *band-limited* if its Fourier transform vanishes outside some bounded interval. Band-limited signals are analogous to trigonometric polynomials and have similar structural properties.

Sampling Theorem. *Let $f \in L^1 \cap L^\infty$ be a continuous function with Fourier transform $\widehat{f}(x) = 0$ for $|x| \geq \pi$, and suppose that $\sum_{n=-\infty}^{\infty} |f(n)| < \infty$. Then f is determined by its values at the integers, and*

$$f(t) = \sum_{n=-\infty}^{\infty} f(n) \frac{\sin \pi(t-n)}{\pi(t-n)}\,, \qquad t \in \mathbb{R}\,. \tag{32}$$

The formula (32) is known to mathematicians as the *cardinal series*. A more general form can be deduced by change of scale when $\widehat{f}(x) = 0$ for $|x| \geq a$, where a is an arbitrary positive number. When t is an integer, the formula is to be interpreted in a limiting sense. Note that for any integer m,

$$\lim_{t \to m} \frac{\sin \pi(t-n)}{\pi(t-n)} = \begin{cases} 0 & \text{for } m \neq n \\ 1 & \text{for } m = n\,. \end{cases}$$

Although the sampling theorem can be obtained as a corollary of the Poisson summation formula, it is simpler to give a direct proof. The following proof was found by Boas [**2**], who also derived the theorem from the Poisson summation formula.

Proof of sampling theorem. Since $\widehat{f}(x) = 0$ for $|x| \geq \pi$, and $\widehat{f}(x)$ is continuous, it is clear that $\widehat{f} \in L^1$. Thus the Fourier transform can be inverted to write

$$f(t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{itx} \widehat{f}(x)\, dx\,, \qquad t \in \mathbb{R}\,.$$

In particular, the Fourier coefficients of the periodic extension of $\widehat{f}$ are

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \widehat{f}(x)\, dx = f(-n)\,.$$

Thus by hypothesis, $\widehat{f}$ has an absolutely convergent Fourier series, which therefore converges uniformly to $\widehat{f}(x)$:

$$\widehat{f}(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} = \sum_{n=-\infty}^{\infty} f(n) e^{-inx}\,, \qquad -\pi \leq x \leq \pi\,.$$

In view of the uniform convergence, the series can be multiplied by the exponential e^{itx} and integrated term by term to obtain

$$\begin{aligned} f(t) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{itx} \widehat{f}(x)\, dx = \sum_{n=-\infty}^{\infty} f(n)\, \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(t-n)x}\, dx \\ &= \sum_{n=-\infty}^{\infty} f(n)\, \frac{1}{\pi} \int_{0}^{\pi} \cos(t-n)x\, dx = \sum_{n=-\infty}^{\infty} f(n) \frac{\sin \pi(t-n)}{\pi(t-n)}\,. \end{aligned}$$

□

The article by Higgins [**5**] may be consulted for further information about the cardinal series. The book by Folland [**3**] offers a broad discussion of Fourier transforms and their applications to problems of physics and engineering.

Siméon-Denis Poisson (1781–1840) was a student at École Polytechnique in Paris, where Laplace and Lagrange recognized his talent and promoted his cause. At age 25 he became a professor there, the successor to Fourier. He made important contributions to mathematical physics (electricity and magnetism), celestial mechanics, and probability theory (the Poisson distribution). The Poisson summation formula was known to Poisson, among others, as early as 1823.

Exercises

1. Verify the formula

$$x = 2\left(\sin x - \tfrac{1}{2}\sin 2x + \tfrac{1}{3}\sin 3x - \dots\right), \qquad -\pi < x < \pi,$$

and conclude that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

2. Calculate the Fourier series

$$\left(\frac{\pi - x}{2}\right)^2 = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}, \qquad 0 \le x \le 2\pi.$$

Note the special cases

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6} \qquad \text{and} \qquad 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}.$$

3. Show that

$$\pi t \csc \pi t = 1 + 2t^2 \sum_{n=1}^{\infty} \frac{(-1)^n}{t^2 - n^2}, \qquad t \neq 0, \pm 1, \pm 2, \dots.$$

Suggestion. Set $x = 0$ in the Fourier expansion (15) of $\cos cx$.

4. Show that

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2nx}{4n^2 - 1} = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{\sin^2 nx}{4n^2 - 1}, \qquad -\infty < x < \infty.$$

Hint. $\frac{1}{2n-1} - \frac{1}{2n+1} = \frac{2}{4n^2-1}$.

5. (*a*) If a function $f(x)$ is periodic with period 2π, and if f has a continuous derivative of order k, show that the Fourier coefficients of f satisfy

$$|a_n| \le C/n^k \qquad \text{and} \qquad |b_n| \le C/n^k$$

for some constant $C > 0$.

(*b*) If a periodic function $f(x)$ has a continuous second derivative, show that its Fourier series converges uniformly to $f(x)$.

6. (*a*) Show that the Fourier coefficients of a continuous periodic piecewise linear function f satisfy

$$|a_n| \le C/n^2 \qquad \text{and} \qquad |b_n| \le C/n^2$$

for some constant $C > 0$. Conclude that the Fourier series converges uniformly to $f(x)$.

(*b*) Show that every continuous periodic function can be approximated uniformly by a continuous piecewise linear function with the same period.

(*c*) Deduce the trigonometric form of the Weierstrass approximation theorem.

7. Prove that

$$\int_{-\pi}^{\pi} |D_n(t)|\, dt \sim \frac{4}{\pi} \log n\,, \qquad n \to \infty\,,$$

where $D_n(t)$ is the Dirichlet kernel.

8. Show that $h(x) = \sin x - x + x^2/\pi$ has the properties $h(0) = h(\pi) = 0$, $h'(x) > 0$ for $0 < x < \pi/2$, and $h'(x) < 0$ for $\pi/2 < x < \pi$. Conclude that

$$\frac{\sin x}{x} > 1 - \frac{x}{\pi}\,, \qquad 0 < x < \pi\,.$$

9. (*a*) Show that the partial sums

$$s_n(x) = \sum_{k=1}^{n} \frac{\sin kx}{k}$$

of Example 3 in Section 8.5 have the property $s_n(x) > 0$ for $0 < x < \pi$.

Hint. Proceed by induction. Assuming that $s_{n-1}(x) > 0$ for $0 < x < \pi$, investigate the sum $s_n(x)$ at a local minimum.

(*b*) Show that $|s_n(x)| \le 1 + \pi/2$ for all $x \in \mathbb{R}$.

Suggestion. Use the Fejér means $\sigma_n(x)$ as indicated at the end of Section 8.7.

10. (*a*) Derive the formula

$$\sum_{k=0}^{n} \cos(2k+1)x = \frac{\sin 2(n+1)x}{2\sin x}\,, \qquad n = 1, 2, \ldots\,.$$

(*b*) For the Fourier series of the square wave function of Example 2, show that the partial sums

$$s_{2n+1}(x) = \frac{4}{\pi} \sum_{k=0}^{n} \frac{1}{2k+1} \sin(2k+1)x$$

have the critical points

$$x = \frac{k\pi}{2(n+1)}\,, \qquad k = 1, 2, \ldots, 2n+1\,,$$

in the interval $(0, \pi)$, which are alternately local maxima and local minima.

(*c*) By a direct method similar to that of Section 8.6, show that

$$\lim_{n\to\infty} s_{2n+1}\left(\frac{\pi}{2(n+1)}\right) = \frac{2}{\pi}\int_0^{\pi} \frac{\sin x}{x}\,dx = 1.1789\ldots\,,$$

so that the partial sums exhibit Gibbs' phenomenon.

11. For the Fourier series of the discontinuous sawtooth function of Example 3, show that the partial sums exhibit an "undershoot" as well as the "overshoot" of the Gibbs phenomenon. Specifically, for the first positive local minimum $\lambda_n = 2\pi/n$, show that

$$\lim_{n\to\infty} s_n(\lambda_n) = \int_0^{2\pi} \frac{\sin x}{x}\,dx = 1.4181\ldots = (0.9028\ldots)\frac{\pi}{2}\,.$$

12. Use the Dirichlet kernel to calculate the integral

$$\int_0^{\infty} \frac{\sin x}{x}\,dx = \frac{\pi}{2}\,.$$

Writing

$$\int_0^{\pi} D_n(x)\,dx = \int_0^{\pi} \sin(n+\tfrac{1}{2})x \left(\frac{1}{2\sin\frac{1}{2}x} - \frac{1}{x}\right) dx + \int_0^{\pi} \frac{\sin(n+\frac{1}{2})x}{x}\,dx\,,$$

show that the first integral on the right-hand side tends to 0 as $n \to \infty$.

13. If $f(x)$ is a continuous function of period 2π, except for a jump discontinuity at a point x_0 where it has one-sided limits $f(x_0+)$ and $f(x_0-)$, show that its Fejér means have the property

$$\lim_{n\to\infty} \sigma_n(x) = \tfrac{1}{2}[f(x_0+) + f(x_0-)]\,.$$

Here no smoothness hypothesis is imposed.

14. The function

$$\widetilde{D}_n(x) = \sum_{k=1}^{n} \sin kx$$

is known as the *conjugate Dirichlet kernel*. Verify the trigonometric identity

$$\cos(k - \tfrac{1}{2})x - \cos(k + \tfrac{1}{2})x = 2\sin\tfrac{1}{2}x\,\sin kx$$

and use it to calculate

$$\widetilde{D}_n(x) = \frac{\cos\frac{1}{2}x - \cos(n+\frac{1}{2})x}{2\sin\frac{1}{2}x}\,.$$

Conclude that $|\widetilde{D}_n(x)| \le \pi/x$ for $0 < x \le \pi$.

15. If $b_{n+1} \le b_n$ and $b_n \to 0$ as $n \to \infty$, and the numbers nb_n are bounded, prove that the partial sums

$$s_n(x) = \sum_{k=1}^{n} b_k \sin kx$$

are uniformly bounded on $0 \le x \le 2\pi$, hence on the entire real line.

Outline of proof. Suppose $nb_n \le M$, $n = 1, 2, \ldots$. Let N be the integer defined by $N \le \pi/x < N + 1$. For $n \le N$, show that $|s_n(x)| \le \pi M$. For $n > N$, split the sum as

$$s_n(x) = \sum_{k=1}^{N} b_k \sin kx + \sum_{k=N+1}^{n} b_k \sin kx$$

and apply Abel summation to the second sum to conclude that $|s_n(x)| \le (\pi + 2)M$.

16. (*a*) If $a_{n+1} \le a_n$ and $a_n \to 0$ as $n \to \infty$, show that the series

$$\tfrac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

converges in the interval $0 < x < 2\pi$ and converges uniformly in $\delta \le x \le 2\pi - \delta$ for each $\delta > 0$.

(*b*) Similarly, if $b_{n+1} \le b_n$ and $b_n \to 0$ as $n \to \infty$, show that the series

$$\sum_{n=1}^{\infty} b_n \sin nx$$

converges for all $x \in \mathbb{R}$ and converges uniformly in $\delta \le x \le 2\pi - \delta$ for each $\delta > 0$.

Hint. Use Abel summation.

(*c*) Integrate the Fourier series

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \frac{\pi - x}{2}\,, \qquad 0 < x < 2\pi\,,$$

of Example 3 in Section 8.5 to calculate (*cf.* Exercise 2)

$$\sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = \left(\frac{x-\pi}{2}\right)^2 - \frac{\pi^2}{12}\,.$$

17. (*a*) By expanding the given function into Fourier series, show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \sin n \sin nx = \begin{cases} \frac{1}{2}(\pi - 1)x\,, & 0 \le x \le 1 \\ \frac{1}{2}(\pi - x)\,, & 1 \le x \le \pi\,. \end{cases}$$

Conclude that

$$\sum_{n=1}^{\infty} \frac{\sin n}{n} = \sum_{n=1}^{\infty} \left(\frac{\sin n}{n}\right)^2 = \frac{\pi - 1}{2}\,.$$

This observation is due to Robert Baillie [**1**].

(*b*) Apply the result of Exercise 2 (see also Exercise 16) to show that

$$\sum_{n=1}^{\infty} \frac{\sin^2 nx}{n^2} = \frac{\pi x - x^2}{2}\,, \qquad 0 \le x \le \pi\,,$$

thus generalizing the result in Part (a), where $x = 1$.

18. Let $f(\theta)$ be a continuous function of period 2π, and let

$$\tfrac{1}{2}\,a_0 + \sum_{n=1}^{\infty} (a_n \cos n\theta + b_n \sin n\theta)$$

be its formal Fourier series.

(*a*) Show that the series

$$u(r,\theta) = \tfrac{1}{2}\,a_0 + \sum_{n=1}^{\infty}(a_n \cos n\theta + b_n \sin n\theta)r^n$$

converges uniformly in θ for each fixed $r \in [0,1)$.

(*b*) For the *Poisson kernel*

$$P(r,\theta) = \frac{1-r^2}{1-2r\cos\theta + r^2}\,,$$

verify the infinite series expansion

$$P(r,\theta) = 1 + 2\sum_{n=1}^{\infty} r^n \cos n\theta\,, \qquad 0 \le r < 1\,.$$

Note that this formula can be viewed either as a Fourier expansion of $P(r,\theta)$ for fixed r or as a Taylor series expansion for fixed θ. *Suggestion:* Show that

$$P(r,\theta) = \operatorname{Re}\left\{\frac{1+re^{i\theta}}{1-re^{i\theta}}\right\}\,.$$

(*c*) Use integral expressions for the Fourier coefficients a_n and b_n to establish the *Poisson formula*

$$u(r,\theta) = \frac{1}{2\pi}\int_0^{2\pi} P(r,\theta-t)f(t)\,dt\,, \qquad 0 \le r < 1\,.$$

(*d*) Deduce from Fejér's theorem that $u(r,\theta) \to f(\theta)$ uniformly in θ as $r \to 1$. (Compare with Exercise 17 in Chapter 6.) Interpret the result as a statement that the Fourier series of a continuous periodic function $f(\theta)$ is uniformly Abel summable to $f(\theta)$. (Apply a uniform version of Frobenius' theorem in Chapter 7.)

19. (*a*) If $\varphi(x)$ is a function of bounded variation over the interval $[-\pi,\pi]$, integrate by parts to show that its Fourier coefficients

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\,\varphi(x)\,dx \qquad \text{and} \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin nx\,\varphi(x)\,dx$$

satisfy $a_n = O(1/n)$ and $b_n = O(1/n)$ as $n \to \infty$.

(*b*) Invoke Littlewood's Tauberian theorem to conclude that the Fourier series of a function of bounded variation is convergent wherever it is Abel summable.

20. (*a*) Suppose that a function $f(x)$ is continuous and periodic with period 2π on the real line, and its Fourier coefficients satisfy $a_n = O(1/n)$ and $b_n = O(1/n)$ as $n \to \infty$. Invoke Hardy's Tauberian theorem to show that the Fourier series converges to $f(x)$ uniformly on $\mathbb{R}$.

(*b*) Conclude that if $f(x)$ is a continuous function of bounded variation over $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$, then the Fourier series converges to $f(x)$ uniformly in $[-\pi, \pi]$.

21. Let $f(x)$ be a continuous function with period 2π, and suppose that all of its Fourier coefficients vanish:

$$\int_{-\pi}^{\pi} f(x)\,dx = 0 \qquad \text{and} \qquad \int_{-\pi}^{\pi} f(x)\cos nx\,dx = \int_{-\pi}^{\pi} f(x)\sin nx\,dx = 0$$

for $n = 1, 2, 3, \ldots$. Show that $f(x) \equiv 0$.

Suggestion. Apply the trigonometric form of the Weierstrass approximation theorem. Compare with Exercise 10 in Chapter 6.

22. Let $f(x)$ be a continuous function with period 2π, and suppose that its Fourier series is absolutely convergent in the sense that $\sum_{n=1}^{\infty}\big(|a_n| + |b_n|\big) < \infty$. Show that the Fourier series converges uniformly to the sum $f(x)$. In other words, show that $s_n(x) \to f(x)$ uniformly on the real line as $n \to \infty$.

23. Suppose that a function $f(x)$ has a continuous derivative on the interval $[0, 2\pi]$ and that $f(0) = f(2\pi)$ and $\int_0^{2\pi} f(x)dx = 0$. Use Parseval's relation to show that

$$\int_0^{2\pi} [f(x)]^2\,dx \le \int_0^{2\pi} [f'(x)]^2\,dx\,,$$

with strict inequality unless $f(x) = A\cos x + B\sin x$ for some constants A and B. The result is known as *Wirtinger's inequality.*

24. Apply Parseval's relation to prove the isoperimetric inequality: Among all simple closed curves of given length, the circle encloses the region of largest area. Let C be a smooth simple closed curve of length 2π, represented parametrically by $(x, y) = (f(s), g(s))$, $0 \le s \le 2\pi$, where s is arclength. Then the coordinate functions have Fourier series expansions

$$x = f(s) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos ns + b_n\sin ns)\,,$$

$$y = g(s) = \frac{c_0}{2} + \sum_{n=1}^{\infty}(c_n\cos ns + d_n\sin ns)\,.$$

(a) Apply Green's theorem to show that the area of the region enclosed by C is

$$A = \int_C x\,dy = \pi \sum_{n=1}^{\infty} n(a_n d_n - b_n c_n)\,.$$

(b) Noting that $(dx/ds)^2 + (dy/ds)^2 = 1$, show by integration that

$$\pi \sum_{n=1}^{\infty} n^2(a_n^2 + b_n^2 + c_n^2 + d_n^2) = 2\pi\,.$$

(c) Combine the results of parts (a) and (b) to deduce that

$$2(\pi - A) = \pi \sum_{n=1}^{\infty} (n^2 - n)(a_n^2 + b_n^2 + c_n^2 + d_n^2) + \pi \sum_{n=1}^{\infty} n[(a_n - d_n)^2 + (b_n + c_n)^2]\,.$$

(d) Conclude that $A < \pi$ unless $a_1 = d_1$, $b_1 = -c_1$, and $a_n = b_n = c_n = d_n = 0$ for every $n > 1$, so that C is a circle.

Note. This argument is due to Hurwitz [**6**]. It is closely related to the proof of Wirtinger's inequality suggested in Exercise 23.

25. Show that

$$\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{6^2}\right)\left(1 - \frac{1}{9^2}\right)\cdots = \frac{3\sqrt{3}}{2\pi} \qquad \text{and}$$
$$\left(1 - \frac{1}{6^2}\right)\left(1 - \frac{1}{12^2}\right)\left(1 - \frac{1}{18^2}\right)\cdots = \frac{3}{\pi}\,.$$

26. Refer to Example 3 in Section 8.9 for the Fourier transform of $f(t) = e^{-a|t|}$, where $a > 0$.

(a) Calculate the Fourier transform of $f'(t)$.

(b) Use the Parseval relation to calculate the integral

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)(x^2 + b^2)}\,dx = \frac{\pi}{a + b}\,, \qquad a > 0\,,\ b > 0\,.$$

27. Calculate the Fourier transform of the function $f(t) = \frac{\sin t}{t}$, and explain your reasoning. Note that f is Riemann integrable but $|f|$ is not. In other words, $f \notin L^1$.

28. For functions $f, g \in L^2$, let

$$\langle f, g\rangle = \int_{-\infty}^{\infty} f(t)\overline{g(t)}\,dt\,,$$

and let $\|f\|^2 = \langle f, f\rangle$. Verify the *polarization identity*

$$4\langle f, g\rangle = \|f+g\|^2 - \|f-g\|^2 + i\|f+ig\|^2 - i\|f-ig\|^2\,,$$

and apply it to derive the general version $2\pi\langle f, g\rangle = \langle \widehat{f}, \widehat{g}\rangle$ of Parseval's relation from the particular form where $f = g$.

29. Apply the Poisson summation formula to calculate the sum

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+t^2} = \frac{\pi}{t}\,\frac{1+e^{-2\pi t}}{1-e^{-2\pi t}}\,, \qquad t>0\,.$$

Use the result to show that $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.

Hint. Subtract $1/t^2$ from both sides of the equation and find the limit as $t \to 0$. Use the Taylor expansion of the exponential function.

30. Use Fourier transforms to find a solution of the ordinary differential equation $y'' + y = g(t)$, where g is an arbitrary continuous function in L^1.

31. Derive the sampling theorem from the Poisson summation formula.

32. Generalize the sampling theorem by showing that if $\widehat{f}(x) = 0$ for $|x| \geq a$, then

$$f(t) = \sum_{n=-\infty}^{\infty} f(n\pi/a)\,\frac{\sin(at-n\pi)}{at-n\pi}\,.$$

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Chapter 9
The Gamma Function

The gamma function is a continuous extension of the factorial function. It is an important function with many remarkable properties. It can be represented as an integral and as an infinite product. In this chapter we develop the basic theory of the gamma function and its relation to the beta function. By way of background we begin with an integral that plays a central role in probability theory.

9.1. Probability integral

The integral

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

is important in probability theory because of its connection with the Gaussian or normal distribution. It is evaluated by a well known device that makes use of area integrals. Since the integrand is an even function, it is equivalent to show that

$$\int_{0}^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}\,. \tag{1}$$

To evaluate the integral in (1), we consider the function

$$f(x,y) = e^{-x^2}e^{-y^2}$$

and the three regions of integration shown in Figure 1. With the notation D_1 and D_2 for for the quarter-disks with respective radii R and $\sqrt{2}R$, and

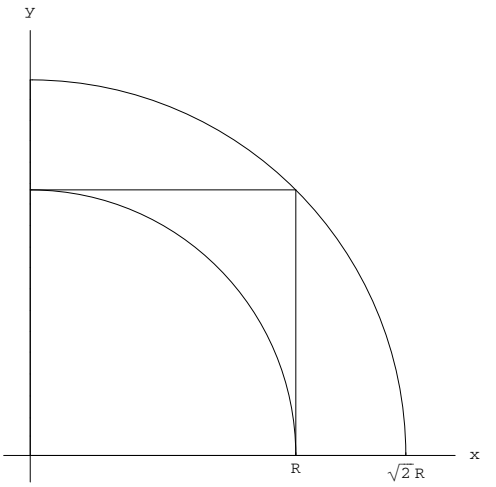


Figure 1. Three regions of integration.

S for the square with side-length R, we observe that $D_1 \subset S \subset D_2$. Since $f(x,y) > 0$, it is therefore clear that

$$\iint_{D_1} f(x,y)\,dA \le \iint_{S} f(x,y)\,dA \le \iint_{D_2} f(x,y)\,dA\,.$$

Now introduce polar coordinates in the regions D_1 and D_2 through the substitutions $x = r\cos\theta$ and $y = r\sin\theta$, and iterate all three integrals to conclude that

$$\int_0^{\pi/2}\int_0^{R} e^{-r^2}\,r\,dr\,d\theta \le \int_0^{R}\int_0^{R} e^{-x^2}e^{-y^2}\,dx\,dy \le \int_0^{\pi/2}\int_0^{\sqrt{2}R} e^{-r^2}\,r\,dr\,d\theta\,.$$

The advantage of the area-integral approach is that the polar-coordinate integrals can be calculated explicitly. These calculations reduce the inequalities to

$$\frac{\pi}{4}\left(1 - e^{-R^2}\right) \le \left(\int_0^{R} e^{-x^2}dx\right)^2 \le \frac{\pi}{4}\left(1 - e^{-2R^2}\right)\,.$$

Now let $R \to \infty$ to obtain

$$\left(\int_0^{\infty} e^{-x^2}dx\right)^2 = \frac{\pi}{4}\,,$$

which is equivalent to (1).

9.2. Gamma function

In the early development of calculus, there were various efforts to analyze the factorial function $n! = 1 \cdot 2 \cdot 3 \cdots n$ and the related binomial coefficients. Stirling's formula

$$n! \sim n^n e^{-n} \sqrt{2\pi n}\,, \qquad n \to \infty\,,$$

was published in 1730. Around the same time, Euler had been searching for a natural extension of the factorial function. The problem was to find a function $f(x)$, defined by an analytic formula for all $x > 0$, to interpolate the values $f(n) = n!$ at the positive integers. In 1729, Euler sketched a suitable construction in a letter to Christian Goldbach (1690–1764), who had proposed the problem along with Daniel Bernoulli (1700–1784), the son of Euler's teacher Johann Bernoulli. In fact, this letter was the first in a long series of mathematical correspondence between Euler and Goldbach, a very talented amateur mathematician who asked thoughtful questions. ("Goldbach's conjecture", that every even number larger than 2 is the sum of two primes, remains unsettled.)

Euler published the details of his construction, but soon found that his solution could be expressed in the equivalent integral form

$$f(x) = \int_0^1 (-\log t)^x \, dt\,.$$

Focusing on the integral representation, he then studied the function and developed its basic properties. Only later did Adrien-Marie Legendre (1752–1833) propose the notation $\Gamma(x)$ and the now standard definition of the gamma function:

$$\Gamma(x) = \int_0^\infty e^{-t}\, t^{x-1}\, dt\,, \qquad x > 0\,. \tag{2}$$

A change of variables shows that $\Gamma(x+1)$ is the same as Euler's function $f(x)$.

One may well wonder what is so special about the gamma function. There are many functions $g(x)$, infinitely differentiable for $x > 0$, with the property $g(n) = n!$ for all integers $n = 1, 2, \ldots$. Why does the gamma function provide the "right" extension of the factorial? One response is that the gamma function not only has the values $\Gamma(n+1) = n!$, but as we shall see presently, it also satisfies the functional equation $\Gamma(x+1) = x\Gamma(x)$ for all $x > 0$. Further justification comes from the Bohr–Mollerup theorem, to be discussed later, but the real case for the gamma function is its frequent appearance in formulas of mathematical analysis.

Integration by parts shows that

$$\Gamma(x+1) = \int_0^\infty e^{-t}\, t^x \, dt = \Big[-e^{-t} t^x\Big]_0^\infty + x \int_0^\infty e^{-t}\, t^{x-1}\, dt = x\Gamma(x)\,.$$

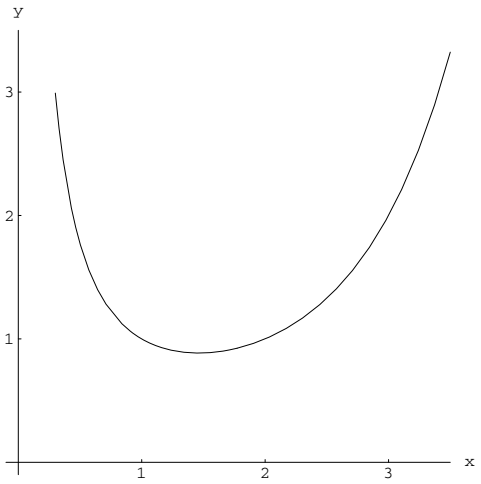


Figure 2. Graph of $y = \Gamma(x)$.

Since $\Gamma(1) = 1$, the formula $\Gamma(x+1) = x\Gamma(x)$ can be iterated to give $\Gamma(2) = 1$, $\Gamma(3) = 2$, and in general $\Gamma(n+1) = n!$ for $n = 1, 2, \ldots$. It is clear that $\Gamma(x) > 0$ for all $x > 0$, because the integrand is positive. A graph of the gamma function is shown in Figure 2.

The graph suggests that $\Gamma(x)$ descends to a minimum value at a point x_0 in the interval $1 < x_0 < 2$, and is increasing for $x > x_0$. It is not difficult to verify this (see Exercise 3). Gauss was interested in the number $x_0 = 1.4616\ldots$ and calculated it to many decimal places. He also calculated $\Gamma(x_0) = 0.8856\ldots$. Another feature suggested by the graph is that $\Gamma(x)$ is a convex function. In fact, we will see later that $\log \Gamma(x)$ is convex, a stronger property. The graph of $\log \Gamma(x)$ is displayed in Figure 3.

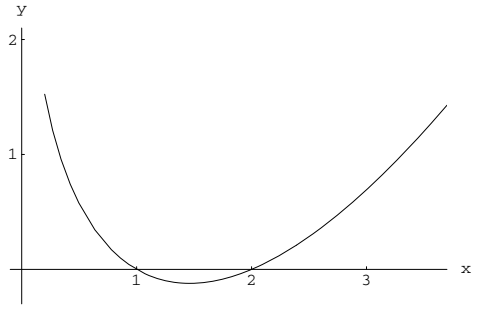


Figure 3. Graph of $y = \log \Gamma(x)$.

With the substitution $t = u^2$, the gamma function takes the form

$$\Gamma(x) = 2\int_0^\infty e^{-u^2} u^{2x-1}\, du\,, \qquad x > 0\,. \tag{3}$$

In view of the formula (1), this shows that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

The functional equation $\Gamma(x+1) = x\Gamma(x)$ can be iterated to show more generally that

$$\Gamma(x+n) = (x+n-1)(x+n-2)\cdots(x+1)x\Gamma(x)\,, \qquad x > 0\,,$$

for $n = 1, 2, \ldots$, or

$$\Gamma(x) = \frac{\Gamma(x+n)}{x(x+1)\cdots(x+n-1)}\,, \qquad x > 0\,.$$

The last formula allows the definition of $\Gamma(x)$ to be extended to the interval $x > -n$, provided that x is not a negative integer or 0. Since n is an arbitrary positive integer, this extends the definition of $\Gamma(x)$ to the entire real line, excluding the singular points $x = 0, -1, -2, \ldots$.

9.3. Beta function

The *beta function* $B(x, y)$ is defined by

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\, dt\,, \qquad x > 0\,,\ y > 0\,. \tag{4}$$

The change of variables $t = 1 - s$ reveals the symmetry property $B(x, y) = B(y, x)$. However, the most important property of the beta function is its expression in terms of the gamma function:

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}\,. \tag{5}$$

The proof of (5) makes use of the alternate expression

$$B(x,y) = 2\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1}\, d\theta\,, \tag{6}$$

which results from the substitution $t = \cos^2\theta$. The argument is very similar to the calculation of the probability integral (1). The proof begins with a comparison of integrals of the positive function

$$g(u,v) = u^{2x-1}\, v^{2y-1}\, e^{-(u^2+v^2)}$$

over the same three regions as before, as displayed in Figure 1. Here, however, x and y are fixed positive parameters, whereas u and v are the variables of integration. After passing to polar coordinates with the substitutions $u = r\cos\theta$ and $v = r\sin\theta$, we arrive at the formulas

$$\begin{aligned} &\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1}\,d\theta \int_0^R e^{-r^2}\, r^{2x+2y-1}\,dr \\ \le\ &\int_0^R e^{-u^2}\, u^{2x-1}\,du \int_0^R e^{-v^2}\, v^{2y-1}\,dv \\ \le\ &\int_0^{\pi/2} (\cos\theta)^{2x-1}(\sin\theta)^{2y-1}\,d\theta \int_0^{\sqrt{2}R} e^{-r^2}\, r^{2x+2y-1}\,dr\,. \end{aligned}$$

Letting $R \to \infty$ and referring to (3) and (6), we conclude that

$$B(x,y)\Gamma(x+y) \le \Gamma(x)\Gamma(y) \le B(x,y)\Gamma(x+y)\,,$$

and (5) follows. Another proof is outlined in the exercises.

9.4. Legendre's duplication formula

Legendre's duplication formula for the gamma function expresses $\Gamma(2x)$ in terms of $\Gamma(x)$ and $\Gamma(x+\frac{1}{2})$. It is

$$\sqrt{\pi}\,\Gamma(2x) = 2^{2x-1}\Gamma(x)\,\Gamma(x+\tfrac{1}{2})\,, \qquad x>0\,. \tag{7}$$

Many proofs have been found. We will give a particularly elementary proof that uses the basic connection (5) between the beta and gamma functions, plus the special formula

$$B(x,x) = 2^{1-2x}B(x,\tfrac{1}{2})\,, \qquad x>0\,. \tag{8}$$

To prove (8), observe first that because of the symmetry of the integrand in (4) when $x = y$, we can write

$$B(x,x) = 2\int_0^{1/2} \big[t(1-t)\big]^{x-1}\,dt\,, \qquad x>0\,.$$

Next observe that $u = 4t(1-t)$ increases from 0 to 1 as t increases from 0 to $\frac{1}{2}$. Make this substitution in the integral, noting that

$$du = 4(1-2t)\,dt \qquad \text{and} \qquad 1-u = (1-2t)^2\,,$$

to arrive at the formula

$$B(x,x) = 2^{1-2x}\int_0^1 u^{x-1}(1-u)^{-1/2}\,du = 2^{1-2x}B(x,\tfrac{1}{2})\,,$$

which proves (8).

Legendre's formula (7) is now deduced from (8) through the relation (5). When (5) is substituted into (8), the formula reduces to

$$\frac{\Gamma(x)^2}{\Gamma(2x)} = 2^{1-2x}\frac{\Gamma(x)\,\Gamma(\frac{1}{2})}{\Gamma(x+\frac{1}{2})}\,.$$

Since $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, this is the Legendre duplication formula.

It may be observed that the special formula (8) is actually a disguised expression of Legendre's formula.

The name of Adrien-Marie Legendre (1752–1833) occurs frequently in topics of classical analysis. In Chapter 13 of this book we will encounter the Legendre differential equation and the associated Legendre polynomials, which are orthogonal over the interval $[-1, 1]$. Chapter 14 contains a discussion of the Legendre relation for elliptic integrals, a field that Legendre cultivated for many years. Yet some of Legendre's greatest contributions lay in celestial mechanics and in number theory. For instance, he developed and applied (before Gauss) the method of least squares for determination of astronomical orbits. He formulated the prime number theorem (*cf.* Chapter 10), the theorem that every arithmetic progression without common factor contains infinitely many primes (proved later by Dirichlet), and the law of quadratic reciprocity, verified by Gauss after Legendre had found a partial proof. Little is known about Legendre's personal life. A portrait traditionally believed to be that of Adrien-Marie Legendre has now been discredited and replaced by a recently discovered caricature portrait, the only true image of Legendre known to exist. An article by Duren [**5**] tells the story of the two portraits.

9.5. Euler's reflection formula

The important relation

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}\,, \qquad 0 < x < 1\,, \tag{9}$$

known as *Euler's reflection formula*, links the gamma function with the sine function. It lies deeper than other properties of the gamma function, and there is no simple derivation. We will base a proof on the formula

$$\cos cx = \frac{2c}{\pi}\sin c\pi \left\{ \frac{1}{2c^2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{c^2 - n^2}\cos nx \right\}\,, \qquad -\pi \le x \le \pi\,, \tag{10}$$

which we derived in Section 8.5 from the convergence theorem for Fourier series. Set $x = 0$ in (10) to obtain the expression

$$\frac{\pi}{\sin c\pi} = 2c\left\{ \frac{1}{2c^2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{c^2 - n^2} \right\}\,, \qquad c \neq 0, \pm 1, \pm 2, \ldots\,. \tag{11}$$

On the other hand, since $\Gamma(1) = 1$, the formula

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}\,, \qquad x>0\,,\ y>0$$

for the beta function gives

$$\Gamma(c)\Gamma(1-c) = B(c,1-c) = \int_0^1 t^{c-1}(1-t)^{-c}\,dt = \int_0^\infty \frac{x^{c-1}}{1+x}\,dx\,, \quad 0<c<1\,,$$

with the substitution $x = \frac{t}{1-t}$. But the substitution $u = 1/x$ reduces part of the last integral to

$$\int_1^\infty \frac{x^{c-1}}{1+x}\,dx = \int_0^1 \frac{u^{-c}}{1+u}\,du\,,$$

so we arrive at the expression

$$\Gamma(c)\Gamma(1-c) = \int_0^1 \frac{x^{c-1}}{1+x}\,dx + \int_0^1 \frac{x^{-c}}{1+x}\,dx\,, \qquad 0<c<1\,.$$

Now introduce the identity

$$\frac{1}{1+x} = 1 - \frac{x}{1+x}$$

into the first integral to obtain

$$\begin{aligned}\Gamma(c)\Gamma(1-c) &= \int_0^1 x^{c-1}\,dx + \int_0^1 \frac{x^{-c}-x^c}{1+x}\,dx \\ &= \frac{1}{c} + \int_0^1 (x^{-c}-x^c)(1-x+x^2-x^3+\dots)\,dx \\ &= \frac{1}{c} + \sum_{n=0}^\infty (-1)^n \int_0^1 (x^{-c}-x^c)x^n\,dx \\ &= \frac{1}{c} + \sum_{n=1}^\infty (-1)^n \frac{2c}{c^2-n^2} = \frac{\pi}{\sin c\pi}\,, \qquad 0<c<1\,,\end{aligned}$$

in view of (11). The term-by-term integration is justified by showing that the tail of the integrated series tends to zero:

$$\lim_{n\to\infty} \int_0^1 \frac{x^{n+1}}{1+x}(x^{-c}-x^c)\,dx = 0\,. \tag{12}$$

Indeed, the factor

$$\frac{x^{1-c}-x^{1+c}}{1+x}$$

is bounded in the interval $[0,1]$, while

$$\int_0^1 x^n\,dx = \frac{1}{n+1} \to 0 \qquad \text{as } n \to \infty\,.$$

This proves (12) and completes the completes the proof of the Euler reflection formula (9).

With the choice $x = \frac{1}{2}$, the reflection formula says that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Another proof of the reflection formula can be based on the infinite product representation of the gamma function, as developed in the next section (see Exercise 16).

9.6. Infinite product representation

Before turning to a representation of the gamma function as an infinite product, we will derive the formula

$$\Gamma(x) = \lim_{n\to\infty} \frac{n!\,n^x}{x(x+1)\cdots(x+n)}\,, \qquad x > 0\,. \tag{13}$$

The existence of the limit is not obvious and is part of the assertion. The result is called the *Gauss product formula* after Carl Friedrich Gauss (1777–1855), although it was known to Euler. In fact, this was Euler's original construction in 1729, but Gauss rediscovered the formula and recognized its importance.

One approach to (13) is to observe that

$$\begin{aligned}\int_0^1 t^{x-1}(1-t)^n\,dt = B(x,n+1) &= \frac{\Gamma(x)\Gamma(n+1)}{\Gamma(x+n+1)} \\ &= \frac{n!\,\Gamma(x)}{(x+n)(x+n-1)\cdots x\Gamma(x)} = \frac{n!}{x(x+1)\ldots(x+n)}\,.\end{aligned}$$

The substitution of t/n for t then leads to the representation

$$\frac{n!\,n^x}{x(x+1)\cdots(x+n)} = \int_0^n t^{x-1}\left(1-\frac{t}{n}\right)^n dt = \int_0^\infty g_n(t)t^{x-1}\,dt\,,$$

where

$$g_n(t) = \begin{cases} \left(1-\frac{t}{n}\right)^n\,, & \text{for } \ 0 \le t \le n \\ 0\,, & \text{for } \ n < t < \infty\,. \end{cases}$$

Because $g_n(t)$ increases to the limit e^{-t} as $n \to \infty$, Dini's theorem (see Section 1.8) ensures that the integrals converge and

$$\lim_{n\to\infty} \frac{n!\,n^x}{x(x+1)\cdots(x+n)} = \int_0^\infty e^{-t}t^{x-1}\,dt = \Gamma(x)\,.$$

This concludes the proof of the Gauss product formula.

It is now a short step to the infinite product representation of the gamma function.

Theorem. *The formula*

$$\frac{1}{\Gamma(x)} = xe^{\gamma x} \prod_{n=1}^{\infty} \left(1 + \frac{x}{n}\right) e^{-x/n}$$

holds for all $x \in \mathbb{R}$ *with* $x \neq 0, -1, -2, \ldots$. *Here*

$$\gamma = \lim_{n\to\infty} \left\{ \sum_{k=1}^{n} \frac{1}{k} - \log n \right\}$$

is Euler's constant.

The infinite product converges for each $x \in \mathbb{R}$ because

$$\begin{aligned} \left(1 + \frac{x}{n}\right) e^{-x/n} &= \left(1 + \frac{x}{n}\right) \left(1 - \frac{x}{n} + \frac{x^2}{2n^2} + \cdots\right) \\ &= 1 - \frac{x^2}{2n^2} + O\left(\frac{1}{n^3}\right) \end{aligned}$$

and the series $\sum 1/n^2$ converges. In fact, the product can be shown to converge uniformly on each bounded subset of $\mathbb{R}$, so that it represents a continuous function on $\mathbb{R}$. Note that the product is equal to zero at the points $x = 0, -1, -2, \ldots$ where the gamma function is infinite.

Proof of theorem. The Gauss product formula (13) allows us to write

$$\begin{aligned} \frac{1}{\Gamma(x)} &= \lim_{n\to\infty} \frac{x(x+1)\cdots(x+n)}{n!\, n^x} \\ &= \lim_{n\to\infty} x \left(1 + \frac{x}{1}\right) \left(1 + \frac{x}{2}\right) \cdots \left(1 + \frac{x}{n}\right) e^{-x \log n} \\ &= \lim_{n\to\infty} xe^{x(1+\frac{1}{2}+\cdots+\frac{1}{n}-\log n)} \prod_{k=1}^{n} \left(1 + \frac{x}{k}\right) e^{-x/k} \\ &= xe^{\gamma x} \prod_{k=1}^{\infty} \left(1 + \frac{x}{k}\right) e^{-x/k}. \end{aligned}$$

□

The proof shows that the Gauss product formula (13) is essentially the same as the infinite product representation of $1/\Gamma(x)$.

9.7. Generalization of Stirling's formula

Implicit in the preceding calculations is an important generalization of Stirling's formula

$$n! \sim \sqrt{2\pi}\, n^{n+\frac{1}{2}} e^{-n}\,, \qquad n \to \infty\,,$$

to the gamma function. Because $\Gamma(n) = (n-1)!$ and $\left(1-\frac{1}{n}\right)^n \to 1/e$, an equivalent form is

$$\Gamma(n) \sim \sqrt{2\pi}\, n^{n-\frac{1}{2}}\, e^{-n}\,, \qquad n \to \infty\,.$$

But the result can be strengthened to

$$\Gamma(x) \sim \sqrt{2\pi}\, x^{x-\frac{1}{2}}\, e^{-x}\,, \qquad x \to \infty\,, \tag{14}$$

where $x \to \infty$ unrestrictedly. This is Stirling's formula for the gamma function.

Stirling's formula (14) is easily deduced from the relation (13), which can be rewritten as

$$\lim_{n\to\infty} \frac{\Gamma(n+a+1)}{n!\, n^a} = 1\,, \qquad a > 0\,,$$

where the convergence is uniform for $0 < a \le 1$, as is evident from the proof of (13). In view of Stirling's formula for $n!$, the last limit implies that

$$\lim_{n\to\infty} \frac{\Gamma(n+a)}{n^{n+a-\frac{1}{2}}\, e^{-n}} = \sqrt{2\pi}\,,$$

or

$$\lim_{n\to\infty} \frac{\Gamma(n+a)}{(n+a)^{n+a-\frac{1}{2}}\, e^{-(n+a)}} = \sqrt{2\pi}\,, \qquad 0 < a \le 1\,,$$

since $\left(1+\frac{a}{n}\right)^n \to e^a$ uniformly in the interval $0 < a \le 1$. This proves Stirling's formula for the gamma function.

9.8. Bohr–Mollerup theorem

Finally, we come to the remarkable fact that the gamma function is completely determined by the relation $\Gamma(x+1) = x\Gamma(x)$, the normalization $\Gamma(1) = 1$, and the fact that $\log\Gamma(x)$ is a convex function. Recall that a function f defined in an interval $I \subset \mathbb{R}$ is *convex* if for each pair of points $a, b \in I$ it satisfies the inequality

$$f(ra + (1-r)b) \le rf(a) + (1-r)f(b)\,, \qquad 0 < r < 1\,.$$

Geometrically, this says that the graph of f lies beneath every chord.

We begin by proving that $\log\Gamma(x)$ is convex, a property evident from its graph in Figure 3. Suppose $0 < a < b$ and $0 < r < 1$. Then by definition

$$\begin{aligned}\Gamma(ra+(1-r)b) &= \int_0^\infty e^{-t}t^{ra+(1-r)b-1}\,dt \\ &= \int_0^\infty \left\{e^{-t}t^{a-1}\right\}^r \left\{e^{-t}t^{b-1}\right\}^{1-r}\,dt \\ &\le \left\{\int_0^\infty e^{-t}t^{a-1}\,dt\right\}^r \left\{\int_0^\infty e^{-t}t^{b-1}\,dt\right\}^{1-r} = \Gamma(a)^r\,\Gamma(b)^{1-r}\,,\end{aligned}$$

where Hölder's inequality has been applied with the conjugate indices $p = 1/r$ and $q = 1/(1-r)$. Taking logarithms, we conclude that

$$\log\Gamma(ra+(1-r)b) \le r\log\Gamma(a) + (1-r)\log\Gamma(b)\,, \qquad 0 < r < 1\,,$$

which shows that $\log\Gamma(x)$ is a convex function. (See also Exercise 26.)

Harald Bohr and Johannes Mollerup [**3**] discovered that the gamma function is actually characterized by its logarithmic convexity. Emil Artin [**2**] gave an elegant presentation of their argument and clarified the role of logarithmic convexity.

Bohr–Mollerup Theorem. *Let $G(x)$ be a positive function on the positive real axis $x > 0$ with the properties $G(x+1) = xG(x)$ and $G(1) = 1$. Suppose further that $\log G(x)$ is a convex function. Then $G(x) \equiv \Gamma(x)$.*

Proof. The hypotheses $G(x+1) = xG(x)$ and $G(1) = 1$ imply that $G(n+1) = n!$ for $n = 1, 2, \ldots$. For any positive integer n and for $0 < x \le 1$, we express

$$n + x = (1-x)n + x(n+1)$$

as a convex combination of n and $n+1$. Then by the convexity hypothesis,

$$\log G(n+x) \le (1-x)\log G(n) + x\log G(n+1)\,,$$

or

$$G(n+x) \le G(n)^{1-x}G(n+1)^x = n!\,n^{x-1}\,.$$

In a similar way, the convex combination

$$n+1 = x(n+x) + (1-x)(n+x+1)$$

produces the inequality

$$n! = G(n+1) \le G(n+x)^x G(n+x+1)^{1-x} = G(n+x)(n+x)^{1-x}\,,$$

since $G(n+x+1) = (n+x)G(n+x)$. When the two inequalities are combined, we find that

$$n!(n+x)^{x-1} \le G(n+x) \le n!\, n^{x-1}\,. \tag{15}$$

But the property $G(x+1) = xG(x)$ can be iterated to give

$$G(x+n) = x(x+1)\cdots(x+n-1)G(x)\,,$$

so the inequality (15) takes the form

$$\frac{n!(n+x)^x}{x(x+1)\cdots(x+n)} \le G(x) \le \frac{n!\,n^x}{x(x+1)\cdots(x+n-1)n}\,. \tag{16}$$

We now let $n \to \infty$ and appeal to the Gauss product formula (13) to conclude from (16) that

$$\Gamma(x) \le G(x) \le \Gamma(x)\,,$$

or $G(x) = \Gamma(x)$ for $0 < x \le 1$. The extension to all $x > 0$ now results from the two identities $G(x+1) = xG(x)$ and $\Gamma(x+1) = x\Gamma(x)$. □

Alternatively, the inequality (16) can be used to establish the existence of the limit in (13) and to show that it is equal to $G(x)$. Since the reasoning applies equally well to $\Gamma(x)$, it then follows that $G(x) = \Gamma(x)$, which gives an independent proof of the Gauss product formula.

Aside from its aesthetic appeal, the Bohr–Mollerup theorem offers an effective way to verify formulas involving the gamma function. As an illustration, we now apply it to establish the *Gauss multiplication formula*

$$\Gamma(mx) = \frac{m^{mx-1/2}}{(2\pi)^{(m-1)/2}}\,\Gamma(x)\,\Gamma\left(x+\frac{1}{m}\right)\Gamma\left(x+\frac{2}{m}\right)\cdots\Gamma\left(x+\frac{m-1}{m}\right) \tag{17}$$

for $x > 0$ and $m = 2, 3, \ldots$. This is a generalization of Legendre's duplication formula (7), which corresponds to the special case $m = 2$. Replacing mx by x in (17), we want to show that the function

$$G(x) = \frac{m^{x-1/2}}{(2\pi)^{(m-1)/2}}\,\Gamma\left(\frac{x}{m}\right)\Gamma\left(\frac{x+1}{m}\right)\cdots\Gamma\left(\frac{x+m-1}{m}\right)$$

is equal to $\Gamma(x)$. In view of the Bohr–Mollerup theorem, it will suffice to show that $\log G(x)$ is convex, $G(x+1) = xG(x)$, and $G(1) = 1$. The logarithmic convexity follows at once from that of the gamma function. To see that $G(x+1) = xG(x)$, it is convenient to write $G(x) = \alpha_m\, g_m(x)$, where

$$\alpha_m = \frac{m^{-1/2}}{(2\pi)^{(m-1)/2}}$$

and

$$g_m(x) = m^x\,\Gamma\left(\frac{x}{m}\right)\Gamma\left(\frac{x+1}{m}\right)\cdots\Gamma\left(\frac{x+m-1}{m}\right).$$

Now observe that

$$g_m(x+1) = m\left[\Gamma\left(\frac{x+m}{m}\right)\Big/\,\Gamma\left(\frac{x}{m}\right)\right] g_m(x) = x g_m(x)\,,$$

since $\Gamma\left(\frac{x+m}{m}\right) = \Gamma\left(\frac{x}{m}+1\right) = \frac{x}{m}\Gamma\left(\frac{x}{m}\right)$. Thus $G(x+1) = xG(x)$.

The difficulty is to show that $G(1) = 1$. In other words, it is to be shown that

$$\alpha_m\, m\,\Gamma\left(\frac{1}{m}\right)\Gamma\left(\frac{2}{m}\right)\cdots\Gamma\left(\frac{m}{m}\right) = 1\,. \tag{18}$$

For this purpose we apply the Gauss product formula (13) to write

$$\Gamma\left(\frac{k}{m}\right) = \lim_{n\to\infty}\frac{n!\,n^{k/m}\,m^{n+1}}{k(k+m)(k+2m)\cdots(k+nm)}\,, \qquad k = 1, 2, \ldots, m\,.$$

The product of these limits is

$$\Gamma\left(\frac{1}{m}\right)\Gamma\left(\frac{2}{m}\right)\cdots\Gamma\left(\frac{m}{m}\right) = \lim_{n\to\infty}\frac{(n!)^m\,n^{(m+1)/2}\,m^{mn+m}}{(m+nm)!}\,. \tag{19}$$

But

$$\frac{(m+nm)!}{(nm)!(nm)^m} = \left(1+\frac{1}{nm}\right)\left(1+\frac{2}{nm}\right)\cdots\left(1+\frac{m}{nm}\right) \to 1$$

as $n\to\infty$, and so the expression (19) reduces to

$$\Gamma\left(\frac{1}{m}\right)\Gamma\left(\frac{2}{m}\right)\cdots\Gamma\left(\frac{m}{m}\right) = \lim_{n\to\infty}\frac{(n!)^m\,m^{mn}}{(nm)!\,n^{(m-1)/2}}\,.$$

Now Stirling's formula $n! \sim n^n e^{-n}\sqrt{2\pi n}$ can be applied to give

$$\frac{(n!)^m}{(nm)!} \sim \frac{(2\pi)^{(m-1)/2}\,n^{(m-1)/2}}{m^{mn+1/2}} \qquad \text{as} \quad n\to\infty\,,$$

so that

$$\Gamma\left(\frac{1}{m}\right)\Gamma\left(\frac{2}{m}\right)\cdots\Gamma\left(\frac{m}{m}\right) = \frac{(2\pi)^{(m-1)/2}}{m^{1/2}} = \frac{1}{m\alpha_m}\,.$$

This proves (18), which shows that $G(1) = 1$. Hence it follows from the Bohr–Mollerup theorem that $G(x) = \Gamma(x)$, which completes the proof of the Gauss multiplication formula.

9.9. A special integral

We close this chapter with a calculation of the integral

$$\int_0^\infty x^{-a} \sin x \, dx = \frac{\pi}{2\,\Gamma(a)} \csc(\pi a/2)\,, \qquad 0 < a < 2\,. \tag{20}$$

This result will be applied in the next chapter to derive a functional equation for the Riemann zeta function. Although the integral can be calculated by contour integration in the complex plane, the method used here is entirely elementary and is based only on standard properties of the gamma function.

Recall first the expression for the Laplace transform

$$F(s) = \int_0^\infty e^{-st} \sin t \, dt = \frac{1}{1+s^2}\,, \qquad s > 0\,, \tag{21}$$

which can be derived via two integrations by parts. Specifically,

$$F(s) = \frac{1}{s}\int_0^\infty e^{-st}\cos t\, dt = \frac{1}{s}\left\{\frac{1}{s} - \frac{1}{s}\int_0^\infty e^{-st}\sin t\, dt\right\}\,,$$

so that $(1+s^2)F(s) = 1$.

Next make a change of variables to obtain

$$\Gamma(a) = \int_0^\infty t^{a-1}e^{-t}\,dt = x^a \int_0^\infty t^{a-1}e^{-xt}\,dt\,, \qquad x > 0\,,\ a > 0\,.$$

Take $\varepsilon > 0$ and use this form of the gamma function to write

$$\begin{aligned}
\int_0^\infty e^{-\varepsilon x} x^{-a} \sin x \, dx &= \frac{1}{\Gamma(a)} \int_0^\infty e^{-\varepsilon x} \sin x \int_0^\infty t^{a-1} e^{-xt}\, dt\, dx \\
&= \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} \int_0^\infty e^{-(\varepsilon+t)x} \sin x \, dx\, dt \\
&= \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} \frac{1}{1+(\varepsilon+t)^2}\, dt\,,
\end{aligned}$$

where the formula (21) has been invoked. To justify the interchange in order of integration, observe that the exponential factor $e^{-\varepsilon x}$ makes the integrand absolutely integrable over the first quadrant of the (x,t) plane.

Now appeal to the integral analogue of Abel's theorem (Section 3.2) to conclude that

$$\begin{aligned}
\int_0^\infty x^{-a} \sin x \, dx &= \lim_{\varepsilon \to 0} \int_0^\infty e^{-\varepsilon x} x^{-a} \sin x \, dx \\
&= \frac{1}{\Gamma(a)} \int_0^\infty \frac{t^{a-1}}{1+t^2}\, dt\,.
\end{aligned}$$

But the last integral is a beta function in disguise. Let $u = t^2$ and refer to Exercise 13 to write

$$\int_0^\infty \frac{t^{a-1}}{1+t^2}\,dt = \frac{1}{2}\int_0^\infty \frac{u^{\frac{a}{2}-1}}{1+u}\,du = \frac{1}{2}\,B(\tfrac{a}{2}, 1-\tfrac{a}{2})\,.$$

Then apply the Euler reflection formula to obtain

$$B(\tfrac{a}{2}, 1-\tfrac{a}{2}) = \Gamma(\tfrac{a}{2})\,\Gamma(1-\tfrac{a}{2}) = \pi\csc(\pi a/2)\,,$$

which verifies the formula (20).

The expression (20) can be recast in other forms. For $0 < a < 1$ it combines with the Euler reflection formula to give

$$\begin{aligned}\int_0^\infty x^{a-1}\sin x\,dx &= \frac{\pi}{2\,\Gamma(1-a)}\,\csc(\pi(1-a)/2)\\ &= \frac{1}{2}\,\Gamma(a)\,\frac{\sin\pi a}{\cos(\pi a/2)} = \Gamma(a)\sin(\pi a/2)\,.\end{aligned}$$

More generally,

$$\int_0^\infty x^{a-1}\sin bx\,dx = b^{-a}\Gamma(a)\sin(\pi a/2)\,, \qquad 0 < a < 1\,,\ b > 0\,.$$

Exercises

1. Prove that $\Gamma(x)$ is continuous for $x > 0$.

2. Show that $\lim_{x\to 0+} x\Gamma(x) = 1$.

3. (*a*) Prove that the function $\Gamma(x)$ has derivatives of all orders at every point $x > 0$, and that its nth derivative is given by

$$\Gamma^{(n)}(x) = \int_0^\infty e^{-t}t^{x-1}(\log t)^n\,dt\,, \qquad n = 1, 2, \ldots\,.$$

(*b*) Show that $\Gamma''(x) > 0$ for $x > 0$, so that the curve $y = \Gamma(x)$ is convex.

(*c*) Show that $\Gamma(x)$ attains a minimum value for $x > 0$ at a point x_0 in the interval $1 < x_0 < 2$. Show further that $\Gamma(x)$ is decreasing in the interval $(0, x_0)$ and increasing in (x_0, ∞).

4. Show that

$$\Gamma\left(n+\tfrac{1}{2}\right) = \frac{(2n)!\sqrt{\pi}}{4^n n!}\,, \qquad n = 1, 2, \ldots\,.$$

5. Verify that Euler's definition

$$\Gamma(x) = \int_0^1 \big(\log(1/t)\big)^{x-1}\, dt$$

is equivalent to the definition (2). Show more generally that

$$\int_0^1 t^{x-1}\big(\log(1/t)\big)^{y-1}\, dt = x^{-y}\,\Gamma(y)\,, \qquad x>0\,,\ y>0\,.$$

6. Calculate the integrals

$$(a) \qquad \int_0^1 \big(\log(1/t)\big)^{1/2}\, dt = \frac{\sqrt{\pi}}{2}$$

$$(b) \qquad \int_0^1 \big(\log(1/t)\big)^{-1/2}\, dt = \sqrt{\pi}$$

$$(c) \qquad \int_0^1 \big(t\log(1/t)\big)^{1/2}\, dt = \frac{\sqrt{2\pi}}{3\sqrt{3}}$$

$$(d) \qquad \int_0^1 \big(t\log(1/t)\big)^{-1/2}\, dt = \sqrt{2\pi}\,.$$

7. Calculate the integral

$$\int_0^1 \sqrt{t(1-t)}\, dt = \frac{\pi}{8}\,.$$

8. Express the function

$$f(x) = \int_0^{\pi/2} (\sin 2\theta)^x\, d\theta\,, \qquad x>-1\,,$$

in terms of the gamma function.

9. Show that

$$\Gamma(\tfrac{1}{6}) = \frac{\sqrt{3}}{\sqrt[3]{2}\,\sqrt{\pi}}\,\Gamma(\tfrac{1}{3})^2\,.$$

10. Calculate the integrals

$$(a) \qquad \int_0^1 \sqrt{1-t^3}\, dt = \frac{\sqrt{3}}{10\pi\sqrt[3]{2}}\,\Gamma(\tfrac{1}{3})^3$$

$$(b) \qquad \int_0^1 \sqrt{1-t^4}\, dt = \frac{1}{6\sqrt{2\pi}}\,\Gamma(\tfrac{1}{4})^2$$

$$(c) \qquad \int_0^1 \frac{1}{\sqrt{1-t^3}}\, dt = \frac{1}{2\pi\sqrt{3}\sqrt[3]{2}}\,\Gamma(\tfrac{1}{3})^3$$

$$(d) \qquad \int_0^1 \frac{1}{\sqrt{1-t^4}}\, dt = \frac{1}{4\sqrt{2\pi}}\,\Gamma(\tfrac{1}{4})^2$$

$$(e) \qquad \int_0^1 \frac{t^2}{\sqrt{1-t^4}}\, dt = \frac{1}{\sqrt{2\pi}}\,\Gamma(\tfrac{3}{4})^2\,.$$

11. Use the beta function to calculate

$$\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta = \int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta = \frac{\sqrt{2}}{\sqrt{\pi}}\, \Gamma(\tfrac{3}{4})^2 .$$

Compare the result with that of Exercise 10(e).

12. Find the area of the region bounded by the hypocycloid $x^{2/3} + y^{2/3} = 1$.

13. Use the substitution $t = \frac{u}{1+u}$ to derive the formula

$$B(x,y) = \int_0^\infty \frac{u^{x-1}}{(1+u)^{x+y}}\, du .$$

14. Apply the formula of the preceding exercise to calculate the integral

$$\int_0^\infty \frac{x^3}{(1+x)^7}\, dx = \frac{1}{60} .$$

15. Calculate the integral

$$\int_0^\infty \frac{x^{-a}}{1+x}\, dx = \frac{\pi}{\sin \pi a} , \qquad 0 < a < 1 .$$

More generally, calculate

$$\int_0^\infty \frac{x^{m-1}}{1+x^n}\, dx = \frac{\pi}{n \sin(m\pi/n)} , \qquad 0 < m < n ,$$

a result known to Euler. (Here m and n need not be integers.)

16. Compare the infinite product representation of $\Gamma(x)$ with that of the sine function to obtain another proof of the Euler reflection formula.

17. Prove that $\Gamma'(1) = -\gamma$, where γ is Euler's constant. Conclude that

$$\gamma = -\int_0^\infty e^{-t} \log t\, dt .$$

Hint. Take logarithmic derivatives in the infinite product formula for $\Gamma(x)$. Justify the term-by-term differentiation.

18. Calculate the integral

$$\int_0^\infty \frac{\log x}{1+e^x}\, dx = -\gamma \log 2 + \sum_{n=1}^\infty (-1)^n \frac{\log n}{n} = -\frac{1}{2}(\log 2)^2 .$$

Hint. Expand $1/(1+e^x) = e^{-x}/(1+e^{-x})$ into geometric series and integrate term by term (justify), then refer to Exercise 13 of Chapter 2 for evaluation of the infinite series.

19. The *Laplace transform* of a function f is the function $F = \mathcal{L}(f)$ defined by the integral

$$F(s) = \int_0^\infty f(t)e^{-st}\,dt\,, \qquad s > s_0\,,$$

which is assumed to converge for all real numbers s larger than some number s_0. For each exponent $a > -1$, show that the Laplace transform of $f(t) = t^a$ is

$$F(s) = \frac{\Gamma(a+1)}{s^{a+1}}\,, \qquad s > 0\,.$$

20. The *convolution* of two functions f and g defined on $(0, \infty)$ is the function $h = f * g$ given by

$$h(t) = \int_0^t f(u)g(t-u)\,du\,, \qquad 0 < t < \infty\,.$$

Show that $\mathcal{L}(f * g) = \mathcal{L}(f)\mathcal{L}(g)$. In other words, show that the Laplace transform of a convolution is the product of transforms.

21. If $f(t) = t^{x-1}$ and $g(t) = t^{y-1}$ for some $x > 0$ and $y > 0$, show that the convolution $h = f * g$ has the form $h(t) = t^{x+y-1}B(x, y)$. By taking Laplace transforms, conclude that

$$B(x, y) = \frac{\Gamma(x)\,\Gamma(y)}{\Gamma(x+y)}\,.$$

22. Use Stirling's formula for the gamma function to show that

$$\lim_{n\to\infty} \frac{\Gamma(n+c-a)\,\Gamma(n+c-b)}{\Gamma(n+c)\,\Gamma(n+c-a-b)} = 1$$

for arbitrary real numbers a, b, and c.

23. Calculate the integral

$$\int_0^\infty x^{-a}\cos x\,dx = \frac{\pi}{2\,\Gamma(a)}\sec(\pi a/2)\,, \qquad 0 < a < 1\,.$$

24. (a) Calculate the integral

$$\int_{-1}^1 (1-x^2)^n\,dx = \frac{2^{2n+1}(n!)^2}{(2n+1)!}\,, \qquad n = 1, 2, \ldots\,,$$

by making the substitution $t = 1 - x^2$ to transform it into a beta function.

(b) Deduce that

$$\int_0^{\pi/2} \sin^{2n+1}\theta\, d\theta = \frac{2^{2n}(n!)^2}{(2n+1)!}\,.$$

(c) Apply the formula (6) for the beta function to reprove (b) and to calculate

$$\int_0^{\pi/2} \sin^{2n}\theta\, d\theta = \frac{(2n)!\,\pi}{2^{2n+1}(n!)^2}\,.$$

(Compare with Exercise 15 in Chapter 2.)

25. Let $v_n(r)$ denote the volume of the n-dimensional sphere

$$B_n(r) = \big\{(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + x_2^2 + \cdots + x_n^2 \le r^2\big\}$$

of radius r, where $n = 1, 2, \ldots$. Then $v_1(r) = 2r$, $v_2(r) = \pi r^2$, and $v_3(r) = \frac{4}{3}\pi r^3$. In general, it is easy to see that $v_n(r) = V_n r^n$, where V_n is the volume of the unit sphere $B_n(1)$. The numbers V_n can be calculated recursively by slicing:

$$V_{n+1} = \int_{-1}^{1} v_n\big(\sqrt{1-x^2}\,\big)\, dx = V_n \int_{-1}^{1} (1-x^2)^{n/2}\, dx\,.$$

Derive the formula

$$V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}\,, \qquad n = 1, 2, \ldots.$$

Show in particular that

$$v_4(r) = \tfrac{1}{2}\pi^2 r^4\,, \quad v_5(r) = \tfrac{8}{15}\pi^2 r^5\,, \quad v_6(r) = \tfrac{1}{6}\pi^3 r^6\,, \quad \text{and} \quad v_7(r) = \tfrac{16}{105}\pi^3 r^7\,.$$

26. The *digamma function* is $\psi(x) = \frac{d}{dx}\{\log\Gamma(x)\} = \Gamma'(x)/\Gamma(x)$.

(a) Apply the functional equation $\Gamma(x+1) = x\Gamma(x)$ to show that $\psi(x+1) = \psi(x) + \frac{1}{x}\,, \quad x > 0$.

(b) Generalize the relation $\Gamma'(1) = -\gamma$ by proving that

$$\psi(n+1) = \big(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{n}\big) - \gamma\,, \qquad n = 1, 2, \ldots.$$

(c) Use the infinite product formula for $1/\Gamma(x)$ to show that

$$\psi(x) = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{x+n-1}\right) - \gamma\,, \qquad x > 0\,.$$

(d) Deduce that

$$\psi'(x) = \sum_{n=0}^{\infty} \frac{1}{(x+n)^2}\,, \qquad x > 0\,,$$

and conclude that the function $\log\Gamma(x)$ is convex.

27. Apply Stirling's formula to verify the Gauss product formula (13) in the special case where x is a positive integer.

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Chapter 10
Two Topics in Number Theory

In this chapter we discuss two unrelated topics that lie at the intersection of analysis with number theory. First we consider the notion of equidistribution, a classical topic that has found applications beyond number theory to such areas as probability, functional analysis, and topological algebra. (The German and French terms are *Gleichverteilung* and *équirépartition*, respectively. Another English term is *uniform distribution*.) Our discussion will focus on a beautiful criterion for equidistribution (modulo 1) due to Hermann Weyl.

Our second topic is the Riemann zeta function, which has intimate connections with the prime numbers. We develop Euler's product formula for the zeta function, then give two relatively elementary derivations of the functional equation, which involves the gamma function. In truth a full appreciation of the functional equation requires some familiarity with complex analysis, which guarantees the uniqueness of an analytic continuation. Otherwise, however, the derivations are entirely self-contained.

10.1. Equidistributed sequences

Loosely speaking, a sequence of points is said to be equidistributed over a given set if each subset receives its proper share of points. To make the notion precise, let us consider the special case of a numerical sequence $\{\alpha_n\}$ in the interval $[0, 1)$. The reason for choosing a half-open interval will become apparent later. For any subinterval $I \subset [0, 1)$, let $\nu(n, I)$ denote the number

of points among $\alpha_1, \alpha_2, \ldots, \alpha_n$ that lie in I. Then the sequence $\{\alpha_n\}$ is defined to be *equidistributed* over $[0,1)$ if

$$\lim_{n\to\infty} \frac{\nu(n,I)}{n} = |I|$$

for every interval $I \subset [0,1)$, where $|I|$ denotes the length of I. The existence of each such limit is part of the requirement.

For example, it is intuitively clear that the sequence

$$\left\{\tfrac{0}{1}, \tfrac{0}{2}, \tfrac{1}{2}, \tfrac{0}{3}, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{0}{4}, \tfrac{1}{4}, \tfrac{2}{4}, \tfrac{3}{4}, \ldots\right\}$$

is equidistributed over $[0,1)$, although the proof requires a bit of effort. At the other extreme, a convergent sequence can never be equidistributed. In fact, it is easy to see that an equidistributed sequence must be everywhere dense; that is, it must have a subsequence that converges to each point of the interval $[0,1]$. If not, then some interval $I \subset [0,1)$ of positive length is free of all but a finite number of points α_n in the sequence, and so $\nu(n,I)/n \to 0$ as $n \to \infty$. Since $|I| \neq 0$, this violates the requirement for equidistribution.

In general it is not easy to determine whether a given sequence is equidistributed. Our main purpose is to develop an effective criterion. Actually, our discussion will focus on a modular notion of equidistribution. Two numbers x and y are said to be *congruent modulo* m, where $m > 0$ is a prescribed number, if their difference $x - y$ is an integer multiple of m. Such a congruence is indicated by writing $x \equiv y (\mathrm{mod}\ m)$. The *integer part* of any real number x is defined to be the greatest integer less than or equal to x, and is denoted by $[x]$. The number $\langle x \rangle = x - [x]$ is called the *fractional part* of x. In particular, $0 \leq \langle x \rangle < 1$ and $x \equiv \langle x \rangle (\mathrm{mod}\ 1)$ for every $x \in \mathbb{R}$. Hence the fractional part of a number x may be viewed as the unique representative in the interval $[0,1)$ of the equivalence class determined by x.

An arbitrary numerical sequence $\{\alpha_n\}$ is said to be *equidistributed modulo* 1 if its sequence $\{\langle \alpha_n \rangle\}$ of fractional parts is uniformly distributed in $[0,1)$. If ξ is a fixed rational number, the sequence defined by $\alpha_n = n\xi$ has periodic fractional parts and therefore is certainly not equidistributed modulo 1. On the other hand, it is a remarkable fact that for any irrational number ξ, the sequence $\{\xi, 2\xi, 3\xi, \ldots\}$ is equidistributed modulo 1. This generalizes a classical result of Kronecker, who showed that the fractional parts of each such sequence are everywhere dense in $[0,1]$. Many elementary proofs are now known. We will obtain a proof later as a corollary of Weyl's criterion.

Meanwhile, however, we digress to give a direct proof of Kronecker's theorem. Since ξ is irrational, the fractional parts $\langle n\xi \rangle$ are all distinct. In

other words, $\langle n\xi\rangle \neq \langle m\xi\rangle$ if $n \neq m$. By the Bolzano–Weierstrass theorem, some subsequence of the sequence $\{\langle n\xi\rangle\}$ converges to a point in the interval $[0,1]$. In particular, for each $\varepsilon > 0$ there exist positive integers m and k such that

$$0 < |\langle (m+k)\xi\rangle - \langle m\xi\rangle| < \varepsilon\,.$$

This says that the point $\langle k\xi\rangle$ is within ε of either 0 or 1. Consequently, the sequence of points

$$\langle k\xi\rangle, \langle 2k\xi\rangle, \langle 3k\xi\rangle, \ldots\,,$$

suitably terminated, forms a chain that moves across the interval $(0,1)$ in one direction (modulo 1) or the other, with distance between successive points always less than ε except for a possible jump near one of the endpoints 0 or 1. As a result, each point of the interval $[0,1]$ is within distance ε of a point $\langle n\xi\rangle$, where n is some multiple of k. Thus the points $\langle n\xi\rangle$ are everywhere dense in the interval $[0,1]$, as Kronecker's theorem asserts. (It should be noted that Kronecker actually obtained a stronger quantitative form of the theorem and gave a generalization to higher dimensions; *cf.* Hardy and Wright [**5**].)

10.2. Weyl's criterion

We now turn to the elegant criterion due to Hermann Weyl [**11**] for equidistribution modulo 1. Weyl's key idea is to express the definition in terms of integrals. For any given interval $I \subset [0,1)$, let f be the periodic extension of its characteristic function. This is the function defined on the interval $[0,1)$ by $f(x) = 1$ for $x \in I$ and $f(x) = 0$ for $x \notin I$, then extended to the whole real line by $f(x+1) = f(x)$, so that it is periodic with period 1. Then the sequence $\{\alpha_n\}$ is equidistributed modulo 1 if and only if

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) = \int_0^1 f(x)\,dx \tag{1}$$

for every interval $I \subset [0,1)$. If the sequence $\{\alpha_n\}$ is equidistributed modulo 1, the relation (1) can be generalized by linearity to periodic extensions of arbitrary step functions on $[0,1)$. Here a step function is understood to be a finite linear combination of characteristic functions of intervals in $[0,1)$. This in turn implies the validity of (1) for each bounded Riemann integrable function f that is periodic with period 1. To see this, recall that by the definition of Riemann integrability, any such function f can be approximated below and above by periodic extensions g and h of step functions on $[0,1)$. More precisely, for each prescribed $\varepsilon > 0$, there are periodic extensions g and h of step functions on $[0,1)$ for which $g(x) \leq f(x) \leq h(x)$ and $0 \leq$

$h(x) - g(x) < \varepsilon$ on $\mathbb{R}$, so that

$$0 \le \int_0^1 h(x)\,dx - \int_0^1 g(x)\,dx < \varepsilon\,.$$

Since the equation (1) holds for both g and h, we see that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} g(\alpha_k) = \int_0^1 g(x)\,dx > \int_0^1 f(x)\,dx - \varepsilon\,,$$

so that

$$\frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) \ge \frac{1}{n} \sum_{k=1}^{n} g(\alpha_k) > \int_0^1 f(x)\,dx - \varepsilon$$

for all n sufficiently large. In a similar way, we find that

$$\frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) \le \frac{1}{n} \sum_{k=1}^{n} h(\alpha_k) < \int_0^1 f(x)\,dx + \varepsilon$$

for all n sufficiently large. But ε was chosen arbitrarily, so the last two inequalities combine to show that (1) holds for f as well. In summary, we have shown that whenever a sequence $\{\alpha_n\}$ is equidistributed modulo 1, it has the property (1) for every periodic function f that is Riemann integrable over $[0, 1)$. In particular, the limit in (1) exists.

The most familiar examples of nonconstant Riemann integrable functions of period 1 are $\sin 2\pi mx$ and $\cos 2\pi mx$ for integers $m > 0$. Applying the formula (1) to these functions, and using Euler's formula $e^{i\theta} = \cos\theta + i \sin\theta$ to simplify the writing, we conclude that if a sequence $\{\alpha_n\}$ is equidistributed modulo 1, then

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} e^{2\pi i m \alpha_k} = 0\,, \qquad m = 1, 2, \ldots\,.$$

Weyl's theorem asserts that the converse is also true.

Weyl's Theorem. *A sequence $\{\alpha_k\}$ of real numbers is equidistributed modulo 1 if and only if*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} e^{2\pi i m \alpha_k} = 0 \tag{2}$$

for each integer $m = 1, 2, \ldots$.

Proof. We have already seen that (2) holds whenever the sequence $\{\alpha_k\}$ is equidistributed modulo 1. For the converse, suppose the sequence $\{\alpha_k\}$ has the property (2). It then follows by linearity that the relation (1) holds for every trigonometric polynomial

$$f(x) = a_0 + \sum_{m=1}^{N} (a_m \cos 2\pi mx + b_m \sin 2\pi mx) \,,$$

since it holds trivially for constant functions. But according to the trigonometric version of the Weierstrass approximation theorem, every continuous function that is periodic with period 1 can be approximated uniformly by such polynomials. In fact, given any $\varepsilon > 0$, we can approximate such a function f above and below by trigonometric polynomials t and u so that

$$t(x) \le f(x) \le u(x) \,, \qquad f(x) < t(x) + \varepsilon \,, \qquad \text{and} \quad f(x) > u(x) - \varepsilon \,.$$

Then we see that

$$\limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) \le \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} u(\alpha_k) = \int_0^1 u(x)\,dx < \int_0^1 f(x)\,dx + \varepsilon \,.$$

Similarly,

$$\liminf_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) \ge \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} t(\alpha_k) = \int_0^1 t(x)\,dx > \int_0^1 f(x)\,dx - \varepsilon \,.$$

Since ε was chosen arbitrarily, it follows that the limit exists and

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(\alpha_k) = \int_0^1 f(x)\,dx \,.$$

Thus the relation (1) holds for every continuous function f that is periodic with period 1.

Finally, for any given interval $I \subset [0,1)$, let f be the periodic extension of the characteristic function of I. It is to be shown that (1) holds for each such function f. But f can be approximated in integral norm above and below by continuous periodic functions, and since we have shown that such functions satisfy (1), an argument similar to the above leads to the conclusion that f satisfies (1) as well. This completes the proof that the sequence $\{\alpha_k\}$ is equidistributed modulo 1. □

As an application, we can now prove that the sequence $\{k\xi\}$ is equidistributed modulo 1 when ξ is irrational. We need to show that the condition

(2) of Weyl's theorem is satisfied for $\alpha_k = k\xi$. But for $\alpha_k = k\xi$ the sum is a geometric series, which can be evaluated as

$$\sum_{k=1}^{n} e^{2\pi imk\xi} = \frac{e^{2\pi im(n+1)\xi} - e^{2\pi im\xi}}{e^{2\pi im\xi} - 1}\,.$$

Therefore, the sum has modulus

$$\left|\frac{e^{2\pi im(n+1)\xi} - e^{2\pi im\xi}}{e^{2\pi im\xi} - 1}\right| \le \frac{2}{|e^{2\pi im\xi} - 1|} = \frac{1}{|\sin \pi m\xi|}\,.$$

Because ξ is irrational, the number $m\xi$ is never an integer, so we have shown that each of the sums in (2) remains bounded as $n \to \infty$. In particular, the condition (2) is easily satisfied for each $m \in \mathbb{N}$, so by Weyl's theorem the sequence $\{n\xi\}$ is equidistributed modulo 1.

As a further application of Weyl's criterion, we now establish a theorem that will provide many more examples of equidistributed sequences. Here is a discrete version of a result known as Fejér's theorem.

Theorem. *Let $\{\alpha_n\}$ be a sequence of real numbers whose differences $\Delta\alpha_n = \alpha_{n+1} - \alpha_n$ are monotonic, and suppose that $\Delta\alpha_n \to 0$ and $n|\Delta\alpha_n| \to \infty$ as $n \to \infty$. Then the sequence $\{\alpha_n\}$ is equidistributed modulo 1.*

To give a proof it suffices to show, according to Weyl's theorem, that the sequence $\{\alpha_n\}$ has the property (2) for each $m \in \mathbb{N}$. For this purpose a simple inequality will be useful.

Lemma. $\left|e^{i\theta} - 1 - i\theta\right| \le \frac{1}{2}\,\theta^2\,, \qquad \theta \in \mathbb{R}\,.$

Proof of lemma. It is enough to take $\theta > 0$. An integration by parts verifies the representation

$$e^{i\theta} - 1 - i\theta = \int_0^\theta (t - \theta)e^{it}\,dt\,,$$

which is actually a version of Taylor's formula with remainder. Hence

$$\left|e^{i\theta} - 1 - i\theta\right| \le \int_0^\theta \left|(t - \theta)e^{it}\right|\,dt = \int_0^\theta (\theta - t)\,dt = \tfrac{1}{2}\,\theta^2\,. \qquad \square$$

Proof of theorem. Since $\{-\alpha_n\}$ is equidistributed modulo 1 if $\{\alpha_n\}$ is, there is no loss of generality in assuming that $\Delta\alpha_n > 0$, so that $\{\Delta\alpha_n\}$

decreases to 0. Fixing a positive integer m, choose $\theta = 2\pi m \Delta\alpha_k$ and apply the lemma to obtain after minor rearrangement

$$\left| \frac{e^{2\pi i m \alpha_{k+1}}}{\Delta\alpha_k} - \frac{e^{2\pi i m \alpha_k}}{\Delta\alpha_k} - 2\pi i m \, e^{2\pi i m \alpha_k} \right| \le 2\pi^2 m^2 \Delta\alpha_k \,, \qquad k = 1, 2, \ldots \,.$$

By the triangle inequality, this implies that

$$\left| \frac{e^{2\pi i m \alpha_{k+1}}}{\Delta\alpha_{k+1}} - \frac{e^{2\pi i m \alpha_k}}{\Delta\alpha_k} - 2\pi i m \, e^{2\pi i m \alpha_k} \right| \le \frac{1}{\Delta\alpha_{k+1}} - \frac{1}{\Delta\alpha_k} + 2\pi^2 m^2 \Delta\alpha_k \,.$$

Another application of the triangle inequality now yields the estimate

$$\begin{aligned} \left| 2\pi i m \sum_{k=1}^{n} e^{2\pi i m \alpha_k} \right| &\le \left| \sum_{k=1}^{n} \left(\frac{e^{2\pi i m \alpha_{k+1}}}{\Delta\alpha_{k+1}} - \frac{e^{2\pi i m \alpha_k}}{\Delta\alpha_k} \right) \right| \\ &\quad + \sum_{k=1}^{n} \left(\frac{1}{\Delta\alpha_{k+1}} - \frac{1}{\Delta\alpha_k} + 2\pi^2 m^2 \Delta\alpha_k \right) \\ &= \left| \frac{e^{2\pi i m \alpha_{n+1}}}{\Delta\alpha_{n+1}} - \frac{e^{2\pi i m \alpha_1}}{\Delta\alpha_1} \right| + \frac{1}{\Delta\alpha_{n+1}} - \frac{1}{\Delta\alpha_1} \\ &\quad + 2\pi^2 m^2 \left(\alpha_{n+1} - \alpha_1 \right) , \end{aligned}$$

since the sums telescope. Dividing the last inequality by n and using the hypothesis that $n\,\Delta\alpha_n \to \infty$, we conclude that (2) holds. To see that $(\alpha_{n+1} - \alpha_1)/n \to 0$, observe that those expressions are arithmetic means of the sequence $\{\Delta\alpha_n\}$, which tends to 0. Because (2) holds for each $m = 1, 2, \ldots$, Weyl's theorem shows that the sequence $\{\alpha_n\}$ is equidistributed modulo 1. □

It is now a short step to a special case which Pólya and Szegő [**7**] attribute to Fejér.

Corollary (Fejér's Theorem). *Let $g(x)$ be continuous on the interval $[1, \infty)$ and differentiable on $(1, \infty)$. Suppose that $g'(x)$ tends monotonically to 0 and $x|g'(x)| \to \infty$ as $x \to \infty$. Then the sequence $\{\alpha_n\} = \{g(n)\}$ is equidistributed modulo 1.*

Proof. By the mean value theorem, $\Delta\alpha_n = g'(x_n)$ for some point $x_n \in (n, n+1)$, and so the sequence $\{\alpha_n\}$ satisfies the hypotheses of the preceding theorem. □

Fejér's theorem shows for instance that the sequences $\{\sqrt{n}\log n\}$, $\{(\log n)^2\}$, and $\{n/\log n\}$ are equidistributed modulo 1. The discrete form of

the theorem is more flexible. For example, it shows that the partial sums of the divergent series $\sum_{n=1}^{\infty} 1/\sqrt{n}$ constitute a sequence that is equidistributed modulo 1.

On the other hand, the sequence $\{\log n\}$ is not equidistributed modulo 1, as can be seen from the following partial converse of Fejér's theorem.

Theorem. *If a sequence $\{\alpha_n\}$ is equidistributed modulo 1, then the differences $\Delta\alpha_n = \alpha_{n+1} - \alpha_n$ have the property*

$$\limsup_{n\to\infty} n|\Delta\alpha_n| = \infty\,.$$

Proof. The proof will invoke Hardy's Tauberian theorem, as discussed in Chapter 7. With the notation $s_n = a_1+a_2+\cdots+a_n$ and $\sigma_n = (s_1+s_2+\cdots+s_n)/n$, Hardy's theorem states that if $a_n = O(1/n)$ and $\sigma_n \to s$ as $n \to \infty$, then $s_n \to s$. Suppose now that the sequence $\{\alpha_n\}$ is equidistributed modulo 1. Then by Weyl's theorem, the arithmetic means

$$\sigma_n = \frac{1}{n}\sum_{k=1}^{n} e^{2\pi i\alpha_k} \to 0 \qquad \text{as} \quad n \to \infty\,.$$

But in view of the inequality $|e^{i\theta} - 1| \le |\theta|$, the assumption that

$$\limsup_{n\to\infty} n|\Delta\alpha_n| < \infty$$

would imply

$$\left|e^{2\pi i\alpha_{n+1}} - e^{2\pi i\alpha_n}\right| \le |\Delta\alpha_n| = O(1/n)\,,$$

and so it would follow from Hardy's theorem that $e^{2\pi i\alpha_n} \to 0$, which is clearly impossible. Therefore, it must be true that $\limsup_{n\to\infty} n|\Delta\alpha_n| = \infty$ whenever the sequence $\{\alpha_n\}$ is equidistributed modulo 1. □

Further information about equidistribution can be found in the book by Pólya and Szegő [**7**]. For a broad discussion of the topic, the reader is referred to the book by Kuipers and Niederreiter [**6**].

10.3. The Riemann zeta function

The Riemann zeta function is defined by

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}\,, \qquad x > 1\,.$$

Euler showed that $\zeta(2) = \pi^2/6$ and he found a general expression for $\zeta(2m)$, where $2m$ is any even integer, in terms of the Bernoulli numbers. (Details will be given in Chapter 11.) Some closely related series are

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^x} = \sum_{n=1}^{\infty} \frac{1}{n^x} - \sum_{k=1}^{\infty} \frac{1}{(2k)^x} = \left(1 - \frac{1}{2^x}\right) \zeta(x)$$

and

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^x} = \sum_{n=1}^{\infty} \frac{1}{n^x} - 2 \sum_{k=1}^{\infty} \frac{1}{(2k)^x} = \left(1 - \frac{1}{2^{x-1}}\right) \zeta(x).$$

The zeta function has intrinsic connections with number theory. For instance, it is not hard to see that

$$\zeta(x)^2 = \sum_{n=1}^{\infty} d(n)\, n^{-x}\,, \qquad x > 1\,,$$

where $d(n)$ is the number of divisors of n. For instance, $d(5) = 2$, $d(9) = 3$, and $d(12) = 6$. Similarly, it can be shown that

$$\zeta(x)\zeta(x-1) = \sum_{n=1}^{\infty} \sigma(n)\, n^{-x}\,, \qquad x > 2\,,$$

where $\sigma(n)$ denotes the sum of the divisors of n. For example,

$$\sigma(12) = 1 + 2 + 3 + 4 + 6 + 12 = 28\,.$$

At a more basic level, the zeta function is an essential tool in the study of prime numbers. An integer k is a *divisor* of n if $n = km$ for some integer m. An integer $p > 1$ is said to be *prime* if it has no positive divisors except 1 and p. The first few primes are

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, \ldots\,.$$

An integer larger than 1 that is not prime is called *composite*. It is not difficult to see that every composite number has a prime divisor.

It was known to the ancient Greeks that there are infinitely many prime numbers. Here is Euclid's elegant proof. Suppose, on the contrary, that there were only a finite number of primes: $p_1, p_2, \ldots, p_n$. Then since the number $N = p_1 p_2 \cdots p_n + 1$ is greater than every prime p_k, it is composite and therefore has a prime divisor. But clearly N is not divisible by any of the primes p_k. This contradiction shows that the number of primes cannot be finite.

Each positive integer has a unique prime factorization. For instance, $60 = 2^2 \cdot 3 \cdot 5$.

Fundamental Theorem of Arithmetic. *Each integer $n \geq 2$ can be represented in one and only one way as a product of primes.*

The existence of a prime factorization is fairly obvious. A formal proof proceeds by induction. Suppose that every integer less than n has a prime factorization. If n is prime, there is nothing to prove. If n is composite, it is the product of two smaller integers, each of which has a prime factorization, by the inductive hypothesis. Hence n has a prime factorization.

The uniqueness of a prime factorization, however, is not at all obvious. Suppose the assertion is false, and let n be the smallest positive integer that has two different representations

$$n = p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

as a product of primes, not necessarily distinct. The same prime factor cannot occur in both representations, because any common factor could be divided out to produce a smaller number with two different prime factorizations, contrary to our assumption that n is the smallest such number. Suppose without loss of generality that p_1 and q_1 are the smallest primes in their respective factorizations. Then $p_1^2 \leq n$ and $q_1^2 \leq n$. Since $p_1 \neq q_1$, one of these two inequalities must be strict, so that $p_1 q_1 < n$. In fact, $p_1 q_1 < n - 1$, because $p_1 \geq 2$ and $q_1 \geq 2$. Thus the number $N = n - p_1 q_1$ lies in the range $2 \leq N < n$, so its prime factorization is unique. But p_1 divides N, so it must occur in that factorization. Similarly, q_1 also occurs in the prime factorization of N. Thus $p_1 q_1$ is a divisor of N, so it is also a divisor of n. Consequently, q_1 divides the number $n/p_1 = p_2 p_3 \cdots p_k$. But this is not possible, because n/p_1 is less than n and therefore has a unique prime factorization, and q_1 does not occur in that factorization. This contradiction proves that the prime factorization of every number is unique.

For further study of elementary number theory, the book by Davenport [**2**] is especially recommended.

The connection of the zeta function with the prime numbers is apparent from its infinite product representation, which goes back to Euler.

Euler's Product Formula.

$$\zeta(x) = \prod_p \frac{1}{1 - p^{-x}}\,, \qquad x > 1\,,$$

where the product extends over all primes p.

Implicit in the statement is the assertion that the infinite product converges for every $x > 1$. The keys to the proof of the product formula are the geometric series expansion

$$\frac{1}{1 - p^{-x}} = 1 + \frac{1}{p^x} + \frac{1}{p^{2x}} + \dots$$

and the fundamental theorem of arithmetic. For each fixed $x > 1$ and integers $M > N \geq 2$ we have

$$\prod_{p \leq N} \left(1 + \frac{1}{p^x} + \frac{1}{p^{2x}} + \cdots + \frac{1}{p^{Mx}}\right) \leq \sum_{n=1}^{\infty} \frac{1}{n^x} = \zeta(x)\,.$$

Let $M \to \infty$ to infer that

$$\prod_{p \leq N} \frac{1}{1 - p^{-x}} \leq \zeta(x)\,,$$

so the infinite product converges and

$$\prod_{p} \frac{1}{1 - p^{-x}} \leq \zeta(x)\,.$$

For the reverse inequality, again choose $M > N \geq 2$ and write

$$\begin{aligned} \sum_{n=1}^{N} \frac{1}{n^x} &\leq \prod_{p \leq N} \left(1 + \frac{1}{p^x} + \frac{1}{p^{2x}} + \cdots + \frac{1}{p^{Mx}}\right) \\ &\leq \prod_{p \leq N} \frac{1}{1 - p^{-x}} \leq \prod_{p} \frac{1}{1 - p^{-x}}\,. \end{aligned}$$

Let $N \to \infty$ to deduce that

$$\zeta(x) \leq \prod_{p} \frac{1}{1 - p^{-x}}\,.$$

The two inequalities now combine to prove Euler's product formula.

The product formula is valid only for $x > 1$, but Euler did not hesitate to take $x = 1$ and "deduce" from the divergence of the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ that the product $\prod_p \left(1 - \frac{1}{p}\right)$ diverges to zero. This in turn implies (see Chapter 5, Theorem 1) that the sum of reciprocals of the primes diverges, which says that the sequence of prime numbers tends to infinity rather slowly.

For a more convincing argument that $\sum_p \frac{1}{p} = \infty$, we need only examine the above proof of Euler's product formula to write

$$\sum_{n=1}^{N} \frac{1}{n} \leq \prod_{p \leq N} \frac{1}{1 - \frac{1}{p}}$$

for each integer $N \geq 2$. Letting $N \to \infty$, we conclude that the product $\prod_p \left(1 - \frac{1}{p}\right)$ diverges to zero.

The problem now arises to determine the density of the primes. If p_n denotes the nth prime, what can be said about the order of magnitude of p_n? More precisely, what is the asymptotic behavior of p_n as $n \to \infty$? To put the question another way, if $\pi(x)$ denotes the number of primes not exceeding x, what is the asymptotic behavior of $\pi(x)$ as $x \to \infty$? In 1792, Gauss conjectured on the basis of extensive numerical data that

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as} \quad x \to \infty\,. \tag{3}$$

Legendre made the same conjecture independently in 1798. Half a century later, Chebyshev proved that

$$A\,\frac{x}{\log x} \leq \pi(x) \leq B\,\frac{x}{\log x}$$

for some positive constants A and B. Then in 1896, Jacques Hadamard and Charles de la Vallée Poussin used results of Riemann on the zeta function to give independent proofs of the asymptotic relation (3), now known as the *prime number theorem*.

An equivalent form of the prime number theorem is that $p_n \sim n \log n$ as $n \to \infty$. To see this, note that there are exactly n primes less than or equal to p_n, which says that $\pi(p_n) = n$. Since $p_n \to \infty$ as $n \to \infty$, the prime number theorem in the form (3) implies that $\pi(p_n) \sim p_n/\log p_n$, so that $p_n \sim n \log p_n$. In other words, $p_n/(n \log p_n) \to 1$, and it follows that

$$\log p_n - \log n - \log\log p_n \to 0 \qquad \text{as} \quad n \to \infty\,.$$

But $\log\log p_n/\log p_n \to 0$, so this shows that $\log p_n \sim \log n$, which allows us to conclude that $p_n \sim n \log n$. A similar argument shows that the asymptotic relation $p_n \sim n \log n$ implies (3). The details are left as an exercise.

Further discussion of prime numbers can be found for instance in the book by Tenenbaum and Mendès France [**8**]. The article by Bateman and Diamond [**1**] gives a nice historical account of the prime number theorem.

10.4. Connection with the gamma function

We now derive the relation

$$\Gamma(x)\zeta(x) = \int_0^\infty \frac{t^{x-1}}{e^t - 1}\,dt\,, \qquad x > 1\,, \tag{4}$$

between the zeta function and the gamma function

$$\Gamma(x) = \int_0^\infty e^{-u} u^{x-1}\, du\,, \qquad x > 0\,.$$

The change of variables $u = nt$ converts the latter integral to

$$\Gamma(x) = n^x \int_0^\infty e^{-nt} t^{x-1}\, dt\,, \qquad x > 0\,,\ n = 1, 2, \ldots.$$

Now take $x > 1$ and sum over n to obtain (formally)

$$\begin{aligned}\Gamma(x)\zeta(x) = \Gamma(x) \sum_{n=1}^\infty n^{-x} &= \sum_{n=1}^\infty \int_0^\infty e^{-nt} t^{x-1}\, dt \\ &= \int_0^\infty \left(\sum_{n=1}^\infty e^{-nt} \right) t^{x-1}\, dt = \int_0^\infty \frac{t^{x-1}}{e^t - 1}\, dt\,,\end{aligned}$$

as claimed. To justify the interchange of summation and integration, we can invoke Dini's theorem (*cf.* Section 1.8, Corollary 3).

Although the relation (4) has been derived under the assumption that $x > 1$, it can be used to extend the definition of $\zeta(x)$ to $0 < x < 1$. Since

$$\int_0^1 t^{x-2}\, dt = \frac{1}{x-1} \qquad \text{for } \ x > 1\,,$$

we can rewrite (4) in the form

$$\Gamma(x)\zeta(x) = \int_0^1 \left(\frac{1}{e^t - 1} - \frac{1}{t} \right) t^{x-1}\, dt + \frac{1}{x-1} + \int_1^\infty \frac{t^{x-1}}{e^t - 1}\, dt\,.$$

Observe now that each of the terms on the right-hand side is defined and continuous for all $x > 0$, except for the singularity of $1/(x-1)$ at $x = 1$. Consequently, since $\Gamma(x)$ is defined and $\Gamma(x) > 0$ for $x > 0$, we have extended the definition of $\zeta(x)$ to all $x > 0$, with a singularity at $x = 1$. Moreover, we see that $\zeta(x) - 1/(x-1)$ is continuous in the interval $(0, \infty)$ because $\Gamma(1) = 1$.

This extension of the zeta function into the interval $(0, 1)$ may appear rather arbitrary, but in the context of functions of a complex variable the extension is found to be uniquely determined. If z is a complex number, the formulas

$$\zeta(z) = \sum_{n=1}^\infty n^{-z} \qquad \text{and} \qquad \Gamma(z) = \int_0^\infty e^{-t} t^{z-1}\, dt$$

define $\zeta(z)$ and $\Gamma(z)$ as analytic functions in the half-planes $\mathrm{Re}\{z\} > 1$ and $\mathrm{Re}\{z\} > 0$, respectively. A generalization of the process just described then extends $\zeta(z)$ to an analytic function in the half-plane $\mathrm{Re}\{z\} > 0$, except for a pole at $z = 1$. But a basic principle of complex analysis says that an analytic extension can be given in at most one way.

In the language of analytic number theory, the vertical strip $0 < \mathrm{Re}\{z\} < 1$ is called the *critical strip* for the zeta function. The Riemann hypothesis, perhaps the most famous unsolved problem in mathematics, is the conjecture that all zeros of $\zeta(z)$ in the critical strip are situated on the line $\mathrm{Re}\{z\} = \frac{1}{2}$, known as the *critical line*. The Riemann hypothesis, if true, would have important implications in number theory. In particular, it would lead to a sharper form of the prime number theorem.

10.5. Functional equation

The functional equation for the Riemann zeta function is

$$\pi^{-x/2}\,\Gamma\Big(\frac{x}{2}\Big)\;\zeta(x) \;=\; \pi^{-(1-x)/2}\,\Gamma\left(\frac{1-x}{2}\right)\;\zeta(1-x)\,. \tag{5}$$

For $0 < x < 1$ this equation exhibits a certain symmetry of the extended function $\zeta(x)$ about the point $x = \frac{1}{2}$. For $x > 1$ it defines a natural extension of the zeta function to the negative real axis, the restriction of its analytic continuation to the left half-plane $\mathrm{Re}\{z\} < 0$. Since the left-hand side is continuous and positive for $x > 1$, the functional equation (5) shows that the same is true for $\Gamma(x/2)\zeta(x)$ when $x < 0$. But the gamma function is continuous and nonvanishing on the negative real line except for its singularities at the negative integers, so the extended zeta function has the same properties except that the singularities of $\Gamma(x/2)$ at the negative even integers must be canceled by zeros of $\zeta(x)$. In other words, the functional equation shows that the zeta function is continuous and nonvanishing on the negative real line except that $\zeta(x) = 0$ at the points $x = -2, -4, -6, \ldots$. These points are sometimes called the "trivial zeros" of the zeta function. It is known that infinitely many zeros lie on the critical line, and that only the trivial zeros lie outside the critical strip. Computer calculations have located millions of zeros on the critical line and none elsewhere in the critical strip, providing overwhelming numerical evidence in favor of the Riemann hypothesis, but no proof has been found.

We will give two distinct proofs of the functional equation. Both are elementary, but a full appreciation of either argument requires a little background in complex analysis; in particular, some familiarity with analytic functions and the uniqueness of an analytic continuation.

The first proof is due to G. H. Hardy [**3, 4**]. It begins with the Fourier series

$$(6) \qquad F(t) = \sum_{n=0}^{\infty} \frac{\sin(2n+1)t}{2n+1} = (-1)^m \frac{\pi}{4}\,, \qquad m\pi < t < (m+1)\pi\,,$$

for $m = 0, 1, 2, \ldots$, which was derived and discussed in Chapter 8. The strategy is to multiply the series (6) by t^{x-1} and integrate term by term, using the general formula

$$(7) \qquad \int_0^{\infty} t^{a-1} \sin bt\, dt = b^{-a}\, \Gamma(a) \sin(\pi a/2)\,, \qquad 0 < a < 1\,,\ b > 0\,,$$

derived at the end of the last chapter (Section 9.9). Deferring a justification of the term-by-term integration, we obtain

$$\begin{aligned} \int_0^{\infty} t^{x-1} F(t)\, dt &= \sum_{n=0}^{\infty} \frac{1}{2n+1} \int_0^{\infty} t^{x-1} \sin(2n+1)t\, dt \\ &= \Gamma(x) \sin(\pi x/2) \sum_{n=0}^{\infty} \frac{1}{(2n+1)^{x+1}} \\ &= \Gamma(x) \sin(\pi x/2) \left(1 - 2^{-x-1}\right) \zeta(x+1)\,, \qquad 0 < x < 1\,. \end{aligned}$$

On the other hand, taking account of the sum of the Fourier series (6), we find that

$$\begin{aligned} \int_0^{\infty} t^{x-1} F(t)\, dt &= \frac{\pi}{4} \sum_{m=0}^{\infty} (-1)^m \int_{m\pi}^{(m+1)\pi} t^{x-1}\, dt \\ &= \frac{\pi^{x+1}}{4x} \left\{ 1 + \sum_{m=1}^{\infty} (-1)^m \left[(m+1)^x - m^x \right] \right\}. \end{aligned}$$

The last series converges for $0 < x < 1$, by Leibniz' alternating series theorem, since the sequence $\{(m+1)^x - m^x\}$ decreases to zero as $m \to \infty$. But for $x < 0$ the same sequence increases to zero, so the series again converges. In fact, the convergence is uniform in each interval $-\infty < x \le b < 1$, so the sum

$$S(x) = 1 + \sum_{m=1}^{\infty} (-1)^m \left[(m+1)^x - m^x \right]$$

is well defined and continuous in the interval $(-\infty, 1)$. Trivially, $S(0) = 1$. Observe now that for $x < -1$ the sum is

$$S(x) = 2\left(1^x - 2^x + 3^x - \ldots\right) = 2\left(1 - 2^{x+1}\right) \zeta(-x)\,.$$

By analytic continuation, this relation $S(x) = 2\left(1 - 2^{x+1}\right)\zeta(-x)$ extends to all $x < 1$ and effectively defines $\zeta(-x)$ for $0 \le x < 1$. In particular, for $0 < x < 1$ the two calculations of the integral can be equated to obtain

$$\frac{\pi^{x+1}}{2x}\left(1 - 2^{x+1}\right)\zeta(-x) = \Gamma(x)\sin(\pi x/2)\left(1 - 2^{-x-1}\right)\zeta(x+1)\,,$$

which reduces to

$$2^x\pi^{x+1}\zeta(-x) = \Gamma(x+1)\sin(\pi x/2)\,\zeta(x+1)\,.$$

Replacing x by $x-1$, we arrive at the functional equation

$$\zeta(1-x) = 2^{1-x}\pi^{-x}\cos(\pi x/2)\Gamma(x)\zeta(x) \tag{8}$$

for the zeta function, an equivalent form of (5). The details of this equivalence are left as an exercise.

It remains to justify the interchange of summation and integration:

$$\int_0^\infty t^{x-1}F(t)\,dt = \sum_{n=0}^\infty \int_0^\infty t^{x-1}\frac{\sin(2n+1)t}{2n+1}\,dt\,, \qquad 0 < x < 1\,. \tag{9}$$

Observe first that the partial sums of the series

$$\sum_{n=0}^\infty \frac{\sin(2n+1)t}{2n+1} = \sum_{n=1}^\infty \frac{\sin nt}{n} - \sum_{n=1}^\infty \frac{\sin 2nt}{2n}$$

are uniformly bounded. (See Chapter 8, Exercise 15.) On the other hand, it can be shown by the technique of Abel summation (*cf.* Chapter 3, Exercise 12) that this series converges uniformly in each closed set that contains no multiple of π (*cf.* Chapter 8, Exercise 16). These two facts allow us to conclude that

$$\int_0^R t^{x-1}F(t)\,dt = \sum_{n=0}^\infty \int_0^R t^{x-1}\,\frac{\sin(2n+1)t}{2n+1}\,dt\,, \qquad 0 < x < 1\,,$$

for arbitrary $R < \infty$. Hence the proof of (9) reduces to showing that

$$\lim_{R\to\infty}\sum_{n=0}^\infty \frac{1}{2n+1}\int_R^\infty t^{x-1}\sin(2n+1)t\,dt = 0\,. \tag{10}$$

But an integration by parts gives

$$\begin{aligned}\left|\int_R^\infty t^{x-1}\sin(2n+1)t\,dt\right| &= \left|\frac{R^{x-1}\cos(2n+1)R}{2n+1}\right.\\ &\qquad \left. + \frac{x-1}{2n+1}\int_R^\infty t^{x-2}\cos(2n+1)t\,dt\right|\\ &\le \frac{R^{x-1}}{2n+1} + \frac{1-x}{2n+1}\int_R^\infty t^{x-2}\,dt = \frac{2\,R^{x-1}}{2n+1}\,,\end{aligned}$$

and (10) follows. This completes the proof of (9) and justifies the term-by-term integration used to derive the functional equation for the zeta function.

Our second proof of the functional equation is quite different and is due to Riemann himself. It relies on the inversion formula $\vartheta(1/t) = \sqrt{t}\,\vartheta(t)$ for Jacobi's theta function

$$\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-n^2\pi t}\,, \qquad t > 0\,,$$

which we derived in Section 8.11 as an application of the Poisson summation formula. The first step is to calculate the integral

$$\int_0^\infty t^{x/2-1} e^{-n^2\pi t}\,dt = \pi^{-x/2}\,\Gamma\left(\frac{x}{2}\right)\,n^{-x}\,, \qquad x > 0\,,$$

directly from the definition of the gamma function. Taking $x > 1$ and summing over n, we then deduce that

$$\pi^{-x/2}\,\Gamma\left(\frac{x}{2}\right)\,\zeta(x) = \sum_{n=1}^{\infty} \int_0^\infty t^{x/2-1} e^{-n^2\pi t}\,dt = \int_0^\infty t^{x/2-1}\psi(t)\,dt\,, \tag{11}$$

where

$$\psi(t) = \sum_{n=1}^{\infty} e^{-n^2\pi t}\,. \tag{12}$$

The interchange of summation and integration is justified by the uniform convergence of the sum (12) in the interval $[0,\infty)$. Observe now that $\vartheta(t) = 2\psi(t) + 1$, so that the formula $\vartheta(1/t) = \sqrt{t}\,\vartheta(t)$ implies

$$\psi(t) = \tfrac{1}{2}\left(t^{-1/2} - 1\right) + t^{-1/2}\,\psi(1/t)\,. \tag{13}$$

Using the relation (13), we can write

$$\begin{aligned}
\int_0^\infty t^{x/2-1}\,\psi(t)\,dt &= \int_0^1 t^{x/2-1}\,\psi(t)\,dt + \int_1^\infty t^{x/2-1}\,\psi(t)\,dt \\
&= \frac{1}{2}\int_0^1 \left(t^{x/2-3/2} - t^{x/2-1}\right)dt + \int_0^1 t^{x/2-3/2}\,\psi(1/t)\,dt \\
&\quad + \int_1^\infty t^{x/2-1}\,\psi(t)\,dt \\
&= \frac{1}{x(x-1)} + \int_1^\infty s^{-x/2-1/2}\,\psi(s)\,ds + \int_1^\infty t^{x/2-1}\,\psi(t)\,dt\,,
\end{aligned}$$

with the substitution $s = 1/t$ in the second integral. Consequently, the equation (11) reduces to

$$\pi^{-x/2}\,\Gamma\left(\frac{x}{2}\right)\,\zeta(x) = \frac{1}{x(x-1)} + \int_1^\infty \left(t^{-x/2-1/2} + t^{x/2-1}\right)\psi(t)\,dt\,, \quad x > 1\,. \tag{14}$$

But it is easy to see that $\psi(t) \le Ce^{-\pi t}$ for some constant $C > 0$ and all $t \ge 1$, and so the last integral in (14) converges for every $x \in \mathbb{R}$. Thus the relation (14) defines a natural extension of the zeta function to the whole real line, except for the singularity at $x = 1$. A simple calculation reveals that the right-hand side of (14) is unchanged if x is replaced by $1 - x$, and so the functional equation (5) follows.

The essence of the proof is to derive the functional equation for the zeta function from Jacobi's inversion formula for the theta function. On the other hand, it is possible to reverse the argument and derive Jacobi's formula from the functional equation. Thus the two relations can be regarded as equivalent.

Readers familiar with complex function theory may wish to consult the book by Titchmarsh [**10**], where both of the preceding proofs of the functional equation can be found, together with several others, in the more natural setting of the complex plane. Further details of Hardy's proof are given in Titchmarsh [**9**], Section 4.45.

Bernhard Riemann had an unusually fertile imagination. Although he published only a few papers, each one introduced new ideas of fundamental importance. Born in Hanover (now part of Germany) in 1826, he entered the University of Göttingen at age 19 but soon transferred to Berlin, where he studied under Dirichlet and Jacobi. Returning to Göttingen in 1849, he became a student of Gauss and completed an inaugural dissertation on geometric aspects of complex function theory, including the Riemann mapping theorem and the concept of a Riemann surface. His *Habilitationschrift* in 1853 focused on functions representable as the sum of a trigonometric series and led to him to formalize the notion of a Riemann integral. In his famous *Habilitationsvortrag* of 1854 he presented the fundamental ideas of Riemannian geometry. When Gauss died in 1855, Dirichlet was appointed his successor, then when Dirichlet died four years later the position passed to Riemann. At that point Riemann produced his celebrated paper on the zeta function, viewing it not as a function of a real variable as Euler had done, but of a complex variable. There he derived the functional equation, stated the Riemann hypothesis, and showed its relevance to the distribution of prime numbers. After several years of ill health, Riemann succumbed to tuberculosis in 1866, two months short of his fortieth birthday.

Exercises

1. Show that the sequence

$$\left\{\tfrac{0}{1}, \tfrac{0}{2}, \tfrac{1}{2}, \tfrac{0}{3}, \tfrac{1}{3}, \tfrac{2}{3}, \tfrac{0}{4}, \tfrac{1}{4}, \tfrac{2}{4}, \tfrac{3}{4}, \dots\right\}$$

is equidistributed over $[0, 1)$.

2. If each of the sequences $\{\alpha_n\}$ and $\{\beta_n\}$ is equidistributed modulo 1, show that the sequence $\{\alpha_1, \beta_1, \alpha_2, \beta_2, \dots\}$ is also equidistributed modulo 1.

3. Give an example of a sequence that is everywhere dense in the interval $[0, 1]$ but is not equidistributed.

4. Show that the sequence $\{\cos n\}$ is everywhere dense in the interval $[-1, 1]$. In particular,

$$\limsup_{n\to\infty} \cos n = 1 \qquad \text{and} \qquad \liminf_{n\to\infty} \cos n = -1\,.$$

5. Since the number e is irrational, the sequence $\{ne\}$ is equidistributed modulo 1. Show, however, that the sequence $\{n!e\}$ is not equidistributed modulo 1.

Hint. Apply an estimate for the difference between e and the partial sums $s_n = 1 + \sum_{k=1}^{n} 1/k!$.

6. Prove that the sequence $\left\{\left(\frac{1+\sqrt{5}}{2}\right)^n\right\}$ is not equidistributed modulo 1. Can you generalize the result?

Hint. Show that the numbers $x_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$ satisfy the relation $x_{n+2} = x_n + x_{n+1}$ with $x_1 = 1$ and $x_2 = 3$.

7. Show that the sequence $\{\alpha_n\}$ is equidistributed modulo 1 if

$$\begin{aligned} &(a) \qquad \alpha_n = n^p\,, \qquad 0 < p < 1\,,\\ &(b) \qquad \alpha_n = (\log n)^p\,, \qquad p > 1\,,\\ &(c) \qquad \alpha_n = n(\log n)^p\,, \qquad p < 0\,. \end{aligned}$$

8. Show that the sequence $\{(\log n)^p\}$ is not equidistributed modulo 1 when $p \le 1$.

9. Prove that a sequence $\{\alpha_n\}$ is equidistributed in the interval $[0, 1]$ if and only if

$$\lim_{n\to\infty} \frac{\alpha_1^k + \alpha_2^k + \cdots + \alpha_n^k}{n} = \frac{1}{k+1}\,, \qquad k = 1, 2, \dots\,.$$

10. Show that

$$\zeta(x)^2 = \sum_{n=1}^{\infty} d(n)\, n^{-x}\,, \qquad x > 1\,,$$

where $d(n)$ is the number of divisors of n.

11. Show that

$$\zeta(x)\zeta(x-1) = \sum_{n=1}^{\infty} \sigma(n)\, n^{-x}\,, \qquad x > 2\,,$$

where $\sigma(n)$ denotes the sum of the divisors of n.

***12.** Show that

$$\frac{\zeta(x-1)}{\zeta(x)} = \sum_{n=1}^{\infty} \varphi(n)\, n^{-x}\,, \qquad x > 2\,,$$

where $\varphi(n)$ denotes the number of integers less than or equal to n that are relatively prime to n. (Two positive integers are said to be relatively prime if their greatest common divisor is 1.)

Discussion. The function φ, introduced by Euler, has remarkable properties. It is multiplicative in the sense that $\varphi(nm) = \varphi(n)\varphi(m)$ whenever n and m are relatively prime. A theorem of Gauss asserts that $\sum_{k|m} \varphi(k) = m$. In other words, the sum of the numbers $\varphi(k)$, extended over all divisors k of m, is equal to m. See for instance Davenport [**2**] for further details.

13. As shown in Section 10.3, the prime number theorem $\pi(x) \sim x/\log x$ implies that the nth prime p_n satisfies the asymptotic relation $p_n \sim n \log n$ as $n \to \infty$. Prove that conversely, $p_n \sim n \log n$ implies $\pi(x) \sim x/\log x$ as $x \to \infty$.

14. The logarithmic integral function

$$\mathrm{Li}(x) = \int_2^x \frac{1}{\log t}\, dt$$

is known to give a better approximation to $\pi(x)$ than does $x/\log x$. Show that $\mathrm{Li}(x) \sim x/\log x$ by proving

$$\mathrm{Li}(x) = \frac{x}{\log x} + O\left(\frac{x}{(\log x)^2}\right).$$

Hint. An integration by parts reduces the problem to showing

$$\int_2^x \frac{1}{(\log t)^2}\, dt = O\left(\frac{x}{(\log x)^2}\right),$$

and the estimate is completed by splitting the integral at $\sqrt{x}$.

15. Show directly that the two forms (5) and (8) of the functional equation for the zeta function are equivalent.

16. Conclude from the functional equation that $\zeta(0) = -\frac{1}{2}$. Show further that $\zeta(-1) = -\frac{1}{12}$, $\zeta(-3) = \frac{1}{120}$, and in general that $\zeta(1-2n) = -\frac{1}{2n} B_{2n}$ for $n = 1, 2, \ldots$. (Apply Euler's formula for $\zeta(2n)$ in terms of the Bernoulli number B_{2n}, quoted in Section 3.7 and proved in Section 11.3.)

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Chapter 11
Bernoulli Numbers

The Bernoulli numbers were introduced by Jacob Bernoulli (1654–1705) in his posthumously published treatise *Ars Conjectandi* (1713) to give a general formula for sums of the form $1^p + 2^p + \cdots + n^p$, where p is a positive integer. Euler used them later to evaluate the infinite series $\sum_{k=1}^{\infty} 1/k^n$ for all even integers n. They have become important in number theory and combinatorics, and they occur in algorithms such as the Euler–Maclaurin summation formula for numerical computation of definite integrals. We will discuss some of these connections later, but our first priority is to define the Bernoulli numbers and to develop an efficient scheme for their calculation.

11.1. Calculation of Bernoulli numbers

The *Bernoulli numbers* B_n are defined by the generating relation

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n \,. \tag{1}$$

Calculations with the power series

$$e^x - 1 = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

(*cf.* Section 3.6 for justification) show that

$$\begin{aligned} \frac{x}{e^x - 1} &= \frac{1}{1 + \frac{1}{2}x + \frac{1}{6}x^2 + \dots} \\ &= 1 - \left(\tfrac{1}{2}x + \tfrac{1}{6}x^2 + \dots\right) + \left(\tfrac{1}{2}x + \tfrac{1}{6}x^2 + \dots\right)^2 - \dots \\ &= 1 - \tfrac{1}{2}x + \tfrac{1}{12}x^2 + \dots \end{aligned}$$

for sufficiently small x, so that $B_0 = 1$, $B_1 = -\frac{1}{2}$, and $B_2 = \frac{1}{6}$. In particular,

$$\frac{x}{e^x - 1} = 1 - \tfrac{1}{2}x + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n \,.$$

Observe now that

$$f(x) = \frac{x}{e^x - 1} + \tfrac{1}{2}x = \tfrac{1}{2}x\,\frac{e^x + 1}{e^x - 1} = 1 + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n \tag{2}$$

is an even function; it has the property $f(-x) = f(x)$. This implies that its power series contains only even powers of x. In other words, $B_3 = B_5 = B_7 = \cdots = 0$.

The Bernoulli numbers of even index are best calculated with the help of a recurrence relation. Multiplying both sides of the generating relation (1) by the function

$$e^x - 1 = \sum_{n=1}^{\infty} \frac{1}{n!} x^n \,,$$

we find that

$$x = \left(\sum_{n=0}^{\infty} \frac{B_n}{n!} x^n \right) \left(\sum_{n=1}^{\infty} \frac{1}{n!} x^n \right) = \sum_{n=1}^{\infty} c_n x^n \,,$$

where

$$c_n = \frac{B_0}{0!\,n!} + \frac{B_1}{1!\,(n-1)!} + \frac{B_2}{2!\,(n-2)!} + \cdots + \frac{B_{n-1}}{(n-1)!\,1!} \,.$$

A comparison of coefficients now shows that $c_1 = 1$ and $c_n = 0$ for all $n \geq 2$. When binomial coefficients are introduced, the relation $c_n = 0$ becomes

$$\sum_{k=0}^{n-1} \binom{n}{k} B_k = 0 \,, \qquad n = 2, 3, \ldots \,. \tag{3}$$

The equations $c_1 = 1$, $c_2 = 0$, $c_3 = 0$ reduce respectively to

$$B_0 = 1 \,, \qquad \tfrac{1}{2}B_0 + B_1 = 0 \,, \qquad \tfrac{1}{3}B_0 + B_1 + B_2 = 0 \,,$$

so that $B_1 = -\frac{1}{2}$ and $B_2 = \frac{1}{6}$. The equation $c_4 = 0$ then reduces to

$$\tfrac{1}{24}B_0 + \tfrac{1}{6}B_1 + \tfrac{1}{4}B_2 + \frac{1}{6}B_3 = \tfrac{1}{24} - \tfrac{1}{12} + \tfrac{1}{24} + \tfrac{1}{6}B_3 = 0 \,,$$

or $B_3 = 0$. The equation $c_5 = 0$ gives

$$\tfrac{1}{5}B_0 + B_1 + 2B_2 + 2B_3 + B_4 = 0\,,$$

so that

$$B_4 = -\tfrac{1}{5} + \tfrac{1}{2} - \tfrac{1}{3} = -\tfrac{1}{30}\,.$$

More generally, the equation $c_{2n+1} = 0$ takes the form

$$\frac{1}{2n+1} - \frac{1}{2} + \sum_{k=1}^{n} \frac{(2n)!}{(2n-2k+1)!\,(2k)!} B_{2k} = 0\,, \tag{4}$$

since $B_{2k+1} = 0$ for $k \geq 1$. The expression (4) is a recurrence relation that allows the successive calculation of the Bernoulli numbers B_{2n} from previously found values of B_2, B_4, ... B_{2n-2}. For $n = 3$ it reduces to

$$\tfrac{1}{7} - \tfrac{1}{2} + 3B_2 + 5B_4 + B_6 = 0\,,$$

which gives

$$B_6 = -\tfrac{1}{7} + \tfrac{1}{2} - \tfrac{1}{2} + \tfrac{1}{6} = \tfrac{1}{42}\,.$$

For $n = 4$ the relation (4) is

$$\tfrac{1}{9} - \tfrac{1}{2} + 4B_2 + 14B_4 + \tfrac{28}{3}B_6 + B_8 = 0\,,$$

and so

$$B_8 = -\tfrac{1}{9} + \tfrac{1}{2} - \tfrac{2}{3} + \tfrac{7}{15} - \tfrac{2}{9} = -\tfrac{1}{30}\,.$$

For convenient reference, here is a summary of results:

$$B_0 = 1\,, \quad B_1 = -\tfrac{1}{2}\,, \quad B_2 = \tfrac{1}{6}\,, \quad B_4 = -\tfrac{1}{30}\,, \quad B_6 = \tfrac{1}{42}\,, \quad B_8 = -\tfrac{1}{30}\,.$$

Similar calculations, increasingly laborious, produce the values

$$B_{10} = \tfrac{5}{66}\,, \quad B_{12} = -\tfrac{691}{2730}\,, \quad B_{14} = \tfrac{7}{6}\,, \quad B_{16} = -\tfrac{3617}{510}\,, \quad \text{etc.}$$

The Bernoulli numbers can be found in books of mathematical tables and in some computer software packages. For instance, *Mathematica* will display B_n as a numerical fraction if one enters the code `BernoulliB[n]` with a specific integer n.

The values of B_{2n} so far calculated suggest that the signs always alternate: $B_2 > 0$, $B_4 < 0$, $B_6 > 0$, $B_8 < 0$, etc. This is true and will be verified by inspection of the formulas for Euler's sums.

11.2. Sums of positive powers

The familiar formula

$$1+2+3+\cdots+n=\frac{n(n+1)}{2}$$

for the sum of the first n integers can be discovered in many ways. Its proof is a popular exercise in mathematical induction. The formulas

$$1^2+2^2+3^2+\cdots+n^2=\frac{n(n+1)(2n+1)}{6}\,,$$
$$1^3+2^3+3^3+\cdots+n^3=\left(\frac{n(n+1)}{2}\right)^2\,,\ \ldots$$

are also easy to verify by induction, but are perhaps more difficult to discover. One method of discovery is to guess by analogy with the integral of x^p that the sum $1^p+2^p+\cdots+n^p$ will be a polynomial in n of degree $p+1$ and to determine the coefficients by interpolation of numerical data. However, this method does not lead to a general formula for the sum. Jacob Bernoulli found such a formula in terms of Bernoulli numbers.

Bernoulli's formula is

$$\sum_{k=1}^{n}k^p=\frac{1}{p+1}\sum_{j=1}^{p+1}\binom{p+1}{j}B_{p-j+1}\,(n+1)^j\,,\qquad p=1,2,\ldots \tag{5}$$

where

$$\binom{p+1}{j}=\frac{(p+1)!}{(p-j+1)!\,j!}$$

is a binomial coefficient. If $p=2$, for instance, the formula (5) reduces to

$$\sum_{k=1}^{n}k^2=B_2(n+1)+B_1(n+1)^2+\tfrac{1}{3}B_0(n+1)^3=\frac{n(n+1)(2n+1)}{6}\,.$$

For a proof of (5), observe first that

$$\sum_{k=0}^{n}e^{kx}=\sum_{k=0}^{n}\sum_{p=0}^{\infty}\frac{1}{p!}(kx)^p=\sum_{p=0}^{\infty}\left(\sum_{k=0}^{n}k^p\right)\frac{x^p}{p!}\,,\qquad x\neq 0\,.$$

On the other hand, the geometric progression has the sum

$$\sum_{k=0}^{n}e^{kx}=\sum_{k=0}^{n}(e^x)^k=\frac{e^{(n+1)x}-1}{e^x-1}\,,\qquad x\neq 0\,.$$

Comparing the two expressions, and recalling the generating relation (1), we see that

$$\sum_{p=0}^{\infty}\left(\sum_{k=0}^{n}k^p\right)\frac{x^{p+1}}{p!}=\frac{x}{e^x-1}\left(e^{(n+1)x}-1\right)$$
$$=\left(\sum_{k=0}^{\infty}\frac{B_k}{k!}x^k\right)\left(\sum_{j=1}^{\infty}\frac{(n+1)^j}{j!}x^j\right)=\sum_{p=0}^{\infty}a_p x^{p+1}\,,$$

where

$$a_p=\sum_{j=1}^{p+1}\frac{1}{(p-j+1)!\,j!}B_{p-j+1}(n+1)^j\,.$$

Equating the coefficients of x^{p+1}, we arrive at the formula

$$\sum_{k=0}^{n}k^p=p!\,a_p=\sum_{j=1}^{p+1}\frac{p!}{(p-j+1)!\,j!}B_{p-j+1}(n+1)^j\,,\qquad p=1,2,\ldots\,.$$

which is equivalent to (5).

11.3. Euler's sums

In Chapter 3 we discussed Euler's solution of the "Basel problem", first proposed by Jacob Bernoulli in the year 1690, to find the sum of the infinite series

$$\sum_{n=1}^{\infty}\frac{1}{n^2}\,.$$

(See Section 3.7.) Euler solved the problem in 1735 with the sensational discovery that the sum of the series is $\pi^2/6$. Shortly thereafter, Euler also found that

$$\sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{90}\,,\qquad\sum_{n=1}^{\infty}\frac{1}{n^6}=\frac{\pi^6}{945}\,,\tag{6}$$

and in general that

$$\sum_{n=1}^{\infty}\frac{1}{n^{2k}}=(-1)^{k+1}\frac{(2\pi)^{2k}}{2(2k)!}B_{2k}\,,\qquad k=1,2,\ldots\,,\tag{7}$$

where B_{2k} is a Bernoulli number. Observe that for $k=1$ the sum (7) reduces to $\sum 1/n^2=\pi^2/6$, since $B_2=1/6$.

Euler's formula $\sum 1/n^2 = \pi^2/6$ has been verified by many methods, some quite elementary, but no proof is really simple. Two elementary proofs were presented in Section 3.7 of this book. Euler based a proof on the infinite product formula for the sine function (*cf.* Section 8.5). His result often appears as a special case of a Fourier expansion, but then it depends on a theorem ensuring that the Fourier series actually converges to the function at the point in question. In fact, Fourier series provide a key to the proof of Euler's general formula (7), since Fourier expansion of the function $\cos cx$ leads to the formula

$$\pi t \cot \pi t = 1 + \sum_{n=1}^{\infty} \frac{2t^2}{t^2 - n^2}\,,$$

as shown in Section 8.5. To understand the relevance of this formula to Euler's sums, expand the summands in geometric series to infer that

$$\begin{aligned} \pi t \cot \pi t &= 1 - 2\sum_{n=1}^{\infty} \frac{(t/n)^2}{1-(t/n)^2} = 1 - 2\sum_{n=1}^{\infty}\sum_{k=1}^{\infty} (t/n)^{2k} \\ &= 1 - 2\sum_{k=1}^{\infty}\sum_{n=1}^{\infty} (t/n)^{2k} = 1 - 2\sum_{k=1}^{\infty}\left(\sum_{n=1}^{\infty}\frac{1}{n^{2k}}\right) t^{2k}\,. \end{aligned} \tag{8}$$

On the other hand, the function $\pi t \cot \pi t$ can be expanded in another way that introduces Bernoulli numbers. Letting $x = 2t$ in the generating relation (2), we obtain the formula

$$1 + \sum_{k=1}^{\infty} B_{2k}\frac{(2t)^{2k}}{(2k)!} = \frac{2t}{e^{2t}-1} + \tfrac{1}{2}(2t) = \frac{t(e^{2t}+1)}{e^{2t}-1} = \frac{t(e^t+e^{-t})}{e^t - e^{-t}}\,. \tag{9}$$

Now recall Euler's formula $e^{it} = \cos t + i \sin t$ and its consequences:

$$\cos t = \frac{e^{it}+e^{-it}}{2}\,, \qquad \sin t = \frac{e^{it}-e^{-it}}{2i}\,.$$

Replacing t by $i\pi t$ in (9), we deduce that

$$\pi t \cot \pi t = 1 + \sum_{k=1}^{\infty} (-1)^k B_{2k} \frac{(2\pi)^{2k}}{(2k)!}\, t^{2k}\,. \tag{10}$$

Equating the coefficients of t^{2k} in the two expansions (8) and (10) for $\pi t \cot \pi t$, we arrive at Euler's formula (7).

A more direct version of the proof can be based instead on the expression for $\pi t \coth \pi t$ obtained in Exercise 29 of Chapter 8 as an application of the

Poisson summation formula. Expand this expression as in (8) to introduce Euler's sums, then replace t by πt in (9) and compare coefficients of t^{2k} to derive the formula (7). A different proof will appear in Section 11.4.

It is clear from the equation (7) that $(-1)^{k+1}B_{2k} > 0$ for all integers $k \geq 1$. This confirms the pattern of alternating signs in the sequence of Bernoulli numbers B_{2k}.

Euler's formula (7) has two simple corollaries:

$$\sum_{m=0}^{\infty} \frac{1}{(2m+1)^{2k}} = (-1)^{k+1} \frac{(2^{2k}-1)\pi^{2k}}{2(2k)!} B_{2k} \tag{11}$$

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^{2k}} = (-1)^{k+1} \frac{(2^{2k-1}-1)\pi^{2k}}{(2k)!} B_{2k} \tag{12}$$

The proofs are left as exercises.

The Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}\,, \qquad \mathrm{Re}\{s\} > 1\,,$$

a function of a complex variable, plays an important role in number theory and was discussed in Chapter 10. Euler's formula evaluates $\zeta(2k)$ for all even integers $2k$. However, the value of the zeta function is not known for any odd integer. In 1973, Roger Apéry proved that $\zeta(3)$ is irrational, but it is still not known whether $\zeta(3)$ is a rational multiple of π^3.

11.4. Bernoulli polynomials

The *Bernoulli polynomials* $b_n(t)$ are defined by the generating relation

$$\frac{xe^{tx}}{e^x - 1} = \sum_{n=0}^{\infty} \frac{b_n(t)}{n!} x^n\,. \tag{13}$$

For $t = 0$ this reduces to the generating relation (1) for the Bernoulli numbers, and so $b_n(0) = B_n$. To see that $b_n(t)$ is a polynomial of degree n, multiply the power series

$$e^{tx} = 1 + \sum_{k=1}^{\infty} \frac{t^k}{k!} x^k$$

by that of (1) and compare with (13) to calculate the coefficients $b_0(t) \equiv 1$ and

$$\frac{b_n(t)}{n!} = \sum_{k=0}^{n} \frac{B_{n-k}}{(n-k)!\,k!}\, t^k\,, \qquad n = 1, 2, \ldots\,,$$

or

$$b_n(t) = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}\, t^k \,. \tag{14}$$

This shows that $b_n(t)$ is a monic polynomial of degree n, since $B_0 = 1$.

The values

$$B_0 = 1, \quad B_1 = -\tfrac{1}{2}, \quad B_2 = \tfrac{1}{6}, \quad B_3 = 0, \quad B_4 = -\tfrac{1}{30}, \quad B_5 = 0, \quad B_6 = \tfrac{1}{42}$$

can be inserted to calculate

$$\begin{aligned} &b_1(t) = t - \tfrac{1}{2}\,, \qquad b_2(t) = t^2 - t + \tfrac{1}{6}\,, \\ &b_3(t) = t^3 - \tfrac{3}{2}t^2 + \tfrac{1}{2}t\,, \qquad b_4(t) = t^4 - 2t^3 + t^2 - \tfrac{1}{30}\,, \\ &b_5(t) = t^5 - \tfrac{5}{2}t^4 + \tfrac{5}{3}t^3 - \tfrac{1}{6}t\,, \qquad b_6(t) = t^6 - 3t^5 + \tfrac{5}{2}t^4 - \tfrac{1}{2}t^2 + \tfrac{1}{42}\,. \end{aligned}$$

We will be interested in the values of $b_n(1)$. Calculations show that $b_1(1) = -B_1$ and $b_n(1) = B_n$ for $n = 2, 3, 4, 5, 6$, which suggests that $b_n(1) = B_n$ for all $n \geq 2$. This is most easily proved by inserting $t = 1$ into the generating relation (13) to obtain

$$\sum_{n=0}^{\infty} \frac{b_n(1)}{n!} x^n = \frac{xe^x}{e^x - 1} = \frac{-x}{e^{-x} - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} (-x)^n \,.$$

Comparison of coefficients then gives $b_n(1) = (-1)^n B_n$. Since $B_n = 0$ for all odd $n \geq 3$, this confirms that $b_n(1) = B_n$ for $n \geq 2$. For another proof, one can apply the recurrence relation (3) directly to the formula (14).

The most important property of the Bernoulli polynomials is the identity

$$b_n'(t) = n b_{n-1}(t)\,, \qquad n = 1, 2, \ldots \,. \tag{15}$$

This again is most easily deduced from the generating relation (13). Term-by-term differentiation of (13) gives

$$\sum_{n=1}^{\infty} \frac{b_n'(t)}{n!} x^n = \frac{x^2 e^{tx}}{e^x - 1} = \sum_{n=0}^{\infty} \frac{b_n(t)}{n!} x^{n+1}\,,$$

and a comparison of coefficients verifies (15). However, the term-by-term differentiation needs to be justified, and this is not so easy. A more pedestrian proof of (15) can be based directly on the representation (14).

In terms of Bernoulli polynomials, the formula (5) for the sum of positive powers can be recast as

$$\sum_{k=1}^{n} k^p = \frac{1}{p+1} \big(b_{p+1}(n+1) - B_{p+1} \big)\,. \tag{16}$$

The proof is left as an exercise.

The Bernoulli polynomials have an elegant Fourier series expansion, most conveniently expressed in complex form as

$$b_n(t) = -\frac{n!}{(2\pi i)^n} \sum_{k\neq 0} \frac{e^{2\pi ikt}}{k^n}\,, \qquad 0 \le t \le 1 \tag{17}$$

for $n = 2, 3, \ldots$, where the sum extends over all integers k, positive and negative, except for $k = 0$. The same formula holds for $n = 1$ but is restricted to the open interval $0 < t < 1$, since the periodic extension of $b_1(t) = t - \frac{1}{2}$ is discontinuous at the endpoints. Recall that $b_n(0) = b_n(1) = B_n$ for every $n \ge 2$.

To verify the expansion (17), write

$$b_n(t) = \sum_{k=-\infty}^{\infty} c_{nk}\, e^{2\pi ikt}\,, \qquad 0 < t < 1\,,$$

where the Fourier coefficients are given by

$$c_{nk} = \int_0^1 b_n(t) e^{-2\pi ikt}\, dt\,.$$

Observe first that

$$\begin{aligned} c_{n0} &= \int_0^1 b_n(t)\, dt = \frac{1}{n+1} \int_0^1 b'_{n+1}(t)\, dt \\ &= \frac{1}{n+1}\big(b_{n+1}(1) - b_{n+1}(0)\big) = \frac{1}{n+1}\big(B_{n+1} - B_{n+1}\big) = 0 \end{aligned}$$

for $n = 1, 2, \ldots$. For $k \neq 0$, an integration by parts gives

$$\begin{aligned} c_{1k} &= \int_0^1 b_1(t) e^{-2\pi ikt}\, dt = \int_0^1 (t - \tfrac{1}{2}) e^{-2\pi ikt}\, dt \\ &= -\frac{1}{2\pi ik}\Big[(t - \tfrac{1}{2}) e^{-2\pi ikt}\Big]_0^1 + \frac{1}{2\pi ik} \int_0^1 e^{-2\pi ikt}\, dt = -\frac{1}{2\pi ik}\,. \end{aligned}$$

For fixed $k \neq 0$, we now proceed by induction on n. Suppose that the formula $c_{nk} = -n!/(2\pi ik)^n$ holds for some $n \ge 1$. Then by (15)

$$\begin{aligned} c_{n+1,k} &= \int_0^1 b_{n+1}(t) e^{-2\pi ikt}\, dt \\ &= -\frac{1}{2\pi ik}\Big[b_{n+1}(t) e^{-2\pi ikt}\Big]_0^1 + \frac{1}{2\pi ik} \int_0^1 b'_{n+1}(t) e^{-2\pi ikt}\, dt \\ &= \frac{n+1}{2\pi ik} \int_0^1 b_n(t) e^{-2\pi ikt}\, dt = \frac{n+1}{2\pi ik}\, c_{nk} = -\frac{(n+1)!}{(2\pi ik)^{n+1}}\,. \end{aligned}$$

This completes the inductive argument and shows that the formula holds for each $k \neq 0$ and all $n \geq 1$. Hence the polynomial $b_n(t)$ has the Fourier expansion (17).

For $n = 2m$ and $t = 0$, the expression (17) reduces to

$$B_{2m} = b_{2m}(0) = -\frac{2(2m)!}{(2\pi i)^{2m}} \sum_{k=1}^{\infty} \frac{1}{k^{2m}}\,,$$

so that

$$\sum_{k=1}^{\infty} \frac{1}{k^{2m}} = (-1)^{m+1}\, \frac{(2\pi)^{2m}}{2(2m)!}\, B_{2m}\,.$$

Thus we have found another proof of Euler's formula (7) for the value of the zeta function at an even integer.

In general, the formula (17) takes the form

$$b_{2m}(t) = (-1)^{m+1}\, \frac{2(2m)!}{(2\pi)^{2m}} \sum_{k=1}^{\infty} \frac{\cos 2\pi kt}{k^{2m}}\,, \qquad 0 \leq t \leq 1\,, \tag{18}$$

for even indices $n = 2m$, and

$$b_{2m+1}(t) = (-1)^{m+1}\, \frac{2(2m+1)!}{(2\pi)^{2m+1}} \sum_{k=1}^{\infty} \frac{\sin 2\pi kt}{k^{2m+1}}\,, \qquad 0 \leq t \leq 1\,, \tag{19}$$

for odd indices $n = 2m+1$, with the proviso that (19) holds only for $0 < t < 1$ when $m = 0$.

11.5. Euler–Maclaurin summation formula

Bernoulli polynomials provide the key to a basic method for numerical calculation and asymptotic analysis of integrals, known as the Euler–Maclaurin summation formula. Euler developed the technique in 1732 and applied it in 1736 to calculate Euler's constant γ accurately to 16 decimal places, a stunning feat in that era of hand calculation. Earlier, before discovering the exact values of $\zeta(2)$ and $\zeta(4)$, he had applied the formula to calculate $\zeta(2)$, $\zeta(3)$, and $\zeta(4)$ to over 15 decimal places. The Scottish mathematician Colin Maclaurin (1698–1746) discovered the method independently and discussed it in his book *A Treatise of Fluxions* (1742).

The basic idea is to improve upon the Riemann sum approximation of a definite integral by taking into account the derivatives of the integrand. This is accomplished through successive integrations by parts. To fix ideas,

suppose a function $f(t)$ is defined on the whole real line and has derivatives of all orders. Introduce the Bernoulli polynomial

$$b_1(t) = t - \tfrac{1}{2} = \tfrac{1}{2}b_2'(t)$$

and integrate by parts to obtain

$$\begin{aligned}\int_0^1 f(t)\,dt = \int_0^1 b_1'(t)f(t)\,dt &= \Big[b_1(t)f(t)\Big]_0^1 - \int_0^1 b_1(t)f'(t)\,dt \\ &= \frac{1}{2}\big[f(0)+f(1)\big] - \frac{1}{2}\int_0^1 b_2'(t)f'(t)\,dt\,.\end{aligned}$$

Now integrate by parts again, invoking the relations $b_2(0) = b_2(1) = B_2$ and $b_2(t) = \frac{1}{3}b_3'(t)$, to write

$$\int_0^1 f(t)\,dt = \frac{1}{2}\big[f(0)+f(1)\big] + \frac{B_2}{2!}\big[f'(0)-f'(1)\big] + \frac{1}{3!}\int_0^1 b_3'(t)f''(t)\,dt\,.$$

The last integral is now transformed in the same way, making repeated use of the relations

$$b_k(t) = \frac{1}{k+1}b_{k+1}'(t) \qquad \text{and} \qquad b_k(0) = b_k(1) = B_k\,,$$

and recalling that $B_k = 0$ for odd indices $k \ge 3$. The result is

$$\begin{aligned}\frac{1}{3!}\int_0^1 b_3'(t)f''(t)\,dt &= \frac{1}{3!}\Big[b_3(t)f''(t)\Big]_0^1 - \frac{1}{4!}\int_0^1 b_4'(t)f'''(t)\,dt \\ &= -\frac{1}{4!}\Big[b_4(t)f'''(t)\Big]_0^1 + \frac{1}{4!}\int_0^1 b_4(t)f^{(4)}(t)\,dt \\ &= \frac{B_4}{4!}\big[f'''(0)-f'''(1)\big] - \frac{1}{5!}\int_0^1 b_5(t)f^{(5)}(t)\,dt\,.\end{aligned}$$

Continuing in this manner, we arrive at the general formula

$$\begin{aligned}\int_0^1 f(t)\,dt = \frac{1}{2}\big[f(0)+f(1)\big] &+ \sum_{k=1}^{\ell}\frac{B_{2k}}{(2k)!}\big[f^{(2k-1)}(0)-f^{(2k-1)}(1)\big] \\ &- \frac{1}{(2\ell+1)!}\int_0^1 b_{2\ell+1}(t)f^{(2\ell+1)}(t)\,dt\,, \qquad \ell = 1,2,\dots\,.\end{aligned} \tag{20}$$

The formula (20) is easily generalized to an arbitrary interval $[j, j+1]$, for $j = 0, \pm1, \pm2, \dots$. The only difference is that the Bernoulli polynomial $b_n(t)$ must be replaced by its periodic extension $P_n(t)$ to the real line, defined

so that $P_n(t) = b_n(t)$ for $0 \le t \le 1$ and $P_n(t+1) = P_n(t)$ for all $t \in \mathbb{R}$. These functions $P_n(t)$ are called the *Bernoulli periodic functions*. They retain the properties $P_n'(t) = nP_{n-1}(t)$ and $P_n(j) = P_n(j+1) = B_n$, and so the formula (20) is generalized to

(21)
$$\int_j^{j+1} f(t)\,dt = \frac{1}{2}\big[f(j) + f(j+1)\big] + \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k)!}\big[f^{(2k-1)}(j) - f^{(2k-1)}(j+1)\big] - \frac{1}{(2\ell+1)!}\int_j^{j+1} P_{2\ell+1}(t) f^{(2\ell+1)}(t)\,dt\,, \qquad \ell = 1, 2, \dots\,.$$

Finally, the formulas (21) for intervals $[j, j+1]$ can be added to obtain a general formula for any interval $[m, n]$ where m and n are integers with $m < n$. It is

(22)
$$\int_m^n f(t)\,dt = \tfrac{1}{2}f(m) + \sum_{j=m+1}^{n-1} f(j) + \tfrac{1}{2}f(n) + \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k)!}\big[f^{(2k-1)}(m) - f^{(2k-1)}(n)\big] - \frac{1}{(2\ell+1)!}\int_m^n P_{2\ell+1}(t) f^{(2\ell+1)}(t)\,dt\,, \qquad \ell = 1, 2, \dots\,.$$

This is the *Euler–Maclaurin summation formula*. It is valid for any function f that has continuous derivatives up to order $2\ell + 1$ on the interval $[m, n]$.

Integration by parts gives the alternate form

(23)
$$\frac{1}{(2\ell)!}\int_m^n P_{2\ell}(t) f^{(2\ell)}(t)\,dt = -\frac{1}{(2\ell+1)!}\int_m^n P_{2\ell+1}(t) f^{(2\ell+1)}(t)\,dt$$

for the remainder. It is often more convenient for estimation because the inequality $|P_n(t)| \le |B_n|$ holds for even indices n (*cf.* Exercise 15).

11.6. Applications of Euler–Maclaurin formula

As a first illustration, consider the problem of calculating Euler's constant

$$\gamma = \lim_{n\to\infty}\left\{\sum_{k=1}^{n}\frac{1}{k} - \log n\right\} = 0.5772156649\dots\,.$$

Apply the Euler–Maclaurin formula (22) with $m = 1$ and $f(t) = 1/t$. The derivatives are

$$f^{(k)}(t) = (-1)^k \frac{k!}{t^{k+1}}\,, \qquad k = 1, 2, \dots\,,$$

and so the formula (22) becomes

$$\sum_{k=1}^{n} \frac{1}{k} - \log n = \frac{1}{2}\left(1 + \frac{1}{n}\right) + \sum_{k=1}^{\ell} \frac{B_{2k}}{2k}\left(1 - \frac{1}{n^{2k}}\right) - \int_1^n P_{2\ell}(t)\, t^{-(2\ell+1)}\, dt\,,$$

with the alternate form (23) of the remainder. Now hold ℓ fixed and let $n \to \infty$ to arrive at the exact formula

$$\gamma = \frac{1}{2} + \sum_{k=1}^{\ell} \frac{B_{2k}}{2k} - \int_1^{\infty} P_{2\ell}(t)\, t^{-(2\ell+1)}\, dt\,,$$

valid for an arbitrary positive integer ℓ. If we choose $\ell = 3$, the formula is

$$\begin{aligned}\gamma &= \tfrac{1}{2} + \tfrac{1}{2}B_2 + \tfrac{1}{4}B_4 + \tfrac{1}{6}B_6 - \int_1^{\infty} P_6(t)\, t^{-7}\, dt \\ &= \tfrac{1}{2} + \tfrac{1}{12} - \tfrac{1}{120} + \tfrac{1}{252} - \int_1^{\infty} P_6(t)\, t^{-7}\, dt \\ &= 0.578968253\ldots - \int_1^{\infty} P_6(t)\, t^{-7}\, dt\,.\end{aligned}$$

Now recall that $P_6(t)$ is the periodic extension of the Bernoulli polynomial

$$b_6(t) = t^6 - 3t^5 + \tfrac{5}{2}t^4 - \tfrac{1}{2}t^2 + \tfrac{1}{42}\,,$$

and $|P_6(t)| \le |B_6| = \frac{1}{42}$. If we calculate the integral over the interval $1 \le t \le 5$, we will have the error estimate

$$\left| \int_5^{\infty} P_6(t)\, t^{-7}\, dt \right| \le B_6 \int_5^{\infty} t^{-7}\, dt = \tfrac{1}{252}\, 5^{-6} < 10^{-6}\,.$$

Thus the problem reduces to the straightforward but laborious calculation of the integral

$$\int_1^5 P_6(t)\, t^{-7}\, dt = \sum_{k=1}^{4} \int_0^1 b_6(t)(t+k)^{-7}\, dt = 0.001752579\ldots\,.$$

Subtracting the two numbers, we calculate

$$\gamma = 0.578968253\ldots - 0.001752579\ldots - \int_5^{\infty} P_6(t)\, t^{-7}\, dt = 0.57721\ldots\,,$$

accurate to 5 decimal places since the error is less than 10^{-6}. (In fact, the error is much smaller, since the calculation gives $\gamma \approx 0.577215674$, correct to 7 decimal places.)

It must be emphasized that γ is not equal to

$$\frac{1}{2} + \sum_{k=1}^{\infty} \frac{B_{2k}}{2k}\,.$$

In fact, that infinite series diverges rapidly (*cf.* Exercise 4).

For a second illustration of the Euler–Maclaurin summation formula, we now develop what is called Stirling's series, a sharpened form of Stirling's formula

$$n! \sim n^n e^{-n} \sqrt{2\pi n}\,, \qquad n \to \infty\,,$$

or equivalently,

$$\lim_{n\to\infty} \left\{ \log n! - \left(n + \tfrac{1}{2}\right) \log n + n \right\} = \log \sqrt{2\pi}\,.$$

Now the integrand is chosen to be $f(t) = \log t$, with derivatives

$$f^{(k)}(t) = (-1)^{k+1} \frac{(k-1)!}{t^k}\,, \qquad k = 1, 2, \ldots\,.$$

The formula (22), with $m = 1$, becomes

$$\int_1^n \log t\, dt = \sum_{k=1}^{n} \log k - \frac{1}{2} \log n + \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k-1)(2k)} \left(1 - \frac{1}{n^{2k-1}}\right) - \frac{1}{2\ell+1} \int_1^n P_{2\ell+1}(t)\, t^{-(2\ell+1)}\, dt\,.$$

Since

$$\int_1^n \log t\, dt = n \log n - n + 1\,,$$

this reduces to

$$\log n! - \left(n + \tfrac{1}{2}\right) \log n + n = 1 + \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k-1)(2k)} \left(\frac{1}{n^{2k-1}} - 1\right) + \frac{1}{2\ell+1} \int_1^n P_{2\ell+1}(t)\, t^{-(2\ell+1)}\, dt\,. \tag{24}$$

But according to Stirling's formula, the left-hand side tends to $\log \sqrt{2\pi}$ as $n \to \infty$, which implies that

$$1 - \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k-1)(2k)} + \frac{1}{2\ell+1} \int_1^{\infty} P_{2\ell+1}(t)\, t^{-(2\ell+1)}\, dt = \log \sqrt{2\pi} \tag{25}$$

for arbitrary integers $\ell = 1, 2, \ldots$. Substituting (25) into (24), we conclude that

$$\log n! = \left(n + \tfrac{1}{2}\right) \log n - n + \log \sqrt{2\pi} + \sum_{k=1}^{\ell} \frac{B_{2k}}{(2k-1)(2k)} \frac{1}{n^{2k-1}} - \frac{1}{2\ell+1} \int_n^{\infty} P_{2\ell+1}(t)\, t^{-(2\ell+1)}\, dt\,.$$

Now observe that the error term is estimated by

$$\left| -\frac{1}{2\ell+1} \int_n^\infty P_{2\ell+1}(t)\, t^{-(2\ell+1)}\, dt \right| \le C \int_n^\infty t^{-(2\ell+1)}\, dt \le C \frac{1}{n^{2\ell}}$$

for some constant C depending only on ℓ. Thus we have arrived at the asymptotic formula

$$\begin{aligned} \log n! = \left(n+\tfrac{1}{2}\right)\log n - n + \log\sqrt{2\pi} + \frac{B_2}{1\cdot 2}\frac{1}{n} + \frac{B_4}{3\cdot 4}\frac{1}{n^3} + \dots \\ + \frac{B_{2k}}{(2k-1)(2k)}\frac{1}{n^{2k-1}} + O\left(\frac{1}{n^{2k}}\right), \qquad n \to \infty, \end{aligned} \tag{26}$$

valid for any positive integer k. (Note the switch of notation, changing ℓ to k.) If $k = 3$, for instance, this says that

$$\log n! - \left(n+\tfrac{1}{2}\right)\log n + n = \log\sqrt{2\pi} + \frac{1}{12\,n} - \frac{1}{360\,n^3} + \frac{1}{1260\,n^5} + O\left(\frac{1}{n^6}\right).$$

In the notation of asymptotic series, the formula (24) can be written

$$\log n! - \left(n+\tfrac{1}{2}\right)\log n + n \simeq \log\sqrt{2\pi} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k-1)(2k)}\frac{1}{n^{2k-1}}. \tag{27}$$

The infinite series in (27) diverges for every fixed value of n, but any partial sum approximates the left-hand side as $n \to \infty$ with error tending to zero at the same rate as the first neglected term.

The formal infinite series in the asymptotic formula (27) is known as *Stirling's series*. The exponential form is

$$\begin{aligned} \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} &\simeq \exp\left\{ \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k-1)(2k)}\frac{1}{n^{2k-1}} \right\} \\ &\simeq 1 + \frac{1}{12\,n} + \frac{1}{288\,n^2} - \frac{139}{51840\,n^3} - \frac{571}{2488320\,n^4} + \dots . \end{aligned} \tag{28}$$

Exercises

1. Use Bernoulli's formula (5) to derive the relation

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2, \qquad n = 1, 2, \dots .$$

2. Use Bernoulli's formula to work out the expression

$$\sum_{k=1}^{n} k^4 = \frac{1}{30}\, n(n+1)(2n+1)(3n^2+3n-1), \qquad n = 1, 2, \dots ,$$

and check it by induction.

3. Use Euler's formula to check the sums (6) for $\zeta(2)$, $\zeta(4)$, and $\zeta(6)$, and to calculate the sum

$$\zeta(8) = \sum_{n=1}^{\infty} \frac{1}{n^8} = \frac{\pi^8}{9450}\,.$$

4. Prove the asymptotic formula

$$|B_{2n}| \sim 4\left(\frac{n}{\pi e}\right)^{2n} \sqrt{\pi n}\,, \qquad n \to \infty\,.$$

Conclude that the generating relation (1) for Bernoulli numbers has radius of convergence 2π, and that $|B_{2n}|(2\pi)^{2n}/(2n)! \to 2$ as $n \to \infty$.

Suggestion. Use Euler's formula (7) for $\zeta(2n)$.

5. Use the formula (14) to give a direct proof of (15).

6. Prove that

$$\frac{b_{p+1}(t+1) - b_{p+1}(t)}{p+1} = t^p\,, \qquad p = 0, 1, 2, \ldots.$$

Suggestion. Prove by induction. Suppose that $b_p(t+1) - b_p(t) = pt^{p-1}$ for some $p \geq 1$. Then observe that

$$\frac{b'_{p+1}(t+1) - b'_{p+1}(t)}{p+1} = b_p(t+1) - b_p(t)\,,$$

and note that $b_{p+1}(1) - b_{p+1}(0) = 0$.

7. Verify the formula (16) for the sum of positive powers.

Suggestion. Apply the formula of Exercise 6.

8. Show that

$$\int_x^{x+1} b_n(t)\,dt = x^n\,, \qquad n = 1, 2, \ldots\,,$$

for all $x \in \mathbb{R}$. Conclude in particular that

$$\int_0^1 b_n(t)\,dt = 0\,, \qquad n = 1, 2, \ldots.$$

9. Deduce the formulas (11) and (12) from Euler's formula (7).

10. Show that the tangent function has the Taylor series expansion

$$\begin{aligned}\tan x &= \sum_{n=1}^{\infty} (-1)^{n+1} \frac{4^n(4^n-1)}{(2n)!} B_{2n}\, x^{2n-1} \\ &= x + \tfrac{1}{3}x^3 + \tfrac{2}{15}x^5 + \tfrac{17}{315}x^7 + \ldots\,, \qquad |x| < \frac{\pi}{2}\,.\end{aligned}$$

Hint. $\tan x = \cot x - 2\cot 2x$. For radius of convergence, use Exercise 4.

11. Verify the Taylor series expansion

$$\frac{x}{\sin x} = 1 + \sum_{n=1}^{\infty} (-1)^{n+1} \frac{4^n - 2}{(2n)!} B_{2n}\, x^{2n}$$
$$= 1 + \tfrac{1}{6}x^2 + \tfrac{7}{360}x^4 + \tfrac{31}{15120}x^6 + \dots\,, \qquad |x| < \pi\,.$$

Hint. $\csc x = \cot x + \tan \frac{x}{2}$.

12. Verify the symmetry property $b_n(1-t) = (-1)^n b_n(t)$ for Bernoulli polynomials. Conclude that $b_n(\frac{1}{2}) = 0$ if n is odd.

13. Apply the Fourier expansion (18) of the Bernoulli polynomial $b_{2n}(t)$ and Euler's formula (12) to show that

$$b_{2n}\big(\tfrac{1}{2}\big) = (2^{1-2n} - 1) B_{2n}\,, \qquad n = 1, 2, \dots\,,$$

a formula that holds trivially for odd indices (*cf.* Exercise 12). Conclude that $|b_{2n}(\frac{1}{2})| < |B_{2n}|$ for $n = 1, 2, \dots$.

14 . Show that $b_{4n+1}(t) < 0$ and $b_{4n+3}(t) > 0$ for $0 < t < \frac{1}{2}$, where $n = 0, 1, 2, \dots$.

Suggestion. Proceed by induction and consider $b_n''(t)$.

15. Prove that $|b_{2n}(t)| < |B_{2n}|$ for $0 < t < 1\,,\ n = 1, 2, \dots$.

Suggestion. Use a computer to plot the polynomials and see what needs to be proved. *Mathematica* will plot the nth Bernoulli polynomial $b_n(t)$ over the interval $0 \le t \le 1$ from the code `Plot[BernoulliB[n,t], {t,0,1}]`.

16. Derive the formula $b_n\big(\frac{1}{2}\big) = (2^{1-n} - 1) B_n$ directly from the generating relation (13), by appeal to the expansion (9) for $t \coth t$.

Hint. $\coth x - \coth 2x = \operatorname{csch} 2x\,.$

17. Calculate the sum

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^3} = \frac{\pi^3}{32}\,.$$

Hint. Think Bernoulli polynomials.

18. Show that Bernoulli polynomials satisfy the relation

$$b_n(t) + (-1)^{n+1} b_n(-t) = -n t^{n-1}\,, \qquad n = 1, 2, \dots\,.$$

19. Show that Bernoulli's formula (5) for the sum of positive powers is a special case of the Euler–Maclaurin summation formula.

20. Carry out the calculations to derive the numerical formula (28) from Stirling's series (27).

21. Use the Euler–Maclaurin summation formula and a hand calculator to compute

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1.6449340\ldots ,$$

and prove that the result is accurate to 7 decimal places.

Suggestion. Take $m = 5$ and $\ell = 3$ in the formula (22), and use the alternate form (23) of the remainder.

*** 22.** There is an explicit formula for the Bernoulli numbers, discovered in 1883 by J. Worpitzky. It is

$$B_n = \frac{(-1)^n n}{2^n - 1} \sum_{k=1}^{n} \frac{1}{2^k} \sum_{j=0}^{k-1} (-1)^j \binom{k-1}{j} (j+1)^{n-1} .$$

For a reference, the paper by Rządkowski [**3**] is recommended.

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Chapter 12
The Cantor Set

In this chapter we explore some of Georg Cantor's ideas about abstract sets, including the notion of cardinality. The centerpiece of our discussion is Cantor's famous "middle-thirds" set, an uncountable set of Lebesgue measure zero that is a valuable source of counterexamples in analysis. For instance, the Cantor set is the basis for construction of the "devil's staircase", described by a continuous nondecreasing function that is not constant, yet has zero derivative at almost every point. The Cantor set also provides the underlying concept for construction of a space-filling curve, as presented at the end of the chapter. We begin, however, with an extended discussion of cardinal numbers.

12.1. Cardinal numbers

Georg Cantor (1845–1918) was the founder of modern set theory. He received his mathematical training in Berlin, where Ernst Kummer, Karl Weierstrass, and Leopold Kronecker were among his teachers. After writing a dissertation in number theory in 1867, he turned to problems of uniqueness of trigonometric series representations. That led to questions about the size of exceptional sets of real numbers, and ultimately to a precise definition of a real number as the limit of a sequence of rational numbers. About the same time (1872), Richard Dedekind published a book giving his definition of real number in terms of what are now called Dedekind cuts. Dedekind and Cantor both realized that the set of real numbers is in some sense "more infinite" than the set of rationals. Cantor transformed that insight into a precise mathematical concept, developing the notion of cardinal number. His

ideas were revolutionary, and they met with considerable resistance at the time, notably from Cantor's former teacher Kronecker, who firmly believed that mathematical proofs should be constructive. The opposition was so intense that Cantor was unable to obtain a proper academic position until 1879, when he became a professor at the University of Halle. He remained in Halle for the rest of his career.

Two abstract sets (not necessarily sets of real numbers) are said to have the same *cardinality*, or the same *cardinal number*, if they are bijectively equivalent; that is, if they can be put in one-to-one correspondence. Thus two finite sets have the same cardinality n if both sets contain exactly n elements. Each set can then be put in one-to-one correspondence with the set $\{1, 2, \ldots, n\}$ of the first n positive integers. A set is said to be *countable* or *denumerable* if either it is finite or it can be put in one-to-one correspondence with the set $\mathbb{N} = \{1, 2, 3, \ldots\}$ of all positive integers. In the latter case it is called *countably infinite*. Clearly, any subset of a countable set is countable. According to Cantor's notation, a countably infinite set has cardinality $\aleph_0$, usually pronounced "aleph-naught" or "aleph-null". (*Aleph* is the first letter of the Hebrew alphabet.) Thus the set $\mathbb{Q}$ of all rational numbers has cardinality $\aleph_0$. Cantor also proved that the set of all algebraic numbers is countable; it too has cardinality $\aleph_0$. Cardinal numbers of infinite sets are sometimes called *transfinite numbers*.

Observe that two infinite sets may have the same cardinality even though one is a proper subset of the other. For instance, the set $\{2, 4, 6, \ldots\}$ of all even integers is in obvious one-to-one correspondence with the set $\mathbb{N}$ through the mapping $n \leftrightarrow 2n$. Cantor introduced the diagonal method to prove that the set $\mathbb{R}$ of all real numbers is not countable. (See Section 1.2.) In other words, its cardinality is larger than $\aleph_0$, and it is in this sense "more infinite" than its subset $\mathbb{Q}$. The cardinality of $\mathbb{R}$ is customarily denoted by $\mathfrak{c}$ and is called the *cardinality of the continuum*.

The notion that $\mathfrak{c}$ is larger than $\aleph_0$ can be formulated in greater generality. The cardinal number of an abstract set A is denoted by $|A|$. If B is another set and there exists an injective mapping $f : A \to B$, then we say that $|A| \leq |B|$. If in fact there exists a injection of A *onto* B, or a bijection, then $|A| = |B|$. If $|A| \leq |B|$ but no such bijection exists, then we say that $|A| < |B|$. In this sense we know that $\aleph_0 < \mathfrak{c}$.

It is worth remarking that the existence of an injective mapping $f : A \to B$ is equivalent to the existence of a surjective mapping $g : B \to A$. In other words, there is a one-to-one mapping of A *into* B if and only if there is a mapping of B *onto* A.

For finite cardinal numbers, or ordinary positive integers n and m, the two relations $n \leq m$ and $m \leq n$ imply $n = m$. This implication can be shown

to extend to transfinite cardinals. Specifically, if A and B are arbitrary sets with the properties $|A| \le |B|$ and $|B| \le |A|$, then $|A| = |B|$. This is the content of the Schröder–Bernstein theorem, which can be stated as follows.

Schröder–Bernstein Theorem. *Let A and B be arbitrary sets. If there exist injective mappings $f : A \to B$ and $g : B \to A$, then there is a bijection $h : A \to B$.*

The result is very useful because it allows us to conclude that two sets have the same cardinality without explicitly constructing a bijection. The theorem is sometimes called the Cantor–Schröder–Bernstein theorem or the Cantor–Bernstein theorem, or simply Bernstein's theorem. After Cantor made the conjecture, Ernst Schröder (1841–1902) and Felix Bernstein (1878–1956) gave independent proofs in 1896 and 1897, respectively. Bernstein presented his proof, at age 19, in Cantor's seminar in Halle.

Proof of Schröder–Bernstein theorem. If g maps B *onto* A, the theorem is proved. Otherwise, the complementary set $E_0 = A \backslash g(B)$ is nonempty. Let

$$E_{n+1} = g\big(f(E_n)\big)\,, \qquad n = 0, 1, 2, \dots\,,$$

and let $E = \cup_{n=0}^{\infty} E_n$. For $x \in A$, define

$$h(x) = \begin{cases} f(x)\,, & \text{for } x \in E \\ g^{-1}(x)\,, & \text{for } x \notin E\,. \end{cases}$$

Note that $x \notin E$ implies $x \notin E_0$, so $x \in g(B)$ and $g^{-1}(x)$ is well defined. We claim that h is the required bijection.

To show that h is a one-to-one mapping, suppose on the contrary that $f(x_1) = g^{-1}(x_2)$ for some elements $x_1 \in E$ and $x_2 \notin E$. But $x_1 \in E$ implies $x_1 \in E_n$ for some index n, so

$$x_2 = g(f(x_1)) \in E_{n+1} \subset E\,,$$

which is a contradiction. Thus h is an injection.

To show that $h(A) = B$, suppose on the contrary that $h(x) \ne y$ for some $y \in B$ and all $x \in A$. By the definition of h, this is equivalent to the two properties

$$\begin{aligned} &(i) \quad y \ne f(x) \qquad \text{for all } x \in E\,, \qquad \text{and} \\ &(ii) \quad y \ne g^{-1}(x) \qquad \text{for all } x \notin E\,. \end{aligned}$$

The property (i) implies that $g(y) \ne g(f(x))$ for all $x \in E$. In particular, $g(y) \ne g(f(x))$ for all $x \in E_n$, where $n = 0, 1, 2, \dots$. Therefore, $g(y) \notin E_{n+1}$

for each $n = 0, 1, 2, \ldots$. But obviously $g(y) \in g(B)$, so it is also clear that $g(y) \notin E_0$. Thus $g(y) \notin E$. On the other hand, the property (ii) implies that $g(y) \neq x$ for all $x \notin E$, and so $g(y) \in E$. This contradicts our earlier conclusion that $g(y) \notin E$, so $h(A) = B$ and the proof is complete. □

As an application of the Schröder–Bernstein theorem, it is easy to show that every interval on the real axis, bounded or unbounded, has the cardinality of the continuum. It is rather awkward to construct a specific bijection between the intervals $(0, 1)$ and $(0, 1]$, for instance, but the Schröder–Bernstein theorem circumvents the problem.

It is also possible to show that any rectangle in the plane, or even the whole plane $\mathbb{R}^2$, has cardinality $\mathfrak{c}$. The same is true for Euclidean space $\mathbb{R}^n$ of any dimension. On the other hand, it is not difficult to find sets with cardinality greater than $\mathfrak{c}$, as we shall now see.

For any set A, the *power set* $\mathcal{P}(A)$ is defined as the set of all subsets of A, including the empty set and the full set A. The power set is essentially the same as the set of all functions f defined on A with values 0 or 1. Such a function corresponds to a particular subset $B \subset A$, where $f(x) = 0$ or 1 is taken to indicate that $x \notin B$ or $x \in B$, respectively. Cantor proved that the power set of any set A has cardinality larger than that of A.

Cantor's Theorem. *The power set of every nonempty set A has cardinality $|\mathcal{P}(A)| > |A|$.*

If A is a finite set with n elements, it is easy to see that $\mathcal{P}(A)$ has 2^n elements, so the main interest of the theorem is for infinite sets. The theorem shows in particular that there exist sets with cardinality greater than that of the continuum, and in fact that there is no largest cardinal number.

Proof of theorem. First note that $|A| \leq |\mathcal{P}(A)|$, since we can construct an injection of A into $\mathcal{P}(A)$ by assigning the singleton subset $\{x\}$ to any element $x \in A$. To show that the cardinality of $\mathcal{P}(A)$ is strictly larger than that of A, let $f : A \to \mathcal{P}(A)$ be an arbitrary injection, and define the set $B \in \mathcal{P}(A)$ by

$$B = \{x \in A : x \notin f(x)\}.$$

Suppose $f(x_0) = B$ for some $x_0 \in A$. If $x_0 \in B$, then by definition $x_0 \notin f(x_0) = B$, a contradiction. If $x_0 \notin B$, then $x_0 \in f(x_0) = B$, which is again a contradiction. Thus there cannot be any point $x \in A$ for which $f(x) = B$. This shows that no injection f of A into $\mathcal{P}(A)$ can have range $f(A) = \mathcal{P}(A)$. Therefore, there is no bijection of A onto $\mathcal{P}(A)$, and the cardinality of $\mathcal{P}(A)$ is larger than that of A. □

As an illustration of Cantor's theorem, it will now be shown that the collection of all subsets of positive integers has the cardinality of the continuum. In symbols, $|\mathcal{P}(\mathbb{N})| = \mathfrak{c}$. To see that $\mathfrak{c} \le |\mathcal{P}(\mathbb{N})|$, define the mapping $f : \mathcal{P}(\mathbb{N}) \to [0,1]$ by $f(\emptyset) = 0$ and

$$f(A) = \sum_{n \in A} \frac{1}{2^n} \qquad \text{for} \quad A \subset \mathbb{N}\,,\ A \neq \emptyset\,.$$

Then $f(A)$ is the number x with binary expansion $x = 0.b_1b_2\ldots$, where $b_n = 1$ if $n \in A$ and $b_n = 0$ if $n \notin A$. Because every number x in $[0,1]$ has a binary expansion, it is clear that f is a surjection, and so $\mathfrak{c} = |[0,1]| \le |\mathcal{P}(\mathbb{N})|$.

On the other hand, f is not an injection because every dyadic rational number has two binary expansions, one ending in a string of zeros, the other in a string of ones. To remedy the situation, we modify f and define a new mapping $g : \mathcal{P}(\mathbb{N}) \to \mathbb{R}$ by

$$g(A) = \sum_{n \in A} \frac{1}{2^n} \qquad \text{if } \mathbb{N} \setminus A \text{ is infinite,} \qquad \text{and}$$

$$g(A) = 1 + \sum_{n \in A} \frac{1}{2^n} \qquad \text{if } \mathbb{N} \setminus A \text{ is finite.}$$

Then $g(A)$ has a binary expansion ending is a string of ones precisely when $\mathbb{N} \setminus A$ is finite. The corresponding binary expansion of the same number $g(A) - 1$ that ends in a string of zeros will occur when A is finite, which is part of the case where $\mathbb{N} \setminus A$ is infinite. These considerations show that g is an injection of $\mathcal{P}(\mathbb{N})$ into the interval $[0,2]$, so that $\mathcal{P}(\mathbb{N})$ has cardinality $|\mathcal{P}(\mathbb{N})| \le |[0,2]| = \mathfrak{c}$. Thus $|\mathcal{P}(\mathbb{N})| = \mathfrak{c}$, by the Schröder–Bernstein theorem.

12.2. Lebesgue measure

The Lebesgue measure of a set of real numbers is a generalization of the length of an interval. An open interval $I = (a,b)$ has length $m(I) = b - a$. A set $E \subset \mathbb{R}$ is said to have *measure zero*, written $m(E) = 0$, if for each $\varepsilon > 0$ there is a finite or countable collection of open intervals I_k such that $E \subset \bigcup_k I_k$ and the intervals have total length $\sum_k m(I_k) < \varepsilon$.

Clearly, every finite set has measure zero. In fact, it is not difficult to show that every countable set has measure zero. To see this, let $E \subset \mathbb{R}$ be a countable set consisting of points $x_1, x_2, \ldots$. Given $\varepsilon > 0$, let I_k be the open interval centered at x_k with length $m(I_k) < \varepsilon/2^k$, for $k = 1, 2, \ldots$. Then

$$E \subset \bigcup_{k=1}^{\infty} I_k \qquad \text{and} \qquad \sum_{k=1}^{\infty} m(I_k) < \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon\,.$$

Thus E has measure zero.

Our subsequent discussion will refer only to the notion of a set of measure zero, but for the sake of completeness we now sketch the more general definition of a measurable set and its Lebesgue measure. For simplicity we will assume that our sets are bounded. Everything depends on the following lemma.

Lemma. *Every bounded open set $E \subset \mathbb{R}$ has a unique representation $E = \bigcup_k I_k$ as the union of a countable collection of disjoint open intervals I_k.*

The proof is left as an exercise. On the basis of the lemma, the measure of a bounded open set $E \subset \mathbb{R}$ can be defined by

$$m(E) = \sum_k m(I_k)\,, \qquad \text{where} \qquad E = \bigcup_k I_k$$

is its canonical representation as a disjoint union of open intervals.

Next let $F \subset \mathbb{R}$ be a bounded closed set. Since F is bounded, it is contained in some bounded open interval J. The complementary set $E = J \setminus F$ is open and bounded, so the measure of F can be defined by

$$m(F) = m(J) - m(E)\,.$$

It has to be checked that the definition is independent of the choice of J.

Finally, let $A \subset \mathbb{R}$ be an arbitrary bounded set of real numbers. The set A is said to be *measurable* if

$$\sup_{F \subset A} m(F) = \inf_{E \supset A} m(E)\,,$$

where the supremum is taken over all closed sets F contained in A and the infimum extends over all bounded open sets E that contain A. If A is measurable, its *Lebesgue measure* $m(A)$ is defined as the common value of the supremum and infimum.

Lebesgue measure has many desirable properties. If A and B are measurable sets with $A \subset B$, it is easily seen that $m(A) \le m(B)$. For arbitrary measurable sets A and B, it is possible to prove that $A \cup B$ is measurable and

$$m(A \cup B) \le m(A) + m(B)\,,$$

with equality if $A \cap B = \emptyset$. More generally, the union of any collection of measurable sets $A_1, A_2, \ldots$ is measurable and

$$m\left(\bigcup_k A_k\right) \le \sum_k m(A_k)\,,$$

with equality if the sets A_k are pairwise disjoint.

Henri Lebesgue (1875–1941) developed this concept of measure in 1901 and used it as a basis for what is now called the Lebesgue integral. Every Riemann integrable function is Lebesgue integrable, and the two integrals are equal. However, the Lebesgue integral applies to a wider class of functions and has significant technical advantages. It has become a standard tool in modern analysis.

12.3. The Cantor set

We are now ready to construct Cantor's famous "middle-thirds" set. Begin with the closed unit interval $[0, 1]$, and remove the open middle third $(\frac{1}{3}, \frac{2}{3})$ to form the set

$$F_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1] \,.$$

From each remaining interval, again extract the open middle third, leaving the set

$$F_2 = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1] \,.$$

Continue in this manner, removing the open middle third from each remaining interval at each stage of the dissection, thus obtaining a nested sequence of closed sets

$$F_1 \supset F_2 \supset F_3 \supset \cdots \,.$$

After n stages of dissection, the remaining set F_n consists of 2^n disjoint closed intervals, each of length 3^{-n}. Figure 1 depicts the first 3 sets F_1, F_2, and F_3.

The Cantor set $\mathcal{C}$ is defined as the intersection

$$\mathcal{C} = \bigcap_{n=1}^{\infty} F_n \,.$$

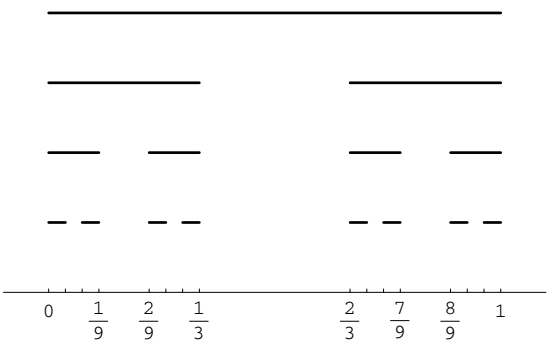


Figure 1. Construction of Cantor set: F_1, F_2, F_3

The nested sets theorem guarantees that $\mathcal{C}$ is not the empty set, but this is also evident by inspection, since each set F_n contains all of the endpoints

$$0,\ 1,\ \tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{1}{9},\ \tfrac{2}{9},\ \tfrac{7}{9},\ \tfrac{8}{9},\ \tfrac{1}{27},\ \tfrac{2}{27},\ \tfrac{7}{27},\ \tfrac{8}{27},\ \ldots$$

of remaining intervals, and so $\mathcal{C}$ must also contain these points. We shall see that the Cantor set contains many other points as well.

It is easy to show that the Cantor set has measure zero. Indeed, F_n has total length $(2/3)^n$, so we can assert that

$$\mathcal{C} \subset F_n \subset \bigcup_{k=1}^{n} I_k$$

for some collection of open intervals I_k with total length

$$\sum_{k=1}^{n} m(I_k) < 2\left(\tfrac{2}{3}\right)^n .$$

But for each $\varepsilon > 0$ we can choose n large enough that $2(2/3)^n < \varepsilon$. This shows that $\mathcal{C}$ has measure zero.

Our next aim is to show that the Cantor set is uncountable. By virtue of its construction, F_n consists precisely of those points x that have *some* ternary expansion

$$x = 0.a_1a_2a_3\cdots = \sum_{k=1}^{\infty} a_k\, 3^{-k}$$

with $a_k = 0$ or 2 for $k = 1, 2, \ldots, n$. Thus a point x belongs to the Cantor set if and only if it has a ternary expansion of the form $x = 0.a_1a_2a_3\cdots$ with $a_k = 0$ or 2 for all k. The endpoints of removed intervals are triadic rationals, numbers of the form $m/3^n$. These points have two distinct ternary expansions, exactly one of which has all digits $a_k = 0$ or 2. For example,

$$\tfrac{7}{9} = 0.21000\cdots = 0.20222\ldots .$$

Given a point $x = 0.a_1a_2a_3\cdots \in \mathcal{C}$ with all $a_k = 0$ or 2, define $b_k = \frac{1}{2}a_k$ and consider the *binary* expansion

$$y = 0.b_1b_2b_3\cdots = \sum_{k=1}^{\infty} b_k\, 2^{-k} .$$

This operation defines a mapping $x \mapsto y$ of the Cantor set onto the interval $[0, 1]$. However, the mapping is not injective. For instance,

$$x = \tfrac{7}{9} = 0.20222\cdots \to y = 0.10111\cdots = \tfrac{3}{4} \qquad \text{and}$$
$$x = \tfrac{8}{9} = 0.22000\cdots \to y = 0.11000\cdots = \tfrac{3}{4} .$$

In any case, at most two points $x \in \mathcal{C}$ correspond in this manner to the same point $y \in [0,1]$. This shows that a proper subset of $\mathcal{C}$ is bijectively equivalent to the interval $[0,1]$, and so has the cardinality of the continuum. In particular, the Cantor set is uncountable, a result that can also be obtained by a diagonal argument similar to the proof that the set of real numbers is uncountable.

To show that the full Cantor set has the cardinality of the continuum, we may apply the Schröder–Bernstein theorem. Since $\mathcal{C} \subset [0,1]$, there is an obvious injection of $\mathcal{C}$ into $[0,1]$. To construct an injective mapping of $[0,1]$ into $\mathcal{C}$, represent an arbitrary point $y \in [0,1]$ by its binary expansion $y = 0.b_1b_2b_3\ldots$, adopting the convention that in cases of ambiguity the expansion shall end in an infinite string of zeros. Let $a_k = 2b_k$ and define

$$x = 0.a_1a_2a_3\cdots = \sum_{k=1}^{\infty} a_k\, 3^{-k}$$

by a ternary expansion with digits a_k. Then $x \in \mathcal{C}$, since all $a_k = 0$ or 2. Thus we have constructed an injective mapping of the interval $[0,1]$ into $\mathcal{C}$, which is a subset of $[0,1]$. By the Schröder–Bernstein theorem, the two sets $\mathcal{C}$ and $[0,1]$ have the same cardinality. In other words the Cantor set has the cardinality of the continuum.

12.4. The Cantor–Scheeffer function

We turn now to the construction of a remarkable function, known as the *Cantor function* although it was discovered in 1883 by Ludwig Scheeffer (1859–1885), a talented young mathematician who was in communication with Cantor, and whose work Cantor promoted. We prefer to call it the *Cantor–Scheeffer function*.

This function $f(x)$ is continuous and nondecreasing on the interval $[0,1]$ and it has the values $f(0) = 0$ and $f(1) = 1$, yet its derivative vanishes almost everywhere; that is, outside a set of measure zero. As we shall see, the exceptional set is the Cantor set.

For a point $x \in \mathcal{C}$ with *ternary* expansion

$$x = 0.a_1a_2a_3\ldots$$

and all digits $a_k = 0$ or 2, the Cantor function f can be defined by the *binary* expansion

$$f(x) = 0.b_1b_2b_3\ldots,$$

where $b_k = \frac{1}{2}a_k$. For points at opposite ends of an interval

$$(x_1, x_2) = (m/3^n, (m+1)/3^n)$$

removed from $[0,1]$ in the construction of the Cantor set, it is not difficult to see that $f(x_1) = f(x_2)$. The definition of f is then extended to the whole interval $[0,1]$ by setting $f(x) \equiv f(x_1)$ in each removed interval $x_1 \le x \le x_2$. The resulting function can then be shown to be continuous and to have the other stated properties.

However, there is an alternate approach that is more geometric and perhaps more transparent. We will produce the function $f(x)$ as the uniform limit of a sequence of continuous nondecreasing functions $f_n(x)$ with values $f_n(0) = 0$ and $f_n(1) = 1$. First define

$$f_1(x) = \begin{cases} \frac{3}{2}x\,, & 0 \le x \le \frac{1}{3} \\ \frac{1}{2}\,, & \frac{1}{3} \le x \le \frac{2}{3} \\ \frac{3}{2}x - \frac{1}{2}\,, & \frac{2}{3} \le x \le 1\,. \end{cases}$$

Note that $f_1(x)$ is constant on the removed interval $(\frac{1}{3}, \frac{2}{3})$ and it increases linearly with derivative $f_1'(x) = \frac{3}{2}$ on each interval of the set F_1. Next define $f_2(x)$ to have the constant values $\frac{1}{4}$ on $(\frac{1}{9}, \frac{2}{9})$ and $\frac{1}{2}$ on $(\frac{1}{3}, \frac{2}{3})$ and $\frac{3}{4}$ on $(\frac{7}{9}, \frac{8}{9})$, and to increase linearly with derivative $f_2'(x) = (\frac{3}{2})^2$ on each interval of the set F_2. Continue in this manner. Thus define $f_3(x)$ by replacing the linear function on each interval of F_2 with a scaled copy of $f_1(x)$. The resulting function $f_3(x)$ will then increase linearly with derivative $f_3'(x) = (\frac{3}{2})^3$ on each interval of F_3 and will be constant on each of the intervening intervals. In general, the function $f_n(x)$ is linear with derivative $f_n'(x) = (\frac{3}{2})^n$ on each interval of the set F_n and has a constant value $j/2^n$ (for some $j = 1, 2, \ldots, 2^n - 1$) on each of the intervening intervals. It is continuous and nondecreasing on the interval $[0,1]$, with the values $f_n(0) = 0$ and $f_n(1) = 1$. Graphs of the functions $f_1(x)$, $f_2(x)$, and $f_3(x)$ are displayed in Figures 2a, 2b, and 2c.

We will now show that the functions $f_n(x)$ constitute a uniform Cauchy sequence on $[0,1]$. For arbitrary $n > m \ge 1$, it is apparent from the construction that

$$|f_n(x) - f_m(x)| \le \frac{1}{2^m}\,, \qquad 0 \le x \le 1\,.$$

This makes clear that $\{f_n(x)\}$ is a uniform Cauchy sequence, so it converges uniformly on $[0,1]$ to a function $f(x)$. Because each function $f_n(x)$ is continuous and nondecreasing, with values $f_n(0) = 0$ and $f_n(1) = 1$, the same properties are bestowed upon the limit function.

Finally, the function $f_m(x)$ is constant on each open interval of the complementary set $[0,1] \setminus F_m$, which has measure $1 - (\frac{2}{3})^m$. But for each $n > m$ the function $f_n(x) = f_m(x)$ on that set, and so $f(x)$ is constant on

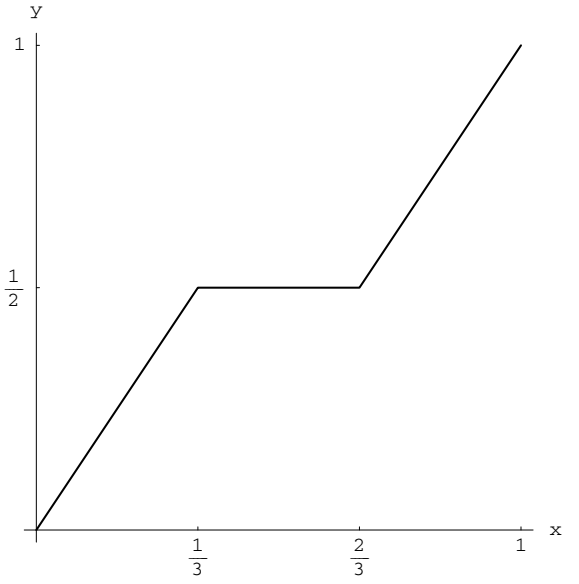


Figure 2a. Graph of $y = f_1(x)$

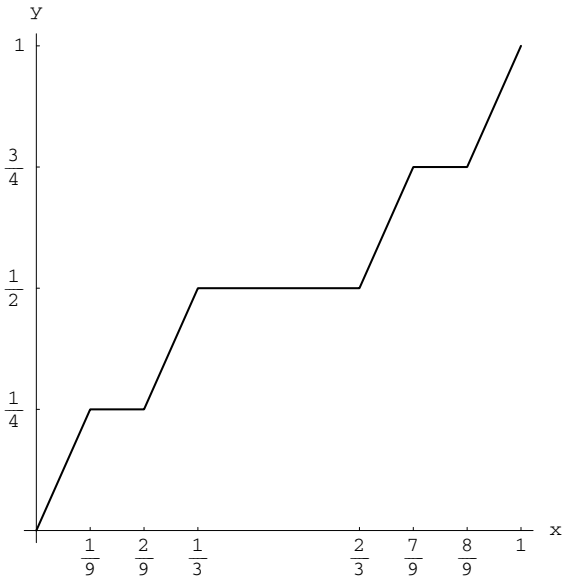


Figure 2b. Graph of $y = f_2(x)$

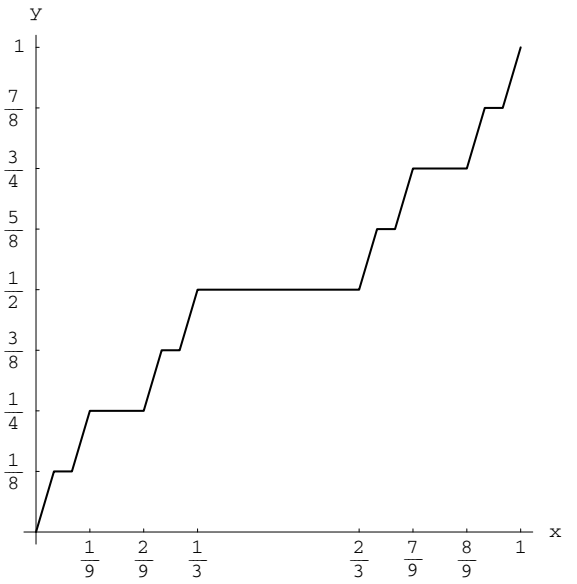


Figure 2c. Graph of $y = f_3(x)$

$[0,1] \setminus F_m$ for each m. Therefore, $f'(x) = 0$ on

$$\bigcup_{m=1}^{\infty} ([0,1] \setminus F_m) = [0,1] \setminus \bigcap_{m=1}^{\infty} F_m = [0,1] \setminus \mathcal{C}\,.$$

Since the Cantor set has measure zero, this shows that $f'(x) = 0$ almost everywhere on the interval $[0,1]$.

The curve in the graph of the Cantor–Scheeffer function is sometimes called the *devil's staircase*. The function was popularized by Lebesgue, who adopted it as an example of a continuous nonconstant monotonic function that is *singular* in the sense that $f'(x) = 0$ almost everywhere. The construction can be modified to produce a continuous singular function that is strictly monotonic; in other words, it is strictly increasing on every open subinterval of $[0,1]$.

12.5. Space-filling curves

A curve in the plane is defined by its parametric representation

$$x = \varphi(t)\,, \quad y = \psi(t)\,, \qquad a \le t \le b\,,$$

where φ and ψ are continuous functions. More precisely, the curve is defined as an equivalence class of such representations, since the actual choice of parameter is unimportant. The reason for such a pedantic definition is that the

more intuitive notion is not entirely adequate. It would seem more natural to view a curve as a continuous image of a line segment, a connected path of points in the plane. However, such an image can have totally unexpected form. In 1890, Giuseppe Peano (1858–1932) astounded the mathematical world by constructing a continuous curve that passes through every point of a square!

Soon after Peano [**4**] produced his example, David Hilbert and E. H. Moore found different constructions. Today such curves are known as *space-filling curves* or *Peano curves*. Lebesgue based another construction on the Cantor set, and I. J. Schoenberg [**6**] modified Lebesgue's construction to produce a very elegant example, which we now proceed to describe.

First define the function

$$g(t) = \begin{cases} 0\,, & 0 \le t \le \frac{1}{3} \\ 3t - 1\,, & \frac{1}{3} \le t \le \frac{2}{3} \\ 1\,, & \frac{2}{3} \le t \le 1\,. \end{cases}$$

Let $g(-t) = g(t)$ and $g(t+2) = g(t)$, so that g is a continuous even function of period 2. Its graph is shown in Figure 3.

The curve is now defined by the parametric representation

$$\begin{aligned} x = \varphi(t) &= \frac{1}{2}\, g(t) + \frac{1}{2^2}\, g(3^2 t) + \frac{1}{2^3}\, g(3^4 t) + \dots \\ y = \psi(t) &= \frac{1}{2}\, g(3t) + \frac{1}{2^2}\, g(3^3 t) + \frac{1}{2^3}\, g(3^5 t) + \dots\,, \end{aligned}$$

where $0 \le t \le 1$. Since $0 \le g(t) \le 1$, the convergence of both series is uniform, and so the functions $\varphi(t)$ and $\psi(t)$ are continuous on the interval $[0, 1]$. Note also that $0 \le \varphi(t) \le 1$ and $0 \le \psi(t) \le 1$, so that the point $(x, y) = (\varphi(t), \psi(t))$ lies in the closed unit square. It is to be shown that the curve meets each point of the square for some value of $t \in [0, 1]$. Suppose,

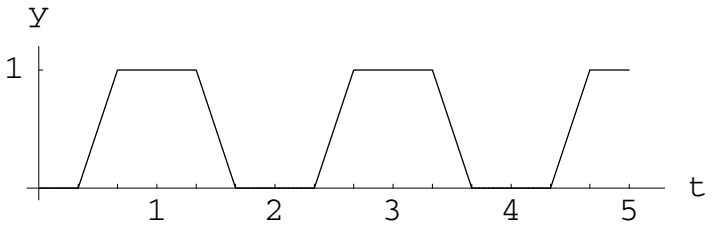


Figure 3. Graph of $y = g(t)$

then, that an arbitrary point (x_0, y_0) is prescribed, where $0 \le x_0 \le 1$ and $0 \le y_0 \le 1$. Represent x_0 and y_0 by their binary expansions

$$x_0 = \frac{1}{2}\, b_1 + \frac{1}{2^2}\, b_3 + \frac{1}{2^3}\, b_5 + \dots$$
$$y_0 = \frac{1}{2}\, b_2 + \frac{1}{2^2}\, b_4 + \frac{1}{2^3}\, b_6 + \dots\,,$$

where $b_k = 0$ or 1. Let $a_k = 2b_k$ and define the number t_0 by the ternary expansion

$$t_0 = \sum_{k=1}^{\infty} a_k\, 3^{-k}\,.$$

It is clear that $0 \le t_0 \le 1$. If $b_1 = 0$, then $0 \le t_0 \le \frac{1}{3}$ and so $g(t_0) = 0$. If $b_1 = 1$, then $\frac{2}{3} \le t_0 \le 1$ and so $g(t_0) = 1$. In either case, $g(t_0) = b_1$. More generally, the same argument shows that

$$g(3^n t_0) = b_{n+1}\,, \qquad n = 0, 1, 2, \dots\,.$$

To see this, observe that $3^n t_0$ differs from $\sum_{k=1}^{\infty} a_{n+k}\, 3^{-k}$ by an even integer, because all of the coefficients a_k are even. Since $g(t+2) = g(t)$, this shows that

$$g(3^n t_0) = g\left(\sum_{j=1}^{\infty} a_{n+j}\, 3^{-j}\right) = b_{n+1}\,, \qquad n = 0, 1, 2, \dots\,.$$

Consequently, the definition of the curve gives

$$\begin{aligned}\varphi(t_0) &= \frac{1}{2}\, g(t_0) + \frac{1}{2^2}\, g(3^2 t_0) + \frac{1}{2^3}\, g(3^4 t_0) + \dots \\ &= \frac{1}{2}\, b_1 + \frac{1}{2^2}\, b_3 + \frac{1}{2^3}\, b_5 + \dots = x_0\end{aligned}$$

and

$$\begin{aligned}\psi(t_0) &= \frac{1}{2}\, g(3t_0) + \frac{1}{2^2}\, g(3^3 t_0) + \frac{1}{2^3}\, g(3^5 t_0) + \dots \\ &= \frac{1}{2}\, b_2 + \frac{1}{2^2}\, b_4 + \frac{1}{2^3}\, b_6 + \dots = y_0\,.\end{aligned}$$

This shows that the curve passes through the arbitrarily prescribed point (x_0, y_0). Therefore, it passes through every point of the unit square.

The coordinate functions $\varphi(t)$ and $\psi(t)$ of Schoenberg's space-filling curve are nowhere differentiable. For an elementary proof, see Sagan [**5**]. Other examples of space-filling curves, but not all, share this property. For instance, the coordinate functions of Lebesgue's example are known to be differentiable almost everywhere.

Exercises

1. (*a*) The *Cartesian product* $A \times B$ of two sets A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. If A and B are countable sets, prove that $A \times B$ is countable.

(*b*) Prove that every countable union of countable sets is countable. More precisely, if each of the sets $E_1, E_2, \ldots$ is countable, then their union $E = \bigcup_{n=1}^{\infty} E_n$ is also countable.

2. (*a*) Apply the Schröder–Bernstein theorem to show that the intervals $(0, 1)$ and $(0, 1]$ have the same cardinality.

(*b*) Find a bijective mapping of the interval $(0, 1)$ to $(0, \infty)$.

(*c*) Find a bijective mapping of the interval $(0, 1)$ to $\mathbb{R}$.

3. A number x is said to be *algebraic* if it is a root of a polynomial with integer coefficients: $a_0 + a_1 x + \cdots + a_n x^n = 0$ for some $n \in \mathbb{N}$ and $a_0, a_1, \ldots, a_n \in \mathbb{Z}$ with $a_n \neq 0$. Every rational number is algebraic, but many algebraic numbers such as $\sqrt{2}$ are irrational. Show that the set of all algebraic numbers is countable.

4. Let A be an arbitrary set and let $\mathcal{P}(A)$ denote its power set, the set of all subsets of A. Show that there is a bijection between $\mathcal{P}(A)$ and the set of all mappings $f : A \to \{0, 1\}$, the set consisting of the two numbers 0 and 1.

5. Prove that every open set $E \subset \mathbb{R}$ has a unique representation $E = \bigcup I_n$ as a union of a countable collection of open intervals I_n.

6. Use a diagonal argument to prove that the Cantor set is uncountable.

7. For any bounded set $E \subset \mathbb{R}$, the *derived set* E' consists of all cluster points of E. We know that E' is always a closed set, although it may be empty. A bounded set $E \subset \mathbb{R}$ is said to be *perfect* if $E = E'$. Thus every closed interval $[a, b]$ is a perfect set. Prove that the Cantor set $\mathcal{C}$ is a perfect set.

8. The Cantor set construction can be generalized to any constant ratio of dissection. Fix a number ξ with $0 < \xi < \frac{1}{2}$. Begin with the unit interval $[0, 1]$ and remove the central open interval $(\xi, 1 - \xi)$ to form the set $F_1 = [0, \xi] \cup [1 - \xi, 1]$. From each of the two intervals comprising F_1 remove the central open interval in the same ratio $\xi : 1 - 2\xi : \xi$ to form the set F_2. Continue in this manner to generate a sequence of sets $F_1, F_2, F_3, \ldots$.

(*a*) Show that F_n is composed of 2^n closed intervals, each of length ξ^n, for $n = 1, 2, \ldots$.

(*b*) Show that the intersection $F = \bigcap_{n=1}^{\infty} F_n$ is an uncountable closed set of measure zero.

9. The Cantor set construction can be further generalized by considering variable ratios of dissection. Fix a sequence of numbers ξ_k with $0 < \xi_k < \frac{1}{2}$. Begin by removing the interval $(\xi_1, 1-\xi_1)$ from the unit interval $[0,1]$ to form the set F_1. From each interval of F_1 remove the central open interval in the ratio $\xi_2 : 1 - 2\xi_2 : \xi_2$ to form the set F_2. Proceeding inductively, construct a nested sequence of closed sets $F_1 \supset F_2 \supset F_3 \supset \cdots$ by removing from each interval of F_{n-1} the central open interval in the ratio $\xi_n : 1 - 2\xi_n : \xi_n$ to form the set F_n.

(*a*) Show that F_n consists of 2^n closed intervals, each of length $\xi_1 \xi_2 \cdots \xi_n$, for $n = 1, 2, \ldots$.

(*b*) Show that the intersection $F = \bigcap_{n=1}^{\infty} F_n$ is a closed set with empty interior; it contains no open intervals.

(*c*) Show that F has measure zero if and only if the infinite product $\prod_{n=1}^{\infty}(2\xi_n)$ diverges to 0.

(*d*) Exhibit a sequence $\{\xi_k\}$ with $0 < \xi_k < \frac{1}{2}$ for which the infinite product $\prod_{n=1}^{\infty}(2\xi_n)$ converges. Thus obtain a closed set of positive measure with empty interior.

10. Show that the Cantor–Scheeffer function satisfies a Lipschitz condition

$$|f(x) - f(t)| \le C\,|x - t|^{\alpha}\,, \qquad \alpha = \log 2/\log 3\,,$$

for $x, t \in [0,1]$, where C is a positive constant.

11. Show that the "devil's staircase" curve given by the graph of the Cantor–Scheeffer function, $y = f(t)$ for $0 \le t \le 1$, has arclength 2. Note the failure of the usual formula

$$\int_0^1 \sqrt{1 + f'(t)^2}\, dt$$

for arclength. Note also the failure of the fundamental theorem of calculus, since

$$\int_0^1 f'(t)\, dt \ne f(1) - f(0)\,.$$

Here both integrals are taken in the Lebesgue sense. Two functions that differ only on a set of measure zero have equal Lebesgue integrals.

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Chapter 13
Differential Equations

Differential equations occur in all branches of science. They provide important mathematical models for classical problems of physics. Many of those models involve linear differential equations of second order, which give rise to special functions such as Bessel functions, Legendre polynomials, and hypergeometric functions. Traditional studies of differential equations emphasize techniques for finding explicit solutions either in closed form or as infinite series. In this chapter we devote some attention, for instance, to the construction of power series solutions at a regular singular point, and we develop formulas for Bessel functions and hypergeometric functions. However, the most important properties of solutions are typically deduced not from explicit formulas but directly from the fact that the functions satisfy the differential equation. This approach comes to full fruition in the Sturm comparison theorem and related results, discussed at the end of the chapter.

13.1. Existence and uniqueness of solutions

Our study of differential equations will begin at the beginning. The questions of existence and uniqueness of solutions have direct importance in applications, and are central to the entire mathematical theory. Here we propose to discuss the existence and uniqueness of solutions to an initial-value problem

$$y' = f(x, y)\,, \qquad y(x_0) = y_0\,. \tag{1}$$

Is there a function $y = \varphi(x)$ that satisfies the differential equation and the initial condition, and is there only one such function?

To be more precise, we will assume that the function $f(x, y)$ is continuous in a rectangle

$$R = \{(x, y) \ : \ |x - x_0| \le a \,, \ |y - y_0| \le b\} \,,$$

where it satisfies a *Lipschitz condition*

$$|f(x, y) - f(x, z)| \le C \, |y - z|$$

for some constant $C > 0$. Then we can prove the following theorem.

Existence and Uniqueness Theorem. *Let $f(x, y)$ be continuous in the rectangle R and satisfy a Lipschitz condition there. Let M be the maximum value of $|f(x, y)|$ on R, and let $\delta = \min\{a, b/M\}$. Then there exists a unique continuously differentiable function $\varphi(x)$ that satisfies the differential equation $\varphi'(x) = f(x, \varphi(x))$ in the interval $|x - x_0| < \delta$ and has a prescribed value $\varphi(x_0) = y_0$.*

Intuitively, the theorem is quite plausible. The differential equation specifies a direction field for the solution curves. In other words, it prescribes the slope of the tangent line as the curve passes through any point of the region. If the direction field is continuous, there would seem to be exactly one way to start at a given point and trace a curve with the specified tangent directions. Nevertheless, the theorem is false without the Lipschitz condition, as a simple example shows. Consider the initial-value problem

$$y' = 3 \, y^{2/3} \,, \qquad y(0) = 0 \,,$$

in some neighborhood of the origin. This problem is easily seen to have two distinct solutions, $y = x^3$ and $y = 0$. The solution is not unique, but the function $f(x, y) = 3 \, y^{2/3}$ does not satisfy a Lipschitz condition as required by the theorem, since

$$|f(x, y) - f(x, 0)| / |y - 0| = 3 \, |y|^{-1/3} \to \infty \qquad \text{as} \ \ y \to 0 \,.$$

We propose to prove the theorem by the method of successive approximations, a technique introduced in 1890 by the French mathematician Émile Picard (1856–1941). His method is constructive in the sense that it provides a mechanism for calculating the solution to any desired accuracy. Another proof, essentially due to Cauchy, was developed and published by Giuseppe Peano in the same year 1890. Peano's proof arrives at the solution through a sequence of polygonal approximations. The Runge–Kutta method, commonly used today for numerical solution of differential equations, is a refinement of the Cauchy–Peano scheme. As a theoretical device, however, Picard's method of iteration has the advantage of elegance and relative simplicity.

Proof of theorem. The first step is to observe that the initial-value problem (1) is equivalent to the integral equation

$$\varphi(x) = y_0 + \int_{x_0}^{x} f(t, \varphi(t))\, dt\,. \tag{2}$$

In other words, any continuously differentiable function φ that solves the initial-value problem (1) must satisfy the integral equation (2), and conversely any continuous solution of the integral equation must be continuously differentiable and satisfy (1). Thus our problem reduces to showing that the integral equation (2) has a continuous solution, and that this solution is unique.

Turning first to the question of existence, we will show that the equation (2) has a continuous solution $\varphi(x)$ in the interval $|x - x_0| < \delta$. The strategy is to obtain the solution as a uniform limit of continuous functions defined inductively by $\varphi_0(x) \equiv y_0$ and

$$\varphi_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, \varphi_n(t))\, dt\,. \qquad |x - x_0| < \delta\,, \tag{3}$$

for $n = 0, 1, 2, \ldots$. For convenience we will restrict attention to the half-interval $[x_0, x_0 + \delta)$. Similar arguments apply to the interval $(x_0 - \delta, x_0]$.

The first step is to show that each of the functions $\varphi_n(x)$ is well defined and continuous on $[x_0, x_0 + \delta)$, and that

$$|\varphi_n(x) - y_0| \le M(x - x_0)\,, \qquad x_0 \le x < x_0 + \delta\,. \tag{4}$$

Obviously the constant function $\varphi_0(x)$ has these properties. Proceeding inductively, suppose that some function $\varphi_n(x)$ has the stated properties. Then since $\delta \le b/M$ it follows that $(x, \varphi_n(x)) \in R$ for all $x \in [x_0, x_0 + \delta)$, so that $\varphi_{n+1}(x)$ is well defined and continuous on this interval and

$$|\varphi_{n+1}(x) - y_0| \le \int_{x_0}^{x} |f(t, \varphi_n(t))|\, dt \le M(x - x_0)\,.$$

Thus every function $\varphi_n(x)$ has the stated properties.

The proof of uniform convergence will rely on the inequality

$$|\varphi_{n+1}(x) - \varphi_n(x)| \le \frac{M\,C^n}{(n+1)!}(x - x_0)^{n+1}\,, \qquad x_0 \le x < x_0 + \delta\,, \tag{5}$$

for $n = 0, 1, 2, \ldots$, where C is the Lipschitz constant of the function $f(x, y)$. For $n = 0$ the inequality (5) is the same as (4). To prove it for general n, we apply the iterative definition (3) to see that

$$\begin{aligned} |\varphi_{n+2}(x) - \varphi_{n+1}(x)| &\le \int_{x_0}^{x} |f(t, \varphi_{n+1}(t)) - f(t, \varphi_n(t))|\, dt \\ &\le C \int_{x_0}^{x} |\varphi_{n+1}(t) - \varphi_n(t))|\, dt\,. \end{aligned}$$

Therefore, if (5) holds for some integer n, then

$$|\varphi_{n+2}(x) - \varphi_{n+1}(x)| \le \frac{M\,C^{n+1}}{(n+1)!} \int_{x_0}^{x} (t-t_0)^{n+1}\,dt$$
$$= \frac{M\,C^{n+1}}{(n+2)!}(x-x_0)^{n+2}\,, \qquad x_0 \le x < x_0+\delta)\,,$$

which shows that (5) holds for the next integer $n+1$ as well. This proves (5) for all n.

In particular, the inequality (5) gives the uniform estimate

$$|\varphi_{n+1}(x) - \varphi_n(x)| \le \frac{M\,C^n\delta^{n+1}}{(n+1)!}\,, \qquad n = 0, 1, 2, \ldots$$

in the interval $[x_0, x_0+\delta)$. From this it follows that $\{\varphi_n(x)\}$ is a uniform Cauchy sequence, hence it converges uniformly on $[x_0, x_0+\delta)$ to some continuous function $\varphi(x)$. Alternatively, the Weierstrass M-test shows that

$$\varphi_{n+1}(x) = y_0 + \sum_{k=0}^{n} \big(\varphi_{k+1}(x) - \varphi_k(x)\big)$$

converges to $\varphi(x)$, uniformly on $[x_0, x_0+\delta)$, as $n \to \infty$. In view of the Lipschitz condition, we can infer that the sequence $\{f(t, \varphi_n(t))\}$ converges uniformly to $f(t, \varphi(t))$, so that

$$\lim_{n\to\infty} \int_{x_0}^{x} f(t, \varphi_n(t))\,dt = \int_{x_0}^{x} f(t, \varphi(t))\,dt\,.$$

Consequently, we can pass to the limit in the iterative relation (3) and conclude that $\varphi(x)$ satisfies the integral equation (2). This proves the existence of a solution of (2), hence of the initial-value problem (1).

Uniqueness is proved in a similar way. If $\varphi(x)$ and $\psi(x)$ are solutions of (2), then

$$|\varphi(x) - \psi(x)| \le \int_{x_0}^{x} |f(t, \varphi(t)) - f(t, \psi(t))|\,dt$$
$$\le C \int_{x_0}^{x} |\varphi(t) - \psi(t)|\,dt\,, \qquad x_0 \le x < x_0 + \delta\,.$$

Since $|\varphi(x) - x_0| < b$ and $|\psi(x) - x_0| < b$, so that $|\varphi(x) - \psi(x)| < 2b$, it follows that

$$|\varphi(x) - \psi(x)| \le 2b\,C(x - x_0)\,, \qquad x_0 \le x < x_0 + \delta\,.$$

Iteration gives

$$|\varphi(x) - \psi(x)| \le 2b\,C^2 \int_{x_0}^{x} (t - x_0)\,dt = \frac{2b\,C^2}{2!}(x - x_0)^2$$

and in general

$$|\varphi(x) - \psi(x)| \le \frac{2b\,C^n}{n!}(x - x_0)^n \le \frac{2b\,C^n\delta^n}{n!}\,, \qquad x_0 \le x < x_0 + \delta\,,$$

for $n = 1, 2, \ldots$. Letting $n \to \infty$, we conclude that $\varphi(x) \equiv \psi(x)$ in the interval $[x_0, x_0 + \delta)$. This proves the uniqueness of a solution of (2), hence of (1), and completes the proof of the theorem. □

It is not difficult to estimate the error committed in approximating the true solution φ by the function φ_n. It can be shown that

$$|\varphi(x) - \varphi_n(x)| \le \frac{M\,e^{C\delta}C^n\delta^{n+1}}{(n+1)!}\,, \qquad |x - x_0| < \delta\,,\ n = 0, 1, 2, \ldots\,.$$

The proof is left as an exercise.

The existence and uniqueness theorem and its proof are readily generalized to initial-value problems for systems of differential equations

$$\mathbf{y}' = \mathbf{f}(x, \mathbf{y})\,, \qquad \mathbf{y}(x_0) = \mathbf{b}\,, \tag{6}$$

where $\mathbf{y}(x) = (y_1(x), \ldots, y_n(x))$ is a vector-valued function of x, $\mathbf{f} = (f_1, \ldots, f_n)$ is a vector-valued function of x and $\mathbf{y}$, and $\mathbf{b} = (b_1, \ldots, b_n)$ is a prescribed point. It is convenient to work with the norm

$$\|\mathbf{y}\| = |y_1| + |y_2| + \cdots + |y_n|\,.$$

The function $\mathbf{f}$ is assumed to be defined and continuous in some rectangular parallelepiped centered at the point $(x_0, \mathbf{b}) = (x_0, b_1, \ldots, b_n)$, and to satisfy a Lipschitz condition

$$\|\mathbf{f}(x, \mathbf{y}) - \mathbf{f}(x, \mathbf{z})\| \le C\,\|\mathbf{y} - \mathbf{z}\|$$

there. Then the theorem asserts the existence of a unique solution $\mathbf{y} = \boldsymbol{\varphi}(x)$ to the initial-value problem (6), valid in some interval $|x - x_0| < \delta$. Picard's iterative proof can be adapted to this more general situation. The problem (6) is converted to an equivalent integral equation

$$\mathbf{y}(x) = \mathbf{b} + \int_{x_0}^{x} \mathbf{f}(t, \mathbf{y}(t))\,dt\,,$$

and a solution $\mathbf{y} = \boldsymbol{\varphi}(x)$ is constructed as a uniform limit of the vector-valued functions $\boldsymbol{\varphi}_k(x)$ defined inductively by $\boldsymbol{\varphi}_0(x) = \mathbf{b}$ and

$$\boldsymbol{\varphi}_{k+1}(x) = \mathbf{b} + \int_{x_0}^{x} \mathbf{f}(t, \boldsymbol{\varphi}_k(t))\, dt\,, \qquad k = 0, 1, 2, \ldots .$$

The details are similar to the scalar case and are left as an exercise.

The last theorem applies in particular to linear systems

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \cdots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \cdots + a_{2n}y_n \\ &\quad\ .\qquad .\qquad . \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \cdots + a_{nn}y_n\,, \end{aligned}$$

where the functions $a_{jk} = a_{jk}(x)$ are continuous in some neighborhood of x_0.

A single linear differential equation of higher order can be viewed as a special case of a first-order linear system. Consider a differential equation of the form

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = 0\,, \tag{7}$$

where each of the functions $a_k(x)$ is continuous in some neighborhood of a point x_0. The initial-value problem is to find a solution of (7) that satisfies a set of initial conditions

$$y(x_0) = b_1\,,\ y'(x_0) = b_2\,,\ \ldots\,,\ y^{(n-1)}(x_0) = b_n \tag{8}$$

for some specified numbers $b_1, b_2, \ldots, b_n$. To view this problem from the perspective of a first-order linear system, set

$$z_1 = y\,,\quad z_2 = y'\,,\ \ldots\,,\ z_n = y^{(n-1)}\,.$$

Then the differential equation (7) is expressed by the linear system

$$\begin{aligned} z_1' &= z_2\,,\quad z_2' = z_3\,,\ \ldots\,,\ z_{n-1}' = z_n\,, \\ z_n' &= -a_n z_1 - a_{n-1}z_2 - \cdots - a_1 z_n\,, \end{aligned}$$

and the initial conditions (8) become $\mathbf{z}(x_0) = \mathbf{b}$, where $\mathbf{z} = (z_1, \ldots, z_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$. Consequently, the existence and uniqueness theorem for first-order linear systems implies the existence and uniqueness of a solution to the linear differential equation (7) under the initial conditions (8).

13.2. Wronskians

Recall that a set of functions $f_1(x), f_2(x), \ldots, f_n(x)$ is said to be *linearly independent* over an open interval $I \subset \mathbb{R}$ if no linear combination

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x)$$

vanishes throughout I, except for the trivial combination with all coefficients $c_k = 0$. It is equivalent to require that no function in the set be expressible as a linear combination of the others. The functions are said to be *linearly dependent* if some nontrivial combination vanishes identically in I.

If all of the functions $f_n(x)$ have derivatives up to order $n-1$ on I, their linear dependence implies that the system of linear equations

$$\begin{aligned} c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) &= 0 \\ c_1 f_1'(x) + c_2 f_2'(x) + \cdots + c_n f_n'(x) &= 0 \\ \cdot \quad \cdot \quad \cdot & \\ c_1 f_1^{(n-1)}(x) + c_2 f_2^{(n-1)}(x) + \cdots + c_n f_n^{(n-1)}(x) &= 0 \end{aligned}$$

has a nontrivial solution $(c_1, c_2, \ldots, c_n)$ for all $x \in I$. Thus the determinant

$$W(x) = W(x; f_1, f_2, \ldots, f_n) = \begin{vmatrix} f_1(x) & f_2(x) & \ldots & f_n(x) \\ f_1'(x) & f_2'(x) & \ldots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \ldots & f_n^{(n-1)}(x) \end{vmatrix}$$

must vanish identically on I. An equivalent statement is that the functions $f_1, \ldots, f_n$ are linearly independent on I if $W(x_0) \neq 0$ for some point $x_0 \in I$. The determinant $W(x)$ is called the *Wronskian* of the functions $f_1, \ldots, f_n$, in honor of the mathematician Josef Wronski (1778–1853).

It may well happen that a set of functions is linearly independent and yet their Wronskian vanishes at some points of the interval. (See Exercise 3 for an example.) This cannot happen, however, if the functions are solutions of the same linear differential equation. Indeed, suppose that $y_1, y_2, \ldots, y_n$ are solutions of the differential equation (7) whose coefficients $a_k(x)$ are continuous on an interval I. Let $W(x) = W(x; y_1, \ldots, y_n)$ denote the Wronskian. If $W(x_0) = 0$ for some point $x_0 \in I$, then the corresponding system of linear equations

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) + \cdots + c_n y_n(x_0) &= 0 \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) + \cdots + c_n y_n'(x_0) &= 0 \\ \cdot \quad \cdot \quad \cdot & \\ c_1 y_1^{(n-1)}(x_0) + c_2 y_2^{(n-1)}(x_0) + \cdots + c_n y_n^{(n-1)}(x_0) &= 0 \end{aligned}$$

has a nontrivial solution $(c_1, c_2, \ldots, c_n)$. This says that the function

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

has the properties

$$y(x_0) = y'(x_0) = \cdots = y^{(n-1)}(x_0) = 0\,.$$

But the differential equation is linear, so y is a solution with the same data at x_0 as the trivial solution $z(x) \equiv 0$. By the uniqueness of the solution to an initial-value problem, it follows that $y(x) = z(x) = 0$ for all $x \in I$. In other words, the functions $y_1, \ldots, y_n$ are linearly dependent on I if their Wronskian vanishes at any point of the interval.

In summary, these results demonstrate a sharp dichotomy for sets of solutions $y_1, \ldots, y_n$ of the same differential equation (7). Either these functions are linearly dependent and their Wronskian vanishes everywhere on the interval I, or they are linearly independent and their Wronskian vanishes nowhere on I. There is no middle ground. The same dichotomy will be apparent from an explicit formula for the Wronskian that we now proceed to derive.

The case $n = 2$ is the most important, and will be considered first. Let y_1 and y_2 be solutions of a differential equation

$$y'' + p(x)y' + q(x)y = 0\,, \qquad x \in I\,, \tag{9}$$

where the coefficient functions p and q are continuous on I. Let

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_1'(x)y_2(x)$$

be the Wronskian of y_1 and y_2. Differentiation gives

$$\begin{aligned} W' &= y_1 y_2'' - y_1'' y_2 \\ &= (py_1' + qy_1)y_2 - y_1(py_2' + qy_2) \\ &= p(y_1'y_2 - y_1y_2') = -pW\,. \end{aligned}$$

This is a linear first-order differential equation for W, namely $W' + pW = 0$. After multiplication by the integrating factor $\exp\left\{\int_{x_0}^x p(t)dt\right\}$ with any conveniently chosen point $x_0 \in I$, the equation reduces to

$$\frac{d}{dx}\left(W(x)\,\exp\left\{\int_{x_0}^x p(t)dt\right\}\right) = 0\,,$$

so that

$$W(x) = W(x_0)\, \exp\left\{ -\int_{x_0}^{x} p(t)dt \right\} . \tag{10}$$

The formula (10) is known as the *Wronskian relation* for the differential equation (9). Its most remarkable feature is its dependence only on the coefficient function p, not on the solutions y_1 and y_2, except for the value of their Wronskian at a single point. The formula shows again that either $W(x) \neq 0$ for all $x \in I$ or $W(x) \equiv 0$ on I.

For a set of solutions $y_1, y_2, \ldots, y_n$ of the general differential equation (7) of order n, a similar calculation shows that the Wronskian $W(x) = W(x; y_1, \ldots, y_n)$ satisfies the equation $W' = -a_1 W$, and so

$$W(x) = W(x_0)\, \exp\left\{ -\int_{x_0}^{x} a_1(t)dt \right\} . \tag{11}$$

Here the calculation involves some general properties of determinants and is left as an exercise.

Finally, it should be noted that the existence and uniqueness theorem leads to an important structural formula for the solutions of a (homogeneous) linear differential equation. Suppose that $y_1, y_2, \ldots, y_n$ is a linearly independent set of solutions to the differential equation (7). Then any linear combination $y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is again a solution, and linear independence implies the non-vanishing of the Wronskian, so that the constant coefficients $c_1, \ldots, c_n$ can be chosen to make the function y satisfy any prescribed initial conditions

$$y(x_0) = b_1\,,\ y'(x_0) = b_2\,,\ \ldots,\ y^{(n-1)}(x_0) = b_n\,.$$

In particular, if $z(x)$ is an arbitrary solution of (7), the coefficients $c_1, \ldots, c_n$ can be chosen so that

$$y(x_0) = z(x_0)\,,\ y'(x_0) = z'(x_0)\,,\ \ldots,\ y^{(n-1)}(x_0) = z^{(n-1)}(x_0)\,.$$

It then follows from the uniqueness property that $z(x) \equiv y(x)$ for this choice of coefficients. In other words, every solution of (7) is a linear combination of the set of linearly independent solutions $y_1, \ldots, y_n$.

In the special case where the equation (7) has constant coefficients, there are well known procedures for explicit construction of a complete set of linearly independent solutions. The general solution is then expressible as some linear combination of these functions. In the case of variable coefficients, the calculation of solutions is generally not so easy. The problem is even more difficult near points where the coefficients have singularities. In the next section we discuss methods for finding solutions in the form of power series.

13.3. Power series solutions

In discussing solutions by power series, we restrict attention to linear differential equations of second order,

$$y'' + p(x)y' + q(x)y = 0\,. \tag{12}$$

A point $x_0 \in \mathbb{R}$ is said to be a *regular point* if in some neighborhood of x_0 both of the functions $p(x)$ and $q(x)$ can be represented as power series centered at x_0. At a regular point it is always possible to find a pair of independent solutions, also represented by power series. To illustrate the procedure, consider the simple example

$$y'' + 2xy' + 2y = 0\,.$$

We look for a solution in the form of a power series $y = \sum_{k=0}^{\infty} c_k x^k$ with positive radius of convergence. Term-by-term differentiation gives

$$y' = \sum_{k=1}^{\infty} k c_k x^{k-1}\,, \qquad y'' = \sum_{k=2}^{\infty} k(k-1) c_k x^{k-2}\,.$$

Putting these expressions into the differential equation, we are led to the requirement that

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k + 2 \sum_{k=0}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k = 0\,.$$

This imposes the equivalent condition that the coefficients of all powers x^k must vanish, so that

$$(k+2)(k+1)c_{k+2} + 2(k+1)c_k = 0\,,$$

or

$$c_{k+2} = -\frac{2}{k+2}\, c_k\,, \qquad k = 0, 1, 2, \ldots\,. \tag{13}$$

It is now clear that the first two coefficients c_0 and c_1 can be chosen arbitrarily, and then the recurrence relation (13) will determine the others. Specifically, we find from (13) that

$$c_2 = -c_0\,, \quad c_4 = -\frac{1}{2} c_2 = \frac{1}{2} c_0\,, \quad c_6 = -\frac{1}{3} c_4 = -\frac{1}{3!}\, c_0\,, \ldots\,;$$
$$c_3 = -\frac{2}{3} c_1\,, \quad c_5 = -\frac{2}{5} c_3 = \frac{2^2}{1 \cdot 3 \cdot 5}\, c_1\,, \ldots\,,$$

and in general

$$c_{2k} = \frac{(-1)^k}{k!}\, c_0 \quad \text{and} \quad c_{2k+1} = \frac{(-1)^k 2^k}{1 \cdot 3 \cdot 5 \cdots (2k+1)}\, c_1\,, \quad k = 1, 2, 3, \ldots\,.$$

Thus the general solution to the differential equation is

$$y = c_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} x^{2k} + c_1 \sum_{k=0}^{\infty} \frac{(-1)^k 2^k}{1 \cdot 3 \cdot 5 \cdots (2k+1)} x^{2k+1}\,.$$

Both series are easily seen to converge on the whole real line, so that the term-by-term differentiation is justified and the series represent actual solutions to the differential equation. In fact, the first series sums to the function e^{-x^2}.

In a similar way, near any regular point x_0 of a differential equation (12) two linearly independent power series solutions can always be constructed. It can be proved that the formal power series so constructed necessarily converge in some neighborhood of x_0, hence represent bonafide solutions.

The method fails if x_0 is a *singular point*; that is, if it is not a regular point of the differential equation (12). However, there may still exist a power series solution at x_0, which can be found by the same device. Consider, for example, Bessel's equation

$$xy'' + y' + xy = 0\,, \qquad \text{or} \qquad y'' + \frac{1}{x} y' + y = 0\,.$$

Here $x_0 = 0$ is a singular point. Nevertheless, we look for a power series solution of the form

$$y(x) = \sum_{k=0}^{\infty} c_k x^k\,.$$

Inserting this series and its derivatives into Bessel's equation, we find that

$$\sum_{k=2}^{\infty} k(k-1) c_k x^{k-1} + \sum_{k=1}^{\infty} k c_k x^{k-1} + \sum_{k=0}^{\infty} c_k x^{k+1} = 0\,,$$

or

$$c_1 + \sum_{k=0}^{\infty} \big[(k+2)^2 c_{k+2} + c_k\big] x^{k+1} = 0\,.$$

Thus the requirements are that $c_1 = 0$ and

$$c_{k+2} = -\frac{1}{(k+2)^2}\, c_k\,, \qquad k = 0, 1, 2, \ldots\,.$$

Since $c_1 = 0$, the recurrence relation implies that $c_k = 0$ for all odd indices k. With the choice $c_0 = 1$, the relation gives

$$c_2 = -\frac{1}{2^2}\,, \qquad c_4 = \frac{1}{2^2 4^2}\,, \qquad c_6 = -\frac{1}{2^2 4^2 6^2}\,, \ldots\,,$$

so that, in standard notation,

$$y(x) = J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 4^2} - \frac{x^6}{2^2 4^2 6^2} + \ldots$$

is a solution of Bessel's equation. This power series converges on the whole real line and is called a *Bessel function of first kind*. Its graph is shown in Figure 1.

Because of the requirement that $c_1 = 0$, the only power series solutions are constant multiples of $J_0(x)$. The search for a second solution leads us to a general discussion of regular singular points and the method of Frobenius. We will now describe this method in detail, applying it to specific equations but omitting a proof of its general validity. For a more complete theoretical discussion, the reader is referred to the book by Coddington and Levinson [**4**].

A singular point x_0 of the differential equation (12) is called a *regular singular point* if $(x-x_0)p(x)$ and $(x-x_0)^2 q(x)$ have power series expansions about x_0:

$$(x-x_0)p(x) = \sum_{k=0}^{\infty} a_k (x-x_0)^k \qquad \text{and} \qquad (x-x_0)^2 q(x) = \sum_{k=0}^{\infty} b_k (x-x_0)^k\,,$$

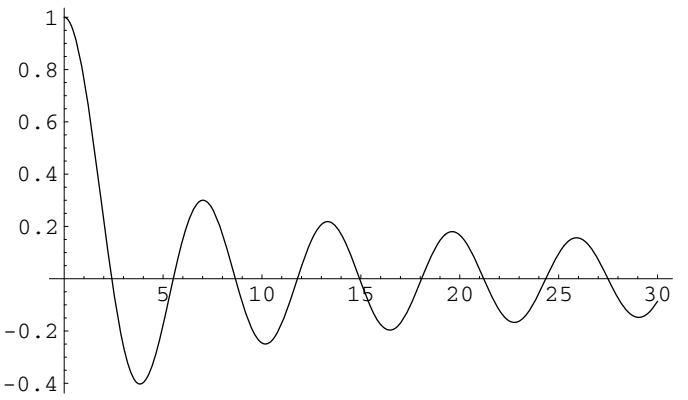


Figure 1. Graph of Bessel's function $J_0(x)$.

where both series converge in some interval $|x - x_0| < \delta$.

The most commonly occurring differential equations of mathematical physics have at worst regular singular points. For instance, the Legendre equation

$$(1 - x^2)y'' - 2xy' + \lambda y = 0\,, \qquad \lambda \text{ a real constant},$$

has regular singular points at $x = 1$ and $x = -1$, and is regular elsewhere on the real axis. Bessel's equation

$$xy'' + y' + xy = 0$$

has a regular singular point at the origin.

For ease of notation, we will suppose that our differential equation has a regular singular point at $x_0 = 0$. The strategy is then to seek a generalized power series solution of the form

$$y = x^\alpha \sum_{k=0}^{\infty} c_k\, x^k\,, \qquad c_0 = 1\,,$$

where the real number α and the coefficients c_k are to be determined. Formal calculations give

$$y' = \sum_{k=0}^{\infty} (k+\alpha)c_k\, x^{k+\alpha-1} \qquad \text{and} \qquad y'' = \sum_{k=0}^{\infty} (k+\alpha)(k+\alpha-1)c_k\, x^{k+\alpha-2}\,,$$

so that

$$py' = \sum_{j=0}^{\infty} a_j x^{j-1} \sum_{k=0}^{\infty} (k+\alpha)c_k\, x^{k+\alpha-1} = \sum_{n=0}^{\infty} \Big\{ \sum_{j+k=n} a_j(k+\alpha)c_k \Big\} x^{n+\alpha-2}$$

and

$$qy = \sum_{j=0}^{\infty} b_j\, x^{j-2} \sum_{k=0}^{\infty} c_k\, x^{k+\alpha} = \sum_{n=0}^{\infty} \Big\{ \sum_{j+k=n} b_j c_k \Big\} x^{n+\alpha-2}\,.$$

Addition of these expressions gives, after some rearrangement of terms,

$$\begin{aligned} y'' + py' + qy =& F(\alpha)x^{\alpha-2} \\ &+ \sum_{n=1}^{\infty} \left\{ F(n+\alpha)c_n + \sum_{k=0}^{n-1} \big[(k+\alpha)a_{n-k} + b_{n-k}\big] c_k \right\} x^{n+\alpha-2}\,, \end{aligned}$$

where $F(\alpha) = \alpha(\alpha-1) + a_0\alpha + b_0$. Therefore, in order that y be a solution to the differential equation, we must require that the coefficient of $x^{n+\alpha-2}$

vanish for every integer $n = 0, 1, 2, \ldots$. Thus the requirements are that $F(\alpha) = 0$ and

$$F(n+\alpha)c_n = -\sum_{k=0}^{n-1} \big[(k+\alpha)a_{n-k} + b_{n-k}\big]c_k\,, \qquad n = 1, 2, \ldots\,. \tag{14}$$

The quadratic equation $F(\alpha) = 0$ is known as the *indicial equation* associated with the given differential equation. Let α_1 and α_2 denote its solutions, real or complex. There are then 4 cases to be treated separately.

Case 1. α_1 and α_2 are real and distinct, and do not differ by an integer. Then set $\alpha = \alpha_1$ in (14). Starting with $c_0 = 1$, this equation determines recursively the coefficients $c_n = c_n(\alpha_1)$, because in this case $F(n+\alpha_1) \neq 0$ for all $n = 1, 2, \ldots$. The same procedure serves to determine coefficients $c_n = c_n(\alpha_2)$. In this manner we have constructed a pair of linearly independent formal (generalized) power series solutions

$$y = x^{\alpha_j} \sum_{n=0}^{\infty} c_n(\alpha_j)\, x^n\,, \qquad j = 1, 2,\,.$$

It can be proved that each of these formal power series actually converges in some neighborhood of the regular singular point $x_0 = 0$, and therefore represents an actual solution to the differential equation.

Case 2. α_1 and α_2 are real and $\alpha_1 = \alpha_2$. Then only one solution can be obtained by the process just described. However, $F(\alpha_1) = F'(\alpha_1) = 0$, since α_1 is a double root of the polynomial $F(\alpha)$. This observation allows us to obtain a second (independent) solution by a process of differentiation with respect to α. For arbitrary real α near α_1, let the functions $c_0(\alpha) = 1$, $c_1(\alpha)$, $c_2(\alpha), \ldots$ be defined recursively through the equation (14). Then each coefficient $c_n(\alpha)$ is a rational function of α, the quotient of two polynomials. Consider the function

$$y = y(x, \alpha) = x^{\alpha} \sum_{n=0}^{\infty} c_n(\alpha)\, x^n\,.$$

Because of the way the coefficients $c_n(\alpha)$ were constructed, we see that

$$y_{xx}(x, \alpha) + p(x)y_x(x, \alpha) + q(x)y(x, \alpha) = F(\alpha)\, x^{\alpha-2}\,,$$

where the subscripts denote partial derivatives. Once more it is clear that $y(x, \alpha_1)$ satisfies the differential equation, but we can say more. Differentiation with respect to α gives

$$(y_\alpha)_{xx} + p(y_\alpha)_x + q\, y_\alpha = [F(\alpha)\log x + F'(\alpha)]\, x^{\alpha-2}\,.$$

Because $F(\alpha_1) = F'(\alpha_1) = 0$, this shows that

$$y_\alpha(x, \alpha_1) = y(x, \alpha_1) \log x + x^{\alpha_1} \sum_{n=0}^{\infty} {c_n}'(\alpha_1) x^n$$

is a formal solution to the differential equation, and it can be shown to be an actual solution; in other words, the power series converges in some open interval about the origin.

Case 3. α_1 and α_2 are real and distinct, and differ by an integer. Number the two roots so that $\alpha_2 = \alpha_1 + m$, where m is a positive integer. Then a solution $y(x, \alpha_2)$ can be constructed by the method described in Case 1. However, if we make a similar attempt to use the relation (14) for recursive determination of the coefficients $c_n(\alpha_1)$, we encounter a major obstacle when $n = m$, since $F(m + \alpha_1) = F(\alpha_2) = 0$. Thus a direct construction fails, and we are in search of a second solution.

It is here that the method of Frobenius comes to the rescue. Consider an expansion of the form

$$y = \varphi(x, \alpha) = (\alpha - \alpha_1) x^\alpha + x^\alpha \sum_{n=1}^{\infty} c_n \, x^n \, .$$

This amounts to replacing $c_0 = 1$ by $(\alpha - \alpha_1)$. With this expression for y, a formal calculation gives, as before,

$$\begin{aligned} y'' + py' + qy &= (\alpha - \alpha_1) F(\alpha) \, x^{\alpha - 2} \\ &\quad + \sum_{n=1}^{\infty} \Bigg\{ F(n + \alpha) c_n + (\alpha a_n + b_n)(\alpha - \alpha_1) \\ &\quad + \sum_{k=1}^{n-1} \big[(k + \alpha) a_{n-k} + b_{n-k} \big] c_k \Bigg\} x^{n + \alpha - 2} \, , \end{aligned}$$

where the last sum is absent for $n = 1$. For α near α_1, we now use the relation

$$F(n + \alpha) c_n = -(\alpha a_n + b_n)(\alpha - \alpha_1) - \sum_{k=1}^{n-1} \big[(k + \alpha) a_{n-k} + b_{n-k} \big] c_k \tag{15}$$

to generate recursively the coefficients $c_n = c_n(\alpha)$ for $n = 1, 2, \ldots$. It is easily seen that for $n = 1, 2, \ldots, m-1$, the functions $c_n(\alpha)$ all carry $(\alpha - \alpha_1)$ as a factor. Since the polynomial $F(\alpha)$ factors as $F(\alpha) = (\alpha - \alpha_2)(\alpha - \alpha_1)$, and $\alpha_2 = \alpha_1 + m$, we see that

$$F(m + \alpha) = (\alpha - \alpha_1)(\alpha + m - \alpha_1) \, ,$$

and so it follows that $c_m(\alpha)$ is well determined at $\alpha = \alpha_1$. Indeed, we have only to divide the equation (15) by $(\alpha - \alpha_1)$ and set $\alpha = \alpha_1$. Consequently, $c_n(\alpha_1)$ is defined by (15) for all n, and

$$\varphi_{xx}(x,\alpha) + p(x)\varphi_x(x,\alpha) + q(x)\varphi(x,\alpha) = (\alpha - \alpha_1)F(\alpha)\, x^{\alpha-2}\,.$$

In particular, $y = \varphi(x,\alpha_1)$ is a solution of the differential equation, but it is not in general independent of $y(x,\alpha_2)$. To obtain an independent solution, differentiate the last equation with respect to α. This shows that $y = \varphi_\alpha(x,\alpha_1)$ is a solution.

Case 4. α_1 and α_2 are (nonreal) complex conjugates: $\alpha_2 = \overline{\alpha_1}$. Then the procedure of Case 1 can be used to generate two complex-valued solutions

$$y(x,\alpha_1) \qquad \text{and} \qquad y(x,\alpha_2) = \overline{y(x,\alpha_1)}\,.$$

For a pair of independent real-valued solutions, take real and imaginary parts.

Bessel's equation $xy'' + y' + xy = 0$ offers an important illustration of the method described in Case 2, the easier case for the method of Frobenius. Here $xp(x) = 1$ and $x^2q(x) = x^2$. Thus $a_0 = 1$ and $a_n = 0$ for all $n \geq 1$; whereas $b_2 = 1$ and $b_n = 0$ for all $n \neq 2$. Therefore, the indicial polynomial is

$$F(\alpha) = \alpha(\alpha - 1) + \alpha = \alpha^2\,,$$

with roots $\alpha_1 = \alpha_2 = 0$. The recurrence relation (14) reduces to $c_1(\alpha) = 0$ and

$$c_n(\alpha) = -(n+\alpha)^{-2}c_{n-2}(\alpha)\,, \qquad n = 2, 3, \ldots\,.$$

It follows recursively that $c_n(\alpha) = 0$ for all odd indices n. Also, since $c_0(\alpha) = 1$, recursion leads to the formula

$$c_{2k}(\alpha) = (-1)^k(2+\alpha)^{-2}(4+\alpha)^{-2}\cdots(2k+\alpha)^{-2}$$

for the coefficients of even index $n = 2k$. Differentiation gives the expression

$$c_{2k}{}'(0) = (-1)^{k+1}2^{-2}4^{-2}\cdots(2k)^{-2}\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{k}\right)\,.$$

These calculations show that the two solutions $y(x,0)$ and $y_\alpha(x,0)$ are

$$J_0(x) = 1 + \sum_{k=1}^{\infty}(-1)^k 2^{-2}4^{-2}\cdots(2k)^{-2}x^{2k}\,,$$

as calculated previously, and

$$Y_0(x) = J_0(x)\log x + \sum_{k=1}^{\infty}(-1)^{k+1}2^{-2}4^{-2}\cdots(2k)^{-2}\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{k}\right)x^{2k}\,.$$

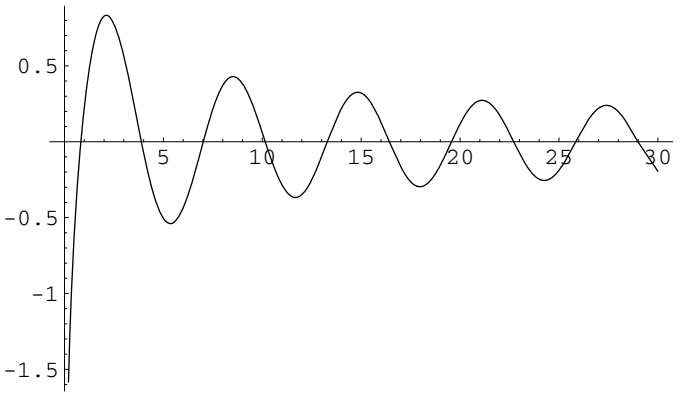


Figure 2. Graph of Bessel's function $Y_0(x)$.

Both power series converge for all $x \in \mathbb{R}$. The functions $J_0(x)$ and $Y_0(x)$ are known as *Bessel functions of first* and *second kinds*, respectively. Figure 2 shows the graph of $Y_0(x)$. It should be remarked that conventions differ and the definition of $Y_0(x)$ is not entirely standard in the literature.

Georg Frobenius (1849–1917) was a student of Weierstrass in Berlin. After serving for many years as a professor in Zürich, he returned to the University of Berlin as one of the leading mathematicians of his day. His work spanned several areas of mathematics but he is perhaps best known for work in algebra, specifically in the theory of groups.

13.4. Bessel functions

Bessel's equation $xy''+y'+xy = 0$ arises in the study of physical problems that involve circular disks or cylinders. As an illustration we now discuss the vibrations of a circular drumhead. Suppose an elastic membrane is clamped along a circular boundary of unit radius as it undergoes vibrations. Denote by $u(r, \theta, t)$ the vertical displacement of the membrane at time t above the point (r, θ), described in polar coordinates. The displacement is governed by the wave equation, which takes the form

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}$$

if the units of time and length are chosen appropriately. The boundary condition $u(1, \theta, t) = 0$ applies, since the membrane is clamped at its circular edge. The motion is then determined by initial position and velocity. If the initial conditions are circularly symmetric (*i.e.*, independent of θ), so that

$$u(r, \theta, 0) = f(r) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(r, \theta, 0) = g(r)\,,$$

then the subsequent motion will also be circularly symmetric. In other words, the vertical displacement is independent of θ and can be written as $u(r,t)$. Thus the wave equation reduces to

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r}\,, \qquad u(1,t) = 0\,.$$

We look for a particular solution of the form $u(r,t) = R(r)T(t)$. Then the wave equation gives $RT'' = [R'' + (1/r)R']T$. Separating the variables, we can write

$$\frac{T''}{T} = \frac{R'' + (1/r)R'}{R}\,.$$

The last equation says that a function of t is identically equal to a function of r, so both functions must be constant. The constant is negative, because otherwise the differential equation for T would have unbounded solutions. Writing the constant as $-\lambda^2$ for $\lambda > 0$, we see that R and T satisfy the differential equations

$$rR'' + R' + \lambda^2 rR = 0\,, \qquad T'' + \lambda^2 T = 0\,.$$

It is easily verified that the solutions of the first equation have the form $R(r) = y(\lambda r)$ where $y(x)$ is a solution of Bessel's equation $xy'' + y' + xy = 0$. We have seen that every solution of Bessel's equation is a linear combination of $J_0(x)$ and $Y_0(x)$, the Bessel functions of first and second kind. But $Y_0(x)$ is unbounded near the origin, where it has a logarithmic singularity. Consequently, the only bounded solutions, hence the only physically relevant ones, are constant multiples of $R(r) = J_0(\lambda r)$. On the other hand, the solutions of the differential equation $T'' + \lambda^2 T = 0$ are linear combinations of $\cos\lambda t$ and $\sin\lambda t$. We conclude that the relevant solutions $u(r,t) = R(r)T(t)$ have the form

$$u(r,t) = J_0(\lambda r)\big(a\cos\lambda t + b\sin\lambda t\big)$$

for arbitrary constants a and b. The boundary condition $u(1,t) = 0$ requires that $J_0(\lambda) = 0$. In other words, λ must be a zero of the Bessel function $J_0(x)$.

As can be surmised from the graph in Figure 1, and will be proved later, the Bessel function $J_0(x)$ has infinitely many positive zeros λ_n, ordered so that $0 < \lambda_1 < \lambda_2 < \dots$. With the boundary condition taken into account, we have arrived at the particular solutions

$$u_n(r,t) = J_0(\lambda_n r)\big(a_n\cos\lambda_n t + b_n\sin\lambda_n t\big)\,, \qquad n = 1,2,\dots\,.$$

By the superposition principle, any sum of such functions is again a solution. We want to choose the coefficients a_n and b_n so that the series

$$u(r,t) = \sum_{n=1}^{\infty} J_0(\lambda_n r)\big(a_n\cos\lambda_n t + b_n\sin\lambda_n t\big) \tag{16}$$

converges and satisfies the prescribed initial conditions $u(r,0) = f(r)$ and $u_t(r,0) = g(r)$. Formally, these requirements reduce to

$$\sum_{n=1}^{\infty} a_n J_0(\lambda_n r) = f(r) \qquad \text{and} \qquad \sum_{n=1}^{\infty} \lambda_n b_n J_0(\lambda_n r) = g(r)\,.$$

Here we encounter an expansion problem similar to that in the theory of Fourier series. The coefficients a_n and b_n are calculated by means of the orthogonality relation

$$\int_0^1 r J_0(\lambda_n r) J_0(\lambda_m r)\, dr = 0\,, \qquad n \neq m\,. \tag{17}$$

For a proof of (17), recall that $y_n(x) = J_0(\lambda_n x)$ satisfies the differential equation

$$\frac{d}{dx}\left\{x y_n'\right\} + \lambda_n^2 x y_n = 0\,.$$

Multiply the equation by y_m and integrate by parts to see that

$$\begin{aligned}
\lambda_n^2 \int_0^1 x y_n(x) y_m(x)\, dx &= -\int_0^1 y_m(x) \frac{d}{dx}\left\{x y_n'\right\} \\
&= -\Big[x y_n'(x) y_m(x)\Big]_0^1 + \int_0^1 x y_n'(x) y_m'(x)\, dx \\
&= \int_0^1 x y_n'(x) y_m'(x)\, dx\,,
\end{aligned}$$

because $y_m(1) = J_0(\lambda_m) = 0$. Now interchange n and m and subtract one equation from the other to conclude that

$$(\lambda_n^2 - \lambda_m^2) \int_0^1 x y_n(x) y_m(x)\, dx = 0\,,$$

which implies (17) since $\lambda_n^2 \neq \lambda_m^2$.

A similar calculation shows that

$$\int_0^1 x J_0(\lambda_n x)^2\, dx = \tfrac{1}{2}\, J_0'(\lambda_n)^2\,.$$

(See Exercise 8 for further details.) Consequently, the equations

$$\tfrac{1}{2}\, J_0'(\lambda_n)^2\, a_n = \int_0^1 r J_0(\lambda_n r) f(r)\, dr \qquad \text{and}$$

$$\tfrac{1}{2}\, J_0'(\lambda_n)^2\, \lambda_n b_n = \int_0^1 r J_0(\lambda_n r) g(r)\, dr$$

determine the "Fourier–Bessel" coefficients a_n and b_n that should be inserted into the series (16) to solve the problem of the vibrating drumhead. A rigorous treatment of the convergence problem for Fourier–Bessel series is beyond the scope of this book, but can be found for instance in Watson [**13**].

A more general form of Bessel's differential equation is

$$x^2y'' + xy' + (x^2 - n^2)y = 0\,,$$

where n is a real parameter. If n is a positive integer, one solution is

$$\begin{aligned} J_n(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(n+k)!} \left(\frac{x}{2}\right)^{n+2k} \\ &= \frac{x^n}{2^n n!} \left\{ 1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2\cdot 4(2n+2)(2n+4)} - \dots \right\} \end{aligned} \tag{18}$$

The graph of $J_1(x)$ is shown in Figure 3. It is an odd function with infinitely many zeros. Bessel functions of negative integer order are defined by $J_{-n}(x) = (-1)^n J_n(x)$.

The integral representations

$$J_n(x) = \frac{1}{\pi} \frac{2^n n!}{(2n)!}\, x^n \int_0^{\pi} \cos(x\cos\theta) \sin^{2n}\theta \, d\theta \tag{19}$$

and

$$J_n(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x\sin\theta - n\theta)\, d\theta \tag{20}$$

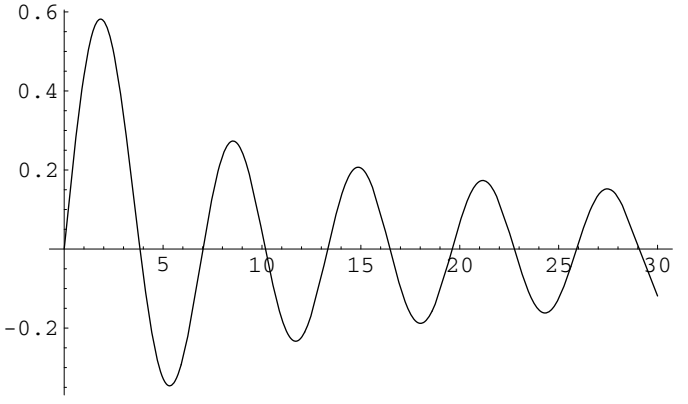


Figure 3. Graph of Bessel's function $J_1(x)$.

are valid for $n = 0, 1, 2, \ldots$, with the usual convention that $0! = 1$. For $n = 0$ the two representations reduce to virtually the same formula

$$J_0(x) = \frac{1}{\pi}\int_0^\pi \cos(x\cos\theta)\,d\theta = \frac{1}{\pi}\int_0^\pi \cos(x\sin\theta)\,d\theta\,.$$

Each of the representations (19) and (20) for the Bessel function $J_n(x)$ can be verified by showing that the right-hand side satisfies Bessel's differential equation and is finite and properly normalized at the origin (see Exercises 10, 13, and 14). The representation (20) is often called *Bessel's formula*. A standard calculation in the theory of functions of a complex variable (contour integration to find the coefficients in a Laurent expansion) produces Bessel's formula as a consequence of the remarkable generating relation

$$e^{\frac{x}{2}(t-\frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x)\,t^n\,. \tag{21}$$

Conversely, the generating relation can be established as a consequence of Bessel's formula. However, the generating relation (21) can also be derived in a more elementary way, without resort to methods of complex analysis. One such derivation proceeds as follows.

Recalling that $J_{-n}(x) = (-1)^n J_n(x)$ and introducing the series (18), we can write

$$\begin{aligned}\sum_{n=-\infty}^{\infty} J_n(x)\,t^n &= \sum_{n=0}^{\infty} J_n(x)\,t^n + \sum_{n=1}^{\infty}(-1)^n J_n(x)\,t^{-n}\\ &= \sum_{n=0}^{\infty}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!(n+k)!}\left(\frac{x}{2}\right)^{n+2k} t^n\\ &\quad + \sum_{n=1}^{\infty}\sum_{k=0}^{\infty}\frac{(-1)^{n+k}}{k!(n+k)!}\left(\frac{x}{2}\right)^{n+2k} t^{-n} = S_1 + S_2\,.\end{aligned}$$

Now set $m = n + 2k$ and rewrite the sums as

$$S_1 = 1 + \sum_{m=1}^{\infty}\sum_{k=0}^{[m/2]}\frac{(-1)^k}{k!(m-k)!}\,t^{m-2k}\left(\frac{x}{2}\right)^m \qquad \text{and}$$

$$S_2 = \sum_{m=1}^{\infty}\sum_{k=0}^{[(m-1)/2]}\frac{(-1)^{m-k}}{k!(m-k)!}\,t^{2k-m}\left(\frac{x}{2}\right)^m\,,$$

where $[\alpha]$ denotes the largest integer less than or equal to α. On the other hand, the binomial theorem gives

$$\left(t-\frac{1}{t}\right)^m = \sum_{k=0}^{m} \frac{m!}{k!(m-k)!}\,(-1)^k t^{m-2k}$$
$$= \sum_{k=0}^{[m/2]} \frac{m!}{k!(m-k)!}\,(-1)^k t^{m-2k} + \sum_{j=0}^{[(m-1)/2]} \frac{m!}{(m-j)!j!}\,(-1)^{m-j} t^{2j-m},$$

with the substitution $k = m - j$ in the second part of the sum. Therefore,

$$S_1 + S_2 = 1 + \sum_{m=1}^{\infty} \frac{1}{m!}\left(t-\frac{1}{t}\right)^m \left(\frac{x}{2}\right)^m = e^{\frac{x}{2}(t-\frac{1}{t})}\,,$$

which is the desired result (21). (For another proof, see Exercise 15.)

It was essentially through the generating relation that the functions now called Bessel functions originally arose in 1817, when Bessel was studying perturbations of astronomical orbits. Other mathematicians had previously encountered special cases of Bessel functions, but Bessel developed their properties in a systematic treatise published in 1824.

Friedrich Wilhelm Bessel (1784–1846) had an unusual life history. He left school at age 14 to become apprentice to a shipping merchant in Bremen. Thereafter he was mainly self-educated. Attracted by the problem of determining longitude at sea, he developed a keen interest in astronomy. His work drew the attention of professional astronomers, and at age 26 he became director of the observatory in Königsberg, where he remained for the rest of his career. Using sophisticated mathematical methods, he made important contributions to astronomy, including improved data on the positions of stars.

13.5. Hypergeometric functions

The hypergeometric differential equation is

$$x(1-x)y'' + [c-(a+b+1)x]y' - ab\,y = 0\,, \tag{22}$$

where a, b, and c are real parameters. There are many important special cases. It has regular singular points at 0 and 1, and (with suitable interpretation) at ∞, and it is in a sense the most general linear differential equation of second order with three regular singular points. In the notation of Section 13.3,

$$xp(x) = \frac{c-(a+b+1)x}{1-x} \qquad \text{and} \qquad x^2 q(x) = -\frac{ab\,x}{1-x}\,,$$

so the indicial equation at the origin is

$$F(\alpha) = \alpha(\alpha-1) + a_0\alpha + b_0 = \alpha^2 + (c-1)\alpha = 0\,,$$

with roots $\alpha_1 = 0$ and $\alpha_2 = 1-c$. In particular, the hypergeometric equation admits a power series solution of the form

$$y = 1 + \sum_{k=1}^{\infty} \gamma_k\, x^k\,, \qquad |x| < 1\,,$$

provided that $\alpha_2 = 1-c$ is not a positive integer. The coefficients γ_k are determined by inserting the series into the equation (22) to obtain after some manipulation

$$\sum_{k=0}^{\infty} \big\{(k+1)(k+c)\gamma_{k+1} - (k+a)(k+b)\gamma_k\big\} x^k = 0\,.$$

Since the coefficient of every power x^k must vanish, this gives the recurrence relation

$$(k+1)(k+c)\gamma_{k+1} = (k+a)(k+b)\gamma_k\,, \qquad k = 0, 1, 2, \ldots\,.$$

Under the assumption that $1-c$ is not a positive integer, or that $c \neq 0, -1, -2, \ldots$, it is clear that $k + c \neq 0$ for all $k = 0, 1, 2, \ldots$. Thus with $\gamma_0 = 1$ the recurrence relation leads to the formula

$$\gamma_k = \frac{a(a+1)\cdots(a+k-1)\,b(b+1)\cdots(b+k-1)}{k!\,c(c+1)\cdots(c+k-1)}\,, \qquad k = 1, 2, \ldots\,.$$

It is convenient to adopt the notation

$$(a)_k = a(a+1)\cdots(a+k-1)\,, \qquad \text{with} \quad (a)_0 = 1\,,$$

for a shifted factorial. Then the solution, known as a *hypergeometric function*, can be written more compactly as

$$y = F(x) = F(a, b; c; x) = 1 + \sum_{k=1}^{\infty} \frac{(a)_k\,(b)_k}{k!\,(c)_k}\, x^k\,. \tag{23}$$

In terms of the gamma function, the shifted factorial is $(a)_k = \Gamma(a+k)/\Gamma(a)$. An application of the ratio test confirms that the power series has radius of convergence equal to 1 unless either a or b is a negative integer, in which case the series terminates and $F(x)$ is a *hypergeometric polynomial*.

Many important functions of mathematical analysis can be expressed as hypergeometric series. For instance, the geometric series is

$$\frac{1}{1-x} = 1 + \sum_{k=1}^{\infty} x^k = F(1,1;1;x)\,, \qquad \text{whereas}$$
$$\log(1+x) = x\,F(1,1;2;-x)\,,$$
$$\sin^{-1} x = x\,F(\tfrac{1}{2},\tfrac{1}{2};\tfrac{3}{2};x^2)\,,$$
$$\tan^{-1} x = x\,F(\tfrac{1}{2},1;\tfrac{3}{2};-x^2)\,.$$

Other examples are complete elliptic integrals and Jacobi polynomials, which include the Legendre polynomials and Chebyshev polynomials as special cases.

Gauss made the first systematic study of hypergeometric functions in a famous paper of 1812, but Euler had found a basic integral representation in 1769.

Euler integral representation. *If $c > b > 0$, then*

$$F(a,b;c;x) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-xt)^{-a}\,dt\,, \qquad |x| < 1\,.$$

Proof. The restriction of parameters to $c > b > 0$ ensures that the integral converges. For fixed x with $|x| < 1$, expand the last factor into binomial series:

$$(1-xt)^{-a} = \sum_{k=0}^{\infty} \binom{-a}{k} (-xt)^k\,,$$

where

$$\binom{-a}{k} = \frac{(-a)(-a-1)\cdots(-a-k+1)}{k!} = \frac{(-1)^k\,(a)_k}{k!}\,.$$

Since the binomial series converges uniformly in the interval $0 \le t \le 1$, the integration can be performed term by term, so that

$$\int_0^1 t^{b-1}(1-t)^{c-b-1}(1-xt)^{-a}\,dt = \sum_{k=0}^{\infty} \int_0^1 t^{b+k-1}(1-t)^{c-b-1}\,dt \binom{-a}{k}(-x)^k$$
$$= \sum_{k=0}^{\infty} B(b+k,c-b)\binom{-a}{k}(-x)^k = \sum_{k=0}^{\infty} \frac{\Gamma(b+k)\,\Gamma(c-b)}{\Gamma(c+k)}\,\frac{(-1)^k(a)_k}{k!}\,(-x)^k$$
$$= \frac{\Gamma(b)\,\Gamma(c-b)}{\Gamma(c)} \sum_{k=0}^{\infty} \frac{(a)_k\,(b)_k}{k!(c)_k}\,x^k = \frac{\Gamma(b)\,\Gamma(c-b)}{\Gamma(c)}\,F(a,b;c;x)\,,$$

which verifies Euler's integral formula. □

Hypergeometric functions satisfy a wide variety of interesting identities. We will give only a few examples here. Much more comprehensive treatments can be found in books on special functions such as Rainville [**11**] or Andrews, Askey, and Roy [**1**].

It is obvious from the definition (23) that $F(b,a;c;x) = F(a,b;c;x)$, but strangely this symmetry is not apparent in Euler's integral representation. The change of variable $t = 1 - s$ in Euler's integral gives

$$\begin{aligned} F(a,b;c;x) &= \frac{\Gamma(c)}{\Gamma(b)\,\Gamma(c-b)} \int_0^1 s^{c-b-1}(1-s)^{b-1}(1-x+xs)^{-a}\,ds \\ &= \frac{\Gamma(c)\,(1-x)^{-a}}{\Gamma(b)\,\Gamma(c-b)} \int_0^1 s^{c-b-1}(1-s)^{b-1}\left(1-\frac{x}{x-1}\,s\right)^{-a} ds \\ &= (1-x)^{-a}\,F\left(a,c-b;c;\frac{x}{x-1}\right), \end{aligned}$$

provided that $c > b > 0$, $|x| < 1$, and $|x/(x-1)| < 1$. This identity is called *Pfaff's transformation*. After interchange of the first two parameters, another application of Pfaff's transformation gives

$$F\left(a,c-b;c;\frac{x}{x-1}\right) = F\left(c-b,a;c;\frac{x}{x-1}\right) = (1-x)^{c-b}F(c-b,c-a;c;x)$$

if $c > a > 0$, with the conclusion that

$$F(a,b;c;x) = (1-x)^{c-a-b}F(c-a,c-b;c;x)\,.$$

The last identity was known to Euler.

The hypergeometric series (23) need not converge at the point $x = 1$. However, it can be shown to converge there if $c > a + b$. For a proof, recall the Gauss product formula (Section 9.6)

$$\Gamma(a) = \lim_{k\to\infty} \frac{k!\,k^a}{a(a+1)\cdots(a+k)}\,, \qquad a > 0\,,$$

which implies that

$$(a)_k \sim \frac{k!\,k^{a-1}}{\Gamma(a)}\,, \qquad k \to \infty\,.$$

This asymptotic relation fails if $a < 0$, but it can be applied to show that $(a)_k \sim M\,k!\,k^{a-1}$ for some constant M depending only on a. Therefore,

$$\frac{(a)_k(b)_k}{k!(c)_k} \sim M\,k^{a+b-c-1}\,, \qquad k \to \infty\,, \tag{24}$$

where the constant M depends only on a, b, and c. In particular, the series (23) converges for $x = 1$ if $c > a + b$.

The actual value of $F(a,b;c;1)$, when the series converges, turns out to be important. Gauss evaluated the sum in his paper of 1812.

Gauss summation formula. *If $c > a + b$, then*

$$F(a,b;c;1) = \sum_{k=0}^{\infty} \frac{(a)_k\,(b)_k}{k!\,(c)_k} = \frac{\Gamma(c)\,\Gamma(c-a-b)}{\Gamma(c-a)\,\Gamma(c-b)}\,.$$

We will give two proofs. The first uses Euler's integral representation and requires the additional assumption that $c > b > 0$. Euler's representation is easily seen to give

$$\begin{aligned}\lim_{x\to 1-} F(a,b;c;x) &= \frac{\Gamma(c)}{\Gamma(b)\,\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-a-b-1}\,dt \\ &= \frac{\Gamma(c)}{\Gamma(b)\,\Gamma(c-b)}\,B(b,c-a-b) = \frac{\Gamma(c)\,\Gamma(c-a-b)}{\Gamma(c-a)\,\Gamma(c-b)}\,.\end{aligned}$$

Because the condition $c > a + b$ ensures that the series converges at $x = 1$, we can infer from Abel's theorem that its sum agrees with its Abel limit, so the Gauss summation formula is proved for $c > b > 0$.

The second proof shows that the condition $c > b > 0$ is superfluous. Assuming only that $c > a + b$, we begin by establishing the identity

$$F(a,b;c;1) = \frac{(c-a)(c-b)}{c(c-a-b)}\,F(a,b;c+1;1)\,. \tag{25}$$

For a proof of (25), denote the terms of the two series by

$$A_k = \frac{(a)_k\,(b)_k}{k!\,(c)_k} \qquad \text{and} \qquad B_k = \frac{(a)_k\,(b)_k}{k!\,(c+1)_k}\,,$$

respectively. Then

$$A_{k+1} = \frac{(a+k)(b+k)}{(k+1)(c+k)}\,A_k \qquad \text{and} \qquad B_k = \frac{c}{c+k}\,A_k\,,$$

and a straightforward calculation confirms that

$$c(c-a-b)A_k = (c-a)(c-b)B_k + c\big[kA_k - (k+1)A_{k+1}\big]\,.$$

Consequently,

$$c(c-a-b)\sum_{k=0}^{n} A_k = (c-a)(c-b)\sum_{k=0}^{n} B_k - c(n+1)A_{n+1}\,,$$

since the last sum telescopes. But because $c > a+b$, the asymptotic relation (24) shows that

$$(n+1)A_{n+1} \sim M\,n^{a+b-c} \to 0 \qquad \text{as} \quad n \to \infty\,,$$

and so (25) follows. Iterate the relation (25) to obtain

$$F(a,b;c;1) = \frac{(c-a)_n\,(c-b)_n}{(c)_n\,(c-a-b)_n}\,F(a,b;c+n;1)\,,$$

or

(26)
$$\frac{\Gamma(c-a)\,\Gamma(c-b)}{\Gamma(c)\,\Gamma(c-a-b)}\,F(a,b;c;1) = \frac{\Gamma(c-a+n)\,\Gamma(c-b+n)}{\Gamma(c+n)\,\Gamma(c-a-b+n)}\,F(a,b;c+n;1)$$

for $n = 1, 2, \ldots$. Now pass to the limit as $n \to \infty$. An application of Stirling's formula for the gamma function (*cf.* Chapter 9, Exercise 22) shows that

$$\lim_{n\to\infty} \frac{\Gamma(c-a+n)\,\Gamma(c-b+n)}{\Gamma(c+n)\,\Gamma(c-a-b+n)} = 1\,.$$

We leave it as an exercise to show that also $F(a,b;c+n;1) \to 1$ as $n \to \infty$. Then the right-hand side of (26) tends to 1 as $n \to \infty$, and the Gauss summation formula is proved.

As a final example, we derive the identity

$$F(a,b;a+b+\tfrac{1}{2};4x(1-x)) = F(2a,2b;a+b+\tfrac{1}{2};x)\,, \qquad \tfrac{1}{2}(1-\sqrt{2}) < x < 1\,,$$

where $a+b+\frac{1}{2}$ is not a negative integer or zero. The proof uses the fact that

$$y(t) = F(a,b;a+b+\tfrac{1}{2};t)$$

satisfies the hypergeometric differential equation

$$t(1-t)y'' + [a+b+\tfrac{1}{2} - (a+b+1)t]y' - ab\,y = 0\,.$$

With the substitution $t = 4x(1-x)$, a calculation then shows, in view of the relation $(1-2x)^2 = 1-t$, that the function

$$Y(x) = y(4x(1-x)) = F(a,b;a+b+\tfrac{1}{2};4x(1-x))$$

satisfies the hypergeometric equation

$$x(1-x)Y'' + [a+b+\tfrac{1}{2} - (2a+2b+1)x]Y' - 4ab\,Y = 0\,.$$

But since $c = a+b+\frac{1}{2}$ is not a negative integer or zero, the only power series solution, up to a constant factor, is $Y(x) = F(2a,2b;a+b+\frac{1}{2};x)$. This gives the desired result. The restriction to the interval $\frac{1}{2}(1-\sqrt{2}) < x < 1$ guarantees that $|4x(1-x)| < 1$ except at $x = \frac{1}{2}$, where the identity is proved by a continuity argument. Note that $a+b+\frac{1}{2} > a+b$, so the hypergeometric series $F(a,b;a+b+\frac{1}{2};t)$ actually converges for $t = 1$.

13.6. Oscillation and comparison theorems

When confronted with a differential equation, it is customary to search for explicit formulas for the solutions. But often the main interest lies in qualitative properties, and explicit formulas may be too complicated to shed light on the behavior of solutions. A case in point is Bessel's function

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 4^2} - \frac{x^6}{2^2 4^2 6^2} + \dots ,$$

which is a solution to Bessel's equation $xy'' + y' + xy = 0$. Numerical calculations lead to the graph shown in Figure 1. The graph suggests that $J_0(x)$ has infinitely many zeros, rather evenly spaced, and it oscillates with decreasing amplitude, tending to zero as $x \to \infty$. However, the power series representation is of little use in confirming such conjectures. On the other hand, we will see that the Sturm comparison theorem and related results allow us to verify all of this information directly from the fact that $J_0(x)$ satisfies Bessel's differential equation.

In fact, the method has broad applications. Many classical problems of mathematical physics lead to linear differential equations of second order. It was Sturm's great insight that such an equation need not be solved explicitly in order to determine the qualitative behavior of its solutions.

Sturm's analysis is clever but entirely elementary. He developed the core of the theory in a famous paper [**12**] of 1836, but his ideas have continued to be refined and extended. Although we propose to discuss only the most basic of Sturm's results, they will reveal some remarkably precise information about Bessel functions.

Consider a linear differential equation with variable coefficients, expressed in "self-adjoint" form

$$(py')' + qy = 0 , \tag{27}$$

where $q(x)$ is continuous, $p(x)$ is continuously differentiable, and $p(x) > 0$ over an interval $[a, b]$. Observe first that each solution of (27) that is not identically zero can have at most a finite number of zeros in the closed bounded interval $[a, b]$. Indeed, if there were infinitely many zeros they would have to cluster at some point in $[a, b]$, and so by Rolle's theorem, the zeros of the derivative would also cluster at the same point. The function would then vanish together with its derivative at this point, and so would vanish identically in $[a, b]$, since it is a solution of a differential equation (27) of second order. Thus each solution of (27) has at most a finite number of zeros in the interval, and all zeros are simple.

It is a familiar fact that the zeros of $\sin x$ and $\cos x$ are interlaced; they occur alternately as x increases. But these two functions are independent

solutions of the differential equation $y'' + y = 0$, a very special form of the equation (27). Sturm discovered that the interlacing phenomenon is actually quite general.

Sturm Separation Theorem. *Let y_1 and y_2 be independent solutions of the differential equation* (27)*, and let x_1 and x_2 be consecutive zeros of y_1 in $[a, b]$. Then y_2 vanishes exactly once in the interval (x_1, x_2).*

To say that x_1 and x_2 are *consecutive zeros* of y_1 means that $y_1(x_1) = y_1(x_2) = 0$ but $y_1(x) \neq 0$ for $x_1 < x < x_2$.

Proof of theorem. Because the two solutions y_1 and y_2 are independent, they cannot vanish simultaneously. Indeed, if both functions were to vanish at some point in $[a, b]$, then every solution would vanish at that point, which is certainly not true. Hence, $y_2(x_1) \neq 0$ and $y_2(x_2) \neq 0$. Suppose now that $y_2(x) \neq 0$ throughout the interval (x_1, x_2). Then the quotient $v = y_1/y_2$ is continuously differentiable in $[x_1, x_2]$ and $v(x_1) = v(x_2) = 0$. By Rolle's theorem, it follows that $v'(\xi) = 0$ at some intermediate point $\xi \in (x_1, x_2)$. But $v' = (y_2y_1' - y_1y_2')/y_2^2$, so this says that the Wronskian of y_1 and y_2 vanishes at the point ξ. This is impossible, however, since the two solutions are independent. Thus the supposition that $y_2(x) \neq 0$ has led to a contradiction, and we conclude that $y_2(x) = 0$ at some point in the interval (x_1, x_2). If y_2 had two zeros in that interval, we could apply the result just established, with the roles of y_1 and y_2 reversed, to see that $y_1(x) = 0$ somewhere between x_1 and x_2, contrary to the hypothesis that these two points are consecutive zeros of y_1. This concludes the proof. □

The next theorem compares zeros of solutions of different but closely related differential equations of the form (27).

Sturm Comparison Theorem. *Let y_1 and y_2 be solutions of the respective differential equations $(py')' + q_1y = 0$ and $(py')' + q_2y = 0$. Let x_1 and x_2 be consecutive zeros of y_1, and suppose that $q_1(x) \leq q_2(x)$ in $[x_1, x_2]$, with strict inequality throughout some subinterval. Then $y_2(x) = 0$ for some point x in the open interval (x_1, x_2).*

In particular, if $y_2(x_1) = 0$, then y_2 vanishes again before y_1 does; it "oscillates faster". The theorem is best illustrated by the two equations $y'' + n^2y = 0$ and $y'' + m^2y = 0$ with $m > n > 0$. Then the solutions are sinusoids with regularly spaced zeros at respective distances π/n and π/m apart.

Proof of theorem. If $y_2(x) \neq 0$ in (x_1, x_2), we may assume without loss of generality that $y_1(x) > 0$ and $y_2(x) > 0$ in (x_1, x_2). Then since $y_1(x_1) =$

$y_1(x_2) = 0$, it follows that $y_1'(x_1) > 0$ and $y_1'(x_2) < 0$; while $y_2(x_1) \geq 0$ and $y_2(x_2) \geq 0$. Observe now that because y_1 and y_2 satisfy their respective differential equations, we have the identity

$$\big[p(y_2y_1' - y_1y_2')\big]' = y_2(py_1')' - y_1(py_2')' = (q_2 - q_1)y_1y_2\,.$$

Integration gives

$$\Big[p(y_2y_1' - y_1y_2')\Big]_{x_1}^{x_2} = \int_{x_1}^{x_2} \big(q_2(x) - q_1(x)\big)y_1(x)y_2(x)\,dx > 0\,,$$

in view of the hypothesis that $q_1(x) \leq q_2(x)$ with strict inequality in some subinterval, and the assumptions that $y_1(x) > 0$ and $y_2(x) > 0$ in (x_1, x_2). On the other hand, since y_1 vanishes at both x_1 and x_2, the left-hand side reduces to

$$\Big[p\,y_2y_1'\Big]_{x_1}^{x_2} = p(x_2)y_2(x_2)y_1'(x_2) - p(x_1)y_2(x_1)y_1'(x_1) \leq 0\,,$$

by virtue of the standing assumption that $p(x) > 0$ and the supposed inequalities $y_2(x_1) \geq 0$, $y_2(x_2) \geq 0$, $y_1'(x_1) > 0$, and $y_1'(x_2) < 0$. This contradicts the previous conclusion that the right-hand side is positive, thereby invalidating our initial supposition that $y_2(x) \neq 0$ throughout the interval (x_1, x_2). Consequently, $y_2(x) = 0$ for some point x in (x_1, x_2), as the theorem asserts. □

We now apply the Sturm comparison theorem to study the behavior of the Bessel function J_0, which is a solution of the differential equation $(xy')' + xy = 0$. Since J_0 is an even function, it is enough to consider $x > 0$. We are going to prove that J_0 has an infinite number of zeros, as suggested by its graph in Figure 1.

In order to study the zeros of J_0, and in fact to prove their existence, we first transform the Bessel equation through the substitution $y(x) = x^\alpha\,u(x)$, where α is a real parameter to be specified shortly. A simple calculation shows that

$$xy'' + y' + xy = x^{\alpha+1}\,u'' + (2\alpha+1)x^\alpha\,u' + \big(\alpha^2x^{\alpha-1} + x^{\alpha+1}\big)\,u = 0\,.$$

Choosing $\alpha = -\frac{1}{2}$ to eliminate the term in u', we find that the differential equation reduces to

$$u'' + \left(1 + \frac{1}{4x^2}\right)u = 0\,. \tag{28}$$

As $x \to \infty$, the equation (28) approaches in some sense the familiar equation $u'' + u = 0$, whose solutions are sinusoids. Thus we can expect the zeros of

u, which are the same as those of $J_0(x) = x^{-1/2}u(x)$, to behave for large x like those of $\sin x$ and $\cos x$.

These notions are made precise by appeal to the Sturm comparison theorem. Since $u(x)$ satisfies the differential equation $u'' + qu = 0$ with

$$q(x) = 1 + 1/(4x^2) > 1 \qquad \text{for all } \ x > 0\,,$$

we can use Sturm's theorem to compare the zeros of $u(x)$ with those of $\cos x$, for instance, which satisfies the differential equation $y'' + y = 0$. The theorem guarantees that $u(x)$ has at least one zero between each pair of zeros of $\cos x$, which lie apart at distance π. We conclude in particular that $J_0(x)$ has infinitely many positive zeros λ_n, which can be ordered by $0 < \lambda_1 < \lambda_2 < \dots$ with $\lambda_n \to \infty$ as $n \to \infty$, since the zeros can have no finite cluster point.

In fact, we can say more. Let $y_1 = \cos(x - \delta)$ be a solution to the differential equation $y'' + y = 0$ which vanishes together with J_0 at some point λ_n. Then the next zero of y_1 will occur at $\lambda_n + \pi$. But by the comparison theorem, the function $y_2(x) = u(x) = \sqrt{x}J_0(x)$ must vanish between successive zeros of y_1. This shows that $\lambda_n < \lambda_{n+1} < \lambda_n + \pi$, or $0 < \lambda_{n+1} - \lambda_n < \pi$ for $n = 1, 2, \dots$.

But we can say even more. By reversing the roles of the two differential equations, we can apply the Sturm comparison theorem to prove that $\lambda_{n+1} - \lambda_n \to \pi$ as $n \to \infty$. For this it will suffice to show that for each $\varepsilon > 0$ there is a number N such that $\lambda_{n+1} - \lambda_n > \pi/(1+\varepsilon)$ for all $n \geq N$. Given $\varepsilon > 0$, choose N so large that

$$1 + 1/(4\lambda_n^2) < (1+\varepsilon)^2 \qquad \text{for all } \ n \geq N\,.$$

This is possible because $\lambda_n \to \infty$, and it follows that

$$1 + 1/(4x^2) < (1+\varepsilon)^2 \qquad \text{for all } \ x \geq \lambda_n \ \text{ if } \ n \geq N\,.$$

Now define $y_2(x) = \cos((1+\varepsilon)x - \delta)$, choosing δ so that $y_2(\lambda_n) = 0$. Then since y_2 satisfies the equation $y'' + (1+\varepsilon)^2 y = 0$, the Sturm comparison theorem ensures that $y_2(x)$ will vanish again before $u(x) = \sqrt{x}J_0(x)$ does. But the zeros of y_2 are spaced at distance $\pi/(1+\varepsilon)$ apart, so this says that

$$\lambda_n < \lambda_n + \pi/(1+\varepsilon) < \lambda_{n+1}\,, \qquad \text{or} \quad \pi/(1+\varepsilon) < \lambda_{n+1} - \lambda_n$$

for all $n \geq N$. Because $\varepsilon > 0$ was chosen arbitrarily, and we have already found that $\lambda_{n+1} - \lambda_n < \pi$ for all n, this implies that $\lambda_{n+1} - \lambda_n \to \pi$ as $n \to \infty$.

Listed below are some approximate numerical values of the zeros λ_n.

$$\begin{array}{lll} \lambda_1 = 2.4048256 & \lambda_2 = 5.5200781 & \lambda_3 = 8.6537279 \\ \lambda_4 = 11.7915344 & \lambda_5 = 14.9309177 & \lambda_6 = 18.0710640 \\ \lambda_7 = 21.2116366 & \lambda_8 = 24.3524715 & \lambda_9 = 27.4934791 \\ \lambda_{10} = 30.6346065 & \lambda_{11} = 33.7758202 & \lambda_{12} = 36.9170984 \end{array}$$

Note that $\lambda_{12} - \lambda_{11} \approx 3.14128$, which is already very close to $\pi \approx 3.14159$. The data suggest that $\lambda_{n+1} - \lambda_n$ increases to π, and this can be verified analytically. (See Exercises 28 and 29.)

Charles-François Sturm (1803–1855) was born in Geneva but worked mainly in Paris, where eventually he secured a professorship. He made important contributions to several areas of science. In 1826 he produced a prize essay on compressible fluids. In 1829 he found a basic algorithm for locating the real roots of a polynomial, a method which Cauchy then extended to complex roots. His seminal work on differential equations led to a fruitful collaboration with Joseph Liouville (1809–1882). Together they produced the celebrated Sturm–Liouville theory of eigenfunction expansions associated with boundary-value problems for linear differential equations of second order. The article by Loria [**10**] gives a more detailed account of Sturm's wide-ranging scientific career.

13.7. Refinements of Sturm's theory

The proof of the Sturm comparison theorem, with slight modification, actually leads to a stronger result. Suppose again that the functions $q_1(x)$ and $q_2(x)$ are continuous on the interval (a, b), and that $p(x)$ is continuously differentiable there with $p(x) > 0$. Under these assumptions, we can state a simple variant of the Sturm comparison theorem.

Comparison Theorem. *Let y_1 and y_2 be solutions of the respective differential equations $(py')' + q_1 y = 0$ and $(py')' + q_2 y = 0$, where $q_1(x) < q_2(x)$ in (a, b). Suppose that $y_1(x_1) = y_2(x_1) \geq 0$ and $y_1'(x_1) = y_2'(x_1) > 0$ at some point $x_1 \in (a, b)$. Then $y_2(x) < y_1(x)$ for all x in each interval $(x_1, x_2) \subset (x_1, b)$ where $y_2(x) \neq 0$.*

Proof. As in the proof of the Sturm comparison theorem, we can write

$$\Big[p(y_2 y_1' - y_1 y_2')\Big]_{x_1}^{x} = \int_{x_1}^{x} \big[q_2(t) - q_1(t)\big] y_1(t) y_2(t)\, dt\,. \tag{29}$$

The hypotheses guarantee that $y_2(t) > 0$ in (x_1, x_2), and so it follows from the Sturm comparison theorem that $y_1(t) > 0$ as well. Thus, in view of the

assumption that $p(x) > 0$, the relation (29) implies that

$$y_2(x)y_1'(x) - y_1(x)y_2'(x) > 0 \qquad \text{for all } \ x \in (x_1, x_2)\,.$$

Therefore, the function y_1/y_2 has positive derivative in this interval. But $y_1(x)/y_2(x) \to 1$ as $x \to x_1+$ (apply l'Hôpital's rule if $y_1(x_1) = y_2(x_1) = 0$), so we conclude that $y_1(x)/y_2(x) > 1$ in (x_1, x_2), as claimed. □

We turn next to a different refinement of Sturm's basic theory that will provide further information about Bessel functions. Using the Sturm comparison theorem, we have already confirmed that the Bessel function $J_0(x)$ has infinitely many zeros with essentially constant spacing, as suggested by its graph in Figure 1. Other features of the graph suggest that the Bessel function oscillates with decreasing amplitude, and that $J_0(x) \to 0$ as $x \to \infty$. To verify these latter properties, we now develop an elementary result known as the Sonin–Pólya oscillation theorem.

Sonin–Pólya Oscillation Theorem. *Let $y = y(x)$ be a solution of the differential equation $(py')' + qy = 0$, where $p(x)$ and $q(x)$ are continuously differentiable on an interval $[a, b]$. Suppose that $p(x) > 0$, $q(x) \neq 0$, and $(pq)'(x) > 0$ on (a, b). Then the successive local maxima (if any) of $|y(x)|$ form a decreasing sequence.*

Proof. Consider the function

$$\phi(x) = y(x)^2 + \frac{p(x)\,y'(x)^2}{q(x)}\,.$$

Writing $\phi = y^2 + (py')^2(pq)^{-1}$ and noting that $(py')' = -qy$, we find

$$\phi' = 2yy' + \frac{2py'}{pq}(py')' - \frac{(pq)'}{(pq)^2}(py')^2 = -\left(\frac{y'}{q}\right)^2 (pq)'\,.$$

Since $(pq)'(x) > 0$ on (a, b) by hypothesis, this shows that $\phi(x)$ is decreasing on the interval $[a, b]$. In particular, if ξ_n are successive local maxima of $|y(x)|$, so that $a < \xi_1 < \xi_2 < \cdots < b$, then $\phi(\xi_1) > \phi(\xi_2) > \ldots$. But $\phi(\xi_n) = y(\xi_n)^2$, because $y'(\xi_n) = 0$ at each local maximum of $|y(x)|$. Thus $|y(\xi_1)| > |y(\xi_2)| > \ldots$, as the theorem asserts. Note that the inequalities extend to the endpoints a or b if $y'(a) = 0$ or $y'(b) = 0$, and if $p(x)/q(x)$ tends to a finite limit as $x \to a+$ or $x \to b-$, respectively. □

Applying the Sonin–Pólya theorem to Bessel's equation $(xy')' + xy = 0$, where $p(x) = q(x) = x$, we see at once that $J_0(x)$ oscillates with strictly decreasing amplitude. More precisely, if $0 < \xi_1 < \xi_2 < \ldots$ are the successive

local maxima of $|J_0(x)|$ in $(0,\infty)$, then $|J_0(\xi_1)| > |J_0(\xi_2)| > \dots$. In fact, it is easy to see from the differential equation that the points ξ_n interlace with the zeros λ_n of J_0, since $J_0''(\xi_n) = -J_0(\xi_n) \neq 0$ at each of these critical points, so that J_0 has a local maximum at ξ_n if $J_0(\xi_n) > 0$ and a local minimum if $J_0(\xi_n) < 0$, as the graph in Figure 1 suggests. Recall that J_0 cannot vanish at a critical point because it is a solution of a linear differential equation of second order. Finally, we note that J_0 has a local maximum the origin, since $J_0'(0) = 0$ and $J_0''(0) = -\frac{1}{2} < 0$. Also $J_0(0) = 1 > 0$, and the Sonin–Pólya theorem extends to show that $J_0(0) > |J_0(\xi_1)|$ despite the fact that the function $p(x) = x$ vanishes at the origin, because $p(x)/q(x) \equiv 1$. In particular, it follows that $|J_0(x)| \leq 1$ for all $x \geq 0$.

A similar elementary argument shows that $J_0(x) \to 0$ as $x \to \infty$. In fact, it may be expected that $J_0(x)$ decays like $1/\sqrt{x}$, because as we have seen, the substitution $u(x) = \sqrt{x}\, y(x)$ transforms Bessel's equation to $u'' + \left(1 + \frac{1}{4x^2}\right) u = 0$, whose solutions behave like sinusoids for large x. To make these notions precise, write the transformed equation as

$$u'' + Qu = 0\,, \qquad \text{where} \quad Q(x) = 1 + \frac{1}{4x^2}\,,$$

and consider the function

$$\psi(x) = u'(x)^2 + Q(x)u(x)^2\,.$$

A simple calculation shows that $\psi'(x) = Q'(x)u(x)^2 \leq 0$, which implies that $\psi(x)$ is nonincreasing over the interval $(0,\infty)$. In particular, if ν_n are the successive local maxima of $|u(x)|$, with $0 < \nu_1 < \nu_2 < \dots$, then

$$Q(\nu_n)u(\nu_n)^2 = \psi(\nu_n) \leq \psi(\nu_1) = Q(\nu_1)u(\nu_1)^2\,, \qquad n = 1, 2, \dots\,,$$

since $u'(\nu_n) = 0$. It follows that $u(x)$ remains bounded as $x \to \infty$, so that $J_0(x) = O(1/\sqrt{x})$ as $x \to \infty$. In particular, $J_0(x) \to 0$ as $x \to \infty$.

A more general form of the preceding argument appears in an article by Duren and Muldoon [**5**]. Our presentation of the Sonin-Pólya theorem is adapted from the book of Kreider, Kuller, and Ostberg [**8**]. For further refinements of the Sturm theory, the reader is referred to Birkhoff and Rota [**2**] and the classic book by Ince [**6**], and to the article by Laforgia and Muldoon [**9**], which gives additional references.

Exercises

1. For the initial-value problem $y' = f(x,y)$, $y(x_0) = y_0$, verify the error estimate

$$|\varphi(x) - \varphi_n(x)| \leq \frac{M\, e^{C\delta} C^n \delta^{n+1}}{(n+1)!}\,, \qquad |x - x_0| < \delta\,,$$

as stated after the proof of the existence and uniqueness theorem in Section 13.1.

2. Carry out the details in the proof of existence and uniqueness of a solution to the initial-value problem (6) for first-order linear systems of differential equations.

3. Show that the functions $f(x) = x$ and $g(x) = x^2$ are linearly independent over the interval $(-1, 1)$, yet their Wronskian vanishes at the origin.

4. Verify the Wronskian relation (11) for solutions to a differential equation (7) of order n.

Hint. The derivative of an $n \times n$ determinant is equal to the sum of n determinants, each involving the derivatives of one row. Loosely speaking, the determinant can be differentiated row by row.

5. Use the method of reduction of order to derive a second solution, independent of $J_0(x)$, for Bessel's equation $xy'' + y' + xy = 0$. Set $y = J_0 v$ and derive a linear differential equation of first order for v. Solve the latter equation to obtain

$$Y_0(x) = J_0(x) \int \frac{1}{x J_0(x)^2}\, dx = J_0(x) \log x + \tfrac{1}{4}x^2 - \tfrac{3}{128}x^4 + \dots .$$

6. (*a*) Find the general power series solution $y = \sum_{k=0}^{\infty} c_k x^k$ of Legendre's equation

$$(1 - x^2)y'' - 2xy' + \lambda y = 0 .$$

Show that the series is convergent for $|x| < 1$.

(*b*) Show that the Legendre equation has a (nontrivial) polynomial solution if and only if $\lambda = n(n + 1)$ for some integer $n = 0, 1, 2, \dots$. When $\lambda = n(n + 1)$, show that the polynomial solution is unique up to a constant factor and has degree n. With standard normalization (choice of constant factor), this solution is called the nth *Legendre polynomial* and is denoted by $P_n(x)$. The standard normalization is to require that $P_n(1) = 1$. Show that $P_n(-x) = (-1)^n P_n(x)$, or in other words that the polynomial is an even function if n is even, odd if n is odd.

(c) Verify that

$$\frac{d}{dx}\left\{(1 - x^2)P_n'(x)\right\} = -n(n + 1)P_n(x) ,$$

and use this relation to establish the orthogonality property

$$\int_{-1}^{1} P_n(x)\, P_m(x)\, dx = 0 , \qquad n \neq m .$$

Note. Legendre's differential equation and Legendre polynomial expansions (orthogonal series expansions analogous to Fourier series) occur naturally when solving Laplace's equation in a spherical region. The book by Jackson [**7**] is a good reference.

7. (*a*) Show that the Legendre polynomial $P_n(x)$ is orthogonal to every polynomial $Q(x)$ of degree lower than n:

$$\int_{-1}^{1} P_n(x)\, Q(x)\, dx = 0\,.$$

(*b*) Show that every zero of $P_n(x)$ in the interval $(-1, 1)$ is simple.

Hint. Remember that $P_n(x)$ satisfies a differential equation.

(*c*) Prove that the Legendre polynomial $P_n(x)$ has exactly n (distinct) zeros in the interval $(-1, 1)$.

Hint. Suppose that $P_n(x) = 0$ in $(-1, 1)$ only at points $x_1, x_2, \ldots, x_k$ for $k < n$, and apply the result of (*a*) to the polynomial $Q(x) = (x - x_1)(x - x_2)\cdots(x - x_k)$.

8. (*a*) For any constant $\lambda > 0$, show that

$$\int_0^1 x\, J_0(\lambda x)^2\, dx = \tfrac{1}{2}\big\{ J_0(\lambda)^2 + J_0'(\lambda)^2 \big\}\,.$$

In particular, if $\lambda_n > 0$ is a zero of $J_0(x)$, conclude that

$$\int_0^1 x\, J_0(\lambda_n x)^2\, dx = \tfrac{1}{2}\, J_0'(\lambda_n)^2\,.$$

Hint. Transform the first integral by substituting $t = \lambda x$. Then show that $y = J_0(t)$ satisfies

$$2ty^2 = \frac{d}{dt}\big\{ t^2(y^2 + y'^2) \big\}\,.$$

(*b*) Generalize the result by proving that

$$\int_0^1 x\, J_n(\lambda x)^2\, dx = \tfrac{1}{2}\big\{ J_n'(\lambda)^2 + \big(1 - \tfrac{n^2}{\lambda^2}\big)\, J_n(\lambda)^2 \big\}$$

for $\lambda > 0$ and $n = 1, 2, \ldots$.

9. If λ and μ are any distinct positive zeros of $J_n(x)$, prove the orthogonality relation

$$\int_0^1 x\, J_n(\lambda x) J_n(\mu x)\, dx = 0\,, \qquad n = 1, 2, \ldots.$$

10. (*a*) For $n = 1, 2, \ldots$, show that Bessel's equation $x^2y'' + xy' + (x^2 - n^2)y = 0$ has a regular singular point at the origin with indicial equation $\alpha^2 - n^2 = 0$.

(*b*) Derive the series expansion (18) for $J_n(x)$ as a solution to Bessel's equation, and show that it is the only solution, up to constant multiples, that is bounded near the origin.

11. (*a*) Show that $J_0'(x) = -J_1(x)$.

(*b*) For $n = 1, 2, \ldots$, show that $\frac{d}{dx}\{x^{-n}J_n(x)\} = -x^{-n}J_{n+1}(x)$.

(*c*) For $n = 1, 2, \ldots$, show that $J_n'(x) = \frac{1}{2}\{J_{n-1}(x) - J_{n+1}(x)\}$.

12. Prove the recurrence relaton

$$J_{n+1}(x) = \frac{2n}{x}\,J_n(x) - J_{n-1}(x)\,, \qquad n = 1, 2, \ldots\,.$$

13. (*a*) Verify the integral representation

$$J_0(x) = \frac{1}{\pi}\int_0^\pi \cos(x\cos\theta)\,d\theta\,.$$

by showing that the integral satisfies Bessel's differential equation. Justify the process of differentiation under the integral sign.

(*b*) Verify the same integral representation for $J_0(x)$ by introducing the Taylor expansion

$$\cos(x\cos\theta) = 1 - \frac{(x\cos\theta)^2}{2!} + \frac{(x\cos\theta)^4}{4!} - \ldots$$

and integrating term by term. Justify the term-by-term integration and apply the formula in Chapter 2, Exercise 15. (See also Section 2.5.)

(*c*) Verify the integral representation (19) for $J_n(x)$, where $n = 1, 2, \ldots$.

14. Verify Bessel's formula (20) for $J_n(x)$, where $n = 1, 2, \ldots$.

Suggestion. If $I_n(x)$ denotes the integral, show by differentiation under the integral sign that

$$x^2I_n''(x)+xI_n'(x)+(x^2-n^2)I_n(x) = -\int_0^\pi \frac{d}{d\theta}\big\{(n+x\cos\theta)\sin(n\theta-x\sin\theta)\big\}\,d\theta\,.$$

15. Write

$$e^{\frac{x}{2}(t-\frac{1}{t})} = e^{\frac{xt}{2}}e^{-\frac{x}{2t}}$$

and multiply the power series expansions of the two exponential functions to recover the generating relation (21) for Bessel functions.

16. Differentiate the generating relation (21) term by term to obtain proofs of the identities

$$J_n'(x) = \tfrac{1}{2}\big\{J_{n-1}(x) - J_{n+1}(x)\big\} \quad \text{and} \quad n\,J_n(x) = \frac{x}{2}\big\{J_{n-1}(x) + J_{n+1}(x)\big\}\,,$$

where $n = 1, 2, \ldots$. Justify the term-by-term differentiation.

17. Show that

$$y = J_{1/2}(x) = \sqrt{\frac{2}{\pi x}}\,\sin x \qquad \text{and} \qquad y = J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}}\,\cos x$$

are solutions of Bessel's equation $x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0$ for $x > 0$.

Suggestion. First transform the differential equation by the substitution $u = \sqrt{x}\,y$.

18. The hypergeometric function $F(a, b; c; x)$ is a solution of the hypergeometric differential equation (22) if c is not a negative integer or zero. If c is not an integer, show that

$$x^{1-c}\,F(a - c + 1, b - c + 1; 2 - c; x)$$

is a second solution, independent of $F(a, b; c; x)$, on the interval $0 < x < 1$.

19. Show that the Legendre polynomials are hypergeometric functions

$$P_n(x) = F\big(-n, n + 1; 1; \tfrac{1-x}{2}\big)\,.$$

Recall that $P_n(x)$ is the polynomial solution of Legendre's differential equation with parameter $\lambda = n(n + 1)$, normalized by $P_n(1) = 1$. (See Exercise 6.)

20. Verify the relations

$$(a) \qquad \log\frac{1+x}{1-x} = 2x\,F\left(\tfrac{1}{2}, 1; \tfrac{3}{2}; x^2\right)\,,$$

$$(b) \qquad (1+x)^{-2a} + (1-x)^{-2a} = 2\,F\left(a, a + \tfrac{1}{2}; \tfrac{1}{2}; x^2\right)\,,$$

$$(c) \qquad \left\{1 + \sqrt{1-x}\,\right\}^{-2a} = 2^{-2a}\,F\left(a, a + \tfrac{1}{2}; 2a + 1; x\right)\,.$$

21. Show that

$$(a) \qquad \frac{d}{dx}\left\{F(a, b; c; x)\right\} = \frac{ab}{c}\,F(a + 1, b + 1; c + 1; x)\,,$$

$$(b) \qquad \frac{d}{dx}\left\{x^a\,F(a, b; c; x)\right\} = ax^{a-1}\,F(a + 1, b; c; x)\,.$$

22. Suppose that $a + b + 1 = 2c$ and let $y(x)$ be a solution of the hypergeometric differential equation (22) on the interval $0 < x < 1$. Show that the function $u(x) = y(1 - x)$ is also a solution.

23. Prove that $F(a, b; c + n; 1) \to 1$ as $n \to \infty$.

24. Obtain the sums

$$(a)\quad F(2a,2b;a+b+\tfrac{1}{2};\tfrac{1}{2}) = \frac{\Gamma(a+b+\frac{1}{2})\,\Gamma(\frac{1}{2})}{\Gamma(a+\frac{1}{2})\,\Gamma(b+\frac{1}{2})}\,, \qquad a+b+\tfrac{1}{2}\neq 0,-1,-2,\ldots\,,$$

$$(b)\quad F(-n,b;c;1) = \frac{(c-b)_n}{(c)_n}\,, \qquad n=1,2,\ldots\,; \quad c\neq 0,-1,-2,\ldots.$$

25. Show that $J_1(x)$, the Bessel function given by the series (18) with $n=1$, has infinitely many positive zeros μ_k, arranged so that $0<\mu_1<\mu_2<\ldots$, and $\mu_{k+1}-\mu_k\to\pi$ as $k\to\infty$. (This property is suggested by the graph in Figure 3.)

26. Show that every solution of Airy's equation $y''+xy=0$ has infinitely many zeros in the interval $(0,\infty)$.

27. (*a*) Let $q(x)$ be continuous and $q(x)<0$ in an interval (a,b). Use the Sturm comparison theorem to show that no nontrivial solution of the differential equation $y''+qy=0$ can have more than one zero in (a,b).

(*b*) Let $y=y(x)$ be a nontrivial solution of the differential equation $y''+p(x)y'+q(x)y=0$ on an interval (a,b) where the coefficient functions $p(x)$ and $q(x)$ are continuous and $q(x)<0$. Show that $y(x)$ has at most one zero on (a,b).

Suggestion. Apply Rolle's theorem.

28. Let $q(x)$ be continuous and $q(x)>0$ in an interval (a,b). Suppose that some nontrivial solution of the differential equation $y''+qy=0$ has zeros at the points x_k, arranged so that $a<x_1<x_2<\cdots<b$. If $q(x)$ is strictly decreasing, use the Sturm comparison theorem to show that

$$x_k-x_{k-1}<x_{k+1}-x_k\,, \qquad k=2,3,\ldots\,.$$

If $q(x)$ is increasing, show that $x_k-x_{k-1}>x_{k+1}-x_k$. This result is due to Sturm and is sometimes called the *Sturm convexity theorem.*

Suggestion. Consider $y(x-x_k+x_{k-1})$ in the interval (x_k,x_{k+1}).

29. (*a*) Let $y=y(x)$ be a nontrivial solution of Bessel's equation $x^2y''+xy'+(x^2-n^2)y=0$. Show that $u(x)=\sqrt{x}\,y(x)$ satisfies the differential equation

$$u''+\left(1+\frac{1-4n^2}{4x^2}\right)u=0\,.$$

(*b*) Conclude that y has infinitely many positive zeros, whose spacing tends to π. More precisely, if $0<x_1<x_2<\ldots$ are the consecutive positive zeros of y, then $x_{k+1}-x_k\to\pi$ as $k\to\infty$.

(*c*) Show further that $x_{k+1} - x_k$ increases to π when $0 \le n < \frac{1}{2}$ and decreases to π when $\frac{1}{2} < n < \infty$.

30. Generalizing the device used to study of zeros of the Bessel function J_0, show that for some choice of function $\varphi(x) > 0$ the transformation $y = \varphi u$ converts a differential equation $y'' + py' + qy = 0$ to an equation of the form $u'' + Qu = 0$. This is known as *Liouville's normal form*. Note that the zeros of $y(x)$ are the same as those of $u(x)$.

31. Show that all real zeros of a hypergeometric polynomial $F(-n, b; c; x)$ are simple, except possibly for zeros at $x = 0$ or 1. Assume, as usual, that the parameter c is not a negative integer or zero.

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Chapter 14
Elliptic Integrals

The study of elliptic integrals dates back at least to the early 18th century, when they arose in connection with problems of mechanics and classical geometry. They comprise a large class of nonelementary integrals whose calculation reduces to standard forms that were tabulated at an early stage. The standard elliptic integrals enjoy some nice transformation properties and have a close, totally unexpected connection with the arithmetic–geometric mean. They also satisfy the Legendre relation, an elegant identity that provides an effective method for numerical calculation of π. These topics will be the focus of this final chapter.

14.1. Standard forms

In the early days of calculus, Leibniz and others embarked on a program to calculate the integrals of all "elementary" functions, those functions expressible as compositions of rational functions, logarithms and exponentials, trigonometric functions, etc. It soon became apparent, however, that an elementary function need not have an elementary integral. One such example arises in calculating the arclength of an ellipse. An example of greater historical importance appears in connection with *Bernoulli's lemniscate*, described in polar coordinates by the equation $r^2 = \cos 2\theta$. This familiar "figure-eight" curve, displayed in Figure 1, was known to ancient Greek geometers and was discussed in detail by Jacob Bernoulli in 1694. Its equation in rectangular coordinates is

$$(x^2 + y^2)^2 = x^2 - y^2 \,.$$

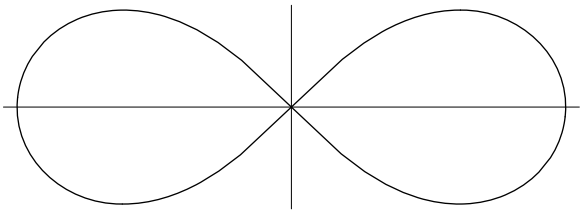


Figure 1. Bernoulli's lemniscate.

The curve can be described geometrically as the locus of points the product of whose distances from the two points $(\pm 1/\sqrt{2}, 0)$ has constant value $\frac{1}{2}$. The arclength of Bernoulli's lemniscate is found to be

$$4\int_0^1 \frac{dx}{\sqrt{1-x^4}}\,.$$

The formula is most easily derived from the polar-coordinate form

$$\begin{aligned} ds^2 = dr^2 + r^2 d\theta^2 &= \frac{\sin^2 2\theta}{r^2} d\theta^2 + r^2 d\theta^2 \\ &= \left(\frac{1-r^4}{r^2} + r^2\right) d\theta^2 = \frac{1}{r^2} d\theta^2\,, \end{aligned}$$

which gives the total arclength

$$4\int_0^{\pi/4} \frac{d\theta}{\sqrt{\cos 2\theta}} = 4\int_0^1 \frac{dr}{\sqrt{1-r^4}}\,.$$

The arclength integrals for the ellipse and lemniscate are examples of what are now called *elliptic integrals*. This term refers to integrals of the form

$$\int R(t, \sqrt{p(t)})\, dt\,,$$

where $R(x, y)$ is a rational function and $p(t)$ is a polynomial of degree 3 or 4 with no repeated factors. Elliptic integrals occur in many problems of mathematics and physics. Every elliptic integral can be reduced to a linear combination of three basic forms. In standard notation, the *complete elliptic integrals of first and second kind* are, respectively,

$$K = K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} = \int_0^1 \frac{dt}{\sqrt{1-t^2}\,\sqrt{1-k^2t^2}}$$

and

$$E = E(k) = \int_0^{\pi/2} \sqrt{1-k^2\sin^2\theta}\, d\theta = \int_0^1 \frac{\sqrt{1-k^2t^2}}{\sqrt{1-t^2}}\, dt\,,$$

for $0 < k < 1$. The parameter k is called the *modulus*, and $k' = \sqrt{1-k^2}$ is known as the *complementary modulus*. Symmetrically related to K and E are K' and E', defined respectively by $K'(k) = K(k')$ and $E'(k) = E(k')$. It is customary to use the notation $K'(k)$ and $E'(k)$ despite possible misinterpretation as derivatives. The corresponding indefinite integrals are called *incomplete elliptic integrals*. The *complete elliptic integral of third kind* is

$$\begin{aligned}\Pi(\nu,k) &= \int_0^{\pi/2} \frac{d\theta}{(1+\nu\sin^2\theta)\sqrt{1-k^2\sin^2\theta}} \\ &= \int_0^1 \frac{dt}{(1+\nu t^2)\sqrt{1-t^2}\,\sqrt{1-k^2t^2}}\,.\end{aligned}$$

It is less common and will not concern us here.

A simple calculation identifies the lemniscate integral as a complete elliptic integral of first kind. Specifically,

$$K(1/\sqrt{2}) = \sqrt{2}\int_0^{\pi/2} \frac{d\theta}{\sqrt{2-\sin^2\theta}} = \sqrt{2}\int_0^1 \frac{dx}{\sqrt{1-x^4}}\,,$$

with the substitution $x = \cos\theta$. The lemniscate integral can also be expressed in terms of the gamma function (see Exercise 2). The arclength of an ellipse is given by a complete elliptic integral of second kind (see Exercise 1).

14.2. Fagnano's duplication formula

The arclength function

$$L(x) = \int_0^x \frac{dt}{\sqrt{1-t^4}}\,, \qquad 0 \le x \le 1\,,$$

for the lemniscate is not elementary, but it has some remarkable properties. The duplication formula

$$2\,L(x) = L\left(\frac{2x\sqrt{1-x^4}}{1+x^4}\right)$$

was discovered in 1718 by Guilio Fagnano (1682–1766), an Italian magistrate (later a marquis) and amateur mathematician. The formula as stated is valid only for x sufficiently small. Analysis of the function

$$g(x) = \frac{2x\sqrt{1-x^4}}{1+x^4}$$

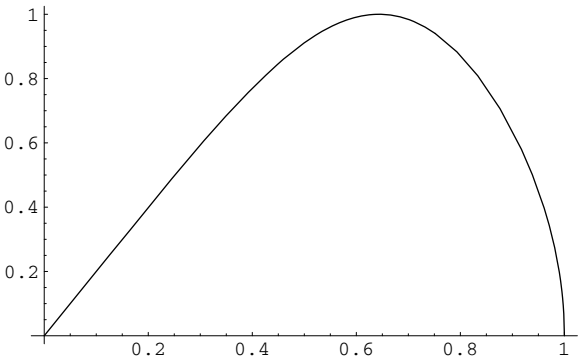


Figure 2. Fagnano's function $y = g(x)$.

shows that it increases to a maximum value 1 at

$$x = x_0 = (3 - 2\sqrt{2})^{1/4} = 0.64359\ldots ,$$

then decreases to 0 at $x = 1$. Its graph is shown in Figure 2.

Fagnano's derivation of the duplication formula begins with the substitution $t^2 = 2u^2/(1 + u^4)$, which gives

$$\sqrt{1 - t^4} = \frac{1 - u^4}{1 + u^4} \qquad \text{and} \qquad \frac{dt}{\sqrt{1 - t^4}} = \sqrt{2}\,\frac{du}{\sqrt{1 + u^4}}\,.$$

Next the substitution $u^2 = 2v^2/(1 - v^4)$ gives

$$\sqrt{1 + u^4} = \frac{1 + v^4}{1 - v^4} \qquad \text{and} \qquad \frac{du}{\sqrt{1 + u^4}} = \sqrt{2}\,\frac{dv}{\sqrt{1 - v^4}}\,.$$

A combination of the two transformations produces the relation $t = g(v)$ and leads to the duplication formula

$$2\int_0^x \frac{dv}{\sqrt{1 - v^4}} = \int_0^{g(x)} \frac{dt}{\sqrt{1 - t^4}}\,, \qquad 0 \le x \le x_0\,.$$

Fagnano's discovery languished in obscurity until 1751, when Euler was asked to examine a copy of Fagnano's collected works. Euler quickly saw that Fagnano's substitution device could be generalized to obtain the addition formula

$$L(x) + L(y) = L\left(\frac{x\sqrt{1 - y^4} + y\sqrt{1 - x^4}}{1 + x^2y^2}\right)$$

for small x and y. He then generalized the addition formula to a large class of elliptic integrals, including all incomplete elliptic integrals of first kind.

To understand the significance of the addition formula, it is instructive to consider an analogy with the arcsine integral

$$\sin^{-1} x = \int_0^x \frac{dt}{\sqrt{1-t^2}}\,, \qquad 0 \le x \le 1\,.$$

If $x = \sin\theta$ and $y = \sin\varphi$, the addition formula

$$\sin(\theta+\varphi) = \sin\theta\cos\varphi + \sin\varphi\cos\theta$$

for the sine function can be written in the form

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

for small x and y, which is similar to Euler's formula for the lemniscate integral.

Clearly, it would be awkward to base a study of trigonometry on the inverse trigonometric functions. It is the sine and cosine that have the elegant addition formulas, plus another key property: periodicity. Yet after Euler's work another half-century elapsed before anyone thought seriously of inverting the incomplete elliptic integrals and basing the theory on their inverse functions, now known as *elliptic functions*. Legendre took some preliminary steps in this direction. Gauss recorded the idea in his diary but did not publish it. In 1827, Abel published the first account of the theory of elliptic functions. He viewed them as functions of a complex variable and found them to be doubly periodic, with two independent complex periods. Around the same time, Carl Jacobi (1804–1851) made similar innovations, which he first published also in 1827. Abel died in 1829 at age 26, but Jacobi gradually developed the theory that revolutionized the subject. For many years, Legendre had labored to produce what was to have been a two-volume treatise [**7**] on the theory of elliptic integrals, complete with extensive numerical tables. The first volume was finally published in 1825 and the second in 1826, but Legendre soon realized that much of his work had been rendered obsolete by the brilliant insights of the young mathematicians Abel and Jacobi. He might well have reacted with dismay, but to his credit Legendre embraced the new ideas with great enthusiasm and presented them in coordination with his earlier work as extensive supplements comprising a third volume of the treatise.

14.3. The arithmetic–geometric mean

We now digress to discuss a classical algorithm for computing a quantity called the arithmetic–geometric mean of two positive numbers. This algorithm would appear to have little to do with elliptic integrals, but it turns out to be very relevant.

Given an arbitrary pair of real numbers a and b with $0 < b < a$, calculate their arithmetic and geometric means

$$a_1 = \tfrac{1}{2}(a+b) \qquad \text{and} \qquad b_1 = \sqrt{ab}\,.$$

Then $0 < b < b_1 < a_1 < a$. Now iterate the process, inductively defining

$$a_{n+1} = \tfrac{1}{2}(a_n + b_n) \qquad \text{and} \qquad b_{n+1} = \sqrt{a_n b_n}\,, \qquad n = 1, 2, \ldots .$$

Then

$$0 < b < b_1 < b_2 < \cdots < b_n < a_n < \cdots < a_2 < a_1 < a\,.$$

Each of the sequences $\{a_n\}$ and $\{b_n\}$ is monotonic and bounded, hence convergent. Let

$$\alpha = \lim_{n\to\infty} a_n\,, \qquad \beta = \lim_{n\to\infty} b_n\,.$$

Observe that

$$\alpha = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \tfrac{1}{2}(a_n + b_n) = \tfrac{1}{2}(\alpha + \beta)\,,$$

which shows that $\alpha = \beta$. The common limit is called the *arithmetic–geometric mean* (AGM) of a and b and is denoted by $M(a, b)$. It is easy to see that the AGM has the properties

$$M(a,b) = M\left(\tfrac{1}{2}(a+b), \sqrt{ab}\right)\,, \qquad M(a,b) = a\,M(1, b/a)\,.$$

The AGM algorithm was discovered by Lagrange around the year 1784. Gauss rediscovered it in 1791, at the age of 14. The algorithm converges rapidly, allowing easy numerical calculation of the AGM.

On May 30, 1799, Gauss made a startling discovery. On that day he recorded a note in his diary to the effect that the two numbers

$$\frac{1}{M(\sqrt{2}, 1)} \qquad \text{and} \qquad \frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{1-x^4}}$$

agree to 11 decimal places. He predicted that "the demonstration of this fact will surely open up an entirely new field of analysis." He soon proved that the two numbers are indeed equal, and he generalized the result to other elliptic integrals. In fact, Lagrange had already discovered and published the result in 1785, but Gauss developed other applications of the AGM over a long period of time, especially in the realm of complex function theory. It is no great exaggeration to say that Gauss single-handedly fulfilled the prophecy he made in 1799.

Here is the unexpected connection between the AGM and elliptic integrals of first kind.

Theorem 1. *For* $0 < k < 1$ *and* $k' = \sqrt{1 - k^2}$,

$$\frac{1}{M(1,k)} = \frac{2}{\pi} K(k') .$$

The theorem provides an explicit formula for the AGM of any two positive numbers. More importantly, it offers an efficient method for the numerical calculation of elliptic integrals of first kind. Elliptic integrals of second kind can be computed by a similar process. The extensive tables of elliptic integrals in Legendre's treatise [**7**] were assembled in this manner.

The proof of the theorem will depend on the fact that the elliptic integral and the AGM have essentially the same transformation property. It is convenient to express this relation in terms of the function

$$I(a,b) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{a^2 \cos^2\theta + b^2 \sin^2\theta}} = \frac{1}{a} K'\left(\frac{b}{a}\right) , \qquad 0 < b < a .$$

Lemma 1. *For* $0 < b < a$,

$$I\left(\tfrac{1}{2}(a+b), \sqrt{ab}\right) = I(a,b) .$$

Deferring the proof of the lemma, let us observe first that the theorem is an immediate consequence.

Proof of Theorem 1. Recall the notation $\{a_n\}$ and $\{b_n\}$ for the sequences of arithmetic and geometric means converging to the common limit $M(a,b)$. In this notation, Lemma 1 says that $I(a,b) = I(a_1, b_1)$. Iterating this identity, we infer that $I(a,b) = I(a_n, b_n)$ for $n = 1, 2, \ldots$. Because the function $I(a,b)$ is continuous, it follows that

$$\begin{aligned} I(a,b) = I(a_n, b_n) &\to I(M(a,b), M(a,b)) \\ &= \frac{1}{M(a,b)} \int_0^{\pi/2} \frac{d\theta}{\sqrt{\cos^2\theta + \sin^2\theta}} = \frac{\pi}{2} \frac{1}{M(a,b)} . \end{aligned}$$

Since $I(1,k) = K'(k) = K(k')$ for $0 < k < 1$, this is the desired result. □

The identity in Lemma 1 is a version of what is called the Landen transformation, so named for the English mathematician John Landen (1719–1790), who introduced it in 1771. It is usually stated, in less symmetric form, as follows.

Lemma 2. *For $0 < k < 1$, the complete elliptic integral of first kind satisfies*

$$(i) \qquad K\left(\frac{2\sqrt{k}}{1+k}\right) = (1+k)K(k)$$

$$(ii) \qquad K\left(\frac{1-k}{1+k}\right) = \frac{1+k}{2}\,K(k')\,.$$

The three identities in Lemmas 1 and 2 are all equivalent. To see this, observe first that the conjugate modulus of

$$\ell = \frac{1-k}{1+k} \qquad \text{is} \qquad \ell' = \frac{2\sqrt{k}}{1+k}\,.$$

Inversion gives

$$k = \frac{1-\ell}{1+\ell}\,, \qquad \frac{1+k}{2} = \frac{1}{1+\ell}\,, \qquad k' = \frac{2\sqrt{\ell}}{1+\ell}\,.$$

Therefore, the identity (ii) of Lemma 2 reduces to

$$K(\ell) = \frac{1}{1+\ell}\,K\left(\frac{2\sqrt{\ell}}{1+\ell}\right)\,,$$

which is (i) with k replaced by ℓ. Thus (i) and (ii) are equivalent.

To see that Lemma 1 is another version of the same identity, let $k = b/a$ and note that the relation

$$I(a,b) = \frac{1}{a}K'\left(\frac{b}{a}\right)$$

implies

$$I\left(\tfrac{1}{2}(a+b), \sqrt{ab}\right) = \frac{2}{a+b}K'\left(\frac{2\sqrt{ab}}{a+b}\right) = \frac{1}{a}\,\frac{2}{1+k}\,K'\left(\frac{2\sqrt{k}}{1+k}\right)\,.$$

But by (ii) in Lemma 2, the last expression is equal to

$$\frac{1}{a}\,\frac{2}{1+k}\,K\left(\frac{1-k}{1+k}\right) = \frac{1}{a}\,K(k') = \frac{1}{a}\,K'(k) = I(a,b)\,.$$

This shows that Lemma 2 implies Lemma 1. The reverse implication is similar.

The validity of Landen's transformation (Lemma 2) can be verified in many ways. One of the most elegant proofs was found in 1796 by the English mathematician and astronomer James Ivory (1765–1842). We give a version of Ivory's proof.

Proof of Lemma 2. Recall the definition of the complete elliptic integral $K(k)$ and expand the integrand in binomial series:

$$\left(1 - k^2 \sin^2\theta\right)^{-1/2} = 1 + \sum_{n=1}^{\infty} \binom{-1/2}{n} (-1)^n k^{2n} \sin^{2n}\theta\,.$$

Integrate term by term, using the formula

$$\int_0^{\pi/2} \sin^{2n}\theta\, d\theta = \frac{(2n)!}{2^{2n}(n!)^2}\frac{\pi}{2} = (-1)^n \binom{-1/2}{n}\frac{\pi}{2}\,,$$

(*cf.* Sections 2.5 and 3.5, and Exercise 15 in Chapter 2) to arrive at the expression

$$K(k) = \frac{\pi}{2}\left\{1 + \sum_{n=1}^{\infty}\binom{-1/2}{n}^2 k^{2n}\right\}.$$

Now write

$$\begin{aligned} 2\,K\left(\frac{2\sqrt{k}}{1+k}\right) &= \int_0^{\pi}\left(1 - \frac{4k}{(1+k)^2}\sin^2\theta\right)^{-1/2} d\theta \\ &= \int_0^{\pi}\left(1 - \frac{2k}{(1+k)^2}(1-\cos 2\theta)\right)^{-1/2} d\theta \\ &= (1+k)\int_0^{\pi}\left(1 + k^2 + 2k\cos 2\theta\right)^{-1/2} d\theta\,. \end{aligned}$$

Since $1 + k^2 + 2k\cos 2\theta = \left(1 + ke^{2i\theta}\right)\left(1 + ke^{-2i\theta}\right)$, the last expression takes the form

$$\begin{aligned} &(1+k)\int_0^{\pi}\left(1 + ke^{2i\theta}\right)^{-1/2}\left(1 + ke^{-2i\theta}\right)^{-1/2} d\theta \\ =\,&(1+k)\sum_{n=0}^{\infty}\sum_{m=0}^{\infty}\binom{-1/2}{n}k^n\binom{-1/2}{m}k^m\int_0^{\pi} e^{2i(n-m)\theta}\, d\theta\,, \end{aligned}$$

with the convention that $\binom{\alpha}{0} = 1$. But $\int_0^{\pi} e^{2i(n-m)\theta}d\theta = 0$ if $n \neq m$, and so a calculation leads to the relation

$$2\,K\left(\frac{2\sqrt{k}}{1+k}\right) = \pi(1+k)\left\{1 + \sum_{n=1}^{\infty}\binom{-1/2}{n}^2 k^{2n}\right\} = 2(1+k)K(k)\,,$$

which is the identity (i) of Lemma 2. □

As a consequence of Lemma 2, one can derive corresponding identities for elliptic integrals of second kind. These relations are also known as Landen transformations.

Corollary. *For $0 < k < 1$, the complete elliptic integral of second kind satisfies*

$$(iii) \qquad E\left(\frac{2\sqrt{k}}{1+k}\right) = \frac{2}{1+k}\,E(k) - (1-k)\,K(k)$$

$$(iv) \qquad E\left(\frac{1-k}{1+k}\right) = \frac{1}{1+k}\,E(k') + \frac{k}{1+k}\,K(k')\,.$$

Derivation of corollary. To derive the relation (iii), the strategy is to differentiate the identity (i) of Lemma 2, using the formula

$$\frac{dK}{dk} = \frac{E - k'^2 K}{k k'^2}\,,$$

which comes from the power series for K and E (see Exercise 9). Define the modulus $\ell = 2\sqrt{k}/(1+k)$ and apply the formulas

$$\ell' = \frac{1-k}{1+k}\,, \qquad \frac{d\ell}{dk} = \frac{1-k}{\sqrt{k}(1+k)^2}$$

to differentiate the composite function $K(2\sqrt{k}/(1+k))$. Specifically,

$$\frac{dK}{d\ell}\,\frac{d\ell}{dk} = \frac{d}{dk}\big\{K(\ell)\big\} = \frac{d}{dk}\Big\{(1+k)K(k)\Big\} = K(k) + (1+k)\frac{dK}{dk}\,,$$

or

$$\frac{E(\ell) - \ell'^2 K(\ell)}{\ell \ell'^2}\,\frac{1-k}{\sqrt{k}(1+k)^2} = K(k) + (1+k)\,\frac{E(k) - k'^2 K(k)}{k k'^2}\,,$$

which reduces to

$$\frac{1+k}{2k(1-k)}\,E(\ell) - \frac{1-k}{2k}\,K(k) = \frac{1}{k(1-k)}\,E(k) - \frac{1}{k}\,K(k)\,,$$

since $K(\ell) = (1+k)K(k)$ by Lemma 2. Further simplification leads to the relation (iii). The identity (iv) can be verified either by algebraic manipulations from (iii) or by differentiating (ii). In fact, as with (i) and (ii), the relations (iii) and (iv) are equivalent forms of the same identity. The details are left as an exercise. □

We are now in position to develop a method for computing complete elliptic integrals of second kind via the AGM algorithm. The integral

$$J(a,b) = \int_0^{\pi/2} \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\;d\theta$$

corresponds to the expression $I(a,b)$ introduced earlier. For $0 < b < a$ and $k = b/a$, we find that $J(a,b) = a\,E(k')$ The substitution $\theta = \frac{\pi}{2} - t$ shows that $J(b,a) = J(a,b)$. Note that the arclength of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

is given by $4J(a,b)$ (see Exercise 1). It will be convenient to use the following analogue of Lemma 1, which is closely related to Landen's transformation for E.

Lemma 3. *For $0 < b < a$,*

$$2\,J\left(\tfrac{1}{2}(a+b), \sqrt{ab}\right) = J(a,b) + ab\,I(a,b)\,.$$

Proof. Let $k = b/a$ and recall the relations $I(a,b) = \frac{1}{a}\,K(k')$ and $J(a,b) = a\,E(k')$. Then by Landen's transformation for E, as given by identity (iv) in the corollary to Lemma 2, we have

$$\begin{aligned}
2\,J\left(\tfrac{1}{2}(a+b), \sqrt{ab}\right) &= 2\left(\tfrac{1}{2}(a+b)\right) E\left(\left(\frac{2\sqrt{ab}}{a+b}\right)'\right) \\
&= a(1+k)E\left(\left(\frac{2\sqrt{k}}{1+k}\right)'\right) \\
&= a(1+k)E\left(\frac{1-k}{1+k}\right) = a\big(E(k') + kK(k')\big) \\
&= J(a,b) + ab\,I(a,b)\,. \qquad \square
\end{aligned}$$

In the notation of the AGM algorithm, let $c_n = \sqrt{a_n^2 - b_n^2}$. It can be shown that the infinite series $\sum_{n=1}^{\infty} 2^n c_n^2$ converges (see Exercise 7). Here is a formula for computing elliptic integrals of second kind.

Theorem 2. *For $0 < k < 1$, let $\{a_n\}$ and $\{b_n\}$ be the sequences generated by the AGM algorithm, beginning with $a = a_0 = 1$ and $b = b_0 = k'$. Let $c_n = \sqrt{a_n^2 - b_n^2}$, so that $c_0 = k$. Then*

$$E(k) = \left(1 - \sum_{n=0}^{\infty} 2^{n-1} c_n^2\right) K(k)\,.$$

Proof. By Lemma 3 and Lemma 1, we have

$$2\,J(a_{n+1},b_{n+1}) - J(a_n,b_n) = a_n b_n\, I(a_n,b_n) = a_n b_n\, I(1,k') = a_n b_n\, K(k)$$

for $n = 0, 1, 2, \ldots$. This can be rewritten in the form

$$2^{n+1}\big[J(a_{n+1},b_{n+1}) - a_{n+1}^2 K(k)\big] - 2^n\big[J(a_n,b_n) - a_n^2 K(k)\big] = 2^{n-1} c_n^2 K(k)\,,$$

since

$$4a_{n+1}^2 - 2a_n^2 + c_n^2 = (a_n+b_n)^2 - 2a_n^2 + (a_n^2 - b_n^2) = 2a_n b_n\,.$$

Now take the sum of these identities as n ranges from 0 to N. Because the left-hand side telescopes, this gives

$$\begin{aligned} &2^{N+1}\big[J(a_{N+1},b_{N+1}) - a_{N+1}^2 K(k)\big] - \big[J(a_0,b_0) - a_0^2 K(k)\big] \\ &\qquad = \left\{\sum_{n=0}^{N} 2^{n-1} c_n^2\right\} K(k)\,, \end{aligned}$$

or

$$\begin{aligned} E(k) &= J(1,k') \\ &= \left\{1 - \sum_{n=0}^{N} 2^{n-1} c_n^2\right\} K(k) + 2^{N+1}\big[J(a_{N+1},b_{N+1}) - a_{N+1}^2 K(k)\big]\,. \end{aligned}$$

It remains to show that the last term tends to 0 as $N \to \infty$. Thus it is to be shown that

$$R_n(k) = 2^n\big[a_n^2 K(k) - J(a_n,b_n)\big] \to 0 \qquad \text{as} \quad n \to \infty\,.$$

But $K(k) = I(1,k') = I(a_n,b_n)$ by Lemma 1, and so

$$R_n(k) = 2^n c_n^2 \int_0^{\pi/2} \frac{\sin^2\theta}{\sqrt{a_n^2\cos^2\theta + b_n^2\sin^2\theta}}\,d\theta\,.$$

Therefore, $0 \le R_n(k) \le 2^n c_n^2 I(a_n,b_n) = 2^n c_n^2 K(k) \to 0$ as $n \to \infty$, since the series $\sum 2^n c_n^2$ converges, and the theorem is proved. □

Much more information about the AGM and its applications can be found in the book by Borwein and Borwein [**2**]. The articles by Cox [**3, 4**] focus on the remarkable connections that Gauss found between the AGM and functions of a complex variable. Almqvist and Berndt [**1**] give an entertaining account of historical developments related to elliptic integrals. Watson [**9**] discusses the early history, with emphasis on the contributions of Fagnano and Landen. The book by Stillwell [**8**] presents similar material in broader historical perspective.

Carl Friedrich Gauss (1777–1855) has been called the prince of mathematicians. He made groundbreaking advances in many scientific fields, including number theory, probability and statistics, celestial mechanics, special functions, differential geometry, electricity and magnetism, and potential theory. Born in Braunschweig to parents of humble means and little education, Gauss was a true prodigy whose conspicuous talent gained him financial support for higher study in Braunschweig and at the University of Göttingen. He spent most of his career in Göttingen, where he was director of the *Sternwarte* (astronomical observatory). For a concise account of Gauss's life and work, the biography by Hall [**6**] is recommended.

14.4. The Legendre relation

In his integral calculus text of 1811, Legendre gave some illustrations of his technique for the reducing elliptic integrals to normal form. His calculations led him to the observe that for the special value

$$k = \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

of the modulus, the combination $P = K'E + E'K - K'K$ of complete elliptic integrals of first and second kind has the value $\pi/2$. Remarking that P also has the value $\pi/2$ for $k = 1/\sqrt{2}$ and as $k \to 0$, Legendre proceeded to compute the derivative. A straightforward calculation, using the formulas

$$\frac{dK}{dk} = \frac{E - k'^2 K}{kk'^2} \qquad \text{and} \qquad \frac{dE}{dk} = \frac{E-K}{k}\,,$$

shows that $dP/dk = 0$, so P is constant. In other words,

$$K'E + E'K - K'K = \frac{\pi}{2}\,, \qquad 0 < k < 1\,.$$

This remarkable identity has become known as the *Legendre relation*.

Legendre seems to have discovered his relation by accident, and his proof is not illuminating. It verifies the Legendre relation but does not explain it. In his treatise [**7**] of 1825, Legendre reproduced the same calculations without adding any new insights. Later, when Abel and Jacobi introduced their theory of elliptic functions, and when Weierstrass developed another version of the theory, the Legendre relation emerged in a natural way. There is, however, a more elementary approach which exploits the fact that K and K' are solutions of the same hypergeometric differential equation and so satisfy a Wronskian relation. It turns out that the Wronskian relation is none other than the Legendre relation.

Here are the details. Recall that the Gauss hypergeometric function is defined by

$$F(a,b;c;x) = 1 + \frac{ab}{c}\,x + \frac{1}{2!}\,\frac{a(a+1)b(b+1)}{c(c+1)}\,x^2 + \dots\,,$$

where a, b, c are real parameters with $c \neq 0, -1, -2, \dots$, and $|x| < 1$. The function $y = F(a,b;c;x)$ satisfies the hypergeometric equation

$$x(1-x)y'' + [c - (a+b+1)x]y' - ab\,y = 0\,.$$

Under the special condition $a + b + 1 = 2c$, it is easily verified that if some function $y(x)$ is a solution, then so is $y(1-x)$. (See Chapter 13, Exercise 22.)

The complete elliptic integrals K and E are hypergeometric functions of $x = k^2$. Specifically,

$$K(k) = \frac{\pi}{2}\,F\big(\tfrac{1}{2},\tfrac{1}{2};1;k^2\big)\,, \qquad E(k) = \frac{\pi}{2}\,F\big(-\tfrac{1}{2},\tfrac{1}{2};1;k^2\big)\,.$$

These formulas can be derived by expanding the integrands in binomial series and integrating term by term. As in the proof of Lemma 2, this process leads to the representation

$$K(k) = \frac{\pi}{2}\left\{1 + \sum_{n=1}^{\infty}\binom{-1/2}{n}^2 k^{2n}\right\}.$$

On the other hand, the coefficient of x^n in the power series for $F(\frac{1}{2},\frac{1}{2};1;x)$ is

$$\frac{\big[(\frac{1}{2})(\frac{3}{2})\cdots(n-\frac{1}{2})\big]^2}{(n!)^2} = \left(\frac{(2n)!}{2^{2n}(n!)^2}\right)^2 = \binom{-1/2}{n}^2, \qquad n = 1, 2, \dots\,,$$

which shows that $K(k) = \frac{\pi}{2}\,F(\frac{1}{2},\frac{1}{2};1;k^2)$. The formula for $E(k)$ can be derived in a similar way.

In fact, both formulas are special cases of the Euler integral representation (*cf.* Section 13.5)

$$F(a,b;c;x) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)}\int_0^1 t^{b-1}(1-t)^{c-b-1}(1-xt)^{-a}\,dt\,,$$

valid for $c > b > 0$. Since $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, the Euler representation reduces to

$$\begin{aligned} F\big(\tfrac{1}{2},\tfrac{1}{2};1;k^2\big) &= \frac{1}{\pi}\int_0^1 t^{-1/2}(1-t)^{-1/2}(1-k^2t)^{-1/2}\,dt \\ &= \frac{2}{\pi}\int_0^1 (1-u^2)^{-1/2}(1-k^2u^2)^{-1/2}\,du = \frac{2}{\pi}\,K(k) \end{aligned}$$

and

$$F\big(-\tfrac{1}{2},\tfrac{1}{2};1;k^2\big) = \frac{1}{\pi}\int_0^1 t^{-1/2}(1-t)^{-1/2}(1-k^2t)^{1/2}\,dt$$
$$= \frac{2}{\pi}\int_0^1 (1-u^2)^{-1/2}(1-k^2u^2)^{1/2}\,du = \frac{2}{\pi}\,E(k)\,,$$

with the substitution $t = u^2$.

Recall now that any pair of solutions y_1 and y_2 of a linear differential equation $y'' + py' + qy = 0$ have a Wronskian $W = y_1y_2' - y_1'y_2$ with the property

$$W(x) = W(x_0)\,\exp\left\{-\int_{x_0}^{x} p(t)\,dt\right\},$$

where x_0 is any convenient base-point. The Wronskian relation was derived in the preceding chapter (Section 13.2). For the hypergeometric equation with $a = b = \frac{1}{2}$ and $c = 1$, it takes the form

$$W(x) = W(\tfrac{1}{2})\,\exp\left\{-\int_{1/2}^{x}\frac{1-2t}{t(1-t)}\,dt\right\},$$

and an easy calculation of the integral gives

$$W(x) = \tfrac{1}{4}\,W(\tfrac{1}{2})\,\frac{1}{x(1-x)}\,, \qquad 0 < x < 1\,.$$

Now choose the particular solutions

$$y_1(x) = K(k) = \frac{\pi}{2}\,F(\tfrac{1}{2},\tfrac{1}{2};1;x)\,,$$
$$y_2(x) = K'(k) = \frac{\pi}{2}\,F(\tfrac{1}{2},\tfrac{1}{2};1;1-x)\,, \qquad x = k^2\,,$$

noting that $K'(k) = K(k') = y_1(1-x)$ is also a solution of the hypergeometric equation because the condition $a+b+1 = 2c$ is satisfied in this case. The differentiation formula for $K(k)$ yields

$$\frac{dy_1}{dx} = \frac{E-(1-x)K}{2x(1-x)}\,, \qquad \frac{dy_2}{dx} = -\frac{E'-xK'}{2x(1-x)}\,,$$

and so the Wronskian W of K and K' is given by

$$-2x(1-x)W(x) = (E-(1-x)K)K' + K(E'-xK')$$
$$= K'E + E'K - K'K\,.$$

But the Wronskian relation says that $x(1-x)W(x)$ is constant, so it follows that Legendre's expression $K'E + E'K - K'K$ is constant.

To show that the constant is $\pi/2$, it is simplest to compute the limit of Legendre's expression as $k \to 0$. An alternative is to calculate $K(1/\sqrt{2}) = K'(1/\sqrt{2})$ and $E(1/\sqrt{2}) = E'(1/\sqrt{2})$ in terms of the gamma function and then show that

$$2K(1/\sqrt{2})E(1/\sqrt{2}) - K(1/\sqrt{2})^2 = \frac{\pi}{2}\,.$$

The details are left as an exercise.

Because the elliptic integrals $K(1/\sqrt{2})$ and $E(1/\sqrt{2})$ can be computed by the rapidly converging AGM algorithm, the Legendre relation offers an effective method for numerical calculation of π. For further information, see the book by Borwein and Borwein [**2**]. For an exposition of the Legendre relation as it occurs in the theory of elliptic functions, see also the paper by Duren [**5**].

Exercises

1. Show that for $0 < b < a$ the arclength of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \text{is} \qquad 4a\,E'(b/a)\,,$$

where $E'(k) = E(k')$ is a complete elliptic integral of second kind. Use the parametric representation $x = a\cos t$, $y = b\sin t$.

2. (a) Evaluate the lemniscate integral

$$\int_0^1 \frac{dt}{\sqrt{1-t^4}} \qquad \text{to show that} \qquad K(1/\sqrt{2}) = \frac{1}{4\sqrt{\pi}}\,\Gamma(\tfrac{1}{4})^2\,.$$

(b) Confirm Gauss's conjecture (made on the basis of numerical evidence) that

$$\frac{1}{M(\sqrt{2},1)} = \frac{2}{\pi}\int_0^1 \frac{dx}{\sqrt{1-x^4}}\,.$$

3. Use the substitution $u = \sqrt{1-t^2}$ to show that

$$\sqrt{2}\,E(1/\sqrt{2}) = \int_0^1 \frac{1+u^2}{\sqrt{1-u^4}}\,du\,.$$

Deduce that

$$E(1/\sqrt{2}) = \frac{1}{8\sqrt{\pi}}\left[\Gamma(\tfrac{1}{4})^2 + 4\,\Gamma(\tfrac{3}{4})^2\right].$$

4. Show that values of $K(1/\sqrt{2})$ and $E(1/\sqrt{2})$ found in the preceding exercises are consistent with the Legendre relation.

Hint. Use the Euler reflection formula for the gamma function.

5. Use a hand calculator to compute the arithmetic–geometric mean $M(2,1)$ to 6 decimal places.

6. In the notation of the AGM algorithm, prove that $a_1 - b_1 < \frac{1}{2}(a-b)$ and conclude that $a_n - b_n < (\frac{1}{2})^n(a-b)$. This shows that the algorithm converges rapidly.

7. In the notation of the AGM algorithm, let $c_n = \sqrt{a_n^2 - b_n^2}$. Show that

$$c_1 = \tfrac{1}{2}(a-b) \le \tfrac{1}{2}\sqrt{a^2-b^2}$$

and in general that $c_{n+1} \le \frac{1}{2}c_n$. Conclude that $c_n^2 \le (\frac{1}{4})^n(a^2-b^2)$, so that the series $\sum_{n=1}^{\infty} 2^n c_n^2$ is convergent.

8. Directly from the definitions of $K(k)$ and $E(k)$, show that

$$\frac{dE}{dk} = \frac{E-K}{k}\,.$$

Justify the process of differentiation under the integral sign.

9. (a) Derive the power series expansion

$$\begin{aligned} E(k) &= \frac{\pi}{2}\left\{1 - \sum_{n=1}^{\infty}(2n-1)\binom{1/2}{n}^2 k^{2n}\right\} \\ &= \frac{\pi}{2}\left\{1 - \sum_{n=1}^{\infty}\frac{1}{2n-1}\left[\frac{(2n)!}{2^{2n}(n!)^2}\right]^2 k^{2n}\right\}\,. \end{aligned}$$

(b) Use the power series expansions of $K(k)$ and $E(k)$ to verify the formula

$$\frac{dK}{dk} = \frac{E - k'^2 K}{k k'^2}\,.$$

Discussion. In contrast with dE/dk in Exercise 8, the expression for dK/dk is not readily calculated by differentiation under the integral sign. The power series calculation is somewhat laborious, but a remarkable combination of terms gives the result.

10. Verify the Landen transformation for complete elliptic integrals of second kind as given by the formula (iv) in the corollary to Lemma 2.

11. As early as 1738, Euler discovered the formula

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} \int_0^1 \frac{x^2\,dx}{\sqrt{1-x^4}} = \frac{\pi}{4}\,.$$

(a) Show that this is a special case of the Legendre relation, with $k = k' = 1/\sqrt{2}$.

(*b*) Without appeal to elliptic integrals, verify Euler's formula by calculating each integral in terms of the gamma function and applying Euler's reflection formula. (See Chapter 9, Exercise 10.)

12. Use the AGM methods of Theorems 1 and 2 to calculate $K(1/\sqrt{2})$ and $E(1/\sqrt{2})$ to at least 6 decimal places. Then use the Legendre relation to calculate π.

13. For $0 < k < 1$ and $k' = \sqrt{1-k^2}$, show that

$$\int_1^{1/k} \frac{ds}{\sqrt{s^2-1}\sqrt{1-k^2s^2}} = \int_0^1 \frac{dt}{\sqrt{1-t^2}\sqrt{1-k'^2t^2}} = K'(k)\,.$$

Suggestion. Let $s = (1-k'^2t^2)^{-1/2}$.

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